

EPICS QT Framework

4.1.6

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Chapter 1

QE framework - EPICS aware Qt Widgets and data access classes

- The EPICS Qt Framework ([QE](#)) is a layered software framework for accessing EPICS data using Channel Access and/or PV Access on a range of platforms.
- The EPICS Qt Framework provides object oriented C++ access to control systems using EPICS (Experimental Physics and Industrial Control System). It is based on Qt, a widely used cross-platform application development framework.
- GUI or console based applications can be written that use [QE](#) at several levels. [QE](#) includes Qt plugin libraries, EPICS aware widgets, data formatting classes, and classes for accessing raw EPICS data in a Qt friendly way.
- The EPICS Qt Framework also includes an application - QEgui - for displaying forms produced by the Qt development tool 'Designer'. Using this application a complete EPICS GUI system can be generated without writing any code. A GUI system produced in this way can interact with existing EPICS display tools such as EDM.
- The EPICS Qt Framework handles much of the complexities of Channel Access and PV Access including initiating and managing a channel. Applications using [QE](#) can interact with Channel Access and PV Access using Qt based classes and data types. Channel Access and PV Access updates are delivered using Qt's signals and slots mechanism.

1.1 Documentation

Support documents can be found in the [documentation](#) section of the epicsqt on GitHub. The framework download (available from [homepage](#)) also includes this documentation as well as full Doxygen generated documentation of all the epicsqt classes and widgets.

1.2 License

The EPICS Qt Framework is distributed under the terms of the [GNU Lesser General Public License](#) 3.0 only.

1.3 Platforms

EPICS Qt Framework might be usable in all environments where you find [Qt](#). It is compatible with Qt $\geq$ 5.12.8. At the Australian Synchrotron, we use Qt 6.8.1.

1.4 Screenshots

- ASguiscreenshots
- otherappscreenshots
- designerscreenshots
- qtcreatorscreenshots

Screenshots are only available in the HTML docs.

1.5 Downloads

Stable releases and development snapshots are available at the [epicsqt project page](#).

1.6 Installation

Read [QE_GettingStarted.pdf](#) in the documentation for setting up an environment for building or using the [epicsqt](#) framework.

To build the framework, open `epicsqt.pro` in QtCreator, ensure shadow build is turned off, and hit build.

The resultant library `libQEPlugin.so` will need to be installed or referenced up according to how it is to be used - see [QE_GettingStarted.pdf](#) for details.

Any Qt specific queries? start at [the Qt Project](#)

1.7 Related Projects

[Qwt](#), The core of a Channel Access aware plotting widget.

1.8 Credits:

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Project admin:

Andrew Starritt < andrews@ansto.gov.au >

Chapter 2

GNU Lesser General Public License

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Chapter 3

ASgui screen shots

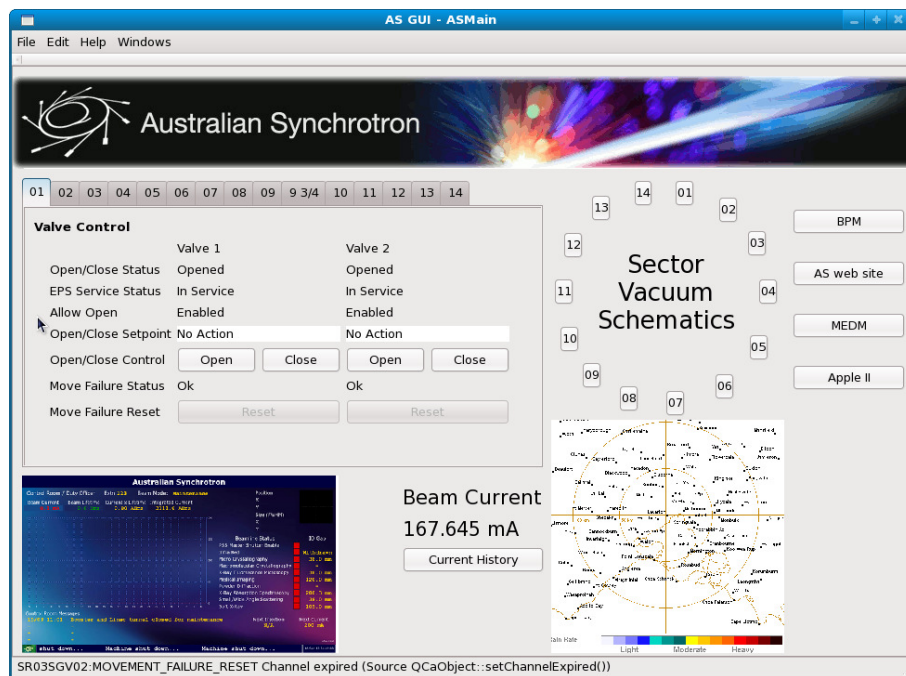


Figure 3.1 Australian Synchrotron mock up

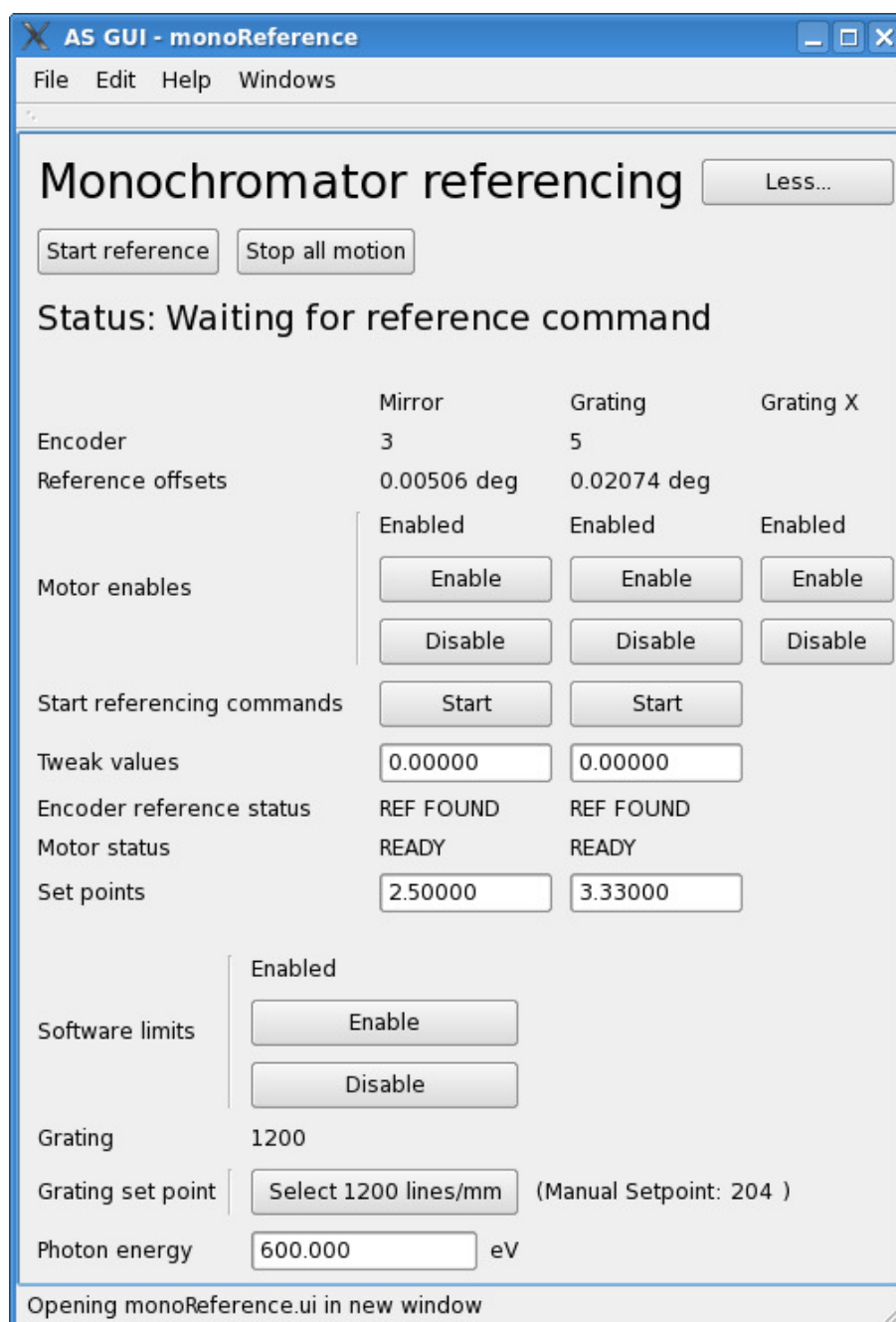


Figure 3.2 Monochromator referencing

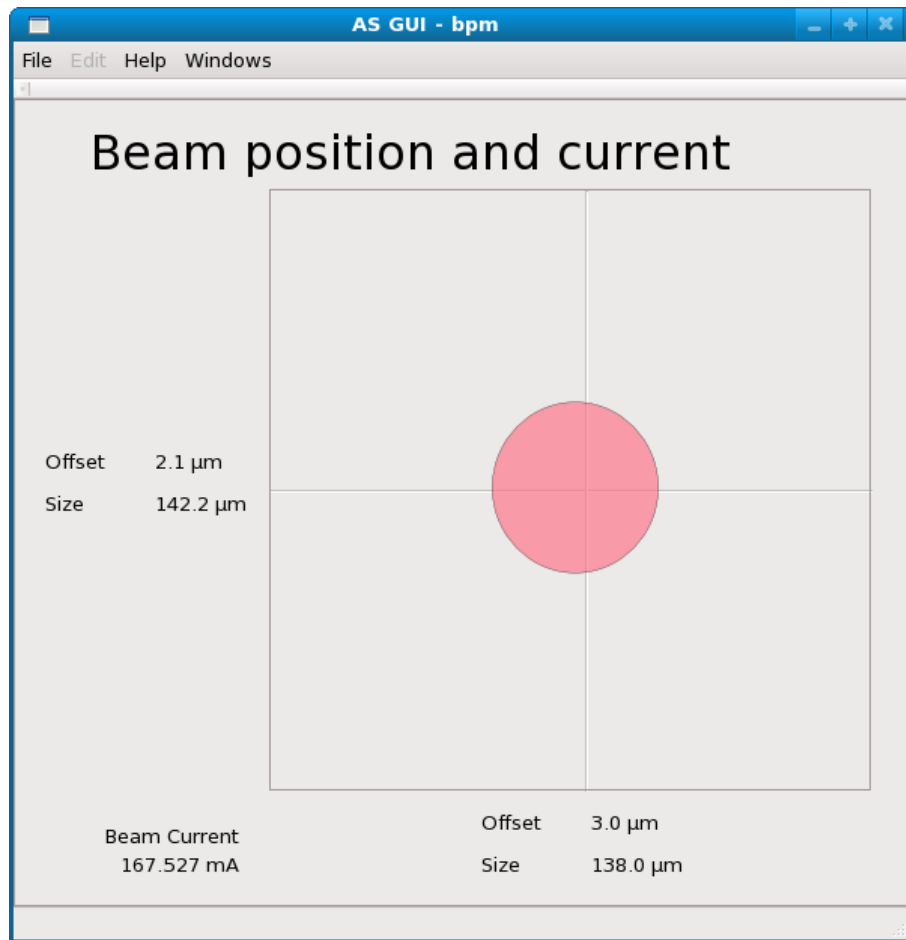


Figure 3.3 Beam position monitor

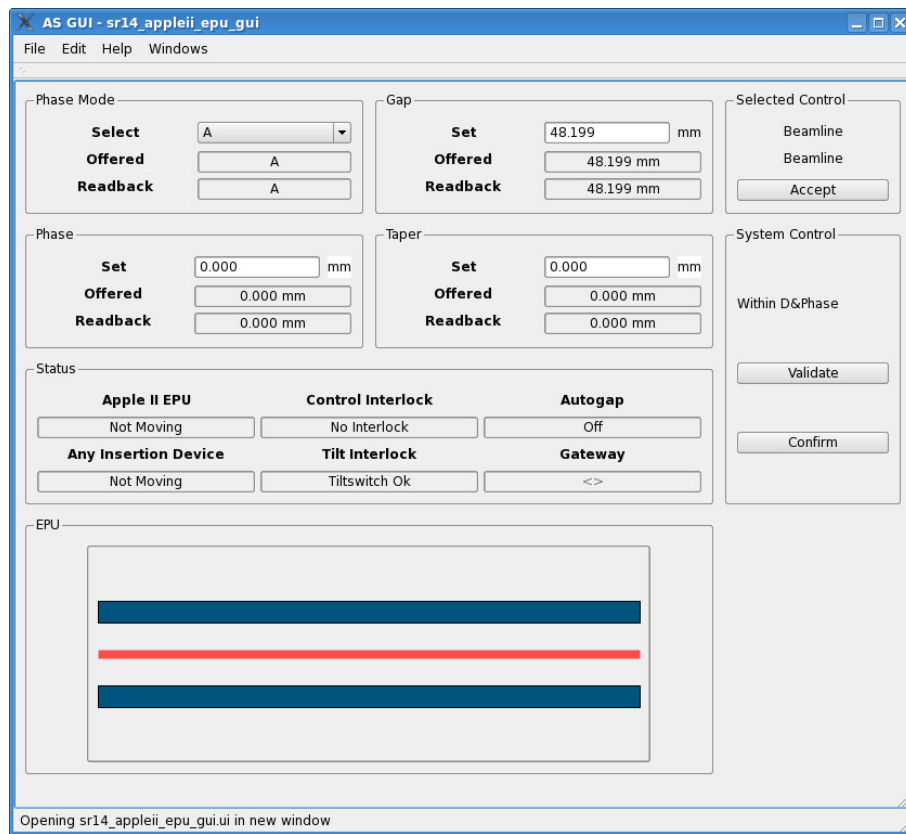


Figure 3.4 Insertion device



Figure 3.5 Injection efficiency monitor

Chapter 4

other applications using epicsqt widgets

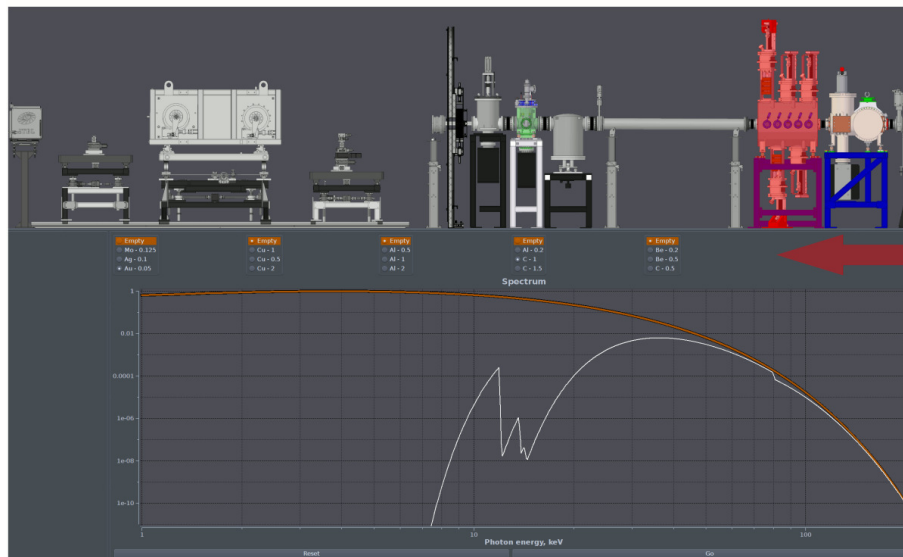


Figure 4.1 Medical Imaging beamline

2B SampleTable Z <@SR08ID01OPI01>

PV name: SR08ID01:MTR32B View mode: Macro

Description: 2B SampleTable Z

Precision: 5 Units: mm

Message: Connection established. clean

User: 6mm Move absolutely Raw: -80932

JOG< UNDO >JOG

LIMIT< Move relatively >LIMIT

< 1mm >

step/10 step/2 step*2 step*10

User: 6mm = Hi limit

Resolution: 0.0001mm/step * 42271.2068mm

Raw: -80932 + Lo Limit

Offset: -2.0932mm -42275.3932mm

Speed Acceleration

Maximum: 1.2mm/s

Normal: 0.8mm/s 1s

Backlash: 0mm/s 1s

Jog: 1mm/s 10s

Backlash: 0mm

Figure 4.2 Motor controller

MotorMx <@SR08ID01OPI01>

- ▲ ▼	DEI Theta Mono	109.5mm	<	0.1	>	UNDO
- ▲ ▼	DEI Mono Z	-0.3mm	<	0.1	>	UNDO
- ▲ ▼	2B Sample Table Y	0mm	<	1	>	UNDO
- ▲ ▼	2B SampleTable Z	6mm	<	1	>	UNDO
- ▲ ▼	2B Detector Table Z	42mm	<	5	>	UNDO
- ▲ ▼	2B Sample Rotate	-1deg	<	1	>	UNDO
- ▲ ▼	2B Detector Table Y	13.9025mm	<	1	>	UNDO
- ▲ ▼	SETUP	0	<	relative	>	STOP
- ▲ ▼	SLW01:LEFT	9.99975mm	<	3456	>	UNDO

Add motor

Figure 4.3 Motor controller

Chapter 5

Qt Designer

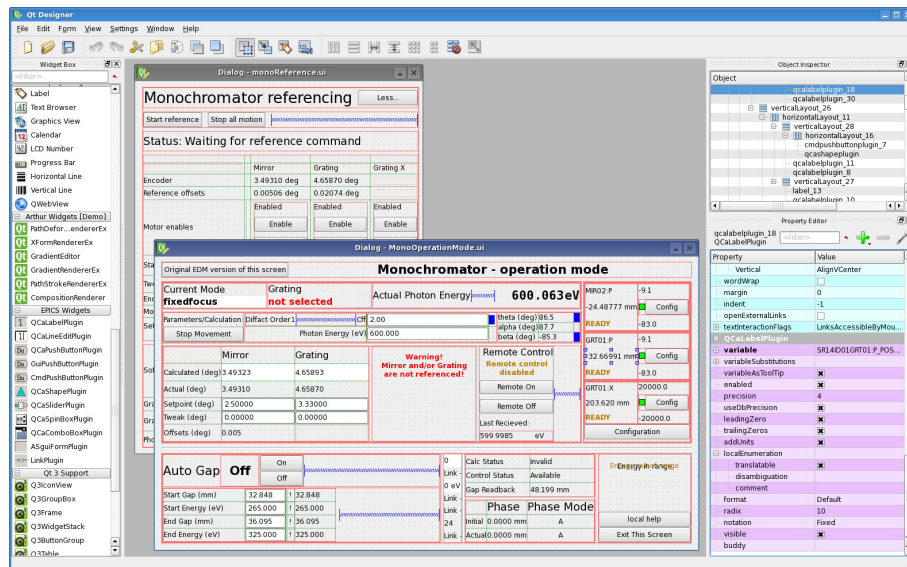


Figure 5.1 Editing multiple GUIs

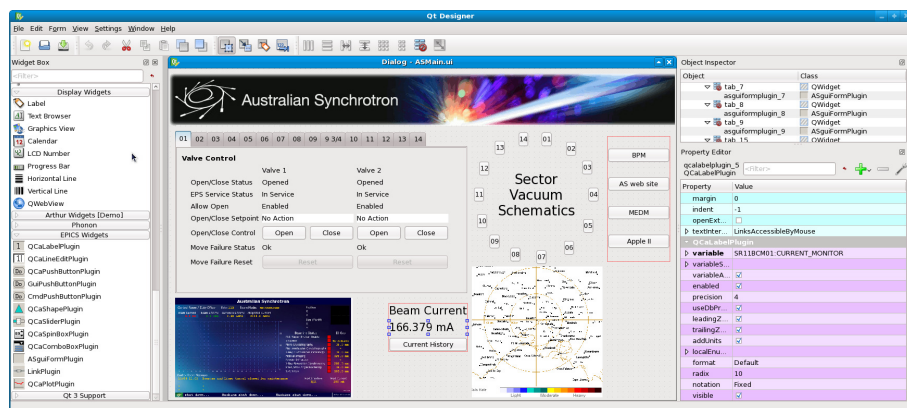


Figure 5.2 Editing a GUI

Chapter 6

Qt Creator

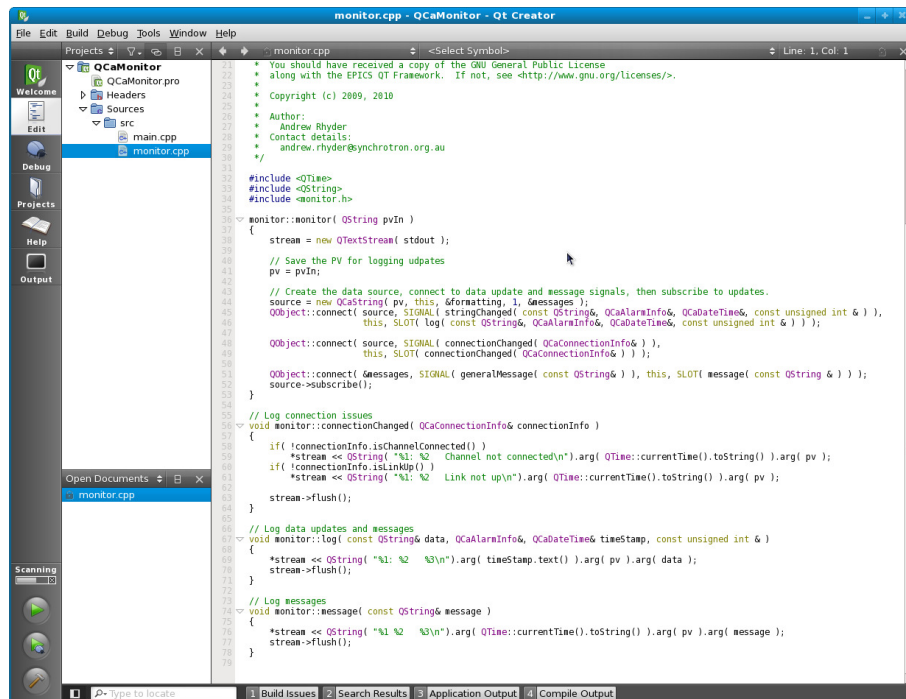


Figure 6.1 Application using epicsqt data source classes

Chapter 7

Hierarchical Index

7.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

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Chapter 8

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Chapter 9

Class Documentation

9.1 `_CopyPaste` Class Reference

Public Member Functions

- `_CopyPaste` (bool pEnable, QString pProgram, QString pParameters, QString pWorkingDirectory, int pTimeOut, bool pStop, bool pLog)
- void `setEnabled` (bool pEnable)
- bool `getEnable` ()
- void `setProgram` (QString pProgram)
- QString `getProgram` ()
- void `setParameters` (QString pParameters)
- QString `getParameters` ()
- void `setWorkingDirectory` (QString pWorkingDirectory)
- QString `getWorkingDirectory` ()
- void `setTimeOut` (int pTimeOut)
- int `getTimeOut` ()
- void `setStop` (bool pStop)
- bool `getStop` ()
- void `setLog` (bool pLog)
- bool `getLog` ()

The documentation for this class was generated from the following files:

- qeframework/widgets/QEScript/QEScript.h
- qeframework/widgets/QEScript/QEScript-wip.cpp
- qeframework/widgets/QEScript/QEScript.cpp

9.2 `_Field` Class Reference

Public Member Functions

- [QEObject](#) * **getWidget** ()
- void **setWidget** (QString *pValue)
- QString **getName** ()
- void **setName** (QString pValue)
- QString **getProcessVariable** ()
- void **setProcessVariable** (QString pValue)
- void **setJoin** (bool pValue)
- bool **getJoin** ()
- int **getType** ()
- void **setType** (int pValue)
- QString **getGroup** ()
- void **setGroup** (QString pValue)
- QString **getVisible** ()
- void **setVisible** (QString pValue)
- QString **getEditable** ()
- void **setEditable** (QString pValue)
- bool **getVisibility** ()
- void **setVisibility** (bool pValue)

Public Attributes

- [QEObject](#) * **qeWidget**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEConfiguredLayout/QEConfiguredLayout.h
- qeframework/widgets/QEConfiguredLayout/QEConfiguredLayout.cpp

9.3 `_Item` Class Reference

Public Member Functions

- void **setName** (QString pValue)
- QString **getName** ()
- void **setSubstitution** (QString pValue)
- QString **getSubstitution** ()
- void **setVisible** (QString pValue)
- QString **getVisible** ()

Public Attributes

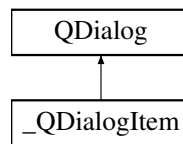
- QList< [_Field](#) * > **fieldList**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEConfiguredLayout/QEConfiguredLayout.h
- qeframework/widgets/QEConfiguredLayout/QEConfiguredLayout.cpp

9.4 _QDialogItem Class Reference

Inheritance diagram for _QDialogItem:



Public Member Functions

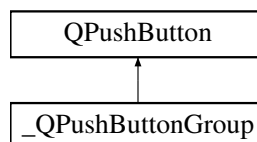
- **_QDialogItem** (QWidget *pParent=0, QString pltemName="", QString pGroupName="", QList< [_Field](#) * > *pCurrentFieldList=0, Qt::WindowFlags pF=Qt::Widget)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEConfiguredLayout/QEConfiguredLayout.h
- qeframework/widgets/QEConfiguredLayout/QEConfiguredLayout.cpp

9.5 _QPushButtonGroup Class Reference

Inheritance diagram for _QPushButtonGroup:



Public Slots

- void **buttonGroupClicked** ()

Public Member Functions

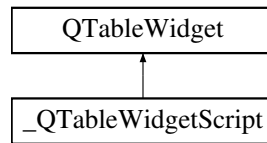
- **_QPushButtonGroup** (QWidget *pParent=0, QString pltemName="", QString pGroupName="", QList< [_Field](#) * > *pCurrentFieldList=0)
- void **mouseReleaseEvent** (QMouseEvent *qMouseEvent)
- void **keyPressEvent** (QKeyEvent *pKeyEvent)
- void **showDialogGroup** ()

The documentation for this class was generated from the following files:

- qeframework/widgets/QEConfiguredLayout/QEConfiguredLayout.h
- qeframework/widgets/QEConfiguredLayout/QEConfiguredLayout.cpp

9.6 `_QTableWidgetScript` Class Reference

Inheritance diagram for `_QTableWidgetScript`:



Public Member Functions

- `_QTableWidgetScript` (`QWidget *pParent=0`)
- void `refreshSize` ()
- void `resizeEvent` (`QResizeEvent *`)
- void `resize` (int w, int h)

The documentation for this class was generated from the following files:

- `qeframework/widgets/QEScript/QEScript.h`
- `qeframework/widgets/QEScript/QEScript-wip.cpp`
- `qeframework/widgets/QEScript/QEScript.cpp`

9.7 `QEFormStateChange::Actions` Class Reference

Public Member Functions

- **Actions** (const `VariableIndicies` index, const `QEFormStateChange` *owner)
- void `doAction` ()

Public Attributes

- const `VariableIndicies` **index**
- const `QEFormStateChange` * **owner**
- `QStringFormatting` **formatting**
- `QCaVariableNamePropertyManager` **vnpm**
- `applicationLauncher` **launcher**
- `QString` **text**

The documentation for this class was generated from the following file:

- `qeframework/widgets/QEFormStateChange/QEFormStateChange.cpp`

9.8 applicationLauncher Class Reference

Public Member Functions

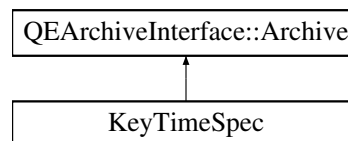
- void **launchImage** (const [VariableNameManager](#) *variableNameManager, QImage image)
- void **launch** (const [VariableNameManager](#) *variableNameManager, QObject *receiver)
- void **launchCommon** (const [VariableNameManager](#) *variableNameManager, QTemporaryFile *tempFile=NULL, QObject *receiver=NULL)
- void **setProgram** (const QString programIn)
- QString **getProgram** () const
- void **setArguments** (const QStringList argumentsIn)
- QStringList **getArguments** () const
- void **setProgramStartupOption** (const [QE::ProgramStartupOptions](#) programStartupOptionIn)
- [QE::ProgramStartupOptions](#) **getProgramStartupOption** () const

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/applicationLauncher.h
- qeframework/widgets/QEWidget/applicationLauncher.cpp

9.9 QEArchiveInterface::Archive Struct Reference

Inheritance diagram for QEArchiveInterface::Archive:



Public Attributes

- int32_t **key**
- int16_t **nameIndex**
- int16_t **pathIndex**

The documentation for this struct was generated from the following file:

- qeframework/archive/QEArchiveInterface.h

9.10 QEArchiveAccess::ArchiverPvInfo Struct Reference

Public Attributes

- QString **hostName**
- int **key**
- QString **path**
- [QCaDateTime](#) **startTime**
- [QCaDateTime](#) **endTime**

The documentation for this struct was generated from the following file:

- qeframework/archive/QEArchiveAccess.h

9.11 arealInfo Class Reference

Public Member Functions

- void **setX1** (long x)
- void **setY1** (long y)
- void **setX2** (long x)
- void **setY2** (long y)
- void **setR** (double r)
- void **setX** (long x)
- void **setY** (long y)
- void **setW** (long w)
- void **setH** (long h)
- void **setPoint1** (QPoint p1In)
- void **setPoint2** (QPoint p2In)
- void **clearX1** ()
- void **clearY1** ()
- void **clearX2** ()
- void **clearY2** ()
- void **clearR** ()
- void **clearX** ()
- void **clearY** ()
- void **clearW** ()
- void **clearH** ()
- bool **getStatus** ()
- QRect **getArea** ()
- QPoint **getPoint1** ()
- QPoint **getPoint2** ()
- bool **hasRotation** ()
- double **getRotation** ()

The documentation for this class was generated from the following file:

- qeframework/widgets/QEImage/QEImage.h

9.12 QEAnalogIndicator::Band Struct Reference

Public Attributes

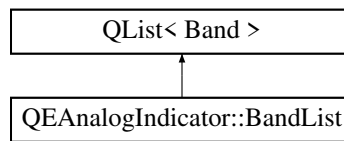
- double **lower**
- double **upper**
- QColor **colour**

The documentation for this struct was generated from the following file:

- qeframework/widgets/QEAnalogIndicator/QEAnalogIndicator.h

9.13 QEAnalogIndicator::BandList Class Reference

Inheritance diagram for QEAnalogIndicator::BandList:

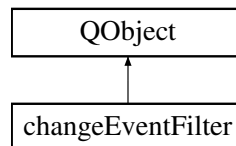


The documentation for this class was generated from the following file:

- qeframework/widgets/QEAnalogIndicator/QEAnalogIndicator.h

9.14 changeEventFilter Class Reference

Inheritance diagram for changeEventFilter:



Public Member Functions

- **changeEventFilter** ([styleManager](#) *managerIn)

Protected Member Functions

- **bool eventFilter** (QObject *obj, QEvent *event)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/styleManager.h
- qeframework/widgets/QEWidget/styleManager.cpp

9.15 componentHostListItem Class Reference

Public Member Functions

- **componentHostListItem** (QWidget *widgetIn, [QE::CreationOptions](#) creationOptionIn, bool hiddenIn, QString titleIn)
- **componentHostListItem** ([componentHostListItem](#) *item)

Public Attributes

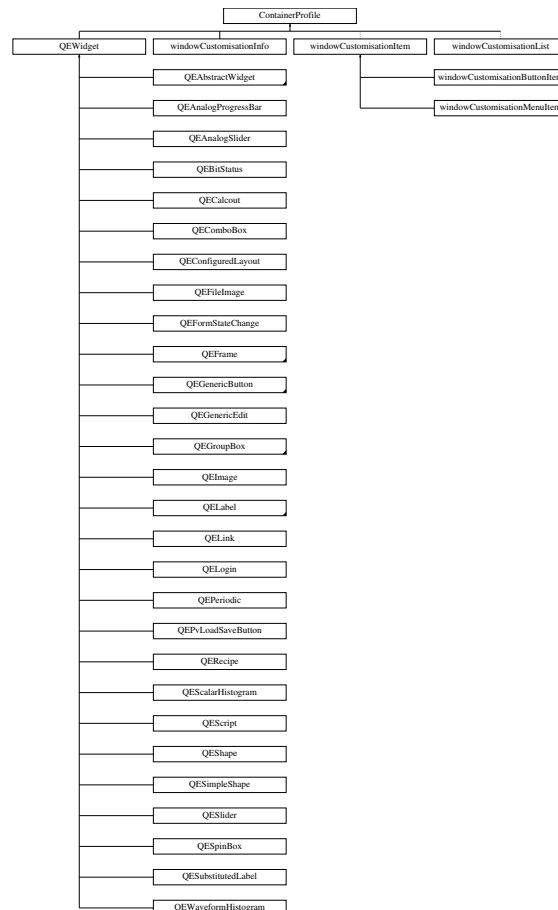
- QWidget * **widget**
- [QE::CreationOptions](#) **creationOption**
- bool **hidden**
- QString **title**

The documentation for this class was generated from the following files:

- qeframework/common/QEActionRequests.h
- qeframework/common/QEActionRequests.cpp

9.16 ContainerProfile Class Reference

Inheritance diagram for ContainerProfile:



Public Member Functions

- void **takeLocalCopy** ()
- void **setupProfile** (QObject *guiLaunchConsumer, const QStringList &pathList, const QString &parentPath, const QString ¯oSubstitutions)

- void **setupLocalProfile** (QObject *guiLaunchConsumer, const QStringList &pathList, const QString &parentPath, const QString ¯oSubstitutions)
- void **publishOwnProfile** ()
- void **updateConsumers** (QObject *guiLaunchConsumer)
- QObject * **replaceGuiLaunchConsumer** (QObject *newGuiLaunchConsumer)
- QObject * **getGuiLaunchConsumer** () const
- QString **getPath** () const
- QStringList **getPathList** () const
- QString **getParentPath** () const
- QString **getMacroSubstitutions** () const
- unsigned int **getMessageFormId** () const

Static Public Member Functions

- static void **addMacroSubstitutions** (QString macroSubstitutions)
- static void **removeMacroSubstitutions** ()
- static void **addPriorityMacroSubstitutions** (QString macroSubstitutions)
- static void **removePriorityMacroSubstitutions** ()
- static void **setPublishedParentPath** (QString publishedParentPath)
- static bool **isProfileDefined** ()
- static bool **areUserLevelPasswordsSet** ()
- static QString **getUserLevelPassword** (QE::UserLevels level)
- static void **setUserLevelPassword** (QE::UserLevels level, QString password)
- static void **addContainedWidget** (QEWWidget *containedWidget)
- static QEWWidget * **getNextContainedWidget** ()
- static void **removeContainedWidget** (QEWWidget *containedWidget)
- static unsigned int **getPublishedMessageFormId** ()
- static void **setPublishedMessageFormId** (unsigned int publishedMessageFormId)
- static QStringList **getEnvPathList** ()
- static bool **setDontActivateYet** (bool dontActivate)
- static bool **getDontActivateYet** ()
- static void **releaseProfile** ()
- static void **setUserLevel** (QE::UserLevels level)
- static QE::UserLevels **getUserLevel** ()
- static PersistenceManager * **getPersistenceManager** ()
- static QChar **platformSeperator** ()
- static QString **getUserLevelName** (QE::UserLevels userLevelValue)
- static QE::UserLevels **getUserLevelValue** (QString userLevelName)

Protected Member Functions

- virtual void **userLevelChangedGeneral** (QE::UserLevels)

Friends

- class **QEProfileUserLevelSlot**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWWidget/ContainerProfile.h
- qeframework/widgets/QEWWidget/ContainerProfile.cpp

9.17 QEArchiveInterface::Context Struct Reference

Public Attributes

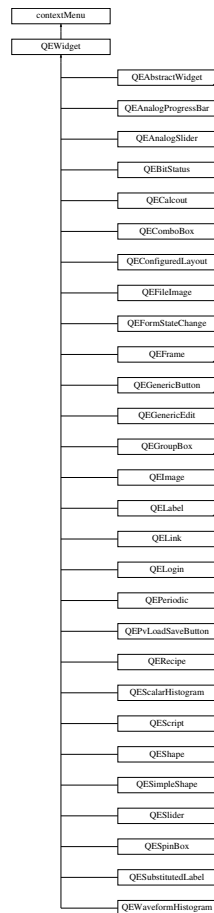
- QEArchiveInterface::Methods **method**
- QObject * **userData**
- unsigned int **requested_element**

The documentation for this struct was generated from the following file:

- qeframework/archive/QEArchiveInterface.h

9.18 contextMenu Class Reference

Inheritance diagram for contextMenu:



Public Types

- enum **contextMenuOptions** {
CM_NOOPTION , **CM_COPY_VARIABLE** , **CM_COPY_DATA** , **CM_PASTE** ,
CM_DRAG_VARIABLE , **CM_DRAG_DATA** , **CM_SHOW_PV_PROPERTIES** , **CM_ADD_TO_↵**
STRIPCHART ,
CM_ADD_TO_SCRATCH_PAD , **CM_ADD_TO_PLOTTER** , **CM_ADD_TO_TABLE** , **CM_SHOW_AS_↵**
HISTOGRAM ,
CM_SHOW_AS_NT_TABLE , **CM_SHOW_AS_NTND_ARRAY** , **CM_SHOW_AS_OPAQUE** , **CM_↵**
GENERAL_PV_EDIT ,
CM_SPECIFIC_WIDGETS_START_HERE }
 • typedef QSet< contextMenuOptions > **ContextMenuOptionSets**

Public Member Functions

- **contextMenu** ([QWidget](#) *qewIn, [QWidget](#) *ownerIn)
- void **setupContextMenu** (const ContextMenuOptionSets &menuSet=contextMenu::defaultMenuSet())
- void **setEditPvUserLevel** (const [QE::UserLevels](#) level)
- bool **getEditPvUserLevel** () const
- void **clearContextMenuRequestHandling** ()
- void **setContextMenuOptions** (const ContextMenuOptionSets &menuSet)
- void **setNumberOfContextMenuItems** (const int numberOfItems)
- bool **isDraggingVariable** ()
- virtual [QMenu](#) * **buildContextMenu** ()
- virtual void **contextMenuTriggered** (int selectedItemNum)
- virtual [QString](#) **copyVariable** ()
- virtual [QVariant](#) **copyData** ()
- virtual void **paste** ([QVariant](#))
- [QAction](#) * **showContextMenuGlobal** (const [QPoint](#) &globalPos)
- [QAction](#) * **showContextMenu** (const [QPoint](#) &pos)
- [QAction](#) * **showContextMenuGlobal** ([QMenu](#) *menu, const [QPoint](#) &globalPos)
- [QAction](#) * **showContextMenu** ([QMenu](#) *menu, const [QPoint](#) &pos)
- void **addMenuItem** ([QMenu](#) *menu, const [QString](#) &title, const bool checkable, const bool checked, const int option)

Static Public Member Functions

- static ContextMenuOptionSets **defaultMenuSet** ()
- static bool **insertBefore** ([QMenu](#) *menu, [QAction](#) *action, const int option)
- static bool **insertAfter** ([QMenu](#) *menu, [QAction](#) *action, const int option)
- static bool **insertSeparatorBefore** ([QMenu](#) *menu, const int option)

Protected Member Functions

- void **setConsumer** ([QObject](#) *consumer)

Friends

- class **QEContextMenuObject**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/contextMenu.h
- qeframework/widgets/QEWidget/contextMenu.cpp

9.19 customisationLog Class Reference

Public Member Functions

- void **add** (const QString message)
- void **add** (const QString message, const QString param1)
- void **add** (const QString message, const QString param1, const QString param2)
- void **startLevel** ()
- void **endLevel** ()
- void **flagError** ()
- const QString **getLog** () const
- bool **getError** () const

The documentation for this class was generated from the following files:

- qeframework/common/windowCustomisation.h
- qeframework/common/windowCustomisation.cpp

9.20 QEPeiodic::elementInfoStruct Struct Reference

Public Attributes

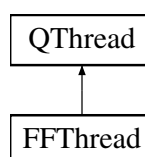
- unsigned int **number**
- double **atomicWeight**
- QString **name**
- QString **symbol**
- double **meltingPoint**
- double **boilingPoint**
- double **density**
- unsigned int **group**
- double **ionizationEnergy**
- unsigned int **tableRow**
- unsigned int **tableCol**
- Category **category**

The documentation for this struct was generated from the following file:

- qeframework/widgets/QEPeiodic/QEPeiodic.h

9.21 FFThread Class Reference

Inheritance diagram for FFThread:



Public Slots

- void **stopGracefully** ()

Signals

- void **updateSignal** (FFBuffer *)

Protected Member Functions

- void **run** ()

Friends

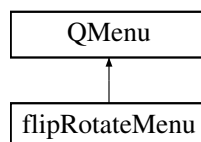
- class **MpegSource**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/mpeg.h
- qeframework/widgets/QEImage/mpeg.cpp

9.22 flipRotateMenu Class Reference

Inheritance diagram for flipRotateMenu:



Public Member Functions

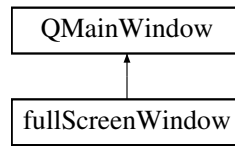
- **flipRotateMenu** (QWidget *parent=0)
- imageContextMenu::imageContextMenuOptions **getFlipRotate** (const QPoint &pos)
- void **setChecked** (const int rotation, const bool flipH, const bool flipV)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/flipRotateMenu.h
- qeframework/widgets/QEImage/flipRotateMenu.cpp

9.23 fullScreenWindow Class Reference

Inheritance diagram for fullScreenWindow:



Signals

- void **fullScreenResize** ()

Public Member Functions

- **fullScreenWindow** (QWidget *parent=0)

Protected Member Functions

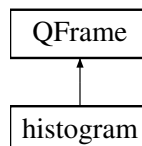
- void **resizeEvent** (QResizeEvent *event)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/fullScreenWindow.h
- qeframework/widgets/QEImage/fullScreenWindow.cpp

9.24 histogram Class Reference

Inheritance diagram for histogram:



Public Member Functions

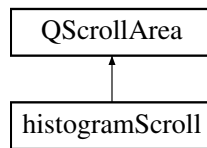
- **histogram** (QWidget *parent, [imageDisplayProperties](#) *idp)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/brightnessContrast.h
- qeframework/widgets/QEImage/brightnessContrast.cpp

9.25 histogramScroll Class Reference

Inheritance diagram for histogramScroll:



Public Member Functions

- **histogramScroll** (QWidget *parent, [imageDisplayProperties](#) *idp)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/brightnessContrast.h
- qeframework/widgets/QEImage/brightnessContrast.cpp

9.26 historicImage Class Reference

Public Member Functions

- **historicImage** (const QByteArray &image, unsigned long dataSize, const [QCaAlarmInfo](#) &alarmInfo, const [QCaDateTime](#) &time)

Public Attributes

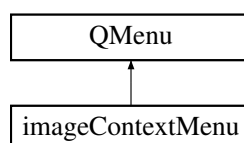
- QByteArray **image**
- unsigned long **dataSize**
- [QCaAlarmInfo](#) **alarmInfo**
- [QCaDateTime](#) **time**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/recording.h
- qeframework/widgets/QEImage/QEImage.cpp

9.27 imageContextMenu Class Reference

Inheritance diagram for imageContextMenu:



Public Types

- enum **imageContextMenuOptions** {
ICM_NONE = contextMenu::CM_SPECIFIC_WIDGETS_START_HERE , **ICM_SAVE** , **ICM_PAUSE** , **ICM_ENABLE_TIME** ,
ICM_ENABLE_FALSE_COLOUR , **ICM_ENABLE_PROFILE_AXES** , **ICM_ENABLE_CURSOR_PIXEL** ,
ICM_ABOUT_IMAGE ,
ICM_ENABLE_VERT1 , **ICM_ENABLE_VERT2** , **ICM_ENABLE_VERT3** , **ICM_ENABLE_VERT4** ,
ICM_ENABLE_VERT5 , **ICM_ENABLE_HOZ1** , **ICM_ENABLE_HOZ2** , **ICM_ENABLE_HOZ3** ,
ICM_ENABLE_HOZ4 , **ICM_ENABLE_HOZ5** , **ICM_ENABLE_AREA1** , **ICM_ENABLE_AREA2** ,
ICM_ENABLE_AREA3 , **ICM_ENABLE_AREA4** , **ICM_ENABLE_LINE** , **ICM_ENABLE_TARGET** ,
ICM_ENABLE_BEAM , **ICM_DISPLAY_BUTTON_BAR** , **ICM_DISPLAY_IMAGE_DISPLAY_PROPERTIES** ,
ICM_DISPLAY_RECORDER ,
ICM_ZOOM_SELECTED , **ICM_ZOOM_FIT** , **ICM_ZOOM_PLUS** , **ICM_ZOOM_MINUS** ,
ICM_ZOOM_10 , **ICM_ZOOM_25** , **ICM_ZOOM_50** , **ICM_ZOOM_75** ,
ICM_ZOOM_100 , **ICM_ZOOM_150** , **ICM_ZOOM_200** , **ICM_ZOOM_300** ,
ICM_ZOOM_400 , **ICM_ROTATE_NONE** , **ICM_ROTATE_RIGHT** , **ICM_ROTATE_LEFT** ,
ICM_ROTATE_180 , **ICM_FLIP_HORIZONTAL** , **ICM_FLIP_VERTICAL** , **ICM_SELECT_PAN** ,
ICM_SELECT_HSLICE1 , **ICM_SELECT_HSLICE2** , **ICM_SELECT_HSLICE3** , **ICM_SELECT_HSLICE4** ,
ICM_SELECT_HSLICE5 , **ICM_SELECT_VSLICE1** , **ICM_SELECT_VSLICE2** , **ICM_SELECT_VSLICE3** ,
ICM_SELECT_VSLICE4 , **ICM_SELECT_VSLICE5** , **ICM_SELECT_AREA1** , **ICM_SELECT_AREA2** ,
ICM_SELECT_AREA3 , **ICM_SELECT_AREA4** , **ICM_SELECT_PROFILE** , **ICM_SELECT_TARGET** ,
ICM_SELECT_BEAM , **ICM_CLEAR_MARKUP** , **ICM_SET_LEGEND** , **ICM_THICKNESS_ONE_MARKUP**
,
ICM_THICKNESS_SELECT_MARKUP , **ICM_COPY_PLOT_DATA** , **ICM_FULL_SCREEN** , **ICM_DISPLAY_HSLICE1** ,
ICM_DISPLAY_HSLICE2 , **ICM_DISPLAY_HSLICE3** , **ICM_DISPLAY_HSLICE4** , **ICM_DISPLAY_HSLICE5**
,
ICM_DISPLAY_VSLICE1 , **ICM_DISPLAY_VSLICE2** , **ICM_DISPLAY_VSLICE3** , **ICM_DISPLAY_VSLICE4**
,
ICM_DISPLAY_VSLICE5 , **ICM_DISPLAY_AREA1** , **ICM_DISPLAY_AREA2** , **ICM_DISPLAY_AREA3** ,
ICM_DISPLAY_AREA4 , **ICM_DISPLAY_PROFILE** , **ICM_DISPLAY_TARGET** , **ICM_DISPLAY_BEAM** ,
ICM_DISPLAY_TIMESTAMP , **ICM_DISPLAY_ELLIPSE** , **ICM_OPTIONS** , **ICM_LOCAL_CONTROL** }

Public Member Functions

- **imageContextMenu** (QWidget *parent=0)
- void **getContextMenuOption** (const QPoint &, imageContextMenuOptions *option, bool *checked)
- void **addMenuItem** (const QString &title, const bool checkable, const bool checked, const imageContextMenuOptions option)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/imageContextMenu.h
- qeframework/widgets/QEImage/imageContextMenu.cpp

9.28 imageDataFormats Class Reference

```
#include <imageDataFormats.h>
```

Static Public Member Functions

- static bool **convertToFormatOption** (const QString &text, [QE::ImageFormatOptions](#) &format)
- static QString **getFormatInformation** (const [QE::ImageFormatOptions](#) format)

9.28.1 Detailed Description

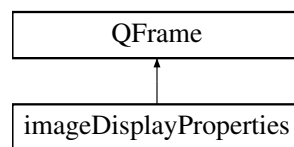
The formats are now defined in QEnums

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/imageDataFormats.h
- qeframework/widgets/QEImage/imageDataFormats.cpp

9.29 imageDisplayProperties Class Reference

Inheritance diagram for imageDisplayProperties:



Classes

- struct [rgbPixel](#)

Signals

- void **brightnessContrastAutoImage** ()
- void **imageDisplayPropertiesChange** ()

Public Member Functions

- void **setBrightnessContrast** (const unsigned int max, const unsigned int min)
- void **setAutoBrightnessContrast** (bool autoBrightnessContrast)
- void **setContrastReversal** (bool contrastReversal)
- void **setLog** (bool log)
- void **setFalseColour** (bool falseColour)
- void **resetBrightnessContrast** ()
- bool **getAutoBrightnessContrast** ()
- bool **getContrastReversal** ()
- bool **getLog** ()
- bool **getFalseColour** ()
- int **getLowPixel** ()
- int **getHighPixel** ()
- void **setStatistics** (unsigned int minPIn, unsigned int maxPIn, unsigned int bitDepth, unsigned int bins↔
In[HISTOGRAM_BINS], [rgbPixel](#) pixelLookup[256])
- void **showStatistics** ()
- void **setHistZoom** (int value)
- int **getHistZoom** ()
- bool **statisticsValid** ()

Public Attributes

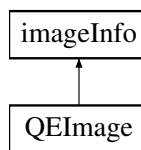
- int **zeroValue**
- int **fullValue**
- bool **defaultFullValue**
- unsigned int **range**
- unsigned int **maxP**
- unsigned int **minP**
- unsigned int **depth**
- unsigned int **bins** [HISTOGRAM_BINS]
- bool **statisticsSet**
- [rgbPixel](#) * **pixelLookup**
- QLabel * **histXLabel**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/brightnessContrast.h
- qeframework/widgets/QEImage/brightnessContrast.cpp

9.30 imageInfo Class Reference

Inheritance diagram for imageInfo:



Public Member Functions

- void **showInfo** (bool show)
- QLayout * **getInfoWidget** ()
- void **infoShow** (const bool show)
- void **infoUpdateTarget** ()
- void **infoUpdateTarget** (const int x, const int y)
- void **infoUpdateBeam** ()
- void **infoUpdateBeam** (const int x, const int y)
- void **infoUpdateVertProfile** ()
- void **infoUpdateVertProfile** (const int x, const unsigned int thickness)
- void **infoUpdateHozProfile** ()
- void **infoUpdateHozProfile** (const int y, const unsigned int thickness)
- void **infoUpdateProfile** ()
- void **infoUpdateProfile** (const QPoint start, const QPoint end, const unsigned int thickness)
- void **infoUpdateRegion** (const unsigned int region)
- void **infoUpdateRegion** (const unsigned int region, const int x1, const int y1, const int x2, const int y2)
- void **infoUpdatePixel** ()
- void **infoUpdatePixel** (const QPoint pos, int value)
- void **infoUpdateZoom** ()
- void **infoUpdateZoom** (int value, const double XStretch, const double YStretch)

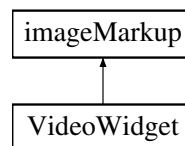
- void **infoUpdatePaused** ()
- void **infoUpdatePaused** (bool paused)
- void **setBriefInfoArea** (const bool briefIn)
- bool **getBriefInfoArea** ()
- void **freshImage** (const QDateTime &time)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/imageInfo.h
- qeframework/widgets/QEImage/imageInfo.cpp

9.31 imageMarkup Class Reference

Inheritance diagram for imageMarkup:



Public Types

- enum **markupIds** {
MARKUP_ID_REGION1 , **MARKUP_ID_REGION2** , **MARKUP_ID_REGION3** , **MARKUP_ID_REGION4** ,
MARKUP_ID_H1_SLICE , **MARKUP_ID_H2_SLICE** , **MARKUP_ID_H3_SLICE** , **MARKUP_ID_H4_SLICE**
 ,
MARKUP_ID_H5_SLICE , **MARKUP_ID_V1_SLICE** , **MARKUP_ID_V2_SLICE** , **MARKUP_ID_V3_SLICE**
 ,
MARKUP_ID_V4_SLICE , **MARKUP_ID_V5_SLICE** , **MARKUP_ID_LINE** , **MARKUP_ID_TARGET** ,
MARKUP_ID_BEAM , **MARKUP_ID_TIMESTAMP** , **MARKUP_ID_ELLIPSE** , **MARKUP_ID_COUNT** ,
MARKUP_ID_NONE }
- enum **beamAndTargetOptions** { **CROSSHAIR1** , **CROSSHAIR2** }

Public Member Functions

- void **setShowTime** (bool visibleIn)
- bool **getShowTime** ()
- markupIds **getMode** ()
- void **setMode** (markupIds modeIn)
- void **setMarkupColor** (markupIds mode, QColor markupColorIn)
- QColor **getMarkupColor** (markupIds mode)
- bool **showMarkupMenu** (const QPoint &pos, const QPoint &globalPos)
- void **markupRegionValueChange** (int areaIndex, QRect area, bool displayMarkups)
- void **markupH1ProfileChange** (int y, bool displayMarkups)
- void **markupH2ProfileChange** (int y, bool displayMarkups)
- void **markupH3ProfileChange** (int y, bool displayMarkups)
- void **markupH4ProfileChange** (int y, bool displayMarkups)
- void **markupH5ProfileChange** (int y, bool displayMarkups)
- void **markupV1ProfileChange** (int x, bool displayMarkups)

- void **markupV2ProfileChange** (int x, bool displayMarkups)
- void **markupV3ProfileChange** (int x, bool displayMarkups)
- void **markupV4ProfileChange** (int x, bool displayMarkups)
- void **markupV5ProfileChange** (int x, bool displayMarkups)
- void **markupLineProfileChange** (QPoint start, QPoint end, bool displayMarkups)
- void **markupTargetValueChange** (QPoint point, bool displayMarkups)
- void **markupBeamValueChange** (QPoint point, bool displayMarkups)
- void **markupEllipseValueChange** (QPoint point1, QPoint point2, double rotation, bool displayMarkups)
- void **markupValueChange** (int markup, bool displayMarkups, QPoint p1, QPoint p2=QPoint(), double rotation=0.0)
- QCursor **getCircleCursor** ()
- QCursor **getTargetCursor** ()
- QCursor **getVLineCursor** ()
- QCursor **getHLineCursor** ()
- QCursor **getLineCursor** ()
- QCursor **getRegionCursor** ()
- virtual void **markupSetCursor** (QCursor cursor)=0
- void **setMarkupLegend** (markupIds mode, QString legend)
- QString **getMarkupLegend** (markupIds mode)
- void **clearMarkup** (markupIds markupId)
- void **showMarkup** (markupIds markupId)
- void **displayMarkup** (markupIds markupId, bool state)
- bool **isMarkupVisible** (markupIds mode)
- double **getZoomScale** ()
- QSize **getImageSize** ()
- void **setImageSize** (const QSize &imageSizeIn)
- beamAndTargetOptions **getTargetOption** ()
- void **setTargetOption** (beamAndTargetOptions option)
- beamAndTargetOptions **getBeamOption** ()
- void **setBeamOption** (beamAndTargetOptions option)
- void **setBeamOrTargetOption** (markupIds item, beamAndTargetOptions option)

Public Attributes

- QVector< [markupItem](#) * > **items**
- QPoint **grabOffset**
- bool **markupAreasStale**
- QFont **legendFont**
- QFontMetrics * **legendFontMetrics**

Protected Member Functions

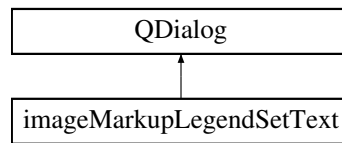
- void **drawMarkups** (QPainter &p, const QRect &rect)
- bool **anyVisibleMarkups** ()
- QCursor **getDefaultMarkupCursor** ()
- void **setMarkupTime** ([QCaDateTime](#) &time)
- bool **markupMousePressEvent** (QMouseEvent *event, bool panning)
- bool **markupMouseReleaseEvent** (QMouseEvent *event, bool panning)
- bool **markupMouseMoveEvent** (QMouseEvent *event, bool panning)
- void **markupResize** (const double scale)
- virtual void **markupChange** (QVector< QRect > &changedAreas)=0
- virtual void **markupAction** (markupIds mode, bool complete, bool clearing, QPoint point1, QPoint point2, unsigned int thickness)=0

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/imageMarkup.h
- qeframework/widgets/QEImage/imageMarkup.cpp

9.32 imageMarkupLegendSetText Class Reference

Inheritance diagram for imageMarkupLegendSetText:



Public Member Functions

- **imageMarkupLegendSetText** (QString existingLegend, QWidget *parent=0)
- QString **getLegend** ()

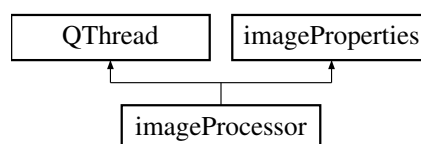
The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/imageMarkupLegendSetText.h
- qeframework/widgets/QEImage/imageMarkupLegendSetText.cpp

9.33 imageProcessor Class Reference

```
#include <imageProcessor.h>
```

Inheritance diagram for imageProcessor:



Signals

- void **imageBuilt** (QImage image, QString error)
An image has been generated from image data and is now ready for presentation.

Public Member Functions

- [imageProcessor](#) ()
Constructor.
- [~imageProcessor](#) ()
Destructor.
- void [setImage](#) (const QByteArray &imageIn, unsigned long dataSize)
Save the image data for analysis processing and display.
- void [buildImage](#) ()
Generate a new image.
- bool [setWidth](#) (unsigned long uValue)
Set the image width.
- bool [setHeight](#) (unsigned long uValue)
Set the image height.
- bool [setNumDimensions](#) (unsigned long uValue)
Set the number of dimensions.
- bool [setDimension0](#) (unsigned long uValue)
Set the first dimension (width if two dimensions, bytes per element if three dimensions)
- bool [setDimension1](#) (unsigned long uValue)
Set the second dimension (height if two dimensions, width if three dimensions)
- bool [setDimension2](#) (unsigned long uValue)
Set the third dimension (unused if two dimensions, height if three dimensions)
- void [setClippingOn](#) (bool clippingOnIn)
Set clipping flag. If true, [setClippingLow\(\)](#) and [setClippingHigh\(\)](#) are used to set clipping values.
- void [setClippingLow](#) (unsigned int value)
Set pixel value below which low clip colour is displayed.
- void [setClippingHigh](#) (unsigned int value)
Set pixel value above which high clip colour is displayed.
- int [getScanOption](#) ()
Determine the way the input pixel data must be scanned to accommodate the required rotate and flip options.
- void [getPixelTranslation](#) ()
Generate a lookup table to convert raw pixel values to display pixel values.
- unsigned int [maxPixelValue](#) ()
Determine the maximum pixel value for the current format.
- unsigned int [rotatedImageBuffWidth](#) ()
Return the image width following any rotation.
- unsigned int [rotatedImageBuffHeight](#) ()
Return the image height following any rotation.
- [imageDisplayProperties::rgbPixel](#) [getFalseColor](#) (const unsigned char value)
Get a false color representation for an entry fro the color lookup table.
- int [getElementCount](#) ()
Determine the element count expected based on the available dimensions.
- bool [validateDimensions](#) ()
Determine if the image dimensional information is valid.
- void [getPixelRange](#) (const QRect &area, unsigned int *min, unsigned int *max)
Determine the range of pixel values an area of the image.
- bool [hasImage](#) ()
Return true if the current image is empty.
- const unsigned char * [getImageDataPtr](#) (QPoint &pos)
Return a pointer to pixel data in the original image data.
- int [getPixelValueFromData](#) (const unsigned char *ptr)

- Return a number representing a pixel intensity given a pointer into an image data buffer.*

 - double [getFloatingPixelValueFromData](#) (const unsigned char *ptr)

Return a floating point number representing a pixel intensity given a pointer into an image data buffer.
- QImage [copyImage](#) ()

Return a QImage based on the current image.
- void [generateVSliceData](#) (QVector< QPointF > &vSliceData, int x, unsigned int thickness)

Generate a series of pixel values from a vertical slice through the current image.
- void [generateHSliceData](#) (QVector< QPointF > &hSliceData, int y, unsigned int thickness)

Generate a series of pixel values from a horizontal slice through the current image.
- void [generateProfileData](#) (QVector< QPointF > &profileData, QPoint point1, QPoint point2, unsigned int thickness)

Generate a series of pseudo pixel values from an arbitrary line between two pixels.
- QRect [rotateFlipToDataRectangle](#) (const QRect &rect)

Transform a rectangle from the image to the original data according to current rotation and flip options.
- QRect [rotateFlipToDataRectangle](#) (const QPoint &pos1, const QPoint &pos2)

Transform a rectangle from the image to the original data according to current rotation and flip options.
- QPoint [rotateFlipToDataPoint](#) (const QPoint &pos)

Transform a point from the image to the original data according to current rotation and flip options.
- QRect [rotateFlipToImageRectangle](#) (const QRect &rect)

Transform a rectangle from the original data to the image according to current rotation and flip options.
- QRect [rotateFlipToImageRectangle](#) (const QPoint &pos1, const QPoint &pos2)

Transform a rectangle from the original data to the image according to current rotation and flip options.
- QPoint [rotateFlipToImagePoint](#) (const QPoint &pos)

Transform a point from the original data to the image according to current rotation and flip options.
- void [run](#) ()

Public Attributes

- QWaitCondition **imageSync**
- QMutex **imageWait**
- QMutex **imageLock**
- bool **finishNow**
- [imagePropertiesCore](#) * **next**

Additional Inherited Members

9.33.1 Detailed Description

This class generates images for presentation from raw image data and formatting information such as brightness, contrast, flip, rotate, canvas size, etc. The work is performed in a dedicated thread .

9.33.2 Member Function Documentation

9.33.2.1 `getPixelValueFromData()`

```
int imageProcessor::getPixelValueFromData (
    const unsigned char * ptr )
```

Return a number representing a pixel intensity given a pointer into an image data buffer.

!! not done - copy of RGB1

!! not done - copy of RGB1

!! not done - copy of RGB1

!! not done - copy of RGB1

!! not done - copy of RGB1

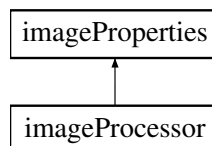
The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/imageProcessor.h
- qeframework/widgets/QEImage/imageProcessor.cpp

9.34 `imageProperties` Class Reference

```
#include <imageProperties.h>
```

Inheritance diagram for `imageProperties`:



Public Member Functions

- [imageProperties](#) ()
Constructor.
- void **setRotation** ([QE::RotationOptions](#) rotationIn)
- [QE::RotationOptions](#) **getRotation** () const
- void **setFlipVert** (bool flipVertIn)
- bool **getFlipVert** () const
- void **setFlipHoz** (bool flipHozIn)
- bool **getFlipHoz** () const
- void **setImageBuffWidth** (unsigned long imageBuffWidthIn)
- void **setImageBuffHeight** (unsigned long imageBuffHeightIn)
- unsigned long **getImageBuffWidth** () const
- unsigned long **getImageBuffHeight** () const
- [QE::ImageFormatOptions](#) **getFormat** () const
- void **setFormat** ([QE::ImageFormatOptions](#) formatIn)
- bool **setFormat** (const QString &text)

- void **setBitDepth** (unsigned int bitDepth)
- unsigned int **getBitDepth** () const
- void **setElementsPerPixel** (unsigned long elementsPerPixelIn)
- unsigned long **getElementsPerPixel** () const
- void **setImageDisplayProperties** ([imageDisplayProperties](#) *imageDisplayPropsIn)
- void **setWidthHeightFromDimensions** ()
// Update the image dimensions (width and height) from the area detector dimension variables.
- void **invalidatePixelLookup** ()
recalculate (when next required) pixel summary information
- QString **getInfoText** () const
Generate textual information regarding the current image.

Protected Attributes

- [imageDisplayProperties](#) * **imageDisplayProps**
- [QE::ImageFormatOptions](#) **formatOption**
- unsigned int **bitDepth**
- unsigned long **imageDataSize**
- unsigned long **elementsPerPixel**
- unsigned long **bytesPerPixel**
- QByteArray **imageData**
- unsigned long **receivedImageSize**
- QString **previousMessageText**
- QImage **image**
- unsigned long **imageBuffWidth**
- unsigned long **imageBuffHeight**
- unsigned long **numDimensions**
- unsigned long **imageDimension0**
- unsigned long **imageDimension1**
- unsigned long **imageDimension2**
- bool **pixelLookupValid**
- [imageDisplayProperties::rgbPixel](#) **pixelLookup** [256]
- int **pixelLow**
- int **pixelHigh**
- bool **clippingOn**
- unsigned int **clippingLow**
- unsigned int **clippingHigh**
- [QE::RotationOptions](#) **rotation**
- bool **flipVert**
- bool **flipHoz**

9.34.1 Detailed Description

This class manages the image attributes required for generating a QImage from a QByteArray holding CA image data. It is used as the base class for the [imageProcessor](#) class. Note, while this class holds and manages all the information needed to process an image, a snapshot of all the information required for processing an image in a separate thread is made by the [imagePropertiesCore](#) class.

9.34.2 Constructor & Destructor Documentation

9.34.2.1 imageProperties()

```
imageProperties::imageProperties ( )
```

Constructor.

Construction. Set all image attributes to sensible defaults.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/imageProperties.h
- qeframework/widgets/QEImage/imageProperties.cpp

9.35 imagePropertiesCore Class Reference

Public Member Functions

- **imagePropertiesCore** (QByteArray imageDataIn, unsigned long imageBuffWidthIn, unsigned long imageBuffHeightIn, int scanOptionIn, unsigned long bytesPerPixelIn, int pixelLowIn, int pixelHighIn, unsigned int bitDepthIn, [imageDisplayProperties::rgbPixel](#) *pixelLookupIn, [QE::ImageFormatOptions](#) formatOptionIn, unsigned long imageDataSizeIn, [imageDisplayProperties](#) *imageDisplayPropsIn, unsigned int rotatedImageBuffWidthIn, unsigned int rotatedImageBuffHeightIn)
- QImage [buildImageCore](#) ()

9.35.1 Member Function Documentation

9.35.1.1 buildImageCore()

```
QImage imagePropertiesCore::buildImageCore ( )
```

!! not done yet - just do the same as RGB1 for the time being and hope

!! not done yet - just do the same as RGB1 for the time being and hope

!! not done yet. do the same as for YUV422

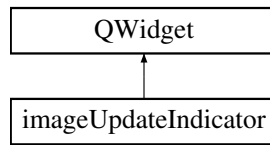
!! not done yet. do the same as for YUV422

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/imageProperties.h
- qeframework/widgets/QEImage/imageProcessor.cpp

9.36 imageUpdateIndicator Class Reference

Inheritance diagram for imageUpdateIndicator:



Public Member Functions

- void **freshImage** ()
- void **paintEvent** (QPaintEvent *)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/imageInfo.h
- qeframework/widgets/QEImage/imageInfo.cpp

9.37 itemCheckInfo Class Reference

Public Member Functions

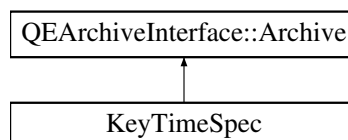
- **itemCheckInfo** (const [itemCheckInfo](#) &other)
- **itemCheckInfo** (const QDomElement &itemElement)
- [itemCheckInfo](#) & **operator=** (const [itemCheckInfo](#) &other)
- QString **getKey** () const
- QString **getValue** () const
- bool **getCheckable** () const

The documentation for this class was generated from the following files:

- qeframework/common/windowCustomisation.h
- qeframework/common/windowCustomisation.cpp

9.38 KeyTimeSpec Struct Reference

Inheritance diagram for KeyTimeSpec:



Public Attributes

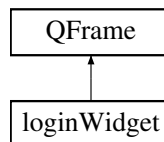
- uint32_t **startTime**
- uint32_t **endTime**

The documentation for this struct was generated from the following file:

- qeframework/archive/QEArchiveManager.cpp

9.39 loginWidget Class Reference

Inheritance diagram for loginWidget:



Public Member Functions

- loginWidget (QELogin *ownerIn)
- QE::UserLevels getUserType ()
- QString getPassword ()
- void clearPassword ()

The documentation for this class was generated from the following files:

- qeframework/widgets/QELogin/QELogin.h
- qeframework/widgets/QELogin/QELogin.cpp

9.40 macroSubstitution Class Reference

```
#include <macroSubstitution.h>
```

Public Member Functions

- macroSubstitution (const QString &key, const QString &value)
- const QString getKey () const
- const QString getValue () const

Friends

- class macroSubstitutionList

9.40.1 Detailed Description

Description:

A class to manage macro substitutions. Macro substitutions are often provided as a string of keys and values.

This class parses such strings, and manages macro substitutions using a list of keys and values.

The documentation for this class was generated from the following files:

- qeframework/common/macroSubstitution.h
- qeframework/common/macroSubstitution.cpp

9.41 macroSubstitutionList Class Reference

Public Member Functions

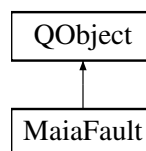
- **macroSubstitutionList** (const QString &string)
- void **addMacroSubstitutions** (const QString &string)
- QString **getString** () const
- QString **substitute** (const QString &string) const
- int **getCount** () const
- const QString **getKey** (const int i) const
- const QString **getValue** (const int i) const
- const QString **getValue** (const QString &key) const
- const [macroSubstitution](#) **getItem** (const int i) const

The documentation for this class was generated from the following files:

- qeframework/common/macroSubstitution.h
- qeframework/common/macroSubstitution.cpp

9.42 MaiaFault Class Reference

Inheritance diagram for MaiaFault:



Public Member Functions

- **MaiaFault** (int faultCode=0, QString faultString=QString(), QObject *parent=0)
- **MaiaFault** (const [MaiaFault](#) &other)
- QString **toString** ()

Public Attributes

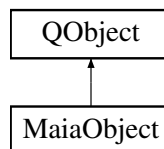
- QMap< QString, QVariant > **fault**

The documentation for this class was generated from the following files:

- qeframework/archive/maiaFault.h
- qeframework/archive/maiaFault.cpp

9.43 MaiaObject Class Reference

Inheritance diagram for MaiaObject:



Public Slots

- void **parseResponse** (QString response, QNetworkReply *reply)

Signals

- void **areponse** (QVariant &, QNetworkReply *reply)
- void **call** (const QString, const QList< QVariant >)
- void **fault** (int, const QString &, QNetworkReply *reply)

Public Member Functions

- **MaiaObject** (QObject *parent=0)
- QString **prepareCall** (QString method, QList< QVariant > args)

Static Public Member Functions

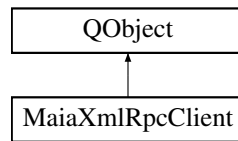
- static QDomElement **toXml** (QVariant arg)
- static QVariant **fromXml** (const QDomElement &elem)
- static QString **prepareResponse** (QVariant arg)

The documentation for this class was generated from the following files:

- qeframework/archive/maiaObject.h
- qeframework/archive/maiaObject.cpp

9.44 MaiaXmlRpcClient Class Reference

Inheritance diagram for MaiaXmlRpcClient:



Signals

- void **sslErrors** (QNetworkReply *reply, const QList< QSslError > &errors)

Public Member Functions

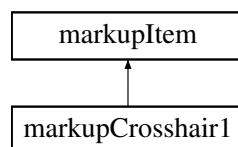
- **MaiaXmlRpcClient** (QObject *parent=0)
- **MaiaXmlRpcClient** (QUrl url, QObject *parent=0)
- **MaiaXmlRpcClient** (QUrl url, QString userAgent, QObject *parent=0)
- void **setUrl** (QUrl url)
- void **setUserAgent** (QString userAgent)
- QNetworkReply * **call** (QString method, QList< QVariant > args, QObject *responseObject, const char *responseSlot, QObject *faultObject, const char *faultSlot)
- void **setSslConfiguration** (const QSslConfiguration &config)
- QSslConfiguration **sslConfiguration** () const

The documentation for this class was generated from the following files:

- qeframework/archive/maiaXmlRpcClient.h
- qeframework/archive/maiaXmlRpcClient.cpp

9.45 markupCrosshair1 Class Reference

Inheritance diagram for markupCrosshair1:



Public Member Functions

- **markupCrosshair1** ([imageMarkup](#) *ownerIn, const bool interactiveIn, const bool reportOnMoveIn, const QString legendIn)
- void **startDrawing** (const QPoint pos)
- void **setArea** ()
- void **drawMarkup** (QPainter &p)
- void **moveTo** (const QPoint pos)
- bool **isOver** (const QPoint point, QCursor *cursor)
- QPoint **origin** ()
- QCursor **cursorForHandle** (const markupItem::markupHandles handle)
- QPoint **getPoint1** ()
- QPoint **getPoint2** ()
- QCursor **defaultCursor** ()
- void **nonInteractiveUpdate** (QPoint p1, QPoint p2, double rotation)

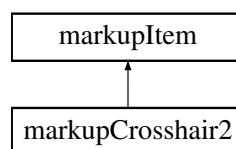
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/markupTarget.h
- qeframework/widgets/QEImage/markupTarget.cpp

9.46 markupCrosshair2 Class Reference

Inheritance diagram for markupCrosshair2:



Public Member Functions

- **markupCrosshair2** ([imageMarkup](#) *ownerIn, const bool interactiveIn, const bool reportOnMoveIn, const QString legendIn)
- void **startDrawing** (const QPoint pos)
- void **setArea** ()
- void **drawMarkup** (QPainter &p)
- void **moveTo** (const QPoint pos)
- bool **isOver** (const QPoint point, QCursor *cursor)
- QPoint **origin** ()
- QCursor **cursorForHandle** (const markupItem::markupHandles handle)
- QPoint **getPoint1** ()
- QPoint **getPoint2** ()
- QCursor **defaultCursor** ()
- void **nonInteractiveUpdate** (QPoint p1, QPoint p2, double rotation)

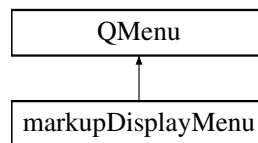
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/markupBeam.h
- qeframework/widgets/QEImage/markupBeam.cpp

9.47 markupDisplayMenu Class Reference

Inheritance diagram for markupDisplayMenu:



Public Member Functions

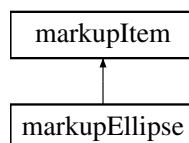
- **markupDisplayMenu** (QWidget *parent=0)
- void **setDisplay** (imageContextMenu::imageContextMenuOptions option, bool state)
- void **setItemText** (imageContextMenu::imageContextMenuOptions option, QString title)
- bool **isDisplayed** (imageContextMenu::imageContextMenuOptions option)
- void **enable** (imageContextMenu::imageContextMenuOptions option, bool state)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/markupDisplayMenu.h
- qeframework/widgets/QEImage/markupDisplayMenu.cpp

9.48 markupEllipse Class Reference

Inheritance diagram for markupEllipse:



Public Member Functions

- **markupEllipse** ([imageMarkup](#) *ownerIn, const bool interactiveIn, const bool reportOnMoveIn, const QString legendIn)
- void **startDrawing** (const QPoint pos)
- void **setArea** ()
- void **drawMarkup** (QPainter &p)
- void **moveTo** (const QPoint pos)
- bool **isOver** (const QPoint point, QCursor *cursor)
- QPoint **origin** ()
- QCursor **cursorForHandle** (const markupItem::markupHandles handle)
- QPoint **getPoint1** ()
- QPoint **getPoint2** ()
- QCursor **defaultCursor** ()
- void **nonInteractiveUpdate** (QPoint p1, QPoint p2, double rotation)

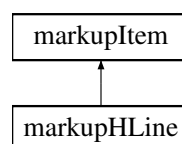
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/markupEllipse.h
- qeframework/widgets/QEImage/markupEllipse.cpp

9.49 markupHLine Class Reference

Inheritance diagram for markupHLine:



Public Member Functions

- **markupHLine** ([imageMarkup](#) *ownerIn, const bool interactiveIn, const bool reportOnMoveIn, const QString legendIn)
- void **startDrawing** (const QPoint pos)
- void **setArea** ()
- void [drawMarkup](#) (QPainter &p)
- void **moveTo** (const QPoint pos)
- bool **isOver** (const QPoint point, QCursor *cursor)
- QPoint **origin** ()
- QCursor **cursorForHandle** (const markupItem::markupHandles handle)
- QPoint **getPoint1** ()
- QPoint **getPoint2** ()
- QCursor **defaultCursor** ()
- void **nonInteractiveUpdate** (QPoint p1, QPoint p2, double rotation)

Additional Inherited Members

9.49.1 Member Function Documentation

9.49.1.1 drawMarkup()

```
void markupHLine::drawMarkup (
    QPainter & p ) [virtual]
```

!! draw the handle in the middle of the existing view, not the entire image

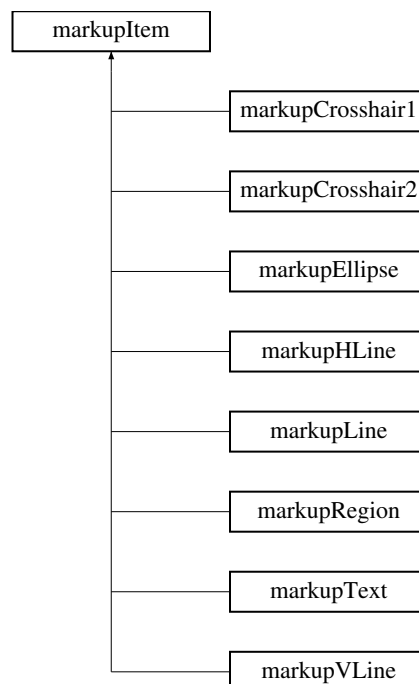
Implements [markupItem](#).

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/markupHLine.h
- qeframework/widgets/QEImage/markupHLine.cpp

9.50 markupItem Class Reference

Inheritance diagram for markupItem:



Public Types

- enum **markupHandles** {
MARKUP_HANDLE_NONE , **MARKUP_HANDLE_START** , **MARKUP_HANDLE_END** , **MARKUP_HANDLE_CENTER** ,
MARKUP_HANDLE_TL , **MARKUP_HANDLE_TR** , **MARKUP_HANDLE_BL** , **MARKUP_HANDLE_BR** ,
MARKUP_HANDLE_T , **MARKUP_HANDLE_B** , **MARKUP_HANDLE_L** , **MARKUP_HANDLE_R** }

Public Member Functions

- void **drawMarkupItem** (QPainter &p)
- QSize **getImageSize** ()
- virtual QPoint **origin** ()=0
- virtual void **moveTo** (const QPoint pos)=0
- virtual void **startDrawing** (const QPoint pos)=0
- virtual bool **isOver** (const QPoint point, QCursor *cursor)=0
- virtual QCursor **cursorForHandle** (const markupItem::markupHandles handle)=0
- virtual QPoint **getPoint1** ()=0
- virtual QPoint **getPoint2** ()=0
- virtual QCursor **defaultCursor** ()=0
- virtual void **nonInteractiveUpdate** (QPoint, QPoint, double)
- void **setThickness** (const unsigned int thicknessIn)
- unsigned int **getThickness** ()
- void **setLegend** (const QString legendIn)
- const QString **getLegend** ()
- void **setColor** (QColor colorIn)
- QColor **getColor** ()

Public Attributes

- QRect **area**
- QRect **scalableArea**
- bool **visible**
- bool **interactive**
- bool **reportOnMove**
- QColor **color**

Protected Types

- enum **isOverOptions** { OVER_LINE , OVER_BORDER , OVER_AREA }
- enum **legendJustification** { ABOVE_RIGHT , BELOW_LEFT , BELOW_RIGHT }

Protected Member Functions

- **markupItem** (imageMarkup *ownerIn, const isOverOptions over, const bool interactiveIn, const bool reportOnMoveIn, const QString legendIn)
- virtual void **setArea** ()=0
- virtual void **drawMarkup** (QPainter &p)=0
- bool **pointIsNear** (QPoint p1, QPoint p)
- const QSize **getLegendSize** ()
- void **addLegendArea** ()
- const QPoint **getLegendTextOrigin** (QPoint posScaled)
- void **setLegendOffset** (QPoint offset, legendJustification just)
- const QPoint **getLegendOffset** ()
- void **drawLegend** (QPainter &p, QPoint posScaled)
- QPoint **limitPointToImage** (const QPoint pos)
- double **getZoomScale** ()

Protected Attributes

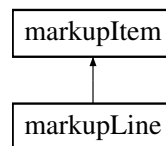
- markupHandles **activeHandle**
- [imageMarkup](#) * **owner**
- unsigned int **thickness**
- unsigned int **maxThickness**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/markupItem.h
- qeframework/widgets/QEImage/markupItem.cpp

9.51 markupLine Class Reference

Inheritance diagram for markupLine:



Public Member Functions

- **markupLine** ([imageMarkup](#) *ownerIn, const bool interactiveIn, const bool reportOnMoveIn, const QString legendIn)
- void **startDrawing** (const QPoint pos)
- void **setArea** ()
- void **drawMarkup** (QPainter &p)
- void **moveTo** (const QPoint pos)
- bool **isOver** (const QPoint point, QCursor *cursor)
- QPoint **origin** ()
- QCursor **cursorForHandle** (const markupItem::markupHandles handle)
- QPoint **getPoint1** ()
- QPoint **getPoint2** ()
- QCursor **defaultCursor** ()
- void **nonInteractiveUpdate** (QPoint p1, QPoint p2, double)

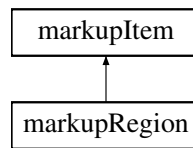
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/markupLine.h
- qeframework/widgets/QEImage/markupLine.cpp

9.52 markupRegion Class Reference

Inheritance diagram for markupRegion:



Public Member Functions

- **markupRegion** ([imageMarkup](#) *ownerIn, const bool interactiveIn, const bool reportOnMoveIn, const QString legendIn)
- void **startDrawing** (const QPoint pos)
- void **setArea** ()
- void **drawMarkup** (QPainter &p)
- void **moveTo** (const QPoint pos)
- bool **isOver** (const QPoint point, QCursor *cursor)
- QPoint **origin** ()
- QCursor **cursorForHandle** (const markupItem::markupHandles handle)
- QPoint **getPoint1** ()
- QPoint **getPoint2** ()
- QCursor **defaultCursor** ()
- void **nonInteractiveUpdate** (QPoint p1, QPoint p2, double rotation)

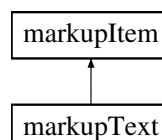
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/markupRegion.h
- qeframework/widgets/QEImage/markupRegion.cpp

9.53 markupText Class Reference

Inheritance diagram for markupText:



Public Member Functions

- **markupText** ([imageMarkup](#) *ownerIn, const bool interactiveIn, const bool reportOnMoveIn, const QString legendIn)
- void **setText** (QString textIn)
- void **startDrawing** (const QPoint pos)
- void **setArea** ()
- void **drawMarkup** (QPainter &p)
- void **moveTo** (const QPoint pos)
- bool **isOver** (const QPoint point, QCursor *cursor)
- QPoint **origin** ()
- QCursor **cursorForHandle** (const markupItem::markupHandles handle)
- QPoint **getPoint1** ()
- QPoint **getPoint2** ()
- QCursor **defaultCursor** ()

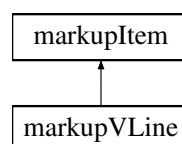
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/markupText.h
- qeframework/widgets/QEImage/markupText.cpp

9.54 markupVLine Class Reference

Inheritance diagram for markupVLine:



Public Member Functions

- **markupVLine** ([imageMarkup](#) *ownerIn, const bool interactiveIn, const bool reportOnMoveIn, const QString legendIn)
- void **startDrawing** (const QPoint pos)
- void **setArea** ()
- void [drawMarkup](#) (QPainter &p)
- void **moveTo** (const QPoint pos)
- bool **isOver** (const QPoint point, QCursor *cursor)
- QPoint **origin** ()
- QCursor **cursorForHandle** (const markupItem::markupHandles handle)
- QPoint **getPoint1** ()
- QPoint **getPoint2** ()
- QCursor **defaultCursor** ()
- void **nonInteractiveUpdate** (QPoint p1, QPoint p2, double rotation)

Additional Inherited Members

9.54.1 Member Function Documentation

9.54.1.1 drawMarkup()

```
void markupVLine::drawMarkup (
    QPainter & p ) [virtual]
```

!! draw the handle in the middle of the existing view, not the entire image

Implements [markupItem](#).

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/markupVLine.h
- qeframework/widgets/QEImage/markupVLine.cpp

9.55 menuItemToBeActivated Class Reference

Public Member Functions

- **menuItemToBeActivated** ([menuItemToBeActivated](#) *other)
- **menuItemToBeActivated** ([windowCustomisationMenuItem](#) *itemIn, QMenu *menuIn)

Public Attributes

- [windowCustomisationMenuItem](#) * **item**
- QMenu * **menu**

The documentation for this class was generated from the following file:

- qeframework/common/windowCustomisation.h

9.56 message_types Class Reference

Public Member Functions

- **message_types** (message_severities severityIn, message_kind_sets kind_setIn=MESSAGE_KIND_↵
STANDARD)
- QString [getSeverityName](#) ()
Function to provide string name for each message type severity.

Public Attributes

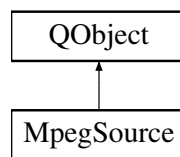
- message_severities **severity**
- message_kind_sets **kind_set**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/UserMessage.h
- qeframework/widgets/QEWidget/UserMessage.cpp

9.57 MpegSource Class Reference

Inheritance diagram for MpegSource:



Signals

- void **aboutToQuit** ()
- void **setDataImage** (const QByteArray &image, unsigned long dataSize, unsigned long elements, unsigned long width, unsigned long height, [QE::ImageFormatOptions](#) format, unsigned int depth)

Public Member Functions

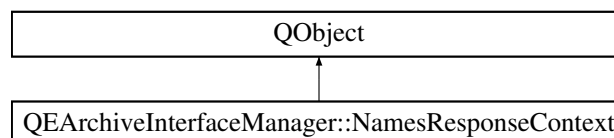
- **MpegSource** (QObject *parent)
- QString **getURL** () const
- void **setURL** (const QString &urlIn)
- void **stopStream** ()
- void **startStream** ()

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/mpeg.h
- qeframework/widgets/QEImage/mpeg.cpp

9.58 QEArchiveInterfaceManager::NamesResponseContext Class Reference

Inheritance diagram for QEArchiveInterfaceManager::NamesResponseContext:



Public Member Functions

- **NamesResponseContext** (const [QEArchiveInterfaceManager](#) *aim, const int instanceIn)

Public Attributes

- const [QEArchiveInterfaceManager](#) * **archiveInterfaceManager**
- const int **instance**

The documentation for this class was generated from the following file:

- qeframework/archive/QEArchiveInterfaceManager.cpp

9.59 NTNDArrayInfo_t Class Reference

Copied from ntndArrayConverter.h.

```
#include <QENTNDArrayConverter.h>
```

Public Attributes

- int **ndims**
- size_t **dims** [ND_ARRAY_MAX_DIMS]
- size_t **nElements**
- size_t **totalBytes**
- int **bytesPerElement**
- NDColorMode_t **colorMode**
- NDDataType_t **dataType**
- std::string **codec**
-

```
struct {
    int dim
    size_t size
    size_t stride
} x
```

-

```
struct {
    int dim
    size_t size
    size_t stride
} y
```

-

```
struct {
    int dim
    size_t size
    size_t stride
} color
```

9.59.1 Detailed Description

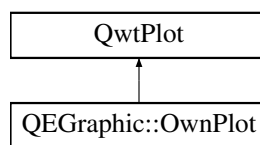
Copied from ntndArrayConverter.h.

The documentation for this class was generated from the following files:

- qeframework/protocol/QENTNDArrayConverter.h
- qeframework/protocol/QENTNDArrayConverter.cpp

9.60 QEGraphic::OwnPlot Class Reference

Inheritance diagram for QEGraphic::OwnPlot:



Public Member Functions

- **OwnPlot** ([QEGraphic](#) *parent=NULL)
- **OwnPlot** (const QwtText &title, [QEGraphic](#) *parent=NULL)

Protected Member Functions

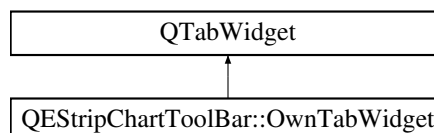
- void **drawCanvas** (QPainter *painter)

The documentation for this class was generated from the following file:

- qeframework/common/QEGraphic.cpp

9.61 QEStripChartToolBar::OwnTabWidget Class Reference

Inheritance diagram for QEStripChartToolBar::OwnTabWidget:



Public Member Functions

- **OwnTabWidget** ([QEStripChartToolBar](#) *parent)

Public Attributes

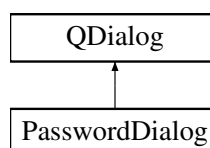
- QPushButton * **pushButtons** [NUMBER_OF_BUTTONS]
- QLabel * **yScaleStatus**
- QLabel * **timeStatus**
- QLabel * **durationStatus**
- QLabel * **numberOfOutstandingRequests**
- QLabel * **timeModeStatus**
- QComboBox * **predefinedComboBox**
- QPushButton * **loadButton**
- QPushButton * **saveAsButton**
- QLabel * **timeRefLabel**
- QLabel * **time1**
- QLabel * **time2**
- QLabel * **timeDeltaLabel**
- QLabel * **timeDelta**
- QLabel * **valueRefLabel**
- QLabel * **value1**
- QLabel * **value2**
- QLabel * **valueDelta1**
- QLabel * **value3**
- QLabel * **value4**
- QLabel * **valueDelta2**

The documentation for this class was generated from the following file:

- qeframework/widgets/QEStripChart/QEStripChartToolBar.cpp

9.62 PasswordDialog Class Reference

Inheritance diagram for PasswordDialog:



Public Member Functions

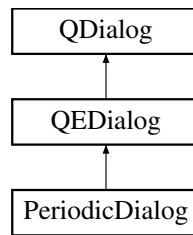
- **PasswordDialog** (QWidget *parent=0)

The documentation for this class was generated from the following files:

- qeframework/common/PasswordDialog.h
- qeframework/common/PasswordDialog.cpp

9.63 PeriodicDialog Class Reference

Inheritance diagram for PeriodicDialog:



Public Slots

- int **exec** (QWidget *targetWidget)

Public Member Functions

- **PeriodicDialog** (QWidget *parent=0)
- void **setColourised** (const bool colouriseIn)
- bool **isColourised** () const
- void **setElement** (QString elementIn, QList< bool > &enabledList)
- QString **getElementName** () const
- QString **getElement** () const
- int **getAtomicNumber** () const

Protected Member Functions

- void **changeEvent** (QEvent *e)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPeriodic/PeriodicDialog.h
- qeframework/widgets/QEPeriodic/PeriodicDialog.cpp

9.64 PersistenceManager Class Reference

Classes

- class [ResourceLocker](#)

Public Member Functions

- `QObject *` **getSaveRestoreObject** ()
- `void` **save** (const QString fileName, const QString rootName, const QString configName, const bool warnUser)
- `void` **restore** (const QString fileName, const QString rootName, const QString configName)
- `bool` **saveWidget** (QWidget *qewidget, const QString fileName, const QString rootName, const QString configName)
- `bool` **restoreWidget** (QWidget *qewidget, const QString fileName, const QString rootName, const QString configName)
- `bool` **isRestoring** () const
- `PMElement` **addNamedConfiguration** (QString name)
- `PMElement` **getNamedConfiguration** (QString name)
- `QStringList` **getConfigNames** (QString fileName, QString rootName)
- `QStringList` **getConfigNames** (QString fileName, QString rootName, bool &hasDefault)
- `void` **deleteConfigs** (const QString fileName, const QString rootName, const QStringList names, const bool warnUser)
- `bool` **isConfigurationPresent** (const QString fileName, const QString rootName, const QString configName)

Static Public Attributes

- static QString **defaultName**

Friends

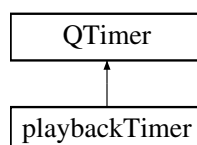
- class **PMElement**
- class **QEWWidget**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWWidget/persistenceManager.h
- qeframework/widgets/QEWWidget/persistenceManager.cpp

9.65 playbackTimer Class Reference

Inheritance diagram for playbackTimer:



Public Member Functions

- `playbackTimer` (recording *recorderIn)
- `void` **timerEvent** (QTimerEvent *event)

Public Attributes

- [recording](#) * **recorder**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/recording.h
- qeframework/widgets/QEImage/recording.cpp

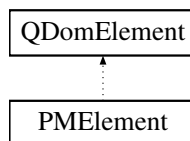
9.66 PMContext Class Reference

The documentation for this class was generated from the following file:

- qeframework/widgets/QEWidget/persistenceManager.h

9.67 PMElement Class Reference

Inheritance diagram for PMElement:



Public Member Functions

- **PMElement** ([PersistenceManager](#) *ownerIn, QDomElement elementIn)
- [PMElement](#) **addElement** (QString name)
- void **addValue** (QString name, bool value)
- void **addValue** (QString name, int value)
- void **addValue** (QString name, double value)
- void **addValue** (QString name, QString value)
- void **addValue** (QString name, const QColor &value)
- void **addAttribute** (QString name, bool value)
- void **addAttribute** (QString name, int value)
- void **addAttribute** (QString name, double value)
- void **addAttribute** (QString name, QString value)
- [PMElement](#) **getElement** (QString name) const
- [PMElement](#) **getElement** (QString name, int i) const
- [PMElement](#) **getElement** (QString name, QString attrName, QString attrValue) const
- [PMElement](#) **getElement** (QString name, QString attrName, int attrValue) const
- [PMElementList](#) **getElementList** (QString name) const
- bool **getValue** (QString name, bool &val) const
- bool **getValue** (QString name, int &val) const
- bool **getValue** (QString name, double &val) const
- bool **getValue** (QString name, QString &val) const
- bool **getValue** (QString name, QColor &val) const

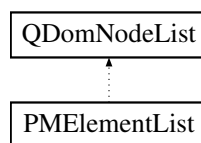
- bool **getAttribute** (QString name, bool &val) const
- bool **getAttribute** (QString name, int &val) const
- bool **getAttribute** (QString name, double &val) const
- bool **getAttribute** (QString name, QString &val) const
- bool **isNull** ()

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/persistenceManager.h
- qeframework/widgets/QEWidget/persistenceManager.cpp

9.68 PMLElementList Class Reference

Inheritance diagram for PMLElementList:



Public Member Functions

- **PMLElementList** ([PersistenceManager](#) *ownerIn, QDomNodeList elementListIn)
- [PMElement](#) **getElement** (int i) const
- int **count** () const

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/persistenceManager.h
- qeframework/widgets/QEWidget/persistenceManager.cpp

9.69 pointInfo Class Reference

Public Member Functions

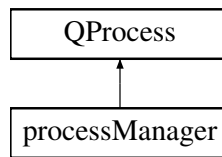
- void **setX** (long x)
- void **setY** (long y)
- void **setPoint** (QPoint pIn)
- void **clearX** ()
- void **clearY** ()
- bool **getStatus** ()
- QPoint **getPoint** ()

The documentation for this class was generated from the following file:

- qeframework/widgets/QEImage/QEImage.h

9.70 processManager Class Reference

Inheritance diagram for processManager:



Public Slots

- void **doRead** ()
- void **doReadToStandardOutput** ()
- void **doReadToStandardError** ()
- void **doFinished** (int exitCode, QProcess::ExitStatus exitStatus)

Signals

- void **processCompleted** ()

Public Member Functions

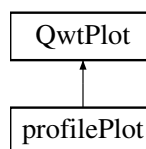
- **processManager** (bool logOutput, bool useStandardIo, QTemporaryFile *tempFileIn)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/applicationLauncher.h
- qeframework/widgets/QEWidget/applicationLauncher.cpp

9.71 profilePlot Class Reference

Inheritance diagram for profilePlot:



Public Types

- enum **plotDirections** { **PROFILEPLOT_LR** , **PROFILEPLOT_RL** , **PROFILEPLOT_TB** , **PROFILEPLOT↔
_BT** }

Public Member Functions

- **profilePlot** (plotDirections plotDirection)
- void **enableProfileAxes** (const bool enable)
- void **setProfile** (QVector< QPointF > *profile, double minX, double maxX, double minY, double maxY, QString title, QPoint start, QPoint end, unsigned int thicknessIn)
- void **clearProfile** ()

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/profilePlot.h
- qeframework/widgets/QEImage/profilePlot.cpp

9.72 ProfilePublisher Class Reference

Public Member Functions

- **ProfilePublisher** ([QEWidget](#) *owner, const QString &prioritySubstitutions)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/ContainerProfile.h
- qeframework/widgets/QEWidget/ContainerProfile.cpp

9.73 PushButtonSpecifications Struct Reference

Public Attributes

- int **side**
- int **gap**
- int **width**
- bool **isIcon**
- const QString **captionOrIcon**
- const QString **toolTip**
- const char * **member**

The documentation for this struct was generated from the following file:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSave.cpp

9.74 QEArchiveAccess::PVDataRequests Struct Reference

Public Attributes

- `QObject *` **userData**
- `int` **metaRequest**
- `int` **key**
- `QString` **pvName**
- `QCaDateTime` **startTime**
- `QCaDateTime` **endTime**
- `int` **count**
- `QEArchiveInterface::How` **how**
- `unsigned int` **element**

The documentation for this struct was generated from the following file:

- `qeframework/archive/QEArchiveAccess.h`

9.75 QEArchiveAccess::PVDataResponses Struct Reference

Public Attributes

- `QObject *` **userData**
- `int` **metaRequest**
- `bool` **isSuccess**
- `QCaDataPointList` **pointsList**
- `QString` **pvName**
- `QString` **supplementary**

The documentation for this struct was generated from the following file:

- `qeframework/archive/QEArchiveAccess.h`

9.76 QEArchiveInterface::PVName Struct Reference

Public Attributes

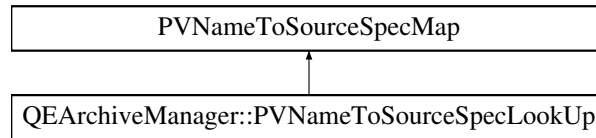
- `QString` **pvName**
- `QCaDateTime` **startTime**
- `QCaDateTime` **endTime**

The documentation for this struct was generated from the following file:

- `qeframework/archive/QEArchiveInterface.h`

9.77 QEArchiveManager::PVNameToSourceSpecLookUp Class Reference

Inheritance diagram for QEArchiveManager::PVNameToSourceSpecLookUp:



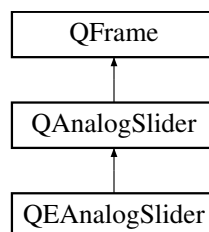
The documentation for this class was generated from the following file:

- qeframework/archive/QEArchiveManager.cpp

9.78 QAnalogSlider Class Reference

```
#include <QAnalogSlider.h>
```

Inheritance diagram for QAnalogSlider:



Public Slots

- void **setValue** (const double [value](#))
- void **setValue** (const int [value](#))

Signals

- void **valueChanged** (const double [value](#))
- void **valueChanged** (const int [value](#))
- void **valueEdited** (const double [value](#))
- void **valueEdited** (const int [value](#))
- void **appliedValue** (const double [value](#))
- void **appliedValue** (const int [value](#))

Public Member Functions

- [QAnalogSlider](#) (QWidget *parent=0)
- virtual [~QAnalogSlider](#) ()
 - Destruction.*
- QSize **sizeHint** () const
- void **setColourBandList** (const [QEColourBandList](#) &bandList)
- [QEColourBandList](#) **getColourBandList** () const
- double **getValue** () const
- void **setPrecision** (const int precision)
 - Access function for [precision](#) - refer to [precision](#) property for details.*
- int **getPrecision** () const
 - Access function for [precision](#) - refer to [precision](#) property for details.*
- void **setMinimum** (const double minimum)
 - Access function for [minimum](#) - refer to [minimum](#) property for details.*
- double **getMinimum** () const
 - Access function for [minimum](#) - refer to [minimum](#) property for details.*
- void **setMaximum** (const double maximum)
 - Access function for [maximum](#) - refer to [maximum](#) property for details.*
- double **getMaximum** () const
 - Access function for [maximum](#) - refer to [maximum](#) property for details.*
- void **setMinorInterval** (const double minorInterval)
 - Access function for [minorInterval](#) - refer to [minorInterval](#) property for details.*
- double **getMinorInterval** () const
 - Access function for [minorInterval](#) - refer to [minorInterval](#) property for details.*
- void **setMajorInterval** (const double majorInterval)
 - Access function for [majorInterval](#) - refer to [majorInterval](#) property for details.*
- double **getMajorInterval** () const
 - Access function for [majorInterval](#) - refer to [majorInterval](#) property for details.*
- void **setLeftText** (const QString &leftText)
 - Access function for [leftText](#) - refer to [leftText](#) property for details.*
- QString **getLeftText** () const
 - Access function for [leftText](#) - refer to [leftText](#) property for details.*
- void **setCentreText** (const QString ¢reText)
 - Access function for [#centreText](#) - refer to [#centreText](#) property for details.*
- QString **getCentreText** () const
 - Access function for [#centreText](#) - refer to [#centreText](#) property for details.*
- void **setRightText** (const QString &rightText)
 - Access function for [#rightText](#) - refer to [#rightText](#) property for details.*
- QString **getRightText** () const
 - Access function for [#rightText](#) - refer to [#rightText](#) property for details.*
- void **setShowSaveRevert** (const bool show)
- bool **getShowSaveRevert** () const
- void **setShowApply** (const bool show)
- bool **getShowApply** () const
- void **setFontColour** (const QColor fontColour)
 - Access function for [fontColour](#) - refer to [fontColour](#) property for details.*
- QColor **getFontColour** () const
 - Access function for [fontColour](#) - refer to [fontColour](#) property for details.*
- void **setIsActive** (const bool value)
- bool **getIsActive** () const
- void **setInvertedAppearance** (const bool invertedAppearance)

- *Access function for [invertedAppearance](#).*
- bool [getInvertedAppearance](#) () const
- *Access function for [invertedAppearance](#).*
- void **setSavedValue** (const double savedValue)
- double **getSavedValue** () const

Protected Slots

- virtual void **applyButtonClicked** (bool)

Protected Member Functions

- [QEAxisPainter](#) * **getAxisPainter** ()
- void **updateAxisAndSlider** ()

Properties

- double [value](#)
Specifies the current value.
- int [precision](#)
- double [minimum](#)
Specifies the minimum allowed value.
- double [maximum](#)
Specifies the maximum allowed value.
- double [minorInterval](#)
- double [majorInterval](#)
- bool [invertedAppearance](#)
- bool [tracking](#)
- QString [leftText](#)
Controls the left, centre and right texts.
- QString **centreText**
- QString **rightText**
- bool [showSaveRevert](#)
Enables/disables the save-revert capability.
- bool [showApply](#)
Enables/disables the apply value capability.
- QColor [fontColour](#)
- bool [isActive](#)
Alternative to isEnabled. Default is true.

9.78.1 Detailed Description

[QAnalogSlider](#) is a non EPICS aware slider that provides an analog equivalent of the QSlider. It is deemed analog as it can be set by/emits floating point (double) values as opposed to integer values. It is also decorated with a scale and text showing the current value; it also provides a local save and restore capability.

9.78.2 Constructor & Destructor Documentation

9.78.2.1 QAnalogSlider()

```
QAnalogSlider::QAnalogSlider (  
    QWidget * parent = 0 ) [explicit]
```

Create with default title.

9.78.3 Property Documentation

9.78.3.1 fontColour

```
QColor QAnalogSlider::fontColour [read], [write]
```

Font colour

9.78.3.2 invertedAppearance

```
bool QAnalogSlider::invertedAppearance [read], [write]
```

Inverted apperance. Default (false) is left ot right.

9.78.3.3 majorInterval

```
double QAnalogSlider::majorInterval [read], [write]
```

Minor scale interval. Only applies for linear scale (not log scale)

9.78.3.4 minorInterval

```
double QAnalogSlider::minorInterval [read], [write]
```

Minor scale interval. Only applies for linear scale (not log scale)

9.78.3.5 precision

```
int QAnalogSlider::precision [read], [write]
```

Precision used for the display and editing of numbers. The default is 2. Strictly speaking, this should be an unsigned int, but the designer int properties editor much 'nicer'.

9.78.3.6 tracking

```
bool QAnalogSlider::tracking [read], [write]
```

Controls when valueChanged signal is emitted. If tracking is enabled (the default), the slider emits the valueChanged () signal while the slider is being dragged. If tracking is disabled, the slider emits the valueChanged () signal only when the user releases the slider.

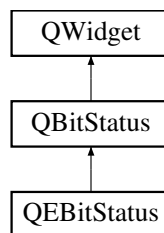
The documentation for this class was generated from the following files:

- qeframework/widgets/QEAnalogSlider/QAnalogSlider.h
- qeframework/widgets/QEAnalogSlider/QAnalogSlider.cpp

9.79 QBitStatus Class Reference

```
#include <QBitStatus.h>
```

Inheritance diagram for QBitStatus:



Public Types

- enum [Shapes](#) { [Rectangle](#) , [Circle](#) }

Public Slots

- void **setValue** (const int value)

Public Member Functions

- **QBitStatus** (QWidget *parent=0)
- virtual QSize **sizeHint** () const
- void **setBorderColour** (const QColor value)
- QColor **getBorderColour** () const
- void **setOnColour** (const QColor value)
- QColor **getOnColour** () const
- void **setOffColour** (const QColor value)
- QColor **getOffColour** () const
- void **setInvalidColour** (const QColor value)
- QColor **getInvalidColour** () const
- void **setClearColour** (const QColor value)
- QColor **getClearColour** () const

- void **setDrawBorder** (const bool value)
- bool **getDrawBorder** () const
- void **setNumberOfBits** (const int value)
- int **getNumberOfBits** () const
- void **setGap** (const int value)
- int **getGap** () const
- void **setShift** (const int value)
- int **getShift** () const
- void **setOnClearMask** (const QString value)
- QString **getOnClearMask** () const
- void **setOffClearMask** (const QString value)
- QString **getOffClearMask** () const
- void **setReversePolarityMask** (const QString value)
- QString **getReversePolarityMask** () const
- void **setIsValid** (const bool value)
- bool **getIsValid** () const
- void **setOrientation** (const Qt::Orientation value)
- Qt::Orientation **getOrientation** () const
- void **setInvertedAppearance** (const bool invertedAppearance)
- bool **getInvertedAppearance** () const
- void **setShape** (const [Shapes](#) value)
- [Shapes](#) **getShape** () const
- void **setUnsetText** (const QString value)
- QString **getUnsetText** () const
- void **setSetText** (const QString value)
- QString **getSetText** () const
- int **getValue** () const

Protected Member Functions

- void **paintEvent** (QPaintEvent *event)
- void **setIsActive** (const bool value)
- bool **getIsActive** () const

Properties

- int **value**
- int **numberOfBits**
- int **shift**
- Qt::Orientation [orientation](#)
- bool **invertedAppearance**
- [Shapes](#) **shape**
- int **gap**
- QString **reversePolarityMask**
- QString **onClearMask**
- QString **offClearMask**
- QColor **boarderColour**
- QColor **invalidColour**
- QColor **onColour**
- QColor **offColour**
- QColor **clearColour**
- bool **drawBorder**
- bool **isValid**
- bool **isActive**
- QString [unsetText](#)
- QString **setText**

9.79.1 Detailed Description

This class is a BitStatus widget based on directly QWidget. It provides a non-EPICS aware bit status widget.

9.79.2 Member Enumeration Documentation

9.79.2.1 Shapes

```
enum QBitStatus::Shapes
```

Specified the displayed bit shape

9.79.3 Property Documentation

9.79.3.1 orientation

```
Qt::Orientation QBitStatus::orientation [read], [write]
```

For orientations/invertedAppearance Default: Horizontal Default: false, i.e. LSB on right if Horizontal, LSB in top if Vertical.

9.79.3.2 unsetText

```
QString QBitStatus::unsetText [read], [write]
```

Both default to ""

The documentation for this class was generated from the following files:

- qeframework/widgets/QEBitStatus/QBitStatus.h
- qeframework/widgets/QEBitStatus/QBitStatus.cpp

9.80 QCaAlarmInfo Class Reference

Public Types

- typedef unsigned short **Status**
- typedef unsigned short **Severity**

Public Member Functions

- **QCaAlarmInfo** (const [QCaAlarmInfo](#) &other)
- **QCaAlarmInfo** (const Status status, const Severity severity)
- **QCaAlarmInfo** (const QEPvNameUri::Protocol protocol, const QString &pvName, const Status status, const Severity severity, const QString &message)
- [QCaAlarmInfo](#) & **operator=** (const [QCaAlarmInfo](#) &other)
- bool **operator==** (const [QCaAlarmInfo](#) &other) const
- bool **operator!=** (const [QCaAlarmInfo](#) &other) const
- QString **statusName** () const
- QString **severityName** () const
- QString **messageText** () const
- bool **isInAlarm** () const
- bool **isMinor** () const
- bool **isMajor** () const
- bool **isInvalid** () const
- bool **isOutOfService** () const
- QString **style** () const
- QString **getStyleColorName** () const
- QString **getColorName** () const
- Severity **getSeverity** () const
- Status **getStatus** () const

Static Public Member Functions

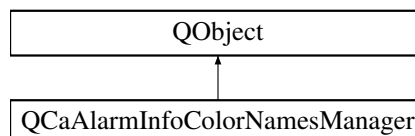
- static Severity **getInvalidSeverity** ()

The documentation for this class was generated from the following files:

- qeframework/data/QCaAlarmInfo.h
- qeframework/data/QCaAlarmInfo.cpp

9.81 QCaAlarmInfoColorNamesManager Class Reference

Inheritance diagram for QCaAlarmInfoColorNamesManager:



Public Types

- enum **ColorNameKinds** { **cnkPrimary** , **cnkOverride** }

Static Public Member Functions

- static void **setStyleColorNames** (const ColorNameKinds kind, const QStringList &styleColorNames)
- static QStringList **getStyleColorNames** (const ColorNameKinds kind)
- static void **setColorNames** (const ColorNameKinds kind, const QStringList &colorNames)
- static QStringList **getColorNames** (const ColorNameKinds kind)
- static QStringList **getDefaultStyleColorNames** ()
- static QStringList **getDefaultColorNames** ()
- static QStringList **getInUseStyleColorNames** ()
- static QStringList **getInUseColorNames** ()
- static void **setOosPvNameList** (const QStringList &pvNameList)
- static QStringList **getOosPvNameList** ()
- static void **clearOosPvNameList** ()

Friends

- class **QCaAlarmInfo**

The documentation for this class was generated from the following files:

- qeframework/data/QCaAlarmInfo.h
- qeframework/data/QCaAlarmInfo.cpp

9.82 QCaConnectionInfo Class Reference

Public Types

- enum **channel_states** { **NEVER_CONNECTED** , **CONNECTED** , **CLOSED** , **CHANNEL_UNKNOWN** }

Public Member Functions

- **QCaConnectionInfo** (const channel_states channelStateIn, const QString &pvName)
- bool **isChannelConnected** () const
- QString **variable** () const

The documentation for this class was generated from the following files:

- qeframework/data/QCaConnectionInfo.h
- qeframework/data/QCaConnectionInfo.cpp

9.83 QCaDataPoint Class Reference

```
#include <QCaDataPoint.h>
```

Public Member Functions

- **QCaDataPoint** (const [QCaDataPoint](#) &other)
- [QCaDataPoint](#) & **operator=** (const [QCaDataPoint](#) &other)
- bool **isDisplayable** () const
- QString **toString** () const
- QString **toString** (const [QCaDateTime](#) &originDateTime) const

Static Public Member Functions

- static bool **registerMetaTypes** ()

Public Attributes

- double **value**
- [QCaDateTime](#) **datetime**
- [QCaAlarmInfo](#) **alarm**

9.83.1 Detailed Description

This class used to hold a single data point. Objects of this type are intended for use QCaStripChart in particular, but also for the interface to the Channel Access and Archive Appliance archives.

The documentation for this class was generated from the following files:

- qeframework/data/QCaDataPoint.h
- qeframework/data/QCaDataPoint.cpp

9.84 QCaDataPointList Class Reference

```
#include <QCaDataPoint.h>
```

Classes

- struct [Statistics](#)

Public Member Functions

- void **reserve** (const int size)
- void **clear** ()
- void **removeLast** ()
- void **removeFirst** ()
- void **append** (const [QCaDataPointList](#) &other)
- void **append** (const [QCaDataPoint](#) &other)
- void **prepend** (const [QCaDataPoint](#) &other)
- void **replace** (const int i, const [QCaDataPoint](#) &t)
- int **count** () const
- void **removeFirstItems** (const int n)
- [QCaDataPoint](#) **value** (const int j) const
- [QCaDataPoint](#) **first** () const
- [QCaDataPoint](#) **last** () const
- void **truncate** (const int position)
- int **indexBeforeTime** (const [QCaDateTime](#) &searchTime, const int defaultIndex) const
- const [QCaDataPoint](#) * **findNearestPoint** (const [QCaDateTime](#) &searchTime) const
- void **resample** (const [QCaDataPointList](#) &source, const double interval, const [QCaDateTime](#) &endTime)
- void **compact** (const [QCaDataPointList](#) &source)
- void **toStream** (QTextStream &target, bool withIndex, bool withRelativeTime) const
- bool **calculateStatistics** ([Statistics](#) &statistics, const bool extendToTimeNow) const
- void **distribute** (double distribution[], const int size, const bool extendToTimeNow, const double first, const double increment) const

9.84.1 Detailed Description

Defines a list of data points.

Note this class originally extended `QList<QCaDataPoint>`, but this way of specifying this class has issues with the Windows Visual Studio Compiler. It has now been modified to include a `QList<QCaDataPoint>` member. The downside of this is that we must now provide list member access functions.

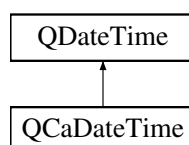
The documentation for this class was generated from the following files:

- qeframework/data/QCaDataPoint.h
- qeframework/data/QCaDataPoint.cpp

9.85 QCaDateTime Class Reference

```
#include <QCaDateTime.h>
```

Inheritance diagram for QCaDateTime:



Public Member Functions

- **QCaDateTime** (const [QCaDateTime](#) &other)
- **QCaDateTime** (const QDateTime &dt)
- **QCaDateTime** (const unsigned long seconds, const unsigned long nanoseconds, const int userTag=0)
- [QCaDateTime](#) & **operator=** (const [QCaDateTime](#) &other)
- QString **text** () const
- QString **ISOText** () const
- [QCaDateTime](#) **addSeconds** (const double seconds) const
- double **secondsTo** (const QDateTime &target) const
- unsigned long **getSeconds** () const

Recover original EPICS time constructor parameters.

- unsigned long **getNanoSeconds** () const
- int **getUserTag** () const

9.85.1 Detailed Description

Extends the Qt datetime object in order to provide nSec precision

9.85.2 Member Function Documentation

9.85.2.1 addSeconds()

```
QCaDateTime QCaDateTime::addSeconds (
    const double seconds ) const
```

Equivalent of addSecs and secsTo in base class, save that we specify the seconds as a floating point number, i.e. take into account fractions of a second (i.e. mSec but not nano sec).

The documentation for this class was generated from the following files:

- qeframework/data/QCaDateTime.h
- qeframework/data/QCaDateTime.cpp

9.86 QCalInfo Class Reference

Class to pass summary information about a QEChannel.

```
#include <QEWidget.h>
```

Public Types

- enum **ACCESS_MODES** { **NO_ACCESS** , **READ_ONLY** , **READ_WRITE** , **UNKNOWN** }

Public Member Functions

- **QCInfo** (QString variableIn, QString typeIn, QString valueIn, QString severityIn, QString statusIn, QString hostIn, int precisionIn, int precisionUserIn, double alarmUserMinIn, double alarmUserMaxIn, double controlLimitLowerIn, double controlLimitUpperIn, double alarmLimitLowerIn, double alarmLimitUpperIn, double warningLimitLowerIn, double warningLimitUpperIn, double driveLimitLowIn, double driveLimitHighIn, bool alarmSensitiveIn, ACCESS_MODES accessModeIn)

Public Attributes

- QString **variable**
- QString **type**
- QString **value**
- QString **severity**
- QString **status**
- QString **host**
- int **precision**
- int **precisionUser**
- double **alarmUserMin**
- double **alarmUserMax**
- double **controlLimitLower**
- double **controlLimitUpper**
- double **alarmLimitLower**
- double **alarmLimitUpper**
- double **warningLimitLower**
- double **warningLimitUpper**
- double **driveLimitLow**
- double **driveLimitHigh**
- bool **alarmSensitive**
- ACCESS_MODES **accessMode**

9.86.1 Detailed Description

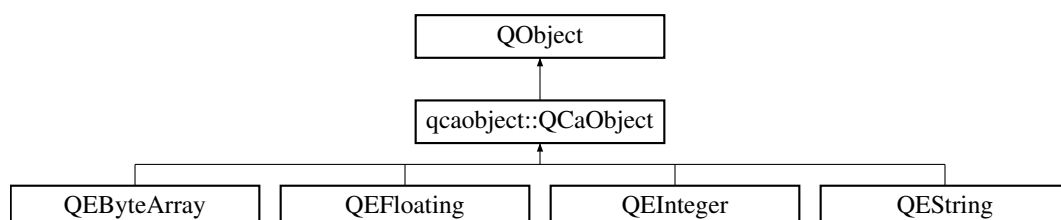
Class to pass summary information about a QEChannel.

The documentation for this class was generated from the following file:

- qeframework/widgets/QEWidget/QEWidget.h

9.87 qcaobject::QCaObject Class Reference

Inheritance diagram for qcaobject::QCaObject:



Public Types

- enum **SignalsToSend** { **SIG_NONE** = 0x00 , **SIG_VARIANT** = 0x01 , **SIG_BYTEARRAY** = 0x02 }
- enum **priorities** { **QE_PRIORITY_LOW** = 0 , **QE_PRIORITY_NORMAL** = 10 , **QE_PRIORITY_HIGH** = 20 }
- typedef quint64 **ObjectIdentity**

Public Slots

- bool **writeData** (const QVariant &value)
- bool **writeDataElement** (const QVariant &elementValue)
- void **resendLastData** ()

Signals

- void **connectionUpdated** (const [QEConnectionUpdate](#) &)
- void **valueUpdated** (const [QEVariantUpdate](#) &)
- void **byteArrayUpdated** (const [QEByteArrayUpdate](#) &)
- void **dataChanged** (const QVariant &, [QCaAlarmInfo](#) &, [QCaDateTime](#) &, const unsigned int &)
- void **byteArrayChanged** (const QByteArray &, unsigned long, [QCaAlarmInfo](#) &, [QCaDateTime](#) &, const unsigned int &)
- void **connectionChanged** ([QCaConnectionInfo](#) &, const unsigned int &)

Public Member Functions

- **QCaObject** (const QString &pvName, QObject *parent, const unsigned int variableIndex, SignalsToSend↵ Flags signalsToSend=SIG_VARIANT, priorities priorityIn=QE_PRIORITY_NORMAL)
- **QCaObject** (const QString &pvName, QObject *parent, const unsigned int variableIndex, [UserMessage](#) *userMessageIn, SignalsToSendFlags signalsToSend=SIG_VARIANT, priorities priorityIn=QE_PRIORITY↵ _NORMAL)
- bool **isCaChannel** () const
- bool **isPvaChannel** () const
- void **setSignalsToSend** (const SignalsToSendFlags signalsToSend)
- QCaObject::SignalsToSendFlags **getSignalsToSend** () const
- bool **subscribe** ()
- bool **singleShotRead** ()
- bool **connectChannel** ()
- void **closeChannel** ()
- bool **dataTypeKnown** () const
- unsigned int **getVariableIndex** () const
- void **setUserMessage** ([UserMessage](#) *userMessageIn)
- void **enableWriteCallbacks** (bool enable)
- bool **isWriteCallbacksEnabled** () const
- void **setRequestedElementCount** (unsigned int elementCount)
- QString **getPvName** () const
- QString **getRecordName** () const
- QString **getEgu** () const
- QStringList **getEnumerations** () const
- unsigned int **getPrecision** () const
- [QCaAlarmInfo](#) **getAlarmInfo** () const
- [QCaDateTime](#) **getDateTime** () const
- QString **getDescription** () const
- double **getDisplayLimitUpper** () const

- double **getDisplayLimitLower** () const
- double **getAlarmLimitUpper** () const
- double **getAlarmLimitLower** () const
- double **getWarningLimitUpper** () const
- double **getWarningLimitLower** () const
- double **getControlLimitUpper** () const
- double **getControlLimitLower** () const
- QString **getHostName** () const
- QString **getFieldType** () const
- unsigned long **getHostElementCount** () const
- unsigned long **getDataElementCount** () const
- Q_DECL_DEPRECATED unsigned long **getElementCount** () const
- bool **getReadAccess** () const
- bool **getWriteAccess** () const
- void **setArrayIndex** (const int index)
- int **getArrayIndex** () const
- void **getLastData** (bool &isDefined, QVariant &value, [QCaAlarmInfo](#) &alarmInfo, [QCaDateTime](#) &timeStamp) const
- bool **getChannellsConnected** () const
- bool **getDatalsAvailable** () const
- bool **getIsMetaDataUpdate** () const
- QString **getStringValue** (const [QE::ArrayActions](#) arrayAction=[QE::Index](#)) const
- bool **getBooleanValue** () const
- long **getIntegerValue** () const
- double **getFloatingValue** () const
- QVector< bool > **getBooleanArray** () const
- QVector< long > **getIntegerArray** () const
- QVector< double > **getFloatingArray** () const
- void **writeStringValue** (const QString &value)
- void **writeBooleanValue** (const bool value)
- void **writeIntegerValue** (const long value)
- void **writeFloatingValue** (const double value)
- void **writeBooleanArray** (const QVector< bool > &array)
- void **writeIntegerArray** (const QVector< long > &array)
- void **writeFloatingArray** (const QVector< double > &array)
- ObjectIdentity **getObjectIdentity** () const

Static Public Member Functions

- static int * **getDisconnectedCountRef** ()
- static int * **getConnectedCountRef** ()
- static ObjectIdentity **nullObjectIdentity** ()

Protected Member Functions

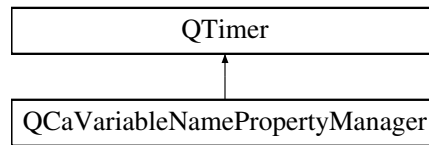
- void **checkDeprecatedSignalUsage** (const char *varSignal)

The documentation for this class was generated from the following files:

- qeframework/data/QCaObject.h
- qeframework/data/QCaObject.cpp

9.88 QCaVariableNamePropertyManager Class Reference

Inheritance diagram for QCaVariableNamePropertyManager:



Signals

- void **newPvNameProperties** (const [QEPvNameProperties](#) &pvNameProperties)
- void **newVariableNameProperty** (QString variable, QString substitutions, unsigned int variableIndex)

Public Member Functions

- QString **getVariableNameProperty** () const
- void **setVariableNameProperty** (QString variableNamePropertyIn)
- QString **getSubstitutionsProperty** () const
- void **setSubstitutionsProperty** (QString substitutionsPropertyIn)
- void **setVariableIndex** (unsigned int variableIndexIn)
- unsigned int **getVariableIndex** () const

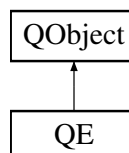
The documentation for this class was generated from the following files:

- qeframework/data/QCaVariableNamePropertyManager.h
- qeframework/data/QCaVariableNamePropertyManager-wip.cpp
- qeframework/data/QCaVariableNamePropertyManager.cpp

9.89 QE Class Reference

```
#include <QEEnums.h>
```

Inheritance diagram for QE:



Public Types

- enum [MessageFilterOptions](#) { **Any** = 0 , **Match** , **None** }
- enum [UserLevels](#) { **User** = 0 , **Scientist** , **Engineer** }
- enum [DisplayAlarmStateOptions](#) { **Always** = 0 , **WhenInAlarm** , **WhenInvalid** , **Never** }
- enum [Formats](#) {
[Default](#) = 0 , [Floating](#) , [Integer](#) , [UnsignedInteger](#) ,
[Time](#) , [LocalEnumeration](#) , [String](#) }
- enum [Separators](#) { [NoSeparator](#) = 0 , [Comma](#) , [Underscore](#) , [Space](#) }
- enum [Notations](#) { [Fixed](#) = 0 , [Scientific](#) , [Automatic](#) }
- enum [ArrayActions](#) { [Append](#) = 0 , [Ascii](#) , [Index](#) }
- enum [Radicies](#) { [Decimal](#) = 0 , [Hexadecimal](#) , [Octal](#) , [Binary](#) }
- enum [UpdateOptions](#) {
[NoUpdate](#) = 0 , [Text](#) = 1 , [Icon](#) = 2 , [TextAndIcon](#) = 3 ,
[State](#) = 4 , [TextAndState](#) = 5 , [IconAndState](#) = 6 , [TextIconAndState](#) = 7 }
- enum [DisabledRecordPolicy](#) { [ignore](#) = 0 , [grayout](#) , [disable](#) }
- enum [ProgramStartupOptions](#) { [NoOutput](#) = 0 , [Terminal](#) , [LogOutput](#) , [StdOutput](#) }
- enum [CreationOptions](#) {
[Open](#) = 0 , [NewTab](#) , [NewWindow](#) , [DockTop](#) ,
[DockBottom](#) , [DockLeft](#) , [DockRight](#) , [DockTopTabbed](#) ,
[DockBottomTabbed](#) , [DockLeftTabbed](#) , [DockRightTabbed](#) , [DockFloating](#) }
- enum [VideoModes](#) { [normal](#) , [reverse](#) }
- enum [DropOption](#) { [DropToVariable](#) = 0 , [DropToText](#) , [DropToTextAndWrite](#) }
- enum [MouseMoveSignals](#) { [signalNone](#) = 0x0000 , [signalStatus](#) = 0x0001 , [signalData](#) = 0x0002 , [signalText](#) = 0x0004 }
- enum [ImageFormatOptions](#) {
[Mono](#) = 0 , [Bayer](#) , [BayerGB](#) , [BayerBG](#) ,
[BayerGR](#) , [BayerRG](#) , [rgb1](#) , [rgb2](#) ,
[rgb3](#) , [yuv444](#) , [yuv422](#) , [yuv421](#) }
- enum [RotationOptions](#) { [NoRotation](#) = 0 , [Rotate90Right](#) , [Rotate90Left](#) , [Rotate180](#) }
- enum [PVLabelMode](#) { [usePvName](#) = 0 , [useAliasName](#) , [useDescription](#) }
- enum [LayoutOptions](#) { **Top** = 0 , **Bottom** , **Left** , **Right** }
- enum [SourceOptions](#) { [SourceText](#) = 0 , [SourceFile](#) }
- enum [GridOrders](#) { [rowMajor](#) = 0 , [colMajor](#) }

Static Public Attributes

- static const int **numberOfImageFormats** = int ([QE::yuv421](#)) + 1

9.89.1 Detailed Description

This class defines common widget property types. This will apply to the EPICS Qt 4 series

This class is not intended to be inherited by any other class. The use of the class, as opposed to a just name space, is to allow moc to generate the associated meta data. The advantage of this is two-fold, firstly it simplifies all the header files in that they all do not need to redefine the enums; and secondly if the widget hierarchy is ever re-organised, the enumeration names in the ui file will remain constant.

Roughly the equivalent of `qnamespace.h`, without the clever namespace/object compile time selection.

9.89.2 Member Enumeration Documentation

9.89.2.1 ArrayActions

enum [QE::ArrayActions](#)

User friendly enumerations for arrayAction property - refer to arrayActions for details.

Enumerator

Append	Interpret each element in the array as a value and append string representations of each element from the array with a space in between each.
Ascii	Interpret each element from the array as a character in a string. Translate all non printing characters to '?' except for trailing zeros (ignore them).
Index	Interpret the element selected by <code>setArrayIndex()</code> as the value.

9.89.2.2 CreationOptions

enum `QE::CreationOptions`

Enumerator

Open	Replace the current GUI with the new GUI.
NewTab	Open new GUI in a new tab.
NewWindow	Open new GUI in a new window.
DockTop	Open new GUI in a top dock window.
DockBottom	Open new GUI in a bottom dock window.
DockLeft	Open new GUI in a left dock window.
DockRight	Open new GUI in a right dock window.
DockTopTabbed	Open new GUI in a top dock window (tabbed with any existing dock in that area)
DockBottomTabbed	Open new GUI in a bottom dock window (tabbed with any existing dock in that area)
DockLeftTabbed	Open new GUI in a left dock window (tabbed with any existing dock in that area)
DockRightTabbed	Open new GUI in a right dock window (tabbed with any existing dock in that area)
DockFloating	Open new GUI in a floating dock window.

9.89.2.3 DisabledRecordPolicy

enum `QE::DisabledRecordPolicy`

Specifies how a (control) widget should behave if/when the underlying record becomes disabled (i.e. DISA and DISV fields become equal).

Enumerator

ignore	No change in appearance - the default.
grayout	Grayed out apperance, i.e. same as when the widget's PV is disconnected.
disable	Widget is disabled.

9.89.2.4 DisplayAlarmStateOptions

```
enum QE::DisplayAlarmStateOptions
```

User friendly enumerations for #displayAlarmStateOption property - refer to #displayAlarmStateOption property and displayAlarmStateOptions enumeration for details.

Enumerator

Always	Always display the alarm state.
WhenInAlarm	Display the alarm state in alarm, i.e. severity $\geq$ MINOR.
WhenInvalid	Display the alarm state invalid, i.e. severity = INVALID.
Never	Never display the alarm state.

9.89.2.5 DropOption

```
enum QE::DropOption
```

Allow drag/drops mode to this widget.

Enumerator

DropToVariable	the dropped text will be used as a new variable name.
DropToText	just set text on the widget
DropToTextAndWrite	write the text to the PV if the PV has been connected otherwise equivalent to DropToText

9.89.2.6 Formats

```
enum QE::Formats
```

User friendly enumerations for format property - refer to formats for details.

Enumerator

Default	Format as best appropriate for the data type.
Floating	Format as a floating point number.
Integer	Format as an integer.
UnsignedInteger	Format as an unsigned integer.
Time	Format as a time, i.e. days hh:mm:ss[.frac].
LocalEnumeration	Format as a selection from the #localEnumeration property.
String	Format as a string.

9.89.2.7 GridOrders

enum [QE::GridOrders](#)

GridOrders specifies how grid elements are layed out.

Row major (example 10, items, 3 cols): 0 1 2 3 4 5 6 7 8 9 - -

Col major (example 10, items, 3 cols): 0 4 8 1 5 9 2 6 - 3 7 -

9.89.2.8 ImageFormatOptions

enum [QE::ImageFormatOptions](#)

User friendly enumerations for #formatOption property - refer to #formatOption property and #formatOptions enumeration for details.

Enumerator

Mono	Grey scale.
Bayer	Colour (Bayer Red Green)
BayerGB	Colour (Bayer Green Blue)
BayerBG	Colour (Bayer Blue Green)
BayerGR	Colour (Bayer Green Red)
BayerRG	Colour (Bayer Red Green)
rgb1	Colour (24 bit RGB)
rgb2	Colour (??? bit RGB)
rgb3	Colour (??? bit RGB)
yuv444	Colour (???)
yuv422	Colour (???)
yuv421	Colour (???)

9.89.2.9 MessageFilterOptions

enum [QE::MessageFilterOptions](#)

Message properties Not all [QE](#) widgets need these properties (only those that do something with messages - like the logging widget)

9.89.2.10 MouseMoveSignals

enum [QE::MouseMoveSignals](#)

Mouse move signal selection options.

Enumerator

signalNone	Can't use literal 0 with new Qt and/or compilers.
signalStatus	signals row, col and value as status text via sendMessage
signalData	signals row, col and value emitted as binary data
signalText	signals row, col and value emitted as as text

9.89.2.11 Notations

```
enum QE::Notations
```

User friendly enumerations for notation property - refer to notations for details.

Enumerator

Fixed	Standard floating point, e.g. 123456.789.
Scientific	Scientific representation, e.g. 1.23456789e6.
Automatic	Automatic choice of standard or scientific notation.

9.89.2.12 ProgramStartupOptions

```
enum QE::ProgramStartupOptions
```

Startup options. Just run the command, run the command within a terminal, or display the output in [QE](#) message system.

Enumerator

NoOutput	Just run the program - was None.
Terminal	Run the program in a terminal (in Windows a command interpreter will also be started, so the program may be a built-in command like 'dir')
LogOutput	Run the program, and log the output in the QE message system.
StdOutput	Run the program, and send doutput to standard output and standard error.

9.89.2.13 PVLabelMode

```
enum QE::PVLabelMode
```

The default is useAliasName, however the default aliasName are undefined, so the effective default is usePvName.

Enumerator

usePvName	display the PV name
useAliasName	display alias if defined otherwise display the PV name
useDescription	display DESCription if defined otherwise display the PV name

9.89.2.14 Radicies

```
enum QE::Radicies
```

Defines allowed radix values.

Enumerator

Decimal	base 10
Hexadecimal	base 16
Octal	base 8
Binary	base 2

9.89.2.15 RotationOptions

```
enum QE::RotationOptions
```

User friendly enumerations for #rotation property

Enumerator

NoRotation	No image rotation.
Rotate90Right	Rotate image 90 degrees clockwise.
Rotate90Left	Rotate image 90 degrees anticlockwise.
Rotate180	Rotate image 180 degrees.

9.89.2.16 Separators

```
enum QE::Separators
```

User friendly enumerations for seprator property - refer to formats for details.

Enumerator

NoSeparator	Use no separator, e.g. 123456.123456789.
-------------	--

Enumerator

Comma	Use ',' as separator, e.g. 123,456.123,456,789.
Underscore	Use '_' as separator, e.g. 123_456.123_456_789.
Space	Use ' ' as separator, e.g. 123 456.123 456 789.

9.89.2.17 SourceOptions

```
enum QE::SourceOptions
```

Enumerator

SourceText	User info held in string/string list property.
SourceFile	User info held in a configuration text file.

9.89.2.18 UpdateOptions

```
enum QE::UpdateOptions
```

User friendly enumerations for button updateOption property. Refer to [QEGenericButton](#) for details.

Enumerator

NoUpdate	Data updates will not update button.
Text	Data updates will update the button text.
Icon	Data updates will update the button icon.
TextAndIcon	Data updates will update the button text and icon.
State	Data updates will update the button state (checked or unchecked)
TextAndState	Data updates will update the button text and state.
IconAndState	Data updates will update the button icon and state.
TextIconAndState	Data updates will update the button - the lot.

9.89.2.19 UserLevels

```
enum QE::UserLevels
```

User friendly enumerations for #userLevelVisibility and #userLevelEnabled properties - refer to #userLevelVisibility and #userLevelEnabled properties and userLevel enumeration for details.

Enumerator

User	User level - least privilaged.
Scientist	User level - more privilaged than user, less than engineer.
Engineer	User level - most privilaged.

9.89.2.20 VideoModes

```
enum QE::VideoModes
```

Allow background mode selection

Enumerator

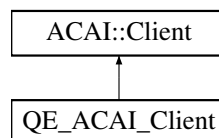
normal	white background
reverse	black background

The documentation for this class was generated from the following files:

- qeframework/common/QEEnums.h
- qeframework/common/QEEnums.cpp

9.90 QE_ACAI_Client Class Reference

Inheritance diagram for QE_ACAI_Client:



Public Member Functions

- **QE_ACAI_Client** (const QString &pvName, [QECaClient](#) *owner)

Protected Member Functions

- void **connectionUpdate** (const bool isConnected)
- void **dataUpdate** (const bool firstUpdate)
- void **putCallbackNotification** (const bool isSuccessful)

The documentation for this class was generated from the following file:

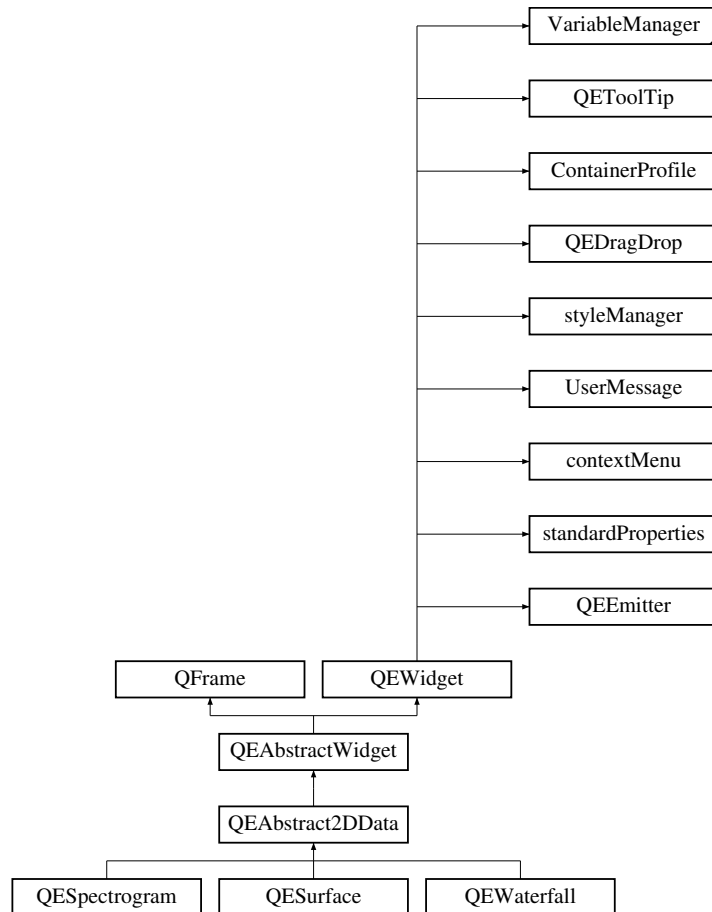
- qeframework/protocol/QECaClient.cpp

9.91 QEAbstract2DData Class Reference

The [QEAbstract2DData](#) class. This is the base class for the [QESpectrogram](#), [QEWaterfall](#) and [QESurface](#) widgets. It provides data management functionality for each sub class, and is not intended to be used as a concrete class.

```
#include <QEAbstract2DData.h>
```

Inheritance diagram for QEAbstract2DData:



Public Types

- enum [DataFormats](#) { [array1D](#) , [array2D](#) }
- enum [DataBinning](#) { [decimate](#) , [mean](#) , [median](#) }
- enum [ScaleModes](#) { [manual](#) , [operatingRange](#) , [dynamic](#) , [displayed](#) }
- enum [Constants](#) { [DATA_PV_INDEX](#) = 0 , [WIDTH_PV_INDEX](#) = 1 }
- enum [OwnContextMenuOptions](#) {
[A2DDCM_NONE](#) = [CM_SPECIFIC_WIDGETS_START_HERE](#) , [A2DDCM_NO_ROTATION](#) , [A2DDCM_ROTATE_90_RIGHT](#) , [A2DDCM_ROTATE_180](#) ,
[A2DDCM_ROTATE_90_LEFT](#) , [A2DDCM_VERTICAL_FLIP](#) , [A2DDCM_HORIZONTAL_FLIP](#) , [A2DDCM_MANUAL_SCALE](#) ,
[A2DDCM_OPERATING_RANGE_SCALE](#) , [A2DDCM_DYNAMIC_SCALE](#) , [A2DDCM_DISPLAYED_SCALE](#) , [A2DDCM_MIN_MAX_DIALOG](#) ,
[A2DDCM_SUB_CLASS_WIDGETS_START_HERE](#) }

Public Slots

- void **setDataWidth** (const int [dataWidth](#))
- void **setDataFormat** (const [DataFormats](#) dataFormat)
- void **setNumberOfSets** (const int [numberOfSets](#))
- void **setVerticalSliceFirst** (const int first)
- void **setVerticalSliceLast** (const int last)
- void **setHorizontalSliceFirst** (const int first)
- void **setHorizontalSliceLast** (const int last)
- void **setVerticalBin** (const int bin)
- void **setHorizontalBin** (const int bin)
- void **setDataBinning** (const [DataBinning](#) option)
- void **setRotation** (const [QE::RotationOptions](#) rotation)
- void **setVerticalFlip** (const bool [verticalFlip](#))
- void **setHorizontalFlip** (const bool [horizontalFlip](#))
- void **setScaleMode** (const [ScaleModes](#))
- void **setMinimum** (const double minimum)
- void **setMinimum** (const int minimum)
- void **setMaximum** (const double maximum)
- void **setMaximum** (const int maximum)
- void **setLogScale** (const bool [logScale](#))

Signals

- void [dbConnectionChanged](#) (const bool &isConnected)
- void [dbValueChanged](#) ()
- void **mouseElementChanged** (const int row, const int col, const double value)
- void **mouseElementChanged** (const QString &text)

Public Member Functions

- [QEAbstract2DDData](#) (QWidget *parent=0)
duplicate
- [QEAbstract2DDData](#) (const QString &dataVariableName, QWidget *parent=0)
- **QEAbstract2DDData** (const QString &dataVariableName, const QString &widthVariableName, QWidget *parent=0)
- virtual [~QEAbstract2DDData](#) ()
Destruction.
- int **getEffectiveDataWidth** () const
- void **setDataPvName** (const QString [dataVariable](#))
- QString **getDataPvName** () const
- void **setWidthPvName** (const QString [widthVariable](#))
- QString **getWidthPvName** () const
- void **setVariableNameSubstitutions** (const QString [variableSubstitutions](#))
- QString **getVariableNameSubstitutions** () const
- void **setMouseMoveSignals** (const QE::MouseMoveSignalFlags flags)
- QE::MouseMoveSignalFlags **getMouseMoveSignals** () const
- int **getDataWidth** () const
- [DataFormats](#) **getDataFormat** () const
- int **getNumberOfSets** () const
- int **getVerticalSliceFirst** () const
- int **getVerticalSliceLast** () const

- int **getHorizontalSliceFirst** () const
- int **getHorizontalSliceLast** () const
- int **getVerticalBin** () const
- int **getHorizontalBin** () const
- [DataBinning](#) **getDataBinning** () const
- [QE::RotationOptions](#) **getRotation** () const
- bool **getVerticalFlip** () const
- bool **getHorizontalFlip** () const
- [ScaleModes](#) **getScaleMode** () const
- double **getMinimum** () const
- double **getMaximum** () const
- bool **getLogScale** () const

Protected Member Functions

- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)
- [QEChannel](#) * **createQcalItem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)
- bool **dataIsAvailable** () const
- double **getValue** (const int displayRow, const int displayCol, const double defaultValue) const
- int **getUpdateCount** () const
- void **getScaleModeMinMaxValues** (double &min, double &max) const
- void **getNumberRowsAndCols** (int &numberRows, int &numberCols) const
- QString **getUnits** () const
- int **getPrecision** () const
- void **setMouseOverElement** (const int displayRow, const int displayCol)
- virtual void **updateDataVisulation** ()

Properties

- QString [dataVariable](#)
- QString [widthVariable](#)
- int [dataWidth](#)
- [DataFormats](#) **dataFormat**
- int [numberOfSets](#)
- int [verticalSliceFirst](#)
- int **verticalSliceLast**
- int **horizontalSliceFirst**
- int **horizontalSliceLast**
- int [verticalBin](#)
- int **horizontalBin**
- [DataBinning](#) **dataBinning**
- [QE::RotationOptions](#) [rotation](#)

- bool [verticalFlip](#)
- bool [horizontalFlip](#)
- [ScaleModes](#) **scaleMode**
- double **minimum**
- double **maximum**
- bool [logScale](#)
- QE::MouseMoveSignalFlags **mouseMoveSignals**
- QString [variableSubstitutions](#)

Additional Inherited Members

9.91.1 Detailed Description

The [QEAbstract2DData](#) class. This is the base class for the QESpectrogram, [QEWaterfall](#) and [QESurface](#) widgets. It provides data management functionality for each sub class, and is not intended to be used as a concrete class.

9.91.2 Member Enumeration Documentation

9.91.2.1 DataBinning

```
enum QEAbstract2DData::DataBinning
```

Enumerator

decimate	default - simple decimation - pick 1 "cental" value
mean	take average of all values
median	take median of all values

9.91.2.2 DataFormats

```
enum QEAbstract2DData::DataFormats
```

Defines whether the source data is treated as a 1D array or a 2D array. When source data is defined as 1D, the widget accumulates data on a FIFO basis, similar to the compress record, upto a maximum of numberOfSets "rows" of data. The accumulated data is then treated as a 2D array. Default: array2D

Enumerator

array2D	default
---------	---------

9.91.2.3 ScaleModes

enum `QEAbstract2DData::ScaleModes`

Enumerator

manual	manually range via min/max properties or dialog - default
operatingRange	range based on HOPR/LOPR values (if available)
dynamic	range adjhusted dynamically (based on current values)
displayed	what is currently displayed - becomes current manual settings and scale mode becomes manual

9.91.3 Constructor & Destructor Documentation

9.91.3.1 QEAbstract2DData() [1/2]

```
QEAbstract2DData::QEAbstract2DData (
    QWidget * parent = 0 ) [explicit]
```

duplicate

Create without a variable. Use `setDataPvName ()` and `setVariableNameSubstitutions()` to define a variable and, optionally, macro substitutions later.

9.91.3.2 QEAbstract2DData() [2/2]

```
QEAbstract2DData::QEAbstract2DData (
    const QString & dataVariableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable (s). A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.91.4 Member Function Documentation

9.91.4.1 dbConnectionChanged

```
void QEAbstract2DData::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applies to every variable.

9.91.4.2 dbValueChanged

```
void QEAAbstract2DData::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on data update 'event' to other widgets.

9.91.4.3 setMouseOverElement()

```
void QEAAbstract2DData::setMouseOverElement (
    const int displayRow,
    const int displayCol ) [protected]
```

TODO: Really need to convert display coordinated back to data coordinates. When binning, choose a mid point row/col.

9.91.5 Property Documentation

9.91.5.1 dataVariable

```
QString QEAAbstract2DData::dataVariable [read], [write]
```

EPICS main data variable name.

9.91.5.2 dataWidth

```
int QEAAbstract2DData::dataWidth [read], [write]
```

If a width variable is NOT specified, the data width may be set as a property. If defined, the widthVariable always overrides this value. Constrained to be ≥ 1 . Default 100

9.91.5.3 horizontalFlip

```
bool QEAAbstract2DData::horizontalFlip [read], [write]
```

If true, flip image horizontally.

9.91.5.4 logScale

```
bool QEAAbstract2DData::logScale [read], [write]
```

If true, use logarithmic scaling/brightness

9.91.5.5 numberOfSets

```
int QEAbstract2DData::numberOfSets [read], [write]
```

Specifies the number of data sets to accumulate when the dataFormat property is set to array1D, otherwise it is ignored. Allowed range: 1 .. 1024 Default: 40

9.91.5.6 rotation

```
QE::RotationOptions QEAbstract2DData::rotation [read], [write]
```

Data rotation option.

9.91.5.7 variableSubstitutions

```
QString QEAbstract2DData::variableSubstitutions [read], [write]
```

Default macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.91.5.8 verticalBin

```
int QEAbstract2DData::verticalBin [read], [write]
```

Data binning sizes and binning options.

9.91.5.9 verticalFlip

```
bool QEAbstract2DData::verticalFlip [read], [write]
```

If true, flip image vertically.

9.91.5.10 verticalSliceFirst

```
int QEAbstract2DData::verticalSliceFirst [read], [write]
```

Data display options - order is slice, rotate then flip.

Slice properties These properties can be negative, which interpreted as: dimension size/number - abs(value)

9.91.5.11 widthVariable

```
QString QEAbstract2DData::widthVariable [read], [write]
```

Data width variable name - needed when data format is 2D, otherwise ignored.

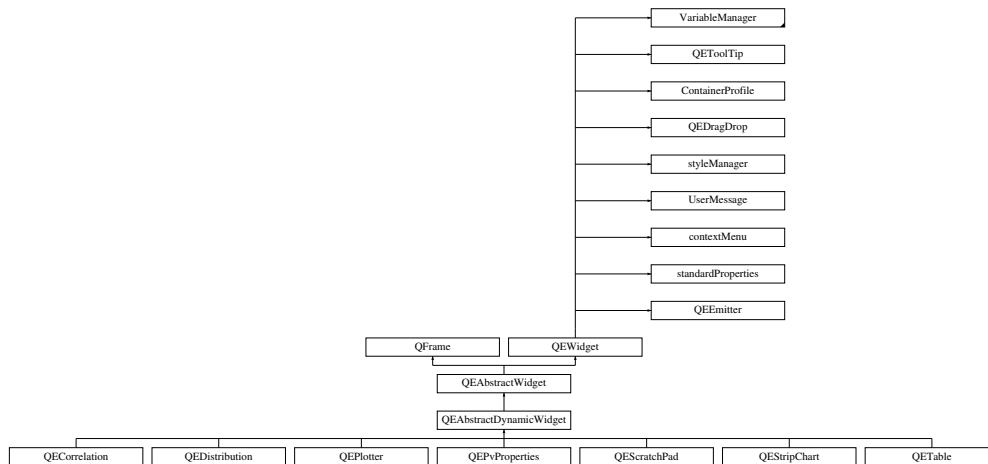
The documentation for this class was generated from the following files:

- qeframework/widgets/QE2DDataVisualisation/QEAbstract2DData.h
- qeframework/widgets/QE2DDataVisualisation/QEAbstract2DData.cpp

9.92 QEAbstractDynamicWidget Class Reference

```
#include <QEAbstractDynamicWidget.h>
```

Inheritance diagram for QEAbstractDynamicWidget:



Public Types

- enum **OwnContextMenuOptions** {
ADWCM_NONE = CM_SPECIFIC_WIDGETS_START_HERE , **ADWCM_LOAD_WIDGET_CONFIG** ,
ADWCM_SAVE_WIDGET_CONFIG , **ADWCM_SELECT_USE_PV_NAME** ,
ADWCM_SELECT_USE_ALIAS_NAME , **ADWCM_SELECT_USE_DESCRIPTION** , **ADWCM_SUB_↵**
CLASS_WIDGETS_START_HERE }

Public Member Functions

- **QEAbstractDynamicWidget** (QWidget *parent=0)
- void **setDefaultDir** (const QString &defaultDir)
- QString **getDefaultDir** () const
- void **setEnabledEditPv** (const bool isEnabled)
- bool **getEnabledEditPv** () const
- void **setPVLabelMode** (const QE::PVLabelMode pvLabelMode)
- QE::PVLabelMode **getPVLabelMode** () const
- void **paste** (QVariant s)
- virtual void **addPvNameList** (const QStringList &pvNameList)
- void **addPvNameSet** (const QString &pvNameSet)
- virtual int **addPvName** (const QString &pvName)=0
- virtual void **clearAllPvNames** ()=0

Protected Slots

- void **loadNamedWidgetConfiguration** (const QString &filename)
- void **saveNamedWidgetConfiguration** (const QString &filename)
- void **loadWidgetConfiguration** ()
- void **saveWidgetConfiguration** ()

Protected Member Functions

- `QMenu *` **buildContextMenu** ()
- `void` **addPVLabelModeContextMenu** (`QMenu *`menu)
- `void` **contextMenuTriggered** (`int` selectedItemNum)
- `QString` **getPersistantName** () `const`
- `virtual void` **pvLabelModeChanged** ()
- `virtual void` **enableEditPvChanged** ()
- `QE::UserLevels` **minimumEditPvUserLevel** () `const`
- `void` **setStandardFormat** (`QELabel *`dataLabel)

Properties

- `QString` **defaultDir**
- `bool` **enableEditPv**
- `QE::PVLabelMode` **pvLabelMode**

Additional Inherited Members

9.92.1 Detailed Description

Provides a common abstract base class for dynamic widgets, i.e. dynamic in the sense that the user can add/removed and modify PVs used by the widget at run time. It has been specifically designed to be a common base class for the [QEStripChart](#), [QEScratchPad](#), [QEPlotter](#) and [QETable](#) widgets. This not only minimises maintainance, but helps ensure we maintain a common look and feel user experiance.

9.92.2 Property Documentation

9.92.2.1 defaultDir

```
QString QEAbstractDynamicWidget::defaultDir [read], [write]
```

Default directory used for loading/saving files. Default to null string which is interpreted as the current directory.

9.92.2.2 enableEditPv

```
bool QEAbstractDynamicWidget::enableEditPv [read], [write]
```

By default, the `contextMenu` class only adds the "Edit PV" menu entry if and only if we are using the engineer user level (provided it has also been included in the widget menu set). When `enableEditPv` set true, the user level required is set to user level user, i.e. always available. Note: this may apply to this widget itself, but is more likely to be applied to the contained dynamic widgets. The default value for this property is false.

9.92.2.3 pvLabelMode

`QE::PVLabelMode` `QEAbstractDynamicWidget::pvLabelMode` [read], [write]

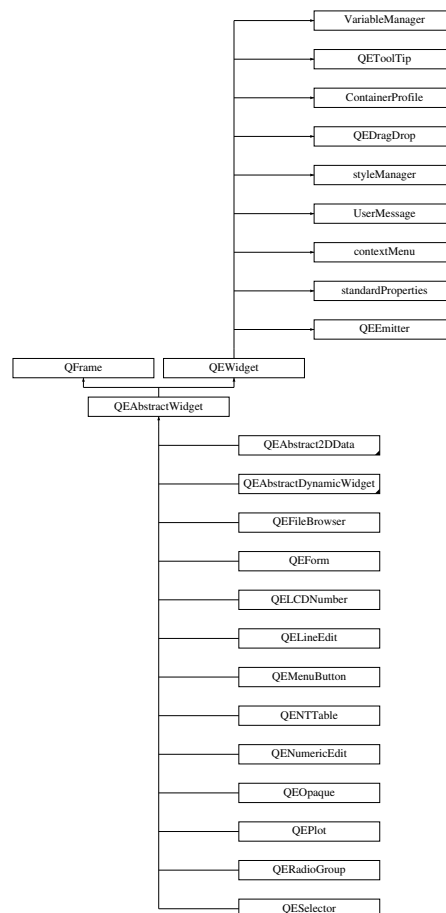
The default is useAliasName, however the default aliasName are undefined, so the effective default is usePvName.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEAbstractWidget/QEAbstractDynamicWidget.h
- qeframework/widgets/QEAbstractWidget/QEAbstractDynamicWidget.cpp

9.93 QEAbstractWidget Class Reference

Inheritance diagram for QEAbstractWidget:



Public Slots

- void `setManagedVisible` (bool v)

Public Member Functions

- `QEAbstractWidget` (`QWidget *parent=0`)

Properties

- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- bool [oosAware](#)

Additional Inherited Members

9.93.1 Member Function Documentation

9.93.1.1 setManagedVisible

```
void QEAbstractWidget::setManagedVisible (  
    bool v ) [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call to this slot, but will only be made visible by a call to this slot if the user level allows.

9.93.2 Property Documentation

9.93.2.1 allowDrop

```
bool QEAbstractWidget::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.93.2.2 defaultStyle

```
QString QEAbstractWidget::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.93.2.3 displayAlarmStateOption

`QE::DisplayAlarmStateOptions` `QEAbstractWidget::displayAlarmStateOption` [read], [write]

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.93.2.4 messageSourceId

`unsigned int` `QEAbstractWidget::messageSourceId` [read], [write]

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.93.2.5 oosAware

`bool` `QEAbstractWidget::oosAware` [read], [write]

Indicates whether the widget should respond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.93.2.6 styleSheet

`QString` `QEAbstractWidget::styleSheet` [read], [write]

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.93.2.7 userLevelEnabled

`QE::UserLevels` `QEAbstractWidget::userLevelEnabled` [read], [write]

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programmatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.93.2.8 userLevelEngineerStyle

`QString` `QEAbstractWidget::userLevelEngineerStyle` [read], [write]

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.93.2.9 userLevelScientistStyle

```
QString QEAbstractWidget::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.93.2.10 userLevelUserStyle

```
QString QEAbstractWidget::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.93.2.11 userLevelVisibility

```
QE::UserLevels QEAbstractWidget::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.93.2.12 variableAsToolTip

```
bool QEAbstractWidget::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.93.2.13 visible

```
bool QEAbstractWidget::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEAbstractWidget/QEAbstractWidget.h
- qeframework/widgets/QEAbstractWidget/QEAbstractWidget.cpp

9.94 QEActionRequests Class Reference

Public Types

- enum **Kinds** {
KindNone , **KindOpenFile** , **KindOpenFiles** , **KindAction** ,
KindWidgetAction , **KindHostComponents** }

Public Member Functions

- QEActionRequests** (const QString &action, const QString &pvName)
- QEActionRequests** (const QString &filename, const QString &config, const [QE::CreationOptions](#) optionIn, const QEFormMapper::FormHandles &formHandle=QEFormMapper::nullHandle())
- QEActionRequests** (const QList< [windowCreationListItem](#) > windowsIn)
- QEActionRequests** (const QString &actionIn, const QString &widgetNameIn, const QStringList &argumentsIn, const bool &initialiseIn, QAction *originator)
- QEActionRequests** (const [componentHostListItem](#) &componentIn)
- QEActionRequests** (const QList< [componentHostListItem](#) > &componentsIn)
- void **setKind** (const Kinds kindIn)
- Kinds **getKind** () const
- void **setArguments** (const QStringList &argumentsIn)
- QStringList **getArguments** () const
- void **setOption** (const [QE::CreationOptions](#) optionIn)
- [QE::CreationOptions](#) **getOption** () const
- void **setAction** (const QString actionIn)
- QString **getAction** () const
- void **setWidgetName** (const QString widgetNameIn)
- QString **getWidgetName** () const
- void **setInitialise** (const bool initialiseIn)
- bool **getInitialise** () const
- void **setOriginator** (QAction *originatorIn)
- QAction * **getOriginator** () const
- void **setFormHandle** (const QEFormMapper::FormHandles &formHandle)
- QEFormMapper::FormHandles **getFormHandle** () const
- QString **getCustomisation** () const
- QList< [windowCreationListItem](#) > **getWindows** () const
- QList< [componentHostListItem](#) > **getComponents** () const

Static Public Member Functions

- static QString **actionPvProperties** ()
- static QString **actionStripChart** ()
- static QString **actionScratchPad** ()
- static QString **actionPlotter** ()
- static QString **actionTable** ()
- static QString **actionShowInHistogram** ()
- static QString **actionShowInHisogram** ()
- static QString **actionGeneralPvEdit** ()
- static QString **actionNTTable** ()
- static QString **actionNTNDAarray** ()
- static QString **actionOpaque** ()
- static bool **isDockCreationOption** (const [QE::CreationOptions](#) createOption)
- static bool **isTabbedDockCreationOption** (const [QE::CreationOptions](#) createOption)

The documentation for this class was generated from the following files:

- qeframework/common/QEActionRequests.h
- qeframework/common/QEActionRequests.cpp

9.95 QEAdaptationParameters Class Reference

```
#include <QEAdaptationParameters.h>
```

Public Member Functions

- **QEAdaptationParameters** (const QString &envPrefix)
- bool **getBool** (const QString &name)
- bool **getBool** (const QString &name, const QChar letter)
- QString **getString** (const QString &name, const QString &defaultValue)
- QString **getString** (const QString &name, const QChar letter, const QString &defaultValue)
- int **getInt** (const QString &name, const int defaultValue)
- int **getInt** (const QString &name, const QChar letter, const int defaultValue)
- double **getFloat** (const QString &name, const double defaultValue)
- double **getFloat** (const QString &name, const QChar letter, const double defaultValue)
- QString **getFilename** (const QString &name, const QString &defaultValue)
- QString **getFilename** (const QString &name, const QChar letter, const QString &defaultValue)

9.95.1 Detailed Description

This class provides general access to adaptation parameters that may be specified in a variety of ways. Specifically it provides a standard and consistent wrapper around the getXxxx () functions out of [QEOptions](#), [QEEnvironmentVariables](#) and [QESettings](#). (where Xxxx is one of Bool, String, Int and Float).

Each get getXxxx () function attempts to extract a value of each source. Where multiple values are available the highest priority value is returned. The priorities are (from lowest to highest):

default_value - as passed into the getXxxx function. environment variable - the name is converted to upper case and is prefixed with default or specified prefix if this has been specified e.g. QE_. setting - read from adaptation section if the adaptation_parameters_file.ini file command line option - command line parameter: -name=value. Note is case sensitive.

If a numeric value is ill-defined, then next lower priority value is used. Numeric validity is defined by QString::toInt () and QString::toDouble (). Do note that QString::toInt () does not accept 0x... hexadecimal or 0... octal numbers. Users may always use getString () and parse such values themselves.

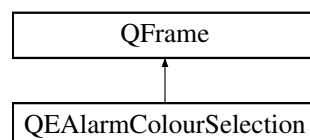
If only command line options, or setting or environment variables to be considered, then use the getXxxx () function out of the [QEOptions](#), [QEEnvironmentVariables](#) or [QESettings](#).

The documentation for this class was generated from the following files:

- qeframework/adaptation_parameters/QEAdaptationParameters.h
- qeframework/adaptation_parameters/QEAdaptationParameters.cpp

9.96 QEAlarmColourSelection Class Reference

Inheritance diagram for QEAlarmColourSelection:



Public Member Functions

- **QEAlarmColourSelection** (QWidget *parent=0)

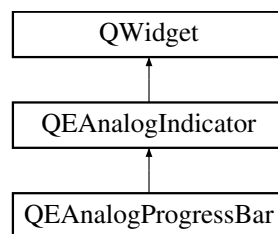
The documentation for this class was generated from the following files:

- qeframework/widgets/QEAlarmColourSelection/QEAlarmColourSelection.h
- qeframework/widgets/QEAlarmColourSelection/QEAlarmColourSelection.cpp

9.97 QEAnalogIndicator Class Reference

```
#include <QEAnalogIndicator.h>
```

Inheritance diagram for QEAnalogIndicator:



Classes

- struct [Band](#)
- class [BandList](#)

Public Types

- enum [Modes](#) { [Bar](#) , [Scale](#) , [Meter](#) }

Public Slots

- void **setValue** (const double [value](#))
- void **setValue** (const int [value](#))
- void **setRange** (const double minimumIn, const double maximumIn)

Public Member Functions

- [QEAnalogIndicator](#) (QWidget *parent=0)
Constructor.
- virtual [~QEAnalogIndicator](#) ()
Destructor.
- virtual QSize [sizeHint](#) () const
Size hint.
- double [getValue](#) () const
Access function for [value](#) property - refer to [value](#) property for details.
- void [setMinimum](#) (const double [value](#))
Access function for [minimum](#) - refer to [minimum](#) property for details.
- double [getMinimum](#) () const
Access function for [minimum](#) - refer to [minimum](#) property for details.
- void [setMaximum](#) (const double [value](#))
Access function for [maximum](#) - refer to [maximum](#) property for details.
- double [getMaximum](#) () const
Access function for [maximum](#) - refer to [maximum](#) property for details.
- void [setOrientation](#) (const Qt::Orientation [value](#))
Access function for [orientation](#) - refer to [orientation](#) property for details.
- Qt::Orientation [getOrientation](#) () const
Access function for [orientation](#) - refer to [orientation](#) property for details.
- void [setInvertedAppearance](#) (const bool [value](#))
Access function for [#invertedAppearance](#) - refer to [#invertedAppearance](#) property for details.
- bool [getInvertedAppearance](#) () const
Access function for [#invertedAppearance](#) - refer to [#invertedAppearance](#) property for details.
- void [setMode](#) (const enum [Modes](#) [mode](#))
Access function for [mode](#) - refer to [mode](#) property for details.
- enum [Modes](#) [getMode](#) () const
Access function for [mode](#) - refer to [mode](#) property for details.
- void [setCentreAngle](#) (const int [angle](#))
Access function for [centreAngle](#) - refer to [centreAngle](#) property for details.
- int [getCentreAngle](#) () const
Access function for [centreAngle](#) - refer to [centreAngle](#) property for details.
- void [setSpanAngle](#) (const int [angle](#))
Access function for [spanAngle](#) - refer to [spanAngle](#) property for details.
- int [getSpanAngle](#) () const
Access function for [spanAngle](#) - refer to [spanAngle](#) property for details.
- void [setAnimationTime](#) (const double [time](#))
Access function for [animationTime](#) - refer to [animationTime](#) property for details.
- double [getAnimationTime](#) () const
Access function for [animationTime](#) - refer to [animationTime](#) property for details.
- void [setMinorInterval](#) (const double [value](#))
Access function for [minorInterval](#) - refer to [minorInterval](#) property for details.
- double [getMinorInterval](#) () const
Access function for [minorInterval](#) - refer to [minorInterval](#) property for details.
- void [setMajorInterval](#) (const double [interval](#))
Access function for [majorInterval](#) - refer to [majorInterval](#) property for details.
- double [getMajorInterval](#) () const
Access function for [majorInterval](#) - refer to [majorInterval](#) property for details.
- void [setLogScaleInterval](#) (const int [interval](#))

- Access function for [logScaleInterval](#) - refer to [logScaleInterval](#) property for details.

 - int [getLogScaleInterval](#) () const
- Access function for [logScaleInterval](#) - refer to [logScaleInterval](#) property for details.

 - void [setBorderColour](#) (const QColor colour)
- Access function for [borderColour](#) - refer to [borderColour](#) property for details.

 - QColor [getBorderColour](#) () const
- Access function for [borderColour](#) - refer to [borderColour](#) property for details.

 - void [setForegroundColour](#) (const QColor colour)
- Access function for [foregroundColour](#) - refer to [foregroundColour](#) property for details.

 - QColor [getForegroundColour](#) () const
- Access function for [foregroundColour](#) - refer to [foregroundColour](#) property for details.

 - void [setBackgroundColour](#) (const QColor colour)
- Access function for [backgroundColour](#) - refer to [backgroundColour](#) property for details.

 - QColor [getBackgroundColour](#) () const
- Access function for [backgroundColour](#) - refer to [backgroundColour](#) property for details.

 - void [setFontColour](#) (const QColor colour)
- Access function for [fontColour](#) - refer to [fontColour](#) property for details.

 - QColor [getFontColour](#) () const
- Access function for [fontColour](#) - refer to [fontColour](#) property for details.

 - void [setShowText](#) (const bool [showText](#))
- Access function for [showText](#) - refer to [showText](#) property for details.

 - bool [getShowText](#) () const
- Access function for [showText](#) - refer to [showText](#) property for details.

 - void [setShowScale](#) (const bool [showScale](#))
- Access function for [showScale](#) - refer to [showScale](#) property for details.

 - bool [getShowScale](#) () const
- Access function for [showScale](#) - refer to [showScale](#) property for details.

 - void [setLogScale](#) (const bool [logScale](#))
- Access function for [logScale](#) - refer to [logScale](#) property for details.

 - bool [getLogScale](#) () const
- Access function for [logScale](#) - refer to [logScale](#) property for details.

Protected Member Functions

- void **paintEvent** (QPaintEvent *event)
- virtual QColor **getBorderPaintColour** () const
- virtual QColor **getBackgroundPaintColour** () const
- virtual QColor **getForegroundPaintColour** () const
- virtual QColor **getFontPaintColour** () const
- virtual QString **getTextImage** () const
- virtual [BandList](#) **getBandList** () const
- void **setIsActive** (const bool [value](#))
- bool **getIsActive** () const

Properties

- double [value](#)
- double [minimum](#)
- double [maximum](#)
- double [minorInterval](#)
- double [majorInterval](#)
- int [logScaleInterval](#)
- bool [showText](#)
- bool [showScale](#)
- bool [logScale](#)
- [Modes](#) [mode](#)
- Qt::Orientation [orientation](#)
- bool **[invertedAppearance](#)**
- int [centreAngle](#)
- int [spanAngle](#)
- double [animationTime](#)
- QColor [borderColour](#)
- QColor [backgroundColour](#)
- QColor [foregroundColour](#)
- QColor [fontColour](#)
- bool [isActive](#)

Alternative to [isEnabled](#). Default is true.

9.97.1 Detailed Description

This class provides a non EPICS aware graphical analog indicator base class. It supports a number of display modes including Bar, Scale and Meter.

When in Bar mode, it mimics QProgressBar and provides an analog progress bar widget.

9.97.2 Member Enumeration Documentation

9.97.2.1 Modes

```
enum QEAnalogIndicator::Modes
```

The type of analog indicator used to represent the value.

Enumerator

Bar	Bar (solid bar from minimum up to current value)
Scale	Scale (diamond marker tracks current value)
Meter	Meter (Needle moving across an arc scale)

9.97.3 Property Documentation

9.97.3.1 animationTime

```
double QEAnalogIndicator::animationTime [read], [write]
```

animationTime in seconds, 0.0 implies instantaneously. Default: 0.0

9.97.3.2 backgroundColour

```
QColor QEAnalogIndicator::backgroundColour [read], [write]
```

Background colour

9.97.3.3 borderColour

```
QColor QEAnalogIndicator::borderColour [read], [write]
```

Border colour

9.97.3.4 centreAngle

```
int QEAnalogIndicator::centreAngle [read], [write]
```

The angle in degrees of the line that Meter indicators are centered around. Zero represents a vertical centerline and angles increment clockwise.

9.97.3.5 fontColour

```
QColor QEAnalogIndicator::fontColour [read], [write]
```

Font colour

9.97.3.6 foregroundColour

```
QColor QEAnalogIndicator::foregroundColour [read], [write]
```

Foreground colour

9.97.3.7 logScale

```
bool QEAnalogIndicator::logScale [read], [write]
```

If set, use a logarithmic scale. If clear, use a linear scale

9.97.3.8 logScaleInterval

```
int QEAnalogIndicator::logScaleInterval [read], [write]
```

Log scale interval.

9.97.3.9 majorInterval

```
double QEAnalogIndicator::majorInterval [read], [write]
```

Minor scale interval. Only applies for linear scale (not log scale)

9.97.3.10 maximum

```
double QEAnalogIndicator::maximum [read], [write]
```

Maximum indicated value.

9.97.3.11 minimum

```
double QEAnalogIndicator::minimum [read], [write]
```

Minimum indicated value.

9.97.3.12 minorInterval

```
double QEAnalogIndicator::minorInterval [read], [write]
```

Minor scale interval. Only applies for linear scale (not log scale)

9.97.3.13 mode

```
Modes QEAnalogIndicator::mode [read], [write]
```

Selects what type of indicator is used (refer to Modes)

9.97.3.14 orientation

```
Qt::Orientation QEAnalogIndicator::orientation [read], [write]
```

The orientation of Bar and Scale indicators For orientation/invertedAppearance Default: Horizontal Default: false, i.e. progress left to right if Horizontal, and progress bototm to top if vertical (aligned with QSlider).

9.97.3.15 showScale

```
bool QEAnalogIndicator::showScale [read], [write]
```

If set, show the scale

9.97.3.16 showText

```
bool QEAnalogIndicator::showText [read], [write]
```

If set, show textual representation of value on the indicator

9.97.3.17 spanAngle

```
int QEAnalogIndicator::spanAngle [read], [write]
```

The span of the Meter scale arc in degrees Typical meters are 180 deg and 270 deg

9.97.3.18 value

```
double QEAnalogIndicator::value [read], [write]
```

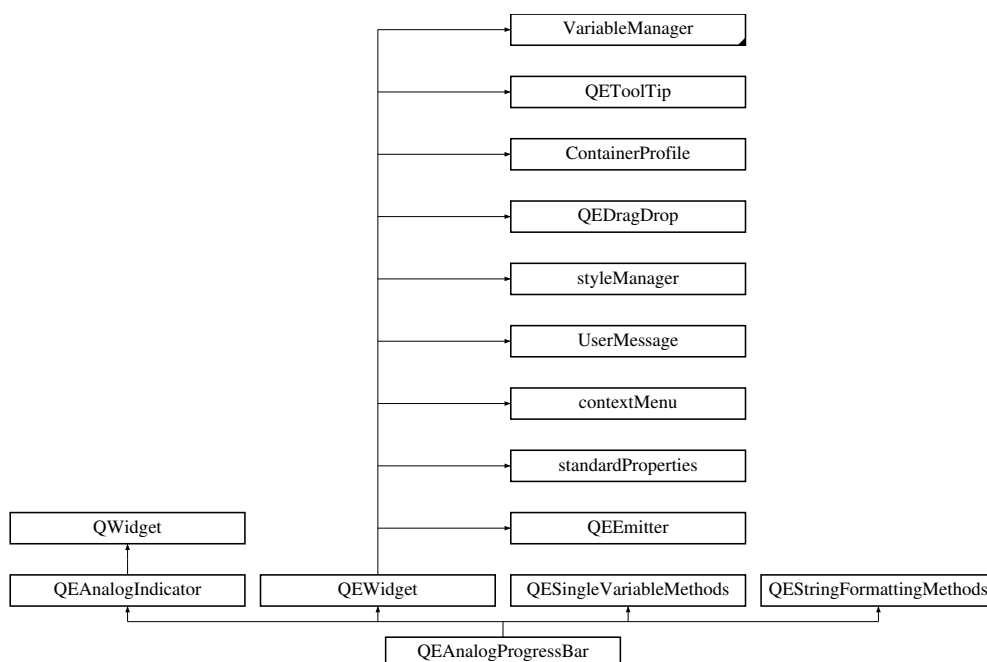
Current indicated value.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEAnalogIndicator/QEAnalogIndicator.h
- qeframework/widgets/QEAnalogIndicator/QEAnalogIndicator.cpp

9.98 QEAnalogProgressBar Class Reference

Inheritance diagram for QEAnalogProgressBar:



Public Types

- enum **AlarmSeverityDisplayModes** { foreground , background }

Public Slots

- void [setManagedVisible](#) (bool v)

Signals

- void [dbValueChanged](#) ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void [dbConnectionChanged](#) (const bool &isConnected)
- void [requestResend](#) ()

Public Member Functions

- void [setFormatProperty](#) (QE::Formats format)
Access function for [format](#) property - refer to [format](#) property for details.
- QE::Formats [getFormatProperty](#) () const
Access function for [format](#) property - refer to [format](#) property for details.
- QEAnalogProgressBar (QWidget *parent=0)
- QEAnalogProgressBar (const QString &variableName, QWidget *parent=0)
- virtual ~QEAnalogProgressBar ()
Destruction.
- void [setUseDbDisplayLimits](#) (bool useDbDisplayLimitsIn)
Access function for [useDbDisplayLimits](#) property - refer to [useDbDisplayLimits](#) property for details.
- bool [getUseDbDisplayLimits](#) () const
Access function for [useDbDisplayLimits](#) property - refer to [useDbDisplayLimits](#) property for details.
- void [setAlarmSeverityDisplayMode](#) (const AlarmSeverityDisplayModes mode)
Access function for #AlarmSeverityDisplayModes property - refer to #AlarmSeverityDisplayModes property for details.
- AlarmSeverityDisplayModes [getAlarmSeverityDisplayMode](#) () const
Access function for #AlarmSeverityDisplayModes property - refer to #AlarmSeverityDisplayModes property for details.

Protected Member Functions

- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- QEChannel * [createQcaltem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **stringFormattingChange** ()
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)
- QColor **getBackgroundPaintColour** () const
- QColor **getForegroundPaintColour** () const
- QString **getTextImage** () const
- BandList [getBandList](#) () const

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- bool [oosAware](#)
- [AlarmSeverityDisplayModes](#) [alarmSeverityDisplayMode](#)
- bool [useDbDisplayLimits](#)
- int **value**
- bool **isActive**
- int [precision](#)
- bool [useDbPrecision](#)
- int [leadingZeros](#)
- bool [addUnits](#)
- bool [forceSign](#)
- QString [localEnumeration](#)
- [QE::Formats](#) [format](#)
- int [radix](#)
- [QE::Separators](#) [separator](#)
- [QE::Notations](#) [notation](#)
- [QE::ArrayActions](#) [arrayAction](#)

Additional Inherited Members

9.98.1 Constructor & Destructor Documentation

9.98.1.1 QEAnalogProgressBar() [1/2]

```
QEAnalogProgressBar::QEAnalogProgressBar (
    QWidget * parent = 0 )
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.98.1.2 QEAnalogProgressBar() [2/2]

```
QEAnalogProgressBar::QEAnalogProgressBar (
    const QString & variableName,
    QWidget * parent = 0 )
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.98.2 Member Function Documentation

9.98.2.1 dbConnectionChanged

```
void QEAnalogProgressBar::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.98.2.2 dbValueChanged

```
void QEAnalogProgressBar::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.98.2.3 requestResend

```
void QEAnalogProgressBar::requestResend ( ) [signal]
```

Internal use only. Used when changing a property value to force a re-display to reflect the new property value.

9.98.2.4 setManagedVisible

```
void QEAnalogProgressBar::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.98.3 Property Documentation

9.98.3.1 addUnits

```
bool QEAnalogProgressBar::addUnits [read], [write]
```

If true (default), add engineering units supplied with the data.

9.98.3.2 alarmSeverityDisplayMode

```
AlarmSeverityDisplayModes QEAnalogProgressBar::alarmSeverityDisplayMode [read], [write]
```

Visualise the EPICS alarm severity

9.98.3.3 allowDrop

```
bool QEAnalogProgressBar::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.98.3.4 arrayAction

```
QE::ArrayActions QEAnalogProgressBar::arrayAction [read], [write]
```

Text formatting option for array data. Default is ASCII. Options are:

- ASCII - treat array as a single text string. For example an array of three characters 'a' 'b' 'c' will be formatted as 'abc'.
- APPEND - treat array as an array of numbers and format a string containing them all with a space between each. For example, an array of three numbers 10, 11 and 12 will be formatted as '10 11 12'.
- INDEX - Extract a single item from the array. The item is then formatted as any other non array data would be. The item selected is determined by the arrayIndex property. For example, if arrayIndex property is 1, an array of three numbers 10, 11 and 12 will be formatted as '11'.

9.98.3.5 arrayIndex

```
int QEAnalogProgressBar::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.98.3.6 defaultStyle

```
QString QEAnalogProgressBar::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.98.3.7 displayAlarmStateOption

`QE::DisplayAlarmStateOptions QEAnalogProgressBar::displayAlarmStateOption [read], [write]`

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.98.3.8 elementsRequired

`int QEAnalogProgressBar::elementsRequired [read], [write]`

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.98.3.9 forceSign

`bool QEAnalogProgressBar::forceSign [read], [write]`

If false (default), no "+" sign, when true always add a sign.

9.98.3.10 format

`QE::Formats QEAnalogProgressBar::format [read], [write]`

Format to apply to data. Default is 'Default' in which case the data type supplied with the data determines how the data is formatted. For all other options, an attempt is made to format the data as requested (whatever its native form).

9.98.3.11 leadingZeros

`int QEAnalogProgressBar::leadingZeros [read], [write]`

The minimum number of leading digits before any decimal/radix point. The default is 1.

9.98.3.12 localEnumeration

```
QString QEAnalogProgressBar::localEnumeration [read], [write]
```

An enumeration list used to data values. Used only when the formatting option is 'local enumeration'. Value is converted to an integer and used to select a string from this list.

Format is:

```
[[<|<=|=|!=|>|=|>]value1|*]: string1 , [[<|<=|=|!=|>|=|>]value2|*]: string2 , [[<|<=|=|!=|>|=|>]value3|*]: string3 , ...
```

Where: < Less than <= Less than or equal = Equal (default if no operator specified) >= Greather than or equal

Greater than

* Always match (used to specify default text)

Values may be numeric or textual Values do not have to be in any order, but first match wins Values may be quoted Strings may be quoted Consecutive values do not have to be present. Operator is assumed to be equality if not present. White space is ignored except within quoted strings. may be included in a string to indicate a line break

Examples are:

```
0:Off,1:On 0 : "Pump Running", 1 : "Pump not running" 0:"", 1:"Warning!\nAlarm" <2:"Value is less than two", =2↵
:"Value is equal to two", >2:"Value is grater than 2" 3:"Beamline Available", *:"" "Pump Off": "OH NO!, the pump is OFF!", "Pump On": "It's OK, the pump is on"
```

The data value is converted to a string if no enumeration for that value is available. For example, if the local enumeration is '0:off,1:on', and a value of 10 is processed, the text generated is '10'. If a blank string is required, this should be explicit. for example, '0:off,1:on,10: ""'

A range of numbers can be covered by a pair of values as in the following example: >=4:"Between 4 and 8", <=8↵
:"Between 4 and 8"

9.98.3.13 messageSourceId

```
unsigned int QEAnalogProgressBar::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.98.3.14 notation

```
QE::Notations QEAnalogProgressBar::notation [read], [write]
```

Notation used for numerical formatting. Default is fixed.

9.98.3.15 oosAware

```
bool QEAnalogProgressBar::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.98.3.16 precision

```
int QEAnalogProgressBar::precision [read], [write]
```

Precision used when formatting floating point numbers. The default is 4. This is only used if useDbPrecision is false.

9.98.3.17 radix

```
int QEAnalogProgressBar::radix [read], [write]
```

Base used for when formatting integers. Default is 10 (duh!)

9.98.3.18 separator

```
QE::Separators QEAnalogProgressBar::separator [read], [write]
```

Seperators used for interger and fixed point formatting. Default is None.

9.98.3.19 styleSheet

```
QString QEAnalogProgressBar::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly setable per se. This also stops transient styles being saved to the ui file.

9.98.3.20 useDbDisplayLimits

```
bool QEAnalogProgressBar::useDbDisplayLimits [read], [write]
```

Use the EPICS database display limits

9.98.3.21 useDbPrecision

```
bool QEAnalogProgressBar::useDbPrecision [read], [write]
```

If true (default), format floating point numbers using the precision supplied with the data. If false, the precision property is used.

9.98.3.22 `userLevelEnabled`

```
QE::UserLevels QEAnalogProgressBar::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.98.3.23 `userLevelEngineerStyle`

```
QString QEAnalogProgressBar::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.98.3.24 `userLevelScientistStyle`

```
QString QEAnalogProgressBar::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.98.3.25 `userLevelUserStyle`

```
QString QEAnalogProgressBar::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.98.3.26 `userLevelVisibility`

```
QE::UserLevels QEAnalogProgressBar::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.98.3.27 variable

```
QString QEAnalogProgressBar::variable [read], [write]
```

EPICS variable name (CA PV)

9.98.3.28 variableAsToolTip

```
bool QEAnalogProgressBar::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.98.3.29 variableSubstitutions

```
QString QEAnalogProgressBar::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.98.3.30 visible

```
bool QEAnalogProgressBar::visible [read], [write]
```

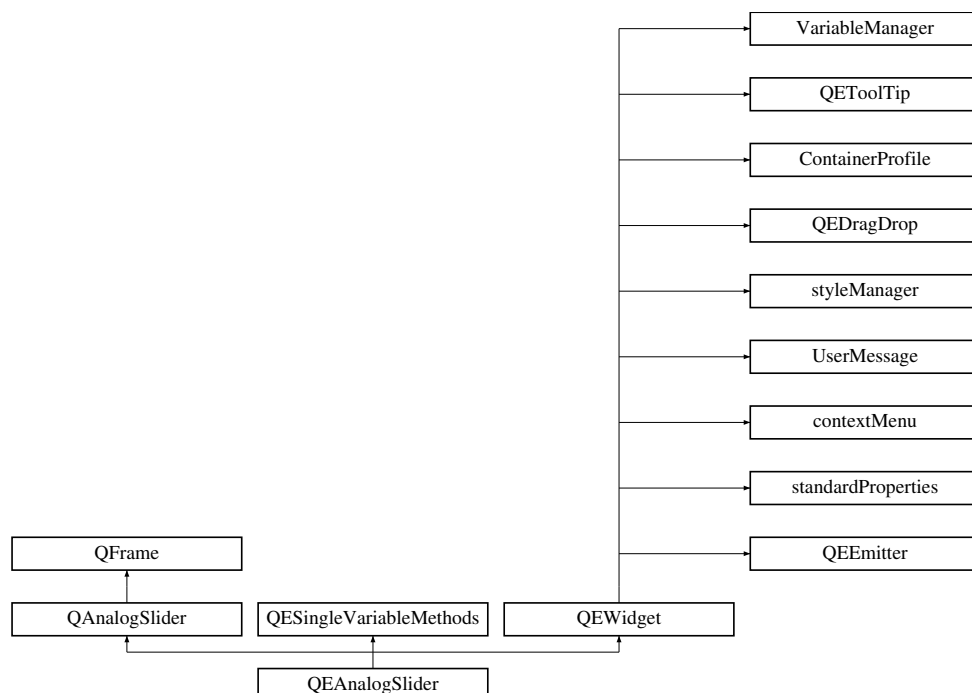
Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEAnalogProgressBar/QEAnalogProgressBar.h
- qeframework/widgets/QEAnalogProgressBar/QEAnalogProgressBar.cpp

9.99 QEAnalogSlider Class Reference

Inheritance diagram for QEAnalogSlider:



Public Slots

- void [setManagedVisible](#) (bool v)
- void **writeNow** ()
- void [setPvValue](#) (const QString &text)
- void **setPvValue** (const int value)
- void **setPvValue** (const double value)
- void **setPvValue** (const bool value)

Signals

- void [dbValueChanged](#) ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void [dbConnectionChanged](#) (const bool &isConnected)

Public Member Functions

- [QEAnalogSlider](#) (QWidget *parent=0)
- [QEAnalogSlider](#) (const QString &variableName, const QString &readbackName="", QWidget *parent=0)
- virtual [~QEAnalogSlider](#) ()
Destruction.
- void [activated](#) ()
Do any post-all-widgets-constructed, i.e. activated stuff.
- void **setVariableNameSubstitutionsProperty** (const QString &substitutions)
- void **setReadbackNameProperty** (const QString &variableName)
- QString **getReadbackNameProperty** () const
- void **setReadbackElementsRequired** (const int [elementsRequired](#))
- int **getReadbackElementsRequired** () const
- void **setReadbackArrayIndex** (const int [arrayIndex](#))
- int **getReadbackArrayIndex** () const
- void **setSubstitutionsProperty** (const QString &substitutions)
- QString **getSubstitutionsProperty** () const
- void **setContinuousWrite** (const bool value)
- bool **getContinuousWrite** () const
- void **setAutoScale** (const bool value)
- bool **getAutoScale** () const
- void **setAlarmColours** (const bool value)
- bool **getAlarmColours** () const

Protected Member Functions

- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- [QEChannel](#) * [createQcaltem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- QString [readbackVariable](#)
- int [readbackElementsRequired](#)
- int [readbackArrayIndex](#)
- bool [continuousWrite](#)
- bool [autoScale](#)
- bool [axisAlarmColours](#)
- double [value](#)
- QString [leftText](#)
- QString [centreText](#)
- QString [rightText](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- bool [oosAware](#)

Additional Inherited Members

9.99.1 Constructor & Destructor Documentation

9.99.1.1 QEAnalogSlider() [1/2]

```
QEAnalogSlider::QEAnalogSlider (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.99.1.2 QEAnalogSlider() [2/2]

```
QEAnalogSlider::QEAnalogSlider (
    const QString & variableName,
    const QString & readbackName = "",
    QWidget * parent = 0 ) [explicit]
```

Create with a variables. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.99.2 Member Function Documentation

9.99.2.1 dbConnectionChanged

```
void QEAnalogSlider::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.99.2.2 dbValueChanged

```
void QEAnalogSlider::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget. Note: this widget emits the numeric enumeration value as opposed to the associated text.

9.99.2.3 setManagedVisible

```
void QEAnalogSlider::setManagedVisible (
    bool v ) [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.99.2.4 setPvValue

```
void QEAnalogSlider::setPvValue (
    const QString & text ) [slot]
```

Update the widget and write to the main associated control process variable. Supplied values are converted to the appropiate widget value type. If conversion not possible or resultant value out of range, no widget/PV update occurs.

9.99.3 Property Documentation

9.99.3.1 allowDrop

```
bool QEAnalogSlider::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.99.3.2 arrayIndex

```
int QEAnalogSlider::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.99.3.3 autoScale

```
bool QEAnalogSlider::autoScale [read], [write]
```

If true (default), display and editing of numbers using the precision, and control limits supplied with the data. If false, the precision, leadingZeros, minimum and maximum properties are used.

9.99.3.4 axisAlarmColours

```
bool QEAnalogSlider::axisAlarmColours [read], [write]
```

If true the background axis colour reflect the alarm and warning values of the (setpoint) variable. If false (default) then axis uses widget colour.

9.99.3.5 continuousWrite

```
bool QEAnalogSlider::continuousWrite [read], [write]
```

If true the widget writes to the PV as the slider is moved. If false (default) a write only occurs when apply button click.

9.99.3.6 defaultStyle

```
QString QEAnalogSlider::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.99.3.7 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEAnalogSlider::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm'. If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.99.3.8 elementsRequired

```
int QEAnalogSlider::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.99.3.9 messageSourceId

```
unsigned int QEAnalogSlider::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.99.3.10 oosAware

```
bool QEAnalogSlider::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.99.3.11 styleSheet

```
QString QEAnalogSlider::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly setable per se. This also stops transient styles being saved to the ui file.

9.99.3.12 userLevelEnabled

```
QE::UserLevels QEAnalogSlider::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.99.3.13 userLevelEngineerStyle

```
QString QEAnalogSlider::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.99.3.14 userLevelScientistStyle

```
QString QEAnalogSlider::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.99.3.15 userLevelUserStyle

```
QString QEAnalogSlider::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.99.3.16 userLevelVisibility

```
QE::UserLevels QEAnalogSlider::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.99.3.17 variable

```
QString QEAnalogSlider::variable [read], [write]
```

EPICS variable name (CA PV)

9.99.3.18 variableAsToolTip

```
bool QEAnalogSlider::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.99.3.19 variableSubstitutions

```
QString QEAnalogSlider::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.99.3.20 visible

```
bool QEAnalogSlider::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

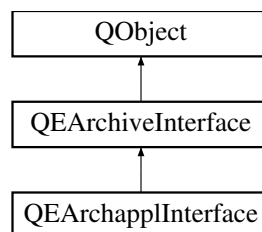
The documentation for this class was generated from the following files:

- qeframework/widgets/QEAnalogSlider/QEAnalogSlider.h
- qeframework/widgets/QEAnalogSlider/QEAnalogSlider.cpp

9.100 QEArchapplInterface Class Reference

```
#include <QEArchapplInterface.h>
```

Inheritance diagram for QEArchapplInterface:



Public Slots

- void **networkManagerResponse** (const [QEArchiveInterface::Context](#) &context, QNetworkReply *reply)
- void **networkManagerFault** (const [QEArchiveInterface::Context](#) &context, const QNetworkReply::NetworkError error)

Public Member Functions

- **QEArchapplInterface** (QUrl url, QObject *parent=0)
- void **namesRequest** (QObject *userData, const int key, QString pattern=".*")
- void **valuesRequest** (QObject *userData, const [QCaDateTime](#) startTime, const [QCaDateTime](#) endTime, const int count, const How how, const QStringList pvNames, const int key=0, const unsigned int requested←_element=0)
- void **infoRequest** (QObject *userData)
- void **archivesRequest** (QObject *userData)

Additional Inherited Members

9.100.1 Detailed Description

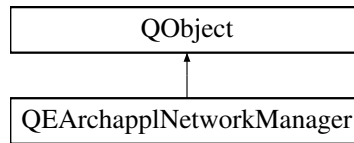
Interface to EPICS Archiver Appliance.

The documentation for this class was generated from the following files:

- qeframework/archive/QEArchapplInterface.h
- qeframework/archive/QEArchapplInterface.cpp

9.101 QEArchapplNetworkManager Class Reference

Inheritance diagram for QEArchapplNetworkManager:



Signals

- void **networkManagerResponse** (const [QEArchiveInterface::Context](#) &context, QNetworkReply *reply)
- void **networkManagerFault** (const [QEArchiveInterface::Context](#) &context, const QNetworkReply::NetworkError error)

Friends

- class **QEArchapplInterface**

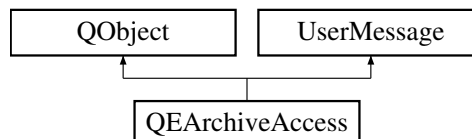
The documentation for this class was generated from the following files:

- qeframework/archive/QEArchapplInterface.h
- qeframework/archive/QEArchapplInterface.cpp

9.102 QEArchiveAccess Class Reference

```
#include <QEArchiveAccess.h>
```

Inheritance diagram for QEArchiveAccess:



Classes

- struct [ArchiverPvInfo](#)
- struct [PVDataRequests](#)
- struct [PVDataResponses](#)
- struct [Status](#)

Public Types

- enum **ArchiverTypes** { **CA** , **ARCHAPPL** , **Error** }
- enum **MetaRequests** { **mrNone** = 0 , **mrSeverity** , **mrStatus** }
- typedef QList< [Status](#) > **StatusList**
- typedef QList< [ArchiverPvInfo](#) > **ArchiverPvInfoLists**

Signals

- void **archiveStatus** (const [QEArchiveAccess::StatusList](#) &statusList)
- void **setArchiveData** (const QObject *userData, const bool isOkay, const [QCaDataPointList](#) &pointsList, const QString &pvName, const QString &supplementary)
- void **reInterrogateArchives** ()
- void **archiveStatusRequest** ()
- void **readArchiveRequest** (const [QEArchiveAccess](#) *, const [QEArchiveAccess::PVDataRequests](#) &)
- void **signalArchiveResponse** (const [QEArchiveAccess::PVDataResponses](#) &response) const

Public Member Functions

- **QEArchiveAccess** (QObject *parent=0)
- ArchiverTypes **getArchiverType** () const
- unsigned int **getMessageSourceId** () const
- void **setMessageSourceId** (unsigned int messageSourceId)
- void **resendStatus** ()
- void **reReadAvailablePVs** ()
- void **readArchive** (QObject *userData, const QString pvName, const [QCaDateTime](#) startTime, const [QCaDateTime](#) endTime, const int count, const [QEArchiveInterface::How](#) how, const unsigned int element=0)

Static Public Member Functions

- static bool **isReady** ()
- static int **getNumberInterfaces** ()
- static QString **getPattern** ()
- static int **getNumberPVs** ()
- static QStringList **getAllPvNames** ()
- static bool **getArchivePvInformation** (const QString &pvName, QString &effectivePvName, [ArchiverPvInfoLists](#) &data)
- static bool **_registerMetaTypes** ()

Friends

- class **QEArchiveManager**

9.102.1 Detailed Description

This class provides the EPICS Qt Framework access to the archives and indirect usage of the underlying [QEArchiveManager](#).

Currently only handles scalar values but can/will be extended to provide array data retrieval.

NOTE: It is the creation of the first object of this class will cause the [QEArchiveManager](#) to initialised if not already done so.

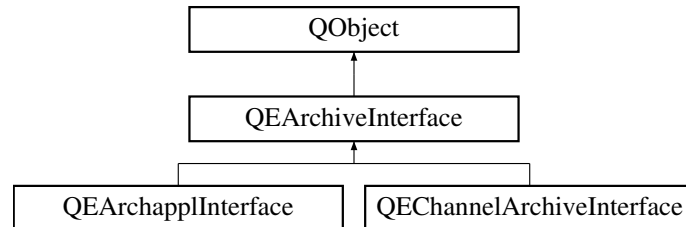
The documentation for this class was generated from the following files:

- qeframework/archive/QEArchiveAccess.h
- qeframework/archive/QEArchiveAccess.cpp

9.103 QEArchiveInterface Class Reference

```
#include <QEArchiveInterface.h>
```

Inheritance diagram for QEArchiveInterface:



Classes

- struct [Archive](#)
- struct [Context](#)
- struct [PVName](#)
- struct [ResponseValues](#)

Public Types

- enum **archiveAlarmSeverity** {
archSevNone = 0 , **archSevMinor** = 1 , **archSevMajor** = 2 , **archSevInvalid** = 3 ,
archSevEstRepeat = 0x0f80 , **archSevDisconnect** = 0x0f40 , **archSevStopped** = 0x0f20 , **archSevRepeat**
= 0x0f10 ,
archSevDisabled = 0x0f08 }
- enum **How** {
Raw = 0 , **SpreadSheet** , **Averaged** , **PlotBinning** ,
Linear }
- enum **Methods** {
Information , **Archives** , **Names** , **Values** ,
Count }
- enum **States** {
Unknown , **Updating** , **Complete** , **InComplete** ,
No_Response , **Error** }
- typedef QList< [QEArchiveInterface::Archive](#) > **ArchiveList**
- typedef QList< [QEArchiveInterface::PVName](#) > **PVNameList**
- typedef std::list< [QEArchiveInterface::ResponseValues](#) > **ResponseValueList**

Signals

- void **pvNamesResponse** (const QObject *, const bool, const QEArchiveInterface::PVNameList &)
- void **valuesResponse** (const QObject *, const bool, const QEArchiveInterface::ResponseValueList &)
- void **infoResponse** (const QObject *, const bool, const int, const QString &)
- void **archivesResponse** (const QObject *, const bool, const QEArchiveInterface::ArchiveList &)
- void **nextRequest** (const int requestIndex)

Public Member Functions

- **QEArchiveInterface** (QObject *parent=0)
- virtual void **setUrl** (QUrl url)
- QUrl **getUrl** () const
- QString **getName** () const
- virtual void **infoRequest** (QObject *userData)=0
- virtual void **archivesRequest** (QObject *userData)=0
- virtual void **namesRequest** (QObject *userData, const int key, QString pattern=".*")=0
- virtual void **valuesRequest** (QObject *userData, const [QCaDateTime](#) startTime, const [QCaDateTime](#) endTime, const int count, const How how, const QStringList pvNames, const int key=0, const unsigned int requested_element=0)=0

Static Public Member Functions

- static QString **alarmSeverityName** (enum archiveAlarmSeverity severity)
- static bool **registerMetaTypes** ()

Static Protected Member Functions

- static [QCaDateTime](#) **convertArchiveToEpics** (const int seconds, const int nanoSecs)
- static void **convertEpicsToArchive** (const [QCaDateTime](#) &datetime, int &seconds, int &nanoSecs)

Protected Attributes

- QUrl **mUrl**

9.103.1 Detailed Description

This virtual class provides the functional interface to the achivers.

There are currently two sub classes: [QEChannelArchiveInterface](#) (for the traditional EPICS CA archiver); and [QEArchapplInterface](#) (for the new [Archive](#) Appliance archiver)

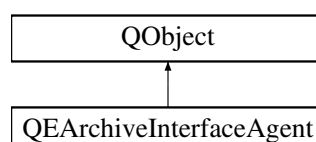
Other archivers could be added.

The documentation for this class was generated from the following files:

- qeframework/archive/QEArchiveInterface.h
- qeframework/archive/QEArchiveInterface.cpp

9.104 QEArchiveInterfaceAgent Class Reference

Inheritance diagram for QEArchiveInterfaceAgent:



Signals

- void **xmlRpcResponse** (const [QEArchiveInterface::Context](#) &, const QVariant &)
- void **xmlRpcFault** (const [QEArchiveInterface::Context](#) &, int, const QString &)

Friends

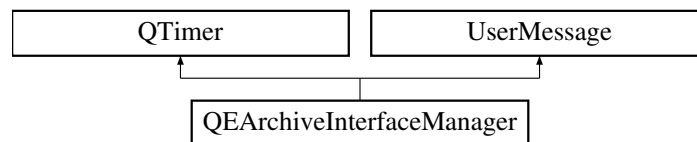
- class **QEChannelArchiveInterface**

The documentation for this class was generated from the following files:

- qeframework/archive/QEChannelArchiveInterface.h
- qeframework/archive/QEChannelArchiveInterface.cpp

9.105 QEArchiveInterfaceManager Class Reference

Inheritance diagram for QEArchiveInterfaceManager:



Classes

- class [NamesResponseContext](#)
- class [ValuesResponseContext](#)

Signals

- void **signalRequestArchives** ()
- void **signalDataRequest** (const [QEArchiveAccess](#) *, const int, const [QEArchiveAccess::PVDataRequests](#) &)
- void **aimPvNamesResponse** ([QEArchiveInterfaceManager](#) *, const [QEArchiveInterface::Archive](#), const [QEArchiveInterface::PVNameList](#) &)
- void **aimDataResponse** (const [QEArchiveAccess](#) *, const [QEArchiveAccess::PVDataResponses](#) &)

Friends

- class **QEArchiveManager**

Additional Inherited Members

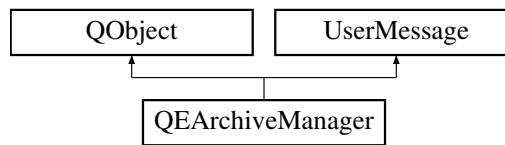
The documentation for this class was generated from the following files:

- qeframework/archive/QEArchiveInterfaceManager.h
- qeframework/archive/QEArchiveInterfaceManager.cpp

9.106 QEArchiveManager Class Reference

```
#include <QEArchiveManager.h>
```

Inheritance diagram for QEArchiveManager:



Classes

- class [PVNameToSourceSpecLookUp](#)

Signals

- void **archiveStatusResponse** (const QEArchiveAccess::StatusList &)
- void **readArchiveResponse** (const [QEArchiveAccess](#) *, const [QEArchiveAccess::PVDataResponses](#) &)

Static Public Member Functions

- static int **getArchiveNameIndex** (const QString &archiveName)
- static QString **getArchiveNameFromIndex** (const int index)
- static int **getPathIndex** (const QString &pathName)
- static QString **getPathFromIndex** (const int index)

Friends

- class **QEArchiveAccess**
- class **QEArchiveInterfaceManager**

Additional Inherited Members

9.106.1 Detailed Description

Archive Manager manages access to the archives, and provides a thick binding around the Archive Interface class. It's main function is to provide a PV Name to interface/archive mapping.

This is a singleton class - the single instance is declared in the .cpp file. It is effectively a private class used exclusively by [QEArchiveAccess](#).

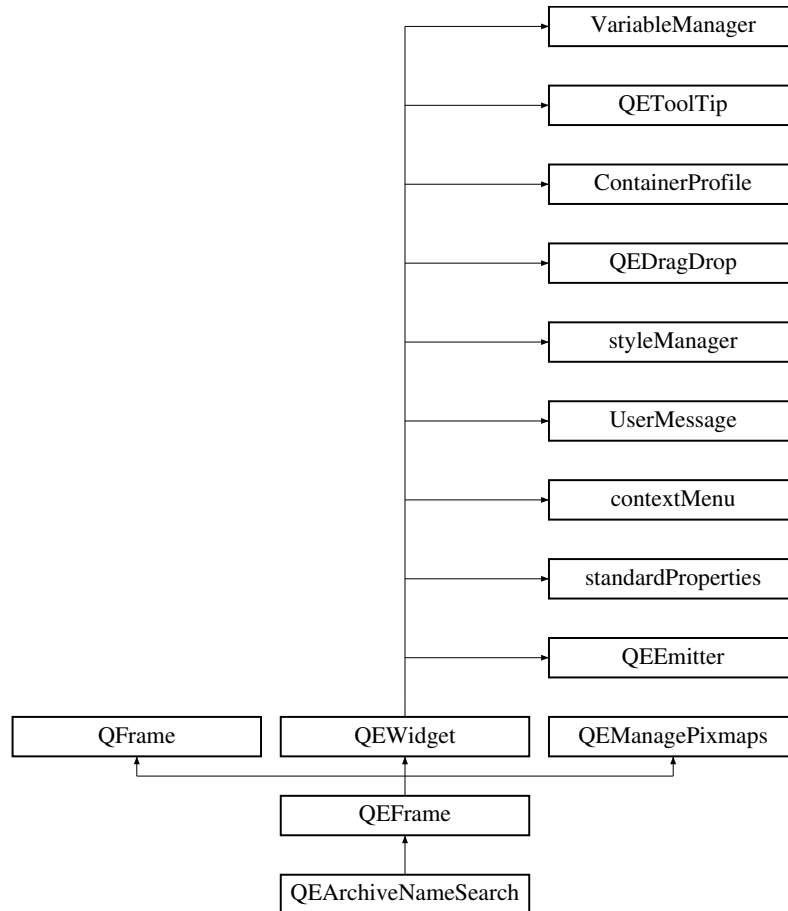
The documentation for this class was generated from the following files:

- qeframework/archive/QEArchiveManager.h
- qeframework/archive/QEArchiveManager.cpp

9.107 QEArchiveNameSearch Class Reference

```
#include <QEArchiveNameSearch.h>
```

Inheritance diagram for QEArchiveNameSearch:



Public Member Functions

- **QEArchiveNameSearch** (QWidget *parent=0)
- QStringList **getSelectedNames** () const
- void **clear** ()

Protected Member Functions

- QSize **sizeHint** () const
- void **mousePressEvent** (QMouseEvent *event)
- QVariant **getDrop** ()
- QString **copyVariable** ()

Additional Inherited Members

9.107.1 Detailed Description

This is a non EPICS aware widget. It provides a simple user means to find archived PV names.

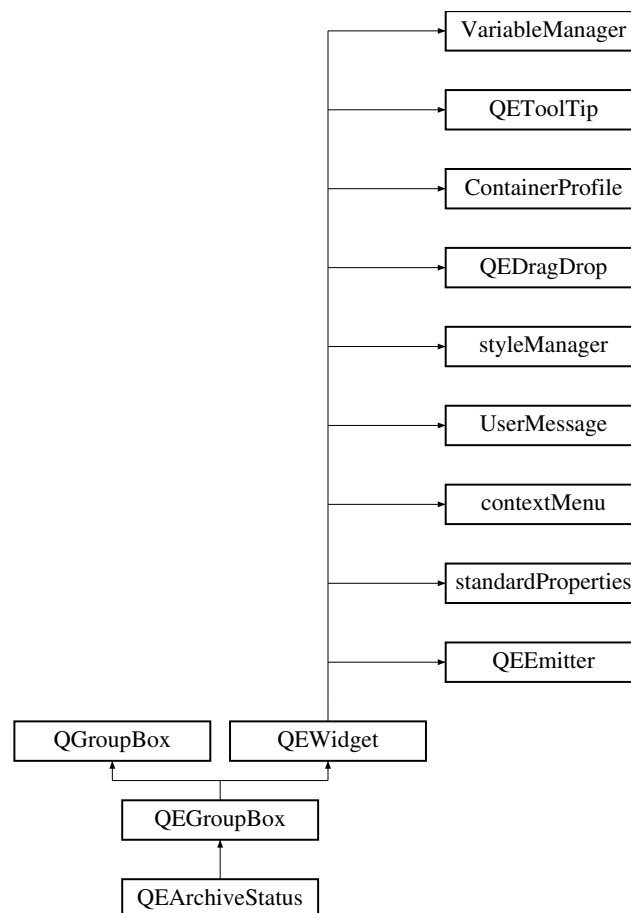
The documentation for this class was generated from the following files:

- qeframework/archive/QEArchiveNameSearch.h
- qeframework/archive/QEArchiveNameSearch.cpp

9.108 QEArchiveStatus Class Reference

```
#include <QEArchiveStatus.h>
```

Inheritance diagram for QEArchiveStatus:



Public Slots

- void **reReadAvailablePVs** ()

Public Member Functions

- **QEArchiveStatus** (QWidget *parent=0)
- QSize **sizeHint** () const

Additional Inherited Members

9.108.1 Detailed Description

This is a non EPICS aware widget. It extracts and displays states information from the archive manager.

The documentation for this class was generated from the following files:

- qeframework/archive/QEArchiveStatus.h
- qeframework/archive/QEArchiveStatus.cpp

9.109 QEAxisIterator Class Reference

```
#include <QEAxisIterator.h>
```

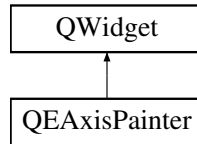
Public Member Functions

- **QEAxisIterator** (const double minimum, const double maximum, const double minorInterval, const int majorMinorRatio, const bool isLogarithmic)
- void **reInitialise** (const double minimum, const double maximum, const double minorInterval, const int majorMinorRatio, const bool isLogarithmic)
- bool **firstValue** (double &value, bool &isMajor, const int maxIterations=10000)
- bool **nextValue** (double &value, bool &isMajor)
 - Returns next value if available indicated by return value being true.*
- double **getMinimum** () const
- double **getMaximum** () const
 - Extracts iterator's minimum value.*
- double **getMinorInterval** () const
 - Extracts iterator's maximum value.*
- int **getMajorMinorRatio** () const
 - Extracts iterator's minor interval.*
- bool **getIsLogarithmic** () const
 - Extracts iterator's major/minor ratio.*

9.110 QEAxisPainter Class Reference

```
#include <QEAxisPainter.h>
```

Inheritance diagram for QEAxisPainter:



Public Types

- enum [TextPositions](#) { [BelowLeft](#) , [AboveRight](#) }
- enum **Constants** { **NUMBER_OF_MARKERS** = 4 }

Public Member Functions

- **QEAxisPainter** (QWidget *parent=0)
- void **setMinimum** (const double minimum)
- double **getMinimum** () const
- void **setMaximum** (const double maximum)
- double **getMaximum** () const
- void **setModulo** (const double modulo)
- double **getModulo** () const
- void **setPrecision** (const int precision)
- int **getPrecision** () const
- void **setMinorInterval** (const double minorInterval)
- double **getMinorInterval** () const
- void **setMajorMinorRatio** (const int majorMinorRatio)
- int **getMajorMinorRatio** () const
- void **setHasAxisLine** (const bool hasAxisLine)
- bool **getHasAxisLine** () const
- void **setLogScale** (const bool isLog)
- bool **getLogScale** () const
- void **setOrientation** (const Qt::Orientations orientation)
- Qt::Orientations **getOrientation** () const
- void **setInvertedAppearance** (const bool invertedAppearance)
- bool **getInvertedAppearance** () const
- void **setTextPosition** (const [TextPositions](#) textPositions)
- [TextPositions](#) **getTextPosition** () const
- void **setPenColour** (const QColor colour)
- QColor **getPenColour** () const
- void **setIndent** (const int topLeftIndent, const int rightBottomIndent)
- int **getTopLeftIndent** () const
- int **getRightBottomIndent** () const
- void **setIndent** (const int indent)
- int **getIndent** () const
- void **setGap** (const int gap)
- int **getGap** () const
- void **setAutoFixedSize** (const bool enabled)

- bool **getAutoFixedSize** () const
- void **setMarkerColour** (const int index, const QColor &colour)
- QColor **getMarkerColour** (const int index) const
- void **setMarkerVisible** (const int index, const bool isVisible)
- bool **getMarkerVisible** (const int index) const
- void **setMarkerValue** (const int index, const double value)
- double **getMarkerValue** (const int index) const
- void **setColourBandList** (const [QEColourBandList](#) &bandList)
- [QEColourBandList](#) **getColourBandList** () const
- void **paint** (QPainter &painter, const int pointSize, const QRect &targetRect)
- void **draw** (QWidget *widget)

Protected Member Functions

- void **paintEvent** (QPaintEvent *event)

9.110.1 Detailed Description

This class provides a support widget for [QAnalogSlider](#), [QEDistribution](#), [QEHistogram](#), and maybe in the future for [QAnalogIndicator](#). It could be promoted to a plugin widget in its own right if ever necessary by adding properties and calling it up in the plugin library.

This class can also be used as a non widget object by constructing it without a parent and calling the paint function with nominated QPainter as parameter or the draw function with the target widget as parameter.

9.110.2 Member Enumeration Documentation

9.110.2.1 TextPositions

```
enum QEAxisPainter::TextPositions
```

The test postion with respect to the axis

Enumerator

BelowLeft	Below when Orientation is horizontal, Left when Orientation is vertical.
AboveRight	Above when Orientation is horizontal, Right when Orientation is vertical.

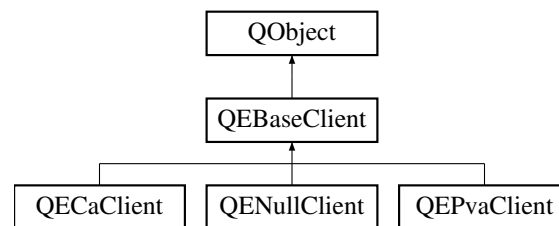
The documentation for this class was generated from the following files:

- qeframework/common/QEAxisPainter.h
- qeframework/common/QEAxisPainter.cpp

9.111 QEBaseClient Class Reference

```
#include <QEBaseClient.h>
```

Inheritance diagram for QEBaseClient:



Public Types

- enum **Type** { **NullType** , **CAType** , **PVAType** }
- enum **ChannelModes** { **None** = 0x00 , **Read** = 0x01 , **Monitor** = 0x02 , **Write** = 0x04 }

Signals

- void **connectionUpdated** (const bool isConnected)
- void **dataUpdated** (const bool firstUpdate)
- void **putCallbackComplete** (const bool isSuccessful)

Public Member Functions

- **QEBaseClient** (const Type type, const QString &pvName, QObject *parent)
- void **setUserMessage** (UserMessage *userMessage)
- UserMessage * **getUserMessage** () const
- Type **getType** () const
- QString **getPvName** () const
- virtual bool **openChannel** (const ChannelModesFlags modes)=0
- virtual void **closeChannel** ()=0
- virtual QVariant **getPvData** () const =0
- virtual bool **putPvData** (const QVariant &value)=0
- virtual bool **getIsConnected** () const =0
- virtual bool **dataIsAvailable** () const =0
- virtual QString **getId** () const =0
- virtual QString **getRemoteAddress** () const =0
- virtual QString **getEgu** () const =0
- virtual int **getPrecision** () const =0
- virtual unsigned int **hostElementCount** () const =0
- virtual unsigned int **dataElementCount** () const =0
- virtual double **getDisplayLimitHigh** () const =0
- virtual double **getDisplayLimitLow** () const =0
- virtual double **getHighAlarmLimit** () const =0
- virtual double **getLowAlarmLimit** () const =0
- virtual double **getHighWarningLimit** () const =0
- virtual double **getLowWarningLimit** () const =0
- virtual double **getControlLimitHigh** () const =0
- virtual double **getControlLimitLow** () const =0
- virtual double **getMinStep** () const =0
- virtual QStringList **getEnumerations** () const =0
- virtual QCaAlarmInfo **getAlarmInfo** () const =0
- virtual QCaDateTime **getTimeStamp** () const =0
- virtual QString **getDescription** () const =0
- virtual bool **getReadAccess** () const =0
- virtual bool **getWriteAccess** () const =0

9.111.1 Detailed Description

This class serves a common base class to both CA abd PVA client class types. While its main function is to be a QObject, i.e. has a parent and therefore is auto deleted when paranet is deleted and also support signals.

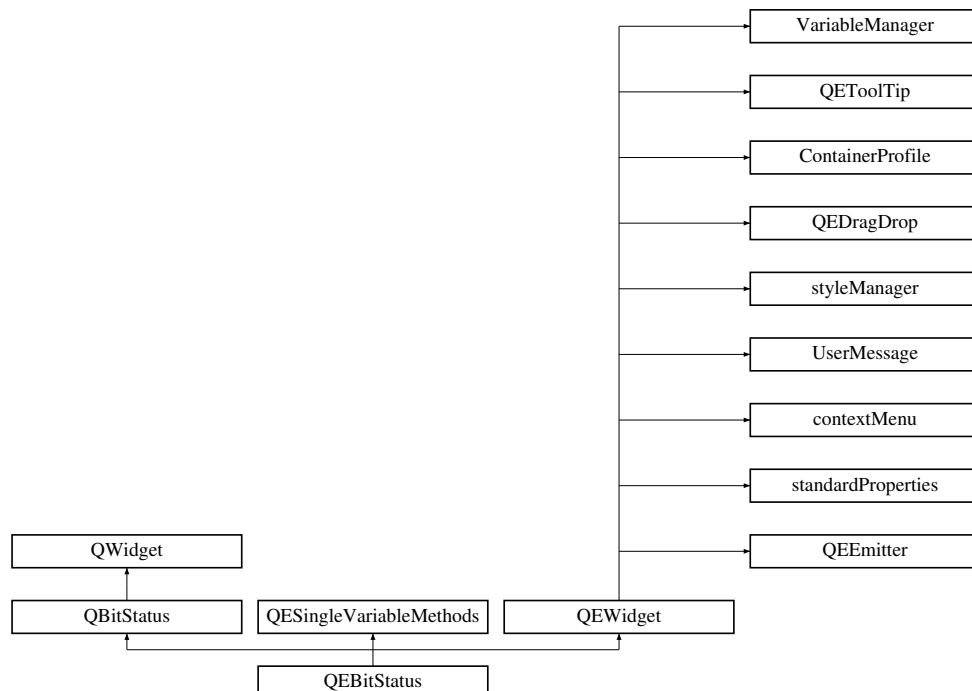
The documentation for this class was generated from the following files:

- qeframework/protocol/QEBaseClient.h
- qeframework/protocol/QEBaseClient.cpp

9.112 QEBitStatus Class Reference

```
#include <QEBitStatus.h>
```

Inheritance diagram for QEBitStatus:



Public Slots

- void [setManagedVisible](#) (bool v)

Signals

- void [dbValueChanged](#) ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void [dbConnectionChanged](#) (const bool &isConnected)

Public Member Functions

- **QEBitStatus** (QWidget *parent=0)
- **QEBitStatus** (const QString &variableName, QWidget *parent=0)

Protected Member Functions

- [QEChannel](#) * [createQcalItem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- bool [oosAware](#)
- double **value**
- bool **isActive**
- bool **isValid**

Additional Inherited Members

9.112.1 Detailed Description

This class is a CA aware Bit Status widget based on the [QBitStatus](#) widget. It is tightly integrated with the base class [QEWidget](#). Refer to QEWidget.cpp for details It provides similar functionality to that provided by the edm/medm/dephi widgets of the same/similar name.

9.112.2 Member Function Documentation

9.112.2.1 dbConnectionChanged

```
void QEBitStatus::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.112.2.2 dbValueChanged

```
void QEBitStatus::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.112.2.3 setManagedVisible

```
void QEBitStatus::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a calll to this slot if the user level allows.

9.112.3 Property Documentation

9.112.3.1 allowDrop

```
bool QEBitStatus::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.112.3.2 arrayIndex

```
int QEBitStatus::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.112.3.3 defaultStyle

```
QString QEBitStatus::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.112.3.4 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEBitStatus::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.112.3.5 elementsRequired

```
int QEBitStatus::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.112.3.6 messageSourceId

```
unsigned int QEBitStatus::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.112.3.7 oosAware

```
bool QEBitStatus::oosAware [read], [write]
```

Indicates whether the widget should respond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.112.3.8 styleSheet

```
QString QEBitStatus::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.112.3.9 userLevelEnabled

```
QE::UserLevels QEBitStatus::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.112.3.10 userLevelEngineerStyle

```
QString QEBitStatus::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.112.3.11 userLevelScientistStyle

```
QString QEBitStatus::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.112.3.12 userLevelUserStyle

```
QString QEBitStatus::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.112.3.13 userLevelVisibility

```
QE::UserLevels QEBitStatus::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.112.3.14 variable

```
QString QEBitStatus::variable [read], [write]
```

EPICS variable name (CA PV)

9.112.3.15 variableAsToolTip

```
bool QEBitStatus::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.112.3.16 variableSubstitutions

```
QString QEBitStatus::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.112.3.17 visible

```
bool QEBitStatus::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

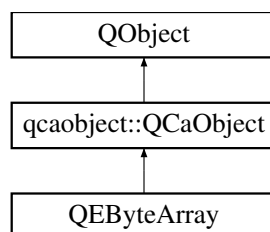
The documentation for this class was generated from the following files:

- qeframework/widgets/QEBitStatus/QEBitStatus.h
- qeframework/widgets/QEBitStatus/QEBitStatus.cpp

9.113 QByteArray Class Reference

```
#include <QByteArray.h>
```

Inheritance diagram for QByteArray:



Public Slots

- void **writeByteArray** (const QByteArray &data)

Public Member Functions

- **QEByteArray** (const QString &pvName, QObject *eventObject, unsigned int variableIndexIn)
- **QEByteArray** (const QString &pvName, QObject *eventObject, unsigned int variableIndexIn, [UserMessage](#) *userMessageIn)

Additional Inherited Members

9.113.1 Detailed Description

This class is essentially just a wrapper class for QEChannel. It sets the QE_PRIORITY_LOW low, and the request signals to SIG_BYTEARRAY. It also provides a writeByteArray slot method.

The documentation for this class was generated from the following files:

- qeframework/data/QEByteArray.h
- qeframework/data/QEByteArray.cpp

9.114 QEByteArrayUpdate Struct Reference

Public Attributes

- QByteArray **array**
- [QCaAlarmInfo](#) **alarmInfo**
- [QCaDateTime](#) **timeStamp**
- unsigned int **dataElementSize**
- unsigned int **variableIndex**
- bool **isMetaUpdate**

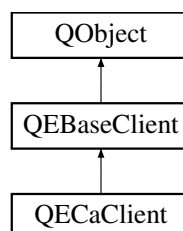
The documentation for this struct was generated from the following file:

- qeframework/data/QCaObject.h

9.115 QECaClient Class Reference

```
#include <QECaClient.h>
```

Inheritance diagram for QECaClient:



Public Member Functions

- **QECaClient** (const QString &pvName, QObject *parent)
- bool **openChannel** (const ChannelModesFlags modes)
- void **closeChannel** ()
- QVariant **getPvData** () const
- bool **putPvData** (const QVariant &value)
- bool **getIsConnected** () const
- bool **dataIsAvailable** () const
- QString **getIid** () const
- QString **getRemoteAddress** () const
- QString **getEgu** () const
- int **getPrecision** () const
- unsigned int **hostElementCount** () const
- unsigned int **dataElementCount** () const
- double **getDisplayLimitHigh** () const
- double **getDisplayLimitLow** () const
- double **getHighAlarmLimit** () const
- double **getLowAlarmLimit** () const
- double **getHighWarningLimit** () const
- double **getLowWarningLimit** () const
- double **getControlLimitHigh** () const
- double **getControlLimitLow** () const
- double **getMinStep** () const
- QStringList **getEnumerations** () const
- [QCaAlarmInfo](#) **getAlarmInfo** () const
- [QCaDateTime](#) **getTimeStamp** () const
- QString **getDescription** () const
- bool **getReadAccess** () const
- bool **getWriteAccess** () const
- void **setPriority** (const unsigned int priority)
- void **setRequestCount** (const unsigned int number)
- void **setUsePutCallback** (const bool enable)
- bool **getUsePutCallback** () const
- unsigned **getDataElementSize** () const
- const void * **getRawDataPointer** (size_t &count, const size_t offset=0) const

Protected Member Functions

- void **connectionUpdate** (const bool isConnected)
- void **dataUpdate** (const bool firstUpdate)
- void **putCallbackNotification** (const bool isSuccessful)

Friends

- class **QE_ACAI_Client**

Additional Inherited Members

9.115.1 Detailed Description

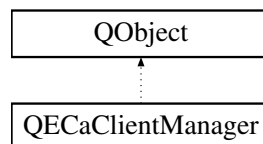
The main purpose of this class is to convert regular call backs from the ACAI client class into Qt signals (specified in [QEBaseClient](#)).

The documentation for this class was generated from the following files:

- qeframework/protocol/QECaClient.h
- qeframework/protocol/QECaClient.cpp

9.116 QECaClientManager Class Reference

Inheritance diagram for QECaClientManager:



Static Public Member Functions

- static void **initialise** ()

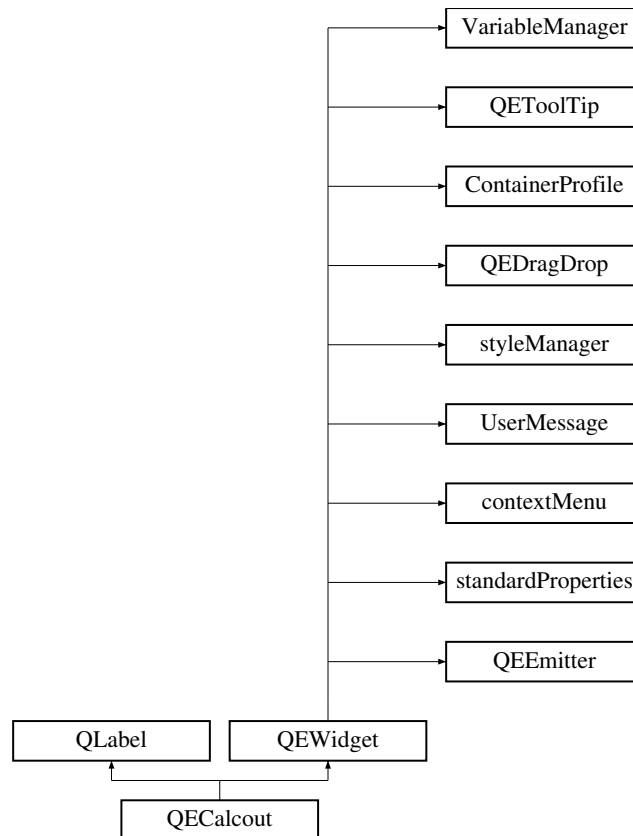
The documentation for this class was generated from the following files:

- qeframework/protocol/QECaClient.h
- qeframework/protocol/QECaClient.cpp

9.117 QECalcout Class Reference

```
#include <QECalcout.h>
```

Inheritance diagram for QECalcout:



Public Types

- enum **DataOptions** { **Use_CALC** , **Use_OCAL** }
- enum **OutputOptions** { **Every_Time** , **On_Change** , **When_Zero** , **When_Non_zero** , **Transition_To_Zero** , **Transition_To_Non_zero** , **Never** }

Public Slots

- void **inpa** (const double [a](#))
- void **inpb** (const double b)
- void **inpc** (const double c)
- void **inpd** (const double d)
- void **inpe** (const double e)
- void **inpf** (const double f)
- void **inpg** (const double g)
- void **inph** (const double h)
- void **inpi** (const double i)
- void **inpj** (const double j)
- void **inpk** (const double k)
- void **inpl** (const double l)
- void **inpa** (const int [a](#))
- void **inpb** (const int b)
- void **inpc** (const int c)
- void **inpd** (const int d)
- void **inpe** (const int e)

- void **inpf** (const int f)
- void **inpg** (const int g)
- void **inph** (const int h)
- void **inpi** (const int i)
- void **inpj** (const int j)
- void **inpk** (const int k)
- void **inpl** (const int l)
- void **inpa** (const bool a)
- void **inpb** (const bool b)
- void **inpc** (const bool c)
- void **inpd** (const bool d)
- void **inpe** (const bool e)
- void **inpf** (const bool f)
- void **inpg** (const bool g)
- void **inph** (const bool h)
- void **inpi** (const bool i)
- void **inpj** (const bool j)
- void **inpk** (const bool k)
- void **inpl** (const bool l)

Signals

- void **out** (const bool &out)
- void **out** (const int &out)
- void **out** (const long &out)
- void **out** (const qlonglong &out)
- void **out** (const double &out)
- void **out** (const QString &out)

Public Member Functions

- **QECalcout** (QWidget *parent=0)
- void **setCalc** (const QString &calc)
- QString **getCalc** () const
- void **setCalcStatus** (const QString &)
- QString **getCalcStatus** () const
- void **setOcal** (const QString &ocal)
- QString **getOcal** () const
- void **setOcalStatus** (const QString &)
- QString **getOcalStatus** () const
- void **setDopt** (const DataOptions dopt)
- DataOptions **getDopt** () const
- void **setOopt** (const OutputOptions oopt)
- OutputOptions **getOopt** () const
- void **setEgu** (const QString &egu)
- QString **getEgu** () const
- void **setPrec** (const int prec)
- int **getPrec** () const
- void **setFormat** (const QChar format)
- QChar **getFormat** () const
- double **geta** () const
- double **getb** () const
- double **getc** () const

- double **getd** () const
- double **gete** () const
- double **getf** () const
- double **getg** () const
- double **geth** () const
- double **geti** () const
- double **getj** () const
- double **getk** () const
- double **getl** () const
- void **process** ()

Properties

- QString [calc](#)
- QString [calcStatus](#)
- QString [ocal](#)
- QString [ocalStatus](#)
- DataOptions **dopt**
- OutputOptions **oopt**
- QString [egu](#)
- int **prec**
- QChar **format**
- double [a](#)
- double **b**
- double **c**
- double **d**
- double **e**
- double **f**
- double **g**
- double **h**
- double **i**
- double **j**
- double **k**
- double **l**
- bool [runVisible](#)

Additional Inherited Members

9.117.1 Detailed Description

This widget provides a calcout-like widget. So much so that where applicable the property names have been chosen to match the calcout record. This widget can be used instead of and/or to complement [QELink](#).

This widget has no PV variables of its own. The inputs, A to L, must be provided either by signals from other ([QE](#)) widgets or preset as 'constants' using the a to l properties.

Under the covers, this widget uses the same calculation engine as the calc/calcout record.

Traditionally, the type of GUI functionality the [QECalcout](#) widget supports has been effected by using EPICS database variables (often CALC records) to determine the state of GUI items. Where the variable is primarily a part of the control system this is appropriate. Where the variable is only present to support the GUI, then this functionality should be embedded in the GUI.

Conversely, while it is tempting to use this widget to quickly and easily do GUI-side calculations, you should ask your self whether this should really be done in an IOC. It can then be archived, alarmed, plotted, and be available to any other EPICS client.

9.117.2 Property Documentation

9.117.2.1 a

```
double QECalcout::a [read], [write]
```

Sets the 'constant' input values.

9.117.2.2 calc

```
QString QECalcout::calc [read], [write]
```

Specifies the calculation to be performed. The syntax is identical to that used by the calc/calcout records. However it is not limited to 80 characters. The default is "0"

9.117.2.3 calcStatus

```
QString QECalcout::calcStatus [read], [write]
```

Shows the state of the calculation expression.

9.117.2.4 egu

```
QString QECalcout::egu [read], [write]
```

Controls the string format for both the out (QString) signal and the widhets own text.

9.117.2.5 ocal

```
QString QECalcout::ocal [read], [write]
```

Specifies the ouput calc to be performed. The syntax is identical to that used by the calc/calcout records. However it is not limited to 80 characters. The default is "0"

9.117.2.6 ocalStatus

```
QString QECalcout::ocalStatus [read], [write]
```

Shows the state of the output calculation expression.

9.117.2.7 runVisible

```
bool QECalcout::runVisible [read], [write]
```

Allows the widget to be visible at run time. Default is false. Note: always visible in designer.

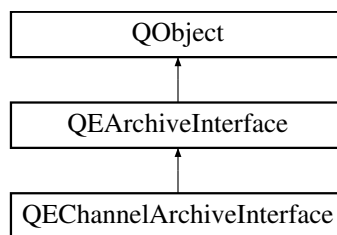
The documentation for this class was generated from the following files:

- qeframework/widgets/QECalcout/QECalcout.h
- qeframework/widgets/QECalcout/QECalcout.cpp

9.118 QEChannelArchiveInterface Class Reference

```
#include <QEChannelArchiveInterface.h>
```

Inheritance diagram for QEChannelArchiveInterface:



Public Member Functions

- **QEChannelArchiveInterface** (QUrl url, QObject *parent=0)
- void **namesRequest** (QObject *userData, const int key, QString pattern=".*")
- void **valuesRequest** (QObject *userData, const [QCaDateTime](#) startTime, const [QCaDateTime](#) endTime, const int count, const How how, const QStringList pvNames, const int key=0, const unsigned int requested←_element=0)
- void **infoRequest** (QObject *userData)
- void **archivesRequest** (QObject *userData)
- void **setUrl** (QUrl url)

Friends

- class **QEArchiveInterfaceAgent**

Additional Inherited Members

9.118.1 Detailed Description

This class provides a thin wrapper around the maiaXmlRpcClient specifically for handling the EPICS Channel Access Archive XML RPC procedure calls. While functionally thin, it does provide the conversion between the types used by the XmlRpcClient (nested structure of QVariants) the types specific archive EpicsQt, e.g. time to/from [QCaDateTime](#) and [QCaDataPoint](#).

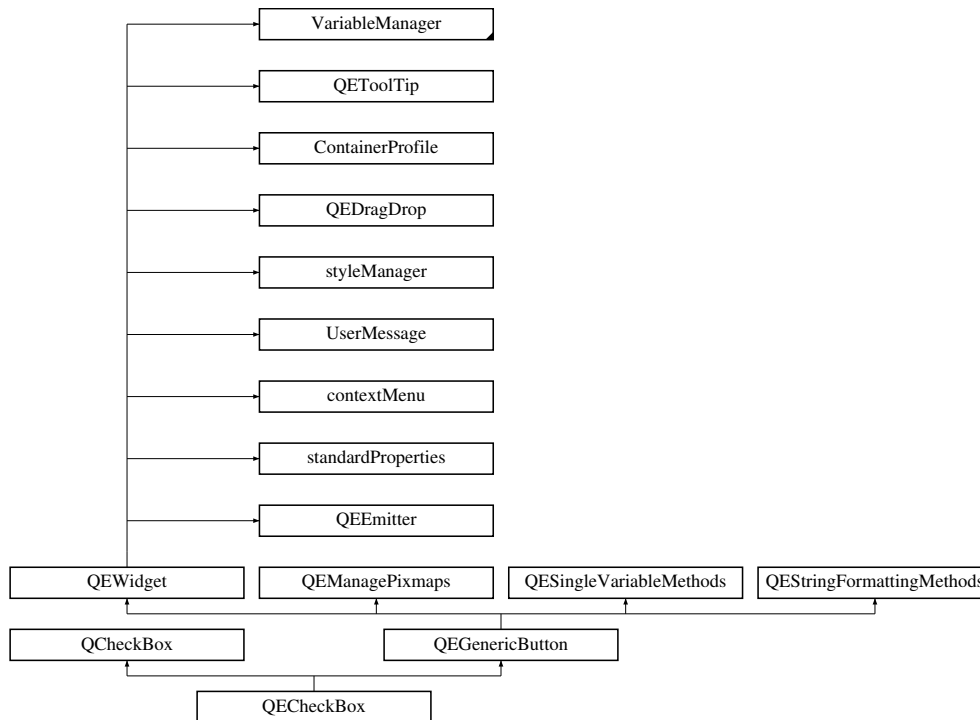
This class uses the libMaia client written by Sebastian Wiedenroth wiedi@frubar.net and Karl Glatz.

The documentation for this class was generated from the following files:

- qeframework/archive/QEChannelArchiveInterface.h
- qeframework/archive/QEChannelArchiveInterface.cpp

9.119 QCheckBox Class Reference

Inheritance diagram for QCheckBox:



Public Slots

- void **writeNow** ()
- void **requestAction** (const [QEActionRequests](#) &request)
- void **setDefaultStyle** (const QString &style)
- void **setManagedVisible** (bool v)

Signals

- void **dbValueChanged** ()
 - void **dbValueChanged** (const QString &out)
 - void **dbValueChanged** (const int &out)
 - void **dbValueChanged** (const long &out)
 - void **dbValueChanged** (const qlonglong &out)
 - void **dbValueChanged** (const double &out)
 - void **dbValueChanged** (const bool &out)
 - void **dbConnectionChanged** (const bool &isConnected)
- Sent when the widget state updated following a channel connection change.*
- void **requestResend** ()
 - void **newGui** (const [QEActionRequests](#) &request)
 - void **pressed** (int value)
 - void **released** (int value)
 - void **clicked** (int value)
 - void **programCompleted** ()
- Program started by button has compelted.*

Public Member Functions

- void [setFormatProperty](#) (QE::Formats format)
Access function for [format](#) property - refer to [format](#) property for details.
- QE::Formats [getFormatProperty](#) () const
Access function for [format](#) property - refer to [format](#) property for details.
- QECheckBox (QWidget *parent=0)
- QECheckBox (const QString &variableName, QWidget *parent=0)
- ~QECheckBox ()
Destructor.

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [arrayIndex](#)
- QString [labelText](#)
- QString [clickText](#)
- QString [clickCheckedText](#)
- bool [writeOnClick](#)
- QString [guiFile](#)
- QString [prioritySubstitutions](#)
- QE::CreationOptions [creationOption](#)
- QString [customisationName](#)
- QString [program](#)
- QStringList [arguments](#)
- QE::ProgramStartupOptions [programStartupOption](#)
- QE::DisabledRecordPolicy [disabledRecordPolicy](#)
- Qt::Alignment [alignment](#)
- QE::UpdateOptions [updateOption](#)
- QPixmap [pixmap0](#)
- QPixmap [pixmap1](#)
- QPixmap [pixmap2](#)
- QPixmap [pixmap3](#)
- QPixmap [pixmap4](#)
- QPixmap [pixmap5](#)
- QPixmap [pixmap6](#)
- QPixmap [pixmap7](#)
- QString [password](#)
- bool [confirmAction](#)
- QString [confirmText](#)
- bool [writeOnPress](#)
- bool [writeOnRelease](#)
- QString [pressText](#)
- QString [releaseText](#)
- bool [subscribe](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)

- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- QE::UserLevels [userLevelVisibility](#)
- QE::UserLevels [userLevelEnabled](#)
- QE::DisplayAlarmStateOptions [displayAlarmStateOption](#)
- bool [oosAware](#)
- int [precision](#)
- bool [useDbPrecision](#)
- int [leadingZeros](#)
- bool [addUnits](#)
- bool [forceSign](#)
- QString [localEnumeration](#)
- QE::Formats [format](#)
- int [radix](#)
- QE::Separators [separator](#)
- QE::Notations [notation](#)
- QE::ArrayActions [arrayAction](#)

Additional Inherited Members

9.119.1 Constructor & Destructor Documentation

9.119.1.1 QECheckBox() [1/2]

```
QECheckBox::QECheckBox (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.119.1.2 QECheckBox() [2/2]

```
QECheckBox::QECheckBox (
    const QString & variableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.119.2 Member Function Documentation

9.119.2.1 clicked

```
void QECheckBox::clicked (
    int value ) [signal]
```

Button has been Clicked. The value emitted is the integer interpretation of the clickText property (or the click↔CheckedText property if the button was checked)

9.119.2.2 dbValueChanged

```
void QECheckBox::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.119.2.3 newGui

```
void QECheckBox::newGui (
    const QEActionRequests & request ) [signal]
```

Internal use only. Request a new GUI is created. Typically, this is caught by the QEGui application.

9.119.2.4 pressed

```
void QECheckBox::pressed (
    int value ) [signal]
```

Button has been Pressed. The value emitted is the integer interpretation of the pressText property

9.119.2.5 released

```
void QECheckBox::released (
    int value ) [signal]
```

Button has been Released The value emitted is the integer interpretation of the releaseText property

9.119.2.6 requestAction

```
void QECheckBox::requestAction (
    const QEActionRequests & request ) [inline], [slot]
```

Default slot used to create a new GUI if there is no slot indicated in the [ContainerProfile](#) class. This slot is typically used when the button is pressed within the Designer preview window to allow the operation of the button to be tested. If an application does not specify a slot to use for creating new windows (through the [ContainerProfile](#) class) a window will still be created through this slot, but it will not respect the window creation options or any other window related application constraints. For example, the QEGui application does provide a slot for creating new GUIs in the [ContainerProfile](#) class which respects the creation options, knows how to add tabs in the application, and extend the application's window menu in the menu bar.

9.119.2.7 requestResend

```
void QECheckBox::requestResend ( ) [signal]
```

Internal use only. Used when changing a property value to force a re-display to reflect the new property value.

9.119.2.8 setDefaultStyle

```
void QECheckBox::setDefaultStyle (
    const QString & style ) [inline], [slot]
```

Update the default style applied to this widget.

9.119.2.9 setManagedVisible

```
void QECheckBox::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call to this slot, but will only be made visible by a call to this slot if the user level allows.

9.119.3 Property Documentation

9.119.3.1 addUnits

```
bool QECheckBox::addUnits [read], [write]
```

If true (default), add engineering units supplied with the data.

9.119.3.2 alignment

```
Qt::Alignment QECheckBox::alignment [read], [write]
```

Set the buttons text alignment. Left justification is particularly useful when displaying quickly changing numeric data updates.

9.119.3.3 allowDrop

```
bool QECheckBox::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.119.3.4 arguments

```
QStringList QCheckBox::arguments [read], [write]
```

Arguments for program specified in the 'program' property.

9.119.3.5 arrayAction

```
QE::ArrayActions QCheckBox::arrayAction [read], [write]
```

Text formatting option for array data. Default is ASCII. Options are:

- ASCII - treat array as a single text string. For example an array of three characters 'a' 'b' 'c' will be formatted as 'abc'.
- APPEND - treat array as an array of numbers and format a string containing them all with a space between each. For example, an array of three numbers 10, 11 and 12 will be formatted as '10 11 12'.
- INDEX - Extract a single item from the array. The item is then formatted as any other non array data would be. The item selected is determined by the arrayIndex property. For example, if arrayIndex property is 1, an array of three numbers 10, 11 and 12 will be formatted as '11'.

9.119.3.6 arrayIndex

```
int QCheckBox::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.119.3.7 clickCheckedText

```
QString QCheckBox::clickCheckedText [read], [write]
```

Text used to compare with text written or read to determine if push button should be marked as checked. Note, must be an exact match following formatting of data updates. When writing values, the 'pressText', 'ReleaseText', or 'clickedtext' must match this property to cause the button to be checked when the write occurs.

Good example: formatting set to display a data value of '1' as 'On', clickCheckedText is 'On', clickText is 'On'. In this example, the push button will be checked when a data update occurs with a value of 1 or when the button is clicked.

Bad example: formatting set to display a data value of '1' as 'On', clickCheckedText is 'On', clickText is '1'. In this example, the push button will be checked when a data update occurs with a value of 1 but, although a valid value will be written when clicked, the button will not be checked when clicked as '1' is not the same as 'On'.

9.119.3.8 clickText

```
QString QCheckBox::clickText [read], [write]
```

Value written when user clicks button if 'writeOnClick' property is true. Note: Set format to Integer when writing numeric values to enum PVs like the bo and mbbo records.

9.119.3.9 confirmAction

```
bool QECheckBox::confirmAction [read], [write]
```

If true, a dialog will be presented asking the user to confirm if the button action should be carried out

9.119.3.10 confirmText

```
QString QECheckBox::confirmText [read], [write]
```

Text used to confirm action if confirmation dialog is presented

9.119.3.11 creationOption

```
QE::CreationOptions QECheckBox::creationOption [read], [write]
```

Creation options when opening a new GUI. Open a new window, open a new tab, or replace the current window. the creation option is supplied when the button generates a newGui signal. Application code connected to this signal should honour this request if possible. When used within the QEGui application, the QEGui application creates a new window, new tab, or replaces the current window as appropriate.

9.119.3.12 customisationName

```
QString QECheckBox::customisationName [read], [write]
```

Window customisation name. This name will be used to select a set of window customisations including menu items and tool bar buttons. Applications such as QEGui can load .xml files containing named sets of window customisations. This property is used to select a set loaded from these files. The selected set of customisations will be applied to the main window containing the new GUI. Customisations are not applied if the GUI is opened as a dock.

9.119.3.13 defaultStyle

```
QString QECheckBox::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.119.3.14 disabledRecordPolicy

```
QE::DisabledRecordPolicy QECheckBox::disabledRecordPolicy [read], [write]
```

Set the widget's disabled record policy, i.e. the action to be taken when the underlying record is disabled, i.e. when the associated record's STAT field value is Disabled. When the policy is ignore, then no special action is taken. This is the default policy. When the policy is grayout, the widget is styled as if disconnected when the record is disabled. When the policy is disable, the widget is disabled when the record is disabled.

9.119.3.15 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QCheckBox::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.119.3.16 forceSign

```
bool QCheckBox::forceSign [read], [write]
```

If false (default), no "+" sign, when true always add a sign.

9.119.3.17 format

```
QE::Formats QCheckBox::format [read], [write]
```

Format to apply to data. Default is 'Default' in which case the data type supplied with the data determines how the data is formatted. For all other options, an attempt is made to format the data as requested (whatever its native form).

9.119.3.18 guiFile

```
QString QCheckBox::guiFile [read], [write]
```

File name of GUI to be presented on button click. File name can be absolute, relative to the path of the [QEForm](#) in which the button widget is located, relative to the any path in the path list published in the [ContainerProfile](#) class, or relative to the current path. See [QEWidget::openQEFile\(\)](#) in `QEWidget.cpp` for details.

9.119.3.19 labelText

```
QString QCheckBox::labelText [read], [write]
```

Button label text (prior to substitution). Macro substitutions will be applied to this text and the result will be set as the button text. Used when data updates are not being represented in the button text. IF NOT LEFT EMPTY, THIS TEXT WILL TAKE PRIORITY OVER THE PUSH BUTTON 'text' PROPERTY! For example, a button in a sub form may have a 'labelText' property of 'Turn Pump On'. When the sub form is used twice in a main form with substitutions PUMPNUM=1 and PUMPNUM=2 respectively, the two identical buttons in the sub forms will have the labels 'Turn Pump 1 On' and 'Turn Pump 2 On' respectively.

9.119.3.20 leadingZeros

```
int QCheckBox::leadingZeros [read], [write]
```

The minimum number of leading digits before any decimal/radix point. The default is 1.

9.119.3.21 localEnumeration

```
QString QCheckBox::localEnumeration [read], [write]
```

An enumeration list used to data values. Used only when the formatting option is 'local enumeration'. Value is converted to an integer and used to select a string from this list.

Format is:

```
[[<|<=|=|!=|>|=|>]value1|*]: string1 , [[<|<=|=|!=|>|=|>]value2|*]: string2 , [[<|<=|=|!=|>|=|>]value3|*]: string3 , ...
```

Where: < Less than <= Less than or equal = Equal (default if no operator specified) >= Greather than or equal

Greater than

* Always match (used to specify default text)

Values may be numeric or textual Values do not have to be in any order, but first match wins Values may be quoted Strings may be quoted Consecutive values do not have to be present. Operator is assumed to be equality if not present. White space is ignored except within quoted strings. may be included in a string to indicate a line break

Examples are:

```
0:Off,1:On 0 : "Pump Running", 1 : "Pump not running" 0:"", 1:"Warning!\nAlarm" <2:"Value is less than two", =2↵
:"Value is equal to two", >2:"Value is grater than 2" 3:"Beamline Available", *:"" "Pump Off": "OH NO!, the pump is OFF!", "Pump On": "It's OK, the pump is on"
```

The data value is converted to a string if no enumeration for that value is available. For example, if the local enumeration is '0:off,1:on', and a value of 10 is processed, the text generated is '10'. If a blank string is required, this should be explicit. for example, '0:off,1:on,10: ""'

A range of numbers can be covered by a pair of values as in the following example: >=4:"Between 4 and 8", <=8↵
:"Between 4 and 8"

9.119.3.22 messageSourceId

```
unsigned int QCheckBox::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.119.3.23 notation

```
QE::Notations QCheckBox::notation [read], [write]
```

Notation used for numerical formatting. Default is fixed.

9.119.3.24 oosAware

```
bool QECheckBox::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.119.3.25 password

```
QString QECheckBox::password [read], [write]
```

Password user will need to enter before any action is taken

9.119.3.26 pixmap0

```
QPixmap QECheckBox::pixmap0 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 0

9.119.3.27 pixmap1

```
QPixmap QECheckBox::pixmap1 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 1

9.119.3.28 pixmap2

```
QPixmap QECheckBox::pixmap2 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 2

9.119.3.29 pixmap3

```
QPixmap QECheckBox::pixmap3 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 3

9.119.3.30 pixmap4

```
QPixmap QECheckBox::pixmap4 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 4

9.119.3.31 pixmap5

```
QPixmap QCheckBox::pixmap5 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 5

9.119.3.32 pixmap6

```
QPixmap QCheckBox::pixmap6 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 6

9.119.3.33 pixmap7

```
QPixmap QCheckBox::pixmap7 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 7

9.119.3.34 precision

```
int QCheckBox::precision [read], [write]
```

Precision used when formatting floating point numbers. The default is 4. This is only used if useDbPrecision is false.

9.119.3.35 pressText

```
QString QCheckBox::pressText [read], [write]
```

Value written when user presses button if 'writeOnPress' property is true

9.119.3.36 prioritySubstitutions

```
QString QCheckBox::prioritySubstitutions [read], [write]
```

Overriding macro substitutions. These macro substitutions take precedence over any existing macro substitutions defined by the variableSubstitutions property, any parent forms, or the application containing the button. These macro substitutions are particularly useful when the button's function is to reload the same form but with different macro substitutions. The variableSubstitutions property cannot be used for this since, although they are added to the list of macro substitutions applied to the new form, they are appended to the list and the existing macro substitutions take precedence.

9.119.3.37 program

```
QString QCheckBox::program [read], [write]
```

Program to run when the button is clicked. No attempt to run a program is made if this property is empty. Example: firefox

9.119.3.38 programStartupOption

[QE::ProgramStartupOptions](#) QECheckBox::programStartupOption [read], [write]

Startup options. Just run the command, run the command within a terminal, or display the output in [QE](#) message system.

9.119.3.39 radix

int QECheckBox::radix [read], [write]

Base used for when formatting integers. Default is 10 (duh!)

9.119.3.40 releaseText

QString QECheckBox::releaseText [read], [write]

Value written when user releases button if 'writeOnRelease' property is true

9.119.3.41 separator

[QE::Separators](#) QECheckBox::separator [read], [write]

Seperators used for interger and fixed point formatting. Default is None.

9.119.3.42 styleSheet

QString QECheckBox::styleSheet [read], [write]

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly setable per se. This also stops transient styles being saved to the ui file.

9.119.3.43 subscribe

bool QECheckBox::subscribe [read], [write]

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.119.3.44 updateOption

[QE::UpdateOptions](#) QECheckBox::updateOption [read], [write]

Update options (text, pixmap, both, or state (checked or unchecked))

9.119.3.45 useDbPrecision

```
bool QECheckBox::useDbPrecision [read], [write]
```

If true (default), format floating point numbers using the precision supplied with the data. If false, the precision property is used.

9.119.3.46 userLevelEnabled

```
QE::UserLevels QECheckBox::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()`. Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.119.3.47 userLevelEngineerStyle

```
QString QECheckBox::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.119.3.48 userLevelScientistStyle

```
QString QECheckBox::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.119.3.49 userLevelUserStyle

```
QString QECheckBox::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.119.3.50 userLevelVisibility

```
QE::UserLevels QECheckBox::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()`. Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.119.3.51 variable

```
QString QECheckBox::variable [read], [write]
```

EPICS variable name (CA PV)

9.119.3.52 variableAsToolTip

```
bool QECheckBox::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.119.3.53 variableSubstitutions

```
QString QECheckBox::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.119.3.54 visible

```
bool QECheckBox::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

9.119.3.55 writeOnClick

```
bool QECheckBox::writeOnClick [read], [write]
```

If true, the 'clickText' property is written when the button is clicked. Default is true

9.119.3.56 writeOnPress

```
bool QECheckBox::writeOnPress [read], [write]
```

If true, the 'pressText' property is written when the button is pressed. Default is false

9.119.3.57 writeOnRelease

```
bool QCheckBox::writeOnRelease [read], [write]
```

If true, the 'releaseText' property is written when the button is released. Default is false

The documentation for this class was generated from the following files:

- qeframework/widgets/QEButton/QCheckBox.h
- qeframework/widgets/QEButton/QCheckBox.cpp

9.120 QEColourBand Struct Reference

Public Attributes

- double **lower**
- double **upper**
- QColor **colour**

The documentation for this struct was generated from the following file:

- qeframework/common/QEColourBandList.h

9.121 QEColourBandList Class Reference

```
#include <QEColourBandList.h>
```

Public Member Functions

- void **clear** ()
- void **append** (const [QEColourBand](#) &item)
- int **count** () const
- [QEColourBand](#) **value** (int j) const
- void **setAlarmColours** (const double dispLower, const double dispUpper, [QEChannel](#) *qca)

9.121.1 Detailed Description

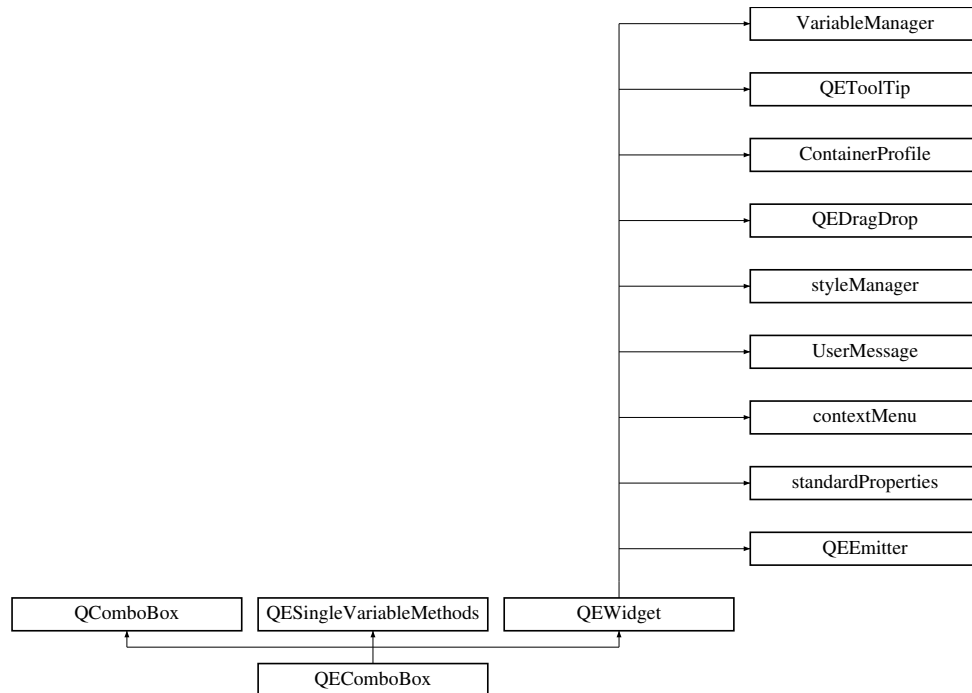
Defines a list of QEColourBands elements. Support class for QAxisPainter

The documentation for this class was generated from the following files:

- qeframework/common/QEColourBandList.h
- qeframework/common/QEColourBandList.cpp

9.122 QEComboBox Class Reference

Inheritance diagram for QEComboBox:



Public Types

- enum **OwnContextMenuOptions** { **QECB_NONE** = CM_SPECIFIC_WIDGETS_START_HERE , **QECB_↔**
APPLY_CURRENT_SELECTION , **QECB_SUB_CLASS_WIDGETS_START_HERE** }

Public Slots

- void **writeNow** ()
- void **setDefaultStyle** (const QString &style)
Update the default style applied to this widget.
- void **setManagedVisible** (bool v)
- void **setPvValue** (const QString &text)
- void **setPvValue** (const int value)
- void **setPvValue** (const double value)
- void **setPvValue** (const bool value)

Signals

- void **dbValueChanged** ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void **dbConnectionChanged** (const bool &isConnected)
Sent when the widget state updated following a channel connection change.
- void **userChange** (const QString &oldValue, const QString &newValue, const QString &lastValue)
Internal use only. Used by [QEConfiguredLayout](#) to be notified when one of its widgets has written something.

Public Member Functions

- **QEComboBox** (QWidget *parent=0)
- **QEComboBox** (const QString &variableName, QWidget *parent=0)
- void **setWriteOnChange** (bool writeOnChangeIn)
- bool **getWriteOnChange** () const
- void **setSubscribe** (bool [subscribe](#))
- bool **getSubscribe** () const
- void **setUseDbEnumerations** (bool [useDbEnumerations](#))
- bool **getUseDbEnumerations** () const
- void **setLocalEnumerations** (const QString &localEnumerations)
- QString **getLocalEnumerations** () const
- void **setAllowFocusUpdate** (bool [allowFocusUpdate](#))
- bool **getAllowFocusUpdate** () const

Protected Member Functions

- bool **eventFilter** (QObject *obj, QEvent *event)
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **setDrop** (QVariant drop)
- QVariant **getDrop** ()
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Protected Attributes

- [QEIntegerFormatting](#) **integerFormatting**
- [QELocalEnumeration](#) **localEnumerations**
- bool [useDbEnumerations](#)
- bool [writeOnChange](#)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- bool [subscribe](#)
- bool [allowFocusUpdate](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)

- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- QE::UserLevels [userLevelVisibility](#)
- QE::UserLevels [userLevelEnabled](#)
- QE::DisplayAlarmStateOptions [displayAlarmStateOption](#)
- bool [oosAware](#)
- QString [localEnumeration](#)

Additional Inherited Members

9.122.1 Member Function Documentation

9.122.1.1 dbValueChanged

```
void QEComboBox::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.122.1.2 setManagedVisible

```
void QEComboBox::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.122.1.3 setPvValue

```
void QEComboBox::setPvValue (
    const QString & text ) [slot]
```

Update the widget and write to the main associated control process variable. Supplied values are converted to the appropriate widget value type. If conversion not possible or resultant value out of range, no widget/PV update occurs.

9.122.2 Member Data Documentation

9.122.2.1 useDbEnumerations

```
bool QEComboBox::useDbEnumerations [read], [write], [protected]
```

Use database enumerations - defaults to true

9.122.2.2 writeOnChange

```
bool QComboBox::writeOnChange [read], [write], [protected]
```

Sets if this widget writes any changes as the user selects values (the QComboBox 'activated' signal is emitted). Default is 'true' (writes any changes when the QComboBox 'activated' signal is emitted).

9.122.3 Property Documentation

9.122.3.1 allowDrop

```
bool QComboBox::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.122.3.2 allowFocusUpdate

```
bool QComboBox::allowFocusUpdate [read], [write]
```

Allow updated while widget has focus - defaults to false

9.122.3.3 arrayIndex

```
int QComboBox::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.122.3.4 defaultStyle

```
QString QComboBox::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.122.3.5 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QComboBox::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.122.3.6 elementsRequired

```
int QEComboBox::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.122.3.7 localEnumeration

```
QString QEComboBox::localEnumeration [read], [write]
```

Enumerations values used when useDbEnumerations is false.

9.122.3.8 messageSourceId

```
unsigned int QEComboBox::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.122.3.9 oosAware

```
bool QEComboBox::oosAware [read], [write]
```

Indicates whether the widget should respond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.122.3.10 styleSheet

```
QString QEComboBox::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.122.3.11 subscribe

```
bool QEComboBox::subscribe [read], [write]
```

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.122.3.12 `userLevelEnabled`

```
QE::UserLevels QComboBox::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.122.3.13 `userLevelEngineerStyle`

```
QString QComboBox::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.122.3.14 `userLevelScientistStyle`

```
QString QComboBox::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.122.3.15 `userLevelUserStyle`

```
QString QComboBox::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.122.3.16 `userLevelVisibility`

```
QE::UserLevels QComboBox::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.122.3.17 variable

```
QString QEComboBox::variable [read], [write]
```

EPICS variable name (CA PV)

9.122.3.18 variableAsToolTip

```
bool QEComboBox::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.122.3.19 variableSubstitutions

```
QString QEComboBox::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.122.3.20 visible

```
bool QEComboBox::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

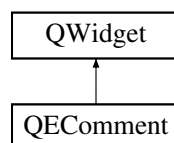
The documentation for this class was generated from the following files:

- qeframework/widgets/QEComboBox/QEComboBox.h
- qeframework/widgets/QEComboBox/QEComboBox.cpp

9.123 QEComment Class Reference

```
#include <QEComment.h>
```

Inheritance diagram for QEComment:



Public Member Functions

- **QEComment** (QWidget *parent=0)
- void **setComment** (const QString &text)
- QString **getComment** () const
- void **setRunVisible** (const bool flag)
- bool **getRunVisible** () const

Protected Member Functions

- QSize **sizeHint** ()
- void **paintEvent** (QPaintEvent *event)

Properties

- QString **comment**
- bool [runVisible](#)

9.123.1 Detailed Description

This widget does nothing per say, save hold a comment string available within designer. Note: the widget itself non-visible by default.

9.123.2 Property Documentation

9.123.2.1 runVisible

```
bool QEComment::runVisible [read], [write]
```

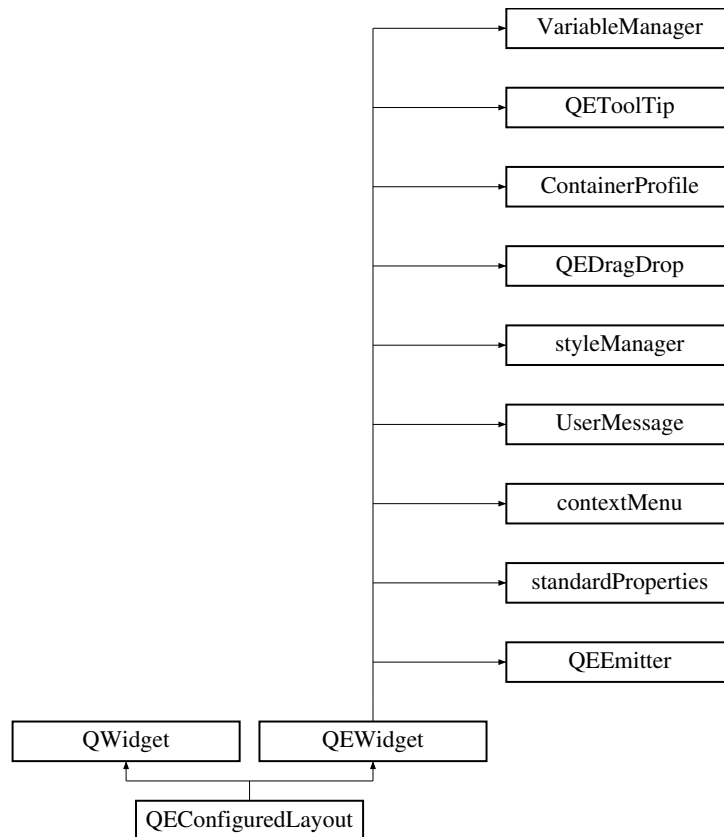
Not sure how usefull it would be to have this widget visible at run time, but let's not second guess the users' needs and desires.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEComment/QEComment.h
- qeframework/widgets/QEComment/QEComment.cpp

9.124 QEConfiguredLayout Class Reference

Inheritance diagram for QEConfiguredLayout:



Public Types

- enum **types** {
LABEL , **LINEEDIT** , **COMBOBOX** , **SPINBOX** ,
BUTTON }

Public Slots

- void [setManagedVisible](#) (bool v)

Public Member Functions

- QEConfiguredLayout** (QWidget *pParent=0, bool pSubscription=true)
- void **setItemDescription** (QString pValue)
- QString **getItemDescription** ()
- void **setShowItemList** (bool pValue)
- bool **getShowItemList** ()
- void **setConfigurationType** ([QE::SourceOptions](#) pValue)
- [QE::SourceOptions](#) **getConfigurationType** ()
- void **setConfigurationFile** (QString pValue)
- QString **getConfigurationFile** ()

- void **setConfigurationText** (QString pValue)
- QString **getConfigurationText** ()
- void **setOptionsLayout** (QE::LayoutOptions pValue)
- QE::LayoutOptions **getOptionsLayout** ()
- void **setCurrentUserType** (int pValue)
- int **getCurrentUserType** ()
- void **refreshFields** ()
- void **userLevelChanged** (QE::UserLevels pValue)

Public Attributes

- QList< [_Item](#) * > **itemList**
- QList< [_Field](#) * > **currentFieldList**

Protected Attributes

- QLabel * **qLabelItemDescription**
- QComboBox * **qComboBoxItemList**
- QVBoxLayout * **qVBoxLayoutFields**
- QScrollArea * **qScrollArea**
- QString **configurationFile**
- QString **configurationText**
- [QE::SourceOptions](#) **configurationType**
- QE::LayoutOptions [optionsLayout](#)

Change the order of the widgets. Valid orders are: TOP, BOTTOM, LEFT and RIG.

- int **currentUserType**
- bool **subscription**

Properties

- QString **itemDescription**
- bool **showItemList**
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)

Additional Inherited Members

9.124.1 Member Function Documentation

9.124.1.1 setManagedVisible

```
void QEConfiguredLayout::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call to this slot, but will only be made visible by a call to this slot if the user level allows.

9.124.2 Property Documentation

9.124.2.1 defaultStyle

```
QString QEConfiguredLayout::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.124.2.2 messageSourceId

```
unsigned int QEConfiguredLayout::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.124.2.3 styleSheet

```
QString QEConfiguredLayout::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.124.2.4 userLevelEnabled

```
QE::UserLevels QEConfiguredLayout::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programmatically through `setUserLevel()`. Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.124.2.5 `userLevelEngineerStyle`

```
QString QEConfiguredLayout::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.124.2.6 `userLevelScientistStyle`

```
QString QEConfiguredLayout::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.124.2.7 `userLevelUserStyle`

```
QString QEConfiguredLayout::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.124.2.8 `userLevelVisibility`

```
QE::UserLevels QEConfiguredLayout::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.124.2.9 `visible`

```
bool QEConfiguredLayout::visible [read], [write]
```

`variableAsToolTip` not applicale to V2 options. Class should ensure this option is initialised to false. `allowDrop` not applicale to V2 options. Class should ensure this option is initialised to false. Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

The documentation for this class was generated from the following files:

- `qeframework/widgets/QEConfiguredLayout/QEConfiguredLayout.h`
- `qeframework/widgets/QEConfiguredLayout/QEConfiguredLayout.cpp`

9.125 QEConnectionUpdate Struct Reference

Public Attributes

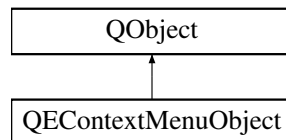
- [QCaConnectionInfo](#) **connectionInfo**
- unsigned int **variableIndex**

The documentation for this struct was generated from the following file:

- qeframework/data/QCaObject.h

9.126 QEContextMenuObject Class Reference

Inheritance diagram for QEContextMenuObject:



Public Slots

- void **contextMenuTriggeredSlot** ([QAction](#) *selectedItem)
- void **showContextMenuSlot** (const [QPoint](#) &pos)

Signals

- void **requestAction** (const [QEActionRequests](#) &)

Public Member Functions

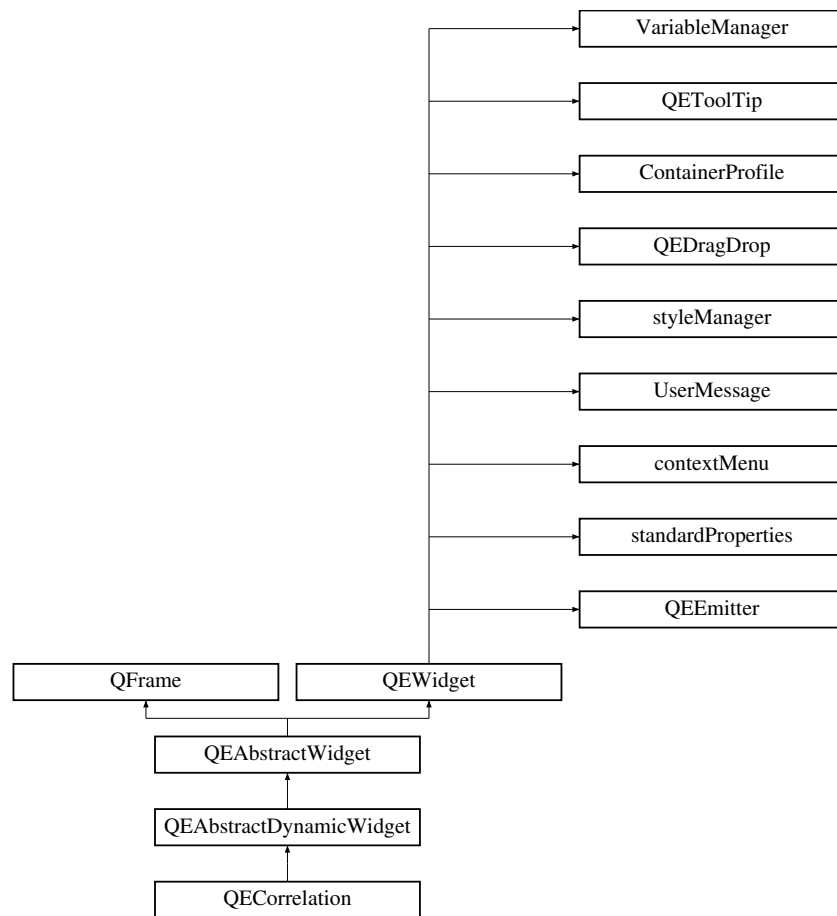
- **QEContextMenuObject** ([contextMenu](#) *menuIn, [QObject](#) *parent)
- void **sendRequestAction** (const [QEActionRequests](#) &request)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/contextMenu.h
- qeframework/widgets/QEWidget/contextMenu.cpp

9.127 QECorrelation Class Reference

Inheritance diagram for QECorrelation:



Public Types

- enum **Constants** { **NUMBER_OF_ITEMS** = 2 }

Public Member Functions

- **QECorrelation** (QWidget *parent=0)
- **QECorrelation** (const QString &xVariableName, QWidget *parent=0)
- **QECorrelation** (const QString &xVariableName, const QString &yVariableName, QWidget *parent=0)
- void **setXVariableName** (const QString &pvName)
- QString **getXVariableName** () const
- void **setYVariableName** (const QString &pvName)
- QString **getYVariableName** () const
- void **setVariableSubstitutions** (const QString &variableNameSubstitutions)
- QString **getVariableSubstitutions** () const
- void **setSampleInterval** (const double interval)
- double **getSampleInterval** () const
- void **setNumberPoints** (const int number)
- int **getNumberPoints** () const

- void **setXLogarithmic** (bool [visible](#))
- bool **getXLogarithmic** () const
- void **setYLogarithmic** (bool [visible](#))
- bool **getYLogarithmic** () const
- int **addPvName** (const QString &pvName)
- void **clearAllPvNames** ()

Protected Member Functions

- bool **eventFilter** (QObject *watched, QEvent *event)
- [QEChannel](#) * **createQcaltem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **mousePressEvent** (QMouseEvent *event)
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **saveConfiguration** ([PersistenceManager](#) *pm)
- void **restoreConfiguration** ([PersistenceManager](#) *pm, [restorePhases](#) restorePhase)

Properties

- QString **variableX**
- QString **variableY**
- QString **variableSubstitutions**
- double **sampleInterval**
- int **numberPoints**
- bool **xLogarithmic**
- bool **yLogarithmic**

Additional Inherited Members

9.127.1 Member Function Documentation

9.127.1.1 **restoreConfiguration()**

```
void QECorrelation::restoreConfiguration (
    PersistenceManager * ,
    restorePhases ) [protected], [virtual]
```

Service a request to restore the [QE](#) widget's configuration. A [QE](#) widget recover any configuration details from the [PersistenceManager](#). For example, a [QEStripChart](#) may restore the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method. This is called twice with an incrementing restorePhase. Most widgets will miss the first call as they don't exist yet (they are created as part of the first phase).

Reimplemented from [QEWidget](#).

9.127.1.2 saveConfiguration()

```
void QECorrelation::saveConfiguration (
    PersistenceManager * ) [protected], [virtual]
```

Service a request to save the [QE](#) widget's current configuration. A widget may save any configuration details through the [PersistenceManager](#). For example, a [QEStripChart](#) may save the variables being plotted. Many [QE](#) widgets do not have any persistent data requirements and do not implement this method.

Reimplemented from [QEWidget](#).

9.127.2 Property Documentation

9.127.2.1 variableSubstitutions

```
QString QECorrelation::variableSubstitutions [read], [write]
```

Default macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'SAMPLE=SAM1, NAME = "Ref foil"' These substitutions are applied to all the variable names.

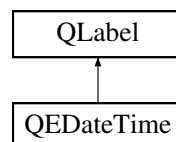
The documentation for this class was generated from the following files:

- qeframework/widgets/QECorrelation/QECorrelation.h
- qeframework/widgets/QECorrelation/QECorrelation.cpp

9.128 QEDateTime Class Reference

```
#include <QEDateTime.h>
```

Inheritance diagram for QEDateTime:



Public Member Functions

- **QEDateTime** (QWidget *parent=0)
- QString **getDateTimeFormat** () const
- void **setDateTimeFormat** (const QString format)
- Qt::TimeSpec **getTimeZone** () const
- void **setTimeZone** (const Qt::TimeSpec zone)
- bool **getShowZone** () const
- void **setShowZone** (const bool showZone)

Protected Member Functions

- QSize **sizeHint** () const
- bool **eventFilter** (QObject *watched, QEvent *event)

Properties

- QString **dateTimeFormat**
- Qt::TimeSpec **timeZone**
- bool **showZone**

9.128.1 Detailed Description

This class is a non EPICS aware label widget based on the Qt label widget and Qt DateTime class. It is only used for displaying data and time of the day.

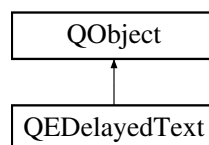
The documentation for this class was generated from the following files:

- qeframework/widgets/QEDateTime/QEDateTime.h
- qeframework/widgets/QEDateTime/QEDateTime.cpp

9.129 QEDelayedText Class Reference

```
#include <QEDelayedText.h>
```

Inheritance diagram for QEDelayedText:



Public Slots

- void **setText** (const QString &)

Signals

- void **textChanged** (const QString &)

Public Member Functions

- **QEDelayedText** (const double delay, QObject *parent=0)
- bool **doubleConnect** (const QObject *sender, const char *signal, const QObject *receiver, const char *member) const

9.129.1 Detailed Description

Provides a delayed text signal, noy unlike that employed by [QCaVariableNamePropertyManager](#)

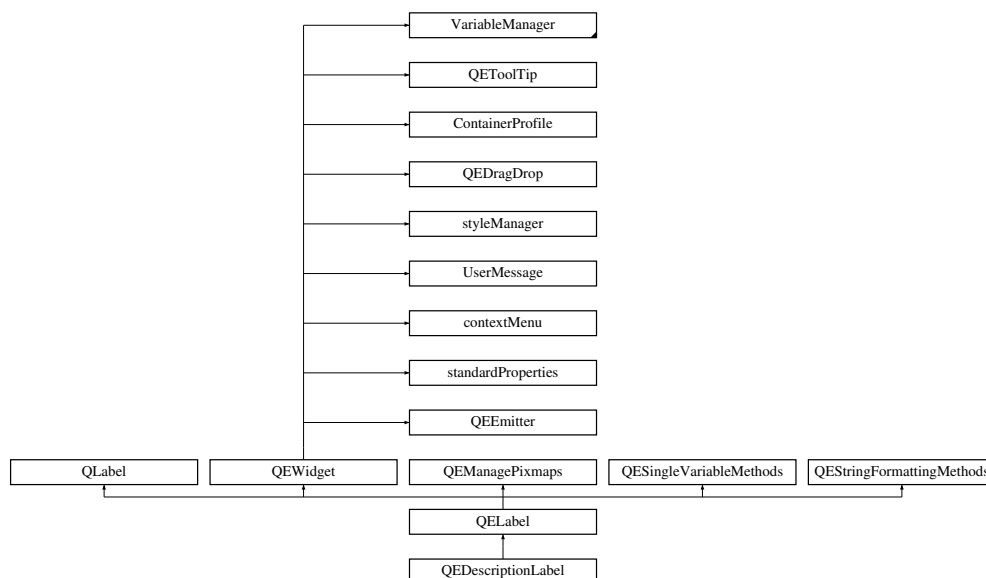
The documentation for this class was generated from the following files:

- qeframework/common/QEDelayedText.h
- qeframework/common/QEDelayedText.cpp

9.130 QEDescriptionLabel Class Reference

```
#include <QEDescriptionLabel.h>
```

Inheritance diagram for QEDescriptionLabel:



Public Member Functions

- [QEDescriptionLabel](#) (QWidget *parent=0)
- [QEDescriptionLabel](#) (const QString &variableName, QWidget *parent=0)
- [~QEDescriptionLabel](#) ()
- void **setEnabledDots** (const bool enable)
- bool **getEnabledDots** () const

Properties

- bool [enableDots](#)

Additional Inherited Members

9.130.1 Detailed Description

This is just a [QELabel](#), but with some different default property values. These are: a) Clear style sheet and default style. b) The indent set to -1, the QLabel default c) Set display alarm option to `DISPLAY_ALARM_STATE_NEVER` d) Set Font size to 8.

Note: there is one additional property: `enableDots`.

9.130.2 Constructor & Destructor Documentation

9.130.2.1 QEDescriptionLabel() [1/2]

```
QEDescriptionLabel::QEDescriptionLabel (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.130.2.2 QEDescriptionLabel() [2/2]

```
QEDescriptionLabel::QEDescriptionLabel (
    const QString & variableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.130.2.3 ~QEDescriptionLabel()

```
QEDescriptionLabel::~QEDescriptionLabel ( )
```

Destructor

9.130.3 Property Documentation

9.130.3.1 enableDots

```
bool QEDescriptionLabel::enableDots [read], [write]
```

When true, text is right padded with trailing " . . . " to right-hand edge of the label. When false, no additional dot and spaces text is added. The default enableDots property value is false.

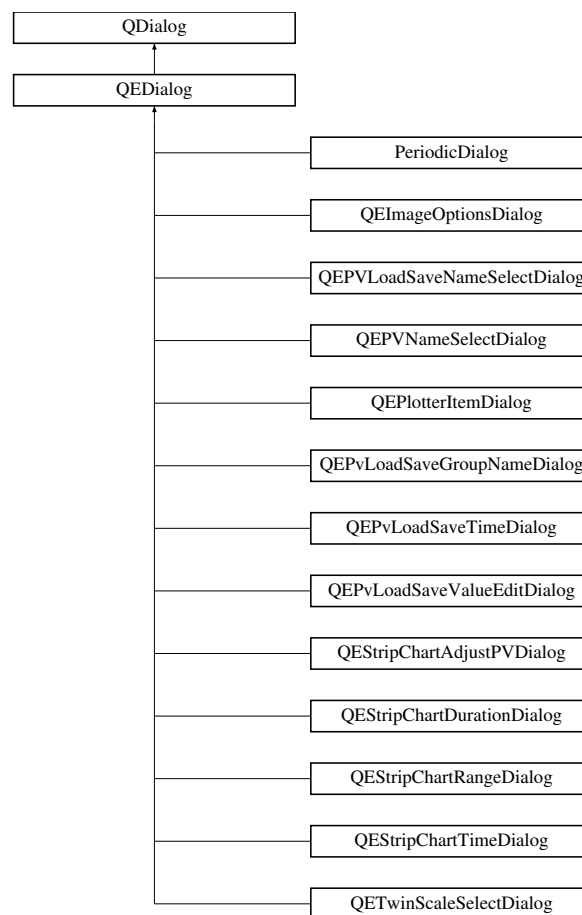
The documentation for this class was generated from the following files:

- qeframework/widgets/QELabel/QEDescriptionLabel.h
- qeframework/widgets/QELabel/QEDescriptionLabel.cpp

9.131 QEDialog Class Reference

```
#include <QEDialog.h>
```

Inheritance diagram for QEDialog:



Public Slots

- virtual int **exec** (QWidget *targetWidget)

Public Member Functions

- **QEDialog** (QWidget *parent=0)

Protected Member Functions

- void **setSourceWidget** (QWidget *sourceWidget)

9.131.1 Detailed Description

This class provides a thin wrapper around the QDialog class, specifically the exec function. There are two issues with the standard QDialog when exec-ed.

a) if the dialog has no parent and there is a single application form open then the dialog is centred over the form - which is good. However if two or more forms are open, it appears in the centre of the screen - okay but not ideal; and

b) when scaling applied (even null scaling) this confuses the dialog and it appears in the top left corner of the screen which is not acceptable.

The overaloded wrapper function allows a widget to be specified, and if specified the dialog is centred over the widget.

The documentation for this class was generated from the following files:

- qeframework/common/QEDialog.h
- qeframework/common/QEDialog.cpp

9.132 QEDisplayRanges Class Reference

Public Member Functions

- **QEDisplayRanges** (const double min, const double max)
- bool **operator==** (const [QEDisplayRanges](#) &other) const
- bool **operator!=** (const [QEDisplayRanges](#) &other) const
- bool **isSimilar** (const [QEDisplayRanges](#) &other, const double tolerance) const
- void **clear** ()
- void **setRange** (const double min, const double max)
- void **merge** (const double d)
- void **merge** (const [QEDisplayRanges](#) &other)
- bool **getIsDefined** () const
- double **getMinimum** () const
- double **getMaximum** () const
- bool **getMinMax** (double &min, double &max) const
- void **adjustLogMinMax** (double &minOut, double &maxOut, double &majorOut) const
- void **adjustMinMax** (const int number, const bool roundToMajor, double &minOut, double &maxOut, double &majorOut) const

The documentation for this class was generated from the following files:

- qeframework/common/QEDisplayRanges.h
- qeframework/common/QEDisplayRanges.cpp

Signals

- void [dbConnectionChanged](#) (const bool &isConnected)
Sent when the widget state updated following a channel connection change.
- void [dbValueChanged](#) ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)

Public Member Functions

- [QEDistribution](#) (QWidget *parent=0)
- [QEDistribution](#) (const QString &variableName, QWidget *parent=0)
- [~QEDistribution](#) ()
- double **getNumberStdDevs** () const
- bool **getIsRectangular** () const
- int **getEdgeWidth** () const
- QColor **getBackgroundColour** () const
- QColor **getEdgeColour** () const
- QColor **getFillColour** () const
- QColor **getGaussianColour** () const
- bool **getShowGaussian** () const
- int **getGaussianWidth** () const
- int **addPvName** (const QString &pvName)
- void **clearAllPvNames** ()

Protected Member Functions

- bool **eventFilter** (QObject *watched, QEvent *event)
- [QEChannel](#) * [createQcItem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant v)
- void [saveConfiguration](#) ([PersistenceManager](#) *pm)
- void [restoreConfiguration](#) ([PersistenceManager](#) *pm, [restorePhases](#) restorePhase)
- void **setReadOut** (const QString &text)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- double [numberStdDevs](#)
- bool [isRectangular](#)
- int [edgeWidth](#)
- bool [showGaussian](#)
- int [gaussianWidth](#)
- QColor [fillColour](#)
- QColor [edgeColour](#)
- QColor [gaussianColour](#)
- QColor [backgroundColour](#)

Additional Inherited Members

9.133.1 Detailed Description

The [QEDistribution](#) class provides a histogram of the distribution of the values of a scaler PV.

To a certain extent, this widget mimics the behaviour of the histogram record in that it bins values.

9.133.2 Constructor & Destructor Documentation

9.133.2.1 QEDistribution() [1/2]

```
QEDistribution::QEDistribution (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.133.2.2 QEDistribution() [2/2]

```
QEDistribution::QEDistribution (
    const QString & variableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.133.2.3 ~QEDistribution()

```
QEDistribution::~QEDistribution ( )
```

Destruction

9.133.3 Member Function Documentation

9.133.3.1 dbValueChanged

```
void QEDistribution::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.133.3.2 restoreConfiguration()

```
void QEDistribution::restoreConfiguration (
    PersistenceManager * ,
    restorePhases ) [protected], [virtual]
```

Service a request to restore the [QE](#) widget's configuration. A [QE](#) widget recover any configuration details from the [PersistenceManager](#). For example, a [QEStripChart](#) may restore the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method. This is called twice with an incrementing restorePhase. Most widgets will miss the first call as they don't exist yet (they are created as part of the first phase).

Reimplemented from [QEWidget](#).

9.133.3.3 saveConfiguration()

```
void QEDistribution::saveConfiguration (
    PersistenceManager * ) [protected], [virtual]
```

Service a request to save the [QE](#) widget's current configuration. A widget may save any configuration details through the [PersistenceManager](#). For example, a [QEStripChart](#) may save the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method.

Reimplemented from [QEWidget](#).

9.133.4 Property Documentation

9.133.4.1 arrayIndex

```
int QEDistribution::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.133.4.2 backgroundColour

QColor QEDistribution::backgroundColour [read], [write]

default : light grey

9.133.4.3 edgeColour

QColor QEDistribution::edgeColour [read], [write]

default: dark blue

9.133.4.4 edgeWidth

int QEDistribution::edgeWidth [read], [write]

Histogram edge width : range 0 to 20, default 2.

9.133.4.5 elementsRequired

int QEDistribution::elementsRequired [read], [write]

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.133.4.6 fillColour

QColor QEDistribution::fillColour [read], [write]

default: light blue

9.133.4.7 gaussianColour

QColor QEDistribution::gaussianColour [read], [write]

default: red

9.133.4.8 gaussianWidth

int QEDistribution::gaussianWidth [read], [write]

gaussian plot width : range 1 to 20, default 2

9.133.4.9 isRectangular

```
bool QEDistribution::isRectangular [read], [write]
```

Selects plot mode - rectangular histogram (default) or smooth

9.133.4.10 numberStdDevs

```
double QEDistribution::numberStdDevs [read], [write]
```

Number of standard deviations to plot: range 0.1 to 9.0, default 3.0

9.133.4.11 showGaussian

```
bool QEDistribution::showGaussian [read], [write]
```

Selects display of gaussian curve - default yes.

9.133.4.12 variable

```
QString QEDistribution::variable [read], [write]
```

EPICS variable name (CA PV)

9.133.4.13 variableSubstitutions

```
QString QEDistribution::variableSubstitutions [read], [write]
```

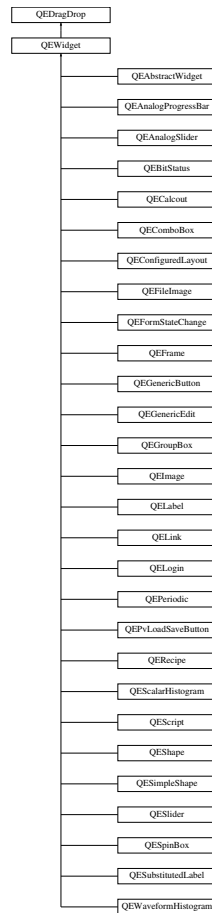
Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEDistribution/QEDistribution.h
- qeframework/widgets/QEDistribution/QEDistribution.cpp

9.134 QEDragDrop Class Reference

Inheritance diagram for QEDragDrop:



Public Member Functions

- **QEDragDrop** ([QEWidget](#) *qewIn, QWidget *ownerIn)
- void **setAllowDrop** (bool allowDropIn)
- bool **getAllowDrop** () const

Protected Member Functions

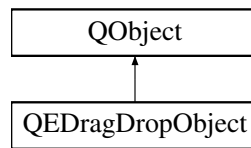
- void **qcaDragEnterEvent** (QDragEnterEvent *event)
- void **qcaDropEvent** (QDropEvent *event, const bool allText=false)
- void **qcaMousePressEvent** (QMouseEvent *event)
- virtual void **setDrop** (QVariant)
- virtual QVariant **getDrop** ()
- void **setDragDropConsumer** (QObject *consumer)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/QEDragDrop.h
- qeframework/widgets/QEWidget/QEDragDrop.cpp

9.135 QEDragDropObject Class Reference

Inheritance diagram for QEDragDropObject:



Signals

- void **requestAction** (const [QEActionRequests](#) &)

Public Member Functions

- **QEDragDropObject** (QObject *parent)
- void **sendRequestAction** (const [QEActionRequests](#) &request)

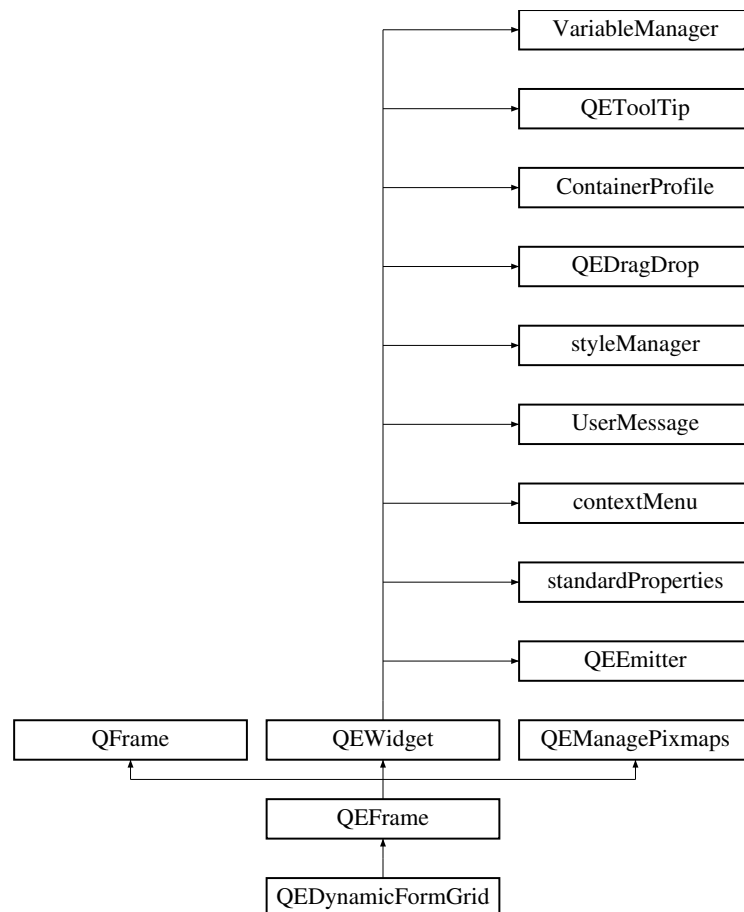
The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/QEDragDrop.h
- qeframework/widgets/QEWidget/QEDragDrop.cpp

9.136 QEDynamicFormGrid Class Reference

```
#include <QEDynamicFormGrid.h>
```

Inheritance diagram for QEDynamicFormGrid:



Public Member Functions

- [QEDynamicFormGrid](#) (`QWidget *parent=0`)
- **QEDynamicFormGrid** (`const QString &uiFileName, QWidget *parent=0`)
- virtual [~QEDynamicFormGrid](#) ()
Destruction.
- void **setUiFile** (`const QString &uiFileName`)
- `QString` **getUiFile** () const
- void **setTitle** (`const QString &title`)
- `QString` **getTitle** () const
- void **setNameSubstitutions** (`const QString &variableSubstitutions`)
- `QString` **getNameSubstitutions** () const
- void **setFormalArg** (`const QString &formalArg`)
- `QString` **getFormalArg** () const
- void **setActualArgList** (`const QStringList &actualArgList`)
- `QStringList` **getActualArgList** () const
- void **setColumns** (`const int n`)
- int **getColumns** () const
- void **setGridOrder** (`const QE::GridOrders go`)
- [QE::GridOrders](#) **getGridOrder** () const
- void **setMargin** (`const int n`)
- int **getMargin** () const
- void **setSpacing** (`const int n`)
- int **getSpacing** () const

Protected Member Functions

- QSize **sizeHint** () const
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **saveConfiguration** (PersistenceManager *pm)
- void **restoreConfiguration** (PersistenceManager *pm, restorePhases restorePhase)

Properties

- QString **uiFile**
- QString **title**
- QString **variableSubstitutions**
- QString **formalArg**
- QStringList **actualArgList**
- int **columns**
- QE::GridOrders **gridOrder**
- int **margin**
- int **spacing**
- bool **variableAsToolTip**
- bool **allowDrop**
- QE::DisplayAlarmStateOptions **displayAlarmStateOption**

Additional Inherited Members

9.136.1 Detailed Description

This class loosely based on [QEFormGrid](#)

9.136.2 Constructor & Destructor Documentation

9.136.2.1 QEDynamicFormGrid()

```
QEDynamicFormGrid::QEDynamicFormGrid (  
    QWidget * parent = 0 ) [explicit]
```

Create a grid widget with default parameters.

9.136.3 Member Function Documentation

9.136.3.1 `restoreConfiguration()`

```
void QEDynamicFormGrid::restoreConfiguration (
    PersistenceManager * ,
    restorePhases ) [protected], [virtual]
```

Service a request to restore the [QE](#) widget's configuration. A [QE](#) widget recover any configuration details from the [PersistenceManager](#). For example, a [QESTripChart](#) may restore the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method. This is called twice with an incrementing `restorePhase`. Most widgets will miss the first call as they don't exist yet (they are created as part of the first phase).

Reimplemented from [QEWidget](#).

9.136.3.2 `saveConfiguration()`

```
void QEDynamicFormGrid::saveConfiguration (
    PersistenceManager * ) [protected], [virtual]
```

Service a request to save the [QE](#) widget's current configuration. A widget may save any configuration details through the [PersistenceManager](#). For example, a [QESTripChart](#) may save the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method.

Reimplemented from [QEWidget](#).

9.136.4 Property Documentation

9.136.4.1 `actualArgList`

```
QStringList QEDynamicFormGrid::actualArgList [read], [write]
```

The set of actual macro values available at run time. Default value: empty list.

9.136.4.2 `allowDrop`

```
bool QEDynamicFormGrid::allowDrop [read], [write]
```

`allowDrop` is added as a non-designable property here only to hide the implementation present in [QEFrame](#)

9.136.4.3 `columns`

```
int QEDynamicFormGrid::columns [read], [write]
```

Specifies the number of columns. This is restricted to the range 1 to 20. Default value: 1.

9.136.4.4 displayAlarmStateOption

`QE::DisplayAlarmStateOptions QEDynamicFormGrid::displayAlarmStateOption [read], [write]`

displayAlarmStateOption is added as a non-designable property here only to hide the implementation present in [QEFrame](#)

9.136.4.5 formalArg

`QString QEDynamicFormGrid::formalArg [read], [write]`

The formal macro name used when loading uiFile Default value: "DEVICE".

9.136.4.6 gridOrder

`QE::GridOrders QEDynamicFormGrid::gridOrder [read], [write]`

Specifies the gridOrder: RowMajor or ColMajor. Default value: RowMajor.

9.136.4.7 margin

`int QEDynamicFormGrid::margin [read], [write]`

Margin applied to/extracted from the internal QGridLayout object. Default value: 2.

9.136.4.8 spacing

`int QEDynamicFormGrid::spacing [read], [write]`

Spacing applied to/extracted from the internal QGridLayout object. Default value: 2.

9.136.4.9 title

`QString QEDynamicFormGrid::title [read], [write]`

The title of the [QEDynamicFormGrid](#). Default value: "".

9.136.4.10 uiFile

`QString QEDynamicFormGrid::uiFile [read], [write]`

GridOrders specifies how grid elements are layed out.

Row major (example 10, items, 3 cols): 0 1 2 3 4 5 6 7 8 9 - -

Col major (example 10, items, 3 cols): 0 4 8 1 5 9 2 6 - 3 7 - The uiFile loaded into each [QEForm](#) element. Default value: "".

9.136.4.11 variableAsToolTip

```
bool QEDynamicFormGrid::variableAsToolTip [read], [write]
```

variableAsToolTip is added as a non-designable property here only to hide the implementation present in [QEFrame](#)

9.136.4.12 variableSubstitutions

```
QString QEDynamicFormGrid::variableSubstitutions [read], [write]
```

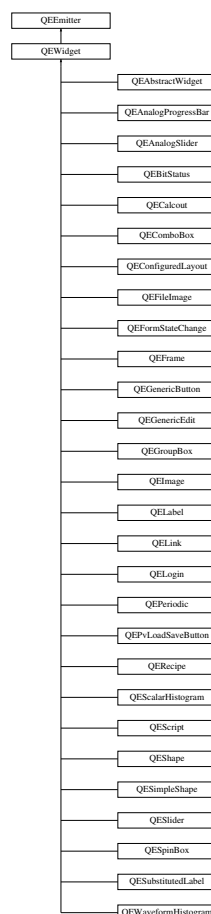
(Default) name substitutions applied to uiFile and title. Called variableSubstitutions for consistency. Default value: "".

The documentation for this class was generated from the following files:

- qeframework/widgets/QEFormGrid/QEDynamicFormGrid.h
- qeframework/widgets/QEFormGrid/QEDynamicFormGrid.cpp

9.137 QEEmitter Class Reference

Inheritance diagram for QEEmitter:



Public Member Functions

- **QEEmitter** ([QWidget](#) *qew, [QWidget](#) *owner)

Protected Member Functions

- void **emitDbConnectionChanged** (const unsigned int variableIndex)
- void **emitDbValueChanged** (const unsigned int variableIndex)
- void **emitDbValueChanged** (const [QString](#) &formattedText, const unsigned int variableIndex)
- bool **isDbValueChangedEmitInProgress** () const

The documentation for this class was generated from the following files:

- qeframework/widgets/QWidget/QEEmitter.h
- qeframework/widgets/QWidget/QEEmitter.cpp

9.138 QEnvironmentVariables Class Reference

```
#include <QEnvironmentVariables.h>
```

Public Member Functions

- **QEnvironmentVariables** (const [QString](#) &prefix)
- bool **getBool** (const [QString](#) &name, const bool defaultValue) const
- [QString](#) **getString** (const [QString](#) &name, const [QString](#) &defaultValue) const
- int **getInt** (const [QString](#) &name, const int defaultValue) const
- double **getFloat** (const [QString](#) &name, const double defaultValue) const

Static Public Member Functions

- static void **setDefaultPrefix** (const [QString](#) &prefix)
- static [QString](#) **getDefaultPrefix** ()

9.138.1 Detailed Description

This class provides a means to access environment variables.

The types of data that may be extracted from an option are bool, [QString](#), int and double; These are accessed by the associated getXxxx () functions where Xxxx is one of Bool, String, Int and Float.

Each of these functions takes a default value of the appropriate type which is returned to the caller if the value is not defined or, as in the case of numerical values, is ill-defined.

Numeric validity is defined by [QString::toInt](#) () and [QString::toDouble](#) (). Do note that [QString::toInt](#) () does not accept 0x... hexadecimal or 0... octal numbers.

The environment variable name is formed by concatenating the prefix and the suffix and converting this to upper case if needed. The prefix can be specified when the object is constructed otherwise the default prefix is used when no prefix is provided. Example prefix could be "QE_".

This class is one of several provided to access configuration data such as from environment variables, command line options and settings. They all provide a similar API.

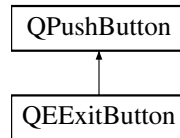
The documentation for this class was generated from the following files:

- qeframework/adaptation_parameters/QEnvironmentVariables.h
- qeframework/adaptation_parameters/QEnvironmentVariables.cpp

9.139 QExitButton Class Reference

```
#include <QExitButton.h>
```

Inheritance diagram for QExitButton:



Public Member Functions

- **QExitButton** (QWidget *parent=0)
- QSize **sizeHint** () const
- void **setExitProgram** (const bool [exitProgram](#))
- bool **getExitProgram** () const

Properties

- bool [exitProgram](#)

9.139.1 Detailed Description

This class provides an an exit button.

9.139.2 Property Documentation

9.139.2.1 exitProgram

```
bool QExitButton::exitProgram [read], [write]
```

This property controls the behaviour of the widget. When `exitProgram` is false (default), it will close the current window. When `exitProgram` is true, it will close the entire application.

The documentation for this class was generated from the following files:

- qframework/widgets/QExitButton/QExitButton.h
- qframework/widgets/QExitButton/QExitButton.cpp

9.140 QExpressionEvaluation Class Reference

```
#include <QExpressionEvaluation.h>
```


Public Types

- enum **InputKinds** { **Normal** , **Primed** }
- typedef double **CalculateArguments**[NumberInputKinds][NumberUserArguments]

Public Member Functions

- **QEEvaluation** (const int numberOfInputs, const bool allowPrimedInput)
- bool **initialise** (const QString &expression)
- QString **getCalcError** () const
- double **evaluate** (const CalculateArguments &userArgs, bool *okay=0) const

Static Public Member Functions

- static void **clear** (CalculateArguments &userArgs)
- static int **indexOf** (const char c)

Static Public Attributes

- static const int **NumberUserArguments** = 26
- static const int **NumberInputKinds** = 2

9.140.1 Detailed Description

Description: The [QEEvaluation](#) object allows the evaluation of expressions similar to that available in calc/calcout records; and in fact "under the covers" it uses the calcRecord's postfix functions out of the Com shared library.

The user initialises the [QEEvaluation](#) object with an expression as per the CALC field of a calc (or calcout) record, e.g. "A + LN (B/C)", and then may evaluate the expression by calling the Evaluate method with an array of Doubles, each element supplying the value for A, B, C etc. The standard CALC functions (SIN, LOG, ABS etc.) are all available.

The [QEEvaluation](#) object provides an extended input name space for the user supplied expressions, i.e. 52 names (A .. Z, A' .. Z') instead of the native 12 inputs (A .. L). However, the MAXIMUM number of inputs that may be used in any one expression is still LIMITED to 12.

The names space is controlled by the allowPrimedInput and numberOfInputs parameters used at construction time.

That is: "B + L' + M + X + Y' " is an allowed expression, but "A + B + C + D + E + F + G + H + I + J + K + L' + M" is invalid because more than 12 inputs specified.

This input name space extension is to allow sensible use with the Strip Chart widget that has 16 available inputs and the Plotter widget that has 16 available inputs plus an X input.

Also, the length of the input string is not limited to the 40 characters of the CALC field, but may use the full 100 characters allowed by the underlying postfix function.

Acknowledgements: [QEEvaluation](#) is a direct crib of TCalculate out of the Delphi OPI framework. The postfix and calcPerform functions were written by Bob Dalesio (12-12-86).

The documentation for this class was generated from the following files:

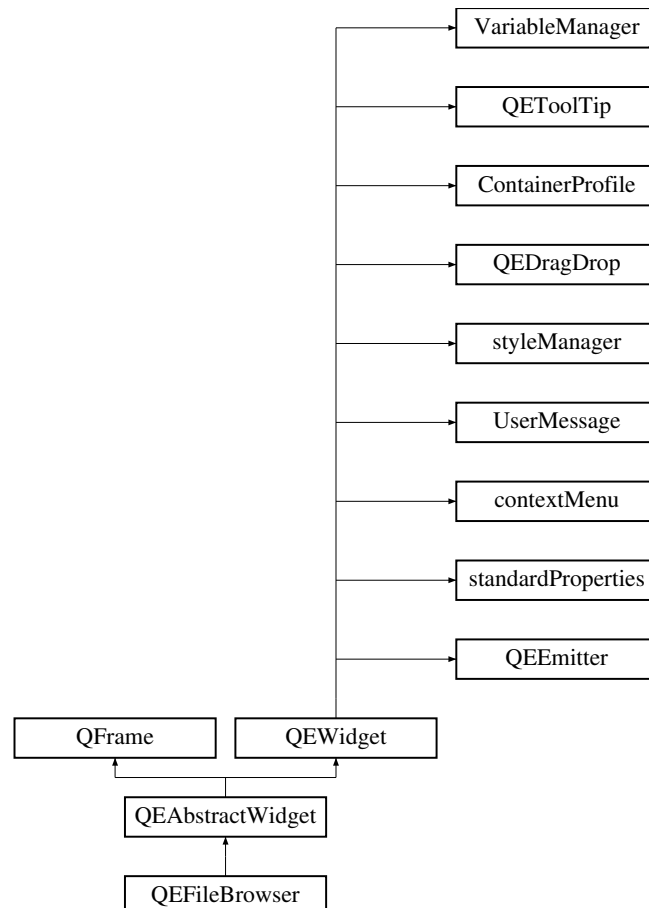
- qeframework/common/QEEvaluation.h
- qeframework/common/QEEvaluation.cpp

9.141 QFileBrowser Class Reference

The [QFileBrowser](#) widget allows the user to browse existing files from a certain directory.

```
#include <QFileBrowser.h>
```

Inheritance diagram for QFileBrowser:



Classes

- class [QEOwnTable](#)

Signals

- void [selected](#) (QString pFilename)

Public Member Functions

- **QFileBrowser** (QWidget *parent=0)
- void **setVariableName** (const QString &pvName)
- QString **getVariableName** () const
- void **setVariableNameSubstitutions** (const QString &substitutions)
- QString **getVariableNameSubstitutions** () const

- void **setDirectoryPath** (const QString &[directoryPath](#))
- QString **getDirectoryPath** () const
- void **setShowDirectoryPath** (bool showPath)
- bool **getShowDirectoryPath** () const
- void **setShowDirectoryBrowser** (bool showBrowser)
- bool **getShowDirectoryBrowser** () const
- void **setShowRefresh** (bool [showRefresh](#))
- bool **getShowRefresh** () const
- void **setShowTable** (bool [showTable](#))
- bool **getShowTable** () const
- void **setShowColumnTime** (bool showTime)
- bool **getShowColumnTime** () const
- void **setShowColumnSize** (bool showSize)
- bool **getShowColumnSize** () const
- void **setShowColumnFilename** (bool showFilename)
- bool **getShowColumnFilename** () const
- void **setShowFileExtension** (bool showExtension)
- bool **getShowFileExtension** () const
- void **setFileFilter** (const QString &[fileFilter](#))
- QString **getFileFilter** () const
- void **setFileDialogDirectoriesOnly** (const bool directoriesOnly)
- bool **getFileDialogDirectoriesOnly** () const
- void **setFileCreationAllowed** (const bool [fileCreationAllowed](#))
- bool **getFileCreationAllowed** () const
- void **setOptionsLayout** (const QE::LayoutOptions layout)
- QE::LayoutOptions **getOptionsLayout** () const
- void **setMargin** (const int [margin](#))
- int **getMargin** () const
- void **updateTable** ()

Protected Member Functions

- bool **event** (QEvent *event)

Protected Attributes

- [QELineEdit](#) * **directoryPathEdit**
- QPushButton * **directoryBrowserButton**
- QPushButton * **refreshButton**
- [QEOwnTable](#) * **browserTable**
- QString **m_fileFilter**
- bool **m_showFileExtension**
- bool **m_fileDialogDirectoriesOnly**
- bool **m_fileCreationAllowed**
- int **m_margin**
- QE::LayoutOptions **m_optionsLayout**

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- QString [directoryPath](#)
- bool [showDirectoryPath](#)
- bool [showDirectoryBrowser](#)
- bool [showRefresh](#)
- bool [showTable](#)
- bool [showColumnTime](#)
Show/hide column containing the time of creation of files.
- bool [showColumnSize](#)
- bool [showColumnFilename](#)
- bool [showFileExtension](#)
- bool [fileDialogDirectoriesOnly](#)
- bool [fileCreationAllowed](#)
- QString [fileFilter](#)
- QE::LayoutOptions [optionsLayout](#)
- int [margin](#)

Additional Inherited Members

9.141.1 Detailed Description

The [QEFileBrowser](#) widget allows the user to browse existing files from a certain directory.

9.141.2 Member Function Documentation

9.141.2.1 selected

```
void QEFileBrowser::selected (
    QString pFilename ) [signal]
```

Signal that is generated every time the user double-clicks a certain file. This signal emits a string that contains the full path and the name of the selected file. This signal may be captured by other widgets that perform further operations (for instance, the [QEImage](#) displays the content of this file if it is a graphical one).

9.141.3 Property Documentation

9.141.3.1 directoryPath

```
QString QEFileBrowser::directoryPath [read], [write]
```

Default directory where to browse files when [QEFileBrowser](#) is launched for the first time.

9.141.3.2 fileCreationAllowed

```
bool QFileDialog::fileCreationAllowed [read], [write]
```

Specifies if the selected can be created. When false, this is the equivalent to open and is the default. When true, this is the equivalent to save. Only applicable when `fileDialogDirectoriesOnly` is set true.

9.141.3.3 fileDialogDirectoriesOnly

```
bool QFileDialog::fileDialogDirectoriesOnly [read], [write]
```

Enable/disable the browsing of directories-only when opening the dialog window. The default is true.

9.141.3.4 fileFilter

```
QString QFileDialog::fileFilter [read], [write]
```

Specify which files to browse. To specify more than one filter, please separate them with a ";". Example: `*.py;*.ui` (this will only display files with an extension `.py` or `.ui`).

9.141.3.5 margin

```
int QFileDialog::margin [read], [write]
```

Margin between contents and bounding container widget.

9.141.3.6 optionsLayout

```
QFileDialog::LayoutOptions QFileDialog::optionsLayout [read], [write]
```

Change the order of the widgets. Valid options are: Top, Bottom, Left and Right. The default is Top.

9.141.3.7 showColumnFilename

```
bool QFileDialog::showColumnFilename [read], [write]
```

Show/hide column containing the name of files.

9.141.3.8 showColumnSize

```
bool QFileDialog::showColumnSize [read], [write]
```

Show/hide column containing the size (in bytes) of files.

9.141.3.9 showDirectoryBrowser

```
bool QFileBrowser::showDirectoryBrowser [read], [write]
```

Show/hide button to open the dialog window to browse for directories and files.

9.141.3.10 showDirectoryPath

```
bool QFileBrowser::showDirectoryPath [read], [write]
```

Show/hide directory path line edit where the user can specify the directory to browse files.

9.141.3.11 showFileExtension

```
bool QFileBrowser::showFileExtension [read], [write]
```

Show/hide the extension of files.

9.141.3.12 showRefresh

```
bool QFileBrowser::showRefresh [read], [write]
```

Show/hide button to refresh the table containing the list of files being browsed.

9.141.3.13 showTable

```
bool QFileBrowser::showTable [read], [write]
```

Show/hide table containing the list of files being browsed.

9.141.3.14 variable

```
QString QFileBrowser::variable [read], [write]
```

EPICS variable name (CA PV). This variable is used for both writing and reading the directory to be used by the widget. An internal [QLineEdit](#) widget is used to manage this PV.

9.141.3.15 variableSubstitutions

```
QString QFileBrowser::variableSubstitutions [read], [write]
```

Default macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

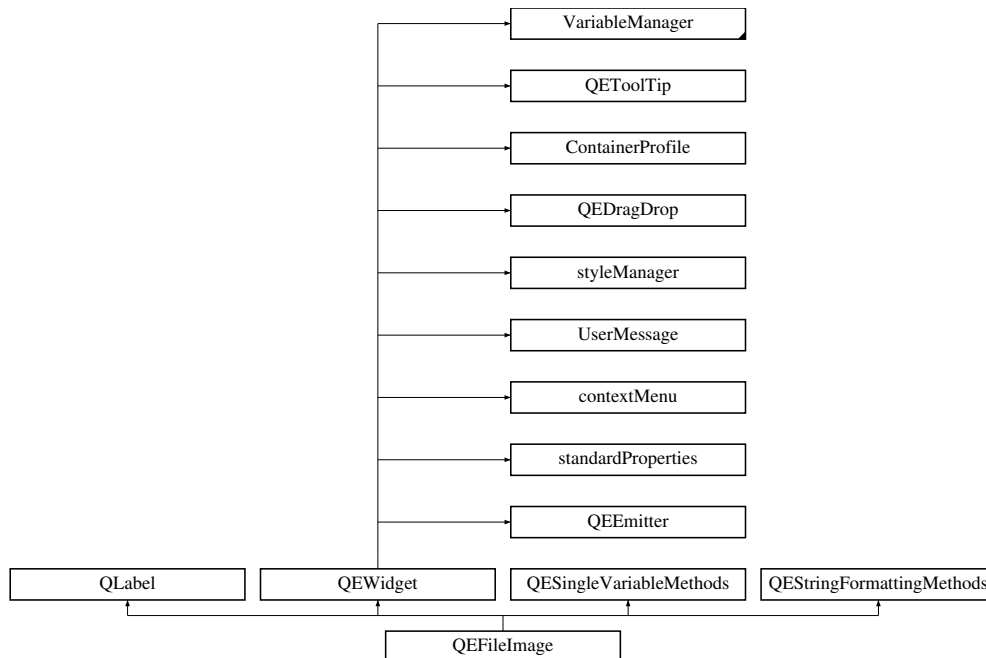
The documentation for this class was generated from the following files:

- qeframework/widgets/QFileBrowser/QFileBrowser.h
- qeframework/widgets/QFileBrowser/QFileBrowser.cpp

9.142 QFileImage Class Reference

```
#include <QFileImage.h>
```

Inheritance diagram for QFileImage:



Public Slots

- void **setThresholdColor** (const QColor [thresholdColor](#))
- void **setThreshold** (const int [threshold](#))
- void **setImageFileName** (const QString &text)
- void **setScaledContents** (bool [scaledContents](#))
- void [setManagedVisible](#) (bool v)

Signals

- void [dbValueChanged](#) ()
- void **dbValueChanged** (const QString &out)
- void [dbConnectionChanged](#) (const bool &isConnected)
Sent when the widget state updated following a channel connection change.
- void [requestResend](#) ()

Public Member Functions

- [QFileImage](#) (QWidget *parent=0)
- [QFileImage](#) (const QString &variableName, QWidget *parent=0)
- [~QFileImage](#) ()
Destructor.
- QColor **getThresholdColor** () const
- int **getThreshold** () const
- QString **getImageFileName** () const
- bool **getScaledContents** () const

Protected Member Functions

- [QEChannel](#) * [createQcaltem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **stringFormattingChange** ()
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- bool [oosAware](#)
- int [threshold](#)
- QColor [thresholdColor](#)
- bool [scaledContents](#)

Additional Inherited Members

9.142.1 Detailed Description

This class is a EPICS aware image widget based on the Qt label widget. When a variable is defined, the label pixmap will be updated with the file specified by the variable. The label will be disabled if the variable is invalid. It is tightly integrated with the base class [QEWidget](#) which provides generic support such as macro substitutions, drag/drop, and standard properties.

9.142.2 Constructor & Destructor Documentation

9.142.2.1 QEFileImage() [1/2]

```
QEFileImage::QEFileImage (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.142.2.2 QEFileImage() [2/2]

```
QEFileImage::QEFileImage (
    const QString & variableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.142.3 Member Function Documentation

9.142.3.1 dbValueChanged

```
void QEFileImage::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.142.3.2 requestResend

```
void QEFileImage::requestResend ( ) [signal]
```

Internal use only. Used when changing a property value to force a re-display to reflect the new property value.

9.142.3.3 setManagedVisible

```
void QEFileImage::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.142.4 Property Documentation

9.142.4.1 allowDrop

```
bool QEFileImage::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.142.4.2 arrayIndex

```
int QEFileImage::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.142.4.3 defaultStyle

```
QString QEFileImage::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.142.4.4 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEFileImage::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.142.4.5 elementsRequired

```
int QEFileImage::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.142.4.6 messageSourceId

```
unsigned int QEFileImage::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.142.4.7 oosAware

```
bool QEFileImage::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.142.4.8 scaledContents

```
bool QEFileImage::scaledContents [read], [write]
```

Flag the pixmap is to be scaled to fit the frame. Similar operation to scaledContents property for a QLabel/QEFrame.

9.142.4.9 styleSheet

```
QString QEFileImage::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly setable per se. This also stops transient styles being saved to the ui file.

9.142.4.10 threshold

```
int QEFileImage::threshold [read], [write]
```

Defines the lightness threshold: range 0 .. 255, or 256 for no thresholding.

9.142.4.11 thresholdColor

```
QColor QEFileImage::thresholdColor [read], [write]
```

Defines to colour to be substituted when the pixel lightness exceeds the given threshold. The default colour is white.

9.142.4.12 userLevelEnabled

```
QE::UserLevels QEFileImage::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through setUserLevel() Widgets that are always accessable should be visible at 'User'. Widgets that are only accessable to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessable to engineers maintaining the facility should be visible at 'Engineer'.

9.142.4.13 `userLevelEngineerStyle`

```
QString QEFileImage::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.142.4.14 `userLevelScientistStyle`

```
QString QEFileImage::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.142.4.15 `userLevelUserStyle`

```
QString QEFileImage::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.142.4.16 `userLevelVisibility`

```
QE::UserLevels QEFileImage::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.142.4.17 `variable`

```
QString QEFileImage::variable [read], [write]
```

EPICS variable name (CA PV)

9.142.4.18 `variableAsToolTip`

```
bool QEFileImage::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.142.4.19 variableSubstitutions

```
QString QEFileImage::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.142.4.20 visible

```
bool QEFileImage::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

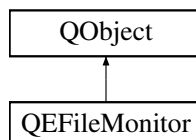
The documentation for this class was generated from the following files:

- qeframework/widgets/QEFileImage/QEFileImage.h
- qeframework/widgets/QEFileImage/QEFileImage.cpp

9.143 QEFileMonitor Class Reference

```
#include <QEFileMonitor.h>
```

Inheritance diagram for QEFileMonitor:



Signals

- void **fileChanged** (const QString &path)
- void **directoryChanged** (const QString &path)

Public Member Functions

- **QEFileMonitor** (QObject *parent=NULL)
- **QEFileMonitor** (const QString &path, QObject *parent=NULL)
- void **setPath** (const QString &path)
- QString **getPath** () const
- void **clearPath** ()

9.143.1 Detailed Description

The [QEFileMonitor](#) class provides a wrapper around a single `QFileSystemWatcher` object which can be used to monitor all file/directories that require watching. This class acts as an agent between the user and the singleton `QFileSystemWatcher`. As each instance of a `QFileSystemWatcher` object creates a new thread, it is far more thread efficient to use a single watcher to monitor many files/directories.

Note: each agent only sets up monitoring for a single file/directory. Multiple agents may/can monitor the same file/directory.

Refer to `QFileSystemWatcher` documentation on limits and monitoring features.

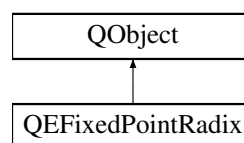
The documentation for this class was generated from the following files:

- `qeframework/common/QEFileMonitor.h`
- `qeframework/common/QEFileMonitor.cpp`

9.144 QEFixedPointRadix Class Reference

```
#include <QEFixedPointRadix.h>
```

Inheritance diagram for `QEFixedPointRadix`:



Public Member Functions

- **QEFixedPointRadix** (`QObject *parent=0`)
- **QEFixedPointRadix** (`const QE::Radices radix, const QE::Separators separator, QObject *parent=0`)
- void **setRadix** (`const QE::Radices radix`)
- [QE::Radices](#) **getRadix** () const
- void **setSeparator** (`const QE::Separators separator`)
- [QE::Separators](#) **getSeparator** () const
- int **getRadixValue** () const
- bool **isRadixDigit** (`const QChar qc`) const
- `QString` **toString** (`const double value, const bool sign, const int zeros, const int precision`) const
- double **toValue** (`const QString &image, bool &okay`) const

9.144.1 Detailed Description

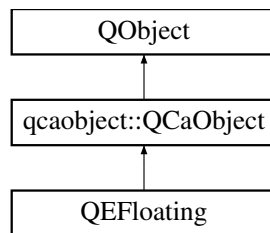
The `QString` class provides general value to/from radix string representation, but only for integer types. This class provides similar functionality for the double type to/from a fixed point radix representation. The class restricts the allowed radix values to 2, 8, 10 and 16. The class allows the for the insertion of 'thousands' separators, typically comma, but space and under score are also allowed.

The documentation for this class was generated from the following files:

- `qeframework/common/QEFixedPointRadix.h`
- `qeframework/common/QEFixedPointRadix.cpp`

9.145 QEFloating Class Reference

Inheritance diagram for QEFloating:



Public Slots

- void **writeFloating** (const double &data)
- void **writeFloatingElement** (const double &data)
- void **writeFloating** (const QVector< double > &data)

Signals

- void **valueUpdated** (const [QEFloatingValueUpdate](#) &value)
- void **arrayUpdated** (const [QEFloatingArrayUpdate](#) &array)
- void **floatingChanged** (const double &value, [QCaAlarmInfo](#) &alarmInfo, [QCaDateTime](#) &timeStamp, const unsigned int &variableIndex)
- void **floatingArrayChanged** (const QVector< double > &values, [QCaAlarmInfo](#) &alarmInfo, [QCaDateTime](#) &timeStamp, const unsigned int &variableIndex)

Public Member Functions

- **QEFloating** (const QString &pvName, QObject *eventObject, [QEFloatingFormatting](#) *floatingFormattingIn, unsigned int variableIndexIn)
- **QEFloating** (const QString &pvName, QObject *eventObject, [QEFloatingFormatting](#) *floatingFormattingIn, unsigned int variableIndexIn, [UserMessage](#) *userMessageIn)

Additional Inherited Members

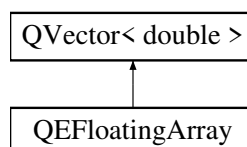
The documentation for this class was generated from the following files:

- qeframework/data/QEFloating.h
- qeframework/data/QEFloating.cpp

9.146 QEFloatingArray Class Reference

```
#include <QEFloatingArray.h>
```

Inheritance diagram for QEFloatingArray:



Public Member Functions

- **QEFloatingArray** (int size)
- **QEFloatingArray** (int size, const double &t)
- **QEFloatingArray** (const QVector< double > &other)
- **QEFloatingArray** & **operator=** (const QVector< double > &other)
- double **minimumValue** (const double &defaultValue=0.0, const bool includeInf=false)
- double **maximumValue** (const double &defaultValue=0.0, const bool includeInf=false)
- **QEFloatingArray** **calcDyByDx** (const QVector< double > &x)
- **QEFloatingArray** **medianFilter** (const int window)

9.146.1 Detailed Description

This class provides short hand for QVector<double> together with some basic double vector operations.

The documentation for this class was generated from the following files:

- qeframework/data/QEFloatingArray.h
- qeframework/data/QEFloatingArray.cpp

9.147 QEFloatingArrayUpdate Struct Reference

Public Attributes

- QVector< double > **values**
- **QCaAlarmInfo** **alarmInfo**
- **QCaDateTime** **timeStamp**
- unsigned int **variableIndex**
- bool **isMetaUpdate**

The documentation for this struct was generated from the following file:

- qeframework/data/QEFloating.h

9.148 QEFloatingFormatting Class Reference

Public Member Functions

- double **formatFloating** (const QVariant &value) const
- double **formatFloating** (const QVariant &value, int arrayIndex) const
- QVector< double > **formatFloatingArray** (const QVariant &value) const
- QVariant **formatValue** (const double &floatingValue) const
- QVariant **formatValue** (const QVector< double > &floatingValue) const

9.148.1 Member Function Documentation

9.148.1.1 formatFloating()

```
double QEFloatingFormatting::formatFloating (
    const QVariant & value ) const
```

This is used to convert the QVariant value received from a QEChannel, which is still based on the data variable type, to a double.

9.148.1.2 formatFloatingArray()

```
QVector< double > QEFloatingFormatting::formatFloatingArray (
    const QVariant & value ) const
```

This is used to convert the QVariant value received from a QEChannel, which is still based on the data variable type, to a double array. Typically used where the input QVariant value is an array of data values, but will work for any QVariant type.

The documentation for this class was generated from the following files:

- qeframework/data/QEFloatingFormatting.h
- qeframework/data/QEFloatingFormatting.cpp

9.149 QEFloatingValueUpdate Struct Reference

Public Attributes

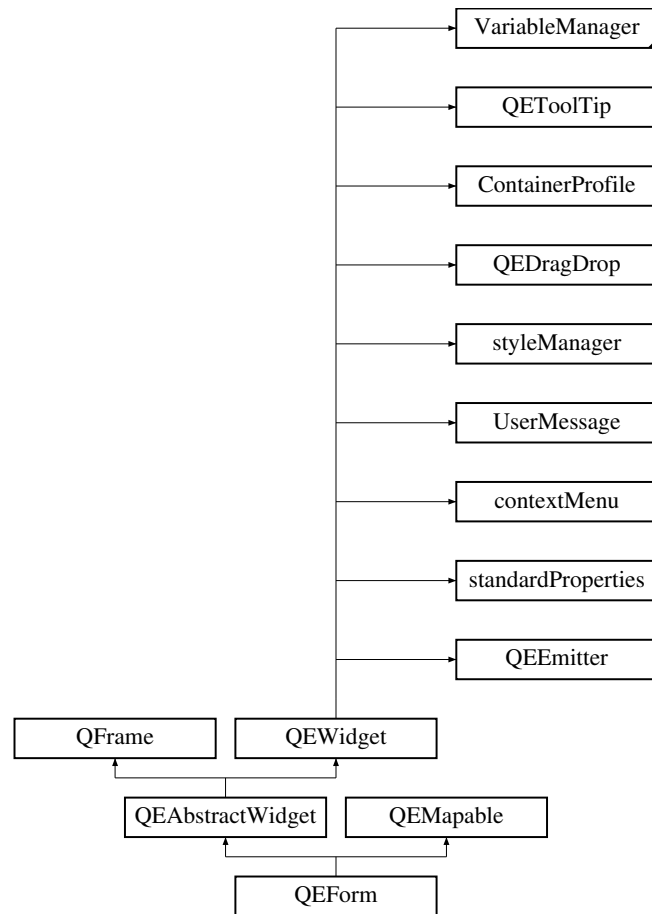
- double **value**
- [QCaAlarmInfo](#) **alarmInfo**
- [QCaDateTime](#) **timeStamp**
- unsigned int **variableIndex**
- bool **isMetaUpdate**

The documentation for this struct was generated from the following file:

- qeframework/data/QEFloating.h

9.150 QForm Class Reference

Inheritance diagram for QForm:



Public Slots

- bool [readUiFile](#) ()
Find a widget within the ui loaded by the [QForm](#). Returns NULL if no UI is loaded yet or if the named widget can't be found.
- void [requestAction](#) (const [QEActionRequests](#) &request)
Read a .ui file and present it within this [QForm](#).
- void [setUiFileName](#) (const QString &uiFileName)
Slot for launching a new gui. Used by [QE](#) buttons and [QForm](#) as the default action for launching a gui.
- void [setUiFileSubstitutions](#) (const QString &uiFileNameSubstitutions)

Signals

- void [formLoaded](#) (bool fileLoaded)

Public Member Functions

- **QForm** (QWidget *parent=0)
- **QForm** (const QString &uiFileNameIn, QWidget *parent=0)
- QSize **sizeHint** () const
- void **setQEGuiTitle** (const QString titleIn)
- QString **getQEGuiTitle** () const
Set the title to be used as the window or form title. (note, also set when reading a .ui file)
- QString **getFullFileName** () const
Get the title to be used as the window or form title.
- QString **getUiFileName** () const
Get the standard, absolute UI file name.
- void **setFileMonitoringIsEnabled** (bool fileMonitoringIsEnabled)
Get the fully substituted file name (Not the uiFile property)
- bool **getFileMonitoringIsEnabled** () const
Set flag indicating if form should take account of file monitoring.
- void **setHandleGuiLaunchRequests** (bool **handleGuiLaunchRequests**)
Get flag indicating if form should take account of file monitoring.
- bool **getHandleGuiLaunchRequests** () const
Set flag indicating form should handle gui form launch requests.
- void **setResizeContents** (bool **resizeContentsIn**)
Get flag indicating form should handle gui form launch requests.
- bool **getResizeContents** () const
Set flag indicating form should resize contents to match form size (otherwise resize form to match contents)
- QString **getContainedFrameworkVersion** () const
Get flag indicating form should resize contents to match form size (otherwise resize form to match contents)
- QString **getUniqueIdentifier** () const
*Get the version of the first **QE** widget (if any) of **QE** widgets by QUILoader.*
- void **setUniqueIdentifier** (QString name)
*Get a unique identifier string for this form. This identifier should be persistent across application runs as it is based on the **QForm**'s position in the widget hierarchy. The same widget will generate the same identifier when opened within the same GUI.*
- void **clearUiFileNames** ()
*Set a unique identifier string for this form. This identifier should be persistent across application runs as it is based on the **QForm**'s position in the widget hierarchy. The same widget will generate the same identifier when opened within the same GUI.*
- int **getDisconnectedCount** () const
- int **getConnectedCount** () const
Return the count of disconnected variables.
- QWidget * **getChild** (QString name) const
Return the count of connected variables.
- void **setUiFileNameProperty** (QString uiFileName)
- QString **getUiFileNameProperty** () const
- void **setVariableNameSubstitutionsProperty** (QString variableNameSubstitutions)
- QString **getVariableNameSubstitutionsProperty** () const
- **QE::MessageFilterOptions** **getMessageFormFilter** () const
- void **setMessageFormFilter** (const **QE::MessageFilterOptions** messageFormFilter)
- **QE::MessageFilterOptions** **getMessageSourceFilter** () const
- void **setMessageSourceFilter** (const **QE::MessageFilterOptions** messageSourceFilter)

Protected Member Functions

- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.

Protected Attributes

- QString **uiFileName**
- QString **fullUiFileName**
- bool [handleGuiLaunchRequests](#)
- bool [resizeContents](#)

Properties

- QString [uiFile](#)
- QString [variableSubstitutions](#)
- unsigned int [messageSourceId](#)
- [QE::MessageFilterOptions](#) [messageFormFilter](#)
- [QE::MessageFilterOptions](#) [messageSourceFilter](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)

Additional Inherited Members

9.150.1 Member Data Documentation

9.150.1.1 [handleGuiLaunchRequests](#)

```
bool QEForm::handleGuiLaunchRequests [read], [write], [protected]
```

If true, the [QEForm](#) widget publishes its own slot for launching new GUIs so all [QE](#) widgets within it will use the [QEForm](#)'s mechanism for launching new GUIs, rather than any mechanism the application may provide (through the [ContainerProfile](#) mechanism)

9.150.1.2 [resizeContents](#)

```
bool QEForm::resizeContents [read], [write], [protected]
```

If set, the [QEForm](#) will resize the top level widget of the .ui file it opens (and set other size and border related properties) to match itself. This is useful if the [QEForm](#) is used as a sub form within a main form (possible another [QEForm](#)) and you want to control the size of the [QEForm](#) being used as a sub form. If clear, the [QEForm](#) will resize itself (and set other size and border related properties) to match the top level widget of the .ui file it opens. This is useful if the [QEForm](#) is used as a sub form within a main form (possible another [QEForm](#)) and you want the main form to resize to match the size of the [QEForm](#) being used as a sub form, or you want the sub form border decorations (such as frame shape and shadow) to be displayed.

9.150.2 Property Documentation

9.150.2.1 allowDrop

```
bool QEForm::allowDrop [read], [write]
```

allowDrop is added as a non-designable property here only to hide the implementation present in [QEAbstractWidget](#)

9.150.2.2 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEForm::displayAlarmStateOption [read], [write]
```

displayAlarmStateOption is added as a non-designable property here only to hide the implementation present in [QEAbstractWidget](#)

9.150.2.3 messageFormFilter

```
QE::MessageFilterOptions QEForm::messageFormFilter [read], [write]
```

Message filter that attempts to match messages sent through the [QE](#) message logging system based on the automatically generated message form ID. This filter will match form ID of the message to the form ID of this QEform as follows:

Note: Any - A message will always be accepted - is not allowed here. If any is selected it will be set to the default which is Match. Match - A message will be accepted if it comes from a [QE](#) widget within this form. None - The message will not be matched based on the form the message comes from. (It may still be accepted based on the message source ID.) Matched messages will be resend with the messageSourceId of this [QEForm](#)

9.150.2.4 messageSourceFilter

```
QE::MessageFilterOptions QEForm::messageSourceFilter [read], [write]
```

!???? is this a valid property. Resending messages based on the source ID is unnecessary as they will be sent on with the same source ID? Message filter that attempts to match messages sent through the [QE](#) message logging system based on the messageSourceId of the widget that generated the message. This filter will match message message source ID of the message to the message source ID of this QEform as follows:

Note: Any - A message will always be accepted - is not allowed here. If any is selected it will be set to the default which is None. Match - A message will be accepted if the message source ID matches this [QEForm](#). None - The message will not be matched based on message source ID (It may still be accepted based on the message form ID.) Matched messages will be resend with the messageSourceId of this [QEForm](#).

9.150.2.5 messageSourceId

```
unsigned int QEForm::messageSourceId [read], [write]
```

Widgets or applications that use messages from the framework have the option of filtering on this ID Messages that the [QEForm](#) widget catches with its message filters will be regenerated using this ID

9.150.2.6 uiFile

```
QString QForm::uiFile [read], [write]
```

File name of the .ui file being presented within the [QForm](#) widget.

9.150.2.7 variableAsToolTip

```
bool QForm::variableAsToolTip [read], [write]
```

`variableAsToolTip` is added as a non-designable property here only to hide the implementation present in [QEAbstractWidget](#)

9.150.2.8 variableSubstitutions

```
QString QForm::variableSubstitutions [read], [write]
```

Macro substitutions to be applied to this widget, and all [QE](#) widgets that are opened when the .ui file is presented. Note, despite the name, the macro substitutions are general macro substitutions, and do not just apply to a variable name (in fact a [QForm](#) widget does not even have a variable name property).

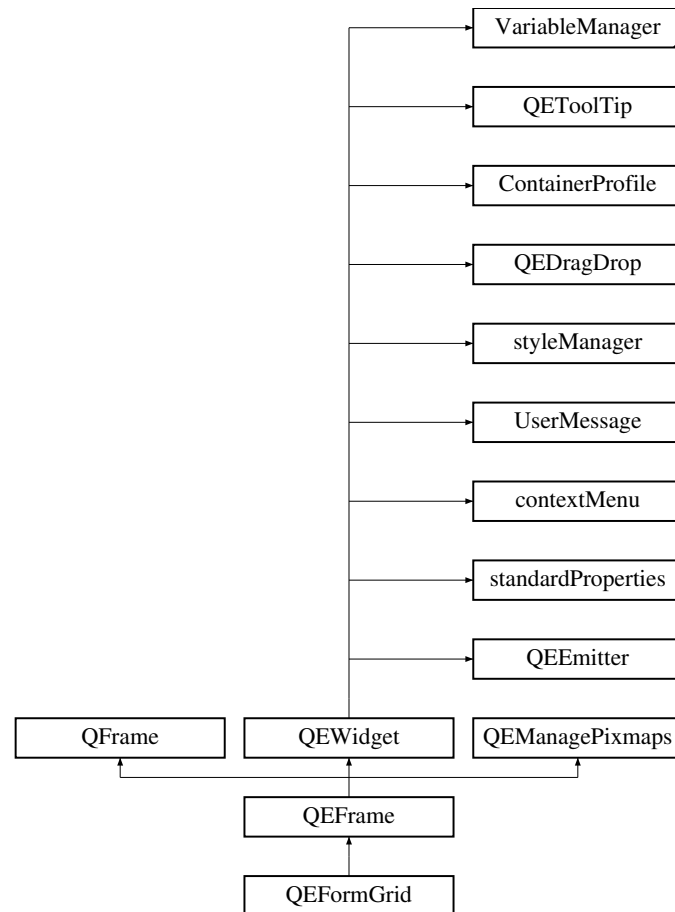
The documentation for this class was generated from the following files:

- qeframework/widgets/QForm/QForm.h
- qeframework/widgets/QForm/QForm.cpp

9.151 QFormGrid Class Reference

```
#include <QFormGrid.h>
```

Inheritance diagram for QFormGrid:



Public Member Functions

- [QEFormGrid](#) (QWidget *parent=0)
- **QEFormGrid** (const QString &uiFile, const int number, const int cols, QWidget *parent=0)
- virtual **~QEFormGrid** ()
Destruction.
- void **setUiFile** (QString uiFileName)
- QString **getUiFile** ()
- void **setGridVariableSubstitutions** (QString variableSubstitutions)
- QString **getGridVariableSubstitutions** ()
- void **setNumber** (int n)
- int **getNumber** ()
- void **setColumns** (int n)
- int **getColumns** ()
- void **setGridOrder** (QE::GridOrders go)
- QE::GridOrders **getGridOrder** ()
- void **setMargin** (int n)
- int **getMargin** ()
- void **setSpacing** (int n)
- int **getSpacing** ()
- int **getRows** ()

Protected Member Functions

- QSize **sizeHint** () const

Properties

- QString [uiFile](#)
- QString **variableSubstitutions**
- int [number](#)
- int [columns](#)
- [QE::GridOrders](#) [gridOrder](#)
- int [margin](#)
- int [spacing](#)
- QString [slotMacroPrefix](#)
- int [slotNumberOffset](#)
- int [slotNumberWidth](#)
- QStringList [slotStrings](#)
- QString [rowMacroPrefix](#)
- int [rowNumberOffset](#)
- int [rowNumberWidth](#)
- QStringList [rowStrings](#)
- QString [colMacroPrefix](#)
- int [colNumberOffset](#)
- int [colNumberWidth](#)
- QStringList [colStrings](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)

Friends

- class **MacroData**

Additional Inherited Members

9.151.1 Detailed Description

This class load a grid QForms.

9.151.2 Constructor & Destructor Documentation

9.151.2.1 QFormGrid()

```
QFormGrid::QFormGrid (  
    QWidget * parent = 0 ) [explicit]
```

Create a grid widget with default parameters.

9.151.3 Property Documentation

9.151.3.1 allowDrop

```
bool QEFormGrid::allowDrop [read], [write]
```

allowDrop is added as a non-designable property here only to hide the implementation present in [QEFrame](#)

9.151.3.2 colMacroPrefix

```
QString QEFormGrid::colMacroPrefix [read], [write]
```

Specified formal slot macro name prefix Default value: COL

9.151.3.3 colNumberOffset

```
int QEFormGrid::colNumberOffset [read], [write]
```

Specifies the COL macro number offset. Default value: 1.

9.151.3.4 colNumberWidth

```
int QEFormGrid::colNumberWidth [read], [write]
```

Specifies the COL macro width format. Default value: 2.

9.151.3.5 colStrings

```
QStringList QEFormGrid::colStrings [read], [write]
```

Specifies the COLNAME macro values applied to each QEForm grid element in a specific column. Indexing of the colStrings is NOT impacted by the colNumberOffset. Default value: ""

9.151.3.6 columns

```
int QEFormGrid::columns [read], [write]
```

Specifies the number of columns. This is restricted to the range 1 to 200. Default value: 1.

9.151.3.7 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEFormGrid::displayAlarmStateOption [read], [write]
```

displayAlarmStateOption is added as a non-designable property here only to hide the implementation present in [QEFrame](#)

9.151.3.8 gridOrder

```
QE::GridOrders QFormGrid::gridOrder [read], [write]
```

Specifies the gridOrder. Default value: RowMajor

9.151.3.9 margin

```
int QFormGrid::margin [read], [write]
```

Margin of the internal QGridLayout object. Default value: 2.

9.151.3.10 number

```
int QFormGrid::number [read], [write]
```

The total number of QForms. This is restricted to the range 1 to 2000. Default value: 4.

9.151.3.11 rowMacroPrefix

```
QString QFormGrid::rowMacroPrefix [read], [write]
```

Specified formal slot macro name prefix Default value: ROW

9.151.3.12 rowNumberOffset

```
int QFormGrid::rowNumberOffset [read], [write]
```

Specifies the ROW macro number offset. Default value: 1.

9.151.3.13 rowNumberWidth

```
int QFormGrid::rowNumberWidth [read], [write]
```

Specifies the COL macro width format. Default value: 2.

9.151.3.14 rowStrings

```
QStringList QFormGrid::rowStrings [read], [write]
```

Specifies the ROWNAME macro values applied to each QForm grid element in a specic row. Indexing of the rowStrings is NOT impacted by the rowNumberOffset. Default value: ""

9.151.3.15 slotMacroPrefix

```
QString QEFormGrid::slotMacroPrefix [read], [write]
```

Specified formal slot macro name prefix Default value: SLOT

9.151.3.16 slotNumberOffset

```
int QEFormGrid::slotNumberOffset [read], [write]
```

Specifies the SLOT macro number offset. Default value: 1.

9.151.3.17 slotNumberWidth

```
int QEFormGrid::slotNumberWidth [read], [write]
```

Specifies the SLOT macro width format. Default value: 2.

9.151.3.18 slotStrings

```
QStringList QEFormGrid::slotStrings [read], [write]
```

Specifies the SLOTNAME macro values applied to each QEForm grid element. Slot number allocation is impacted by the gridOrder property. Indexing of the slotStrings is NOT impacted by the slotNumberOffset. Default value: ""

9.151.3.19 spacing

```
int QEFormGrid::spacing [read], [write]
```

Spacing of the internal QGridLayout object. Default value: 2.

9.151.3.20 uiFile

```
QString QEFormGrid::uiFile [read], [write]
```

GridOrders specifies how grid elements are layed out.

Row major (example 10, items, 3 cols): 0 1 2 3 4 5 6 7 8 9 - -

Col major (example 10, items, 3 cols): 0 4 8 1 5 9 2 6 - 3 7 - The uiFile loaded into each [QEForm](#) element. Default value: ""

9.151.3.21 variableAsToolTip

```
bool QEFormGrid::variableAsToolTip [read], [write]
```

variableAsToolTip is added as a non-designable property here only to hide the implementation present in [QEFrame](#)

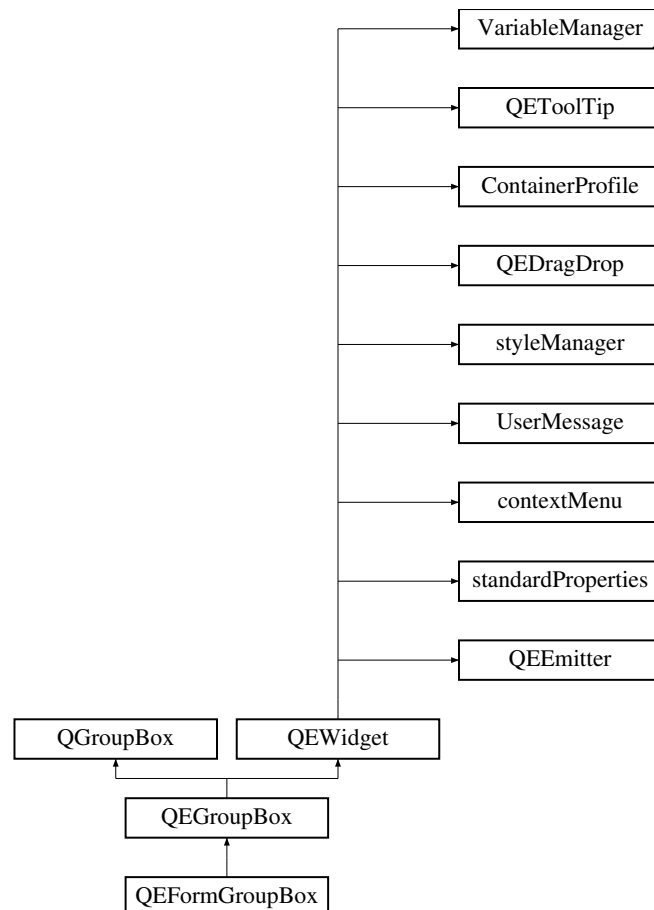
The documentation for this class was generated from the following files:

- qeframework/widgets/QEFormGrid/QEFormGrid.h
- qeframework/widgets/QEFormGrid/QEFormGrid.cpp

9.152 QEFormGroupBox Class Reference

```
#include <QEFormGroupBox.h>
```

Inheritance diagram for QEFormGroupBox:



Public Slots

- void **setUiFileName** (const QString &uiFileName)
- void **setUiFileSubstitutions** (const QString &uiFileNameSubstitutions)

Public Member Functions

- **QFormGroupBox** (QWidget *parent=0)
- QSize **sizeHint** () const
- **QE_EXPOSE_INTERNAL_OBJECT_FUNCTIONS** (form, QString, getUiFileNameProperty, setUiFile↵NameProperty)
- **QE_EXPOSE_INTERNAL_OBJECT_FUNCTIONS** (form, QString, getVariableNameSubstitutionsProperty, setVariableNameSubstitutionsProperty)
- **QE_EXPOSE_INTERNAL_OBJECT_FUNCTIONS** (form, bool, getHandleGuiLaunchRequests, set↵HandleGuiLaunchRequests)
- **QE_EXPOSE_INTERNAL_OBJECT_FUNCTIONS** (form, bool, getResizeContents, setResizeContents)
- **QE_EXPOSE_INTERNAL_OBJECT_FUNCTIONS** (form, unsigned int, [getMessageSourceId](#), [setMessageSourceId](#))
- **QE_EXPOSE_INTERNAL_OBJECT_FUNCTIONS** (form, [QE::MessageFilterOptions](#), getMessageForm↵Filter, setMessageFormFilter)
- **QE_EXPOSE_INTERNAL_OBJECT_FUNCTIONS** (form, [QE::MessageFilterOptions](#), getMessage↵SourceFilter, setMessageSourceFilter)
- void **setHideInnerFrameShape** (const bool [useFormTitle](#))
- bool **getHideInnerFrameShape** () const
- void **setUseFormTitle** (const bool [useFormTitle](#))
- bool **getUseFormTitle** () const

Protected Member Functions

- void **showEvent** (QShowEvent *event)
- void [saveConfiguration](#) ([PersistenceManager](#) *pm)
- void [restoreConfiguration](#) ([PersistenceManager](#) *pm, [restorePhases](#) restorePhase)

Properties

- bool **checkable**
- QString [uiFile](#)
- QString **variableSubstitutions**
- bool **handleGuiLaunchRequests**
- bool **resizeContents**
- unsigned int **formMessageSourceId**
- [QE::MessageFilterOptions](#) **messageFormFilter**
- [QE::MessageFilterOptions](#) **messageSourceFilter**
- bool [hideInnerFrameShape](#)
- bool [useFormTitle](#)

Additional Inherited Members

9.152.1 Detailed Description

This widget is a group box that contains a [QForm](#) widget, which allows arbitrary ui files to be loaded - see [QForm](#) doco and comments for details. The group box is set checkable, and when un checked the group box's height is set to all is tilte to be seen, but non of the contents. When checked the group box will expand to accomodate the loaded ui file.

Note: this widget takes control/management of it's minimum and maximum width and heights, so setting in designer will have effectively no impact.

9.152.2 Member Function Documentation

9.152.2.1 restoreConfiguration()

```
void QEFormGroupBox::restoreConfiguration (
    PersistenceManager * ,
    restorePhases ) [protected], [virtual]
```

Service a request to restore the [QE](#) widget's configuration. A [QE](#) widget recover any configuration details from the [PersistenceManager](#). For example, a [QEStriptChart](#) may restore the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method. This is called twice with an incrementing restorePhase. Most widgets will miss the first call as they don't exist yet (they are created as part of the first phase).

Reimplemented from [QEWidget](#).

9.152.2.2 saveConfiguration()

```
void QEFormGroupBox::saveConfiguration (
    PersistenceManager * ) [protected], [virtual]
```

Service a request to save the [QE](#) widget's current configuration. A widget may save any configuration details through the [PersistenceManager](#). For example, a [QEStriptChart](#) may save the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method.

Reimplemented from [QEWidget](#).

9.152.3 Property Documentation

9.152.3.1 hideInnerFrameShape

```
bool QEFormGroupBox::hideInnerFrameShape [read], [write]
```

When true, the frame shape of the inner frame widget is set to NoShape provided a unique QFrame (or QE-Frame/other sub classes) is found at the top level of the loaded form.

9.152.3.2 uiFile

```
QString QEFormGroupBox::uiFile [read], [write]
```

The following propeties are directly copied form QEForm - see [QEForm](#) documentation and code comments for details, the the exception of the messageSourceId porperty which has been rebagded to formMessageSourceId to avoid conflict with [QEFormGroupBox](#) own messageSourceId porperty.

9.152.3.3 useFormTitle

```
bool QFormGroupBox::useFormTitle [read], [write]
```

When true, the form title is extracted from the loaded form and used to set the group box title.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEGroupBox/QFormGroupBox.h
- qeframework/widgets/QEGroupBox/QFormGroupBox.cpp

9.153 QFormMapper Class Reference

```
#include <QFormMapper.h>
```

Public Types

- typedef quint64 **FormHandles**

Static Public Member Functions

- static FormHandles **nullHandle** ()
- static FormHandles **allocateHandle** ()
- static QWidget * **findWidget** (const FormHandles &handle)
- static QWidget * **findWidget** (const FormHandles &handle, const QString &className)

Friends

- class **QEMapable**

9.153.1 Detailed Description

The [QFormMapper](#) mapper class is a singleton class that allows a user allocated form handle to be associated with an [QForm](#), specifically the form handle is included in the action request (along with the ui filename, macros etc.) emitted and then received up by the QEGui (or other) display manager. When the [QForm](#) object is created the user allocated handle is associated with the [QForm](#) object. This allows the originator of the request to find the [QForm](#) that was created as a result of the request.

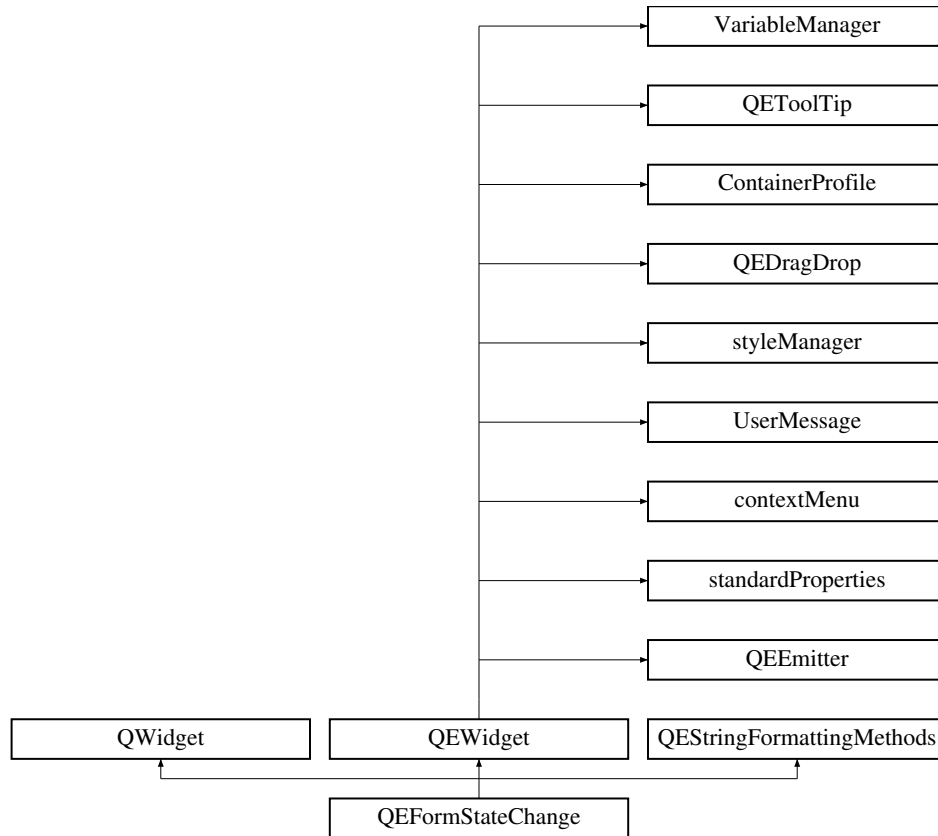
The documentation for this class was generated from the following files:

- qeframework/common/QFormMapper.h
- qeframework/common/QFormMapper.cpp

9.154 QFormStateChange Class Reference

```
#include <QFormStateChange.h>
```

Inheritance diagram for QFormStateChange:



Classes

- class [Actions](#)

Public Types

- enum **VariableIndicies** { viOpen = 0 , viClose , NUMBER_OF_VARIABLES }

Public Member Functions

- **QFormStateChange** (QWidget *parent=0)
- **QFormStateChange** (const QString &openVariableName, QWidget *parent=0)
- **QFormStateChange** (const QString &openVariableName, const QString &closeVariableName, QWidget *parent=0)
- void **setOpenVariableNameProperty** (const QString value)
- QString **getOpenVariableNameProperty** () const
- void **setOpenText** (const QString text)
- QString **getOpenText** () const
- void **setOpenProgram** (const QString program)

- QString **getOpenProgram** () const
- void **setOpenArguments** (const QStringList arguments)
- QStringList **getOpenArguments** () const
- void **setCloseVariableNameProperty** (const QString text)
- QString **getCloseVariableNameProperty** () const
- void **setCloseText** (const QString value)
- QString **getCloseText** () const
- void **setCloseProgram** (const QString program)
- QString **getCloseProgram** () const
- void **setCloseArguments** (const QStringList arguments)
- QStringList **getCloseArguments** () const
- void **setVariableNameSubstitutionsProperty** (const QString value)
- QString **getVariableNameSubstitutionsProperty** () const
- void **setFormatProperty** (const [QE::Formats](#) format)
- [QE::Formats](#) **getFormatProperty** () const

Protected Member Functions

- QSize **sizeHint** ()
- void **paintEvent** (QPaintEvent *event)
- [QEChannel](#) * **createQcItem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **stringFormattingChange** ()

Properties

- QString [variableOpen](#)
The variable to write to on open.
- QString [variableClose](#)
The variable to write to on close. May be the same as the open variable.
- QString [variableSubstitutions](#)
The default variable substitutions - same for both open and close variables.
- [QE::Formats](#) [format](#)
- QString **openText**
- QString [openProgram](#)
value to write on open
- QStringList [openArguments](#)
program to run
- QString [closeText](#)
program arguments
- QString [closeProgram](#)
value to write on close
- QStringList [closeArguments](#)
program to run
- bool [runVisible](#)
program arguments

Additional Inherited Members

9.154.1 Detailed Description

This widget is capable of responding to window/form open and close events. On open and/or close the widget can write to a PV and/or execute an arbitrary local program/script. This is not unlike automatically clicking a [QEPushButton](#) on open/close, except that there is no option to open another ui file. Note: the widget itself non-visible by default.

9.154.2 Property Documentation

9.154.2.1 format

`QE::Formats QEFormStateChange::format [read], [write]`

Format to apply to data. Default is 'Default' in which case the data type supplied with the data determines how the data is formatted. For all other options, an attempt is made to format the data as requested (whatever its native form). NOTE: The same format value applies to both the open and close PV formatting

9.154.2.2 runVisible

`bool QEFormStateChange::runVisible [read], [write]`

program arguments

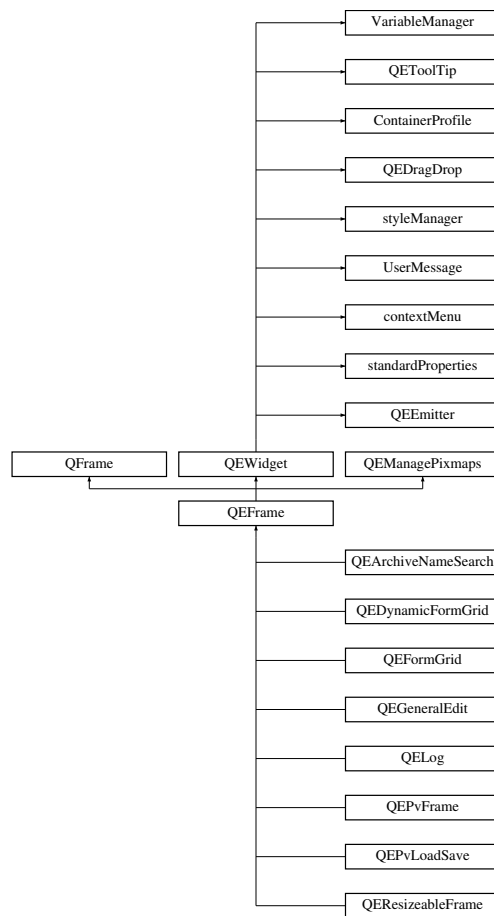
Not sure how usefull it would be to have this widget visible at run time, but let's not second guess the users' needs and desires.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEFormStateChange/QEFormStateChange.h
- qeframework/widgets/QEFormStateChange/QEFormStateChange.cpp

9.155 QEFrame Class Reference

Inheritance diagram for QEFrame:



Public Slots

- void **setSelectPixmap** (const int index)
- void **setManagedVisible** (bool v)

Public Member Functions

- **QEFrame** (QWidget *parent=0)
- QSize **sizeHint** () const
- void **setScaledContents** (bool scaledContentsIn)
- bool **getScaledContents** () const
- int **getSelectedPixmap** () const

Protected Member Functions

- void **paintEvent** (QPaintEvent *event)
- void **pixmapUpdated** (const int index)

Properties

- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- bool [oosAware](#)
- bool [scaledContents](#)
- QPixmap [pixmap0](#)
- QPixmap [pixmap1](#)
- QPixmap [pixmap2](#)
- QPixmap [pixmap3](#)
- QPixmap [pixmap4](#)
- QPixmap [pixmap5](#)
- QPixmap [pixmap6](#)
- QPixmap [pixmap7](#)
- QPixmap [pixmap8](#)
- QPixmap [pixmap9](#)
- QPixmap [pixmap10](#)
- QPixmap [pixmap11](#)
- QPixmap [pixmap12](#)
- QPixmap [pixmap13](#)
- QPixmap [pixmap14](#)
- QPixmap [pixmap15](#)

Additional Inherited Members

9.155.1 Member Function Documentation

9.155.1.1 [setManagedVisible](#)

```
void QEFrame::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.155.2 Property Documentation

9.155.2.1 allowDrop

```
bool QEFrame::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.155.2.2 defaultStyle

```
QString QEFrame::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.155.2.3 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEFrame::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.155.2.4 messageSourceId

```
unsigned int QEFrame::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.155.2.5 oosAware

```
bool QEFrame::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.155.2.6 pixmap0

```
QPixmap QEFrame::pixmap0 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 0.

9.155.2.7 pixmap1

```
QPixmap QFrame::pixmap1 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 1.

9.155.2.8 pixmap10

```
QPixmap QFrame::pixmap10 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 10.

9.155.2.9 pixmap11

```
QPixmap QFrame::pixmap11 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 11.

9.155.2.10 pixmap12

```
QPixmap QFrame::pixmap12 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 12.

9.155.2.11 pixmap13

```
QPixmap QFrame::pixmap13 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 13.

9.155.2.12 pixmap14

```
QPixmap QFrame::pixmap14 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 14.

9.155.2.13 pixmap15

```
QPixmap QFrame::pixmap15 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 15.

9.155.2.14 pixmap2

```
QPixmap QFrame::pixmap2 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 2.

9.155.2.15 pixmap3

```
QPixmap QEFrame::pixmap3 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 3.

9.155.2.16 pixmap4

```
QPixmap QEFrame::pixmap4 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 4.

9.155.2.17 pixmap5

```
QPixmap QEFrame::pixmap5 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 5.

9.155.2.18 pixmap6

```
QPixmap QEFrame::pixmap6 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 6.

9.155.2.19 pixmap7

```
QPixmap QEFrame::pixmap7 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 7.

9.155.2.20 pixmap8

```
QPixmap QEFrame::pixmap8 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 8.

9.155.2.21 pixmap9

```
QPixmap QEFrame::pixmap9 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 9.

9.155.2.22 scaledContents

```
bool QEFrame::scaledContents [read], [write]
```

Flag the pixmap for the background is to be scaled to fit the frame. Similar operation to scaledContents property for a QLabel.

9.155.2.23 `styleSheet`

```
QString QEFrame::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.155.2.24 `userLevelEnabled`

```
QE::UserLevels QEFrame::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.155.2.25 `userLevelEngineerStyle`

```
QString QEFrame::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.155.2.26 `userLevelScientistStyle`

```
QString QEFrame::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.155.2.27 `userLevelUserStyle`

```
QString QEFrame::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.155.2.28 userLevelVisibility

```
QE::UserLevels QEFrame::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programmatically through `setUserLevel()`. Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.155.2.29 variableAsToolTip

```
bool QEFrame::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.155.2.30 visible

```
bool QEFrame::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEFrame/QEFrame.h
- qeframework/widgets/QEFrame/QEFrame.cpp

9.156 QEFrameworkVersion Class Reference**Static Public Member Functions**

- static unsigned int **getMajor** ()
- static unsigned int **getMinor** ()
- static unsigned int **getRelease** ()
- static const QString **getStage** ()
- static const QString **getString** ()
- static const QString **getDateTime** ()
- static const QString **getAttributes** ()
- static const QString **getQtVersionStr** ()
- static const QString **getQwtVersionStr** ()
- static const QString **getEpicsVersionStr** ()
- static const QString **getAcaiVersionStr** ()

The documentation for this class was generated from the following files:

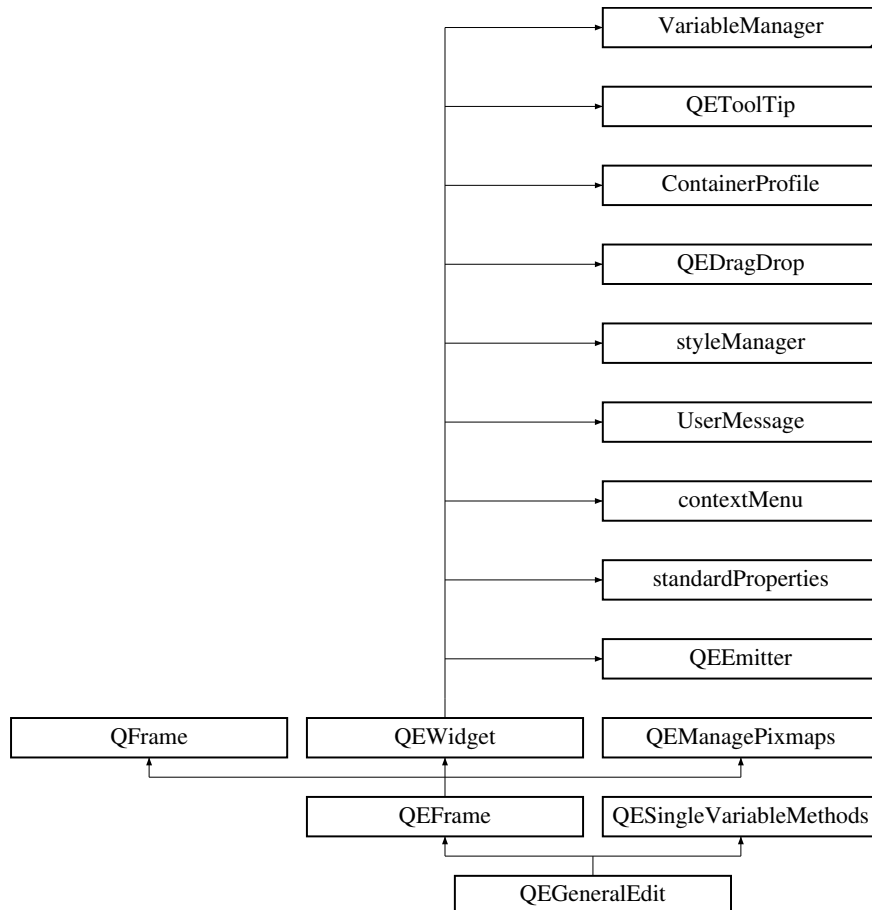
- qeframework/common/QEFrameworkVersion.h
- qeframework/common/QEFrameworkVersion.cpp

9.157 QGeneralEdit Class Reference

The [QGeneralEdit](#) class This class provides a general PV edit widget, presenting one off a [QLineEdit](#), a [QNumericEdit](#) or a [QERadioGroup](#) for string, numerical and enumeration data kinds respectively.

```
#include <QGeneralEdit.h>
```

Inheritance diagram for QGeneralEdit:



Signals

- void [dbValueChanged](#) ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void [dbConnectionChanged](#) (const bool &isConnected)

Public Member Functions

- [QEGeneralEdit](#) (QWidget *parent=0)
- [QEGeneralEdit](#) (const QString &variableName, QWidget *parent=0)
- virtual [~QEGeneralEdit](#) ()
Destruction.
- void **setArrayIndex** (const int [arrayIndex](#))
- void **setButtonStyle** (const [QRadioGroup::ButtonStyles](#) style)
- [QRadioGroup::ButtonStyles](#) **getButtonStyle** () const
- void **setButtonOrder** (const [QE::GridOrders](#) order)
- [QE::GridOrders](#) **getButtonOrder** () const
- void **setUseApplyButton** (const bool [useApplyButton](#))
- bool **getUseApplyButton** () const

Protected Member Functions

- QSize **sizeHint** () const
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- [QEChannel](#) * **createQcItem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- [QRadioGroup::ButtonStyles](#) [buttonStyle](#)
- [QE::GridOrders](#) [buttonOrder](#)
- bool [useApplyButton](#)

Additional Inherited Members

9.157.1 Detailed Description

The [QEGeneralEdit](#) class This class provides a general PV edit widget, presenting one off a [QELineEdit](#), a [QENumericEdit](#) or a [QERadioGroup](#) for string, numerical and enumeration data kinds respectively.

This widget is intended for use within a qegui predefined form, displayed in response to context menu request to edit an arbitrary PV.

9.157.2 Constructor & Destructor Documentation

9.157.2.1 QEGeneralEdit() [1/2]

```
QEGeneralEdit::QEGeneralEdit (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.157.2.2 QEGeneralEdit() [2/2]

```
QEGeneralEdit::QEGeneralEdit (
    const QString & variableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.157.3 Member Function Documentation

9.157.3.1 dbConnectionChanged

```
void QEGeneralEdit::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.157.3.2 dbValueChanged

```
void QEGeneralEdit::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.157.4 Property Documentation

9.157.4.1 arrayIndex

```
int QEGeneralEdit::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.157.4.2 buttonOrder

```
QE::GridOrders QEGeneralEdit::buttonOrder [read], [write]
```

Button order for enumerations edits.

9.157.4.3 buttonStyle

```
QRadioGroup::ButtonStyles QEGeneralEdit::buttonStyle [read], [write]
```

Button style for enumerations edits.

9.157.4.4 elementsRequired

```
int QEGeneralEdit::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.157.4.5 useApplyButton

```
bool QEGeneralEdit::useApplyButton [read], [write]
```

For numeric and string edits, the change can be applied on enter/lose focus, or applied via explicitly clicking an apply button. Setting this property false uses the former, true the latter. The default property value is false.

9.157.4.6 variable

```
QString QEGeneralEdit::variable [read], [write]
```

EPICS variable name (CA PV)

9.157.4.7 variableSubstitutions

```
QString QEGeneralEdit::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

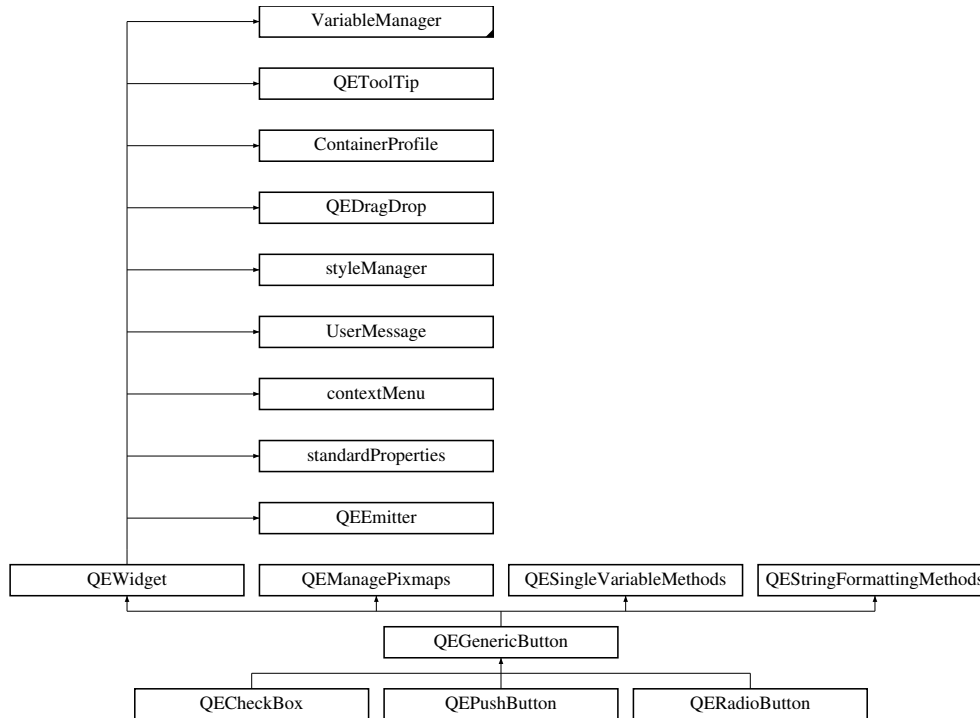
The documentation for this class was generated from the following files:

- qeframework/widgets/QEGeneralEdit/QEGeneralEdit.h
- qeframework/widgets/QEGeneralEdit/QEGeneralEdit.cpp

9.158 QEGenericButton Class Reference

```
#include <QEGenericButton.h>
```

Inheritance diagram for QEGenericButton:



Public Types

- enum **VariableAllocation** { **VAR_PRIMARY** = 0 , **VAR_READBACK** , **NUMBER_OF_VARIABLES** }

Public Member Functions

- **QEGenericButton** (QAbstractButton *owner)
- void **setSubscribe** (bool [subscribe](#))
- bool **getSubscribe** ()
- void **setUpdateOption** ([QE::UpdateOptions](#) updateOptionIn)
- [QE::UpdateOptions](#) **getUpdateOption** ()
- void **setTextAlignment** (Qt::Alignment alignment)
- Qt::Alignment **getTextAlignment** ()
- void **setPassword** (QString password)
- QString **getPassword** ()
- void **setConfirmAction** (bool confirmRequiredIn)
- bool **getConfirmAction** ()
- void **setConfirmText** (QString confirmTextIn)
- QString **getConfirmText** ()
- void **setWriteOnPress** (bool writeOnPress)
- bool **getWriteOnPress** ()
- void **setWriteOnRelease** (bool writeOnRelease)
- bool **getWriteOnRelease** ()

- void **setWriteOnClick** (bool writeOnClick)
- bool **getWriteOnClick** ()
- void **setPressText** (QString pressText)
- QString **getPressText** ()
- void **setReleaseText** (QString releaseTextIn)
- QString **getReleaseText** ()
- void **setClickText** (QString clickTextIn)
- QString **getClickText** ()
- void **setClickCheckedText** (QString clickCheckedTextIn)
- QString **getClickCheckedText** ()
- void **setProgram** (QString program)
- QString **getProgram** ()
- void **setArguments** (QStringList arguments)
- QStringList **getArguments** ()
- void **setProgramStartupOption** ([QE::ProgramStartupOptions](#) programStartupOptionIn)
- [QE::ProgramStartupOptions](#) **getProgramStartupOption** ()
- void **setGuiName** (QString guiName)
- QString **getGuiName** ()
- void **setPrioritySubstitutions** (QString prioritySubstitutionsIn)
- QString **getPrioritySubstitutions** ()
- void **setCustomisationName** (QString customisationNameIn)
- QString **getCustomisationName** ()
- void **setCreationOption** ([QE::CreationOptions](#) creationOption)
- [QE::CreationOptions](#) **getCreationOption** ()
- void **setLabelTextProperty** (QString labelTextIn)
- QString **getLabelTextProperty** ()
- void **setDisabledRecordPolicy** (const [QE::DisabledRecordPolicy](#) disabledRecordPolicy)
- [QE::DisabledRecordPolicy](#) **getDisabledRecordPolicy** () const
- void **writeClickedNow** (const bool checked=false)

Protected Member Functions

- void **useGenericNewVariableName** (const [QEPvNameProperties](#) &pvNameProperties)
- void **connectionChanged** (const [QEConnectionUpdate](#) &update)
- void **setGenericButtonText** (const [QEStringValueUpdate](#) &update)
- void **userPressed** ()
- void **userReleased** ()
- void **userClicked** (bool checked)
- void **processWriteNow** (const bool checked)
- virtual [QE::UpdateOptions](#) **getDefaultUpdateOption** ()=0
- void **setup** ()
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **calcStyleOption** ()

Protected Attributes

- [applicationLauncher](#) **programLauncher**
- [QESingleVariableMethods](#) * **altReadback**

Additional Inherited Members

9.158.1 Detailed Description

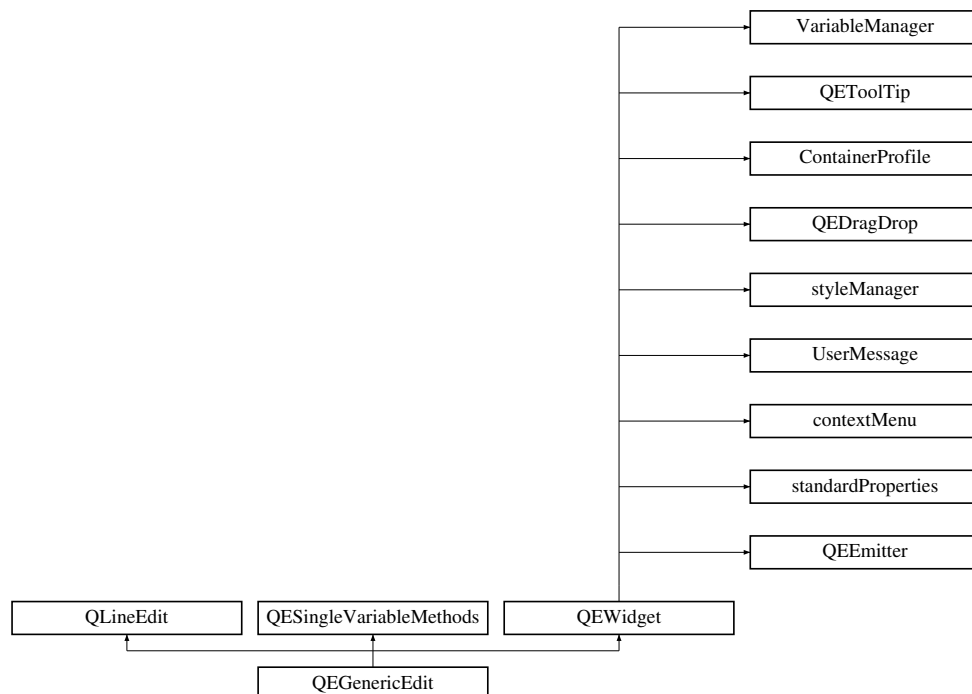
Class common to all [QE](#) buttons

The documentation for this class was generated from the following files:

- qeframework/widgets/QEButton/QEGenericButton.h
- qeframework/widgets/QEButton/QEGenericButton.cpp

9.159 QEGenericEdit Class Reference

Inheritance diagram for QEGenericEdit:



Public Slots

- void [setDefaultStyle](#) (const QString &style)
Update the default style applied to this widget.
- void [setManagedVisible](#) (bool v)
- void [writeNow](#) ()

Signals

- void [userChange](#) (const QVariant &oldValue, const QVariant &newValue, const QVariant &lastValue)
Internal use only. Used by [QEConfiguredLayout](#) to be notified when one of its widgets has written something.
- void [requestResend](#) ()
Internal use only. Used when changing a property value to force a re-display to reflect the new property value.

Public Member Functions

- void **setDropOption** (QE::DropOption option)
- QE::DropOption **getDropOption** ()
- QEGenericEdit (QWidget *parent=0)
- QEGenericEdit (const QString &variableName, QWidget *parent=0)
- void **setWriteOnLoseFocus** (bool writeOnLoseFocus)
- bool **getWriteOnLoseFocus** ()
- void **setWriteOnEnter** (bool writeOnEnter)
- bool **getWriteOnEnter** ()
- void **setWriteOnFinish** (bool writeOnFinish)
- bool **getWriteOnFinish** ()
- void **setConfirmWrite** (bool confirmWrite)
- bool **getConfirmWrite** ()
- void **setAllowFocusUpdate** (bool allowFocusUpdate)
- bool **getAllowFocusUpdate** () const
Returns 'true' if this widget configured to allow updates while it has focus.
- void **setSubscribe** (bool subscribe)
- bool **getSubscribe** ()
- void **writeValue** (QEChannel *qca, QVariant newValue)

Protected Member Functions

- void **setDataIfNoFocus** (const QEVariantUpdate &update)
- bool **getIsConnected** () const
- bool **getIsFirstUpdate** () const
- virtual void **setValue** (const QVariant &value)=0
- virtual QVariant **getValue** ()=0
- virtual bool **writeData** (const QVariant &value, QString &message)=0

Protected Attributes

- QVariant **lastValue**
- QVariant **lastUserValue**
- bool **messageDialogPresent**
- bool **writeFailMessageDialogPresent**
- bool **isConnected**

Properties

- QString **text**
- QString **variable**
- QString **variableSubstitutions**
- int **arrayIndex**
- bool **subscribe**
- bool **writeOnLoseFocus**
- bool **writeOnEnter**
- bool **writeOnFinish**
- bool **confirmWrite**
- bool **allowFocusUpdate**

Allow updated while widget has focus - defaults to false.

- [QE::DropOption dropOption](#)
Allow drag/drops mode to this widget.
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels userLevelVisibility](#)
- [QE::UserLevels userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions displayAlarmStateOption](#)
- bool [oosAware](#)

Additional Inherited Members

9.159.1 Constructor & Destructor Documentation

9.159.1.1 QEGenericEdit() [1/2]

```
QEGenericEdit::QEGenericEdit (
    QWidget * parent = 0 )
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.159.1.2 QEGenericEdit() [2/2]

```
QEGenericEdit::QEGenericEdit (
    const QString & variableName,
    QWidget * parent = 0 )
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.159.2 Member Function Documentation

9.159.2.1 getConfirmWrite()

```
bool QEGenericEdit::getConfirmWrite ( )
```

Returns 'true' if this widget will ask for confirmation (using a dialog box) prior to writing data.

9.159.2.2 getSubscribe()

```
bool QEGenericEdit::getSubscribe ( )
```

Returns 'true' if this widget subscribes for data updates and displays current data.

9.159.2.3 getWriteOnEnter()

```
bool QEGenericEdit::getWriteOnEnter ( )
```

Returns 'true' if this widget writes any changes when the user presses 'enter'.

9.159.2.4 getWriteOnFinish()

```
bool QEGenericEdit::getWriteOnFinish ( )
```

Returns 'true' if this widget writes any changes when the user finished editing (the QLineEdit 'editingFinished' signal is emitted).

9.159.2.5 getWriteOnLoseFocus()

```
bool QEGenericEdit::getWriteOnLoseFocus ( )
```

Returns 'true' if this widget automatically writes any changes when it loses focus.

9.159.2.6 setAllowFocusUpdate()

```
void QEGenericEdit::setAllowFocusUpdate (
    bool allowFocusUpdate )
```

Sets if this widget configured to allow updates while it has focus. Default is 'false'.

9.159.2.7 setConfirmWrite()

```
void QEGenericEdit::setConfirmWrite (
    bool confirmWrite )
```

Sets if this widget will ask for confirmation (using a dialog box) prior to writing data. Default is 'false' (will not ask for confirmation (using a dialog box) prior to writing data).

9.159.2.8 setManagedVisible

```
void QEGenericEdit::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.159.2.9 setSubscribe()

```
void QEGenericEdit::setSubscribe (
    bool subscribe )
```

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.159.2.10 setWriteOnEnter()

```
void QEGenericEdit::setWriteOnEnter (
    bool writeOnEnter )
```

Sets if this widget writes any changes when the user presses 'enter'. Note, the current value will be written even if the user has not changed it. Default is 'true' (writes any changes when the user presses 'enter').

9.159.2.11 setWriteOnFinish()

```
void QEGenericEdit::setWriteOnFinish (
    bool writeOnFinish )
```

Sets if this widget writes any changes when the user finished editing (the QLineEdit 'editingFinished' signal is emitted). No writing occurs if no changes were made. Default is 'true' (writes any changes when the QLineEdit 'editingFinished' signal is emitted).

9.159.2.12 setWriteOnLoseFocus()

```
void QEGenericEdit::setWriteOnLoseFocus (
    bool writeOnLoseFocus )
```

Sets if this widget automatically writes any changes when it loses focus. Default is 'false' (does not write any changes when it loses focus).

9.159.3 Property Documentation

9.159.3.1 allowDrop

```
bool QEGenericEdit::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.159.3.2 arrayIndex

```
int QEGenericEdit::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.159.3.3 confirmWrite

```
bool QEGenericEdit::confirmWrite [read], [write]
```

Sets if this widget will ask for confirmation (using a dialog box) prior to writing data. Default is 'false' (will not ask for confirmation (using a dialog box) prior to writing data).

9.159.3.4 defaultStyle

```
QString QEGenericEdit::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.159.3.5 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEGenericEdit::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.159.3.6 dropOption

```
QE::DropOption QEGenericEdit::dropOption [read], [write]
```

Allow drag/drops mode to this widget.

Option: DropToText - just set text on the widget Option: DropToTextAndWrite - write the text to the PV if the PV has been connected otherwise equivalent to DropToText

9.159.3.7 messageSourceId

```
unsigned int QEGenericEdit::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.159.3.8 oosAware

```
bool QEGenericEdit::oosAware [read], [write]
```

Indicates whether the widget should respond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.159.3.9 `styleSheet`

```
QString QEGenericEdit::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.159.3.10 `subscribe`

```
bool QEGenericEdit::subscribe [read], [write]
```

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.159.3.11 `userLevelEnabled`

```
QE::UserLevels QEGenericEdit::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.159.3.12 `userLevelEngineerStyle`

```
QString QEGenericEdit::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.159.3.13 `userLevelScientistStyle`

```
QString QEGenericEdit::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.159.3.14 `userLevelUserStyle`

```
QString QEGenericEdit::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.159.3.15 userLevelVisibility

```
QE::UserLevels QEGenericEdit::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programmatically through `setUserLevel()`. Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.159.3.16 variable

```
QString QEGenericEdit::variable [read], [write]
```

EPICS variable name (CA PV)

9.159.3.17 variableAsToolTip

```
bool QEGenericEdit::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.159.3.18 variableSubstitutions

```
QString QEGenericEdit::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.159.3.19 visible

```
bool QEGenericEdit::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

9.159.3.20 writeOnEnter

```
bool QEGenericEdit::writeOnEnter [read], [write]
```

Sets if this widget writes any changes when the user presses 'enter'. Note, the current value will be written even if the user has not changed it. Default is 'true' (writes any changes when the user presses 'enter').

9.159.3.21 writeOnFinish

```
bool QEGenericEdit::writeOnFinish [read], [write]
```

Sets if this widget writes any changes when the user finished editing (the QLineEdit 'editingFinished' signal is emitted). No writing occurs if no changes were made. Default is 'true' (writes any changes when the QLineEdit 'editingFinished' signal is emitted).

9.159.3.22 writeOnLoseFocus

```
bool QEGenericEdit::writeOnLoseFocus [read], [write]
```

Sets if this widget automatically writes any changes when it loses focus. Default is 'false' (does not write any changes when it loses focus).

The documentation for this class was generated from the following files:

- qeframework/widgets/QLineEdit/QEGenericEdit.h
- qeframework/widgets/QLineEdit/QEGenericEdit.cpp

9.160 QEGlobalStyle Class Reference

```
#include <QEGlobalStyle.h>
```

Public Member Functions

- [QEGlobalStyle](#) ()
- [~QEGlobalStyle](#) ()
- void [apply](#) ()

9.160.1 Detailed Description

This class sets the QApplication instance style sheet, and hence is a global style that is applicable to all widgets. As this functionality is invoked from the [QEWidget](#) constructor, this style applies equally within designer, qegui or any other display managers using at least one [QEWidget](#).

9.160.2 Constructor & Destructor Documentation

9.160.2.1 QEGlobalStyle()

```
QEGlobalStyle::QEGlobalStyle ( )
```

Constructor

9.160.2.2 ~QEGlobalStyle()

```
QEGlobalStyle::~QEGlobalStyle ( )
```

Destructor

9.160.3 Member Function Documentation

9.160.3.1 apply()

```
void QEGlobalStyle::apply ( )
```

Checks for the existence of the QE_GLOBAL_STYLE_SHEET environment variable and if defined appends the specified style to the application style sheet. If the (trimmed) value of the environment variable starts with "file://" then the rest of the environment variable will be interpreted as file name path.

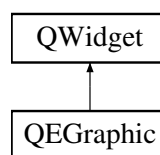
The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/QEGlobalStyle.h
- qeframework/widgets/QEWidget/QEGlobalStyle.cpp

9.161 QEGraphic Class Reference

```
#include <QEGraphic.h>
```

Inheritance diagram for QEGraphic:



Classes

- class [OwnPlot](#)

Signals

- void **mouseMove** (const QPointF &posn)
- void **wheelRotate** (const QPointF &posn, const int delta)
- void **areaDefinition** (const QPointF &from, const QPointF &to)
- void **lineDefinition** (const QPointF &from, const QPointF &to)
- void **crosshairsMove** (const QPointF &posn)
- void **markupMove** (const QEGraphicNames::Markups markup, const QPointF &position)

Public Member Functions

- **QEGraphic** (QWidget *parent=0)
- **QEGraphic** (const QString &title, QWidget *parent=0)
- void **setTitle** (const QString &title)
- QString **getTitle** () const
- void **releaseCurves** ()
- void **attachOwnCurve** (QwtPlotCurve *curve)
- void **setBackgroundColour** (const QColor colour)
- void **setGridPen** (const QPen &pen)
- void **setGridPens** (const QPen &majorPen, const QPen &minorPen, const bool enableMajorX, const bool enableMajorY, const bool enableMinorX, const bool enableMinorY)
- void **setAvailableMarkups** (const QEGraphicNames::MarkupFlags graphicMarkupsSet)
- QEGraphicNames::MarkupFlags **getAvailableMarkups** () const
- void **setMarkupVisible** (const QEGraphicNames::Markups markup, const bool isVisible)
- bool **getMarkupVisible** (const QEGraphicNames::Markups markup) const
- void **setMarkupEnabled** (const QEGraphicNames::Markups markup, const bool isEnabled)
- bool **getMarkupEnabled** (const QEGraphicNames::Markups markup) const
- void **setMarkupSelected** (const QEGraphicNames::Markups markup, const bool selected)
- bool **getMarkupsSelected** (const QEGraphicNames::Markups markup) const
- void **setMarkupPosition** (const QEGraphicNames::Markups markup, const QPointF &position)
- QPointF **getMarkupPosition** (const QEGraphicNames::Markups markup) const
- void **setMarkupData** (const QEGraphicNames::Markups markup, const QVariant &text)
- QVariant **getMarkupData** (const QEGraphicNames::Markups markup) const
- void **plotCurveData** (const QEGraphicNames::DoubleVector &xData, const QEGraphicNames::DoubleVector &yData, const QwtPlot::Axis yAxis=QwtPlot::yLeft)
- void **drawText** (const QPointF &posn, const QString &text, const QEGraphicNames::TextPositions option, bool isCentred=true)
- void **drawText** (const QPoint &posn, const QString &text, const QEGraphicNames::TextPositions option, bool isCentred=true)
- void **setXRange** (const double min, const double max, const QEGraphicNames::AxisMajorIntervalModes mode, const double value, const bool immediate)
- void **getXRange** (double &min, double &max) const
- void **setYRange** (const double min, const double max, const QEGraphicNames::AxisMajorIntervalModes mode, const double value, const bool immediate, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft)
- void **getYRange** (double &min, double &max, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft) const
- void **setAxisTitle** (const QwtPlot::Axis selectedAxis, const QString &title)
- QString **getAxisTitle** (const QwtPlot::Axis selectedAxis) const
- void **setAxisAutoScale** (const QwtPlot::Axis selectedAxis, const bool enabled)
- bool **getAxisAutoScale** (const QwtPlot::Axis selectedAxis) const
- void **replot** ()
- QPointF **getRealMousePosition** () const
- bool **globalPosIsOverCanvas** (const QPoint &globalPos) const
- bool **getSlopesDefined** (QPointF &slope) const
- void **setAxisEnableX** (const bool enable)
- bool **getAxisEnableX** () const
- void **setAxisEnableY** (const bool enable, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft)
- bool **getAxisEnableY** (const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft) const
- void **setXScale** (const double scale)
- double **getXScale** () const
- void **setXOffset** (const double offset)
- double **getXOffset** () const
- void **setXLogarithmic** (const bool isLog)
- bool **getXLogarithmic** () const
- void **setYScale** (const double scale, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft)

- double [getYScale](#) (const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft) const
- void [setYOffset](#) (const double offset, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft)
- double [getYOffset](#) (const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft) const
- void [setYLogarithmic](#) (const bool isLog, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft)
- bool [getYLogarithmic](#) (const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft) const
- void [setYColor](#) (const QColor color, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft)
- void [setCurvePen](#) (const QPen &pen)
- QPen [getCurvePen](#) () const
- void [setCurveBrush](#) (const QBrush &brush)
- QBrush [getCurveBrush](#) () const
- void [setTextFont](#) (const QFont &font)
- QFont [getTextFont](#) () const
- void [setTextPointSize](#) (const int pointSize)
- int [getTextPointSize](#) () const
- void [setCurveRenderHint](#) (const QwtPlotItem::RenderHint hint, const bool on=true)
- QwtPlotItem::RenderHint [getCurveRenderHint](#) () const
- bool [getCurveRenderHintOn](#) () const
- void [setCurveStyle](#) (const QwtPlotCurve::CurveStyle style)
- QwtPlotCurve::CurveStyle [getCurveStyle](#) ()
- QPoint [pixelDistance](#) (const QPointF &from, const QPointF &to) const
- QPointF [realOffset](#) (const QPoint &offset, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft) const
- QPointF [realOffset](#) (const QPointF &offset, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft) const
- QPointF [pointToReal](#) (const QPoint &pos, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft) const
- QPointF [pointToReal](#) (const QPointF &pos, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft) const
- QPoint [realToPoint](#) (const QPointF &pos, const QwtPlot::Axis selectedYAxis=QwtPlot::yLeft) const
- void [enableAxis](#) (int axisId, bool tf=true)
- void [setAxisScale](#) (int axisId, double min, double max, double step=0)
- void [installCanvasEventFilter](#) (QObject *filter)
- bool [isCanvasObject](#) (QObject *obj) const
- QRect [getEmbeddedCanvasGeometry](#) () const
- QwtPlot * [getEmbeddedQwtPlot](#) () const
- void [saveConfiguration](#) (PMElement &parentElement)
- void [restoreConfiguration](#) (PMElement &parentElement)

Protected Member Functions

- void [canvasMousePress](#) (QMouseEvent *mouseEvent)
- void [canvasMouseRelease](#) (QMouseEvent *mouseEvent)
- void [canvasMouseMove](#) (QMouseEvent *mouseEvent, const bool isButtonAction)
- bool [eventFilter](#) (QObject *obj, QEvent *event)

Friends

- class [OwnPlot](#)
- class [QEGraphicMarkup](#)
- class [QEGraphicAreaMarkup](#)
- class [QEGraphicLineMarkup](#)
- class [QEGraphicCrosshairsMarkup](#)
- class [QEGraphicHorizontalMarkup](#)
- class [QEGraphicVerticalMarkup](#)

9.161.1 Detailed Description

Provides a basic wrapper around QwtPlot, which:

- a) Allocates and attaches curves and grids, and releases these on delete, and releases curves on request;
- b) Interprets mouse events with real world co-ordinates;
- c) Provides a log scale mode (X and/or Y);
- d) Provides a consistant means to round down/up min/max values and the selection of a major interval value, e.g.: 2.1 .. 7.83 (user min/max) => 2.0 .. 8.0, 1.0 (display min/max, major);
- e) Standardised mouse and wheel zooming;
- f) Smart axis re-scaling;
- g) Provides markups; and
- h) Provides wrapper functions to hide QWT version API changes.

9.161.2 Member Function Documentation

9.161.2.1 drawText()

```
void QEGraphic::drawText (
    const QPointF & posn,
    const QString & text,
    const QEGraphicNames::TextPositions option,
    bool isCentred = true )
```

Draw text at position specifying centre of bottom left corner. Position may be real world coordinates or pixel coordinates.

Parameters

<i>posn</i>	The postion to draw text
<i>text</i>	The text to draw
<i>option</i>	Specifies if position is in cavas pixels or real world co-ordinates
<i>pointSize</i>	Defines text point size. Honors QEScaling parameters.
<i>isCentred</i>	When true text is centered about point, otherwise point defines bottom left corner of the generated text

9.161.2.2 getAxisEnableY()

```
bool QEGraphic::getAxisEnableY (
    const QwtPlot::Axis selectedYAxis = QwtPlot::yLeft ) const
```

Is left/right Y axis enabled/disabled

Parameters

<i>selectedYAxis</i>	Left/right Y axis. If not provided left axis is used by default
----------------------	---

Returns

Boolean signaling if axis is enabled/disabled

9.161.2.3 getXRange()

```
void QEGraphic::getXRange (
    double & min,
    double & max ) const
```

Returns the range of X axis

Parameters

<i>min</i>	Minimum X value
<i>max</i>	Maximum X value

9.161.2.4 getYLogarithmic()

```
bool QEGraphic::getYLogarithmic (
    const QwtPlot::Axis selectedYAxis = QwtPlot::yLeft ) const
```

Is the scale set to be logarithmic or not on left/right axis

Parameters

<i>selectedYAxis</i>	Left/right Y axis. If not provided left axis is used by default
----------------------	---

Returns

Boolean flag defining if the scale set is logarithmic

9.161.2.5 getYOffset()

```
double QEGraphic::getYOffset (
    const QwtPlot::Axis selectedYAxis = QwtPlot::yLeft ) const
```

Returns the offset set on left/right Y axis.

Parameters

<i>selectedYAxis</i>	Left/right Y axis. If not provided left axis is used by default.
----------------------	--

Returns

Offset set

9.161.2.6 getYRange()

```
void QEGraphic::getYRange (
    double & min,
    double & max,
    const QwtPlot::Axis selectedYAxis = QwtPlot::yLeft ) const
```

Returns the range of either left or right Y axis

Parameters

<i>min</i>	Reference to variable to be used to store minimum value
<i>max</i>	Reference to variable to be used to store maximumSize() value
<i>selectedYAxis</i>	Range for which Y axis is required. If not provided left axis is used by default

9.161.2.7 getYScale()

```
double QEGraphic::getYScale (
    const QwtPlot::Axis selectedYAxis = QwtPlot::yLeft ) const
```

Returns the scale set on left/right Y axis.

Parameters

<i>selectedYAxis</i>	Left/right Y axis. If not provided left axis is used by default.
----------------------	--

Returns

Scale set

9.161.2.8 plotCurveData()

```
void QEGraphic::plotCurveData (
    const QEGraphicNames::DoubleVector & xData,
```

```
const QEGraphicNames::DoubleVector & yData,
const QwtPlot::Axis yAxis = QwtPlot::yLeft )
```

Allocates a curve, sets current curve attributes and attaches to plot.

Parameters

<i>xData</i>	Vector of X axis values
<i>yData</i>	Vector of Y axis values
<i>yAxis</i>	Which Y axis should the yData be plotted against. If not provided left axis is used by default

9.161.2.9 setAxisEnableY()

```
void QEGraphic::setAxisEnableY (
    const bool enable,
    const QwtPlot::Axis selectedYAxis = QwtPlot::yLeft )
```

Enables or disables left or right Y axis

Parameters

<i>enable</i>	Enable/disable Y axis
<i>selectedYAxis</i>	Left/right Y axis. If not provided left axis is used by default

9.161.2.10 setYColor()

```
void QEGraphic::setYColor (
    const QColor color,
    const QwtPlot::Axis selectedYAxis = QwtPlot::yLeft )
```

Sets the color of left/right Y axis

Parameters

<i>color</i>	Color to be set
<i>selectedYAxis</i>	Left/right Y axis. If not provided left axis is used by default

9.161.2.11 setYLogarithmic()

```
void QEGraphic::setYLogarithmic (
    const bool isLog,
    const QwtPlot::Axis selectedYAxis = QwtPlot::yLeft )
```


Sets the scale on Y axis to be logarithmic or not

Parameters

<i>isLog</i>	Boolean flag defining if the scale is to be logarithmic
<i>selectedYAxis</i>	Left/right Y axis. If not provided left axis is used by default

9.161.2.12 setYOffset()

```
void QEGraphic::setYOffset (
    const double offset,
    const QwtPlot::Axis selectedYAxis = QwtPlot::yLeft )
```

Sets offset on either left or right Y axis

Parameters

<i>offset</i>	Offset to be set
<i>selectedYAxis</i>	Left/right Y axis. If not provided left axis is used by default

9.161.2.13 setYScale()

```
void QEGraphic::setYScale (
    const double scale,
    const QwtPlot::Axis selectedYAxis = QwtPlot::yLeft )
```

Sets scale on either left or right Y axis

Parameters

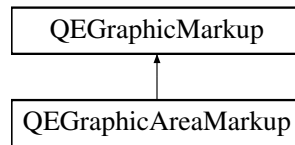
<i>scale</i>	Scale to be set
<i>selectedYAxis</i>	Left/right Y axis. If not provided left axis is used by default

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphic.h
- qeframework/common/QEGraphic.cpp

9.162 QEGraphicAreaMarkup Class Reference

Inheritance diagram for QEGraphicAreaMarkup:



Public Member Functions

- **QEGraphicAreaMarkup** ([QEGraphic](#) *owner)
- void **mousePress** (const QPointF &realMousePosition, const Qt::MouseButton button)
- void **mouseRelease** (const QPointF &realMousePosition, const Qt::MouseButton button)
- void **mouseMove** (const QPointF &realMousePosition)

Protected Member Functions

- void **plotMarkup** ()

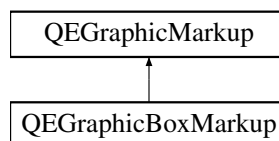
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicMarkup.h
- qeframework/common/QEGraphicMarkup.cpp

9.163 QEGraphicBoxMarkup Class Reference

Inheritance diagram for QEGraphicBoxMarkup:



Public Member Functions

- **QEGraphicBoxMarkup** ([QEGraphic](#) *owner)
- bool **isOver** (const QPointF &point, int &distance) const
- void **setSelected** (const bool selected)

Protected Member Functions

- void **plotMarkup** ()

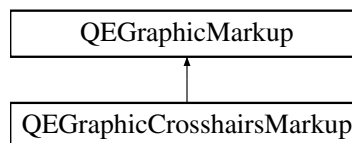
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicMarkup.h
- qeframework/common/QEGraphicMarkup.cpp

9.164 QEGraphicCrosshairsMarkup Class Reference

Inheritance diagram for QEGraphicCrosshairsMarkup:



Public Member Functions

- **QEGraphicCrosshairsMarkup** ([QEGraphic](#) *owner)
- void **setVisible** (const bool visible)
- bool **isOver** (const QPointF &point, int &distance) const
- void **mousePress** (const QPointF &realMousePosition, const Qt::MouseButton button)
- void **mouseRelease** (const QPointF &realMousePosition, const Qt::MouseButton button)
- void **mouseMove** (const QPointF &realMousePosition)

Protected Member Functions

- void **plotMarkup** ()

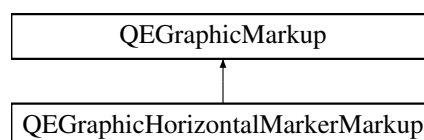
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicMarkup.h
- qeframework/common/QEGraphicMarkup.cpp

9.165 QEGraphicHorizontalMarkerMarkup Class Reference

Inheritance diagram for QEGraphicHorizontalMarkerMarkup:



Public Member Functions

- **QEGraphicHorizontalMarkerMarkup** (QEGraphicNames::Markups markup, [QEGraphic](#) *owner)
- void **setVisible** (const bool visible)

Protected Member Functions

- void **plotMarkup** ()

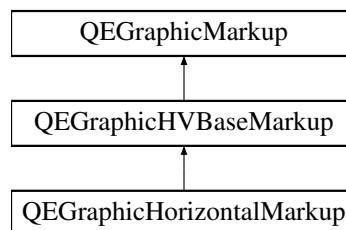
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicMarkup.h
- qeframework/common/QEGraphicMarkup.cpp

9.166 QEGraphicHorizontalMarkup Class Reference

Inheritance diagram for QEGraphicHorizontalMarkup:



Public Member Functions

- **QEGraphicHorizontalMarkup** (const QEGraphicNames::Markups markup, [QEGraphic](#) *owner)
- bool **isOver** (const QPointF &point, int &distance) const
- void **relocate** ()

Protected Member Functions

- void **getLine** (double &xmin, double &xmax, double &ymin, double &ymax)
- void **getShape** (QPoint shape[])

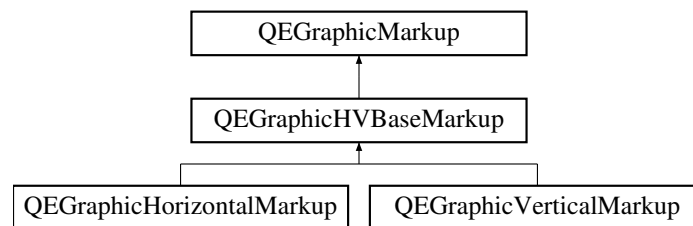
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicMarkup.h
- qeframework/common/QEGraphicMarkup.cpp

9.167 QEGraphicHVBBaseMarkup Class Reference

Inheritance diagram for QEGraphicHVBBaseMarkup:



Public Member Functions

- **QEGraphicHVBBaseMarkup** (QEGraphicNames::Markups markup, [QEGraphic](#) *owner)
- void **setInUse** (const bool inUse)
- void **mousePress** (const QPointF &realMousePosition, const Qt::MouseButton button)
- void **mouseRelease** (const QPointF &realMousePosition, const Qt::MouseButton button)
- void **mouseMove** (const QPointF &realMousePosition)

Protected Member Functions

- void **setColours** (const unsigned int baseRGB)
- void **plotMarkup** ()
- virtual void **getLine** (double &xmin, double &xmax, double &ymin, double &ymax)=0
- virtual void **getShape** (QPoint shape[])=0

Protected Attributes

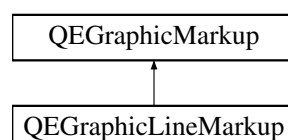
- QColor **brushDisabled**
- QColor **brushEnabled**
- QColor **brushSelected**

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicMarkup.h
- qeframework/common/QEGraphicMarkup.cpp

9.168 QEGraphicLineMarkup Class Reference

Inheritance diagram for QEGraphicLineMarkup:



Public Member Functions

- **QEGraphicLineMarkup** ([QEGraphic](#) *owner)
- **QPointF** **getSlope** () const
- void **mousePress** (const QPointF &realMousePosition, const Qt::MouseButton button)
- void **mouseRelease** (const QPointF &realMousePosition, const Qt::MouseButton button)
- void **mouseMove** (const QPointF &realMousePosition)

Protected Member Functions

- void **plotMarkup** ()

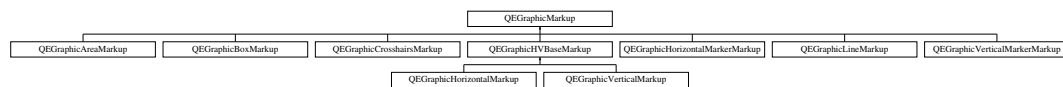
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicMarkup.h
- qeframework/common/QEGraphicMarkup.cpp

9.169 QEGraphicMarkup Class Reference

Inheritance diagram for QEGraphicMarkup:



Public Member Functions

- **QEGraphicMarkup** (const QEGraphicNames::Markups markup, [QEGraphic](#) *owner)
- QEGraphicNames::Markups **getMarkup** () const
- void **setCurrentPosition** (const QPointF ¤tPosition)
- QPointF **getCurrentPosition** () const
- void **setData** (const QVariant &data)
- QVariant **getData** () const
- void **saveConfiguration** ([PMElement](#) &parentElement)
- void **restoreConfiguration** ([PMElement](#) &parentElement)
- virtual void **setInUse** (const bool inUse)
- virtual bool **isInUse** () const
- virtual void **setVisible** (const bool visible)
- virtual bool **isVisible** () const
- virtual void **setEnabled** (const bool enabled)
- virtual bool **isEnabled** () const
- virtual void **setSelected** (const bool selected)
- virtual bool **isSelected** () const
- virtual void **mousePress** (const QPointF &realMousePosition, const Qt::MouseButton button)
- virtual void **mouseRelease** (const QPointF &realMousePosition, const Qt::MouseButton button)
- virtual void **mouseMove** (const QPointF &realMousePosition)
- virtual QCursor **getCursor** () const
- virtual bool **isOver** (const QPointF &point, int &distance) const
- virtual void **relocate** ()
- void **plot** ()

Protected Member Functions

- [QEGraphic](#) * **getOwner** () const
- bool **isOverHere** (const QPointF &here, const QPointF &point, int &distance) const
- void **plotCurve** (const QEGraphicNames::DoubleVector &xData, const QEGraphicNames::DoubleVector &yData)
- QFontMetrics **getFontMetrics** ()
- void **emitCurrentPosition** ()
- virtual void **plotMarkup** ()=0

Protected Attributes

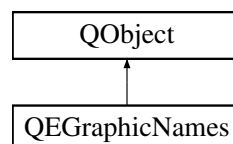
- QPointF **position**
- QVariant **data**
- QPen **pen**
- QBrush **brush**
- QwtPlotCurve::CurveStyle **curveStyle**
- QCursor **cursor**
- Qt::MouseButton **activationButton**
- bool **inUse**
- bool **visible**
- bool **enabled**
- bool **selected**
- const QEGraphicNames::Markups **markup**
- [QEGraphic](#) * **owner**

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicMarkup.h
- qeframework/common/QEGraphicMarkup.cpp

9.170 QEGraphicNames Class Reference

Inheritance diagram for QEGraphicNames:



Public Types

- enum **TextPositions** { **RealWorldPosition** , **PixelPosition** }
- enum **AxisMajorIntervalModes** { **SelectByValue** , **SelectBySize** , **UserInterval** }
- enum **Markups** {
None = 0x000000 , **Area** = 0x000001 , **Line** = 0x000002 , **CrossHair** = 0x000004 ,
HorizontalLine_1 = 0x000010 , **HorizontalLine_2** = 0x000020 , **HorizontalLine_3** = 0x000040 ,
HorizontalLine_4 = 0x000080 ,
VerticalLine_1 = 0x000100 , **VerticalLine_2** = 0x000200 , **VerticalLine_3** = 0x000400 , **VerticalLine_4** =
0x000800 ,
Box = 0x001000 , **HorizontalMarker_1** = 0x002000 , **HorizontalMarker_2** = 0x004000 , **VerticalMarker_1**
= 0x008000 ,
VerticalMarker_2 = 0x010000 }
- typedef QVector< double > **DoubleVector**
- typedef QMap< Markups, [QEGraphicMarkup](#) * > **QEGraphicMarkupsSets**
- typedef QList< Markups > **MarkupLists**

Static Public Member Functions

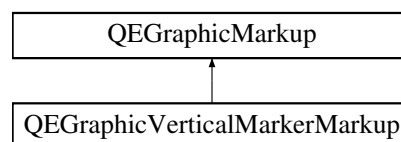
- static QString **markupToString** (const Markups value)
- static Markups **stringToMarkup** (const QString &image, bool *ok=0)
- static QEGraphicMarkupsSets * **createGraphicMarkupsSet** ([QEGraphic](#) *owner)
- static void **cleanGraphicMarkupsSet** (QEGraphicMarkupsSets &markupsSet)
- static void **saveConfiguration** (QEGraphicMarkupsSets &markupsSet, [PMElement](#) &parentElement)
- static void **restoreConfiguration** (QEGraphicMarkupsSets &markupsSet, [PMElement](#) &parentElement)

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicNames.h
- qeframework/common/QEGraphicNames.cpp

9.171 QEGraphicVerticalMarkerMarkup Class Reference

Inheritance diagram for QEGraphicVerticalMarkerMarkup:



Public Member Functions

- **QEGraphicVerticalMarkerMarkup** (QEGraphicNames::Markups markup, [QEGraphic](#) *owner)
- void **setVisible** (const bool visible)

Protected Member Functions

- void **plotMarkup** ()

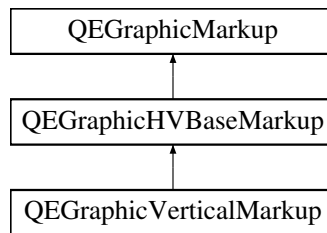
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicMarkup.h
- qeframework/common/QEGraphicMarkup.cpp

9.172 QEGraphicVerticalMarkup Class Reference

Inheritance diagram for QEGraphicVerticalMarkup:



Public Member Functions

- **QEGraphicVerticalMarkup** (const QEGraphicNames::Markups markup, [QEGraphic](#) *owner)
- bool **isOver** (const QPointF &point, int &distance) const
- void **relocate** ()

Protected Member Functions

- void **getLine** (double &xmin, double &xmax, double &ymin, double &ymax)
- void **getShape** (QPoint shape[])

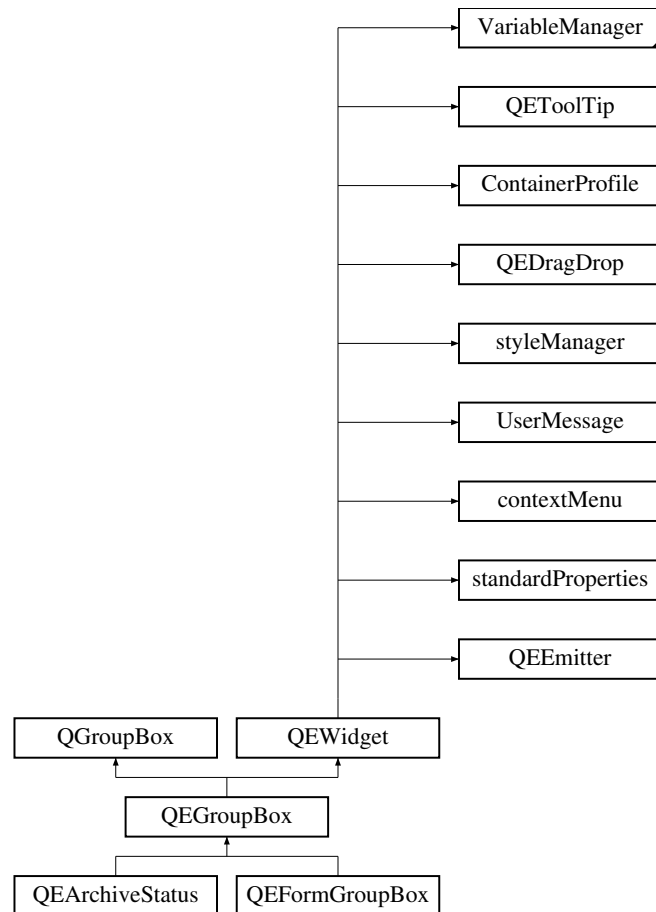
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/QEGraphicMarkup.h
- qeframework/common/QEGraphicMarkup.cpp

9.173 QEGroupBox Class Reference

Inheritance diagram for QEGroupBox:



Public Slots

- void **setManagedVisible** (bool v)

Public Member Functions

- **QGroupBox** (QWidget *parent=0)
- **QGroupBox** (const QString &title, QWidget *parent=0)
- QSize **sizeHint** () const

Protected Member Functions

- virtual void **setSubstitutionsProperty** (const QString macroSubstitutions)
- virtual QString **getSubstitutionsProperty** () const
- virtual void **setSubstitutedTitleProperty** (const QString [ownSubstitutedTitle](#))
- virtual QString **getSubstitutedTitleProperty** () const

Properties

- QString [ownSubstitutedTitle](#)
- QString **title**
- QString [textSubstitutions](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)

Additional Inherited Members

9.173.1 Constructor & Destructor Documentation

9.173.1.1 QEGroupBox()

```
QEGroupBox::QEGroupBox (
    QWidget * parent = 0 ) [explicit]
```

displayAlarmStateOption not applicale to V2 options. Class should ensure this option is initialised to DISPLAY_↔
ALARM_STATE_NEVER.

9.173.2 Member Function Documentation

9.173.2.1 setManagedVisible

```
void QEGroupBox::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call
this slot, by will only be made visible by a calll to this slot if the user level allows.

9.173.3 Property Documentation

9.173.3.1 defaultStyle

```
QString QEGroupBox::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.173.3.2 messageSourceId

```
unsigned int QEGroupBox::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.173.3.3 ownSubstitutedTitle

```
QString QEGroupBox::ownSubstitutedTitle [read], [write]
```

Group box title text to be substituted. This text will be copied to the group box title text after applying any macro substitutions from the variableSubstitutions property. The ownSubstitutedTitle property is deprecated and included for backward compatabilty only and not presented on designer. The title property hides the parent title property.

9.173.3.4 styleSheet

```
QString QEGroupBox::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly setable per se. This also stops transient styles being saved to the ui file.

9.173.3.5 textSubstitutions

```
QString QEGroupBox::textSubstitutions [read], [write]
```

Text substitutions. These substitutions are applied to the 'substitutedTitle' property prior to copying it to the label text.

9.173.3.6 userLevelEnabled

```
QE::UserLevels QEGroupBox::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through setUserLevel() Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.173.3.7 userLevelEngineerStyle

```
QString QEGroupBox::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.173.3.8 userLevelScientistStyle

```
QString QEGroupBox::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.173.3.9 userLevelUserStyle

```
QString QEGroupBox::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.173.3.10 userLevelVisibility

```
QE::UserLevels QEGroupBox::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.173.3.11 visible

```
bool QEGroupBox::visible [read], [write]
```

`variableAsToolTip` not applicale to V2 options. Class should ensure this option is initialised to false. `allowDrop` not applicale to V2 options. Class should ensure this option is initialised to false. Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

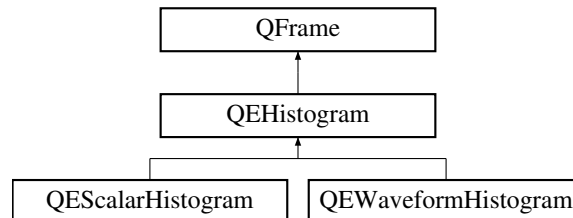
The documentation for this class was generated from the following files:

- `qeframework/widgets/QEGroupBox/QEGroupBox.h`
- `qeframework/widgets/QEGroupBox/QEGroupBox.cpp`

9.174 QEHistogram Class Reference

```
#include <QEHistogram.h>
```

Inheritance diagram for QEHistogram:



Public Types

- typedef QVector< double > **DataArray**

Public Slots

- void **setColour** (const int index, const QColor &value)
- void **setValue** (const int index, const double value)
- void **setValues** (const DataArray &values)

Signals

- void **mouseIndexChanged** (const int index)
- void **mouseIndexPressed** (const int index, const Qt::MouseButton button)

Public Member Functions

- **QEHistogram** (QWidget *parent=0)
- virtual QSize **sizeHint** () const
- int **count** () const
- int **indexOfPosition** (const int x, const int y) const
- int **indexOfPosition** (const QPoint &p) const
- QRect **positionOfIndex** (const int index) const
- void **clearValue** (const int index)
- void **clearColour** (const int index)
- void **clear** ()
- double **value** (const int index) const
- DataArray **values** () const

Protected Member Functions

- virtual void **onMouseIndexChanged** (const int index)
- virtual void **onMouseIndexPressed** (const int index, const Qt::MouseButton button)
- bool **eventFilter** (QObject *obj, QEvent *event)

Properties

- bool **autoBarGapWidths**
- int **barWidth**
- int **gap**
- int **margin**
- bool **autoScale**
- double **minimum**
- double **maximum**
- double **baseLine**
- bool **drawAxes**
- bool **showScale**
- int **precision**
- bool **showGrid**

This is the value grid selection.

- bool **logScale**
- QColor **backgroundColour**

Where possible I spell colour properly.

- QColor **secondBgColour**
- int **secondBgSize**
- bool **showSecondBg**
- QColor **barColour**
- bool **drawBorder**
- Qt::Orientation **orientation**
- int **testSize**

9.174.1 Detailed Description

The [QEHistogram](#) class is a non-EPICS aware histogram widget. The value of, i.e. the length of each bar, and colour may be set independently.

9.174.2 Property Documentation

9.174.2.1 orientation

```
Qt::Orientation QEHistogram::orientation [read], [write]
```

orientation horizontal (default) or vertical. Horizontal means each element displayed horizontally from left to right with the bar representing the value increasing vertically from bottom to top.

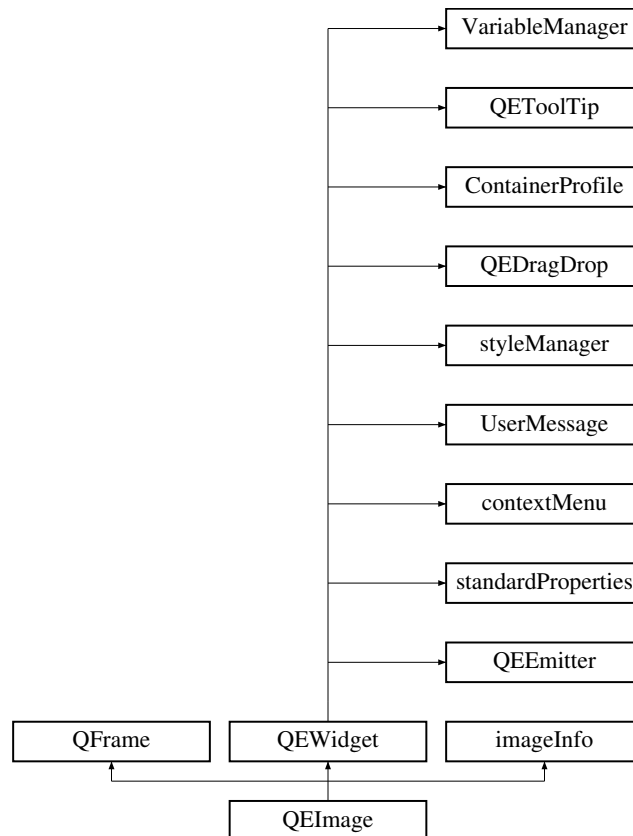
The documentation for this class was generated from the following files:

- qeframework/widgets/QEHistogram/QEHistogram.h
- qeframework/widgets/QEHistogram/QEHistogram.cpp

9.175 QEImage Class Reference

```
#include <QEImage.h>
```

Inheritance diagram for QEImage:



Public Types

- enum `selectOptions` { `SO_NONE`, `SO_PANNING`, `SO_VSLICE1`, `SO_VSLICE2`, `SO_VSLICE3`, `SO_VSLICE4`, `SO_VSLICE5`, `SO_HSLICE1`, `SO_HSLICE2`, `SO_HSLICE3`, `SO_HSLICE4`, `SO_HSLICE5`, `SO_AREA1`, `SO_AREA2`, `SO_AREA3`, `SO_AREA4`, `SO_PROFILE`, `SO_TARGET`, `SO_BEAM` }
- enum `imageUses` { `IMAGE_USE_DISPLAY`, `IMAGE_USE_SAVE`, `IMAGE_USE_DISPLAY_AND_SAVE` }
- enum `resizeOptions` { `RESIZE_OPTION_ZOOM`, `RESIZE_OPTION_FIT` }
- enum `ellipseVariableDefinitions` { `BOUNDING_RECTANGLE`, `CENTRE_AND_SIZE` }
- enum `EllipseVariableDefinitions` { `BoundingBoxRectangle` = `BOUNDING_RECTANGLE`, `CenterAndSize` = `CENTRE_AND_SIZE` }
- enum `TargetOptions` { `DottedFullCrosshair` = `VideoWidget::CROSSHAIR1`, `SolidSmallCrosshair` = `VideoWidget::CROSSHAIR2` }
- enum `ResizeOptions` { `Zoom` = `QEImage::RESIZE_OPTION_ZOOM`, `Fit` = `QEImage::RESIZE_OPTION_FIT` }
- typedef `VideoWidget::CustomisePaintHandlers` **CustomisePaintHandlers**

Public Slots

- void **setShowTime** (const bool pValue)
Access function for [showTime](#) property - refer to [showTime](#) property for details.
- void **setUseFalseColour** (const bool pValue)
Access function for [useFalseColour](#) property - refer to [useFalseColour](#) property for details.
- void **setDataImage** (const QByteArray &imageIn, unsigned long dataSize, unsigned long elements, unsigned long width, unsigned long height, [QE::ImageFormatOptions](#) format, unsigned int depth)
- void **setPvalImage** (const [QVariantUpdate](#) &update)
- void **setByteArrayImage** (const [QEByteArrayUpdate](#) &update)
- void **setImage** (const QByteArray &image, unsigned long dataSize, const [QCaAlarmInfo](#) &, const [QCaDateTime](#) &, const unsigned int &)
- void **connectionUpdated** (const [QEConnectionUpdate](#) &update)
- void **setManagedVisible** (bool v)
- void **brightnessContrastAutoImageRequest** ()
- void **imageDisplayPropertiesChanged** ()
- void **setImageFile** (QString name)
- void **setSelectPanMode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectVSliceMode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectHSliceMode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectArea1Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectArea2Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectArea3Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectArea4Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectProfileMode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectTargetMode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectBeamMode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectVSlice1Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectVSlice2Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectVSlice3Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectVSlice4Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectVSlice5Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectHSlice1Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectHSlice2Mode** ()
Framework use only. Slot to allow external setting of selection menu options.
- void **setSelectHSlice3Mode** ()

- Framework use only. Slot to allow external setting of selection menu options.*

 - void [setSelectHSlice4Mode](#) ()
- Framework use only. Slot to allow external setting of selection menu options.*

 - void [setSelectHSlice5Mode](#) ()
- Framework use only. Slot to allow external setting of selection menu options.*

 - void [pauseClicked](#) ()
- Framework use only. Slot to allow external setting of selection menu options.*

 - void [saveClicked](#) ()
- Framework use only. Slot to allow external setting of selection menu options.*

 - void [targetClicked](#) ()
- Framework use only. Slot to allow external setting of selection menu options.*

 - void [imageDisplayPropsDestroyed](#) (QObject *)

Framework use only. Slot to catch deletion of components (such as profile plots) that have been passed to the application for presentation.
- Framework use only. Slot to catch deletion of components (such as profile plots) that have been passed to the application for presentation.*

 - void [vSliceDisplayDestroyed](#) (QObject *)

Framework use only. Slot to catch deletion of components (such as profile plots) that have been passed to the application for presentation.
- Framework use only. Slot to catch deletion of components (such as profile plots) that have been passed to the application for presentation.*

 - void [hSliceDisplayDestroyed](#) (QObject *)

Framework use only. Slot to catch deletion of components (such as profile plots) that have been passed to the application for presentation.
- Framework use only. Slot to catch deletion of components (such as profile plots) that have been passed to the application for presentation.*

 - void [profileDisplayDestroyed](#) (QObject *)

Framework use only. Slot to catch deletion of components (such as profile plots) that have been passed to the application for presentation.
- Framework use only. Slot to catch deletion of components (such as profile plots) that have been passed to the application for presentation.*

 - void [recorderDestroyed](#) (QObject *)

Framework use only. Slot to catch deletion of components (such as profile plots) that have been passed to the application for presentation.
- Show the arbitrary line (profile) markup - refer to [enableProfileSelection](#) property and #displayMarkups property for details.*

 - void [showProfile](#) ()
- Show or hide the arbitrary line (profile) markup. Note that when hiding if its PV changes it will reshow unless Display↔Markups has been set to off - refer to [enableProfileSelection](#) property and #displayMarkups property for details.*

 - void [showProfile](#) (bool show)
- Hide the arbitrary line (profile) markup but note that if its PV changes it will reshow unless DisplayMarkups has been set to off - refer to [enableProfileSelection](#) property and #displayMarkups property for details.*

 - void [hideProfile](#) ()
- Show the area1 markup - refer to [enableArea1Selection](#) property and #displayMarkups property for details.*

 - void [showArea1](#) ()
- show or hide the area1 markup. Note that when hiding if its PV changes it will reshow unless DisplayMarkups has been set to off - refer to [enableArea1Selection](#) property and #displayMarkups property for details*

 - void [showArea1](#) (bool show)
- Hide the area1 markup but note that if its PV changes it will reshow unless DisplayMarkups has been set to off - refer to [enableArea1Selection](#) property and #displayMarkups property for details.*

 - void [hideArea1](#) ()
- Set markup display to on to show all markups that change either due to user or PV activity, even if their set↔Display????Selection is off - refer to #displayMarkups property for details.*

 - void [setDisplayMarkupsOn](#) ()
- Set markup display to on to or off show all markups that change either due to user or PV activity, even if their set↔Display????Selection is off - refer to #displayMarkups property for details.*

 - void [setDisplayMarkupsOn](#) (bool on)
- Set markup display to off to stop PV controlled pvs from showing even if they change, unless their set↔Display????Selection is on - refer to #displayMarkups property for details.*

 - void [setDisplayMarkupsOff](#) ()

Signals

- void [dbValueChanged](#) (const QString &out)
- void [requestResend](#) ()
Internal use only. Used when changing a property value to force a re-display to reflect the new property value.
- void **componentHostRequest** (const [QEActionRequests](#) &request)

Public Member Functions

- [QEImage](#) (QWidget *parent=0)
- [QEImage](#) (const QString &variableName, QWidget *parent=0)
- [~QEImage](#) ()
Destructor.
- [selectOptions](#) **getSelectionOption** ()
- void [setBitDepth](#) (int bitDepthIn)
Access function for [bitDepth](#) property - refer to [bitDepth](#) property for details.
- int [getBitDepth](#) ()
Access function for [bitDepth](#) property - refer to [bitDepth](#) property for details.
- void [setFormatOption](#) (QE::ImageFormatOptions formatOption)
Access function for [formatOption](#) property - refer to [formatOption](#) property for details.
- [QE::ImageFormatOptions](#) [getFormatOption](#) ()
Access function for [formatOption](#) property - refer to [formatOption](#) property for details.
- void [setResizeOption](#) (resizeOptions resizeOptionIn)
Access function for [#resizeOption](#) property - refer to [#resizeOption](#) property for details.
- [resizeOptions](#) [getResizeOption](#) ()
Access function for [#resizeOption](#) property - refer to [#resizeOption](#) property for details.
- void [setZoom](#) (int zoomIn)
Access function for [zoom](#) property - refer to [zoom](#) property for details.
- int [getZoom](#) ()
Access function for [zoom](#) property - refer to [zoom](#) property for details.
- void [setXStretch](#) (double XStretchIn)
Access function for [XStretch](#) property - refer to [XStretch](#) property for details.
- double [getXStretch](#) ()
Access function for [XStretch](#) property - refer to [XStretch](#) property for details.
- void [setYStretch](#) (double YStretchIn)
Access function for [YStretch](#) property - refer to [YStretch](#) property for details.
- double [getYStretch](#) ()
Access function for [YStretch](#) property - refer to [YStretch](#) property for details.
- void [setRotation](#) (QE::RotationOptions rotationIn)
Access function for [rotation](#) property - refer to [rotation](#) property for details.
- [QE::RotationOptions](#) [getRotation](#) ()
Access function for [rotation](#) property - refer to [rotation](#) property for details.
- void [setHorizontalFlip](#) (bool flipHozIn)
Access function for [horizontalFlip](#) property - refer to [horizontalFlip](#) property for details.
- bool [getHorizontalFlip](#) ()
Access function for [horizontalFlip](#) property - refer to [horizontalFlip](#) property for details.
- void [setVerticalFlip](#) (bool flipVertIn)
Access function for [verticalFlip](#) property - refer to [verticalFlip](#) property for details.
- bool [getVerticalFlip](#) ()
Access function for [verticalFlip](#) property - refer to [verticalFlip](#) property for details.
- void [setInitialHozScrollPos](#) (int initialHosScrollPosIn)

- Access function for [initialHosScrollPos](#) property - refer to [initialHosScrollPos](#) property for details.

 - int [getInitialHozScrollPos](#) ()

Access function for [initialHosScrollPos](#) property - refer to [initialHosScrollPos](#) property for details.
- void [setInitialVertScrollPos](#) (int initialVertScrollPosIn)

Access function for [initialVertScrollPos](#) property - refer to [initialVertScrollPos](#) property for details.
- int [getInitialVertScrollPos](#) ()

Access function for [initialVertScrollPos](#) property - refer to [initialVertScrollPos](#) property for details.
- void [setDisplayButtonBar](#) (bool displayButtonBarIn)

Access function for [displayButtonBar](#) property - refer to [displayButtonBar](#) property for details.
- bool [getDisplayButtonBar](#) ()

Access function for [displayButtonBar](#) property - refer to [displayButtonBar](#) property for details.
- bool [getShowTime](#) () const

Access function for [showTime](#) property - refer to [showTime](#) property for details.
- bool [getUseFalseColour](#) () const

Access function for [useFalseColour](#) property - refer to [useFalseColour](#) property for details.
- void [setEnableProfileAxes](#) (const bool value)

Access function for [enableProfileAxes](#) property - refer to [enableProfileAxes](#) property for details.
- bool [getEnableProfileAxes](#) () const

Access function for [enableProfileAxes](#) property - refer to [enableProfileAxes](#) property for details.
- void [setVertSlice1MarkupColor](#) (QColor pValue)

Access function for [#vertSliceColor](#) property - refer to [#vertSliceColor](#) property for details.
- QColor [getVertSlice1MarkupColor](#) ()

Access function for [#vertSliceColor](#) property - refer to [#vertSliceColor](#) property for details.
- void [setVertSlice2MarkupColor](#) (QColor pValue)

Access function for [vertSlice2Color](#) property - refer to [vertSlice2Color](#) property for details.
- QColor [getVertSlice2MarkupColor](#) ()

Access function for [vertSlice2Color](#) property - refer to [vertSlice2Color](#) property for details.
- void [setVertSlice3MarkupColor](#) (QColor pValue)

Access function for [vertSlice3Color](#) property - refer to [vertSlice3Color](#) property for details.
- QColor [getVertSlice3MarkupColor](#) ()

Access function for [vertSlice3Color](#) property - refer to [vertSlice3Color](#) property for details.
- void [setVertSlice4MarkupColor](#) (QColor pValue)

Access function for [vertSlice4Color](#) property - refer to [vertSlice4Color](#) property for details.
- QColor [getVertSlice4MarkupColor](#) ()

Access function for [vertSlice4Color](#) property - refer to [vertSlice4Color](#) property for details.
- void [setVertSlice5MarkupColor](#) (QColor pValue)

Access function for [vertSlice5Color](#) property - refer to [vertSlice5Color](#) property for details.
- QColor [getVertSlice5MarkupColor](#) ()

Access function for [vertSlice5Color](#) property - refer to [vertSlice5Color](#) property for details.
- void [setHozSlice1MarkupColor](#) (QColor pValue)

Access function for [#hozSliceColor](#) property - refer to [#hozSliceColor](#) property for details.
- QColor [getHozSlice1MarkupColor](#) ()

Access function for [#hozSliceColor](#) property - refer to [#hozSliceColor](#) property for details.
- void [setHozSlice2MarkupColor](#) (QColor pValue)

Access function for [hozSlice2Color](#) property - refer to [hozSlice2Color](#) property for details.
- QColor [getHozSlice2MarkupColor](#) ()

Access function for [hozSlice2Color](#) property - refer to [hozSlice2Color](#) property for details.
- void [setHozSlice3MarkupColor](#) (QColor pValue)

Access function for [hozSlice3Color](#) property - refer to [hozSlice3Color](#) property for details.
- QColor [getHozSlice3MarkupColor](#) ()

Access function for [hozSlice3Color](#) property - refer to [hozSlice3Color](#) property for details.

- void [setHozSlice4MarkupColor](#) (QColor pValue)
Access function for [hozSlice4Color](#) property - refer to [hozSlice4Color](#) property for details.
- QColor [getHozSlice4MarkupColor](#) ()
Access function for [hozSlice4Color](#) property - refer to [hozSlice4Color](#) property for details.
- void [setHozSlice5MarkupColor](#) (QColor pValue)
Access function for [hozSlice5Color](#) property - refer to [hozSlice5Color](#) property for details.
- QColor [getHozSlice5MarkupColor](#) ()
Access function for [hozSlice5Color](#) property - refer to [hozSlice5Color](#) property for details.
- void [setProfileMarkupColor](#) (QColor pValue)
Access function for [profileColor](#) property - refer to [profileColor](#) property for details.
- QColor [getProfileMarkupColor](#) ()
Access function for [profileColor](#) property - refer to [profileColor](#) property for details.
- void [setAreaMarkupColor](#) (QColor pValue)
Access function for [areaColor](#) property - refer to [areaColor](#) property for details.
- QColor [getAreaMarkupColor](#) ()
Access function for [areaColor](#) property - refer to [areaColor](#) property for details.
- void [setTargetMarkupColor](#) (QColor pValue)
Access function for [targetColor](#) property - refer to [targetColor](#) property for details.
- QColor [getTargetMarkupColor](#) ()
Access function for [targetColor](#) property - refer to [targetColor](#) property for details.
- void [setBeamMarkupColor](#) (QColor pValue)
Access function for [beamColor](#) property - refer to [beamColor](#) property for details.
- QColor [getBeamMarkupColor](#) ()
Access function for [beamColor](#) property - refer to [beamColor](#) property for details.
- void [setTimeMarkupColor](#) (QColor pValue)
Access function for [timeColor](#) property - refer to [timeColor](#) property for details.
- QColor [getTimeMarkupColor](#) ()
Access function for [timeColor](#) property - refer to [timeColor](#) property for details.
- void [setEllipseMarkupColor](#) (QColor markupColor)
Access function for [ellipseColor](#) property - refer to [ellipseColor](#) property for details.
- QColor [getEllipseMarkupColor](#) ()
Access function for [ellipseColor](#) property - refer to [ellipseColor](#) property for details.
- void [setDisplayCursorPixelInfo](#) (bool displayCursorPixelInfo)
Access function for [displayCursorPixelInfo](#) property - refer to [displayCursorPixelInfo](#) property for details.
- bool [getDisplayCursorPixelInfo](#) ()
Access function for [displayCursorPixelInfo](#) property - refer to [displayCursorPixelInfo](#) property for details.
- void [setContrastReversal](#) (bool contrastReversalIn)
Access function for [contrastReversal](#) property - refer to [contrastReversal](#) property for details.
- bool [getContrastReversal](#) ()
Access function for [contrastReversal](#) property - refer to [contrastReversal](#) property for details.
- void [setLog](#) (bool log)
Access function for [logBrightness](#) property - refer to [logBrightness](#) property for details.
- bool [getLog](#) ()
Access function for [logBrightness](#) property - refer to [logBrightness](#) property for details.
- void [setEnableVertSlice1Selection](#) (bool enableVSliceSelection)
Access function for [enableVertSlice1Selection](#) property - refer to [enableVertSlice1Selection](#) property for details.
- bool [getEnableVertSlice1Selection](#) ()
Access function for [enableVertSlice1Selection](#) property - refer to [enableVertSlice1Selection](#) property for details.
- void [setEnableVertSlice2Selection](#) (bool enableVSliceSelection)
Access function for [enableVertSlice2Selection](#) property - refer to [enableVertSlice2Selection](#) property for details.
- bool [getEnableVertSlice2Selection](#) ()

- [illegible]

- `bool getEnableProfileSelection ()`
Access function for [enableProfileSelection](#) property - refer to [enableProfileSelection](#) property for details.
- `void setEnableTargetSelection (bool enableTargetSelectionIn)`
Access function for [enableTargetSelection](#) property - refer to [enableTargetSelection](#) property for details.
- `bool getEnableTargetSelection ()`
Access function for [enableTargetSelection](#) property - refer to [enableTargetSelection](#) property for details.
- `void setEnableBeamSelection (bool enableBeamSelectionIn)`
Access function for [enableBeamSelection](#) property - refer to [enableBeamSelection](#) property for details.
- `bool getEnableBeamSelection ()`
Access function for [enableBeamSelection](#) property - refer to [enableBeamSelection](#) property for details.
- `void setEnableImageDisplayProperties (bool enableImageDisplayPropertiesIn)`
Access function for [enableImageDisplayProperties](#) property - refer to [enableImageDisplayProperties](#) property for details.
- `bool getEnableImageDisplayProperties ()`
Access function for [enableImageDisplayProperties](#) property - refer to [enableImageDisplayProperties](#) property for details.
- `void setEnableRecording (bool enableRecordingIn)`
Access function for [enableRecording](#) property - refer to [enableRecording](#) property for details.
- `bool getEnableRecording ()`
Access function for [enableRecording](#) property - refer to [enableRecording](#) property for details.
- `void setAutoBrightnessContrast (bool autoBrightnessContrastIn)`
Access function for [autoBrightnessContrast](#) property - refer to [autoBrightnessContrast](#) property for details.
- `bool getAutoBrightnessContrast ()`
Access function for [autoBrightnessContrast](#) property - refer to [autoBrightnessContrast](#) property for details.
- `void setExternalControls (bool externalControlsIn)`
Access function for [externalControls](#) property - refer to [externalControls](#) property for details.
- `bool getExternalControls ()`
Access function for [externalControls](#) property - refer to [externalControls](#) property for details.
- `void setFullContextMenu (bool fullContextMenuIn)`
Access function for [#fullContextMenu](#) property - refer to [#fullContextMenu](#) property for details.
- `bool getFullContextMenu ()`
Access function for [#fullContextMenu](#) property - refer to [#fullContextMenu](#) property for details.
- `void setEnableProfilePresentation (bool enableProfilePresentationIn)`
Access function for [#enableProfilePresentation](#) property - refer to [#enableProfilePresentation](#) property for details.
- `bool getEnableProfilePresentation ()`
Access function for [#enableProfilePresentation](#) property - refer to [#enableProfilePresentation](#) property for details.
- `void setEnableHozSlicePresentation (bool enableHozSlicePresentationIn)`
Access function for [#enableHozSlicePresentation](#) property - refer to [#enableHozSlicePresentation](#) property for details.
- `bool getEnableHozSlicePresentation ()`
Access function for [#enableHozSlicePresentation](#) property - refer to [#enableHozSlicePresentation](#) property for details.
- `void setEnableVertSlicePresentation (bool enableVertSlicePresentationIn)`
Access function for [#enableVertSlicePresentation](#) property - refer to [#enableVertSlicePresentation](#) property for details.
- `bool getEnableVertSlicePresentation ()`
Access function for [#enableVertSlicePresentation](#) property - refer to [#enableVertSlicePresentation](#) property for details.
- `void setDisplayVertSlice1Selection (bool displayVSLiceSelection)`
Access function for [displayVertSlice1Selection](#) property - refer to [displayVertSlice1Selection](#) property for details.
- `bool getDisplayVertSlice1Selection ()`
Access function for [displayVertSlice1Selection](#) property - refer to [displayVertSlice1Selection](#) property for details.
- `void setDisplayVertSlice2Selection (bool displayVSLiceSelection)`
Access function for [displayVertSlice2Selection](#) property - refer to [displayVertSlice2Selection](#) property for details.
- `bool getDisplayVertSlice2Selection ()`

- [illegible]

- `bool getDisplayProfileSelection ()`
Access function for `displayProfileSelection` property - refer to `displayProfileSelection` property for details.
- `void setDisplayTargetSelection (bool displayTargetSelection)`
Access function for `displayTargetSelection` property - refer to `displayTargetSelection` property for details.
- `bool getDisplayTargetSelection ()`
Access function for `displayTargetSelection` property - refer to `displayTargetSelection` property for details.
- `void setDisplayBeamSelection (bool displayBeamSelection)`
Access function for `displayBeamSelection` property - refer to `displayBeamSelection` property for details.
- `bool getDisplayBeamSelection ()`
Access function for `displayBeamSelection` property - refer to `displayBeamSelection` property for details.
- `void setDisplayEllipse (bool displayEllipse)`
Access function for `displayEllipse` property - refer to `displayEllipse` property for details.
- `bool getDisplayEllipse ()`
Access function for `displayEllipse` property - refer to `displayEllipse` property for details.
- `ellipseVariableDefinitions getEllipseVariableDefinition ()`
Access function for `ellipseVariableDefinition` property - refer to `ellipseVariableDefinition` property for details.
- `void setEllipseVariableDefinition (ellipseVariableDefinitions def)`
Access function for `ellipseVariableDefinition` property - refer to `ellipseVariableDefinition` property for details.
- `void setDisplayMarkups (bool displayMarkupsIn)`
Access function for `#displayMarkups` property - refer to `#displayMarkups` property for details.
- `bool getDisplayMarkups ()`
Access function for `#displayMarkups` property - refer to `#displayMarkups` property for details.
- `void setName (QString nameIn)`
Access function for `name` property - refer to `#name` property for details.
- `QString getName ()`
Access function for `name` property - refer to `#name` property for details.
- `void setProgram1 (QString program)`
Access function for `program1` property - refer to `program1` property for details.
- `QString getProgram1 ()`
Access function for `program1` property - refer to `program1` property for details.
- `void setProgram2 (QString program)`
Access function for `program2` property - refer to `program2` property for details.
- `QString getProgram2 ()`
Access function for `program2` property - refer to `program2` property for details.
- `void setArguments1 (QStringList arguments)`
Access function for `arguments1` property - refer to `arguments1` property for details.
- `QStringList getArguments1 ()`
Access function for `arguments1` property - refer to `arguments1` property for details.
- `void setArguments2 (QStringList arguments)`
Access function for `arguments2` property - refer to `arguments2` property for details.
- `QStringList getArguments2 ()`
Access function for `arguments2` property - refer to `arguments2` property for details.
- `void setProgramStartupOption1 (QE::ProgramStartupOptions programStartupOption)`
Access function for `programStartupOption1` property - refer to `programStartupOption1` property for details.
- `QE::ProgramStartupOptions getProgramStartupOption1 ()`
Access function for `programStartupOption1` property - refer to `programStartupOption1` property for details.
- `void setProgramStartupOption2 (QE::ProgramStartupOptions programStartupOption)`
Access function for `programStartupOption2` property - refer to `programStartupOption2` property for details.
- `QE::ProgramStartupOptions getProgramStartupOption2 ()`
Access function for `programStartupOption2` property - refer to `programStartupOption2` property for details.
- `QString getHozSlice1Legend ()`

- Access function for [hozSlice1Legend](#) property - refer to [hozSlice1Legend](#) property for details.

 - void [setHozSlice1Legend](#) (QString legend)

Access function for [hozSlice1Legend](#) property - refer to [hozSlice1Legend](#) property for details.

- QString [getHozSlice2Legend](#) ()

Access function for [hozSlice2Legend](#) property - refer to [hozSlice2Legend](#) property for details.

 - void [setHozSlice2Legend](#) (QString legend)

Access function for [hozSlice2Legend](#) property - refer to [hozSlice2Legend](#) property for details.

- QString [getHozSlice3Legend](#) ()

Access function for [hozSlice3Legend](#) property - refer to [hozSlice3Legend](#) property for details.

 - void [setHozSlice3Legend](#) (QString legend)

Access function for [hozSlice3Legend](#) property - refer to [hozSlice3Legend](#) property for details.

- QString [getHozSlice4Legend](#) ()

Access function for [hozSlice4Legend](#) property - refer to [hozSlice4Legend](#) property for details.

 - void [setHozSlice4Legend](#) (QString legend)

Access function for [hozSlice4Legend](#) property - refer to [hozSlice4Legend](#) property for details.

- QString [getHozSlice5Legend](#) ()

Access function for [hozSlice5Legend](#) property - refer to [hozSlice5Legend](#) property for details.

 - void [setHozSlice5Legend](#) (QString legend)

Access function for [hozSlice5Legend](#) property - refer to [hozSlice5Legend](#) property for details.

- QString [getVertSlice1Legend](#) ()

Access function for [vertSlice1Legend](#) property - refer to [vertSlice1Legend](#) property for details.

 - void [setVertSlice1Legend](#) (QString legend)

Access function for [vertSlice1Legend](#) property - refer to [vertSlice1Legend](#) property for details.

- QString [getVertSlice2Legend](#) ()

Access function for [vertSlice2Legend](#) property - refer to [vertSlice2Legend](#) property for details.

 - void [setVertSlice2Legend](#) (QString legend)

Access function for [vertSlice2Legend](#) property - refer to [vertSlice2Legend](#) property for details.

- QString [getVertSlice3Legend](#) ()

Access function for [vertSlice3Legend](#) property - refer to [vertSlice3Legend](#) property for details.

 - void [setVertSlice3Legend](#) (QString legend)

Access function for [vertSlice3Legend](#) property - refer to [vertSlice3Legend](#) property for details.

- QString [getVertSlice4Legend](#) ()

Access function for [vertSlice4Legend](#) property - refer to [vertSlice4Legend](#) property for details.

 - void [setVertSlice4Legend](#) (QString legend)

Access function for [vertSlice4Legend](#) property - refer to [vertSlice4Legend](#) property for details.

- QString [getVertSlice5Legend](#) ()

Access function for [vertSlice5Legend](#) property - refer to [vertSlice5Legend](#) property for details.

 - void [setVertSlice5Legend](#) (QString legend)

Access function for [vertSlice5Legend](#) property - refer to [vertSlice5Legend](#) property for details.

- QString [getprofileLegend](#) ()

Access function for [profileLegend](#) property - refer to [profileLegend](#) property for details.

 - void [setProfileLegend](#) (QString legend)

Access function for [profileLegend](#) property - refer to [profileLegend](#) property for details.

- QString [getAreaSelection1Legend](#) ()

Access function for [areaSelection1Legend](#) property - refer to [areaSelection1Legend](#) property for details.

 - void [setAreaSelection1Legend](#) (QString legend)

Access function for [areaSelection1Legend](#) property - refer to [areaSelection1Legend](#) property for details.

- QString [getAreaSelection2Legend](#) ()

Access function for [areaSelection2Legend](#) property - refer to [areaSelection2Legend](#) property for details.

 - void [setAreaSelection2Legend](#) (QString legend)

Access function for [areaSelection2Legend](#) property - refer to [areaSelection2Legend](#) property for details.

- `QString getAreaSelection3Legend ()`
Access function for [areaSelection3Legend](#) property - refer to [areaSelection3Legend](#) property for details.
- `void setAreaSelection3Legend (QString legend)`
Access function for [areaSelection3Legend](#) property - refer to [areaSelection3Legend](#) property for details.
- `QString getAreaSelection4Legend ()`
Access function for [areaSelection4Legend](#) property - refer to [areaSelection4Legend](#) property for details.
- `void setAreaSelection4Legend (QString legend)`
Access function for [areaSelection4Legend](#) property - refer to [areaSelection4Legend](#) property for details.
- `QString getTargetLegend ()`
Access function for [targetLegend](#) property - refer to [targetLegend](#) property for details.
- `void setTargetLegend (QString legend)`
Access function for [targetLegend](#) property - refer to [targetLegend](#) property for details.
- `QString getBeamLegend ()`
Access function for [beamLegend](#) property - refer to [beamLegend](#) property for details.
- `void setBeamLegend (QString legend)`
Access function for [beamLegend](#) property - refer to [beamLegend](#) property for details.
- `QString getEllipseLegend ()`
Access function for [ellipseLegend](#) property - refer to [ellipseLegend](#) property for details.
- `void setEllipseLegend (QString legend)`
Access function for [ellipseLegend](#) property - refer to [ellipseLegend](#) property for details.
- `bool getFullScreen ()`
Access function for [#fullScreen](#) property - refer to [#fullScreen](#) property for details.
- `void setFullScreen (bool fullScreenIn)`
Access function for [#fullScreen](#) property - refer to [#fullScreen](#) property for details.
- `void setSubstitutedUrl (QString urlIn)`
Access function for [URL](#) property - refer to [URL](#) property for deta.
- `QString getSubstitutedUrl ()`
Access function for [URL](#) property - refer to [URL](#) property for deta.
- `void setCustomisePaintHandler (CustomisePaintHandlers paintExtraHandler, QObject *context=0)`
- `CustomisePaintHandlers getCustomisePaintHandler () const`
- `void resetBrightnessContrast ()`
- `void setVariableNameSubstitutionsProperty (QString variableNameSubstitutions)`
Property access function for [variableSubstitutions](#) property. This has special behaviour to work well within designer.
- `QString getVariableNameSubstitutionsProperty ()`
Property access function for [variableSubstitutions](#) property. This has special behaviour to work well within designer.
- `void setBitDepthProperty (unsigned int bitDepth)`
Access function for [bitDepth](#) property - refer to [bitDepth](#) property for details.
- `unsigned int getBitDepthProperty ()`
Access function for [bitDepth](#) property - refer to [bitDepth](#) property for details.
- `EllipseVariableDefinitions getEllipseVariableDefinitionProperty ()`
Access function for [#EllipseVariableDefinition](#) property - refer to [#EllipseVariableDefinition](#) property for details.
- `void setEllipseVariableDefinitionProperty (EllipseVariableDefinitions variableUsage)`
Access function for [EllipseVariableDefinitions](#) property - refer to [EllipseVariableDefinitions](#) property for details.
- `TargetOptions getTargetOptionProperty ()`
Access function for [targetOption](#) property - refer to [targetOption](#) property for details.
- `void setTargetOptionProperty (TargetOptions option)`
Access function for [targetOption](#) property - refer to [targetOption](#) property for details.
- `TargetOptions getBeamOptionProperty ()`
Access function for [beamOption](#) property - refer to [beamOption](#) property for details.
- `void setBeamOptionProperty (TargetOptions option)`
Access function for [beamOption](#) property - refer to [beamOption](#) property for details.

- void **setResizeOptionProperty** ([ResizeOptions](#) resizeMode)
Access function for #resizeOption property - refer to #resizeOption property for details.
- [ResizeOptions](#) **getResizeOptionProperty** ()
Access function for #resizeOption property - refer to #resizeOption property for details.

Protected Types

- enum **variableIndexes** {
IMAGE_VARIABLE , **FORMAT_VARIABLE** , **BIT_DEPTH_VARIABLE** , **DATA_TYPE_VARIABLE** ,
WIDTH_VARIABLE , **HEIGHT_VARIABLE** , **NUM_DIMENSIONS_VARIABLE** , **DIMENSION_0_VARIABLE** ,
DIMENSION_1_VARIABLE , **DIMENSION_2_VARIABLE** , **ROI1_X_VARIABLE** , **ROI1_Y_VARIABLE** ,
ROI1_W_VARIABLE , **ROI1_H_VARIABLE** , **ROI2_X_VARIABLE** , **ROI2_Y_VARIABLE** ,
ROI2_W_VARIABLE , **ROI2_H_VARIABLE** , **ROI3_X_VARIABLE** , **ROI3_Y_VARIABLE** ,
ROI3_W_VARIABLE , **ROI3_H_VARIABLE** , **ROI4_X_VARIABLE** , **ROI4_Y_VARIABLE** ,
ROI4_W_VARIABLE , **ROI4_H_VARIABLE** , **TARGET_X_VARIABLE** , **TARGET_Y_VARIABLE** ,
BEAM_X_VARIABLE , **BEAM_Y_VARIABLE** , **TARGET_TRIGGER_VARIABLE** , **CLIPPING_ONOFF_VARIABLE** ,
CLIPPING_LOW_VARIABLE , **CLIPPING_HIGH_VARIABLE** , **PROFILE_H1_VARIABLE** , **PROFILE_H1_THICKNESS_VARIABLE** ,
PROFILE_H2_VARIABLE , **PROFILE_H2_THICKNESS_VARIABLE** , **PROFILE_H3_VARIABLE** ,
PROFILE_H3_THICKNESS_VARIABLE ,
PROFILE_H4_VARIABLE , **PROFILE_H4_THICKNESS_VARIABLE** , **PROFILE_H5_VARIABLE** ,
PROFILE_H5_THICKNESS_VARIABLE ,
PROFILE_V1_VARIABLE , **PROFILE_V1_THICKNESS_VARIABLE** , **PROFILE_V2_VARIABLE** ,
PROFILE_V2_THICKNESS_VARIABLE ,
PROFILE_V3_VARIABLE , **PROFILE_V3_THICKNESS_VARIABLE** , **PROFILE_V4_VARIABLE** ,
PROFILE_V4_THICKNESS_VARIABLE ,
PROFILE_V5_VARIABLE , **PROFILE_V5_THICKNESS_VARIABLE** , **LINE_PROFILE_X1_VARIABLE** ,
LINE_PROFILE_Y1_VARIABLE ,
LINE_PROFILE_X2_VARIABLE , **LINE_PROFILE_Y2_VARIABLE** , **LINE_PROFILE_THICKNESS_VARIABLE** , **PROFILE_H_ARRAY** ,
PROFILE_V_ARRAY , **PROFILE_LINE_ARRAY** , **ELLIPSE_X_VARIABLE** , **ELLIPSE_Y_VARIABLE** ,
ELLIPSE_W_VARIABLE , **ELLIPSE_H_VARIABLE** , **ELLIPSE_R_VARIABLE** , **QEIMAGE_NUM_VARIABLES** }

Protected Member Functions

- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **redisplayAllMarkups** ()
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **setDrop** (QVariant drop)
- QVariant **getDrop** ()
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant v)
- void **resizeEvent** (QResizeEvent *)

Protected Attributes

- [QStringFormatting](#) **stringFormatting**
- [QEIntegerFormatting](#) **integerFormatting**
- [QEFloatingFormatting](#) **floatingFormatting**
- [resizeOptions](#) **resizeOption**
- int [zoom](#)
Zoom percentage. Used when #resizeOption is [Zoom](#).
- double [XStretch](#)
Stretch X factor. Used when generating canvas in which fully processed image is presented.
- double [YStretch](#)
Stretch Y factor. Used when generating canvas in which fully processed image is presented.
- int **initialHozScrollPos**
- int [initialVertScrollPos](#)
- bool [displayButtonBar](#)

Properties

- QString [imageVariable](#)
- QString [formatVariable](#)
- QString [bitDepthVariable](#)
- QString [dataTypeVariable](#)
- QString [widthVariable](#)
- QString [heightVariable](#)
- QString [dimensionsVariable](#)
- QString [dimension1Variable](#)
- QString [dimension2Variable](#)
- QString [dimension3Variable](#)
- QString [regionOfInterest1XVariable](#)
- QString [regionOfInterest1YVariable](#)
- QString [regionOfInterest1WVariable](#)
- QString [regionOfInterest1HVariable](#)
- QString [regionOfInterest2XVariable](#)
- QString [regionOfInterest2YVariable](#)
- QString [regionOfInterest2WVariable](#)
- QString [regionOfInterest2HVariable](#)
- QString [regionOfInterest3XVariable](#)
- QString [regionOfInterest3YVariable](#)
- QString [regionOfInterest3WVariable](#)
- QString [regionOfInterest3HVariable](#)
- QString [regionOfInterest4XVariable](#)
- QString [regionOfInterest4YVariable](#)
- QString [regionOfInterest4WVariable](#)
- QString [regionOfInterest4HVariable](#)
- QString [targetXVariable](#)
- QString [targetYVariable](#)
- QString [beamXVariable](#)
- QString [beamYVariable](#)
- QString [targetTriggerVariable](#)
- QString [clippingOnOffVariable](#)
- QString [clippingLowVariable](#)
- QString [clippingHighVariable](#)
- QString **profileHozVariable**

- QString [profileHoz1Variable](#)
- QString **profileHozThicknessVariable**
- QString [profileHoz1ThicknessVariable](#)
- QString [profileHoz2Variable](#)
- QString [profileHoz2ThicknessVariable](#)
- QString [profileHoz3Variable](#)
- QString [profileHoz3ThicknessVariable](#)
- QString [profileHoz4Variable](#)
- QString [profileHoz4ThicknessVariable](#)
- QString [profileHoz5Variable](#)
- QString [profileHoz5ThicknessVariable](#)
- QString **profileVertVariable**
- QString [profileVert1Variable](#)
- QString **profileVertThicknessVariable**
- QString [profileVert1ThicknessVariable](#)
- QString [profileVert2Variable](#)
- QString [profileVert2ThicknessVariable](#)
- QString [profileVert3Variable](#)
- QString [profileVert3ThicknessVariable](#)
- QString [profileVert4Variable](#)
- QString [profileVert4ThicknessVariable](#)
- QString [profileVert5Variable](#)
- QString [profileVert5ThicknessVariable](#)
- QString [lineProfileX1Variable](#)
- QString [lineProfileY1Variable](#)
- QString [lineProfileX2Variable](#)
- QString [lineProfileY2Variable](#)
- QString [lineProfileThicknessVariable](#)
- QString [profileHozArrayVariable](#)
- QString [profileVertArrayVariable](#)
- QString [lineProfileArrayVariable](#)
- QString [ellipseXVariable](#)
- QString [ellipseYVariable](#)
- QString [ellipseWVariable](#)
- QString [ellipseHVariable](#)
- QString [ellipseRotationVariable](#)
- QString [variableSubstitutions](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- QE::UserLevels [userLevelVisibility](#)
- QE::UserLevels [userLevelEnabled](#)
- QE::DisplayAlarmStateOptions [displayAlarmStateOption](#)
- bool [oosAware](#)
- QE::ImageFormatOptions [formatOption](#)
- int [bitDepth](#)
- bool **enableVertSliceSelection**
- bool [enableVertSlice1Selection](#)
- bool [enableVertSlice2Selection](#)

- bool [enableVertSlice3Selection](#)
- bool [enableVertSlice4Selection](#)
- bool [enableVertSlice5Selection](#)
- bool **enableHozSliceSelection**
- bool [enableHozSlice1Selection](#)
- bool [enableHozSlice2Selection](#)
- bool [enableHozSlice3Selection](#)
- bool [enableHozSlice4Selection](#)
- bool [enableHozSlice5Selection](#)
- bool [enableProfileSelection](#)
- bool [enableArea1Selection](#)
- bool [enableArea2Selection](#)
- bool [enableArea3Selection](#)
- bool [enableArea4Selection](#)
- bool [enableTargetSelection](#)
- bool [enableBeamSelection](#)
- QString **hozSliceLegend**
- QString [hozSlice1Legend](#)
Name of horizontal slice 1 markup.
- QString [hozSlice2Legend](#)
Name of horizontal slice 2 markup.
- QString [hozSlice3Legend](#)
Name of horizontal slice 3 markup.
- QString [hozSlice4Legend](#)
Name of horizontal slice 4 markup.
- QString [hozSlice5Legend](#)
Name of horizontal slice 5 markup.
- QString **vertSliceLegend**
- QString [vertSlice1Legend](#)
Name of vertical slice 1 markup.
- QString [vertSlice2Legend](#)
Name of vertical slice 2 markup.
- QString [vertSlice3Legend](#)
Name of vertical slice 3 markup.
- QString [vertSlice4Legend](#)
Name of vertical slice 4 markup.
- QString [vertSlice5Legend](#)
Name of vertical slice 5 markup.
- QString [profileLegend](#)
Name of arbitrary profile markup.
- QString [areaSelection1Legend](#)
Name of area selection 1 markup.
- QString [areaSelection2Legend](#)
Name of area selection 2 markup.
- QString [areaSelection3Legend](#)
Name of area selection 3 markup.
- QString [areaSelection4Legend](#)
Name of area selection 4 markup.
- QString [targetLegend](#)
Name of target markup.
- QString [beamLegend](#)
Name of beam markup.

- QString [ellipseLegend](#)
Name of ellipse markup.
- bool **displayVertSliceSelection**
- bool [displayVertSlice1Selection](#)
- bool [displayVertSlice2Selection](#)
- bool [displayVertSlice3Selection](#)
- bool [displayVertSlice4Selection](#)
- bool [displayVertSlice5Selection](#)
- bool **displayHozSliceSelection**
- bool [displayHozSlice1Selection](#)
- bool [displayHozSlice2Selection](#)
- bool [displayHozSlice3Selection](#)
- bool [displayHozSlice4Selection](#)
- bool [displayHozSlice5Selection](#)
- bool [displayProfileSelection](#)
- bool [displayArea1Selection](#)
- bool [displayArea2Selection](#)
- bool [displayArea3Selection](#)
- bool [displayArea4Selection](#)
- bool [displayTargetSelection](#)
- bool [displayBeamSelection](#)
- bool [displayEllipse](#)
- [EllipseVariableDefinitions](#) [ellipseVariableDefinition](#)
Definition of how ellipse variables are to be used.
- [TargetOptions](#) [targetOption](#)
Definition of target markup options.
- [TargetOptions](#) [beamOption](#)
Definition of beam markup options.
- bool [displayCursorPixelInfo](#)
- bool [contrastReversal](#)
- bool [logBrightness](#)
- bool [showTime](#)
- bool [useFalseColour](#)
- bool [enableProfileAxes](#)
- QColor **vertSliceColor**
- QColor [vertSlice1Color](#)
- QColor [vertSlice2Color](#)
- QColor [vertSlice3Color](#)
- QColor [vertSlice4Color](#)
- QColor [vertSlice5Color](#)
- QColor **hozSliceColor**
- QColor [hozSlice1Color](#)
- QColor [hozSlice2Color](#)
- QColor [hozSlice3Color](#)
- QColor [hozSlice4Color](#)
- QColor [hozSlice5Color](#)
- QColor [profileColor](#)
- QColor [areaColor](#)
- QColor [beamColor](#)
- QColor [targetColor](#)
- QColor [timeColor](#)
- QColor [ellipseColor](#)
- [ResizeOptions](#) [resizeOption](#)
- [QE::RotationOptions](#) [rotation](#)

- bool [verticalFlip](#)
- bool [horizontalFlip](#)
- int [initialHosScrollPos](#)
- bool [enableImageDisplayProperties](#)
If true, the local Image Display Properties controls are displayed.
- bool [enableRecording](#)
If true, the recording controls are displayed.
- bool [autoBrightnessContrast](#)
- bool [externalControls](#)
- bool [briefInfoArea](#)
- QString [program1](#)
- QStringList [arguments1](#)
- [QE::ProgramStartupOptions](#) [programStartupOption1](#)
- QString [program2](#)
- QStringList [arguments2](#)
- [QE::ProgramStartupOptions](#) [programStartupOption2](#)
- QString [URL](#)

Additional Inherited Members

9.175.1 Detailed Description

This class is a EPICS aware image widget. When image related variables are defined the image will be displayed. Many PVs may be defined to allow user interaction, such as selecting regions of interest. It is tightly integrated with the base class [QEWidget](#) which provides generic support such as macro substitutions, drag/drop, and standard properties.

9.175.2 Member Enumeration Documentation

9.175.2.1 ellipseVariableDefinitions

```
enum QEImage::ellipseVariableDefinitions
```

Options for the use of ellipse markup variables.

Enumerator

<code>BOUNDING_RECTANGLE</code>	Variables define bounding rectagle of ellipse.
---------------------------------	--

9.175.2.2 EllipseVariableDefinitions

```
enum QEImage::EllipseVariableDefinitions
```

User friendly enumerations for [ellipseVariableDefinition](#) property - refer to [ellipseVariableDefinition](#) property for details.

Enumerator

BoundingBox	Refer to BOUNDING_RECTANGLE for details.
CenterAndSize	Refer to CENTRE_AND_SIZE for details.

9.175.2.3 resizeModeOptions

```
enum QImage::resizeOptions
```

Image resize options

Enumerator

RESIZE_OPTION_ZOOM	Zoom to selected percentage.
RESIZE_OPTION_FIT	Zoom to fit the current window size.

9.175.2.4 ResizeOptions

```
enum QImage::ResizeOptions
```

User friendly enumerations for #resizeOption property

Enumerator

Zoom	Zoom to selected percentage.
Fit	Zoom to fit the current window size.

9.175.2.5 selectOptions

```
enum QImage::selectOptions
```

Internal use only. Selection options. What will happen when the user interacts with the image area

Enumerator

SO_NONE	Do nothing.
SO_PANNING	User is panning.
SO_VSLICE1	Select the vertical slice 1 point.
SO_VSLICE2	Select the vertical slice 2 point.
SO_VSLICE3	Select the vertical slice 3 point.
SO_VSLICE4	Select the vertical slice 4 point.

Enumerator

SO_VSLICE5	Select the vertical slice 5 point.
SO_HSLICE1	Select the horizontal slice 1 point.
SO_HSLICE2	Select the horizontal slice 2 point.
SO_HSLICE3	Select the horizontal slice 3 point.
SO_HSLICE4	Select the horizontal slice 4 point.
SO_HSLICE5	Select the horizontal slice 5 point.
SO_AREA4	User is selecting an area (for region of interest)
SO_PROFILE	Select an arbitrary line across the image (to determine a profile)
SO_TARGET	Mark the target point.
SO_BEAM	Mark the current beam location.

9.175.2.6 TargetOptions

```
enum QEImage::TargetOptions
```

User friendly enumerations for #targetOptions property - refer to #targetOptions property for details.

Enumerator

DottedFullCrosshair	Refer to CROSSHAIR1 for details.
SolidSmallCrosshair	Refer to CROSSHAIR2 for details.

9.175.3 Constructor & Destructor Documentation

9.175.3.1 QEImage() [1/2]

```
QEImage::QEImage (
    QWidget * parent = 0 )
```

Create without a variable. Use setVariableName'n'Property() - where 'n' is a number from 0 to 40 - and set↔SubstitutionsProperty() to define variables and, optionally, macro substitutions later. Note, each variable property is named by function (such as imageVariable and widthVariable) but given a numeric get and set property access function such as setVariableName22Property(). Refer to the property definitions to determine what 'set' and 'get' function is used for each variable, or use Qt library functions to set or get the variable names by name.

9.175.3.2 QEImage() [2/2]

```
QEImage::QEImage (
    const QString & variableName,
    QWidget * parent = 0 )
```

Create with a variable. A connection is automatically established. The variable is set up as the first variable. This is consistent with other widgets, but will not result in an updating image as the width and height variables are required as a minimum.

9.175.4 Member Function Documentation

9.175.4.1 brightnessContrastAutoImageRequest

```
void QEImage::brightnessContrastAutoImageRequest ( ) [slot]
```

Allows auto brightness control.

9.175.4.2 dbValueChanged

```
void QEImage::dbValueChanged (
    const QString & out ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.175.4.3 imageDisplayPropertiesChanged

```
void QEImage::imageDisplayPropertiesChanged ( ) [slot]
```

Allows brightness or contrast or contrast reversal has changed

9.175.4.4 setDataImage

```
void QEImage::setDataImage (
    const QByteArray & imageIn,
    unsigned long dataSize,
    unsigned long elements,
    unsigned long width,
    unsigned long height,
    QE::ImageFormatOptions format,
    unsigned int depth ) [slot]
```

!! Should the format, bit depth, width and height be clobered like this? !! (especially where we are altering properties, like bitDepth). !! Perhaps CA delivered and MPEG delivered images should maintain their own attributes?

!! should also set format as delivered with image from mpeg source???

9.175.4.5 setImageFile

```
void QEImage::setImageFile (
    QString name ) [slot]
```

!! memcpy will be more efficient.

9.175.4.6 setManagedVisible

```
void QEImage::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.175.5 Member Data Documentation

9.175.5.1 displayButtonBar

```
bool QEImage::displayButtonBar [read], [write], [protected]
```

If true, a button bar will be displayed above the image. If not displayed, all buttons in the button bar are still available in the right click menu.

9.175.5.2 initialVertScrollPos

```
int QEImage::initialVertScrollPos [read], [write], [protected]
```

Sets the initial position of the vertical scroll bar, if present. Used to set up an initial view when zoomed in.

9.175.6 Property Documentation

9.175.6.1 allowDrop

```
bool QEImage::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.175.6.2 areaColor

```
QColor QEImage::areaColor [read], [write]
```

Used to select the color of the area selection markups.

9.175.6.3 arguments1

```
QStringList QEImage::arguments1 [read], [write]
```

Arguments for program specified in the 'program1' property.

9.175.6.4 arguments2

```
QStringList QEImage::arguments2 [read], [write]
```

Arguments for program specified in the 'program2' property.

9.175.6.5 autoBrightnessContrast

```
bool QEImage::autoBrightnessContrast [read], [write]
```

If true, auto set local brightness and contrast when any area is selected. The brightness and contrast is set to use the full range of pixels in the selected area.

9.175.6.6 beamColor

```
QColor QEImage::beamColor [read], [write]
```

Used to select the color of the beam marker.

9.175.6.7 beamXVariable

```
QString QEImage::beamXVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the selected beam X position.

9.175.6.8 beamYVariable

```
QString QEImage::beamYVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the selected beam Y position.

9.175.6.9 bitDepth

```
int QEImage::bitDepth [read], [write]
```

Bit depth. Note, EPICS data type size will typically be adequate for the number of bits required (one byte for up to 8 bits, 2 bytes for up to 16 bits, etc), but can be larger (for example, 4 bytes for 24 bits) and may be larger than necessary (4 bytes for 8 bits).

9.175.6.10 bitDepthVariable

```
QString QEImage::bitDepthVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read the bit depth of the image.

9.175.6.11 briefInfoArea

```
bool QEImage::briefInfoArea [read], [write]
```

If true, the information area will be brief (one row)

9.175.6.12 clippingHighVariable

```
QString QEImage::clippingHighVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector clipping high level.

9.175.6.13 clippingLowVariable

```
QString QEImage::clippingLowVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector clipping low level.

9.175.6.14 clippingOnOffVariable

```
QString QEImage::clippingOnOffVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector clipping on/off command.

9.175.6.15 contrastReversal

```
bool QEImage::contrastReversal [read], [write]
```

If true, the image will undergo contrast reversal.

9.175.6.16 dataTypeVariable

```
QString QEImage::dataTypeVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to infer the bit depth of the image.

9.175.6.17 defaultStyle

```
QString QEImage::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.175.6.18 dimension1Variable

```
QString QEImage::dimension1Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read the first area detector dimension of the image. If there are 2 dimensions, this will be the image width. If there are 3 dimensions, this will be the number of elements per pixel.

9.175.6.19 dimension2Variable

```
QString QEImage::dimension2Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read the second area detector dimension of the image. If there are 2 dimensions, this will be the image height. If there are 3 dimensions, this will be the image width.

9.175.6.20 dimension3Variable

```
QString QEImage::dimension3Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read the third area detector dimension of the image. If there are 3 dimensions, this will be the image height.

9.175.6.21 dimensionsVariable

```
QString QEImage::dimensionsVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read the number of area detector dimensions of the image. If used, this will be 2 (one element per pixel arranged by width and height) or 3 (multiple elements per pixel arranged by pixel, width and height)

9.175.6.22 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEImage::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.175.6.23 displayArea1Selection

```
bool QEImage::displayArea1Selection [read], [write]
```

If true, selected area 1 will be displayed on the image. Note, this property is ignored unless the [enableArea1Selection](#) property is true.

9.175.6.24 displayArea2Selection

```
bool QEImage::displayArea2Selection [read], [write]
```

If true, selected area 2 will be displayed on the image. Note, this property is ignored unless the [enableArea2Selection](#) property is true.

9.175.6.25 displayArea3Selection

```
bool QEImage::displayArea3Selection [read], [write]
```

If true, selected area 3 will be displayed on the image. Note, this property is ignored unless the [enableArea3Selection](#) property is true.

9.175.6.26 displayArea4Selection

```
bool QEImage::displayArea4Selection [read], [write]
```

If true, selected area 4 will be displayed on the image. Note, this property is ignored unless the [enableArea4Selection](#) property is true.

9.175.6.27 displayBeamSelection

```
bool QEImage::displayBeamSelection [read], [write]
```

If true, beam selection will be displayed on the image. Note, this property is ignored unless the [enableBeamSelection](#) property is true.

9.175.6.28 displayCursorPixelInfo

```
bool QEImage::displayCursorPixelInfo [read], [write]
```

If true, an area will be presented under the image with textual information about the pixel under the cursor, and for other selections such as selected areas.

9.175.6.29 displayEllipse

```
bool QEImage::displayEllipse [read], [write]
```

If true, the ellipse markup will be displayed on the image.

9.175.6.30 displayHozSlice1Selection

```
bool QEImage::displayHozSlice1Selection [read], [write]
```

If true, the selected horizontal slice will be displayed on the image. Note, this property is ignored unless the [enableHozSlice1Selection](#) property is true.

9.175.6.31 displayHozSlice2Selection

```
bool QEImage::displayHozSlice2Selection [read], [write]
```

If true, the selected horizontal slice will be displayed on the image. Note, this property is ignored unless the [enableHozSlice2Selection](#) property is true.

9.175.6.32 displayHozSlice3Selection

```
bool QEImage::displayHozSlice3Selection [read], [write]
```

If true, the selected horizontal slice will be displayed on the image. Note, this property is ignored unless the [enableHozSlice3Selection](#) property is true.

9.175.6.33 displayHozSlice4Selection

```
bool QEImage::displayHozSlice4Selection [read], [write]
```

If true, the selected horizontal slice will be displayed on the image. Note, this property is ignored unless the [enableHozSlice4Selection](#) property is true.

9.175.6.34 displayHozSlice5Selection

```
bool QEImage::displayHozSlice5Selection [read], [write]
```

If true, the selected horizontal slice will be displayed on the image. Note, this property is ignored unless the [enableHozSlice5Selection](#) property is true.

9.175.6.35 displayProfileSelection

```
bool QEImage::displayProfileSelection [read], [write]
```

If true, the selected arbitrary line will be displayed on the image. Note, this property is ignored unless the [enableProfileSelection](#) property is true.

9.175.6.36 displayTargetSelection

```
bool QEImage::displayTargetSelection [read], [write]
```

If true, target selection will be displayed on the image. Note, this property is ignored unless the [enableTargetSelection](#) property is true.

9.175.6.37 displayVertSlice1Selection

```
bool QEImage::displayVertSlice1Selection [read], [write]
```

If true, the selected vertical slice 1 will be displayed on the image. Note, this property is ignored unless the [enableVertSlice1Selection](#) property is true.

9.175.6.38 displayVertSlice2Selection

```
bool QEImage::displayVertSlice2Selection [read], [write]
```

If true, the selected vertical slice 2 will be displayed on the image. Note, this property is ignored unless the [enableVertSlice2Selection](#) property is true.

9.175.6.39 displayVertSlice3Selection

```
bool QEImage::displayVertSlice3Selection [read], [write]
```

If true, the selected vertical slice 3 will be displayed on the image. Note, this property is ignored unless the [enableVertSlice3Selection](#) property is true.

9.175.6.40 displayVertSlice4Selection

```
bool QEImage::displayVertSlice4Selection [read], [write]
```

If true, the selected vertical slice 4 will be displayed on the image. Note, this property is ignored unless the [enableVertSlice4Selection](#) property is true.

9.175.6.41 displayVertSlice5Selection

```
bool QEImage::displayVertSlice5Selection [read], [write]
```

If true, the selected vertical slice 5 will be displayed on the image. Note, this property is ignored unless the [enableVertSlice5Selection](#) property is true.

9.175.6.42 ellipseColor

```
QColor QEImage::ellipseColor [read], [write]
```

Used to select the color of the ellipse marker.

9.175.6.43 ellipseHVariable

```
QString QEImage::ellipseHVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read an ellipse height

9.175.6.44 ellipseRotationVariable

```
QString QEImage::ellipseRotationVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read an ellipse height

9.175.6.45 ellipseWVariable

```
QString QEImage::ellipseWVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read an ellipse width.

9.175.6.46 ellipseXVariable

```
QString QEImage::ellipseXVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read an ellipse X (center or top left corner of bounding rectangle depending on property ellipseDefinition).

9.175.6.47 ellipseYVariable

```
QString QEImage::ellipseYVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read an ellipse Y (center or top left corner of bounding rectangle depending on property ellipseDefinition).

9.175.6.48 enableArea1Selection

```
bool QEImage::enableArea1Selection [read], [write]
```

If true, the user will be able to select area 1. These are used for selection of Region of Interests, and for zooming to area 1

9.175.6.49 enableArea2Selection

```
bool QEImage::enableArea2Selection [read], [write]
```

If true, the user will be able to select area 2. These are used for selection of Region of Interests, and for zooming to area 2

9.175.6.50 enableArea3Selection

```
bool QEImage::enableArea3Selection [read], [write]
```

If true, the user will be able to select area 3. These are used for selection of Region of Interests, and for zooming to area 3

9.175.6.51 enableArea4Selection

```
bool QEImage::enableArea4Selection [read], [write]
```

If true, the user will be able to select area 4. These are used for selection of Region of Interests, and for zooming to area 4

9.175.6.52 enableBeamSelection

```
bool QEImage::enableBeamSelection [read], [write]
```

If true, the user will be able to select points on the image to mark a beam position. This can be used for automatic beam positioning.

9.175.6.53 enableHozSlice1Selection

```
bool QEImage::enableHozSlice1Selection [read], [write]
```

If true, the option to select a horizontal slice through the image will be available to the user. This will be used to generate a horizontal pixel profile, and write the position of the slice to the optional variable specified by the [profileHoz1Variable](#) property. The profile will only be presented to the user if [enableHozSlicePresentation](#) property is true.

9.175.6.54 enableHozSlice2Selection

```
bool QEImage::enableHozSlice2Selection [read], [write]
```

If true, the option to select a second horizontal slice through the image will be available to the user. This will be used to write the position of the slice to the optional variable specified by the [profileHoz2Variable](#) property.

9.175.6.55 enableHozSlice3Selection

```
bool QEImage::enableHozSlice3Selection [read], [write]
```

If true, the option to select a third horizontal slice through the image will be available to the user. This will be used to write the position of the slice to the optional variable specified by the [profileHoz3Variable](#) property.

9.175.6.56 enableHozSlice4Selection

```
bool QEImage::enableHozSlice4Selection [read], [write]
```

If true, the option to select a fourth horizontal slice through the image will be available to the user. This will be used to write the position of the slice to the optional variable specified by the [profileHoz4Variable](#) property.

9.175.6.57 enableHozSlice5Selection

```
bool QEImage::enableHozSlice5Selection [read], [write]
```

If true, the option to select a fifth horizontal slice through the image will be available to the user. This will be used to write the position of the slice to the optional variable specified by the [profileHoz5Variable](#) property.

9.175.6.58 enableProfileAxes

```
bool QEImage::enableProfileAxes [read], [write]
```

If true, the show axes on horizontal, vertical and arbitray line profiles.

9.175.6.59 enableProfileSelection

```
bool QEImage::enableProfileSelection [read], [write]
```

If true, the option to select an arbitrary line through any part of the image will be available to the user. This will be used to generate a pixel profile.

9.175.6.60 enableTargetSelection

```
bool QEImage::enableTargetSelection [read], [write]
```

If true, the user will be able to select points on the image to mark a target position. This can be used for automatic beam positioning.

9.175.6.61 enableVertSlice1Selection

```
bool QEImage::enableVertSlice1Selection [read], [write]
```

If true, the option to select a vertical slice through the image will be available to the user. This will be used to generate a horizontal pixel profile, and write the position of the slice to the optional variable specified by the [profileVert1Variable](#) property. The profile will only be presented to the user if `#enableVertSlicePresentation` property is true.

9.175.6.62 enableVertSlice2Selection

```
bool QEImage::enableVertSlice2Selection [read], [write]
```

If true, the option to select a second vertical slice through the image will be available to the user. This will be used to write the position of the slice to the optional variable specified by the [profileVert2Variable](#) property.

9.175.6.63 enableVertSlice3Selection

```
bool QEImage::enableVertSlice3Selection [read], [write]
```

If true, the option to select a third vertical slice through the image will be available to the user. This will be used to write the position of the slice to the optional variable specified by the [profileVert3Variable](#) property.

9.175.6.64 enableVertSlice4Selection

```
bool QEImage::enableVertSlice4Selection [read], [write]
```

If true, the option to select a fourth vertical slice through the image will be available to the user. This will be used to write the position of the slice to the optional variable specified by the [profileVert4Variable](#) property.

9.175.6.65 enableVertSlice5Selection

```
bool QEImage::enableVertSlice5Selection [read], [write]
```

If true, the option to select a fifth vertical slice through the image will be available to the user. This will be used to write the position of the slice to the optional variable specified by the [profileVert5Variable](#) property.

9.175.6.66 externalControls

```
bool QEImage::externalControls [read], [write]
```

If true, image controls and views such as brightness controls and profile plots are hosted by the application as dock windows, toolbars, etc. Refer to the [ContainerProfile](#) class and the [windowCustomisation](#) class to see how this class asks an application to act as a host.

9.175.6.67 formatOption

```
QE::ImageFormatOptions QEImage::formatOption [read], [write]
```

Video format. EPICS data type size will typically be adequate for the number of bits required (one byte for 8 bits, 2 bytes for 12 and 16 bits), but can be larger (4 bytes for 24 bits.)

9.175.6.68 formatVariable

```
QString QEImage::formatVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read the format of the image.

9.175.6.69 heightVariable

```
QString QEImage::heightVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to read the height of the image.

9.175.6.70 horizontalFlip

```
bool QEImage::horizontalFlip [read], [write]
```

If true, flip image horizontally.

9.175.6.71 hozSlice1Color

```
QColor QEImage::hozSlice1Color [read], [write]
```

Used to select the color of the horizontal slice 1 markup.

9.175.6.72 hozSlice2Color

```
QColor QEImage::hozSlice2Color [read], [write]
```

Used to select the color of the horizontal slice 2 markup.

9.175.6.73 hozSlice3Color

```
QColor QEImage::hozSlice3Color [read], [write]
```

Used to select the color of the horizontal slice 3 markup.

9.175.6.74 hozSlice4Color

```
QColor QEImage::hozSlice4Color [read], [write]
```

Used to select the color of the horizontal slice 4 markup.

9.175.6.75 hozSlice5Color

```
QColor QEImage::hozSlice5Color [read], [write]
```

Used to select the color of the horizontal slice 5 markup.

9.175.6.76 imageVariable

```
QString QEImage::imageVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used as the source the image waveform.

9.175.6.77 initialHosScrollPos

```
int QEImage::initialHosScrollPos [read], [write]
```

Sets the initial position of the horizontal scroll bar, if present. Used to set up an initial view when zoomed in.

9.175.6.78 lineProfileArrayVariable

```
QString QEImage::lineProfileArrayVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector arbitrary line profile array.

9.175.6.79 lineProfileThicknessVariable

```
QString QEImage::lineProfileThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector arbitrary line profile end Y.

9.175.6.80 lineProfileX1Variable

```
QString QEImage::lineProfileX1Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector arbitrary line profile start X.

9.175.6.81 lineProfileX2Variable

```
QString QEImage::lineProfileX2Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector arbitrary line profile end X.

9.175.6.82 lineProfileY1Variable

```
QString QEImage::lineProfileY1Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector arbitrary line profile start Y.

9.175.6.83 lineProfileY2Variable

```
QString QEImage::lineProfileY2Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector arbitrary line profile end Y.

9.175.6.84 logBrightness

```
bool QEImage::logBrightness [read], [write]
```

If true, the image will be displayed using a logarithmic brightness scale.

9.175.6.85 messageSourceId

```
unsigned int QEImage::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.175.6.86 oosAware

```
bool QEImage::oosAware [read], [write]
```

Indicates whether the widget should respond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.175.6.87 profileColor

```
QColor QEImage::profileColor [read], [write]
```

Used to select the color of the arbitrary profile line markup.

9.175.6.88 profileHoz1ThicknessVariable

```
QString QEImage::profileHoz1ThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector first horizontal profile thickness.

9.175.6.89 profileHoz1Variable

```
QString QEImage::profileHoz1Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector first horizontal profile.

9.175.6.90 profileHoz2ThicknessVariable

```
QString QEImage::profileHoz2ThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector second horizontal profile thickness.

9.175.6.91 profileHoz2Variable

```
QString QEImage::profileHoz2Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector second horizontal profile.

9.175.6.92 profileHoz3ThicknessVariable

```
QString QEImage::profileHoz3ThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector third horizontal profile thickness.

9.175.6.93 profileHoz3Variable

```
QString QEImage::profileHoz3Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector third horizontal profile.

9.175.6.94 profileHoz4ThicknessVariable

```
QString QEImage::profileHoz4ThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector fourth horizontal profile thickness.

9.175.6.95 profileHoz4Variable

```
QString QEImage::profileHoz4Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector fourth horizontal profile.

9.175.6.96 profileHoz5ThicknessVariable

```
QString QEImage::profileHoz5ThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector fifth horizontal profile thickness.

9.175.6.97 profileHoz5Variable

```
QString QEImage::profileHoz5Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector fifth horizontal profile.

9.175.6.98 profileHozArrayVariable

```
QString QEImage::profileHozArrayVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector horizontal profile array.

9.175.6.99 profileVert1ThicknessVariable

```
QString QEImage::profileVert1ThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector first vertical profile.

9.175.6.100 profileVert1Variable

```
QString QEImage::profileVert1Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector first vertical profile.

9.175.6.101 profileVert2ThicknessVariable

```
QString QEImage::profileVert2ThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector second vertical profile.

9.175.6.102 profileVert2Variable

```
QString QEImage::profileVert2Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector second vertical profile.

9.175.6.103 profileVert3ThicknessVariable

```
QString QEImage::profileVert3ThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector third vertical profile.

9.175.6.104 profileVert3Variable

```
QString QEImage::profileVert3Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector third vertical profile.

9.175.6.105 profileVert4ThicknessVariable

```
QString QEImage::profileVert4ThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector fourth vertical profile.

9.175.6.106 profileVert4Variable

```
QString QEImage::profileVert4Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector fourth vertical profile.

9.175.6.107 profileVert5ThicknessVariable

```
QString QEImage::profileVert5ThicknessVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector fifth vertical profile.

9.175.6.108 profileVert5Variable

```
QString QEImage::profileVert5Variable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector fifth vertical profile.

9.175.6.109 profileVertArrayVariable

```
QString QEImage::profileVertArrayVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the areadetector vertical profile array.

9.175.6.110 program1

```
QString QEImage::program1 [read], [write]
```

Program to run when a request is made to pass on the current image to the first external application. No attempt to run a program is made if this property is empty. Example: paint.exe

9.175.6.111 program2

```
QString QEImage::program2 [read], [write]
```

Program to run when a request is made to pass on the current image to the second external application. No attempt to run a program is made if this property is empty. Example: paint.exe

9.175.6.112 programStartupOption1

```
QE::ProgramStartupOptions QEImage::programStartupOption1 [read], [write]
```

Startup options for the program specified in the 'program1' property. Just run the command, run the command within a terminal, or display the output in [QE](#) message system.

9.175.6.113 programStartupOption2

```
QE::ProgramStartupOptions QEImage::programStartupOption2 [read], [write]
```

Startup options for the program specified in the 'program2' property. Just run the command, run the command within a terminal, or display the output in [QE](#) message system.

9.175.6.114 regionOfInterest1HVariable

```
QString QEImage::regionOfInterest1HVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the first region of interest height.

9.175.6.115 regionOfInterest1WVariable

```
QString QEImage::regionOfInterest1WVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the first region of interest width.

9.175.6.116 regionOfInterest1XVariable

```
QString QEImage::regionOfInterest1XVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the first region of interest X position.

9.175.6.117 regionOfInterest1YVariable

```
QString QEImage::regionOfInterest1YVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the first region of interest Y position.

9.175.6.118 regionOfInterest2HVariable

```
QString QEImage::regionOfInterest2HVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the second region of interest height.

9.175.6.119 regionOfInterest2WVariable

```
QString QEImage::regionOfInterest2WVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the second region of interest width.

9.175.6.120 regionOfInterest2XVariable

```
QString QEImage::regionOfInterest2XVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the second region of interest X position.

9.175.6.121 regionOfInterest2YVariable

```
QString QEImage::regionOfInterest2YVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the second region of interest Y position.

9.175.6.122 regionOfInterest3HVariable

```
QString QEImage::regionOfInterest3HVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the third region of interest height.

9.175.6.123 regionOfInterest3WVariable

```
QString QEImage::regionOfInterest3WVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the third region of interest width.

9.175.6.124 regionOfInterest3XVariable

```
QString QEImage::regionOfInterest3XVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the third region of interest X position.

9.175.6.125 regionOfInterest3YVariable

```
QString QEImage::regionOfInterest3YVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the third region of interest Y position.

9.175.6.126 regionOfInterest4HVariable

```
QString QEImage::regionOfInterest4HVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the fourth region of interest height.

9.175.6.127 regionOfInterest4WVariable

```
QString QEImage::regionOfInterest4WVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the fourth region of interest width.

9.175.6.128 regionOfInterest4XVariable

```
QString QEImage::regionOfInterest4XVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the fourth region of interest X position.

9.175.6.129 regionOfInterest4YVariable

```
QString QEImage::regionOfInterest4YVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the fourth region of interest Y position.

9.175.6.130 resizeMode

```
ResizeOptions QEImage::resizeOption [read], [write]
```

Resize option. Zoom to zoom to the percentage given by the [zoom](#) property, or fit to the window size.

9.175.6.131 rotation

```
QE::RotationOptions QEImage::rotation [read], [write]
```

Image rotation option.

9.175.6.132 showTime

```
bool QEImage::showTime [read], [write]
```

If true, the image timestamp will be written in the top left of the image.

9.175.6.133 styleSheet

```
QString QEImage::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.175.6.134 targetColor

```
QColor QImage::targetColor [read], [write]
```

Used to select the color of the target marker.

9.175.6.135 targetTriggerVariable

```
QString QImage::targetTriggerVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write a 'trigger' to initiate movement of the target into the beam as defined by the target and beam X and Y positions.

9.175.6.136 targetXVariable

```
QString QImage::targetXVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the selected target X position.

9.175.6.137 targetYVariable

```
QString QImage::targetYVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to write the selected target Y position.

9.175.6.138 timeColor

```
QColor QImage::timeColor [read], [write]
```

Used to select the color of the timestamp.

9.175.6.139 URL

```
QString QImage::URL [read], [write]
```

MPEG stream URL. If this is specified, this will be used as the source of the image in preference to variables (variables defining the image data, width, and height will be ignored)

9.175.6.140 useFalseColour

```
bool QImage::useFalseColour [read], [write]
```

If true, the apply false colour to the image.

9.175.6.141 userLevelEnabled

```
QE::UserLevels QEImage::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.175.6.142 userLevelEngineerStyle

```
QString QEImage::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.175.6.143 userLevelScientistStyle

```
QString QEImage::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.175.6.144 userLevelUserStyle

```
QString QEImage::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.175.6.145 userLevelVisibility

```
QE::UserLevels QEImage::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.175.6.146 variableAsToolTip

```
bool QImage::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.175.6.147 variableSubstitutions

```
QString QImage::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'CAM=1, NAME = "Image 1"' These substitutions are applied to all the variable names.

9.175.6.148 verticalFlip

```
bool QImage::verticalFlip [read], [write]
```

If true, flip image vertically.

9.175.6.149 vertSlice1Color

```
QColor QImage::vertSlice1Color [read], [write]
```

Used to select the color of the vertical slice 1 markup.

9.175.6.150 vertSlice2Color

```
QColor QImage::vertSlice2Color [read], [write]
```

Used to select the color of the vertical slice 2 markup.

9.175.6.151 vertSlice3Color

```
QColor QImage::vertSlice3Color [read], [write]
```

Used to select the color of the vertical slice 3 markup.

9.175.6.152 vertSlice4Color

```
QColor QImage::vertSlice4Color [read], [write]
```

Used to select the color of the vertical slice 4 markup.

9.175.6.153 vertSlice5Color

```
QColor QEImage::vertSlice5Color [read], [write]
```

Used to select the color of the vertical slice 5 markup.

9.175.6.154 visible

```
bool QEImage::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

9.175.6.155 widthVariable

```
QString QEImage::widthVariable [read], [write]
```

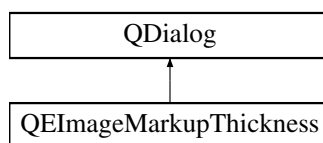
EPICS variable name (CA PV). This variable is used to read the width of the image.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/QEImage.h
- qeframework/widgets/QEImage/QEImage.cpp

9.176 QEImageMarkupThickness Class Reference

Inheritance diagram for QEImageMarkupThickness:

**Public Member Functions**

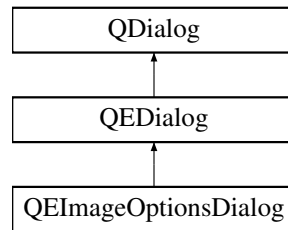
- **QEImageMarkupThickness** (QWidget *parent=0)
- void **setThickness** (unsigned int thicknessIn)
- unsigned int **getThickness** ()

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/QEImageMarkupThickness.h
- qeframework/widgets/QEImage/QEImageMarkupThickness.cpp

9.177 QEImageOptionsDialog Class Reference

Inheritance diagram for QEImageOptionsDialog:



Signals

- void **optionChange** (imageContextMenu::imageContextMenuOptions option, bool checked)

Public Member Functions

- **QEImageOptionsDialog** (QWidget *parent=0)
- void **initialise** ()
- void **optionSet** (imageContextMenu::imageContextMenuOptions option, bool checked)
- bool **optionGet** (imageContextMenu::imageContextMenuOptions option)

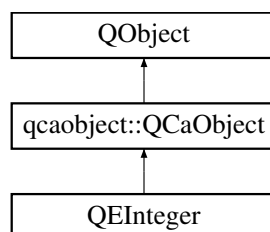
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/QEImageOptionsDialog.h
- qeframework/widgets/QEImage/QEImageOptionsDialog.cpp

9.178 QEInteger Class Reference

Inheritance diagram for QEInteger:



Public Slots

- void **writelnInteger** (const long &data)
- void **writelnIntegerElement** (const long &data)
- void **writelnInteger** (const QVector< long > &data)

Signals

- void **valueUpdated** (const [QEIntegerValueUpdate](#) &value)
- void **arrayUpdated** (const [QEIntegerArrayUpdate](#) &array)
- void **integerChanged** (const long &value, [QCaAlarmInfo](#) &alarmInfo, [QCaDateTime](#) &timeStamp, const unsigned int &variableIndex)
- void **integerArrayChanged** (const QVector< long > &values, [QCaAlarmInfo](#) &alarmInfo, [QCaDateTime](#) &timeStamp, const unsigned int &variableIndex)

Public Member Functions

- **QEInteger** (const QString &pvName, QObject *eventObject, [QEIntegerFormatting](#) *integerFormattingIn, unsigned int variableIndexIn)
- **QEInteger** (const QString &pvName, QObject *eventObject, [QEIntegerFormatting](#) *integerFormattingIn, unsigned int variableIndexIn, [UserMessage](#) *userMessageIn)

Additional Inherited Members

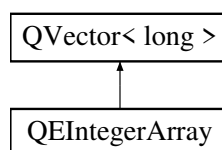
The documentation for this class was generated from the following files:

- qeframework/data/QEInteger.h
- qeframework/data/QEInteger.cpp

9.179 QEIntegerArray Class Reference

```
#include <QEIntegerArray.h>
```

Inheritance diagram for QEIntegerArray:



Public Member Functions

- **QEIntegerArray** (int size)
- **QEIntegerArray** (int size, const long &t)
- **QEIntegerArray** (const QVector< long > &other)
- [QEIntegerArray](#) & **operator=** (const QVector< long > &other)
- long **minimumValue** (const long &defaultValue=0)
- long **maximumValue** (const long &defaultValue=0)

9.179.1 Detailed Description

This class provides short hand for `QVector<long>` together with some basic long vector operations.

The documentation for this class was generated from the following files:

- `qeframework/data/QEIntegerArray.h`
- `qeframework/data/QEIntegerArray.cpp`

9.180 QEIntegerArrayUpdate Struct Reference

Public Attributes

- `QVector< long > values`
- `QCaAlarmInfo alarmInfo`
- `QCaDateTime timeStamp`
- `unsigned int variableIndex`
- `bool isMetaUpdate`

The documentation for this struct was generated from the following file:

- `qeframework/data/QEInteger.h`

9.181 QEIntegerFormatting Class Reference

```
#include <QEIntegerFormatting.h>
```

Public Member Functions

- `long formatInteger (const QVariant &value) const`
- `long formatInteger (const QVariant &value, int arrayIndex) const`
- `QVector< long > formatIntegerArray (const QVariant &value) const`
- `QVariant formatValue (const long &integerValue) const`
- `QVariant formatValue (const QVector< long > &integerValue) const`

9.181.1 Detailed Description

This class holds formatting instructions and uses them to convert to/from an integer and a `QVariant` of any type. It is generally set up with it's formatting instructions and then passed to a `QEInteger` class that will sink and source integer data to widgets or other code. It is used to convert data to and from a `QEChannel` (which sources and sinks data in the form of a `QVariant` where the `QVariant` reflects the underlying variable data type) and the `QEInteger` class. An example of a requirement for integer data is a combo box which must determine an integer index to select a menu option.

9.181.2 Member Function Documentation

9.181.2.1 formatInteger()

```
long QEIntegerFormatting::formatInteger (
    const QVariant & value ) const
```

This is used to convert the QVariant value received from a QEChannel, which is still based on the data variable type, to an integer.

9.181.2.2 formatIntegerArray()

```
QVector< long > QEIntegerFormatting::formatIntegerArray (
    const QVariant & value ) const
```

This is used to convert the QVariant value received from a QEChannel, which is still based on the data variable type, to an integer array. Typically used where the input QVariant value is an array of data values, but will work for any QVariant type.

9.181.2.3 formatValue()

```
QVariant QEIntegerFormatting::formatValue (
    const long & integerValue ) const
```

Given an integer value, format it as a data value of the specified type, according to the formatting instructions held by the class. This is used when writing integer data to a QEChannel.

The documentation for this class was generated from the following files:

- qeframework/data/QEIntegerFormatting.h
- qeframework/data/QEIntegerFormatting.cpp

9.182 QEIntegerValueUpdate Struct Reference

Public Attributes

- long **value**
- [QCaAlarmInfo](#) **alarmInfo**
- [QCaDateTime](#) **timeStamp**
- unsigned int **variableIndex**
- bool **isMetaUpdate**

The documentation for this struct was generated from the following file:

- qeframework/data/QEInteger.h

Public Member Functions

- [QELabel](#) (QWidget *parent=0)
- [QELabel](#) (const QString &variableName, QWidget *parent=0)
- [~QELabel](#) ()
Destructor.
- void [setFormatProperty](#) (QE::Formats format)
Access function for [format](#) property - refer to [format](#) property for details.
- [QE::Formats getFormatProperty](#) () const
Access function for [format](#) property - refer to [format](#) property for details.

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- QE::UserLevels [userLevelVisibility](#)
- QE::UserLevels [userLevelEnabled](#)
- QE::DisplayAlarmStateOptions [displayAlarmStateOption](#)
- bool [oosAware](#)
- int [precision](#)
- bool [useDbPrecision](#)
- int [leadingZeros](#)
- bool [addUnits](#)
- bool [forceSign](#)
- QString [localEnumeration](#)
- QE::Formats [format](#)
- int [radix](#)
- QE::Separators [separator](#)
- QE::Notations [notation](#)
- QE::ArrayActions [arrayAction](#)
- QString **text**
- UpdateOptions [updateOption](#)
- QPixmap [pixmap0](#)
- QPixmap [pixmap1](#)
- QPixmap [pixmap2](#)
- QPixmap [pixmap3](#)
- QPixmap [pixmap4](#)
- QPixmap [pixmap5](#)
- QPixmap [pixmap6](#)
- QPixmap [pixmap7](#)
- QPixmap [pixmap8](#)
- QPixmap [pixmap9](#)

- QPixmap [pixmap10](#)
- QPixmap [pixmap11](#)
- QPixmap [pixmap12](#)
- QPixmap [pixmap13](#)
- QPixmap [pixmap14](#)
- QPixmap [pixmap15](#)

Additional Inherited Members

9.183.1 Detailed Description

This class is a EPICS aware label widget based on the Qt label widget. When a variable is defined, the label text (or optionally the background pixmap) will be updated. The label will be disabled if the variable is invalid. It is tightly integrated with the base class [QEWidget](#) which provides generic support such as macro substitutions, drag/drop, and standard properties.

9.183.2 Member Enumeration Documentation

9.183.2.1 UpdateOptions

```
enum QELabel::UpdateOptions
```

Options for updating the label. The formatted text is used to update the label text, or select a background pixmap.

Enumerator

Text	Update the label text.
Picture	Update the label background pixmap.

9.183.3 Constructor & Destructor Documentation

9.183.3.1 QELabel() [1/2]

```
QELabel::QELabel (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.183.3.2 QELabel() [2/2]

```
QELabel::QELabel (
    const QString & variableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.183.4 Member Function Documentation

9.183.4.1 dbValueChanged

```
void QELabel::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.183.4.2 setManagedVisible

```
void QELabel::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.183.5 Property Documentation

9.183.5.1 addUnits

```
bool QELabel::addUnits [read], [write]
```

If true (default), add engineering units supplied with the data.

9.183.5.2 allowDrop

```
bool QELabel::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.183.5.3 arrayAction

```
QE::ArrayActions QELabel::arrayAction [read], [write]
```

Text formatting option for array data. Default is ASCII. Options are:

- ASCII - treat array as a single text string. For example an array of three characters 'a' 'b' 'c' will be formatted as 'abc'.
- APPEND - treat array as an array of numbers and format a string containing them all with a space between each. For example, an array of three numbers 10, 11 and 12 will be formatted as '10 11 12'.
- INDEX - Extract a single item from the array. The item is then formatted as any other non array data would be. The item selected is determined by the arrayIndex property. For example, if arrayIndex property is 1, an array of three numbers 10, 11 and 12 will be formatted as '11'.

9.183.5.4 arrayIndex

```
int QELabel::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.183.5.5 defaultStyle

```
QString QELabel::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.183.5.6 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QELabel::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.183.5.7 elementsRequired

```
int QELabel::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.183.5.8 forceSign

```
bool QELabel::forceSign [read], [write]
```

If false (default), no "+" sign, when true always add a sign.

9.183.5.9 format

```
QE::Formats QELabel::format [read], [write]
```

Format to apply to data. Default is 'Default' in which case the data type supplied with the data determines how the data is formatted. For all other options, an attempt is made to format the data as requested (whatever its native form).

9.183.5.10 leadingZeros

```
int QELabel::leadingZeros [read], [write]
```

The minimum number of leading digits before any decimal/radix point. The default is 1.

9.183.5.11 localEnumeration

```
QString QELabel::localEnumeration [read], [write]
```

An enumeration list used to data values. Used only when the formatting option is 'local enumeration'. Value is converted to an integer and used to select a string from this list.

Format is:

```
[[<|<=|=|=|>|=|>]value1[*]: string1 , [[<|<=|=|=|>|=|>]value2[*]: string2 , [[<|<=|=|=|>|=|>]value3[*]: string3 , ...
```

Where: < Less than <= Less than or equal = Equal (default if no operator specified) >= Greather than or equal

Greater than

* Always match (used to specify default text)

Values may be numeric or textual Values do not have to be in any order, but first match wins Values may be quoted Strings may be quoted Consecutive values do not have to be present. Operator is assumed to be equality if not present. White space is ignored except within quoted strings. may be included in a string to indicate a line break

Examples are:

```
0:Off,1:On 0 : "Pump Running", 1 : "Pump not running" 0:"" , 1:"Warning!\nAlarm" <2:"Value is less than two", =2↵
:"Value is equal to two", >2:"Value is grater than 2" 3:"Beamline Available", *:"" "Pump Off": "OH NO!, the pump is OFF!", "Pump On": "It's OK, the pump is on"
```

The data value is converted to a string if no enumeration for that value is available. For example, if the local enumeration is '0:off,1:on', and a value of 10 is processed, the text generated is '10'. If a blank string is required, this should be explicit. for example, '0:off,1:on,10:'''

A range of numbers can be covered by a pair of values as in the following example: >=4:"Between 4 and 8", <=8↵
:"Between 4 and 8"

9.183.5.12 messageSourceId

```
unsigned int QELabel::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.183.5.13 notation

```
QE::Notations QELabel::notation [read], [write]
```

Notation used for numerical formatting. Default is fixed.

9.183.5.14 oosAware

```
bool QELabel::oosAware [read], [write]
```

Indicates whether the widget should respond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.183.5.15 pixmap0

```
QPixmap QELabel::pixmap0 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 0.

9.183.5.16 pixmap1

```
QPixmap QELabel::pixmap1 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 1.

9.183.5.17 pixmap10

```
QPixmap QELabel::pixmap10 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 10.

9.183.5.18 pixmap11

```
QPixmap QELabel::pixmap11 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 11.

9.183.5.19 pixmap12

```
QPixmap QELabel::pixmap12 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 12.

9.183.5.20 pixmap13

```
QPixmap QELabel::pixmap13 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 13.

9.183.5.21 pixmap14

```
QPixmap QELabel::pixmap14 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 14.

9.183.5.22 pixmap15

```
QPixmap QELabel::pixmap15 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 15.

9.183.5.23 pixmap2

```
QPixmap QELabel::pixmap2 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 2.

9.183.5.24 pixmap3

```
QPixmap QELabel::pixmap3 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 3.

9.183.5.25 pixmap4

```
QPixmap QELabel::pixmap4 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 4.

9.183.5.26 pixmap5

```
QPixmap QELabel::pixmap5 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 5.

9.183.5.27 pixmap6

```
QPixmap QELabel::pixmap6 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 6.

9.183.5.28 pixmap7

```
QPixmap QELabel::pixmap7 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 7.

9.183.5.29 pixmap8

```
QPixmap QELabel::pixmap8 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 8.

9.183.5.30 pixmap9

```
QPixmap QELabel::pixmap9 [read], [write]
```

Pixmap displayed when updateOption property is 'Picture' and data is interpreted as 9.

9.183.5.31 precision

```
int QELabel::precision [read], [write]
```

Precision used when formatting floating point numbers. The default is 4. This is only used if useDbPrecision is false.

9.183.5.32 radix

```
int QELabel::radix [read], [write]
```

Base used for when formatting integers. Default is 10 (duh!)

9.183.5.33 separator

```
QE::Separators QELabel::separator [read], [write]
```

Seperators used for interger and fixed point formatting. Default is None.

9.183.5.34 `styleSheet`

```
QString QELabel::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.183.5.35 `updateOption`

```
UpdateOptions QELabel::updateOption [read], [write]
```

Determines if data updates the label text, or the label pixmap. For both options all normal string formatting is applied. If Text, the formatted text is simply presented as the label text. If Picture, the FORMATTED text is then interpreted as an integer and used to select one of the pixmaps specified by properties pixmap0 through to pixmap7.

9.183.5.36 `useDbPrecision`

```
bool QELabel::useDbPrecision [read], [write]
```

If true (default), format floating point numbers using the precision supplied with the data. If false, the precision property is used.

9.183.5.37 `userLevelEnabled`

```
QE::UserLevels QELabel::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.183.5.38 `userLevelEngineerStyle`

```
QString QELabel::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.183.5.39 `userLevelScientistStyle`

```
QString QELabel::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.183.5.40 `userLevelUserStyle`

```
QString QELabel::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.183.5.41 `userLevelVisibility`

```
QE::UserLevels QELabel::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.183.5.42 `variable`

```
QString QELabel::variable [read], [write]
```

EPICS variable name (CA PV)

9.183.5.43 `variableAsToolTip`

```
bool QELabel::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.183.5.44 `variableSubstitutions`

```
QString QELabel::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.183.5.45 `visible`

```
bool QELabel::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

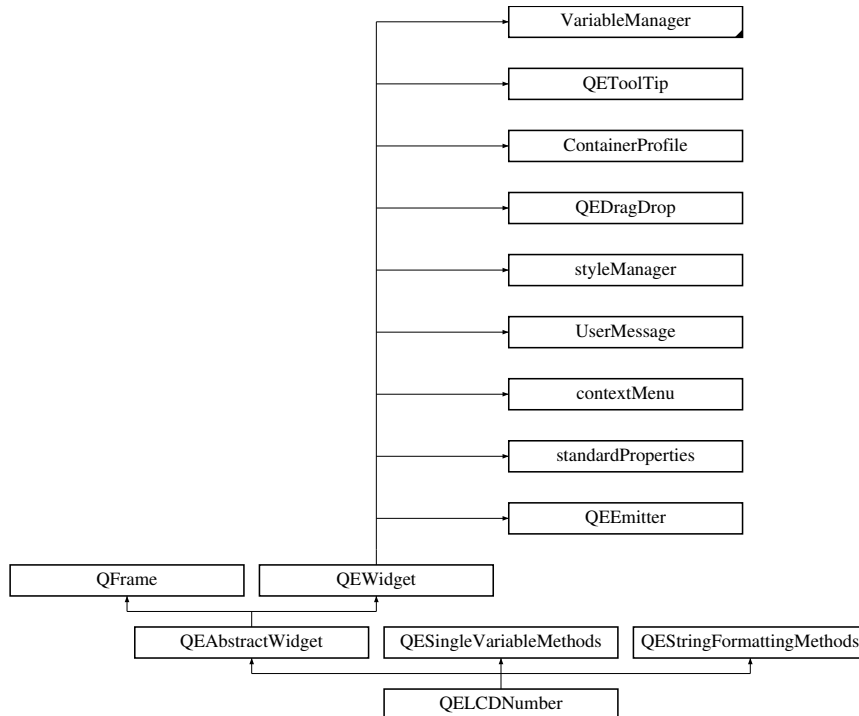
The documentation for this class was generated from the following files:

- `qeframework/widgets/QELabel/QELabel.h`
- `qeframework/widgets/QELabel/QELabel.cpp`

9.184 QELCDNumber Class Reference

```
#include <QELCDNumber.h>
```

Inheritance diagram for QELCDNumber:



Signals

- void [dbValueChanged](#) ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void [dbConnectionChanged](#) (const bool &isConnected)
- void [requestResend](#) ()

Public Member Functions

- [QELCDNumber](#) (QWidget *parent=0)
- [QELCDNumber](#) (const QString &variableName, QWidget *parent=0)
- virtual [~QELCDNumber](#) ()
Destruction.
- QSize [sizeHint](#) () const
Size hint for designer.
- void [setNotationProperty](#) (QE::Notations notation)
Access function for [notation](#) property - refer to [notation](#) property for details.
- QE::Notations [getNotationProperty](#) () const
Access function for [notation](#) property - refer to [notation](#) property for details.

Protected Member Functions

- [QEChannel](#) * [createQcaltem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void [stringFormattingChange](#) ()
- void [dragEnterEvent](#) (QDragEnterEvent *event)
- void [dropEvent](#) (QDropEvent *event)
- void [mousePressEvent](#) (QMouseEvent *event)
- QString [copyVariable](#) ()
- QVariant [copyData](#) ()
- void [paste](#) (QVariant v)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- bool [smallDecimalPoint](#)
- QLCDNumber::SegmentStyle [segmentStyle](#)
- int [precision](#)
- bool [useDbPrecision](#)
- [QE::Notations](#) [notation](#)

Additional Inherited Members

9.184.1 Detailed Description

This class provides an EPICS aware extention to the QLCDNumber widget.

9.184.2 Constructor & Destructor Documentation

9.184.2.1 QELCDNumber() [1/2]

```
QLCDNumber::QLCDNumber (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.184.2.2 QELCDNumber() [2/2]

```
QELCDNumber::QELCDNumber (
    const QString & variableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.184.3 Member Function Documentation

9.184.3.1 dbConnectionChanged

```
void QELCDNumber::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.184.3.2 dbValueChanged

```
void QELCDNumber::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.184.3.3 requestResend

```
void QELCDNumber::requestResend ( ) [signal]
```

Internal use only. Used when changing a property value to force a re-display to reflect the new property value.

9.184.4 Property Documentation

9.184.4.1 arrayIndex

```
int QELCDNumber::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.184.4.2 elementsRequired

```
int QELCDNumber::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.184.4.3 notation

```
QE::Notations QELCDNumber::notation [read], [write]
```

Notation used for numerical formatting. Default is fixed.

9.184.4.4 precision

```
int QELCDNumber::precision [read], [write]
```

Precision used when formatting floating point numbers. The default is 4. This is only used if useDbPrecision is false.

9.184.4.5 useDbPrecision

```
bool QELCDNumber::useDbPrecision [read], [write]
```

If true (default), format floating point numbers using the precision supplied with the data. If false, the precision property is used.

9.184.4.6 variable

```
QString QELCDNumber::variable [read], [write]
```

EPICS variable name (CA PV)

9.184.4.7 variableSubstitutions

```
QString QELCDNumber::variableSubstitutions [read], [write]
```

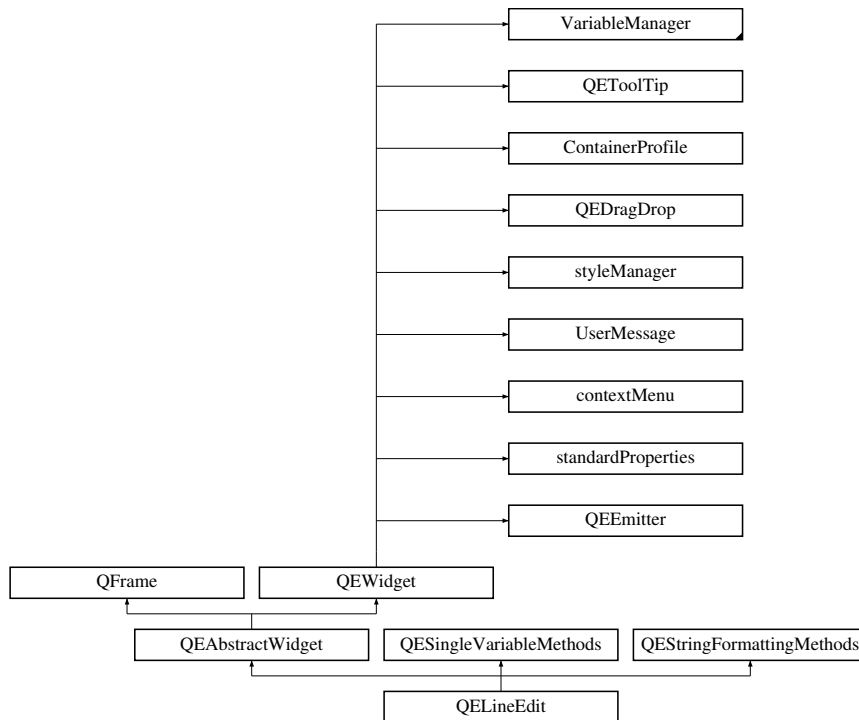
Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

The documentation for this class was generated from the following files:

- qeframework/widgets/QELCDNumber/QELCDNumber.h
- qeframework/widgets/QELCDNumber/QELCDNumber.cpp

9.185 QLEdit Class Reference

Inheritance diagram for QLEdit:



Public Slots

- void **writeNow** ()
- void **setDefaultStyle** (const QString &style)
Update the default style applied to this widget.
- void **setPvValue** (const QString &text)
- void **setPvValue** (const int value)
- void **setPvValue** (const double value)
- void **setPvValue** (const bool value)

Signals

- void **dbValueChanged** ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void **dbConnectionChanged** (const bool &isConnected)
- void **userChange** (const QString &oldValue, const QString &newValue, const QString &lastValue)
- void **requestResend** ()
- void **textChanged** (const QString &)
From interbal widget.
- void **textEdited** (const QString &)

Public Member Functions

- void **setFormatProperty** ([QE::Formats](#) format)
Access function for [format](#) property - refer to [format](#) property for details.
- [QE::Formats](#) **getFormatProperty** () const
Access function for [format](#) property - refer to [format](#) property for details.
- [QELineEdit](#) (QWidget *parent=0)
- [QELineEdit](#) (const QString &variableName, QWidget *parent=0)
- virtual [~QELineEdit](#) ()
Destruction.
- QString **getValue** () const
- void **setValue** (const QString &value, const bool isUserUpdate=false)
- void **setUseApplyButton** (const bool)
- bool **getUseApplyButton** () const
- void **setApplyButtonText** (const QString &text)
- QString **getApplyButtonText** () const
- void **setSubscribe** (const bool)
- bool **getSubscribe** () const
- void **setWriteOnLoseFocus** (const bool)
- bool **getWriteOnLoseFocus** () const
- void **setWriteOnEnter** (const bool)
- bool **getWriteOnEnter** () const
- void **setWriteOnFinish** (const bool)
- bool **getWriteOnFinish** () const
- void **setConfirmWrite** (const bool)
- bool **getConfirmWrite** () const
- void **setAllowFocusUpdate** (const bool)
- bool **getAllowFocusUpdate** () const
- void **setDropOption** ([QE::DropOption](#) option)
- [QE::DropOption](#) **getDropOption** () const

Protected Member Functions

- bool **eventFilter** (QObject *watched, QEvent *event)
- void **focusInEvent** (QFocusEvent *event)
- void **resizeEvent** (QResizeEvent *event)
- QMenu * **getDefaultContextMenu** ()
Flag indicating this widget is running inside Qt's 'designer'.
- void **stringFormattingChange** ()
- void **activated** ()
Do any post-all-widgets-constructed, i.e. activated stuff.
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- [QEChannel](#) * **createQcaltem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Properties

- QString [variable](#)
 - QString [variableSubstitutions](#)
 - int [elementsRequired](#)
 - int [arrayIndex](#)
 - QString [inputMask](#)
- Refer to QLElineEdit documentation.*
- int [maxLength](#)
 - bool [frame](#)
 - QLElineEdit::EchoMode **echoMode**
 - Qt::Alignment [alignment](#)
 - bool **readOnly**
 - QString **placeholderText**
 - Qt::CursorMoveStyle **cursorMoveStyle**
 - bool **clearButtonEnabled**
 - int [precision](#)
 - bool [useDbPrecision](#)
 - int [leadingZeros](#)
 - bool [addUnits](#)
 - bool [forceSign](#)
 - QString [localEnumeration](#)
 - QE::Formats [format](#)
 - int [radix](#)
 - QE::Separators [separator](#)
 - QE::Notations [notation](#)
 - QE::ArrayActions [arrayAction](#)
 - bool [subscribe](#)
 - bool [useApplyButton](#)
 - QString [applyButtonText](#)
 - bool [writeOnLoseFocus](#)
 - bool [writeOnEnter](#)
 - bool [writeOnFinish](#)
 - bool [confirmWrite](#)
 - bool [allowFocusUpdate](#)
 - QE::DropOption [dropOption](#)

Allow drag/drops mode to this widget.

Additional Inherited Members

9.185.1 Constructor & Destructor Documentation

9.185.1.1 QLElineEdit() [1/2]

```
QLElineEdit::QLElineEdit (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.185.1.2 QLELineEdit() [2/2]

```
QLELineEdit::QLELineEdit (
    const QString & variableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.185.2 Member Function Documentation

9.185.2.1 dbConnectionChanged

```
void QLELineEdit::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.185.2.2 dbValueChanged

```
void QLELineEdit::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.185.2.3 requestResend

```
void QLELineEdit::requestResend ( ) [signal]
```

Internal use only. Used when changing a property value to force a re-display to reflect the new property value.

9.185.2.4 setPvValue

```
void QLELineEdit::setPvValue (
    const QString & text ) [slot]
```

Update the widget and write to the main associated control process variable. Supplied values are converted to the appropriate widget value type. If conversion not possible or resultant value out of range, no widget/PV update occurs.

9.185.2.5 userChange

```
void QLELineEdit::userChange (
    const QString & oldValue,
    const QString & newValue,
    const QString & lastValue ) [signal]
```

Internal use only. Used by [QEConfiguredLayout](#) to be notified when one of its widgets has written something.

9.185.3 Property Documentation

9.185.3.1 addUnits

```
bool QELineEdit::addUnits [read], [write]
```

If true (default), add engineering units supplied with the data.

9.185.3.2 alignment

```
Qt::Alignment QELineEdit::alignment [read], [write]
```

This property holds the alignment of the line edit. Both horizontal and vertical alignment is allowed here, Qt::AlignLeft will map to Qt::AlignLeft. By default, this property contains a combination of Qt::AlignLeft and Qt::AlignVCenter.

9.185.3.3 allowFocusUpdate

```
bool QELineEdit::allowFocusUpdate [read], [write]
```

Allow updated while widget has focus - defaults to false.

9.185.3.4 applyButtonText

```
QString QELineEdit::applyButtonText [read], [write]
```

The text displayed on the apply button. This default text is "A".

9.185.3.5 arrayAction

```
QE::ArrayActions QELineEdit::arrayAction [read], [write]
```

Text formatting option for array data. Default is ASCII. Options are:

- ASCII - treat array as a single text string. For example an array of three characters 'a' 'b' 'c' will be formatted as 'abc'.
- APPEND - treat array as an array of numbers and format a string containing them all with a space between each. For example, an array of three numbers 10, 11 and 12 will be formatted as '10 11 12'.
- INDEX - Extract a single item from the array. The item is then formatted as any other non array data would be. The item selected is determined by the arrayIndex property. For example, if arrayIndex property is 1, an array of three numbers 10, 11 and 12 will be formatted as '11'.

9.185.3.6 arrayIndex

```
int QELineEdit::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.185.3.7 confirmWrite

```
bool QELineEdit::confirmWrite [read], [write]
```

Sets if this widget will ask for confirmation (using a dialog box) prior to writing data. Default is 'false' (will not ask for confirmation (using a dialog box) prior to writing data). Note: writeOnChange and confirmWrite are mutually exclusive.

9.185.3.8 dropOption

```
QE::DropOption QELineEdit::dropOption [read], [write]
```

Allow drag/drops mode to this widget.

Option: DropToText - just set text on the widget Option: DropToTextAndWrite - write the text to the PV if the PV has been connected otherwise equivalent to DropToText

9.185.3.9 elementsRequired

```
int QELineEdit::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.185.3.10 forceSign

```
bool QELineEdit::forceSign [read], [write]
```

If false (default), no "+" sign, when true always add a sign.

9.185.3.11 format

```
QE::Formats QELineEdit::format [read], [write]
```

Format to apply to data. Default is 'Default' in which case the data type supplied with the data determines how the data is formatted. For all other options, an attempt is made to format the data as requested (whatever its native form).

9.185.3.12 frame

```
bool QELineEdit::frame [read], [write]
```

This property holds whether the numeric edit draws itself with a frame. If enabled (the default) the numeric edit draws itself inside a frame, otherwise the line edit draws itself without any frame.

9.185.3.13 leadingZeros

```
int QELineEdit::leadingZeros [read], [write]
```

The minimum number of leading digits before any decimal/radix point. The default is 1.

9.185.3.14 localEnumeration

```
QString QELineEdit::localEnumeration [read], [write]
```

An enumeration list used to data values. Used only when the formatting option is 'local enumeration'. Value is converted to an integer and used to select a string from this list.

Format is:

```
[[<|<=|=|!=|>|=|>]value1|*]: string1 , [[<|<=|=|!=|>|=|>]value2|*]: string2 , [[<|<=|=|!=|>|=|>]value3|*]: string3 , ...
```

Where: < Less than <= Less than or equal = Equal (default if no operator specified) >= Greather than or equal

Greater than

* Always match (used to specify default text)

Values may be numeric or textual Values do not have to be in any order, but first match wins Values may be quoted Strings may be quoted Consecutive values do not have to be present. Operator is assumed to be equality if not present. White space is ignored except within quoted strings. may be included in a string to indicate a line break

Examples are:

```
0:Off,1:On 0 : "Pump Running", 1 : "Pump not running" 0:"", 1:"Warning!\nAlarm" <2:"Value is less than two", =2↵
:"Value is equal to two", >2:"Value is grater than 2" 3:"Beamline Available", *:"" "Pump Off": "OH NO!, the pump is OFF!", "Pump On": "It's OK, the pump is on"
```

The data value is converted to a string if no enumeration for that value is available. For example, if the local enumeration is '0:off,1:on', and a value of 10 is processed, the text generated is '10'. If a blank string is required, this should be explicit. for example, '0:off,1:on,10:'''

A range of numbers can be covered by a pair of values as in the following example: >=4:"Between 4 and 8", <=8↵
:"Between 4 and 8"

9.185.3.15 maxLength

```
int QELineEdit::maxLength [read], [write]
```

Defines the maximum input text length.

9.185.3.16 notation

`QE::Notations QELineEdit::notation [read], [write]`

Notation used for numerical formatting. Default is fixed.

9.185.3.17 precision

`int QELineEdit::precision [read], [write]`

Precision used when formatting floating point numbers. The default is 4. This is only used if `useDbPrecision` is false.

9.185.3.18 radix

`int QELineEdit::radix [read], [write]`

Base used for when formatting integers. Default is 10 (duh!)

9.185.3.19 separator

`QE::Separators QELineEdit::separator [read], [write]`

Seperators used for interger and fixed point formatting. Default is None.

9.185.3.20 subscribe

`bool QELineEdit::subscribe [read], [write]`

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.185.3.21 useApplyButton

`bool QELineEdit::useApplyButton [read], [write]`

When set true, internal apply button is exposed and used to trigger write. All other write are options set false and inhibited from being set true.

9.185.3.22 useDbPrecision

`bool QELineEdit::useDbPrecision [read], [write]`

If true (default), format floating point numbers using the precision supplied with the data. If false, the precision property is used.

9.185.3.23 variable

```
QString QELineEdit::variable [read], [write]
```

EPICS variable name (CA PV)

9.185.3.24 variableSubstitutions

```
QString QELineEdit::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.185.3.25 writeOnEnter

```
bool QELineEdit::writeOnEnter [read], [write]
```

Sets if this widget writes any changes when the user presses 'enter'. Note, the current value will be written even if the user has not changed it. Default is 'true' (writes any changes when the user presses 'enter').

9.185.3.26 writeOnFinish

```
bool QELineEdit::writeOnFinish [read], [write]
```

Sets if this widget writes any changes when the user finished editing (the underlying QLineEdit 'editingFinished' signal is emitted). No writing occurs if no changes were made. Default is 'true' (writes any changes when the QLineEdit 'editingFinished' signal is emitted).

9.185.3.27 writeOnLoseFocus

```
bool QELineEdit::writeOnLoseFocus [read], [write]
```

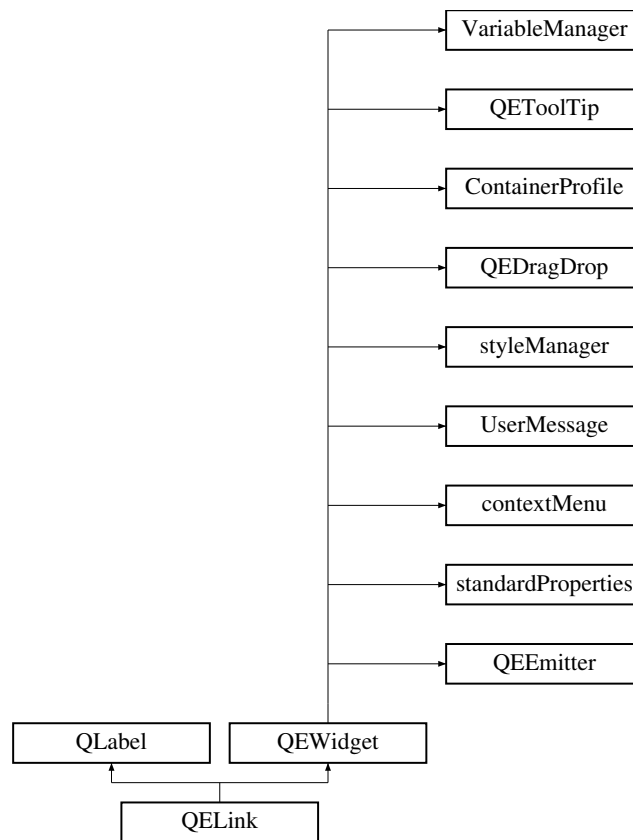
Sets if this widget automatically writes any changes when it loses focus. Default is 'false' (does not write any changes when it loses focus).

The documentation for this class was generated from the following files:

- qeframework/widgets/QELineEdit/QELineEdit.h
- qeframework/widgets/QELineEdit/QELineEdit.cpp

9.186 QELink Class Reference

Inheritance diagram for QELink:



Public Types

- enum **ConditionNames** {
Equal = 0 , **NotEqual** , **GreaterThan** , **GreaterThanOrEqual** ,
LessThan , **LessThanOrEqual** , **Lookup** }

Public Slots

- void **in** (const bool &in)
- void **in** (const int &in)
- void **in** (const long &in)
- void **in** (const qlonglong &in)
- void **in** (const double &in)
- void **in** (const QString &in)
- void **autoFillBackground** (const bool &enable)

Signals

- void **out** (const bool &out)
- void **out** (const int &out)
- void **out** (const long &out)
- void **out** (const qlonglong &out)
- void **out** (const double &out)
- void **out** (const QString &out)

Public Member Functions

- **QELink** (QWidget *parent=0)
- void **setCondition** (const ConditionNames conditionIn)
- ConditionNames **getCondition** ()
- void **setComparisonValue** (const QString &comparisonValue)
- QString **getComparisonValue** () const
- void **setSignalTrue** (const bool signalTrue)
- bool **getSignalTrue** () const
- void **setSignalFalse** (const bool signalFalse)
- bool **getSignalFalse** () const
- void **setOutTrueValue** (const QString &outTrueValue)
- QString **getOutTrueValue** () const
- void **setOutFalseValue** (const QString &outFalseValue)
- QString **getOutFalseValue** () const
- void **setLookupValues** (const QStringList &lookupValues)
- QStringList **getLookupValues** () const

Protected Attributes

- QVariant **comparisonValue**
- QVariant **outTrueValue**
- QVariant **outFalseValue**

Properties

- ConditionNames **condition**
- QString **comparisonValue**
- bool **signalTrue**
- bool **signalFalse**
- QString **outTrueValue**
- QString **outFalseValue**
- QStringList **lookupValues**
- bool **runVisible**

Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QELink/QELink.h
- qeframework/widgets/QELink/QELink.cpp

9.187 QELocalEnumeration Class Reference

```
#include <QELocalEnumeration.h>
```

Public Member Functions

- [QELocalEnumeration](#) ()
- [QELocalEnumeration](#) (const QString &localEnumeration)
- void [setLocalEnumeration](#) (const QString &localEnumeration)
- QString [getLocalEnumeration](#) () const
- bool [isDefined](#) () const
- QString [valueToText](#) (const QVariant &value, bool &match) const
- QVariant [textToValue](#) (const QString &text, bool &ok) const
- int [textToInt](#) (const QString &text, bool &ok) const
- double [textToDouble](#) (const QString &text, bool &ok) const

9.187.1 Detailed Description

This class allows a user defined two-way value to enumeration map. The map is define using a single string, typically a widget property string. This may then be used to replace the enumeration values provided by EPICS and/or provide an enueration set of more that 16 values. See [setLocalEnumeration\(\)](#) for the use of 'localEnumeration'.

This functionality that this class provided was formerly embedded within [QStringFormatting](#).

9.187.2 Constructor & Destructor Documentation

9.187.2.1 QELocalEnumeration() [1/2]

```
QELocalEnumeration::QELocalEnumeration ( )
```

Constructors

9.187.2.2 QELocalEnumeration() [2/2]

```
QELocalEnumeration::QELocalEnumeration (
    const QString & localEnumeration )
```

Constructor with localEnumeration

9.187.3 Member Function Documentation

9.187.3.1 getLocalEnumeration()

```
QString QELocalEnumeration::getLocalEnumeration ( ) const
```

Get the local enumeration strings. See [setLocalEnumeration\(\)](#) for the use of 'localEnumeration'.

9.187.3.2 isDefined()

```
bool QELocalEnumeration::isDefined ( ) const
```

Evaluates: `getLocalEnumeration.count() > 0`

9.187.3.3 setLocalEnumeration()

```
void QELocalEnumeration::setLocalEnumeration (
    const QString & localEnumeration )
```

Parse the local enumeration string.

Format is:

```
[[<|<=|=|!=|>|=|>]value1|*]: string1 , [[<|<=|=|!=|>|=|>]value2|*]: string2 , [[<|<=|=|!=|>|=|>]value3|*]: string3
, ...
```

Where: < Less than <= Less than or equal = Equal (default if no operator specified) >= Greather than or equal

Greater than

* Always match (used to specify default text)

Values may be numeric or textual Values do not have to be in any order, but first match wins Values may be quoted Strings may be quoted Consecutive values do not have to be present. Operator is assumed to be equality if not present. White space is ignored except within quoted strings. may be included in a string to indicate a line break

Examples are:

```
0:Off,1:On 0 : "Pump Running", 1 : "Pump not running" 0:"" , 1:"Warning!\nAlarm" <2:"Value is less than two", =2↵
:"Value is equal to two", >2:"Value is grater than 2" 3:"Beamline Available", *:"" "Pump Off": "OH NO!, the pump is
OFF!", "Pump On": "It's OK, the pump is on"
```

The data value is converted to a string if no enumeration for that value is available. For example, if the local enumeration is '0:off,1:on', and a value of 10 is processed, the text generated is '10'. If a blank string is required, this should be explicit. for example, '0:off,1:on,10: ""'

```
A range of numbers can be covered by a pair of values as in the following example: >=4:"Between 4 and 8", <=8↵
:"Between 4 and 8"
```

Will completely re-initialises the object.

9.187.3.4 textToDouble()

```
double QELocalEnumeration::textToDouble (
    const QString & text,
    bool & ok ) const
```

Generate a double value given a string, using formatting defined within this class. If the value can be formatted the formatted value is returned and 'ok' is true. If the value can't be formatted then 0.0 is returned and 'ok' is false.

9.187.3.5 textToInt()

```
int QELocalEnumeration::textToInt (
    const QString & text,
    bool & ok ) const
```

Generate an integer value given a string, using formatting defined within this class. If the value can be formatted the formatted value is returned and 'ok' is true. If the value can't be formatted then 0 is returned and 'ok' is false.

9.187.3.6 textToValue()

```
QVariant QELocalEnumeration::textToValue (
    const QString & text,
    bool & ok ) const
```

Generate a value given a string, using formatting defined within this class. If the value can be formatted the formatted value is returned and 'ok' is true. If the value can't be formatted an error string is returned and 'ok' is false

9.187.3.7 valueToText()

```
QString QELocalEnumeration::valueToText (
    const QVariant & value,
    bool & match ) const
```

Format a variant value using local enumeration list. If the value is numeric, then the value is compared to the numeric interpretation of the enumeration values, if the value is textual, then the value is compared to the textual enumeration values.

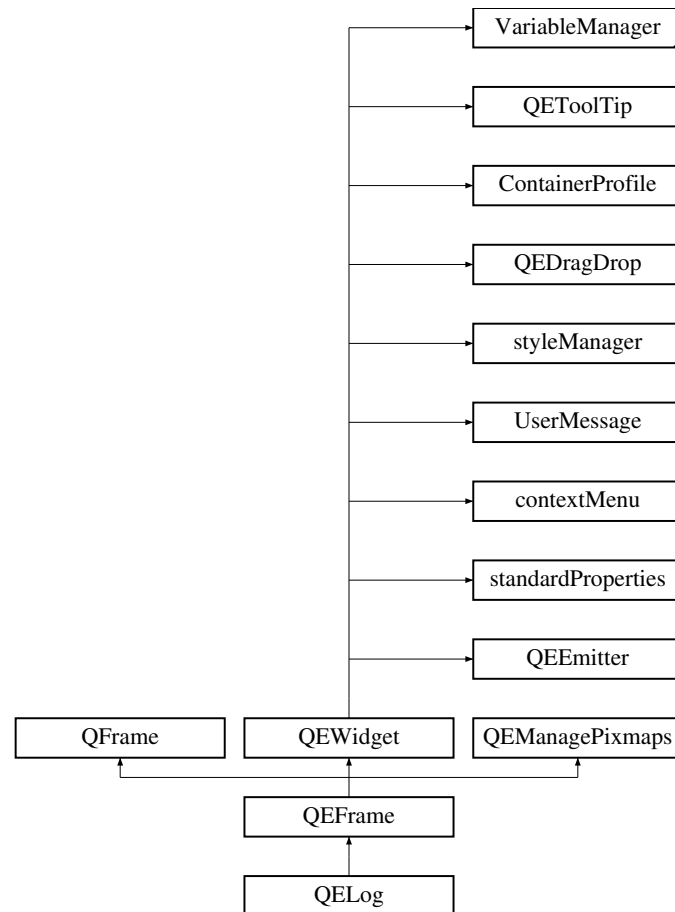
The documentation for this class was generated from the following files:

- qeframework/data/QELocalEnumeration.h
- qeframework/data/QELocalEnumeration.cpp

9.188 QELog Class Reference

```
#include <QELog.h>
```

Inheritance diagram for QELog:



Classes

- class [UserMessageReceiver](#)

Public Member Functions

- **QELog** (QWidget *parent=0)
- void **setShowColumnTime** (bool pValue)
- bool **getShowColumnTime** () const
- void **setShowColumnType** (bool pValue)
- bool **getShowColumnType** () const
- void **setShowColumnMessage** (bool pValue)
- bool **getShowColumnMessage** () const
- void **setShowMessageFilter** (bool pValue)
- bool **getShowMessageFilter** () const
- void **setShowClear** (bool pValue)
- bool **getShowClear** () const
- void **setShowSave** (bool pValue)
- bool **getShowSave** () const
- void **setOptionsLayout** (QE::LayoutOptions pValue)
- QE::LayoutOptions **getOptionsLayout** () const
- void **setScrollToBottom** (bool pValue)
- bool **getScrollToBottom** () const
- void **setMaster** (const bool isMaster)

- bool **getMaster** () const
- void **setInfoColor** (QColor pValue)
- QColor **getInfoColor** () const
- void **setWarningColor** (QColor pValue)
- QColor **getWarningColor** () const
- void **setErrorColor** (QColor pValue)
- QColor **getErrorColor** () const
- void **setMargin** (int margin)
- int **getMargin** () const
- void **setMessageFormFilter** (QE::MessageFilterOptions messageFormFilter)
- QE::MessageFilterOptions **getMessageFormFilter** () const
- void **setMessageSourceFilter** (QE::MessageFilterOptions messageSourceFilter)
- QE::MessageFilterOptions **getMessageSourceFilter** () const
- void **clearLog** ()
- void **addLog** (int pType, QString pMessage, const QDateTime &dateTime)
- void **refreshLog** ()

Static Public Member Functions

- static void **createUserMessageReceiver** ()

Protected Member Functions

- void **resizeEvent** (QResizeEvent *resizeEvent)
- void **newMessage** (QString msg, [message_types](#) type)
Virtual function to pass messages to derived classes (typically logging widgets or application windows)

Properties

- bool **showColumnTime**
- bool **showColumnType**
- bool **showColumnMessage**
- bool **showMessageFilter**
- bool **showClear**
- bool **showSave**
- bool **scrollToBottom**
- bool **master**
- QE::LayoutOptions [optionsLayout](#)
- int **margin**
- QColor **infoColor**
- QColor **warningColor**
- QColor **errorColor**
- QE::MessageFilterOptions [messageFormFilter](#)
- QE::MessageFilterOptions **messageSourceFilter**

Friends

- class **UserMessageReceiver**

Additional Inherited Members

9.188.1 Detailed Description

The [QELog](#) class provides the means to display messages to the user.

9.188.2 Property Documentation

9.188.2.1 messageFormFilter

`QE::MessageFilterOptions QELog::messageFormFilter [read], [write]`

Message properties Not all [QE](#) widgets need these properties (only those that do something with messages - like the logging widget)

9.188.2.2 optionsLayout

`QE::LayoutOptions QELog::optionsLayout [read], [write]`

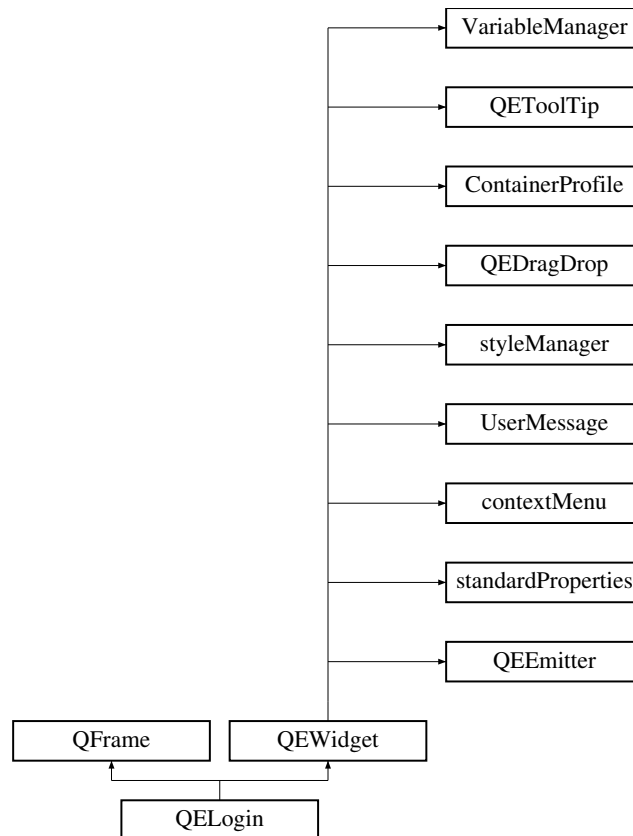
Provides the location of filter check boxes and save/clear buttons.

The documentation for this class was generated from the following files:

- qeframework/widgets/QELog/QELog.h
- qeframework/widgets/QELog/QELog.cpp

9.189 QELogin Class Reference

Inheritance diagram for QELogin:



Signals

- void **login** ()

Public Member Functions

- **QELogin** (QWidget *pParent=0)
- bool **login** ([QE::UserLevels](#) level, QString password)
- QString **getPriorityUserPassword** ()
- QString **getPriorityScientistPassword** ()
- QString **getPriorityEngineerPassword** ()
- void **setUserPassword** (QString pValue)
- QString **getUserPassword** ()
- void **setScientistPassword** (QString pValue)
- QString **getScientistPassword** ()
- void **setEngineerPassword** (QString pValue)
- QString **getEngineerPassword** ()
- void **setCompactStyle** (bool compactStyle)
- bool **getCompactStyle** ()
- void **setStatusOnly** (bool statusOnlyIn)
- bool **getStatusOnly** ()
- QString **getUserTypeName** ([QE::UserLevels](#) type)

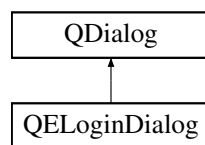
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QELogin/QELogin.h
- qeframework/widgets/QELogin/QELogin.cpp

9.190 QELoginDialog Class Reference

Inheritance diagram for QELoginDialog:



Public Member Functions

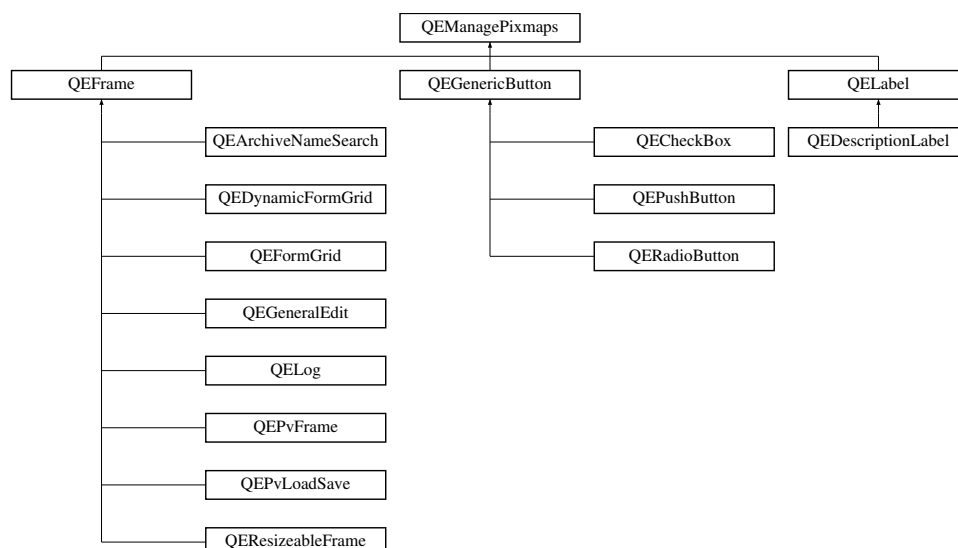
- **QELoginDialog** ([QELogin](#) *ownerIn)

The documentation for this class was generated from the following files:

- qeframework/widgets/QELogin/QELogin.h
- qeframework/widgets/QELogin/QELogin.cpp

9.191 QEManagePixmaps Class Reference

Inheritance diagram for QEManagePixmaps:



Public Types

- enum **Constants** { **NUM_PIXMAPS_MANAGED** = 16 }

Public Member Functions

- void **setDataPixmap** (const QPixmap &Pixmap, const int index)
- QPixmap **getDataPixmap** (const int index) const
- QPixmap **getDataPixmap** (const QString &value) const

Protected Member Functions

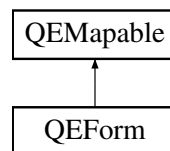
- virtual void **pixmapUpdated** (const int index)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/managePixmaps.h
- qeframework/widgets/QEWidget/managePixmaps.cpp

9.192 QEMapable Class Reference

Inheritance diagram for QEMapable:



Public Member Functions

- **QEMapable** (QWidget *ownerIn)
- bool **setFormHandle** (const QEFormMapper::FormHandles &formHandle)
- QEFormMapper::FormHandles **getFormHandle** () const

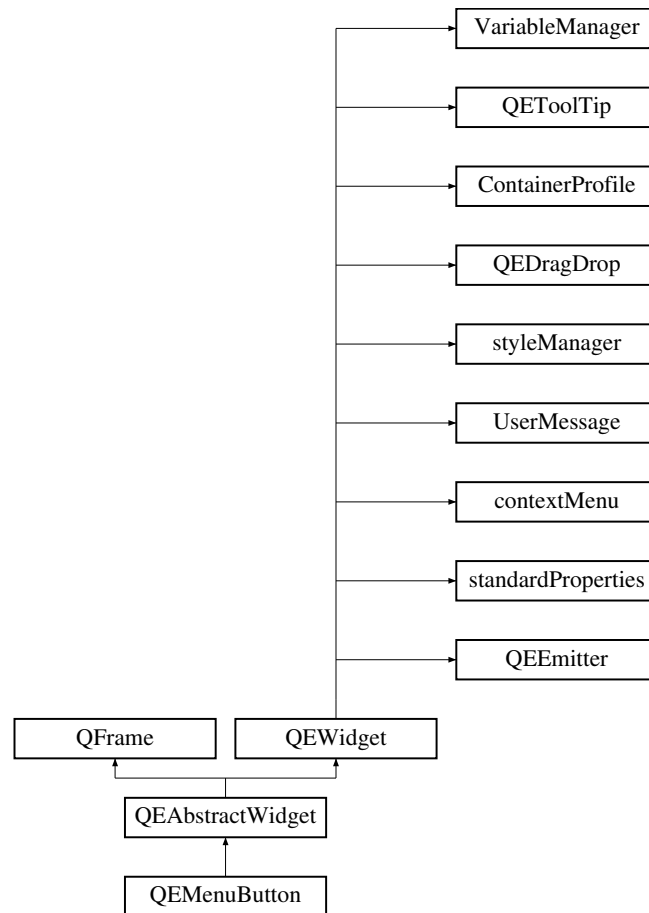
The documentation for this class was generated from the following files:

- qeframework/common/QEFormMapper.h
- qeframework/common/QEFormMapper.cpp

9.193 QEMenuButton Class Reference

```
#include <QEMenuButton.h>
```

Inheritance diagram for QEMenuButton:



Signals

- void `newGui` (const `QEActionRequests` &request)

Public Member Functions

- **QEMenuButton** (QWidget *parent=0)
- void **setSubstitutionsProperty** (const QString &substitutions)
- QString **getSubstitutionsProperty** () const
- void **setButtonTextProperty** (const QString &buttonText)
- QString **getButtonTextProperty** () const
- void **setIcon** (const QIcon &icon)
- QIcon **getIcon** () const
- void **setIconSize** (const QSize &size)
- QSize **getIconSize** () const
- void **setMenuString** (const QString &s)
- QString **getMenuString** () const

Protected Member Functions

- `QSize` **sizeHint** () const
- `bool` **eventFilter** (QObject *watched, QEvent *event)

Protected Attributes

- `QMenu *` **buttonMainMenu**
- `QString` **theMenuString**

Properties

- `bool` **allowDrop**
- `bool` **variableAsToolTip**
- `QE::DisplayAlarmStateOptions` **displayAlarmStateOption**
- `QString` **defaultSubstitutions**
- `QString` **labelText**
Button label text.
- `QIcon` **icon**
- `QSize` **iconSize**
- `QString` **menuEntries**
Specifies the menu entry values, encoded as an XML string.

Additional Inherited Members

9.193.1 Detailed Description

This widget provides a menu attached to a button. Each menu/submenu can trigger same items as a [QEPushButton](#), i.e: a) write a value to a PV (click only); and/or b) run a nominated program (with arguments); and/or c) open a specified ui file.

9.193.2 Member Function Documentation

9.193.2.1 newGui

```
void QEMenuButton::newGui (
    const QEActionRequests & request ) [signal]
```

Internal use only. Request a new GUI is created. Typically, this is caught by the QEGui or bespoke display manager application's requestAction slot.

9.193.3 Property Documentation

9.193.3.1 defaultSubstitutions

QString QEMenuButton::defaultSubstitutions [read], [write]

Default macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to program names and arguments etc.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEMenuButton/QEMenuButton.h
- qeframework/widgets/QEMenuButton/QEMenuButton.cpp

9.194 QEMenuButtonData Class Reference

Public Member Functions

- QVariant **toVariant** () const
- bool **setValue** (const QVariant &data)

Static Public Member Functions

- static QString **psoToString** (const [QE::ProgramStartupOptions](#) value)
- static [QE::ProgramStartupOptions](#) **stringToPso** (const QString &image)
- static QString **optionToString** (const [QE::CreationOptions](#) value)
- static [QE::CreationOptions](#) **stringToOption** (const QString &image)
- static QString **formatToString** (const [QE::Formats](#) value)
- static [QE::Formats](#) **stringToFormat** (const QString &image)
- static QString **join** (const QStringList &x)
- static QStringList **split** (const QString &x)

Public Attributes

- bool **separator**
- QString **programName**
- QStringList **programArguments**
- [QE::ProgramStartupOptions](#) **programStartupOption**
- QString **uiFilename**
- QString **prioritySubstitutions**
- [QE::CreationOptions](#) **creationOption**
- QString **customisationName**
- QString **variable**
- QString **variableValue**
- [QE::Formats](#) **format**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEMenuButton/QEMenuButtonData.h
- qeframework/widgets/QEMenuButton/QEMenuButtonData.cpp

9.195 QMenuItem Class Reference

Public Member Functions

- **QMenuItem** (const QString &name, const bool isSubMenuContainer, [QMenuButton](#) *owner, [QMenuItem](#) *parent)
- QString **getName** () const
- bool **getIsSubMenuContainer** () const
- QDomElement **createDomElement** (QDomDocument &doc) const
- bool **extractFromDomElement** (const QDomElement &element)
- QAction * **constructAction** (QMenu *parent)
- QMenu * **constructMenu** (QMenu *parent)
- int **columnCount** () const
- [QMenuItem](#) * **getChild** (const int position) const
- [QMenuItem](#) * **getParent** () const
- int **childCount** () const
- int **childPosition** () const
- QVariant **getData** (const int column) const
- bool **insertChild** (const int position, [QMenuItem](#) *child)
- bool **removeChildren** (const int position, const int count)

Public Attributes

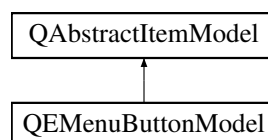
- [QMenuButton](#) * **owner**
- QString **name**
- bool **isSubMenuContainer**
- [QMenuButtonData](#) **data**

The documentation for this class was generated from the following files:

- qeframework/widgets/QMenuButton/QMenuItem.h
- qeframework/widgets/QMenuButton/QMenuItem.cpp

9.196 QMenuButtonModel Class Reference

Inheritance diagram for QMenuButtonModel:



Public Member Functions

- **QMenuButtonModel** ([QMenuButton](#) *parent)
- bool **parseXml** (const QString &xml)
- QString **serialiseXml** () const
- bool **constructMenu** (QMenu *rootMenu)
- [QMenuButtonItem](#) * **indexToItem** (const QModelIndex &index) const
- QModelIndex **getIndex** (const [QMenuButtonItem](#) *item) const
- bool **addItemToModel** ([QMenuButtonItem](#) *item, [QMenuButtonItem](#) *parentItem, const int position=-1)
- bool **removeItemFromModel** ([QMenuButtonItem](#) *item)
- void **modelUpdated** ()
- void **itemUpdated** (const [QMenuButtonItem](#) *item)

Static Public Member Functions

- static QString **emptyXml** ()

Protected Member Functions

- QVariant **data** (const QModelIndex &index, int role) const
- QVariant **headerData** (int section, Qt::Orientation orientation, int role=Qt::DisplayRole) const
- QModelIndex **index** (int row, int column, const QModelIndex &parent=QModelIndex()) const
- QModelIndex **parent** (const QModelIndex &child) const
- int **rowCount** (const QModelIndex &parent=QModelIndex()) const
- int **columnCount** (const QModelIndex &parent=QModelIndex()) const
- Qt::ItemFlags **flags** (const QModelIndex &index) const
- bool **setData** (const QModelIndex &index, const QVariant &value, int role=Qt::EditRole)
- bool **setHeaderData** (int section, Qt::Orientation orientation, const QVariant &value, int role=Qt::EditRole)
- bool **insertRows** (int position, int rows, const QModelIndex &parent=QModelIndex())
- bool **removeRows** (int position, int rows, const QModelIndex &parent=QModelIndex())
- Qt::DropActions **supportedDropActions** () const
- QStringList **mimeTypes** () const
- QMimeData * **mimeData** (const QModelIndexList &indexes) const
- bool **dropMimeData** (const QMimeData *data, Qt::DropAction action, int row, int column, const QModelIndex &parent)

The documentation for this class was generated from the following files:

- qeframework/widgets/QMenuButton/QMenuButtonModel.h
- qeframework/widgets/QMenuButton/QMenuButtonModel.cpp

9.197 QENTNDArrayData Class Reference

```
#include <QENTNDArrayData.h>
```

Public Member Functions

- **QENTNDArrayData** (const [QENTNDArrayData](#) &other)
- [QENTNDArrayData](#) & **operator=** (const [QENTNDArrayData](#) &other)
- QByteArray **getData** () const
- QString **getColourMode** () const
- QString **getDataType** () const
- int **getNumberDimensions** () const
- int **getDimensionSize** (const int dimension) const
- int **getBytesPerPixel** () const
- QString **getDescription** () const
- int **getUniqueld** () const
- void **clear** ()
- QVariant **toVariant** () const
- bool **assignFromVariant** (const QVariant &item)

Static Public Member Functions

- static bool **isAssignableVariant** (const QVariant &item)

9.197.1 Detailed Description

Defines an image class type, specifically to support the epics:nt/NTNDArray type. This type is registered as a QVariant type, and can be set/get like this:

```
QENTNDArrayData arrayData; QVariant variant;
```

```
variant.setValue (arrayData); or variant = arrayData.toVariant(); – defined in this module
```

```
arrayData = variant.value<QENTNDArrayData>(); or arrayData = qvariant_cast<QENTNDArrayData>(variant); or  
arrayData.assignFromVariant (variant); – defined in this module
```

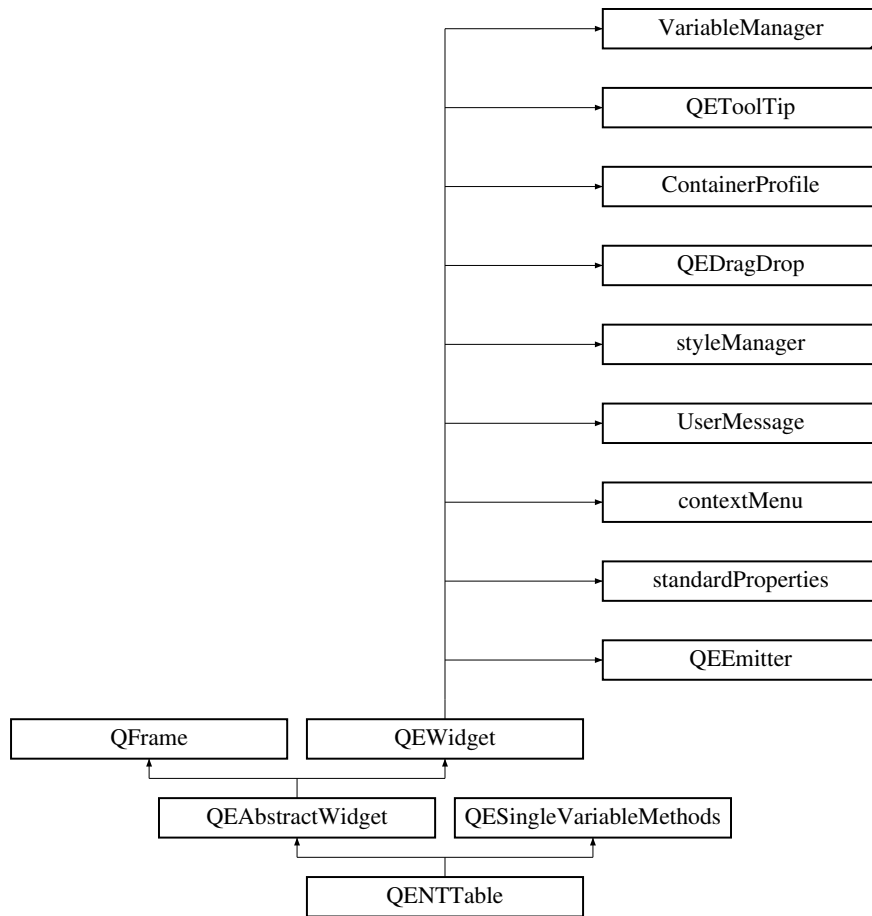
Much of this class was based on ntndArrayConverter out of areaDetector

The documentation for this class was generated from the following files:

- qeframework/protocol/QENTNDArrayData.h
- qeframework/protocol/QENTNDArrayData.cpp

9.198 QENTTable Class Reference

Inheritance diagram for QENTTable:



Public Slots

- void **setSelection** (int value)

Signals

- void **selectionChanged** (int value)
- void **pvNameSetChanged** (const QStringList &pvNameSet)
- void **titlesChanged** (const QStringList &pvNameSet)
- void **dbValueChanged** (const [QENTTableData](#) &out)
- void **dbValueChanged** (const QVariant &out)

Public Member Functions

- [QENTTable](#) (QWidget *parent=0)
- [QENTTable](#) (const QString &variableName, QWidget *parent=0)
Create with a variable.
- virtual [~QENTTable](#) ()

Destruction.

- void **setColumnWidthMinimum** (const int columnWidthMinimum)
- int **getColumnWidthMinimum** () const
- void **setDisplayMaximum** (const int [displayMaximum](#))
- int **getDisplayMaximum** () const
- void **setOrientation** (const Qt::Orientation [orientation](#))
- Qt::Orientation **getOrientation** () const
- void **setShowPvName** (const bool showName)
- bool **getShowPvName** () const
- QStringList **getTitles** () const
- int **getSelection** () const

Protected Types

- enum **OwnContextMenuOptions** { **CM_HORIZONTAL_TABLE** = CM_SPECIFIC_WIDGETS_START_↵
HERE , **CM_VERTICAL_TABLE** }

Protected Member Functions

- QSize **sizeHint** () const
- bool **eventFilter** (QObject *watched, QEvent *event)
- void **resizeEvent** (QResizeEvent *event)
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- [QEChannel](#) * **createQcaltem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)
- void **mousePressEvent** (QMouseEvent *event)
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- bool [showPvName](#)
- int [colWidthMinimum](#)
- int [displayMaximum](#)
- Qt::Orientation [orientation](#)
- bool [showGrid](#)
- Qt::PenStyle [gridStyle](#)

Additional Inherited Members

9.198.1 Constructor & Destructor Documentation

9.198.1.1 QENTTable()

```
QENTTable::QENTTable (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use setVariableName functions.

9.198.2 Member Function Documentation

9.198.2.1 dbValueChanged

```
void QENTTable::dbValueChanged (
    const QENTTableData & out ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget. Note: this widget emits the numeric enumeration value as opposed to the associated text.

9.198.3 Property Documentation

9.198.3.1 colWidthMinimum

```
int QENTTable::colWidthMinimum [read], [write]
```

Specified the minimum allow column width. The widget will shrink/expand the width of each column to as to exactly fit the with of the widget. However, columns will not shrink to less than the value provided by this property. Defaults to 80.

9.198.3.2 displayMaximum

```
int QENTTable::displayMaximum [read], [write]
```

The maximum number of array elements that will be displayed irrespective of the number of elements that the EPICS variable contains. Defaults to 4096.

9.198.3.3 gridStyle

```
Qt::PenStyle QENTTable::gridStyle [read], [write]
```

Sets table grid style. Defaults to SolidLine.

9.198.3.4 orientation

```
Qt::Orientation QENTTable::orientation [read], [write]
```

Determines if the variable values are displayed in rows (orientation is horizontal) or in columns (orientation is vertical). The default is vertical.

9.198.3.5 showGrid

```
bool QENTTable::showGrid [read], [write]
```

Controls if table grid is displayed. Default to true.

9.198.3.6 showPvName

```
bool QENTTable::showPvName [read], [write]
```

Controls if PV name displayed. Default is false.

9.198.3.7 variable

```
QString QENTTable::variable [read], [write]
```

EPICS variable name (PVA PV)

9.198.3.8 variableSubstitutions

```
QString QENTTable::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

The documentation for this class was generated from the following files:

- qeframework/widgets/QETable/QENTTable.h
- qeframework/widgets/QETable/QENTTable.cpp

9.199 QENTTableData Class Reference

```
#include <QENTTableData.h>
```

Public Member Functions

- **QENTTableData** (const [QENTTableData](#) &other)
- [QENTTableData](#) & **operator=** (const [QENTTableData](#) &other)
- void **clear** ()
- QStringList **getLabels** () const
- int **getRowCount** () const
- int **getColCount** () const
- QVariantList **getRowData** (const int row) const
- QVariantList **getColData** (const int col) const
- QVariant **getItem** (const int row, const int col) const
- QVariant **toVariant** () const
- bool **assignFromVariant** (const QVariant &item)

Static Public Member Functions

- static bool **isAssignableVariant** (const QVariant &item)
- static bool **registerMetaType** ()

9.199.1 Detailed Description

Defines a table data class type, specifically to support the epics:nt/NTTable type. This type is registered as a QVariant type, and can be set/get like this:

```
QENTableData table; QVariant variant;
```

```
variant.setValue (table); or variant = table.toVariant(); – defined in this module
```

```
table = variant.value<QENTTableData>(); or table = qvariant_cast<QENTTableData>(variant); or table.assign←  
FromVariant (variant); – defined in this module
```

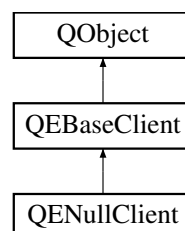
The documentation for this class was generated from the following files:

- qeframework/protocol/QENTTableData.h
- qeframework/protocol/QENTTableData.cpp

9.200 QENullClient Class Reference

```
#include <QENullClient.h>
```

Inheritance diagram for QENullClient:



Public Member Functions

- **QENullClient** (const QString &pvName, QObject *parent)
- bool **openChannel** (const ChannelModesFlags modes)
- void **closeChannel** ()
- QVariant **getPvData** () const
- bool **putPvData** (const QVariant &value)
- bool **getIsConnected** () const
- bool **dataIsAvailable** () const
- QString **getId** () const
- QString **getRemoteAddress** () const
- QString **getEgu** () const
- int **getPrecision** () const
- unsigned int **hostElementCount** () const
- unsigned int **dataElementCount** () const
- double **getDisplayLimitHigh** () const
- double **getDisplayLimitLow** () const
- double **getHighAlarmLimit** () const
- double **getLowAlarmLimit** () const
- double **getHighWarningLimit** () const
- double **getLowWarningLimit** () const
- double **getControlLimitHigh** () const
- double **getControlLimitLow** () const
- double **getMinStep** () const
- QStringList **getEnumerations** () const
- [QCaAlarmInfo](#) **getAlarmInfo** () const
- [QCaDateTime](#) **getTimeStamp** () const
- QString **getDescription** () const
- bool **getReadAccess** () const
- bool **getWriteAccess** () const

Additional Inherited Members

9.200.1 Detailed Description

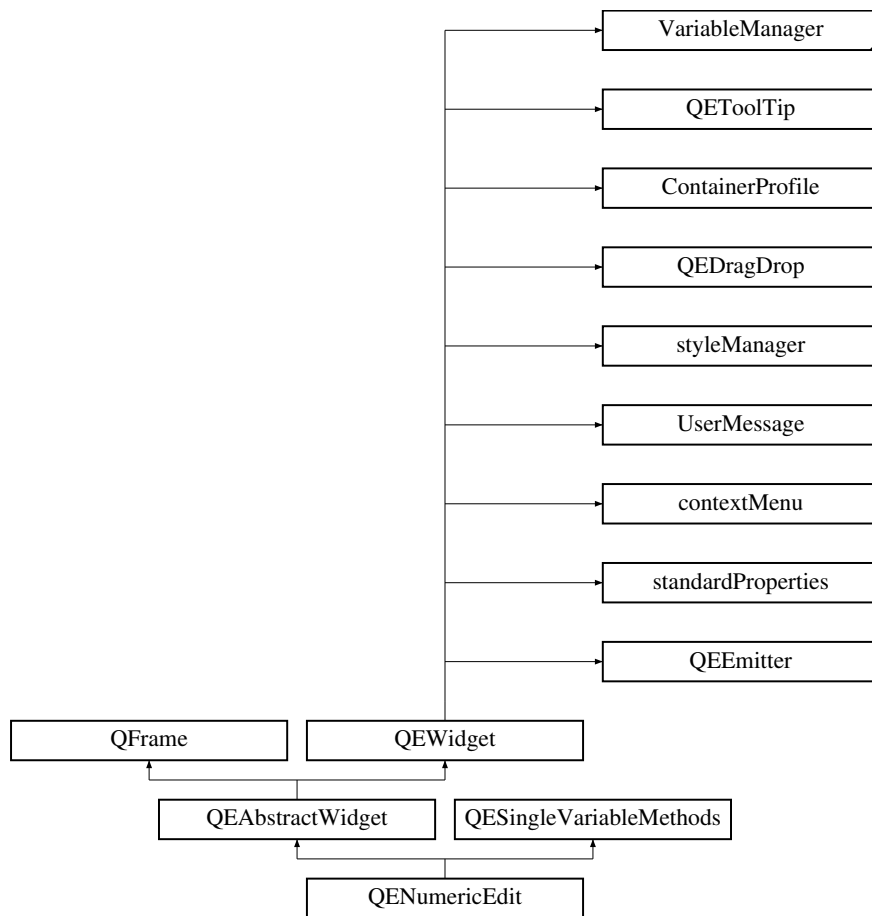
Dummy/null client created when an invalid/unknown protocol is specified.

The documentation for this class was generated from the following files:

- qeframework/protocol/QENullClient.h
- qeframework/protocol/QENullClient.cpp

9.201 QENumericEdit Class Reference

Inheritance diagram for QENumericEdit:



Public Slots

- void **writeNow** ()
- void **setDefaultStyle** (const QString &style)
Update the default style applied to this widget.
- void **setPvValue** (const QString &text)
- void **setPvValue** (const int value)
- void **setPvValue** (const double value)
- void **setPvValue** (const bool value)

Signals

- void **dbValueChanged** ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void **dbConnectionChanged** (const bool &isConnected)

Public Member Functions

- [QNumericEdit](#) (QWidget *parent=0)
- [QNumericEdit](#) (const QString &variableName, QWidget *parent=0)
- virtual [~QNumericEdit](#) ()
- Destruction.*
- double **getValue** () const
- void **setValue** (const double value, const bool isUserUpdate=false)
- void **setAutoScale** (const bool)
- bool **getAutoScale** () const
- void **setAddUnits** (const bool)
- bool **getAddUnits** () const
- void **setUseApplyButton** (const bool)
- bool **getUseApplyButton** () const
- void **setApplyButtonText** (const QString &text)
- QString **getApplyButtonText** () const
- void **setSubscribe** (const bool)
- bool **getSubscribe** () const
- void **setWriteOnLoseFocus** (const bool)
- bool **getWriteOnLoseFocus** () const
- void **setWriteOnEnter** (const bool)
- bool **getWriteOnEnter** () const
- void **setWriteOnFinish** (const bool)
- bool **getWriteOnFinish** () const
- void **setWriteOnChange** (const bool)
- bool **getWriteOnChange** () const
- void **setConfirmWrite** (const bool)
- bool **getConfirmWrite** () const
- void **setAllowFocusUpdate** (const bool)
- bool **getAllowFocusUpdate** () const
- void **setLeadingZeros** (const int value)
- int **getLeadingZeros** () const
- void **setPrecision** (const int value)
- int **getPrecision** () const
- void **setMinimum** (const double value)
- double **getMinimum** () const
- void **setMaximum** (const double value)
- double **getMaximum** () const
- QString **getCleanText** () const

Protected Member Functions

- bool **eventFilter** (QObject *watched, QEvent *event)
- void **focusInEvent** (QFocusEvent *event)
- void **resizeEvent** (QResizeEvent *event)
- QMenu * [getDefaultContextMenu](#) ()
- Flag indicating this widget is running inside Qt's 'designer'.*
- void [activated](#) ()
- Do any post-all-widgets-constructed, i.e. activated stuff.*
- void [establishConnection](#) (unsigned int variableIndex)
- Create a CA connection and initiates updates if required.*
- [QEChannel](#) * [createQcalItem](#) (unsigned int variableIndex)
- Function to create a appropriate superclass of QEChannel to stream data updates.*
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- bool [frame](#)
- Qt::Alignment [alignment](#)
- QString [cleanText](#)
- bool [autoScale](#)
- QE::Notations [notation](#)
- QE::Radices [radix](#)
- QE::Separators [separator](#)
- int [leadingZeros](#)
- int [precision](#)
- bool [forceSign](#)
- bool [wrapValue](#)
- double [minimum](#)
- double [maximum](#)
- bool [addUnits](#)
- bool [subscribe](#)
- bool [useApplyButton](#)
- QString [applyButtonText](#)
- bool [writeOnLoseFocus](#)
- bool [writeOnEnter](#)
- bool [writeOnFinish](#)
- bool [writeOnChange](#)
- bool [confirmWrite](#)
- bool [allowFocusUpdate](#)

Additional Inherited Members

9.201.1 Constructor & Destructor Documentation

9.201.1.1 QENumericEdit() [1/2]

```
QENumericEdit::QENumericEdit (  
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.201.1.2 QENumericEdit() [2/2]

```
QENumericEdit::QENumericEdit (  
    const QString & variableName,  
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.201.2 Member Function Documentation

9.201.2.1 dbConnectionChanged

```
void QENumericEdit::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.201.2.2 dbValueChanged

```
void QENumericEdit::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.201.2.3 setPvValue

```
void QENumericEdit::setPvValue (
    const QString & text ) [slot]
```

Update the widget and write to the main associated control process variable. Supplied values are converted to the appropiate widget value type. If conversion not possible or resultant value out of range, no widget/PV update occurs.

9.201.3 Property Documentation

9.201.3.1 addUnits

```
bool QENumericEdit::addUnits [read], [write]
```

If true (default is false), add engineering units supplied with the data.

9.201.3.2 alignment

```
Qt::Alignment QENumericEdit::alignment [read], [write]
```

This property holds the alignment of the numeric edit. Both horizontal and vertical alignment is allowed here, Qt::AlignJustify will map to Qt::AlignLeft. By default, this property contains a combination of Qt::AlignRight and Qt::AlignVCenter.

9.201.3.3 allowFocusUpdate

```
bool QENumericEdit::allowFocusUpdate [read], [write]
```

Allow updated while widget has focus - defaults to false.

9.201.3.4 applyButtonText

```
QString QENumericEdit::applyButtonText [read], [write]
```

The text displayed on the apply button. This default text is "A".

9.201.3.5 arrayIndex

```
int QENumericEdit::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.201.3.6 autoScale

```
bool QENumericEdit::autoScale [read], [write]
```

If true (default), display and editing of numbers using the PV's precision and control limits supplied with the data. If false, the precision, leadingZeros, minimum and maximum properties values are used.

9.201.3.7 cleanText

```
QString QENumericEdit::cleanText [read]
```

This property holds the displayed text. Not a property available to designer.

9.201.3.8 confirmWrite

```
bool QENumericEdit::confirmWrite [read], [write]
```

Sets if this widget will ask for confirmation (using a dialog box) prior to writing data. Default is 'false' (will not ask for confirmation (using a dialog box) prior to writing data). Note: writeOnChange and confirmWrite are mutually exclusive.

9.201.3.9 elementsRequired

```
int QENumericEdit::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.201.3.10 forceSign

```
bool QENumericEdit::forceSign [read], [write]
```

When true, there is always a sign character "+" or "-". When false (default), no "+" sign when range of values is non-negative.

9.201.3.11 frame

```
bool QENumericEdit::frame [read], [write]
```

This property holds whether the numeric edit draws itself with a frame. If enabled (the default) the numeric edit draws itself inside a frame, otherwise the line edit draws itself without any frame.

9.201.3.12 leadingZeros

```
int QENumericEdit::leadingZeros [read], [write]
```

Specifies the number of leading zeros. The default is 3. This is only used if autoScale is false. When autoScale is true the PV's control range is used to determine the number of required leading zeros. Strictly speaking, this should be an unsigned int, but designer properties editor much 'nicer' with ints.

9.201.3.13 maximum

```
double QENumericEdit::maximum [read], [write]
```

Specifies the maximum allowed value. This is only used if autoScale is false.

9.201.3.14 minimum

```
double QENumericEdit::minimum [read], [write]
```

Specifies the minimum allowed value. This is only used if autoScale is false.

9.201.3.15 notation

```
QE::Notations QENumericEdit::notation [read], [write]
```

Notation used for formatting/editing. Default is fixed. The widget does not allow Automatic

9.201.3.16 precision

```
int QENumericEdit::precision [read], [write]
```

Precision used for the display and editing of numbers. The default is 2. This is only used if autoScale is false. When autoScale is true the PV's precision is used. Strictly speaking, this should be an unsigned int, but designer properties editor much 'nicer' with ints.

9.201.3.17 radix

`QE::Radices QENumericEdit::radix [read], [write]`

Specify radix, default is Decimal.

9.201.3.18 separator

`QE::Separators QENumericEdit::separator [read], [write]`

Specify digit 'thousands' separator character, default is NoSeparator.

9.201.3.19 subscribe

`bool QENumericEdit::subscribe [read], [write]`

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.201.3.20 useApplyButton

`bool QENumericEdit::useApplyButton [read], [write]`

When set true, internal apply button is exposed and used to trigger write. All other write are options set false and inhibited from being set true.

9.201.3.21 variable

`QString QENumericEdit::variable [read], [write]`

EPICS variable name (CA PV)

9.201.3.22 variableSubstitutions

`QString QENumericEdit::variableSubstitutions [read], [write]`

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.201.3.23 wrapValue

`bool QENumericEdit::wrapValue [read], [write]`

When set true (default is false) the numeric edit widget will wrap around from the max/min to min/max value.

9.201.3.24 writeOnChange

```
bool QENumericEdit::writeOnChange [read], [write]
```

If true the widget writes to the PV as value are changes. If false (default) a write only occurs when as per writeOnLoseFocus, writeOnEnter and/or writeOnFinish values. Note: writeOnChange and confirmWrite are mutually exclusive.

9.201.3.25 writeOnEnter

```
bool QENumericEdit::writeOnEnter [read], [write]
```

Sets if this widget writes any changes when the user presses 'enter'. Note, the current value will be written even if the user has not changed it. Default is 'true' (writes any changes when the user presses 'enter').

9.201.3.26 writeOnFinish

```
bool QENumericEdit::writeOnFinish [read], [write]
```

Sets if this widget writes any changes when the user finished editing (the underlying QLineEdit 'editingFinished' signal is emitted). No writing occurs if no changes were made. Default is 'true' (writes any changes when the QLineEdit 'editingFinished' signal is emitted).

9.201.3.27 writeOnLoseFocus

```
bool QENumericEdit::writeOnLoseFocus [read], [write]
```

Sets if this widget automatically writes any changes when it loses focus. Default is 'false' (does not write any changes when it loses focus).

The documentation for this class was generated from the following files:

- qeframework/widgets/QENumericEdit/QENumericEdit.h
- qeframework/widgets/QENumericEdit/QENumericEdit.cpp

9.202 QEOneToOne< D, C > Class Template Reference

```
#include <QEOneToOne.h>
```

Public Member Functions

- void **clear** ()
- bool **insertF** (const D &dkey, const C &ckey)
- bool **insertI** (const C &ckey, const D &dkey)
- bool **containsF** (const D &key) const
- bool **containsI** (const C &key) const
- C **valueF** (const D &key) const
- D **valueI** (const C &key) const
- C **valueF** (const D &key, const C &defaultValue) const
- D **valueI** (const C &key, const D &defaultValue) const
- bool **removeF** (const D &dkey)
- bool **removeI** (const C &ckey)

9.202.1 Detailed Description

```
template<class D, class C>
class QEOneToOne< D, C >
```

This template provides a one-to-one mapping from one type, the Domain type, to a another, possibly the same, type, the CoDomain type. Despite the terminology used here (i.e. domain/codomain), the association between the two types is symmetrical, therefore `QEOneToOne<T1, T2>` provides the effectively the same functionality as `QEOneToOne<T2, T1>`

Most functions are available in both forward (domain to co-domain) and in the inverse (co-domain to domain) modes of operation. These are named as `<func>F` and `<func>I`. We cannot use on overloading as user may require one to one mapping between the same type, e.g. an int to int association.

The undelying mappings are provided by a QHash. This may change.

9.202.2 Member Function Documentation

9.202.2.1 clear()

```
template<class D , class C >
void QEOneToOne< D, C >::clear ( ) [inline]
```

Clear all mapping relationships.

9.202.2.2 containsF()

```
template<class D , class C >
bool QEOneToOne< D, C >::containsF (
    const D & key ) const [inline]
```

Indicates if assoication contains specified key.

9.202.2.3 insertF()

```
template<class D , class C >
bool QEOneToOne< D, C >::insertF (
    const D & dkey,
    const C & ckey ) [inline]
```

Insert a key-key mapping into the association. Returns true if successfull. Returns false on fail. The insertF function fails if the association currently contains the domain key dkey and/or the codomain key ckey.

9.202.2.4 removeF()

```
template<class D , class C >
bool QEOneToOne< D, C >::removeF (
    const D & dkey ) [inline]
```

Removes the association containing specified key.

9.202.2.5 valueF() [1/2]

```
template<class D , class C >
C QEOneToOne< D, C >::valueF (
    const D & key ) const [inline]
```

Extracts the b/a value associated with a/b key. If the association does not contain a/b key, then returns A ()/ B() as default.

9.202.2.6 valueF() [2/2]

```
template<class D , class C >
C QEOneToOne< D, C >::valueF (
    const D & key,
    const C & defaultValue ) const [inline]
```

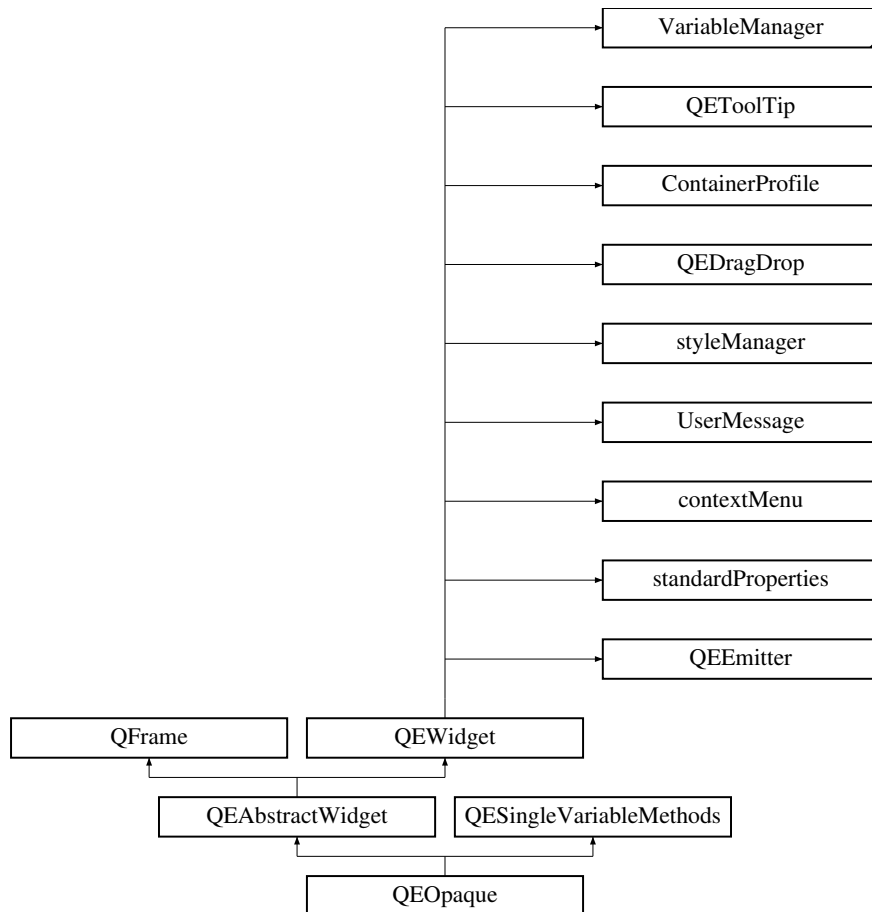
Extracts the value associated with key. If the association does not contain the key, then returns specified default value.

The documentation for this class was generated from the following file:

- qeframework/common/QEOneToOne.h

9.203 QEOpaque Class Reference

Inheritance diagram for QEOpaque:



Signals

- void [dbValueChanged](#) (const [QEOpaqueData](#) &out)
- void **dbValueChanged** (const QVariant &out)

Public Member Functions

- [QEOpaque](#) (QWidget *parent=0)
- [QEOpaque](#) (const QString &variableName, QWidget *parent=0)
Create with a variable.
- virtual [~QEOpaque](#) ()
Destruction.
- void **setShowPvName** (const bool showName)
- bool **getShowPvName** () const

Protected Member Functions

- QSize **sizeHint** () const
- bool **eventFilter** (QObject *watched, QEvent *event)
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- [QEChannel](#) * [createQcaltem](#) (unsigned int variableIndex)

Function to create a appropriate superclass of QEChannel to stream data updates.

- void **mousePressEvent** (QMouseEvent *event)
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- bool [showPvName](#)

Additional Inherited Members

9.203.1 Constructor & Destructor Documentation

9.203.1.1 QEOpaque()

```
QEOpaque::QEOpaque (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use setVariableName functions.

9.203.2 Member Function Documentation

9.203.2.1 dbValueChanged

```
void QEOpaque::dbValueChanged (
    const QEOpaqueData & out ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget. Note: this widget emits the numeric enumeration value as opposed to the associated text.

9.203.3 Property Documentation

9.203.3.1 showPvName

```
bool QEOpaque::showPvName [read], [write]
```

Controls if PV name displayed. Default is false.

9.203.3.2 variable

```
QString QEOpaque::variable [read], [write]
```

EPICS variable name (PVA PV)

9.203.3.3 variableSubstitutions

```
QString QEOpaque::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEOpaque/QEOpaque.h
- qeframework/widgets/QEOpaque/QEOpaque.cpp

9.204 QEOpaqueData Class Reference

```
#include <QEOpaqueData.h>
```

Public Member Functions

- **QEOpaqueData** (const [QEOpaqueData](#) &other)
- [QEOpaqueData](#) & **operator=** (const [QEOpaqueData](#) &other)
- void **clear** ()
- QString **getText** () const
- QVariant **toVariant** () const
- bool **assignFromVariant** (const QVariant &item)

Static Public Member Functions

- static bool **isAssignableVariant** (const QVariant &item)
- static bool **registerMetaType** ()

9.204.1 Detailed Description

Defines a data class type, specifically to support the unknown PV Access data types such as non-normative types. This type is essentially just a QStringList containing "stream << opaque".

This type is registered as a QVariant type, and can be set/get like this:

```
QEOpaqueData opaque; QVariant var;
```

```
var.setValue (opaque); – or var = opaque.toVariant(); – defined in this module
```

```
opaque = var.value<QEOpaqueData>(); – or opaque = qvariant_cast<QEOpaqueData>(var); – or opaque.←←
assignFromVariant (var); – defined in this module
```

The documentation for this class was generated from the following files:

- qeframework/protocol/QEOpaqueData.h
- qeframework/protocol/QEOpaqueData.cpp

9.205 QEOptions Class Reference

```
#include <QEOptions.h>
```

Public Member Functions

- **QEOptions** (const QStringList &args)
- int **getParameterCount** ()
- QString **getParameter** (const int i)
- bool **isSpecified** (const QString &option, const QChar letter)
- bool **isSpecified** (const QString &option)
- bool **isSpecified** (const QChar letter)
- bool **getBool** (const QString &option, const QChar letter)
- bool **getBool** (const QString &option)
- bool **getBool** (const QChar letter)
- QString **getString** (const QString &option, const QChar letter, const QString &defaultValue)
- QString **getString** (const QString &option, const QString &defaultValue)
- QString **getString** (const QChar letter, const QString &defaultValue)
- int **getInt** (const QString &option, const QChar letter, const int defaultValue)
- int **getInt** (const QString &option, const int defaultValue)
- int **getInt** (const QChar letter, const int defaultValue)
- double **getFloat** (const QString &option, const QChar letter, const double defaultValue)
- double **getFloat** (const QString &option, const double defaultValue)
- double **getFloat** (const QChar letter, const double defaultValue)

9.205.1 Detailed Description

This class provides a basic means to parse command line options or any other arbitrary list of strings. Options are named and the name format may be either the short format (single letter) and/or the long format.

The types of data that may be extracted from an option are bool, QString, int and double; These are accessed by the associated getXxx () functions where Xxx is one of Bool, String, Int and Float.

Each of these function takes a default value of the appropriate type which is returned to the caller if the value is not defined or, as in the case of numerical values, is ill-defined.

Numeric validity is defined by QString::toInt () and QString::toDouble () Do note that QString::toInt () does not accept 0x... hexadecimal or 0... octal numbers.

Examples: program -v – boolean or flag (short format) program –verbose – boolean or flag (long format) program -n=31 – integer (short format) program -n 31 – integer (alternate short format) program –number=31 – integer (long format) program –number 31 – integer (alternate long format) program –output="fred.txt" – string (long format) program –output fred.txt – string (alternate long format) program –output= – specifies a null string

Note: Currently it provides no means to "compress" multiple flags, e.g. to interpret: program -cap as program -c -a -p

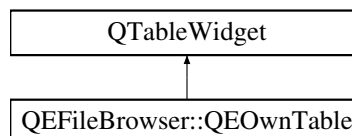
This class is one of several provided to access configuration data such as from environment variables, command line options and settings. They all provide a similar API.

The documentation for this class was generated from the following files:

- qeframework/adaptation_parameters/QEOptions.h
- qeframework/adaptation_parameters/QEOptions.cpp

9.206 QFileBrowser::QOwnTable Class Reference

Inheritance diagram for QFileBrowser::QOwnTable:



Public Member Functions

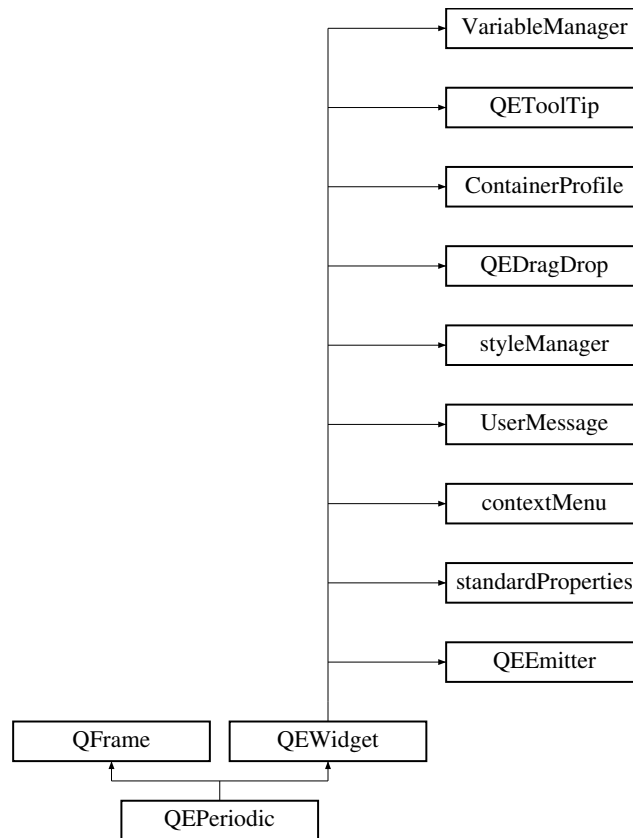
- **QOwnTable** (QWidget *parent=0)
- void **refreshSize** ()
- void **resizeEvent** (QResizeEvent *)

The documentation for this class was generated from the following file:

- qeframework/widgets/QFileBrowser/QFileBrowser.cpp

9.207 QEPeriodic Class Reference

Inheritance diagram for QEPeriodic:



Classes

- struct [elementInfoStruct](#)
- struct [userInfoStructArray](#)

Public Types

- enum **Category** {
Hydrogen = 0 , **AlkaliMetal** , **AlkalineEarthMetal** , **TransitionMetal** ,
PostTransitionMetal , **Metalloid** , **ReactiveNonMetal** , **NobleGas** ,
Lanthanide , **Actinide** , **UnknownProperties** , **NUMBER_OF_CATEGORIES** }
- enum **variableTypes** {
VARIABLE_TYPE_NUMBER , **VARIABLE_TYPE_ATOMIC_WEIGHT** , **VARIABLE_TYPE_MELTING** ↔
POINT , **VARIABLE_TYPE_BOILING_POINT** ,
VARIABLE_TYPE_DENSITY , **VARIABLE_TYPE_GROUP** , **VARIABLE_TYPE_IONIZATION_ENERGY** ,
VARIABLE_TYPE_USER_VALUE_1 ,
VARIABLE_TYPE_USER_VALUE_2 }
- enum **presentationOptions** { **PRESENTATION_BUTTON_AND_LABEL** , **PRESENTATION_BUTTON** ↔
ONLY , **PRESENTATION_LABEL_ONLY** }
- enum **PresentationOptions** { **buttonAndLabel** = QEPeriodic::PRESENTATION_BUTTON_AND_LABEL ,
buttonOnly = QEPeriodic::PRESENTATION_BUTTON_ONLY , **labelOnly** = QEPeriodic::PRESENTATION ↔
_LABEL_ONLY }

- enum **VariableTypes** {
Number = QEPeiodic::VARIABLE_TYPE_NUMBER , **atomicWeight** = QEPeiodic::VARIABLE_TYPE_↵
 _ATOMIC_WEIGHT , **meltingPoint** = QEPeiodic::VARIABLE_TYPE_MELTING_POINT , **boilingPoint** =
 QEPeiodic::VARIABLE_TYPE_BOILING_POINT ,
density = QEPeiodic::VARIABLE_TYPE_DENSITY , **group** = QEPeiodic::VARIABLE_TYPE_GROUP ,
ionizationEnergy = QEPeiodic::VARIABLE_TYPE_IONIZATION_ENERGY , **userValue1** = QEPeiodic↵
 ::VARIABLE_TYPE_USER_VALUE_1 ,
userValue2 = QEPeiodic::VARIABLE_TYPE_USER_VALUE_2 }

Public Slots

- void **setElement** (const QString symbol)
- void **setAtomicNumber** (const int atomicNumber)

Signals

- void **userElementChanged** (const QString &symbol)
Sent when the element is changed by the user selecting an element.
- void **userAtomicNumberChanged** (const int atomicNumber)
- void **dbValueChanged** (const double &out)
- void **dbElementChanged** (const QString &out)
- void **dbAtomicNumberChanged** (const int atomicNumber)
- void **requestResend** ()

Public Member Functions

- **QEPeiodic** (QWidget *parent=0)
- **QEPeiodic** (const QString &variableName, QWidget *parent=0)
- void **setSubscribe** (bool [subscribe](#))
- bool **getSubscribe** () const
- void **setPresentationOption** (presentationOptions presentationOptionIn)
- presentationOptions **getPresentationOption** () const
- void **setVariableType1** (variableTypes variableType1In)
- variableTypes **getVariableType1** () const
- void **setVariableType2** (variableTypes variableType2In)
- variableTypes **getVariableType2** () const
- void **setVariableTolerance1** (double variableTolerance1In)
- double **getVariableTolerance1** () const
- void **setVariableTolerance2** (double variableTolerance2In)
- double **getVariableTolerance2** () const
- void **setUserInfo** (QString userInfo)
- QString **getUserInfo** () const
- void **setUserInfoText** (QString userInfo)
- QString **getUserInfoText** () const
- void **setUserInfoFile** (QString userInfoFileIn)
- QString **getUserInfoFile** () const
- void **setColourised** (const bool colouriseIn)
- bool **isColourised** () const
- void **setUserInfoSourceOption** ([QE::SourceOptions](#) userInfoSourceOptionIn)
- [QE::SourceOptions](#) **getUserInfoSourceOption** () const
- void **updateUserInfoSource** ()
- bool **getElementValues** (QString symbol, double *value1, double *value2) const

- QString **getSelectedSymbol** () const
- int **getSelectedAtomicNumber** () const
- void **setVariableNameSubstitutionsProperty** (QString variableNameSubstitutions)
Property access function for [variableSubstitutions](#) property. This has special behaviour to work well within designer.
- QString **getVariableNameSubstitutionsProperty** ()
Property access function for [variableSubstitutions](#) property. This has special behaviour to work well within designer.
- void **setPresentationOptionProperty** (PresentationOptions presentationOption)
- PresentationOptions **getPresentationOptionProperty** ()
- void **setVariableType1Property** (VariableTypes variableType)
- void **setVariableType2Property** (VariableTypes variableType)
- VariableTypes **getVariableType1Property** ()
- VariableTypes **getVariableType2Property** ()

Static Public Member Functions

- static QColor **categoryColour** (const Category cat)

Public Attributes

- [userInfoStruct](#) **userInfo** [NUM_ELEMENTS]

Static Public Attributes

- static const [elementInfoStruct](#) **elementInfo** [NUM_ELEMENTS]

Protected Types

- enum **variableIndexes** {
WRITE_VARIABLE_1 , **WRITE_VARIABLE_2** , **READ_VARIABLE_1** , **READ_VARIABLE_2** ,
QEPERIODIC_NUM_VARIABLES }

Protected Member Functions

- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- void **setDrop** (QVariant drop)
- QVariant **getDrop** ()
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Protected Attributes

- [QEFloatingFormatting](#) **floatingFormatting**
- bool **localEnabled**
- bool [allowDrop](#)
- variableTypes **variableType1**
- variableTypes **variableType2**
- double **variableTolerance1**
- double **variableTolerance2**

Properties

- QString [writeButtonVariable1](#)
- QString [writeButtonVariable2](#)
- QString [readbackLabelVariable1](#)
- QString [readbackLabelVariable2](#)
- QString [variableSubstitutions](#)
- bool [subscribe](#)
- bool [variableAsToolTip](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- PresentationOptions **presentationOption**
- VariableTypes **variableType1**
- VariableTypes **variableType2**
- QString **userInfo**
- bool [colourised](#)

9.207.1 Member Function Documentation

9.207.1.1 dbElementChanged

```
void QEPeiodic::dbElementChanged (
    const QString & out ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.207.1.2 dbValueChanged

```
void QEPeiodic::dbValueChanged (
    const double & out ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.207.1.3 requestResend

```
void QEPeriodic::requestResend ( ) [signal]
```

Internal use only. Used when changing a property value to force a re-display to reflect the new property value.

9.207.2 Member Data Documentation

9.207.2.1 allowDrop

```
bool QEPeriodic::allowDrop [read], [write], [protected]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.207.3 Property Documentation

9.207.3.1 colourised

```
bool QEPeriodic::colourised [read], [write]
```

This property sets element colourised enable/disable on the PeridicDialog.

9.207.3.2 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEPeriodic::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'Never' widget will never indicate the alarm state of any variable data it is displaying. If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm'. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.207.3.3 messageSourceId

```
unsigned int QEPeriodic::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.207.3.4 readbackLabelVariable1

```
QString QEPeiodic::readbackLabelVariable1 [read], [write]
```

EPICS variable name (CA PV). This variable is used to read the value to the first of two positioners to determine which (if any) element is currently selected.

9.207.3.5 readbackLabelVariable2

```
QString QEPeiodic::readbackLabelVariable2 [read], [write]
```

EPICS variable name (CA PV). This variable is used to read the value to the second of two positioners to determine which (if any) element is currently selected.

9.207.3.6 subscribe

```
bool QEPeiodic::subscribe [read], [write]
```

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.207.3.7 userLevelEnabled

```
QE::UserLevels QEPeiodic::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.207.3.8 userLevelEngineerStyle

```
QString QEPeiodic::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.207.3.9 userLevelScientistStyle

```
QString QEPeiodic::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.207.3.10 `userLevelUserStyle`

```
QString QEP periodic::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.207.3.11 `userLevelVisibility`

```
QE::UserLevels QEP periodic::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programmatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.207.3.12 `variableAsToolTip`

```
bool QEP periodic::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.207.3.13 `variableSubstitutions`

```
QString QEP periodic::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'SAMPLE=SAM1, NAME = "Ref foil"' These substitutions are applied to all the variable names.

9.207.3.14 `visible`

```
bool QEP periodic::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is useful if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

9.207.3.15 `writeButtonVariable1`

```
QString QEP periodic::writeButtonVariable1 [read], [write]
```

EPICS variable name (CA PV). This variable is used to write a value to the first of two positioners that will position the select element.

9.207.3.16 writeButtonVariable2

QString QEPERiodic::writeButtonVariable2 [read], [write]

EPICS variable name (CA PV). This variable is used to write a value to the second of two positioners that will position the select element.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPERiodic/QEPERiodic.h
- qeframework/widgets/QEPERiodic/QEPERiodic.cpp

9.208 QEPERiodicComponentData Class Reference

Public Attributes

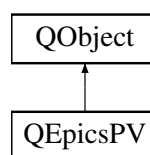
- unsigned int **variableIndex1**
- double **lastData1**
- bool **haveLastData1**
- unsigned int **variableIndex2**
- double **lastData2**
- bool **haveLastData2**

The documentation for this class was generated from the following file:

- qeframework/widgets/QEPERiodic/QEPERiodic.h

9.209 QEpicsPV Class Reference

Inheritance diagram for QEpicsPV:



Public Slots

- const QVariant & **set** (QVariant value, int delay=-1)
- void **setPV** (const QString &_pvName="")

Signals

- void **connectionChanged** (bool connected)
- void **connected** ()
- void **disconnected** ()
- void **valueChanged** (const QVariant &value)
- void **valueUpdated** (const QVariant &value)
- void **valueInitd** (const QVariant &value)

Public Member Functions

- **QEpicsPV** (const QString &_pvName, QObject *parent=0)
- **QEpicsPV** (QObject *parent=0)
- const QVariant & **get** () const
- void **needUpdated** () const
- const QVariant & **getUpdated** (int delay=defaultDelay) const
- bool **isConnected** () const
- const QStringList & **getEnum** () const
- const QString & **pv** () const
- const QVariant & **getReady** (int delay=defaultDelay) const

Static Public Member Functions

- static void **setDebugLevel** (unsigned level=0)
- static QVariant **get** (const QString &_pvName, int delay=defaultDelay)
- static QVariant **set** (QString &_pvName, const QVariant &value, int delay=-1)

The documentation for this class was generated from the following files:

- qeframework/data/qepicspv.h
- qeframework/data/qepicspv.cpp

9.210 QEPlatform Class Reference

```
#include <QEPlatform.h>
```

Static Public Member Functions

- static QMetaType::Type **metaType** (const QVariant &v)
- static QPoint **positionOf** (QDropEvent *event)
- static QPoint **positionOf** (QMouseEvent *event)
- static QDateTime **constructEpoch** (const int year)
- static void **setTimeZone** (QDateTime &dateTime, const Qt::TimeSpec timeSpec)
- static QDateTime **toTimeZone** (const QDateTime &dateTime, const Qt::TimeSpec timeSpec)
- static bool **isNaN** (const double x)

This function test if the specified double floating point number is 'Not a Number'.

- static bool **isInf** (const double x)

This function test if the specified double floating point number is +/-Infinity.

9.210.1 Detailed Description

Geneal purpose platform specific functions used to hide compiler, environment and version specific features. This locates all the #indef and the like tests into one centralised and consistent location. We use a class of static methods as opposed to a set of regular functions. This provide a name space, but also allows inheritance if needs be.

9.210.2 Member Function Documentation

9.210.2.1 constructEpoch()

```
QDateTime QEPlatform::constructEpoch (
    const int year ) [static]
```

Construct an epoch date time, always Jan 1st 00:00:00

9.210.2.2 metaType()

```
QMetaType::Type QEPlatform::metaType (
    const QVariant & v ) [static]
```

Returns the meta type type of a variant, e.g. QMetaType::QStringList, as opposed to returning an int.

9.210.2.3 positionOf()

```
QPoint QEPlatform::positionOf (
    QDropEvent * event ) [static]
```

Returns the position associated with various types of event.

9.210.2.4 setTimeZone()

```
void QEPlatform::setTimeZone (
    QDateTime & dateTime,
    const Qt::TimeSpec timeSpec ) [static]
```

Changes the time zone to UTC or LocalTime only.

9.210.2.5 toTimeZone()

```
QDateTime QEPlatform::toTimeZone (
    const QDateTime & dateTime,
    const Qt::TimeSpec timeSpec ) [static]
```

This converts the actual time to UTC or LocalTime.

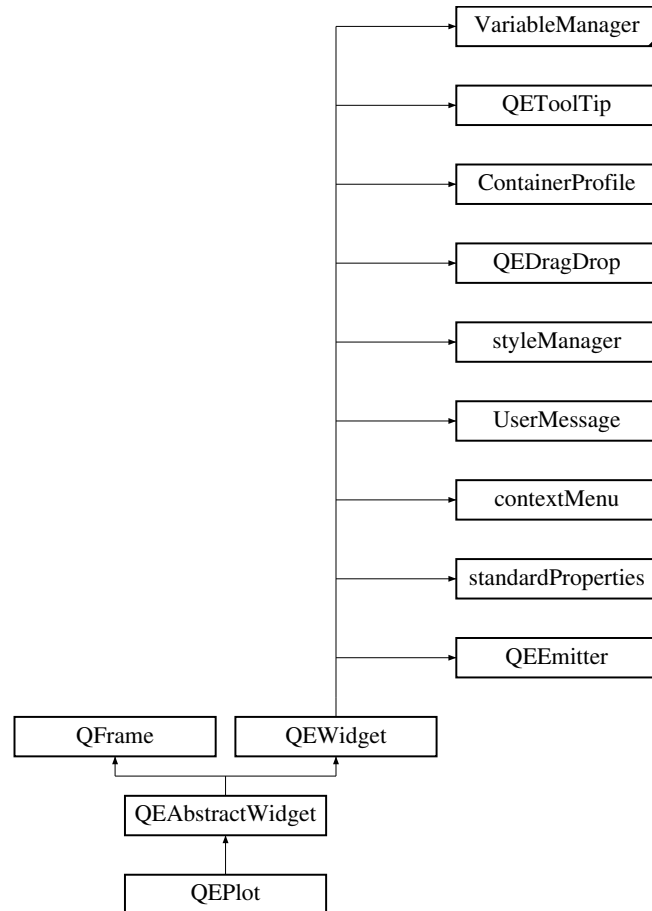
The documentation for this class was generated from the following files:

- qeframework/common/QEPlatform.h
- qeframework/common/QEPlatform.cpp

9.211 QEPlot Class Reference

```
#include <QEPlot.h>
```

Inheritance diagram for QEPlot:



Classes

- class [Trace](#)

Public Types

- enum **Constants** { **QEPlot_NUM_PLOTS** = 8 , **QEPlot_NUM_VARIABLES** = 16 }
- enum **TraceStyles** { **Lines** = 1 , **Sticks** , **Steps** , **Dots** }
- enum **SelectedYAxis** { **Left** = 1 , **Right** }
- enum **OwnContextMenuOptions** { **PLOT_CM_NONE** = CM_SPECIFIC_WIDGETS_START_HERE , **PLOT_CM_ARCHIVE_BACKFILL** , **PLOT_CM_SUB_CLASS_WIDGETS_START_HERE** }
- typedef [QEAbstractWidget](#) **ParentWidgetClass**

Public Slots

- void **setTraceVisible1** (const bool [visible](#))
- void **setTraceVisible2** (const bool [visible](#))
- void **setTraceVisible3** (const bool [visible](#))
- void **setTraceVisible4** (const bool [visible](#))
- void **setTraceVisible5** (const bool [visible](#))
- void **setTraceVisible6** (const bool [visible](#))
- void **setTraceVisible7** (const bool [visible](#))
- void **setTraceVisible8** (const bool [visible](#))

Signals

- void [dbValueChanged](#) (const double &out)
- void [dbValueChanged](#) (const QVector< double > &out)
- void **mouseMove** (const QPointF &posn)
- void **mouseMove** (const QString &text)

Public Member Functions

- **QEPlot** (QWidget *parent=0)
- **QEPlot** (const QString &variable1Name, QWidget *parent=0)
- QSize **sizeHint** () const
- void **setVariableNameSubstitutionsProperty** (const QString &variableNameSubstitutions)
- QString **getVariableNameSubstitutionsProperty** () const
- void **setYMin** (const double yMin)
- double **getYMin** () const
- void **setYMax** (double yMax)
- double **getYMax** () const
- void **setAutoScale** (const bool yAxisAutoScale)
- bool **getAutoScale** () const
- void **setLogScale** (const bool logScale)
- bool **getLogScale** () const
- void **setArchiveBackfill** (const bool archiveBackfill)
- bool **getArchiveBackfill** () const
- void **setAxisEnableX** (const bool axisEnableX)
- bool **getAxisEnableX** () const
- void **setAxisEnableY** (const bool axisEnableY)
- bool **getAxisEnableY** () const
- void **setSelectedYAxis** (const SelectedYAxis selectedYAxis)
- SelectedYAxis **getSelectedYAxis** () const
- void **setTitle** (const QString &title)
- QString **getTitle** () const
- void **setBackgroundColor** (const QColor backgroundColor)
- QColor **getBackgroundColor** () const
- void **setVariableNameIndexProperty** (const QString &variableName, const unsigned int variableIndex)
- QString **getVariableNameIndexProperty** (const unsigned int variableIndex) const
- void **setSizeVariableNameIndexProperty** (const QString &variableName, const unsigned int variableIndex)
- QString **getSizeVariableNameIndexProperty** (const unsigned int variableIndex) const
- void **setTraceStyle** (const TraceStyles traceStyle, const unsigned int variableIndex)
- TraceStyles **getTraceStyle** (const unsigned int variableIndex) const
- void **setTraceWidth** (const int traceWidth, const unsigned int variableIndex)
- int **getTraceWidth** (const unsigned int variableIndex) const

- void **setTraceColor** (const QColor traceColor, const unsigned int variableIndex)
- QColor **getTraceColor** (const unsigned int variableIndex) const
- void **setTraceLegend** (const QString &traceLegend, const unsigned int variableIndex)
- QString **getTraceLegend** (const unsigned int variableIndex) const
- void **setTraceVisible** (const bool traceVisible, const unsigned int variableIndex)
- bool **getTraceVisible** (const unsigned int variableIndex) const
- bool **getTraceVisible1** () const
- bool **getTraceVisible2** () const
- bool **getTraceVisible3** () const
- bool **getTraceVisible4** () const
- bool **getTraceVisible5** () const
- bool **getTraceVisible6** () const
- bool **getTraceVisible7** () const
- bool **getTraceVisible8** () const
- void **setXUnit** (const QString &xUnit)
- QString **getXUnit** () const
- void **setYUnit** (const QString &yUnit)
- QString **getYUnit** () const
- void **setGridEnableMajorX** (const bool gridEnableMajorX)
- bool **getGridEnableMajorX** () const
- void **setGridEnableMajorY** (const bool gridEnableMajorY)
- bool **getGridEnableMajorY** () const
- void **setGridEnableMinorX** (const bool gridEnableMinorX)
- bool **getGridEnableMinorX** () const
- void **setGridEnableMinorY** (const bool gridEnableMinorY)
- bool **getGridEnableMinorY** () const
- void **setGridMajorColor** (const QColor gridMajorColor)
- QColor **getGridMajorColor** () const
- void **setGridMinorColor** (const QColor gridMinorColor)
- QColor **getGridMinorColor** () const
- void **setXStart** (const double xStart)
- double **getXStart** () const
- void **setXIncrement** (const double xIncrement)
- double **getXIncrement** () const
- void **setXFirst** (const double xFirst)
- double **getXFirst** () const
- void **setXLast** (const double xLast)
- double **getXLast** () const
- void **setTimeSpan** (const int timeSpan)
- int **getTimeSpan** () const
- void **setTickRate** (const int tickRate)
- int **getTickRate** () const
- void **setMargin** (const int margin)
- int **getMargin** () const
- void **setMouseMoveSignals** (const QE::MouseMoveSignalFlags flags)
- QE::MouseMoveSignalFlags **getMouseMoveSignals** () const
- void **setFullLengthArraySubscriptions** (const bool useFullLengthArraySubscriptions)
- bool **getFullLengthArraySubscriptions** () const
- [QEGraphic](#) * **getGraphic** () const

Protected Member Functions

- bool **eventFilter** (QObject *watched, QEvent *event)
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)
- [QEChannel](#) * **createQcItem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Protected Attributes

- [QEFloatingFormatting](#) **floatingFormatting**
- [QEIntegerFormatting](#) **integerFormatting**

Properties

- QString **variable1**
- QString **variable2**
- QString **variable3**
- QString **variable4**
- QString **variable5**
- QString **variable6**
- QString **variable7**
- QString **variable8**
- QString **sizeVariable1**
- QString **sizeVariable2**
- QString **sizeVariable3**
- QString **sizeVariable4**
- QString **sizeVariable5**
- QString **sizeVariable6**
- QString **sizeVariable7**
- QString **sizeVariable8**
- QColor **traceColor1**
- QColor **traceColor2**
- QColor **traceColor3**
- QColor **traceColor4**
- QColor **traceColor5**
- QColor **traceColor6**
- QColor **traceColor7**
- QColor **traceColor8**
- TraceStyles **traceStyle1**
- TraceStyles **traceStyle2**
- TraceStyles **traceStyle3**
- TraceStyles **traceStyle4**
- TraceStyles **traceStyle5**

- TraceStyles **traceStyle6**
- TraceStyles **traceStyle7**
- TraceStyles **traceStyle8**
- bool **traceVisible1**
- bool **traceVisible2**
- bool **traceVisible3**
- bool **traceVisible4**
- bool **traceVisible5**
- bool **traceVisible6**
- bool **traceVisible7**
- bool **traceVisible8**
- int **traceWidth1**
- int **traceWidth2**
- int **traceWidth3**
- int **traceWidth4**
- int **traceWidth5**
- int **traceWidth6**
- int **traceWidth7**
- int **traceWidth8**
- QString **traceLegend1**
- QString **traceLegend2**
- QString **traceLegend3**
- QString **traceLegend4**
- QString **traceLegend5**
- QString **traceLegend6**
- QString **traceLegend7**
- QString **traceLegend8**
- double **yMin**
- double **yMax**
- bool **autoScale**
- bool **logScale**
- bool **archiveBackfill**
- bool **axisEnableX**
- bool **axisEnableY**
- SelectedYAxis **selectedYAxis**
- bool **gridEnableMajorX**
- bool **gridEnableMajorY**
- bool **gridEnableMinorX**
- bool **gridEnableMinorY**
- QColor **gridMajorColor**
- QColor **gridMinorColor**
- QString **title**
- QColor **backgroundColor**
- QString **xUnit**
- QString **yUnit**
- double **xStart**

Waveform related properties.

- double **xIncrement**
- double **xFirst**
- double **xLast**
- int **timeSpan**

Plot time span/duration (specified in seconds), constrained to be ≥ 1 .

- int **tickRate**
- int **margin**
- QE::MouseMoveSignalFlags **mouseMoveSignals**
- bool **fullLengthArraySubscriptions**
- QString **variableSubstitutions**

Additional Inherited Members

9.211.1 Detailed Description

This class provides the means to display a number of PVs either as a mini strip chart for scalar PVs or as a mini waveform plotter for array PVs.

NOTE: Mixing scalar and array PVs is not recommended.

9.211.2 Member Function Documentation

9.211.2.1 dbValueChanged [1/2]

```
void QEPlot::dbValueChanged (
    const double & out ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.211.2.2 dbValueChanged [2/2]

```
void QEPlot::dbValueChanged (
    const QVector< double > & out ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.211.3 Property Documentation

9.211.3.1 fullLengthArraySubscriptions

```
bool QEPlot::fullLengthArraySubscriptions [read], [write]
```

Older servers (3.14 era as I recall) do not support zero length subscriptions. Enable this property to work with ancient IOCs. Note: This applies to all arrays, no per plot control is provided.

9.211.3.2 sizeVariable1

```
QString QEPlot::sizeVariable1 [read], [write]
```

These variables is used to read effective waveforms sizes, e.g. wavefom.NORD, and control how many points are plotted. If not specified/connected then the whole array is used for display purposes. Note: Only applicable to waveforms, still connects but otherwise ignored for scalars.

9.211.3.3 tickRate

```
int QEPlot::tickRate [read], [write]
```

Tick rate (specified in mSec), constrained to be ≥ 20 and ≤ 2000 . Would be better called tick interval

9.211.3.4 variable1

```
QString QEPlot::variable1 [read], [write]
```

EPICS variable name (CA PV). These variables is used to read updating values or waveforms.

9.211.3.5 variableSubstitutions

```
QString QEPlot::variableSubstitutions [read], [write]
```

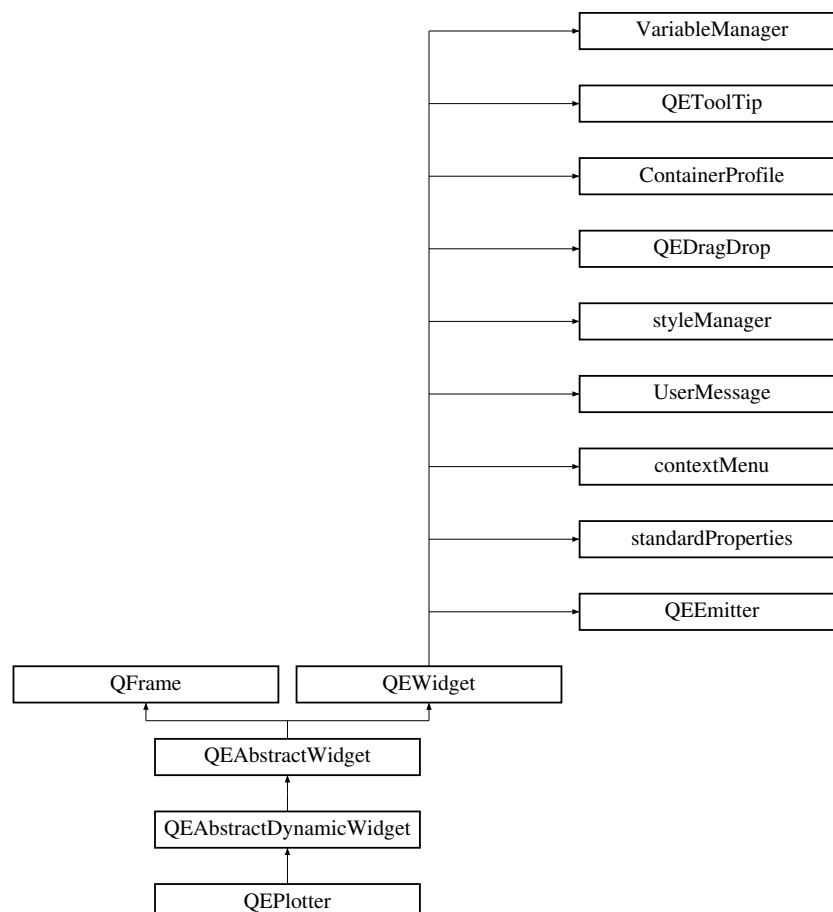
Default macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'SAMPLE=SAM1, NAME = "Ref foil"' These substitutions are applied to all the variable names.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPlot/QEPlot.h
- qeframework/widgets/QEPlot/QEPlot.cpp

9.212 QEPlotter Class Reference

Inheritance diagram for QEPlotter:



Public Types

- enum **Constants** {
PVS_PER_SLOT = 2 , **NUMBER_OF_PLOTS** = 16 , **NUMBER_OF_SLOTS** = 17 , **NONE_SLOT_VI_BASE**
= PVS_PER_SLOT * NUMBER_OF_SLOTS ,
TOTAL_VI_NUMBER = NONE_SLOT_VI_BASE + 2 }

Public Slots

- void **setXYDataPV** (const int slot, const QString &pvName)
 - void **setXYAlias** (const int slot, const QString &alias)
 - void **setXRange** (const double xMinimum, const double xMaximum)
 - void **setYRange** (const double yMinimum, const double yMaximum)
 - void **setDataPvNameSet** (const QStringList &pvNameSet)
 - void **setAliasNameSet** (const QStringList &aliasNameSet)
 - void **setPlotterEntry** (const int slot, const QString &pvName, const QString &alias)
 - void **menuSelected** (const QEPlotterNames::MenuActions action, const int slot)
 - void **usePvNameProperties** (const [QEPvNameProperties](#) &pvNameProperties)
- New style slot functions.*
- void **dataConnectionUpdated** (const [QEConnectionUpdate](#) &update)
 - void **dataArrayUpdated** (const [QEFloatingArrayUpdate](#) &update)
 - void **sizeConnectionUpdated** (const [QEConnectionUpdate](#) &update)
 - void **sizeValueUpdated** (const [QEIntegerValueUpdate](#) &update)
 - void **markerConnectionUpdated** (const [QEConnectionUpdate](#) &update)
 - void **markerValueUpdated** (const [QEFloatingValueUpdate](#) &update)
 - void **setNewVariableName** (QString variableName, QString variableNameSubstitutions, unsigned int variableIndex)
 - void **dataConnectionChanged** ([QCaConnectionInfo](#) &connectionInfo, const unsigned int &variableIndex)
 - void **dataArrayChanged** (const QVector< double > &values, [QCaAlarmInfo](#) &alarmInfo, [QCaDateTime](#) &timeStamp, const unsigned int &variableIndex)
 - void **sizeConnectionChanged** ([QCaConnectionInfo](#) &connectionInfo, const unsigned int &variableIndex)
 - void **sizeValueChanged** (const long &value, [QCaAlarmInfo](#) &alarmInfo, [QCaDateTime](#) &timeStamp, const unsigned int &variableIndex)
 - void **markerConnectionChanged** ([QCaConnectionInfo](#) &connectionInfo, const unsigned int &variableIndex)
 - void **markerValueChanged** (const double &value, [QCaAlarmInfo](#) &alarmInfo, [QCaDateTime](#) &timeStamp, const unsigned int &variableIndex)

Signals

- void **crosshairIndexChanged** (int value)
- void **coordinateSelected** (QPointF xyvalue)
- void **xCoordinateSelected** (double xvalue)
- void **yCoordinateSelected** (double yvalue)
- void **pvDataNameSetChanged** (const QStringList &nameSet)
- void **alaisNameSetChanged** (const QStringList &nameSet)
- void **requestAction** (const [QEActionRequests](#) &)

Public Member Functions

- **QEPlotter** (QWidget *parent=0)
- QSize **sizeHint** () const
- void **setXMarkerPV** (const QString &pvName)
- QString **getXMarkerPV** () const
- void **setYMarkerPV** (const QString &pvName)
- QString **getYMarkerPV** () const
- QString **getXYDataPV** (const int slot) const
- void **setXYSizePV** (const int slot, const QString &pvName)
- QString **getXYSizePV** (const int slot) const
- QString **getXYAlias** (const int slot) const
- void **setXYColour** (const int slot, const QColor &colour)
- QColor **getXYColour** (const int slot) const
- void **setXYLineVisible** (const int slot, const bool isVisible)
- bool **getXYLineVisible** (const int slot) const
- void **setXYLineBold** (const int slot, const bool isBold)
- bool **getXYLineBold** (const int slot) const
- void **setXYLineDashed** (const int slot, const bool isDashed)
- bool **getXYLineDashed** (const int slot) const
- void **setXYLineHasDots** (const int slot, const bool hasDots)
- bool **getXYLineHasDots** (const int slot) const
- void **setEnabledContextMenu** (bool enable)
- bool **getEnabledContextMenu** () const
- void **setMenuEmitText** (const QString &text)
- QString **getMenuEmitText** () const
- void **setToolBarVisible** (bool [visible](#))
- bool **getToolBarVisible** () const
- void **setPvItemsVisible** (bool [visible](#))
- bool **getPvItemsVisible** () const
- void **setStatusVisible** (bool [visible](#))
- bool **getStatusVisible** () const
- void **setAxisEnableX** (bool axisEnableX)
- bool **getAxisEnableX** () const
- void **setAxisEnableY** (bool axisEnableY)
- bool **getAxisEnableY** () const
- void **setVideoMode** (const [QE::VideoModes](#) mode)
- [QE::VideoModes](#) **getVideoMode** () const
- void **setXLogarithmic** (bool [visible](#))
- bool **getXLogarithmic** () const
- void **setYLogarithmic** (bool [visible](#))
- bool **getYLogarithmic** () const
- void **setXMinimum** (const double xMinimum)
- double **getXMinimum** () const
- void **setXMaximum** (const double xMaximum)
- double **getXMaximum** () const
- void **setYMinimum** (const double yMinimum)
- double **getYMinimum** () const
- void **setYMaximum** (const double yMaximum)
- double **getYMaximum** () const
- void **setFullLengthArraySubscriptions** (const bool useFullLengthArraySubscriptions)
- bool **getFullLengthArraySubscriptions** () const
- void **setIsPaused** (bool paused)
- bool **getIsPaused** () const
- int **addPvName** (const QString &pvName)

- void **clearAllPvNames** ()
- int **getCrosshairIndex** () const
- QStringList **getDataPvNameSet** () const
- QStringList **getAliasNameSet** () const
- [QEGraphic](#) * **getGraphic** () const

Protected Member Functions

- void **pvLabelModeChanged** ()
- [QEChannel](#) * **createQcItem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **activated** ()
Do any post-all-widgets-constructed, i.e. activated stuff.
- bool **eventFilter** (QObject *obj, QEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **saveConfiguration** ([PersistenceManager](#) *pm)
- void **restoreConfiguration** ([PersistenceManager](#) *pm, [restorePhases](#) restorePhase)
- int **findSlot** (QObject *obj)
- QString **getXYExpandedDataPV** (const int slot) const
- QString **getXYExpandedSizePV** (const int slot) const

Properties

- bool **enableContextMenu**
- bool **toolBarsVisible**
- bool **pvItemsVisible**
- bool **statusVisible**
- [QE::VideoModes](#) **videoMode**
- bool **xLogarithmic**
- bool **yLogarithmic**
- bool **axisEnableX**
- bool **axisEnableY**
- double **xMinimum**
- double **xMaximum**
- double **yMinimum**
- double **yMaximum**
- QString **contextMenuEmitLegend**
- QString **xMarkerVariable**
- QString **yMarkerVariable**
- QString **DataVariableX**
- QString **DataVariableA**
- QString **DataVariableB**
- QString **DataVariableC**
- QString **DataVariableD**
- QString **DataVariableE**
- QString **DataVariableF**

- QString **DataVariableG**
- QString **DataVariableH**
- QString **DataVariableI**
- QString **DataVariableJ**
- QString **DataVariableK**
- QString **DataVariableL**
- QString **DataVariableM**
- QString **DataVariableN**
- QString **DataVariableO**
- QString **DataVariableP**
- QString **SizeVariableX**
- QString **SizeVariableA**
- QString **SizeVariableB**
- QString **SizeVariableC**
- QString **SizeVariableD**
- QString **SizeVariableE**
- QString **SizeVariableF**
- QString **SizeVariableG**
- QString **SizeVariableH**
- QString **SizeVariableI**
- QString **SizeVariableJ**
- QString **SizeVariableK**
- QString **SizeVariableL**
- QString **SizeVariableM**
- QString **SizeVariableN**
- QString **SizeVariableO**
- QString **SizeVariableP**
- QString **AliasNameX**
- QString **AliasNameA**
- QString **AliasNameB**
- QString **AliasNameC**
- QString **AliasNameD**
- QString **AliasNameE**
- QString **AliasNameF**
- QString **AliasNameG**
- QString **AliasNameH**
- QString **AliasNameI**
- QString **AliasNameJ**
- QString **AliasNameK**
- QString **AliasNameL**
- QString **AliasNameM**
- QString **AliasNameN**
- QString **AliasNameO**
- QString **AliasNameP**
- QColor **ColourA**
- QColor **ColourB**
- QColor **ColourC**
- QColor **ColourD**
- QColor **ColourE**
- QColor **ColourF**
- QColor **ColourG**
- QColor **ColourH**
- QColor **ColourI**
- QColor **ColourJ**
- QColor **ColourK**

- QColor **ColourL**
- QColor **ColourM**
- QColor **ColourN**
- QColor **ColourO**
- QStringList [aliasNames](#)
- bool [fullLengthArraySubscriptions](#)
- QString [variableSubstitutions](#)

Friends

- class **DataSets**

Additional Inherited Members

9.212.1 Member Function Documentation

9.212.1.1 `restoreConfiguration()`

```
void QEPlotter::restoreConfiguration (
    PersistenceManager * ,
    restorePhases ) [protected], [virtual]
```

Service a request to restore the [QE](#) widget's configuration. A [QE](#) widget recover any configuration details from the [PersistenceManager](#). For example, a [QEStripChart](#) may restore the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method. This is called twice with an incrementing restorePhase. Most widgets will miss the first call as they don't exist yet (they are created as part of the first phase).

Reimplemented from [QEWidget](#).

9.212.1.2 `saveConfiguration()`

```
void QEPlotter::saveConfiguration (
    PersistenceManager * ) [protected], [virtual]
```

Service a request to save the [QE](#) widget's current configuration. A widget may save any configuration details through the [PersistenceManager](#). For example, a [QEStripChart](#) may save the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method.

Reimplemented from [QEWidget](#).

9.212.1.3 setNewVariableName

```
void QEPlotter::setNewVariableName (
    QString variableName,
    QString variableNameSubstitutions,
    unsigned int variableIndex ) [slot]
```

These slots are public, therefore keep but flag as deprecated. They will eventually be deleted.

9.212.2 Property Documentation

9.212.2.1 aliasNames

```
QStringList QEPlotter::aliasNames [read], [write]
```

Allows specification of alias as a 'single property', which is also a slot.

9.212.2.2 fullLengthArraySubscriptions

```
bool QEPlotter::fullLengthArraySubscriptions [read], [write]
```

Older servers (3.14 era as I recall) do not support zero length subscriptions. Enable this property to work with ancient IOCs. Note: This applies to all arrays, no per plot control is provided.

9.212.2.3 variableSubstitutions

```
QString QEPlotter::variableSubstitutions [read], [write]
```

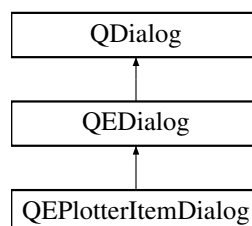
Default macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'SAMPLE=SAM1, NAME = "Ref foil"' These substitutions are applied to all the variable names.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPlotter/QEPlotter.h
- qeframework/widgets/QEPlotter/QEPlotter.cpp

9.213 QEPlotterItemDialog Class Reference

Inheritance diagram for QEPlotterItemDialog:



Public Member Functions

- **QEPlotterItemDialog** (QWidget *parent=0)
- void **setFieldInformation** (const QString dataIn, const QString aliasIn, const QString sizeIn)
- void **getFieldInformation** (QString &dataOut, QString &aliasOut, QString &sizeOut)

Additional Inherited Members

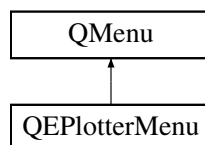
The documentation for this class was generated from the following files:

- qeframework/widgets/QEPlotter/QEPlotterItemDialog.h
- qeframework/widgets/QEPlotter/QEPlotterItemDialog.cpp

9.214 QEPlotterMenu Class Reference

```
#include <QEPlotterMenu.h>
```

Inheritance diagram for QEPlotterMenu:



Signals

- void **selected** (const QEPlotterNames::MenuActions action, const int slot)

Public Member Functions

- **QEPlotterMenu** (QWidget *parent=0)
- **QEPlotterMenu** (const int slot, QWidget *parent=0)
- void **setActionChecked** (const QEPlotterNames::MenuActions action, const bool checked)
- void **setActionEnabled** (const QEPlotterNames::MenuActions action, const bool enabled)
- void **setActionVisible** (const QEPlotterNames::MenuActions action, const bool visible)
- void **setActionText** (const QEPlotterNames::MenuActions action, const QString &caption)
- void **setCheckedStates** (const bool isDisplayed, const bool isBold, const bool isDashed, const bool show↔ Dots)

9.214.1 Detailed Description

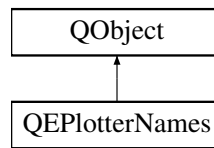
[QEPlotter](#) PV item specific context menu.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPlotter/QEPlotterMenu.h
- qeframework/widgets/QEPlotter/QEPlotterMenu.cpp

9.215 QEPlotterNames Class Reference

Inheritance diagram for QEPlotterNames:



Public Types

- enum **MenuActions** {
PLOTTER_FIRST = contextMenu::CM_SPECIFIC_WIDGETS_START_HERE , **PLOTTER_SHOW_HIDE**↔
_CROSSHAIRS , **PLOTTER_SHOW_HIDE_TOOLBAR** , **PLOTTER_SHOW_HIDE_PV_ITEMS** ,
PLOTTER_SHOW_HIDE_STATUS , **PLOTTER_EMIT_COORDINATES** , **PLOTTER_SAVE** , **PLOTTER**↔
PREV ,
PLOTTER_NEXT , **PLOTTER_NORMAL_VIDEO** , **PLOTTER_REVERSE_VIDEO** , **PLOTTER_LINEAR**↔
Y_SCALE ,
PLOTTER_LOG_Y_SCALE , **PLOTTER_MANUAL_Y_RANGE** , **PLOTTER_CURRENT_Y_RANGE** ,
PLOTTER_DYNAMIC_Y_RANGE ,
PLOTTER_NORAMLISED_Y_RANGE , **PLOTTER_FRACTIONAL_Y_RANGE** , **PLOTTER_LINEAR_X**↔
SCALE , **PLOTTER_LOG_X_SCALE** ,
PLOTTER_MANUAL_X_RANGE , **PLOTTER_CURRENT_X_RANGE** , **PLOTTER_DYNAMIC_X_RANGE** ,
PLOTTER_MANUAL_XY_RANGE ,
PLOTTER_PLAY , **PLOTTER_PAUSE** , **PLOTTER_LOAD_CONFIG** , **PLOTTER_SAVE_CONFIG** ,
PLOTTER_COPY_VARIABLE , **PLOTTER_COPY_DATA** , **PLOTTER_PASTE** , **PLOTTER_DRAG**↔
VARIABLE ,
PLOTTER_DRAG_DATA , **PLOTTER_SELECT_USE_PV_NAME** , **PLOTTER_SELECT_USE_ALIAS**↔
NAME , **PLOTTER_SELECT_USE_DESCRIPTION** ,
PLOTTER_LINE_BOLD , **PLOTTER_LINE_DASHED** , **PLOTTER_LINE_DOTS** , **PLOTTER_LINE**↔
VISIBLE ,
PLOTTER_LINE_NO_MEDIAN_FILTER , **PLOTTER_LINE_MEDIAN_3_FILTER** , **PLOTTER_LINE**↔
MEDIAN_5_FILTER , **PLOTTER_LINE_COLOUR** ,
PLOTTER_DATA_SELECT , **PLOTTER_DATA_DIALOG** , **PLOTTER_DATA_CLEAR** , **PLOTTER**↔
SCALE_TO_MIN_MAX ,
PLOTTER_SCALE_TO_ZERO_MAX , **PLOTTER_COPY_DATA_VARIABLE** , **PLOTTER_COPY_DATA**↔
DATA , **PLOTTER_PASTE_DATA_PV** ,
PLOTTER_SHOW_DATA_PV_PROPERTIES , **PLOTTER_ADD_DATA_PV_TO_STRIPCHART** , **PLOTTER**↔
_ADD_DATA_PV_TO_SCRATCH_PAD , **PLOTTER_GENERAL_DATA_PV_EDIT** ,
PLOTTER_COPY_SIZE_VARIABLE , **PLOTTER_COPY_SIZE_DATA** , **PLOTTER_PASTE_SIZE_PV** ,
PLOTTER_SHOW_SIZE_PV_PROPERTIES ,
PLOTTER_ADD_SIZE_PV_TO_STRIPCHART , **PLOTTER_ADD_SIZE_PV_TO_SCRATCH_PAD** ,
PLOTTER_GENERAL_SIZE_PV_EDIT , **PLOTTER_LAST** }
 enum **ScaleModes** { **smFixed** , **smNormalised** , **smFractional** , **smDynamic** }

Public Member Functions

- **QEPlotterNames** (QObject *parent=0)

The documentation for this class was generated from the following file:

- qeframework/widgets/QEPlotter/QEPlotterNames.h

9.216 QEPlotterPushButtonSpecifications Struct Reference

Public Attributes

- int **gap**
- int **width**
- bool **isIcon**
- const QString **captionOrIcon**
- QEPlotterNames::MenuActions **action**
- const QString **toolTip**

The documentation for this struct was generated from the following file:

- qeframework/widgets/QEPlotter/QEPlotterToolBar.cpp

9.217 QEPlotterState Class Reference

Public Member Functions

- void **saveConfiguration** ([PMElement](#) &parentElement)
- void **restoreConfiguration** ([PMElement](#) &parentElement)

Public Attributes

- bool **isXLogarithmic**
- bool **isYLogarithmic**
- bool **isReverse**
- bool **isPaused**
- double **xMinimum**
- double **xMaximum**
- double **yMinimum**
- double **yMaximum**
- QEPlotterNames::ScaleModes **xScaleMode**
- QEPlotterNames::ScaleModes **yScaleMode**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPlotter/QEPlotterState.h
- qeframework/widgets/QEPlotter/QEPlotterState.cpp

9.218 QEPlotterStateList Class Reference

Public Member Functions

- void **clear** ()
- void **push** (const QEPlotterState &state)
- bool **prev** (QEPlotterState &state)
- bool **next** (QEPlotterState &state)
- bool **prevAvailable** ()
- bool **nextAvailable** ()

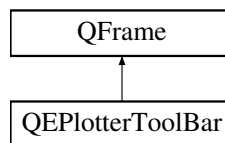
The documentation for this class was generated from the following files:

- qeframework/widgets/QEPlotter/QEPlotterState.h
- qeframework/widgets/QEPlotter/QEPlotterState.cpp

9.219 QEPlotterToolBar Class Reference

```
#include <QEPlotterToolBar.h>
```

Inheritance diagram for QEPlotterToolBar:



Signals

- void **selected** (const QEPlotterNames::MenuActions action, const int slot)

Public Member Functions

- QEPlotterToolBar (QWidget *parent=0)
- void **setEnabled** (const QEPlotterNames::MenuActions action, const bool enabled)

Static Public Attributes

- static const int **designHeight** = 32

Protected Member Functions

- void **resizeEvent** (QResizeEvent *event)

9.219.1 Detailed Description

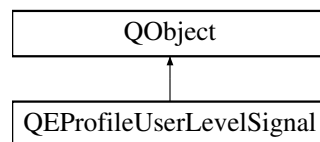
This class holds all the [QEPlotter](#) tool bar widget.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPlotter/QEPlotterToolBar.h
- qeframework/widgets/QEPlotter/QEPlotterToolBar.cpp

9.220 QEProfileUserLevelSignal Class Reference

Inheritance diagram for QEProfileUserLevelSignal:



Signals

- void [userChanged](#) ([QE::UserLevels](#) level)
Internal use only. Send when the user level has changed.

Public Member Functions

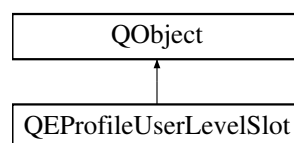
- void **setLevel** ([QE::UserLevels](#) level)
- [QE::UserLevels](#) **getLevel** () const

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/ContainerProfile.h
- qeframework/widgets/QEWidget/ContainerProfile.cpp

9.221 QEProfileUserLevelSlot Class Reference

Inheritance diagram for QEProfileUserLevelSlot:



Public Slots

- void **userChanged** ([QE::UserLevels](#) level)

Public Member Functions

- void **setOwner** ([ContainerProfile](#) *owner)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/ContainerProfile.h
- qeframework/widgets/QEWidget/ContainerProfile.cpp

9.222 QEPublishedProfile Class Reference

Public Attributes

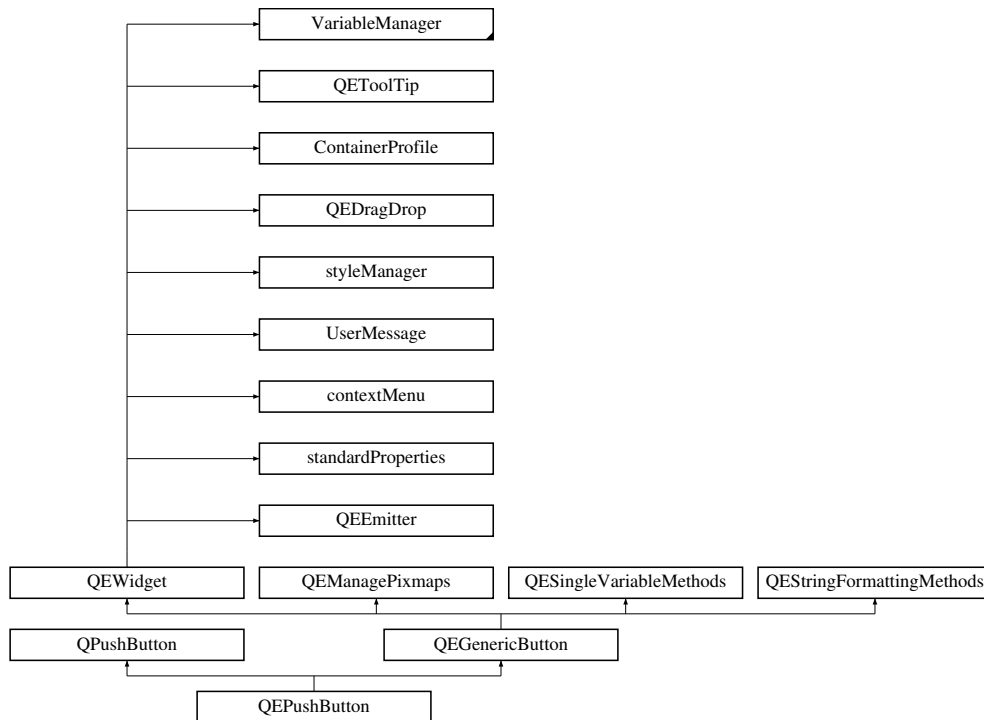
- QObject * **guiLaunchConsumer**
- QStringList **pathList**
- QString **parentPath**
- QList< QString > **macroSubstitutions**
- unsigned int **messageFormId**
- QList< [QEWidget](#) * > **containedWidgets**
- [QEProfileUserLevelSignal](#) **userSignal**
- QString **userLevelPassword**
- QString **scientistLevelPassword**
- QString **engineerLevelPassword**
- bool **profileDefined**
- [PersistenceManager](#) **persistenceManager**
- bool **dontActivateYet**
- bool **userLevelPasswordsSet**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/ContainerProfile.h
- qeframework/widgets/QEWidget/ContainerProfile.cpp

9.223 QEPushButton Class Reference

Inheritance diagram for QEPushButton:



Public Slots

- void **writeNow** ()
- void **requestAction** (const [QEActionRequests](#) &request)
- void **setDefaultStyle** (const QString &style)
- void **setManagedVisible** (bool v)

Signals

- void **dbValueChanged** ()
 - void **dbValueChanged** (const QString &out)
 - void **dbValueChanged** (const int &out)
 - void **dbValueChanged** (const long &out)
 - void **dbValueChanged** (const qlonglong &out)
 - void **dbValueChanged** (const double &out)
 - void **dbValueChanged** (const bool &out)
 - void **dbConnectionChanged** (const bool &isConnected)
- Sent when the widget state updated following a channel connection change.*
- void **requestResend** ()
 - void **newGui** (const [QEActionRequests](#) &request)
 - void **pressed** (int value)
 - void **released** (int value)
 - void **clicked** (int value)
 - void **programCompleted** ()
- Program started by button has compelted.*

Public Member Functions

- void **setVariableNameSubstitutionsProperty** (const QString &substitutions)
- void **setAltReadbackProperty** (const QString &variableName)
- QString **getAltReadbackProperty** () const
- void **setAltReadbackArrayIndex** (const int [arrayIndex](#))
- int **getAltReadbackArrayIndex** () const
- void **setFormatProperty** (QE::Formats [format](#))
 - Access function for [format](#) property - refer to [format](#) property for details.*
- QE::Formats **getFormatProperty** () const
 - Access function for [format](#) property - refer to [format](#) property for details.*
- QEPushButton (QWidget *parent=0)
- QEPushButton (const QString &variableName, QWidget *parent=0)
- ~QEPushButton ()
 - Destructor.*

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [arrayIndex](#)
- QString [altReadbackVariable](#)
- int [altReadbackArrayIndex](#)
- QString [labelText](#)
- QString [clickText](#)
- QString [clickCheckedText](#)
- bool [writeOnClick](#)
- QString [guiFile](#)
- QString [prioritySubstitutions](#)
- QE::CreationOptions [creationOption](#)
- QString [customisationName](#)
- QString [program](#)
- QStringList [arguments](#)
- QE::ProgramStartupOptions [programStartupOption](#)
- QE::DisabledRecordPolicy [disabledRecordPolicy](#)
- Qt::Alignment [alignment](#)
- QE::UpdateOptions [updateOption](#)
- QPixmap [pixmap0](#)
- QPixmap [pixmap1](#)
- QPixmap [pixmap2](#)
- QPixmap [pixmap3](#)
- QPixmap [pixmap4](#)
- QPixmap [pixmap5](#)
- QPixmap [pixmap6](#)
- QPixmap [pixmap7](#)
- QString [password](#)
- bool [confirmAction](#)
- QString [confirmText](#)
- bool [writeOnPress](#)
- bool [writeOnRelease](#)
- QString [pressText](#)
- QString [releaseText](#)
- bool [subscribe](#)

- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- QE::UserLevels [userLevelVisibility](#)
- QE::UserLevels [userLevelEnabled](#)
- QE::DisplayAlarmStateOptions [displayAlarmStateOption](#)
- bool [oosAware](#)
- int [precision](#)
- bool [useDbPrecision](#)
- int [leadingZeros](#)
- bool [addUnits](#)
- bool [forceSign](#)
- QString [localEnumeration](#)
- QE::Formats [format](#)
- int [radix](#)
- QE::Separators [separator](#)
- QE::Notations [notation](#)
- QE::ArrayActions [arrayAction](#)

Additional Inherited Members

9.223.1 Constructor & Destructor Documentation

9.223.1.1 QEPushButton() [1/2]

```
QEPushButton::QEPushButton (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.223.1.2 QEPushButton() [2/2]

```
QEPushButton::QEPushButton (
    const QString & variableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.223.2 Member Function Documentation

9.223.2.1 clicked

```
void QEPushButton::clicked (
    int value ) [signal]
```

Button has been Clicked. The value emitted is the integer interpretation of the clickText property (or the click↔CheckedText property if the button was checked)

9.223.2.2 dbValueChanged

```
void QEPushButton::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.223.2.3 newGui

```
void QEPushButton::newGui (
    const QEAActionRequests & request ) [signal]
```

Internal use only. Request a new GUI is created. Typically, this is caught by the QEGui application.

9.223.2.4 pressed

```
void QEPushButton::pressed (
    int value ) [signal]
```

Button has been Pressed. The value emitted is the integer interpretation of the pressText property

9.223.2.5 released

```
void QEPushButton::released (
    int value ) [signal]
```

Button has been Released The value emitted is the integer interpretation of the releaseText property

9.223.2.6 requestAction

```
void QEPushButton::requestAction (
    const QEAActionRequests & request ) [inline], [slot]
```

Default slot used to create a new GUI if there is no slot indicated in the [ContainerProfile](#) class. This slot is typically used when the button is pressed within the Designer preview window to allow the operation of the button to be tested. If an application does not specify a slot to use for creating new windows (through the [ContainerProfile](#) class) a window will still be created through this slot, but it will not respect the window creation options or any other window related application constraints. For example, the QEGui application does provide a slot for creating new GUIs in the [ContainerProfile](#) class which respects the creation options, knows how to add tabs in the application, and extend the application's window menu in the menu bar.

9.223.2.7 requestResend

```
void QEPushButton::requestResend ( ) [signal]
```

Internal use only. Used when changing a property value to force a re-display to reflect the new property value.

9.223.2.8 setDefaultStyle

```
void QEPushButton::setDefaultStyle (
    const QString & style ) [inline], [slot]
```

Update the default style applied to this widget.

9.223.2.9 setManagedVisible

```
void QEPushButton::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call to this slot, but will only be made visible by a call to this slot if the user level allows.

9.223.3 Property Documentation

9.223.3.1 addUnits

```
bool QEPushButton::addUnits [read], [write]
```

If true (default), add engineering units supplied with the data.

9.223.3.2 alignment

```
Qt::Alignment QEPushButton::alignment [read], [write]
```

Set the buttons text alignment. Left justification is particularly useful when displaying quickly changing numeric data updates.

9.223.3.3 allowDrop

```
bool QEPushButton::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.223.3.4 altReadbackVariable

```
QString QEPushButton::altReadbackVariable [read], [write]
```

EPICS variable name (CA PV). This variable is used to provide a readback value when different to the variable written to by a button press.

9.223.3.5 arguments

```
QStringList QEPushButton::arguments [read], [write]
```

Arguments for program specified in the 'program' property.

9.223.3.6 arrayAction

```
QE::ArrayActions QEPushButton::arrayAction [read], [write]
```

Text formatting option for array data. Default is ASCII. Options are:

- ASCII - treat array as a single text string. For example an array of three characters 'a' 'b' 'c' will be formatted as 'abc'.
- APPEND - treat array as an array of numbers and format a string containing them all with a space between each. For example, an array of three numbers 10, 11 and 12 will be formatted as '10 11 12'.
- INDEX - Extract a single item from the array. The item is then formatted as any other non array data would be. The item selected is determined by the arrayIndex property. For example, if arrayIndex property is 1, an array of three numbers 10, 11 and 12 will be formatted as '11'.

9.223.3.7 arrayIndex

```
int QEPushButton::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.223.3.8 clickCheckedText

```
QString QEPushButton::clickCheckedText [read], [write]
```

Text used to compare with text written or read to determine if push button should be marked as checked. Note, must be an exact match following formatting of data updates. When writing values, the 'pressText', 'ReleaseText', or 'clickedtext' must match this property to cause the button to be checked when the write occurs.

Good example: formatting set to display a data value of '1' as 'On', clickCheckedText is 'On', clickText is 'On'. In this example, the push button will be checked when a data update occurs with a value of 1 or when the button is clicked.

Bad example: formatting set to display a data value of '1' as 'On', clickCheckedText is 'On', clickText is '1'. In this example, the push button will be checked when a data update occurs with a value of 1 but, although a valid value will be written when clicked, the button will not be checked when clicked as '1' is not the same as 'On'.

9.223.3.9 clickText

```
QString QEPushButton::clickText [read], [write]
```

Value written when user clicks button if 'writeOnClick' property is true. Note: Set format to Integer when writing numeric values to enum PVs like the bo and mbbo records.

9.223.3.10 confirmAction

```
bool QEPushButton::confirmAction [read], [write]
```

If true, a dialog will be presented asking the user to confirm if the button action should be carried out

9.223.3.11 confirmText

```
QString QEPushButton::confirmText [read], [write]
```

Text used to confirm action if confirmation dialog is presented

9.223.3.12 creationOption

```
QE::CreationOptions QEPushButton::creationOption [read], [write]
```

Creation options when opening a new GUI. Open a new window, open a new tab, or replace the current window. the creation option is supplied when the button generates a newGui signal. Application code connected to this signal should honour this request if possible. When used within the QEGui application, the QEGui application creates a new window, new tab, or replaces the current window as appropriate.

9.223.3.13 customisationName

```
QString QEPushButton::customisationName [read], [write]
```

Window customisation name. This name will be used to select a set of window customisations including menu items and tool bar buttons. Applications such as QEGui can load .xml files containing named sets of window customisations. This property is used to select a set loaded from these files. The selected set of customisations will be applied to the main window containing the new GUI. Customisations are not applied if the GUI is opened as a dock.

9.223.3.14 defaultStyle

```
QString QEPushButton::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.223.3.15 disabledRecordPolicy

`QE::DisabledRecordPolicy QEPushButton::disabledRecordPolicy [read], [write]`

Set the widget's disabled record policy, i.e. the action to be taken when the under lying record is disabled, i.e. when the associated record's STAT field value is Disabled. When the policy is ignore, then no special action is taken. This is the default policy. When the policy is grayout, the widget is style is set as if disconnected when the record is disabled. When the policy is disable, the widget is disabled when the record is disabled.

9.223.3.16 displayAlarmStateOption

`QE::DisplayAlarmStateOptions QEPushButton::displayAlarmStateOption [read], [write]`

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.223.3.17 forceSign

`bool QEPushButton::forceSign [read], [write]`

If false (default), no "+" sign, when true always add a sign.

9.223.3.18 format

`QE::Formats QEPushButton::format [read], [write]`

Format to apply to data. Default is 'Default' in which case the data type supplied with the data determines how the data is formatted. For all other options, an attempt is made to format the data as requested (whatever its native form).

9.223.3.19 guiFile

`QString QEPushButton::guiFile [read], [write]`

File name of GUI to be presented on button click. File name can be absolute, relative to the path of the [QEForm](#) in which the button widget is located, relative to the any path in the path list published in the [ContainerProfile](#) class, or relative to the current path. See [QEWidget::openQEFile\(\)](#) in [QEWidget.cpp](#) for details.

9.223.3.20 labelText

`QString QEPushButton::labelText [read], [write]`

Button label text (prior to substitution). Macro substitutions will be applied to this text and the result will be set as the button text. Used when data updates are not being represented in the button text. IF NOT LEFT EMPTY, THIS TEXT WILL TAKE PRIORITY OVER THE PUSH BUTTON 'text' PROPERTY! For example, a button in a sub form may have a 'labelText' property of 'Turn Pump On'. When the sub form is used twice in a main form with substitutions PUMPNUM=1 and PUMPNUM=2 respectively, the two identical buttons in the sub forms will have the labels 'Turn Pump 1 On' and 'Turn Pump 2 On' respectively.

9.223.3.21 leadingZeros

```
int QEPushButton::leadingZeros [read], [write]
```

The minimum number of leading digits before any decimal/radix point. The default is 1.

9.223.3.22 localEnumeration

```
QString QEPushButton::localEnumeration [read], [write]
```

An enumeration list used to data values. Used only when the formatting option is 'local enumeration'. Value is converted to an integer and used to select a string from this list.

Format is:

```
[[<|<=|=|!=|>|=|>]value1|*]: string1 , [[<|<=|=|!=|>|=|>]value2|*]: string2 , [[<|<=|=|!=|>|=|>]value3|*]: string3 , ...
```

Where: < Less than <= Less than or equal = Equal (default if no operator specified) >= Greather than or equal

Greater than

* Always match (used to specify default text)

Values may be numeric or textual Values do not have to be in any order, but first match wins Values may be quoted Strings may be quoted Consecutive values do not have to be present. Operator is assumed to be equality if not present. White space is ignored except within quoted strings. may be included in a string to indicate a line break

Examples are:

```
0:Off,1:On 0 : "Pump Running", 1 : "Pump not running" 0:"", 1:"Warning!\nAlarm" <2:"Value is less than two", =2↵
:"Value is equal to two", >2:"Value is grater than 2" 3:"Beamline Available", *:"" "Pump Off": "OH NO!, the pump is OFF!", "Pump On": "It's OK, the pump is on"
```

The data value is converted to a string if no enumeration for that value is available. For example, if the local enumeration is '0:off,1:on', and a value of 10 is processed, the text generated is '10'. If a blank string is required, this should be explicit. for example, '0:off,1:on,10: ""'

A range of numbers can be covered by a pair of values as in the following example: >=4:"Between 4 and 8", <=8↵
:"Between 4 and 8"

9.223.3.23 messageSourceId

```
unsigned int QEPushButton::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.223.3.24 notation

```
QE::Notations QEPushButton::notation [read], [write]
```

Notation used for numerical formatting. Default is fixed.

9.223.3.25 oosAware

```
bool QEPushButton::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.223.3.26 password

```
QString QEPushButton::password [read], [write]
```

Password user will need to enter before any action is taken

9.223.3.27 pixmap0

```
QPixmap QEPushButton::pixmap0 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 0

9.223.3.28 pixmap1

```
QPixmap QEPushButton::pixmap1 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 1

9.223.3.29 pixmap2

```
QPixmap QEPushButton::pixmap2 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 2

9.223.3.30 pixmap3

```
QPixmap QEPushButton::pixmap3 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 3

9.223.3.31 pixmap4

```
QPixmap QEPushButton::pixmap4 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 4

9.223.3.32 pixmap5

```
QPixmap QEPushButton::pixmap5 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 5

9.223.3.33 pixmap6

```
QPixmap QEPushButton::pixmap6 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 6

9.223.3.34 pixmap7

```
QPixmap QEPushButton::pixmap7 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 7

9.223.3.35 precision

```
int QEPushButton::precision [read], [write]
```

Precision used when formatting floating point numbers. The default is 4. This is only used if useDbPrecision is false.

9.223.3.36 pressText

```
QString QEPushButton::pressText [read], [write]
```

Value written when user presses button if 'writeOnPress' property is true

9.223.3.37 prioritySubstitutions

```
QString QEPushButton::prioritySubstitutions [read], [write]
```

Overriding macro substitutions. These macro substitutions take precedence over any existing macro substitutions defined by the variableSubstitutions property, any parent forms, or the application containing the button. These macro substitutions are particularly usefull when the button's function is to reload the same form but with different macro substitutions. The variableSubstitutions property cannot be used for this since, although they are added to the list of macro substitutions applied to the new form, they are appended to the list and the existing macro substitutions take precedence.

9.223.3.38 program

```
QString QEPushButton::program [read], [write]
```

Program to run when the button is clicked. No attempt to run a program is made if this property is empty. Example: firefox

9.223.3.39 programStartupOption

```
QE::ProgramStartupOptions QEPushButton::programStartupOption [read], [write]
```

Startup options. Just run the command, run the command within a terminal, or display the output in [QE](#) message system.

9.223.3.40 radix

```
int QEPushButton::radix [read], [write]
```

Base used for when formatting integers. Default is 10 (duh!)

9.223.3.41 releaseText

```
QString QEPushButton::releaseText [read], [write]
```

Value written when user releases button if 'writeOnRelease' property is true

9.223.3.42 separator

```
QE::Separators QEPushButton::separator [read], [write]
```

Seperators used for interger and fixed point formatting. Default is None.

9.223.3.43 styleSheet

```
QString QEPushButton::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly setable per se. This also stops transient styles being saved to the ui file.

9.223.3.44 subscribe

```
bool QEPushButton::subscribe [read], [write]
```

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.223.3.45 updateOption

`QE::UpdateOptions QEPushButton::updateOption [read], [write]`

Update options (text, pixmap, both, or state (checked or unchecked))

9.223.3.46 useDbPrecision

`bool QEPushButton::useDbPrecision [read], [write]`

If true (default), format floating point numbers using the precision supplied with the data. If false, the precision property is used.

9.223.3.47 userLevelEnabled

`QE::UserLevels QEPushButton::userLevelEnabled [read], [write]`

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.223.3.48 userLevelEngineerStyle

`QString QEPushButton::userLevelEngineerStyle [read], [write]`

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.223.3.49 userLevelScientistStyle

`QString QEPushButton::userLevelScientistStyle [read], [write]`

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.223.3.50 userLevelUserStyle

`QString QEPushButton::userLevelUserStyle [read], [write]`

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.223.3.51 userLevelVisibility

```
QE::UserLevels QEPushButton::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.223.3.52 variable

```
QString QEPushButton::variable [read], [write]
```

EPICS variable name (CA PV)

9.223.3.53 variableAsToolTip

```
bool QEPushButton::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.223.3.54 variableSubstitutions

```
QString QEPushButton::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.223.3.55 visible

```
bool QEPushButton::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

9.223.3.56 writeOnClick

```
bool QEPushButton::writeOnClick [read], [write]
```

If true, the 'clickText' property is written when the button is clicked. Default is true

9.223.3.57 writeOnPress

```
bool QEPushButton::writeOnPress [read], [write]
```

If true, the 'pressText' property is written when the button is pressed. Default is false

9.223.3.58 writeOnRelease

```
bool QEPushButton::writeOnRelease [read], [write]
```

If true, the 'releaseText' property is written when the button is released. Default is false

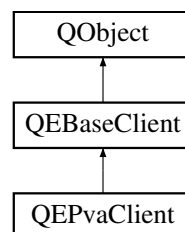
The documentation for this class was generated from the following files:

- qeframework/widgets/QEButton/QEPushButton.h
- qeframework/widgets/QEButton/QEPushButton.cpp

9.224 QEPvaClient Class Reference

```
#include <QEPvaClient.h>
```

Inheritance diagram for QEPvaClient:



Public Member Functions

- **QEPvaClient** (const QString &pvName, QObject *parent)
- bool **openChannel** (const ChannelModesFlags modes)
- void **closeChannel** ()
- QVariant **getPvData** () const
- bool **putPvData** (const QVariant &value)
- bool **getIsConnected** () const
- bool **dataIsAvailable** () const
- QString **getId** () const
- QString **getRemoteAddress** () const
- QString **getEgu** () const
- int **getPrecision** () const
- unsigned int **hostElementCount** () const
- unsigned int **dataElementCount** () const
- double **getDisplayLimitHigh** () const
- double **getDisplayLimitLow** () const
- double **getHighAlarmLimit** () const
- double **getLowAlarmLimit** () const
- double **getHighWarningLimit** () const
- double **getLowWarningLimit** () const
- double **getControlLimitHigh** () const
- double **getControlLimitLow** () const
- double **getMinStep** () const
- QStringList **getEnumerations** () const
- [QCaAlarmInfo](#) **getAlarmInfo** () const
- [QCaDateTime](#) **getTimeStamp** () const
- QString **getDescription** () const
- bool **getReadAccess** () const
- bool **getWriteAccess** () const

Friends

- class **QEPvaClientReference**
- class **QEPvaPutRequesterInterface**

Additional Inherited Members

9.224.1 Detailed Description

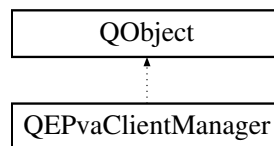
The [QEPvaClient](#) class

The documentation for this class was generated from the following files:

- qeframework/protocol/QEPvaClient.h
- qeframework/protocol/QEPvaClient.cpp

9.225 QEPvaClientManager Class Reference

Inheritance diagram for QEPvaClientManager:



Static Public Member Functions

- static void **initialise** ()

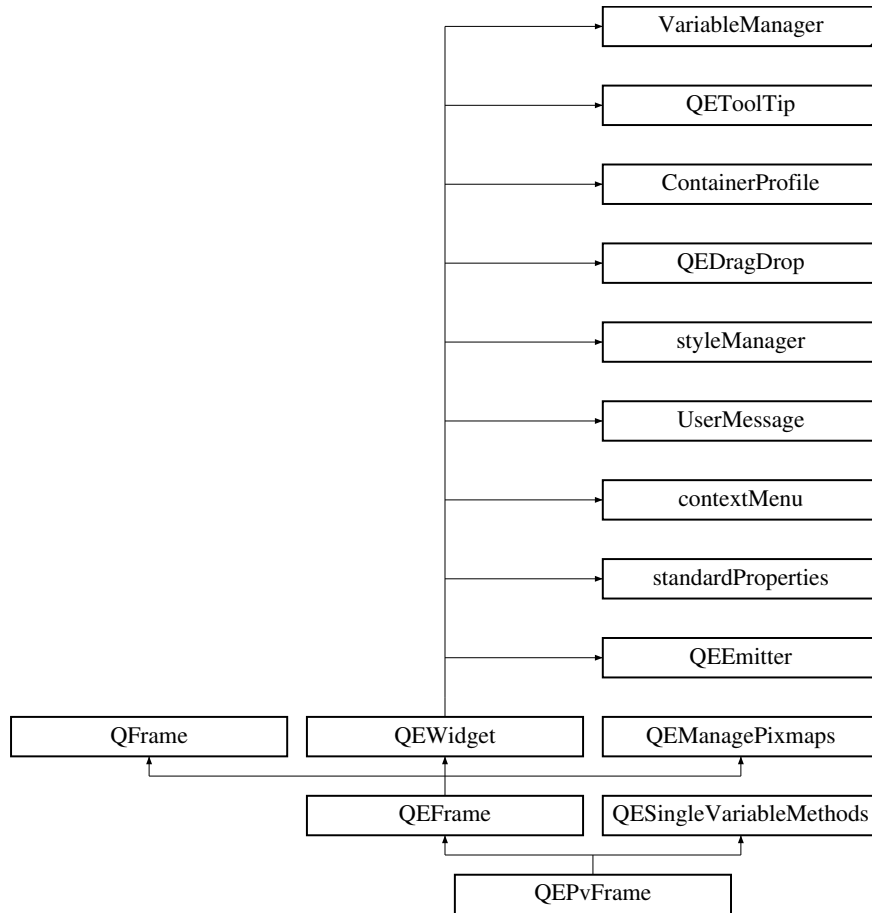
The documentation for this class was generated from the following files:

- qeframework/protocol/QEPvaClient.h
- qeframework/protocol/QEPvaClient.cpp

9.226 QEPvFrame Class Reference

```
#include <QEPvFrame.h>
```

Inheritance diagram for QEPvFrame:



Signals

- void [dbValueChanged](#) ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void [dbConnectionChanged](#) (const bool &isConnected)

Public Member Functions

- **QEPvFrame** (QWidget *parent=0)
- **QEPvFrame** (const QString &variableName, QWidget *parent=0)

Protected Member Functions

- [QEChannel](#) * [createQcaltem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- QString [copyVariable](#) ()
- QVariant [copyData](#) ()

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)

Additional Inherited Members

9.226.1 Detailed Description

The [QEPvFrame](#) class provides an extension to the [QEFrame](#) class in that it allows the value of a nominated PV to select one of 8 pixmaps to use as frame background.

9.226.2 Member Function Documentation

9.226.2.1 dbConnectionChanged

```
void QEPvFrame::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.226.2.2 dbValueChanged

```
void QEPvFrame::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.226.3 Property Documentation

9.226.3.1 arrayIndex

```
int QEPvFrame::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.226.3.2 elementsRequired

```
int QEPvFrame::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.226.3.3 variable

```
QString QEPvFrame::variable [read], [write]
```

EPICS variable name (CA PV)

9.226.3.4 variableSubstitutions

```
QString QEPvFrame::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEFrame/QEPvFrame.h
- qeframework/widgets/QEFrame/QEPvFrame.cpp

9.227 QEPvInfo Class Reference

Public Attributes

- QString **pvName**
- QString **value**
- [QE::Formats](#) **format**

The documentation for this class was generated from the following file:

- qeframework/common/windowCustomisation.h

Static Public Attributes

- static const int **NumberOfButtons** = 15

Protected Member Functions

- [QEChannel](#) * [createQcItem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **resizeEvent** (QResizeEvent *)
- bool **eventFilter** (QObject *obj, QEvent *event)

Properties

- QString [configurationFileLeft](#)
- QString **configurationFileRight**
- QString [defaultSubstitutions](#)
- QString [defaultDir](#)
- bool [confirmAction](#)

Friends

- class **QEPvLoadSaveCompare**
- class **QEPvLoadSaveModel**

Additional Inherited Members

9.228.1 Constructor & Destructor Documentation

9.228.1.1 QEPvLoadSave()

```
QEPvLoadSave::QEPvLoadSave (
    QWidget * parent = 0 )
```

Create without a nominated config file.

9.228.2 Property Documentation

9.228.2.1 configurationFileLeft

QString QEPvLoadSave::configurationFileLeft [read], [write]

configurationFile

9.228.2.2 confirmAction

bool QEPvLoadSave::confirmAction [read], [write]

If true, a dialog will be presented asking the user to confirm if the PV write actions should be carried out. Defaults to true.

9.228.2.3 defaultDir

QString QEPvLoadSave::defaultDir [read], [write]

Default directory used for loading/saving files. Default to null string which is interpreted as the current directory.

9.228.2.4 defaultSubstitutions

QString QEPvLoadSave::defaultSubstitutions [read], [write]

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

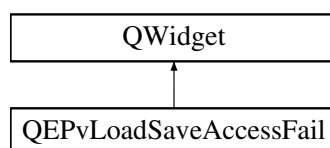
The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSave.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSave.cpp

9.229 QEPvLoadSaveAccessFail Class Reference

```
#include <QEPvLoadSaveAccessFail.h>
```

Inheritance diagram for QEPvLoadSaveAccessFail:



Public Member Functions

- **QEPvLoadSaveAccessFail** ([QEPvLoadSave](#) *owner, QWidget *parent=0)
- void **clear** ()
- void **addPVName** (const QString &pvName)
- void **addPVNames** (const QStringList &pvNameList)

9.229.1 Detailed Description

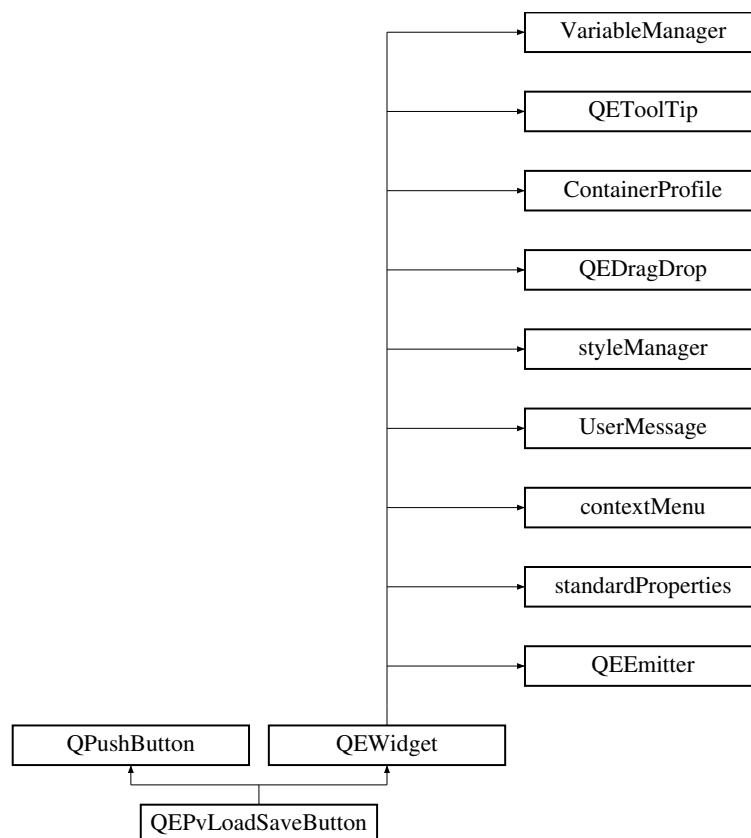
This widget displays a list of PVs that failed to extracted, applied or read from the archive,

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveAccessFail.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveAccessFail.cpp

9.230 QEPvLoadSaveButton Class Reference

Inheritance diagram for QEPvLoadSaveButton:



Public Types

- enum **Actions** { **LoadToPVs** , **SaveToFile** }

Public Slots

- void [setManagedVisible](#) (bool v)

Public Member Functions

- [QEPvLoadSaveButton](#) (QWidget *parent=NULL)
- virtual [~QEPvLoadSaveButton](#) ()
Destruction.
- void **setConfigurationFile** (QString [configurationFile](#))
- QString **getConfigurationFile** ()
- void **setSubstitutions** (QString configurationFileSubstitutions)
- QString **getSubstitutions** () const
- void **setAction** (Actions loadOrSaveIn)
- Actions **getAction** () const
- void **setConfirmAction** (bool confirmRequiredIn)
- bool **getConfirmAction** () const
- void **setConfirmText** (QString confirmTextIn)
- QString **getConfirmText** () const
- void **setShowProgressDialog** (bool showProgressDialogIn)
- bool **getShowProgressDialog** () const

Properties

- QString [configurationFile](#)
- QString [defaultSubstitutions](#)
- Actions [action](#)
- bool [confirmAction](#)
- QString [confirmText](#)
- bool [showProgressDialog](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)

Additional Inherited Members

9.230.1 Constructor & Destructor Documentation

9.230.1.1 QEPvLoadSaveButton()

```
QEPvLoadSaveButton::QEPvLoadSaveButton (
    QWidget * parent = NULL )
```

displayAlarmStateOption not applicable to V2 options. Class should ensure this option is initialised to DISPLAY_↔ALARM_STATE_NEVER. Create without a nominated config file.

9.230.2 Member Function Documentation

9.230.2.1 setManagedVisible

```
void QEPvLoadSaveButton::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call to this slot, but will only be made visible by a call to this slot if the user level allows.

9.230.3 Property Documentation

9.230.3.1 action

```
Actions QEPvLoadSaveButton::action [read], [write]
```

If true, a dialog will be presented asking the user to confirm if the PV write actions should be carried out. Defaults to true.

9.230.3.2 configurationFile

```
QString QEPvLoadSaveButton::configurationFile [read], [write]
```

configurationFile

9.230.3.3 confirmAction

```
bool QEPvLoadSaveButton::confirmAction [read], [write]
```

If true, a dialog will be presented asking the user to confirm if the PV write actions should be carried out. Defaults to true.

9.230.3.4 confirmText

QString QEPvLoadSaveButton::confirmText [read], [write]

Text used to confirm action if confirmation dialog is presented

9.230.3.5 defaultStyle

QString QEPvLoadSaveButton::defaultStyle [read], [write]

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.230.3.6 defaultSubstitutions

QString QEPvLoadSaveButton::defaultSubstitutions [read], [write]

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.230.3.7 messageSourceId

unsigned int QEPvLoadSaveButton::messageSourceId [read], [write]

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.230.3.8 showProgressDialog

bool QEPvLoadSaveButton::showProgressDialog [read], [write]

If true, progress dialog will be shown while the data is being read or written

9.230.3.9 styleSheet

QString QEPvLoadSaveButton::styleSheet [read], [write]

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.230.3.10 userLevelEnabled

[QE::UserLevels](#) QEPvLoadSaveButton::userLevelEnabled [read], [write]

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through setUserLevel() Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.230.3.11 userLevelEngineerStyle

```
QString QEPvLoadSaveButton::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.230.3.12 userLevelScientistStyle

```
QString QEPvLoadSaveButton::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.230.3.13 userLevelUserStyle

```
QString QEPvLoadSaveButton::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.230.3.14 userLevelVisibility

```
QE::UserLevels QEPvLoadSaveButton::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.230.3.15 visible

```
bool QEPvLoadSaveButton::visible [read], [write]
```

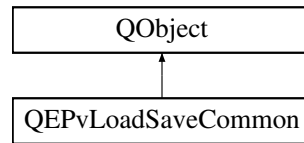
variableAsToolTip not applicale to V2 options. Class should ensure this option is initialised to false. allowDrop not applicale to V2 options. Class should ensure this option is initialised to false. Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSaveButton/QEPvLoadSaveButton.h
- qeframework/widgets/QEPvLoadSaveButton/QEPvLoadSaveButton.cpp

9.231 QEPvLoadSaveCommon Class Reference

Inheritance diagram for QEPvLoadSaveCommon:



Classes

- struct [StatusSummary](#)

Public Types

- enum **ActionKinds** {
NullAction = 0 , **Apply** , **Extract** , **ReadArchive** ,
Update }
- enum **ColumnKinds** {
NodeName = 0 , **LoadSave** , **Live** , **Delta** ,
NUMBER_OF_COLUMNS }
- typedef QMap< QString, double > **PvNameValueMaps**

Static Public Member Functions

- static void **clear** ([StatusSummary](#) &item)
- static [StatusSummary](#) **merge** (const [StatusSummary](#) &a, const [StatusSummary](#) &b)
- static PvNameValueMaps **merge** (const PvNameValueMaps &a, const PvNameValueMaps &b)

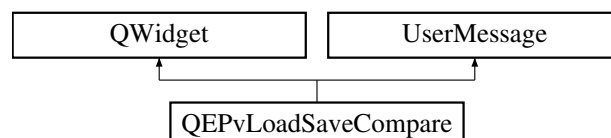
The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveCommon.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveCommon.cpp

9.232 QEPvLoadSaveCompare Class Reference

```
#include <QEPvLoadSaveCompare.h>
```

Inheritance diagram for QEPvLoadSaveCompare:



Public Member Functions

- **QEPvLoadSaveCompare** ([QEPvLoadSave](#) *owner, const int side, QWidget *parent=0)
- void **processModelData** ()

9.232.1 Detailed Description

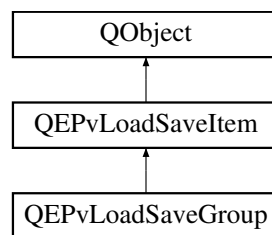
This widget displays graphically the difference between two PV data sets from the left/right hand side of a [QEPvLoadSave](#).

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveCompare.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveCompare.cpp

9.233 QEPvLoadSaveGroup Class Reference

Inheritance diagram for QEPvLoadSaveGroup:



Public Member Functions

- **QEPvLoadSaveGroup** (const QString &groupName, [QEPvLoadSaveItem](#) *parent=0)
- QVariant **getData** (int column) const
- bool **getIsGroup** () const
- [QEPvLoadSaveItem](#) * **clone** ([QEPvLoadSaveItem](#) *parent)
- void **actionConnect** (QObject *actionCompleteObject, const char *actionSetReadOutSlot, const char *actionCompleteSlot, const char *actionInCompleteSlot)
- void **extractPVDData** ()
- void **applyPVDData** ()
- void **readArchiveData** (const [QCaDateTime](#) &dateTime)
- void **abortAction** ()
- int **leafCount** () const
- void **sortChildItems** ()
- [QEPvLoadSaveCommon::StatusSummary](#) **getStatusSummary** () const
- [QEPvLoadSaveCommon::PvNameValueMaps](#) **getPvNameValueMap** () const

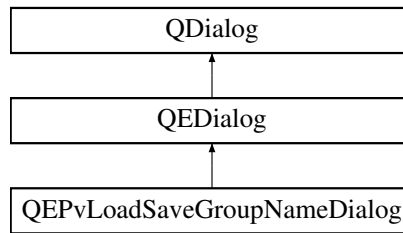
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveItem.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveItem.cpp

9.234 QEPvLoadSaveGroupNameDialog Class Reference

Inheritance diagram for QEPvLoadSaveGroupNameDialog:



Public Member Functions

- **QEPvLoadSaveGroupNameDialog** (QWidget *parent=0)
- void **setGroupName** (QString pvNameIn)
- QString **getGroupName** ()
- bool **isClear** ()

Additional Inherited Members

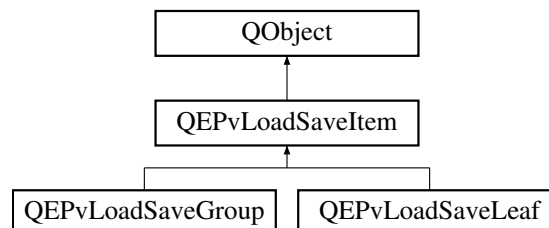
The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveGroupNameDialog.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveGroupNameDialog.cpp

9.235 QEPvLoadSaveItem Class Reference

```
#include <QEPvLoadSaveItem.h>
```

Inheritance diagram for QEPvLoadSaveItem:



Public Member Functions

- int **columnCount** () const
- [QEPvLoadSaveItem](#) * **getChild** (int position) const
- [QEPvLoadSaveItem](#) * **getParent** () const
- int **childCount** () const
- int **childPosition** () const
- virtual QVariant **getData** (int column) const
- bool **insertChild** (int position, [QEPvLoadSaveItem](#) *getChild)
- bool **removeChildren** (int position, int count)
- void **appendChild** ([QEPvLoadSaveItem](#) *getChild)
- [QEPvLoadSaveItem](#) * **getNamedChild** (const QString &searchName)
- virtual [QEPvLoadSaveItem](#) * **clone** ([QEPvLoadSaveItem](#) *parent)
- virtual void **actionConnect** (QObject *actionCompleteObject, const char *actionSetReadOutSlot, const char *actionCompleteSlot, const char *actionInCompleteSlot)
- QStringList **getNodePath** ()
- virtual void **setNodeName** (const QString &nodeName)
- QString **getNodeName** () const
- void **setNodeValue** (const QVariant &value)
- QVariant **getNodeValue** () const
- int **getElementCount** () const
- virtual bool **getIsPV** () const
- virtual bool **getIsGroup** () const
- virtual void **extractPVDData** ()
- virtual void **applyPVDData** ()
- virtual void **readArchiveData** (const [QCaDateTime](#) &dateTime)
- virtual void **abortAction** ()
- virtual int **leafCount** () const
- virtual [QEPvLoadSaveCommon::StatusSummary](#) **getStatusSummary** () const
- virtual [QEPvLoadSaveCommon::PvNameValueMaps](#) **getPvNameValueMap** () const

Static Public Member Functions

- static void **initReadArchiveData** ()

Protected Member Functions

- **QEPvLoadSaveItem** (const QString &nodeName, const QVariant &value, [QEPvLoadSaveItem](#) *parent=0)

Protected Attributes

- QList< [QEPvLoadSaveItem](#) * > **childItems**
- [QEPvLoadSaveItem](#) * **parentItem**
- QVariant **value**
- QVariant **liveValue**
- QString **nodeName**

Static Protected Attributes

- static int **readArchiveCount** = 0

9.235.1 Detailed Description

This class is based on the TreeItem example specified in: <http://qt-project.org/doc/qt-4.8/itemviews-editabletreemodel.html>

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A major difference is that it is derived from QObject because each leaf item is associated with a PV and needs slots to receive value data. A consequence of this is that this class must be exposed in a header file even though it is essentially a QEPVLoadSave private class. Also there is no itemData variant array - these values calculated as an when needed. Lastly some function name changes such as parent => getParent (as parent already inherited from QObject) and some changes just to follow my preferred style.

Note: although QObjects provide a parent/child linking, this class manages it's own parent child relationships as per the TreeItem example.

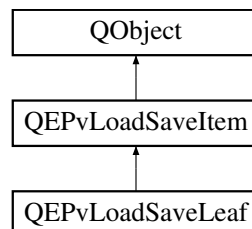
QEPVLoadSaveItem are created in one of two flavours: a/ node - used for groups b/ leaf - used for PVs. both of which inherited from the QEPVLoadSaveItem base class

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPVLoadSave/QEPVLoadSaveItem.h
- qeframework/widgets/QEPVLoadSave/QEPVLoadSaveItem.cpp

9.236 QEPVLoadSaveLeaf Class Reference

Inheritance diagram for QEPVLoadSaveLeaf:



Signals

- void **setReadOut** (const QString &text)
- void **reportActionComplete** (const QEPVLoadSaveItem *item, const QEPVLoadSaveCommon::ActionKinds action, const bool actionSuccessful)
- void **reportActionInComplete** (const QEPVLoadSaveItem *item, const QEPVLoadSaveCommon::ActionKinds action)

Public Member Functions

- **QEPvLoadSaveLeaf** (const QString &setPointPvName, const QString &readBackPvName, const QString &archiverPvName, const QVariant &value, [QEPvLoadSaveItem](#) *parent=0)
- void **setPvNames** (const QString &setPointPvName, const QString &readBackPvName, const QString &archiverPvName)
- QString **copyVariables** () const
- void **setSetPointPvName** (const QString &pvName)
- QString **getSetPointPvName** () const
- void **setReadBackPvName** (const QString &pvName)
- QString **getReadBackPvName** () const
- void **setArchiverPvName** (const QString &pvName)
- QString **getArchiverPvName** () const
- QVariant **getData** (int column) const
- void **setNodeName** (const QString &nodeName)
- bool **getIsPV** () const
- [QEPvLoadSaveItem](#) * **clone** ([QEPvLoadSaveItem](#) *parent)
- QEPvLoadSaveCommon::PvNameValueMaps **getPvNameValueMap** () const
- void **actionConnect** (QObject *actionCompleteObject, const char *actionSetReadOutSlot, const char *actionCompleteSlot, const char *actionInCompleteSlot)
- void **extractPVDData** ()
- void **applyPVDData** ()
- void **readArchiveData** (const [QCaDateTime](#) &dateTime)
- void **abortAction** ()
- int **leafCount** () const
- [QEPvLoadSaveCommon::StatusSummary](#) **getStatusSummary** () const

Additional Inherited Members

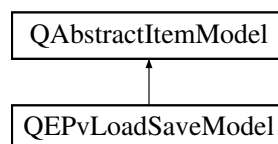
The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveItem.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveItem.cpp

9.237 QEPvLoadSaveModel Class Reference

```
#include <QEPvLoadSaveModel.h>
```

Inheritance diagram for QEPvLoadSaveModel:



Signals

- void **reportActionComplete** (const [QEPvLoadSaveItem](#) *item, QEPvLoadSaveCommon::ActionKinds, bool)
- void **reportActionInComplete** (const [QEPvLoadSaveItem](#) *item, QEPvLoadSaveCommon::ActionKinds action)

Public Member Functions

- **QEPvLoadSaveModel** (QTreeView *treeView, [QEPvLoadSave](#) *parent=NULL)
- QVariant **data** (const QModelIndex &index, int role) const
- QVariant **headerData** (int section, Qt::Orientation orientation, int role=Qt::DisplayRole) const
- QModelIndex **index** (int row, int column, const QModelIndex &parent=QModelIndex()) const
- QModelIndex **parent** (const QModelIndex &child) const
- int **rowCount** (const QModelIndex &parent=QModelIndex()) const
- int **columnCount** (const QModelIndex &parent=QModelIndex()) const
- Qt::ItemFlags **flags** (const QModelIndex &index) const
- bool **setData** (const QModelIndex &index, const QVariant &value, int role=Qt::EditRole)
- bool **setHeaderData** (int section, Qt::Orientation orientation, const QVariant &value, int role=Qt::EditRole)
- bool **insertRows** (int position, int rows, const QModelIndex &parent=QModelIndex())
- bool **removeRows** (int position, int rows, const QModelIndex &parent=QModelIndex())
- void **setHeading** (const QString &heading)
- QString **getHeading** () const
- void **setupModelData** ([QEPvLoadSaveItem](#) *rootItem, const QString &heading)
- void **modelUpdated** ()
- void **itemUpdated** (const [QEPvLoadSaveItem](#) *item, const QEPvLoadSaveCommon::ColumnKinds kind, const bool updateParent)
- bool **addItemToModel** ([QEPvLoadSaveItem](#) *item, [QEPvLoadSaveItem](#) *parentItem)
- bool **removeItemFromModel** ([QEPvLoadSaveItem](#) *item)
- bool **mergeItemInToModel** ([QEPvLoadSaveItem](#) *item)
- void **extractPVDData** ()
- void **applyPVDData** ()
- void **readArchiveData** (const [QCaDateTime](#) &dateTime)
- void **abortAction** ()
- void **sort** ([QEPvLoadSaveGroup](#) *group)
- int **leafCount** () const
- QEPvLoadSaveCommon::PvNameValueMaps **getPvNameValueMap** () const
- [QEPvLoadSaveItem](#) * **getRootItem** ()
- [QEPvLoadSaveItem](#) * **getSelectedItem** ()
- QModelIndex **getRootIndex** ()
- [QEPvLoadSaveItem](#) * **indexToItem** (const QModelIndex &index) const

Protected Member Functions

- bool **eventFilter** (QObject *obj, QEvent *event)

9.237.1 Detailed Description

This class is based on the TreeModel example specified in: <http://qt-project.org/doc/qt-4.8/itemviews-editabletreemodel.html>

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Note on naming: the example's root item that provide header info is referred to the core item. It only ever has one child which is the visible (on the QTreeView) which is the user root item, named "ROOT".

Note: we only re-size the number of rows (children). The number of columns is fixed.

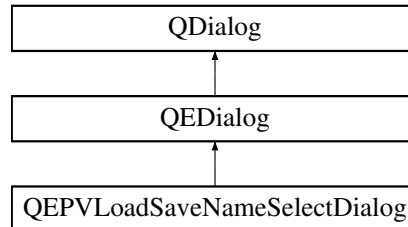
The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveModel.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveModel.cpp

9.238 QEPVLoadSaveNameSelectDialog Class Reference

```
#include <QEPVLoadSaveNameSelectDialog.h>
```

Inheritance diagram for QEPVLoadSaveNameSelectDialog:



Public Member Functions

- **QEPVLoadSaveNameSelectDialog** (QWidget *parent=0)
- void **setPvNames** (const QString &setPointPvName, const QString &readBackPvName, const QString &archiverPvName)
- void **getPvNames** (QString &setPointPvName, QString &readBackPvName, QString &archiverPvName)

Protected Member Functions

- void **closeEvent** (QCloseEvent *e)

Additional Inherited Members

9.238.1 Detailed Description

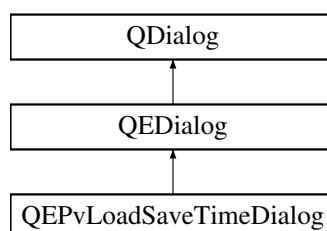
Manager class for the QEPVLoadSaveNameSelectDialog.ui compiled form. This is a copy of [QEPVNameSelectDialog](#) that allows three PV names to be specified, i.e.: a read/get PV name; a write/put PV name; and an archive/fetch PV name.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPVLoadSaveNameSelectDialog.h
- qeframework/widgets/QEPvLoadSave/QEPVLoadSaveNameSelectDialog.cpp

9.239 QEPvLoadSaveTimeDialog Class Reference

Inheritance diagram for QEPvLoadSaveTimeDialog:



Public Member Functions

- **QEPvLoadSaveTimeDialog** (QWidget *parent=0)
- void **setMaximumDateTime** (const QDateTime &datetime)
- void **setDateTime** (const QDateTime &datetime)
- QDateTime **getDateTime** ()

Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveTimeDialog.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveTimeDialog.cpp

9.240 QEPvLoadSaveUtilities Class Reference

Static Public Member Functions

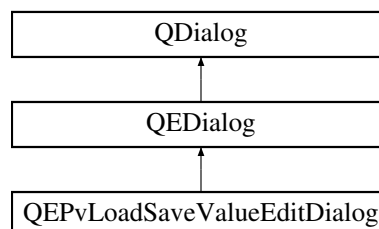
- static [QEPvLoadSaveItem](#) * **readTree** (const QString &filename, const QString ¯oString, QString &errorMessage)
- static bool **writeTree** (const QString &filename, const [QEPvLoadSaveItem](#) *root, QString &errorMessage)
- static QString **mergePvNames** (const QString &setPoint, const QString &readBack, const QString &archiver)
- static bool **splitPvNames** (const QString &mergedName, QString &setPoint, QString &readBack, QString &archiver)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveUtilities.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveUtilities.cpp

9.241 QEPvLoadSaveValueEditDialog Class Reference

Inheritance diagram for QEPvLoadSaveValueEditDialog:



Public Member Functions

- **QEPvLoadSaveValueEditDialog** (QWidget *parent=0)
- void **setPvName** (const QString &pvName)
- void **setValue** (const QVariant &valueList)
- QVariant **getValue** () const

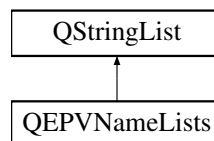
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveValueEditDialog.h
- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveValueEditDialog.cpp

9.242 QEPVNameLists Class Reference

Inheritance diagram for QEPVNameLists:



Public Member Functions

- void **prependOrMoveToFirst** (const QString &item)
- void **saveConfiguration** ([PMElement](#) &parentElement)
- void **restoreConfiguration** ([PMElement](#) &parentElement)

Static Public Member Functions

- static void **constructor** ()

The documentation for this class was generated from the following file:

- qeframework/widgets/QEStripChart/QEStripChart.cpp

9.243 QEPvNameProperties Struct Reference

Public Attributes

- QString **pvName**
- QString **substitutions**
- unsigned int **index**

The documentation for this struct was generated from the following file:

- qeframework/data/QCaVariableNamePropertyManager.h

9.244 QEPvNameSearch Class Reference

```
#include <QEPvNameSearch.h>
```

Public Member Functions

- **QEPvNameSearch** (const [QEPvNameSearch](#) &other)
- **QEPvNameSearch** (const QStringList &pvNameList)
- void **clear** ()
- int **count** () const
- void **setPvNameList** (const QStringList &pvNameList)
- void **addPvNameList** (const QStringList &pvNameList)
- QStringList **getAllPvNames** () const
- QStringList **getMatchingPvNames** (const QRegularExpression &re, const bool exactMatch) const
- QStringList **getMatchingPvNames** (const QString &str, const Qt::CaseSensitivity cs) const

9.244.1 Detailed Description

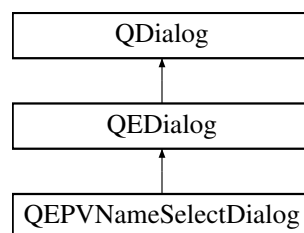
Provides a basic name search capability. Extracted from archiver manager in order to provide a more flexibility, i.e. using sets of PV names which can be sourced from anywhere, not just the archiver.

The documentation for this class was generated from the following files:

- qeframework/common/QEPvNameSearch.h
- qeframework/common/QEPvNameSearch.cpp

9.245 QEPVNameSelectDialog Class Reference

Inheritance diagram for QEPVNameSelectDialog:



Public Member Functions

- **QEPVNameSelectDialog** (QWidget *parent=0)
- void **setPvName** (QString pvNameIn)
- QString **getPvName** ()

Static Public Member Functions

- static void **setPvNameList** (const QStringList &pvNameList)
- static QStringList **getPvNameList** ()

Protected Member Functions

- void **closeEvent** (QCloseEvent *e)

Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/QEPVNameSelectDialog.h
- qeframework/common/QEPVNameSelectDialog.cpp

9.246 QEPvNameUri Class Reference

```
#include <QEPvNameUri.h>
```

Public Types

- enum **Protocol** { **undefined** = 0 , **ca** , **pva** , **NUMBER_OF_PROTOCOLS** }

Public Member Functions

- **QEPvNameUri** (const QString &pvName, const Protocol protocol=ca)
- **QEPvNameUri** (const QString &uri, const bool strict)
- QString **encodeUri** () const
- bool **decodeUri** (const QString &uri, const bool strict)
- void **setProtocol** (const Protocol protocol)
- Protocol **getProtocol** () const
- void **setPvName** (const QString &pvName)
- QString **getPvName** () const

Static Public Member Functions

- static QString **protocollImage** (const Protocol protocol)

9.246.1 Detailed Description

This class create a PV name uniform resource identifier. This includes the protocol, i.e. Channel Access or PV access. The format is, for example:

```
"ca://SR11BCM01:CURRENT_MONITOR" "pva://SR11BCM01:CURRENT_MONITOR"
```

where "ca://" specifies the Channel Access protocol and "pva://" specifies the PV Access protocol and "SR11BCM01:CURRENT_MONITOR" is the PV name.

The default provider, when not specified as indicated above, may be specified by the environment variable QE_DEFAULT_PROVIDER, and can be defined as either "CA" or "PVA" - case insensitive. If the environment variable is not defined or is ill-defined the default default provider is Channel Access.

The documentation for this class was generated from the following files:

- qeframework/protocol/QEPvNameUri.h
- qeframework/protocol/QEPvNameUri.cpp

9.247 QEPvNameUriList Class Reference

Public Member Functions

- void **clear** ()
- void **append** (const [QEPvNameUri](#) &other)
- int **count** () const
- [QEPvNameUri](#) **value** (const int j) const

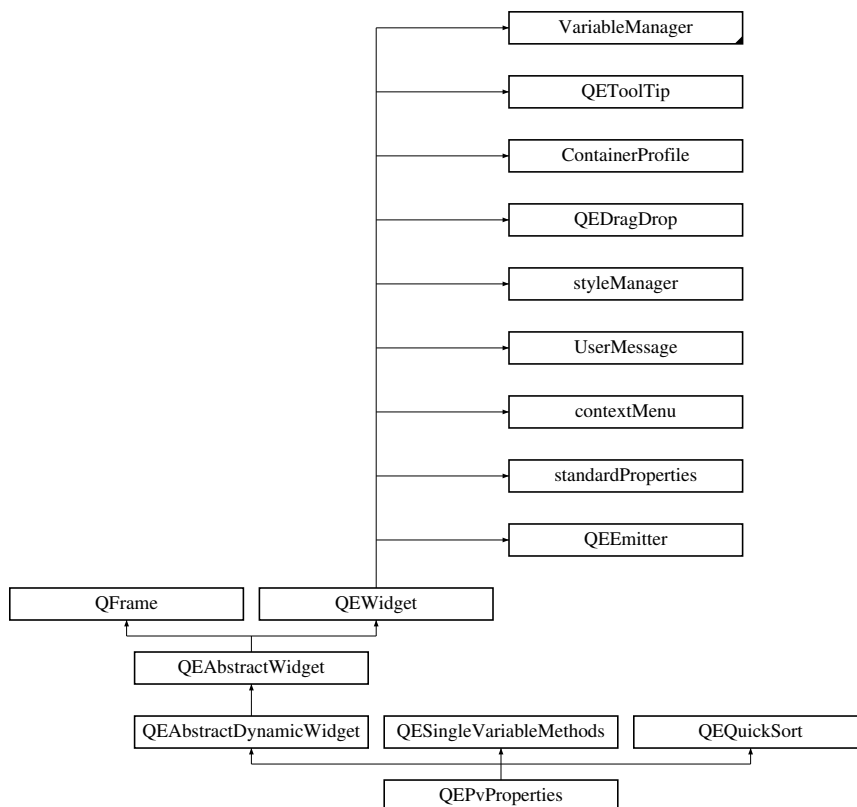
The documentation for this class was generated from the following files:

- qeframework/protocol/QEPvNameUri.h
- qeframework/protocol/QEPvNameUri.cpp

9.248 QEPvProperties Class Reference

```
#include <QEPvProperties.h>
```

Inheritance diagram for QEPvProperties:



Public Types

- enum **OwnContextMenuOptions** {
PVPROP_NONE = QEAbstractDynamicWidget::ADWCM_SUB_CLASS_WIDGETS_START_HERE ,
PVPROP_SORT_FIELD_NAMES , **PVPROP_RESET_FIELD_NAMES** , **PVPROP_PROCESS_RECORD** ,
PVPROP_SUB_CLASS_WIDGETS_START_HERE }

Signals

- void **setCurrentBoxIndex** (int index)

Public Member Functions

- **QEPvProperties** (QWidget *parent=0)
- **QEPvProperties** (const QString &variableName, QWidget *parent=0)
- QSize **sizeHint** () const
- int **addPvName** (const QString &pvName)
- void **addPvNameList** (const QStringList &pvNameList)
- void **clearAllPvNames** ()

Protected Member Functions

- void **resizeEvent** (QResizeEvent *event)
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)
- bool **itemLessThan** (const int a, const int b, QObject *context=NULL) const
- void **swapItems** (const int a, const int b, QObject *context=NULL)
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- **QEChannel** * **createQcalItem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **mousePressEvent** (QMouseEvent *event)
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **saveConfiguration** ([PersistenceManager](#) *pm)
- void **restoreConfiguration** ([PersistenceManager](#) *pm, [restorePhases](#) restorePhase)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **scaleBy** (const int m, const int d)
- void **enableEditPvChanged** ()

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)

Additional Inherited Members

9.248.1 Detailed Description

The [QEPvProperties](#) class allows user to view all the displayable fields of the associated IOC record.

9.248.2 Member Function Documentation

9.248.2.1 `restoreConfiguration()`

```
void QEPvProperties::restoreConfiguration (
    PersistenceManager * ,
    restorePhases ) [protected], [virtual]
```

Service a request to restore the [QE](#) widget's configuration. A [QE](#) widget recover any configuration details from the [PersistenceManager](#). For example, a [QEStripChart](#) may restore the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method. This is called twice with an incrementing `restorePhase`. Most widgets will miss the first call as they don't exist yet (they are created as part of the first phase).

Reimplemented from [QEWidget](#).

9.248.2.2 `saveConfiguration()`

```
void QEPvProperties::saveConfiguration (
    PersistenceManager * ) [protected], [virtual]
```

Service a request to save the [QE](#) widget's current configuration. A widget may save any configuration details through the [PersistenceManager](#). For example, a [QEStripChart](#) may save the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method.

Reimplemented from [QEWidget](#).

9.248.2.3 `scaleBy()`

```
void QEPvProperties::scaleBy (
    const int ,
    const int ) [protected], [virtual]
```

Any [QEWidget](#) that requires additional scaling, i.e. above and beyond the standard scaling applied to size, minimum size, maximum size and font size, may override this function in order to perform any bespoke scaling need by the widget (for example see [QEShape](#)). The scaling is defined using a rational number specified by two integers (m, d). The first (m) parameter is the multiplier and the second (d) parameter is the divisor. For example, if m = 4 and d = 5, then an 80% scaling should be applied. And if m = 5 and d = 4, and a 125% scaling is required.

Reimplemented from [QEWidget](#).

9.248.3 Property Documentation

9.248.3.1 arrayIndex

```
int QEPvProperties::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.248.3.2 elementsRequired

```
int QEPvProperties::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.248.3.3 variable

```
QString QEPvProperties::variable [read], [write]
```

EPICS variable name (CA PV)

9.248.3.4 variableSubstitutions

```
QString QEPvProperties::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

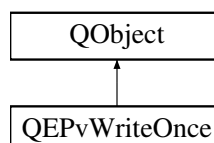
The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvProperties/QEPvProperties.h
- qeframework/widgets/QEPvProperties/QEPvProperties.cpp

9.249 QEPvWriteOnce Class Reference

```
#include <QEPvWriteOnce.h>
```

Inheritance diagram for QEPvWriteOnce:



Public Member Functions

- **QEPvWriteOnce** (const QString &pvName, const QString &value, const [QE::Formats](#) format=[QE::Default](#), QObject *parent=0)
- bool **writeNow** (const QString ¯oSubstitutions)

Public Attributes

- QString **pvName**
- QString **value**
- [QE::Formats](#) **format**

9.249.1 Detailed Description

This class provides a basic write and forget class. It has been refactored out of [QEMenuButton](#) so that it can be used by [windowCustomisation](#). NOTE: The object deletes itself once the write is complete or times out.

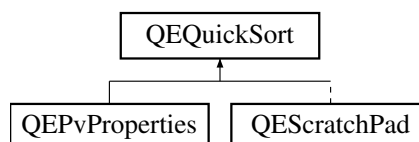
The documentation for this class was generated from the following files:

- qeframework/common/QEPvWriteOnce.h
- qeframework/common/QEPvWriteOnce.cpp

9.250 QEQuickSort Class Reference

```
#include <QEQuickSort.h>
```

Inheritance diagram for QEQuickSort:



Public Member Functions

- void **sort** (const int first, const int last, QObject *context=NULL)

Protected Member Functions

- virtual bool **itemLessThan** (const int a, const int b, QObject *context) const =0
- virtual void **swapItems** (const int a, const int b, QObject *context)=0

9.250.1 Detailed Description

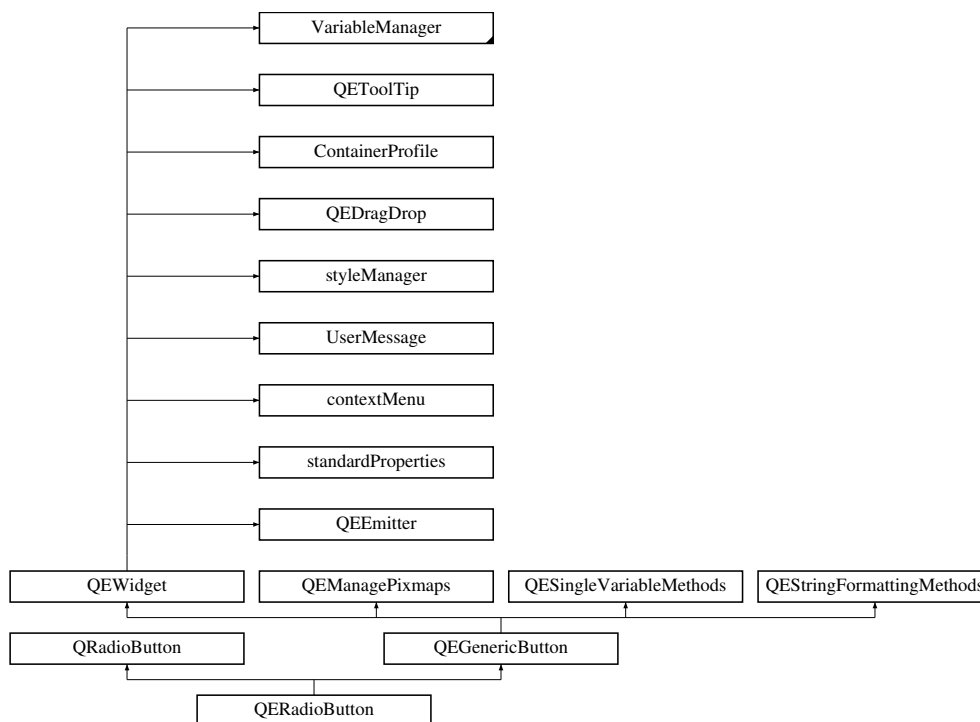
This base class provides the means to add quick sort functionality to a class. Alas the algorithms qSort API does not accept a class instance LessThan function, so we roll out own. The items to be sorted must be indexable via consecutive integers.

The documentation for this class was generated from the following files:

- qeframework/common/QEQuickSort.h
- qeframework/common/QEQuickSort.cpp

9.251 QERadioButton Class Reference

Inheritance diagram for QERadioButton:



Public Slots

- void **writeNow** ()
- void **requestAction** (const [QEActionRequests](#) &request)
- void **setDefaultStyle** (const QString &style)
- void **setManagedVisible** (bool v)

Signals

- void [dbValueChanged](#) ()
 - void **dbValueChanged** (const QString &out)
 - void **dbValueChanged** (const int &out)
 - void **dbValueChanged** (const long &out)
 - void **dbValueChanged** (const qlonglong &out)
 - void **dbValueChanged** (const double &out)
 - void **dbValueChanged** (const bool &out)
 - void [dbConnectionChanged](#) (const bool &isConnected)
- Sent when the widget state updated following a channel connection change.*
- void [requestResend](#) ()
 - void [newGui](#) (const [QEActionRequests](#) &request)
 - void [pressed](#) (int value)
 - void [released](#) (int value)
 - void [clicked](#) (int value)
 - void [programCompleted](#) ()
- Program started by button has compelled.*

Public Member Functions

- void [setFormatProperty](#) ([QE::Formats](#) format)
- Access function for [format](#) property - refer to [format](#) property for details.*
- [QE::Formats](#) [getFormatProperty](#) () const
- Access function for [format](#) property - refer to [format](#) property for details.*
- [QERadioButton](#) (QWidget *parent=0)
 - [QERadioButton](#) (const QString &variableName, QWidget *parent=0)
 - [~QERadioButton](#) ()
- Destructor.*

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [arrayIndex](#)
- QString [labelText](#)
- QString [clickText](#)
- QString [clickCheckedText](#)
- bool [writeOnClick](#)
- QString [guiFile](#)
- QString [prioritySubstitutions](#)
- [QE::CreationOptions](#) [creationOption](#)
- QString [customisationName](#)
- QString [program](#)
- QStringList [arguments](#)
- [QE::ProgramStartupOptions](#) [programStartupOption](#)
- [QE::DisabledRecordPolicy](#) [disabledRecordPolicy](#)
- Qt::Alignment [alignment](#)
- [QE::UpdateOptions](#) [updateOption](#)
- QPixmap [pixmap0](#)
- QPixmap [pixmap1](#)
- QPixmap [pixmap2](#)

- QPixmap [pixmap3](#)
- QPixmap [pixmap4](#)
- QPixmap [pixmap5](#)
- QPixmap [pixmap6](#)
- QPixmap [pixmap7](#)
- QString [password](#)
- bool [confirmAction](#)
- QString [confirmText](#)
- bool [writeOnPress](#)
- bool [writeOnRelease](#)
- QString [pressText](#)
- QString [releaseText](#)
- bool [subscribe](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- QE::UserLevels [userLevelVisibility](#)
- QE::UserLevels [userLevelEnabled](#)
- QE::DisplayAlarmStateOptions [displayAlarmStateOption](#)
- bool [oosAware](#)
- int [precision](#)
- bool [useDbPrecision](#)
- int [leadingZeros](#)
- bool [addUnits](#)
- bool [forceSign](#)
- QString [localEnumeration](#)
- QE::Formats [format](#)
- int [radix](#)
- QE::Separators [separator](#)
- QE::Notations [notation](#)
- QE::ArrayActions [arrayAction](#)

Additional Inherited Members

9.251.1 Constructor & Destructor Documentation

9.251.1.1 QERadioButton() [1/2]

```
QERadioButton::QERadioButton (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.251.1.2 QERadioButton() [2/2]

```
QERadioButton::QERadioButton (
    const QString & variableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.251.2 Member Function Documentation

9.251.2.1 clicked

```
void QERadioButton::clicked (
    int value ) [signal]
```

Button has been Clicked. The value emitted is the integer interpretation of the clickText property (or the click↔CheckedText property if the button was checked)

9.251.2.2 dbValueChanged

```
void QERadioButton::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.251.2.3 newGui

```
void QERadioButton::newGui (
    const QEActionRequests & request ) [signal]
```

Internal use only. Request a new GUI is created. Typically, this is caught by the QEGui application.

9.251.2.4 pressed

```
void QERadioButton::pressed (
    int value ) [signal]
```

Button has been Pressed. The value emitted is the integer interpretation of the pressText property

9.251.2.5 released

```
void QERadioButton::released (
    int value ) [signal]
```

Button has been Released The value emitted is the integer interpretation of the releaseText property

9.251.2.6 requestAction

```
void QERadioButton::requestAction (
    const QEActionRequests & request ) [inline], [slot]
```

Default slot used to create a new GUI if there is no slot indicated in the [ContainerProfile](#) class. This slot is typically used when the button is pressed within the Designer preview window to allow the operation of the button to be tested. If an application does not specify a slot to use for creating new windows (through the [ContainerProfile](#) class) a window will still be created through this slot, but it will not respect the window creation options or any other window related application constraints. For example, the QEGui application does provide a slot for creating new GUIs in the [ContainerProfile](#) class which respects the creation options, knows how to add tabs in the application, and extend the application's window menu in the menu bar.

9.251.2.7 requestResend

```
void QERadioButton::requestResend ( ) [signal]
```

Internal use only. Used when changing a property value to force a re-display to reflect the new property value.

9.251.2.8 setDefaultStyle

```
void QERadioButton::setDefaultStyle (
    const QString & style ) [inline], [slot]
```

Update the default style applied to this widget.

9.251.2.9 setManagedVisible

```
void QERadioButton::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.251.3 Property Documentation

9.251.3.1 addUnits

```
bool QERadioButton::addUnits [read], [write]
```

If true (default), add engineering units supplied with the data.

9.251.3.2 alignment

```
Qt::Alignment QERadioButton::alignment [read], [write]
```

Set the buttons text alignment. Left justification is particularly useful when displaying quickly changing numeric data updates.

9.251.3.3 allowDrop

```
bool QERadioButton::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.251.3.4 arguments

```
QStringList QERadioButton::arguments [read], [write]
```

Arguments for program specified in the 'program' property.

9.251.3.5 arrayAction

```
QE::ArrayActions QERadioButton::arrayAction [read], [write]
```

Text formatting option for array data. Default is ASCII. Options are:

- ASCII - treat array as a single text string. For example an array of three characters 'a' 'b' 'c' will be formatted as 'abc'.
- APPEND - treat array as an array of numbers and format a string containing them all with a space between each. For example, an array of three numbers 10, 11 and 12 will be formatted as '10 11 12'.
- INDEX - Extract a single item from the array. The item is then formatted as any other non array data would be. The item selected is determined by the arrayIndex property. For example, if arrayIndex property is 1, an array of three numbers 10, 11 and 12 will be formatted as '11'.

9.251.3.6 arrayIndex

```
int QERadioButton::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.251.3.7 clickCheckedText

```
QString QERadioButton::clickCheckedText [read], [write]
```

Text used to compare with text written or read to determine if push button should be marked as checked. Note, must be an exact match following formatting of data updates. When writing values, the 'pressText', 'ReleaseText', or 'clickedtext' must match this property to cause the button to be checked when the write occurs.

Good example: formatting set to display a data value of '1' as 'On', clickCheckedText is 'On', clickText is 'On'. In this example, the push button will be checked when a data update occurs with a value of 1 or when the button is clicked.

Bad example: formatting set to display a data value of '1' as 'On', clickCheckedText is 'On', clickText is '1'. In this example, the push button will be checked when a data update occurs with a value of 1 but, although a valid value will be written when clicked, the button will not be checked when clicked as '1' is not the same as 'On'.

9.251.3.8 clickText

```
QString QERadioButton::clickText [read], [write]
```

Value written when user clicks button if 'writeOnClick' property is true. Note: Set format to Integer when writing numeric values to enum PVs like the bo and mbbo records.

9.251.3.9 confirmAction

```
bool QERadioButton::confirmAction [read], [write]
```

If true, a dialog will be presented asking the user to confirm if the button action should be carried out

9.251.3.10 confirmText

```
QString QERadioButton::confirmText [read], [write]
```

Text used to confirm action if confirmation dialog is presented

9.251.3.11 creationOption

```
QE::CreationOptions QERadioButton::creationOption [read], [write]
```

Creation options when opening a new GUI. Open a new window, open a new tab, or replace the current window. the creation option is supplied when the button generates a newGui signal. Application code connected to this signal should honour this request if possible. When used within the QEGui application, the QEGui application creates a new window, new tab, or replaces the current window as appropriate.

9.251.3.12 customisationName

```
QString QERadioButton::customisationName [read], [write]
```

Window customisation name. This name will be used to select a set of window customisations including menu items and tool bar buttons. Applications such as QEGui can load .xml files containing named sets of window customisations. This property is used to select a set loaded from these files. The selected set of customisations will be applied to the main window containing the new GUI. Customisations are not applied if the GUI is opened as a dock.

9.251.3.13 defaultStyle

```
QString QERadioButton::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.251.3.14 disabledRecordPolicy

`QE::DisabledRecordPolicy QERadioButton::disabledRecordPolicy [read], [write]`

Set the widget's disabled record policy, i.e. the action to be taken when the under lying record is disabled, i.e. when the associated record's STAT field value is Disabled. When the policy is ignore, then no special action is taken. This is the default policy. When the policy is grayout, the widget is style is set as if disconnected when the record is disabled. When the policy is disable, the widget is disabled when the record is disabled.

9.251.3.15 displayAlarmStateOption

`QE::DisplayAlarmStateOptions QERadioButton::displayAlarmStateOption [read], [write]`

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.251.3.16 forceSign

`bool QERadioButton::forceSign [read], [write]`

If false (default), no "+" sign, when true always add a sign.

9.251.3.17 format

`QE::Formats QERadioButton::format [read], [write]`

Format to apply to data. Default is 'Default' in which case the data type supplied with the data determines how the data is formatted. For all other options, an attempt is made to format the data as requested (whatever its native form).

9.251.3.18 guiFile

`QString QERadioButton::guiFile [read], [write]`

File name of GUI to be presented on button click. File name can be absolute, relative to the path of the [QEForm](#) in which the button widget is located, relative to the any path in the path list published in the [ContainerProfile](#) class, or relative to the current path. See [QEWidget::openQEFile\(\)](#) in [QEWidget.cpp](#) for details.

9.251.3.19 labelText

`QString QERadioButton::labelText [read], [write]`

Button label text (prior to substitution). Macro substitutions will be applied to this text and the result will be set as the button text. Used when data updates are not being represented in the button text. IF NOT LEFT EMPTY, THIS TEXT WILL TAKE PRIORITY OVER THE PUSH BUTTON 'text' PROPERTY! For example, a button in a sub form may have a 'labelText' property of 'Turn Pump On'. When the sub form is used twice in a main form with substitutions PUMPNUM=1 and PUMPNUM=2 respectively, the two identical buttons in the sub forms will have the labels 'Turn Pump 1 On' and 'Turn Pump 2 On' respectively.

9.251.3.20 leadingZeros

```
int QERadioButton::leadingZeros [read], [write]
```

The minimum number of leading digits before any decimal/radix point. The default is 1.

9.251.3.21 localEnumeration

```
QString QERadioButton::localEnumeration [read], [write]
```

An enumeration list used to data values. Used only when the formatting option is 'local enumeration'. Value is converted to an integer and used to select a string from this list.

Format is:

```
[[<|<=|=|!=|>|=|>]value1|*]: string1 , [[<|<=|=|!=|>|=|>]value2|*]: string2 , [[<|<=|=|!=|>|=|>]value3|*]: string3 , ...
```

Where: < Less than <= Less than or equal = Equal (default if no operator specified) >= Greather than or equal

Greater than

* Always match (used to specify default text)

Values may be numeric or textual Values do not have to be in any order, but first match wins Values may be quoted Strings may be quoted Consecutive values do not have to be present. Operator is assumed to be equality if not present. White space is ignored except within quoted strings. may be included in a string to indicate a line break

Examples are:

```
0:Off,1:On 0 : "Pump Running", 1 : "Pump not running" 0:"", 1:"Warning!\nAlarm" <2:"Value is less than two", =2↵
:"Value is equal to two", >2:"Value is grater than 2" 3:"Beamline Available", *:"" "Pump Off": "OH NO!, the pump is OFF!", "Pump On": "It's OK, the pump is on"
```

The data value is converted to a string if no enumeration for that value is available. For example, if the local enumeration is '0:off,1:on', and a value of 10 is processed, the text generated is '10'. If a blank string is required, this should be explicit. for example, '0:off,1:on,10: ""'

A range of numbers can be covered by a pair of values as in the following example: >=4:"Between 4 and 8", <=8↵
:"Between 4 and 8"

9.251.3.22 messageSourceId

```
unsigned int QERadioButton::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.251.3.23 notation

```
QE::Notations QERadioButton::notation [read], [write]
```

Notation used for numerical formatting. Default is fixed.

9.251.3.24 oosAware

```
bool QERadioButton::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.251.3.25 password

```
QString QERadioButton::password [read], [write]
```

Password user will need to enter before any action is taken

9.251.3.26 pixmap0

```
QPixmap QERadioButton::pixmap0 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 0

9.251.3.27 pixmap1

```
QPixmap QERadioButton::pixmap1 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 1

9.251.3.28 pixmap2

```
QPixmap QERadioButton::pixmap2 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 2

9.251.3.29 pixmap3

```
QPixmap QERadioButton::pixmap3 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 3

9.251.3.30 pixmap4

```
QPixmap QERadioButton::pixmap4 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 4

9.251.3.31 pixmap5

```
QPixmap QERadioButton::pixmap5 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 5

9.251.3.32 pixmap6

```
QPixmap QERadioButton::pixmap6 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 6

9.251.3.33 pixmap7

```
QPixmap QERadioButton::pixmap7 [read], [write]
```

Pixmap to display if updateOption is Icon or TextAndIcon and data value translates to an index of 7

9.251.3.34 precision

```
int QERadioButton::precision [read], [write]
```

Precision used when formatting floating point numbers. The default is 4. This is only used if useDbPrecision is false.

9.251.3.35 pressText

```
QString QERadioButton::pressText [read], [write]
```

Value written when user presses button if 'writeOnPress' property is true

9.251.3.36 prioritySubstitutions

```
QString QERadioButton::prioritySubstitutions [read], [write]
```

Overriding macro substitutions. These macro substitutions take precedence over any existing macro substitutions defined by the variableSubstitutions property, any parent forms, or the application containing the button. These macro substitutions are particularly usefull when the button's function is to reload the same form but with different macro substitutions. The variableSubstitutions property cannot be used for this since, although they are added to the list of macro substitutions applied to the new form, they are appended to the list and the existing macro substitutions take precedence.

9.251.3.37 program

```
QString QERadioButton::program [read], [write]
```

Program to run when the button is clicked. No attempt to run a program is made if this property is empty. Example: firefox

9.251.3.38 programStartupOption

```
QE::ProgramStartupOptions QERadioButton::programStartupOption [read], [write]
```

Startup options. Just run the command, run the command within a terminal, or display the output in [QE](#) message system.

9.251.3.39 radix

```
int QERadioButton::radix [read], [write]
```

Base used for when formatting integers. Default is 10 (duh!)

9.251.3.40 releaseText

```
QString QERadioButton::releaseText [read], [write]
```

Value written when user releases button if 'writeOnRelease' property is true

9.251.3.41 separator

```
QE::Separators QERadioButton::separator [read], [write]
```

Seperators used for interger and fixed point formatting. Default is None.

9.251.3.42 styleSheet

```
QString QERadioButton::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly setable per se. This also stops transient styles being saved to the ui file.

9.251.3.43 subscribe

```
bool QERadioButton::subscribe [read], [write]
```

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.251.3.44 updateOption

[QE::UpdateOptions](#) QERadioButton::updateOption [read], [write]

Update options (text, pixmap, both, or state (checked or unchecked))

9.251.3.45 useDbPrecision

bool QERadioButton::useDbPrecision [read], [write]

If true (default), format floating point numbers using the precision supplied with the data. If false, the precision property is used.

9.251.3.46 userLevelEnabled

[QE::UserLevels](#) QERadioButton::userLevelEnabled [read], [write]

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through setUserLevel() Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.251.3.47 userLevelEngineerStyle

QString QERadioButton::userLevelEngineerStyle [read], [write]

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.251.3.48 userLevelScientistStyle

QString QERadioButton::userLevelScientistStyle [read], [write]

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.251.3.49 userLevelUserStyle

QString QERadioButton::userLevelUserStyle [read], [write]

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.251.3.50 userLevelVisibility

```
QE::UserLevels QERadioButton::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.251.3.51 variable

```
QString QERadioButton::variable [read], [write]
```

EPICS variable name (CA PV)

9.251.3.52 variableAsToolTip

```
bool QERadioButton::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.251.3.53 variableSubstitutions

```
QString QERadioButton::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.251.3.54 visible

```
bool QERadioButton::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

9.251.3.55 writeOnClick

```
bool QERadioButton::writeOnClick [read], [write]
```

If true, the 'clickText' property is written when the button is clicked. Default is true

9.251.3.56 writeOnPress

```
bool QERadioButton::writeOnPress [read], [write]
```

If true, the 'pressText' property is written when the button is pressed. Default is false

9.251.3.57 writeOnRelease

```
bool QERadioButton::writeOnRelease [read], [write]
```

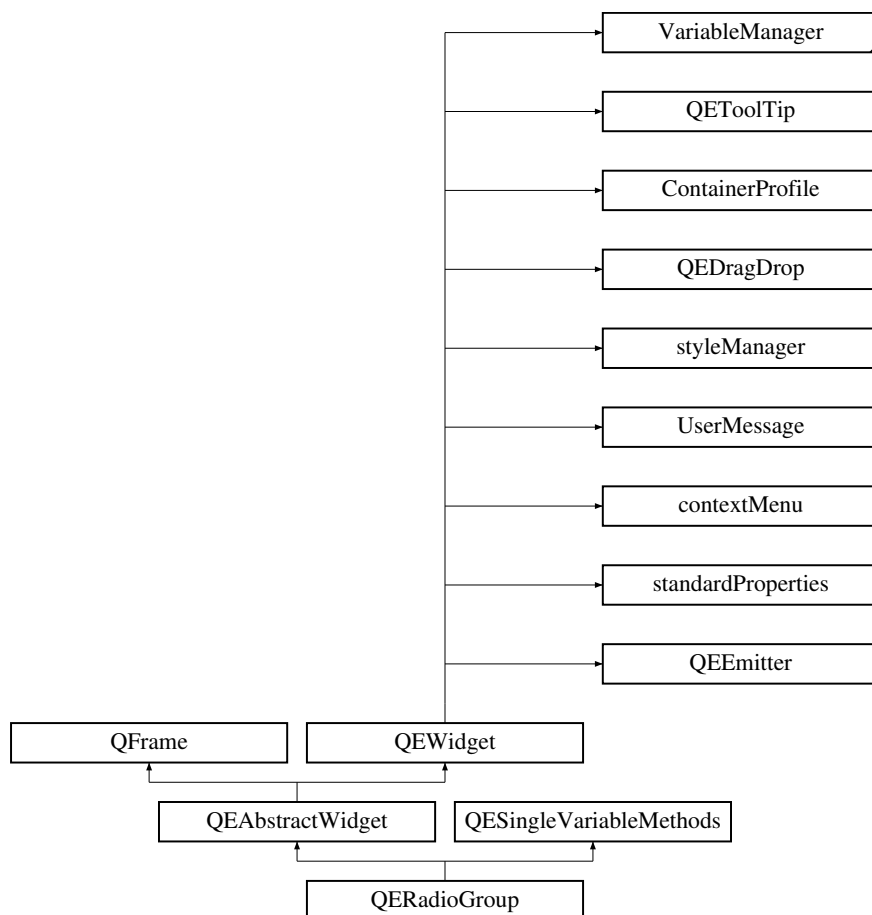
If true, the 'releaseText' property is written when the button is released. Default is false

The documentation for this class was generated from the following files:

- qeframework/widgets/QEButton/QERadioButton.h
- qeframework/widgets/QEButton/QERadioButton.cpp

9.252 QERadioGroup Class Reference

Inheritance diagram for QERadioGroup:



Public Types

- enum **OwnContextMenuOptions** { **QERG_NONE** = CM_SPECIFIC_WIDGETS_START_HERE , **QERG_↵**
_APPLY_CURRENT_SELECTION , **QERG_SUB_CLASS_WIDGETS_START_HERE** }

Public Slots

- void [writeNow](#) ()
write the value (of the underlying QComboBox object) into the PV immediately
- void [setDefaultStyle](#) (const QString &style)
Update the default style applied to this widget.
- void [setPvValue](#) (const QString &text)
- void **setPvValue** (const int value)
- void **setPvValue** (const double value)
- void **setPvValue** (const bool value)

Signals

- void [dbValueChanged](#) ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void [dbConnectionChanged](#) (const bool &isConnected)

Public Member Functions

- [QERadioGroup](#) (QWidget *parent=0)
- [QERadioGroup](#) (const QString &variableName, QWidget *parent=0)
- [QERadioGroup](#) (const QString &title, const QString &variableName, QWidget *parent=0)
- virtual [~QERadioGroup](#) ()
Destruction.
- int **getCurrentIndex** () const
- void **setVariableNameSubstitutionsProperty** (const QString &substitutions)
- void **setSubstitutedTitleProperty** (const QString &[substitutedTitle](#))
- QString **getSubstitutedTitleProperty** () const
- void **setUseDbEnumerations** (bool [useDbEnumerations](#))
- bool **getUseDbEnumerations** () const
- void **setLocalEnumerations** (const QString &localEnumerations)
- QString **getLocalEnumerations** () const

Protected Member Functions

- QSize **sizeHint** () const
- bool **eventFilter** (QObject *watched, QEvent *event)
- void [activated](#) ()
Do any post-all-widgets-constructed, i.e. activated stuff.
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- [QEChannel](#) * [createQcaltem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- QString [substitutedTitle](#)
- QString [title](#)
- int [columns](#)
- int [spacing](#)
- bool [useDbEnumerations](#)
- QString [localEnumeration](#)
- [QERadioGroup::ButtonStyles](#) [buttonStyle](#)
- [QE::GridOrders](#) [buttonOrder](#)

Additional Inherited Members

9.252.1 Constructor & Destructor Documentation

9.252.1.1 QERadioGroup() [1/3]

```
QERadioGroup::QERadioGroup (  
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use [setVariableNameProperty\(\)](#) and [setSubstitutionsProperty\(\)](#) to define a variable and, optionally, macro substitutions later.

9.252.1.2 QERadioGroup() [2/3]

```
QERadioGroup::QERadioGroup (  
    const QString & variableName,  
    QWidget * parent = 0 ) [explicit]
```

Create with a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.252.1.3 QERadioGroup() [3/3]

```
QERadioGroup::QERadioGroup (  
    const QString & title,  
    const QString & variableName,  
    QWidget * parent = 0 ) [explicit]
```

Create with a group title and a variable. A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.252.2 Member Function Documentation

9.252.2.1 dbConnectionChanged

```
void QERadioGroup::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.252.2.2 dbValueChanged

```
void QERadioGroup::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.252.2.3 setPvValue

```
void QERadioGroup::setPvValue (
    const QString & text ) [slot]
```

Update the widget and write to the main associated control process variable. Supplied values are converted to the appropiate widget value type. If conversion not possible or resultant value out of range, no widget/PV update occurs.

9.252.3 Property Documentation

9.252.3.1 arrayIndex

```
int QERadioGroup::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.252.3.2 buttonOrder

```
QE::GridOrders QERadioGroup::buttonOrder [read], [write]
```

Allows selection of buttom order (rowMajor (default) or colMajor)

9.252.3.3 buttonStyle

```
QRadioGroup::ButtonStyles QERadioGroup::buttonStyle [read], [write]
```

Allows selection of buttom style (Radio (default) or Push)

9.252.3.4 columns

```
int QERadioGroup::columns [read], [write]
```

Number of colums - defaults to two.

9.252.3.5 elementsRequired

```
int QERadioGroup::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.252.3.6 localEnumeration

```
QString QERadioGroup::localEnumeration [read], [write]
```

Enumrations values used when useDbEnumerations is false.

9.252.3.7 spacing

```
int QERadioGroup::spacing [read], [write]
```

Interay layout margins and spacing - defaults to 4.

9.252.3.8 substitutedTitle

```
QString QERadioGroup::substitutedTitle [read], [write]
```

Group box title text to be substituted. This text will be copied to the group box title text after applying any macro substitutions from the variableSubstitutions property. The former is depricated and included for backward compatabilty only and not presented on designer.

9.252.3.9 useDbEnumerations

```
bool QERadioGroup::useDbEnumerations [read], [write]
```

Use database enumerations - defaults to true. False implies use of local enumeration.

9.252.3.10 variable

```
QString QERadioGroup::variable [read], [write]
```

EPICS variable name (CA PV)

9.252.3.11 variableSubstitutions

QString QERadioGroup::variableSubstitutions [read], [write]

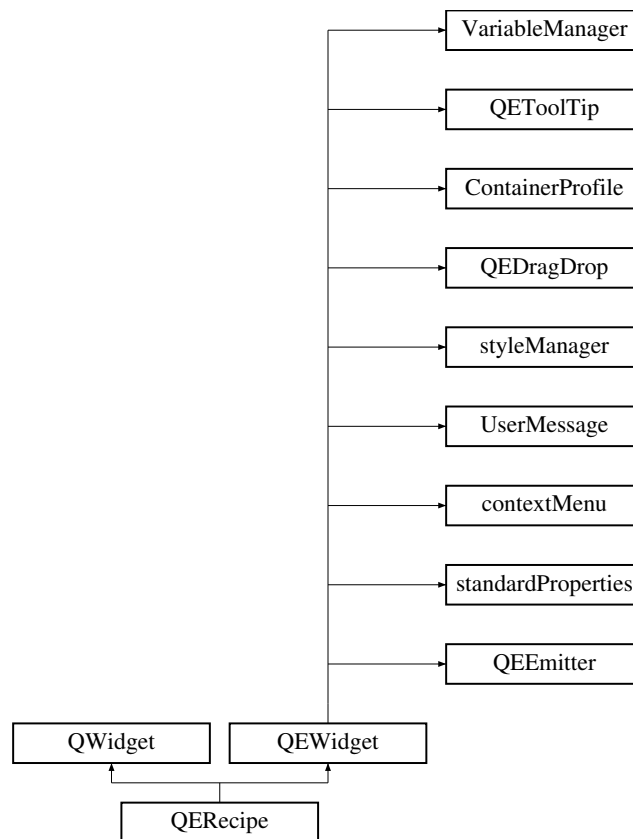
Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are are also used for other purposes.

The documentation for this class was generated from the following files:

- qeframework/widgets/QERadioGroup/QERadioGroup.h
- qeframework/widgets/QERadioGroup/QERadioGroup.cpp

9.253 QERecipe Class Reference

Inheritance diagram for QERecipe:



Public Member Functions

- **QERecipe** (QWidget *pParent=0)
- void **setRecipeDescription** (QString pValue)
- QString **getRecipeDescription** ()
- void **setShowRecipeList** (bool pValue)
- bool **getShowRecipeList** ()
- void **setShowNew** (bool pValue)

- bool **getShowNew** ()
- void **setShowSave** (bool pValue)
- bool **getShowSave** ()
- void **setShowDelete** (bool pValue)
- bool **getShowDelete** ()
- void **setShowApply** (bool pValue)
- bool **getShowApply** ()
- void **setShowRead** (bool pValue)
- bool **getShowRead** ()
- void **setShowFields** (bool pValue)
- bool **getShowFields** ()
- void **setConfigurationType** (QE::SourceOptions pValue)
- QE::SourceOptions **getConfigurationType** ()
- void **setConfigurationFile** (QString pValue)
- QString **getConfigurationFile** ()
- void **setRecipeFile** (QString pValue)
- QString **getRecipeFile** ()
- void **setConfigurationText** (QString pValue)
- QString **getConfigurationText** ()
- void **setOptionsLayout** (QE::LayoutOptions pValue)
- QE::LayoutOptions **getOptionsLayout** ()
- void **setCurrentUserType** (QE::UserLevels pValue)
- QE::UserLevels **getCurrentUserType** ()
- bool **saveRecipeList** ()
- void **refreshRecipeList** ()
- void **refreshButton** ()
- void **userLevelChanged** (QE::UserLevels pValue)
- void **setCurrentUserTypeProperty** (QE::UserLevels pUserType)
- QE::UserLevels **getCurrentUserTypeProperty** ()

Protected Attributes

- QLabel * **qLabelRecipeDescription**
- QComboBox * **qComboBoxRecipeList**
- QPushButton * **qPushButtonNew**
- QPushButton * **qPushButtonSave**
- QPushButton * **qPushButtonDelete**
- QPushButton * **qPushButtonApply**
- QPushButton * **qPushButtonRead**
- QEConfiguredLayout * **qEConfiguredLayoutRecipeFields**
- QDomDocument **document**
- QString **recipeFile**
- QString **filename**
- QE::LayoutOptions **optionsLayout**
- QE::UserLevels **currentUserType**

Properties

- QString **recipeDescription**
- bool **showRecipeList**
- bool **showNew**
- bool **showSave**
- bool **showDelete**
- bool **showApply**
- bool **showRead**
- bool **showFields**
- [QE::SourceOptions](#) **configurationType**
- QString **configurationFile**
- QString **configurationText**

Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QERecipe/QERecipe.h
- qeframework/widgets/QERecipe/QERecipe.cpp

9.254 QERecordFieldName Class Reference

Static Public Member Functions

- static QString **recordName** (const QString &pvName)
- static QString **fieldName** (const QString &pvName)
- static QString **fieldPvName** (const QString &pvName, const QString &field)
- static QString **rtypePvName** (const QString &pvName)
- static bool **pvNamelsValid** (const QString &pvName)
- static bool **extractPvName** (const QString &item, QString &pvName)

The documentation for this class was generated from the following files:

- qeframework/common/QERecordFieldName.h
- qeframework/common/QERecordFieldName.cpp

9.255 QERecordSpec Class Reference

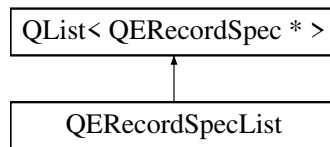
Public Member Functions

- **QERecordSpec** (const QString &recordType)
- void **append** (const QString item)
- int **size** () const
- QString **getRecordType** () const
- QString **getFieldName** (const int index) const
- bool **fieldMayUseCharArray** (const int index) const
- bool **fieldIsLinkField** (const int index) const
- QString **getDescription** (const int index) const

The documentation for this class was generated from the following files:

- qeframework/widgets/QEPvProperties/QEPvPropertiesUtilities.h
- qeframework/widgets/QEPvProperties/QEPvPropertiesUtilities.cpp

Inheritance diagram for QERecordSpecList:



- **QERRecordSpec** * **find** (const QString &recordType) const
- void **appendOrReplace** (QERRecordSpec *recordSpec)
- bool **processRecordSpecFile** (const QString &filename)

- qeframework/widgets/QEPvProperties/QEPvPropertiesUtilities.h
- qeframework/widgets/QEPvProperties/QEPvPropertiesUtilities.cpp

```
#include <QEResizableFrame.h>
```

[illegible]

Public Types

- enum **GrabbingEdges** { **TopEdge** , **LeftEdge** , **BottomEdge** , **RightEdge** }

Public Member Functions

- **QEResizeableFrame** (QWidget *parent=0)
- **QEResizeableFrame** (GrabbingEdges [grabbingEdge](#), int minimum, int maximum, QWidget *parent=0)
Construct widget specifying min and max allowed heights.
- QWidget * [widget](#) () const
Returns a ref to the resizable frame's widget, or 0 if there is none.
- void [setWidget](#) (QWidget *[widget](#))
- QWidget * [takeWidget](#) ()
Removes the resizable frame's widget, and passes ownership management of the widget to the caller.
- void [setGrabberToolTip](#) (const QString &tip)
Set the tool tip for the internal grabber object.
- void [setAllowedMinimum](#) (const int minimum)
(Re)set allowed limits.
- int [getAllowedMinimum](#) () const
- void [setAllowedMaximum](#) (const int maximum)
- int [getAllowedMaximum](#) () const
- void [setGrabbingEdge](#) (const GrabbingEdges edge)
- GrabbingEdges [getGrabbingEdge](#) () const

Protected Member Functions

- bool [eventFilter](#) (QObject *obj, QEvent *event)

Properties

- GrabbingEdges [grabbingEdge](#)
- int [allowedMinimum](#)
- int [allowedMaximum](#)

Additional Inherited Members

9.257.1 Detailed Description

The [QEResizeableFrame](#) provides a frame capable of holding another widget together with a grabber widget that allows the frame to be re-sized, and hence contained widget to be resized. The class currently only supports vertical or horizontal resizing, but not both.

9.257.2 Member Function Documentation

9.257.2.1 setWidget()

```
void QEResizeableFrame::setWidget (
    QWidget * widget )
```

Sets the resizable frame's widget. The widget becomes a child of the resizable frame, and will be destroyed when the resizable frame is deleted or when a new widget is set. Any existing widget is deleted - use `takeWidget` first if needs be.

9.257.3 Property Documentation

9.257.3.1 allowedMaximum

```
int QEResizeableFrame::allowedMaximum [read], [write]
```

Set the maximum allowed size (defaults to 100).

9.257.3.2 allowedMinimum

```
int QEResizeableFrame::allowedMinimum [read], [write]
```

Set the minimum allowed size (defaults to 10).

9.257.3.3 grabbingEdge

```
GrabbingEdges QEResizeableFrame::grabbingEdge [read], [write]
```

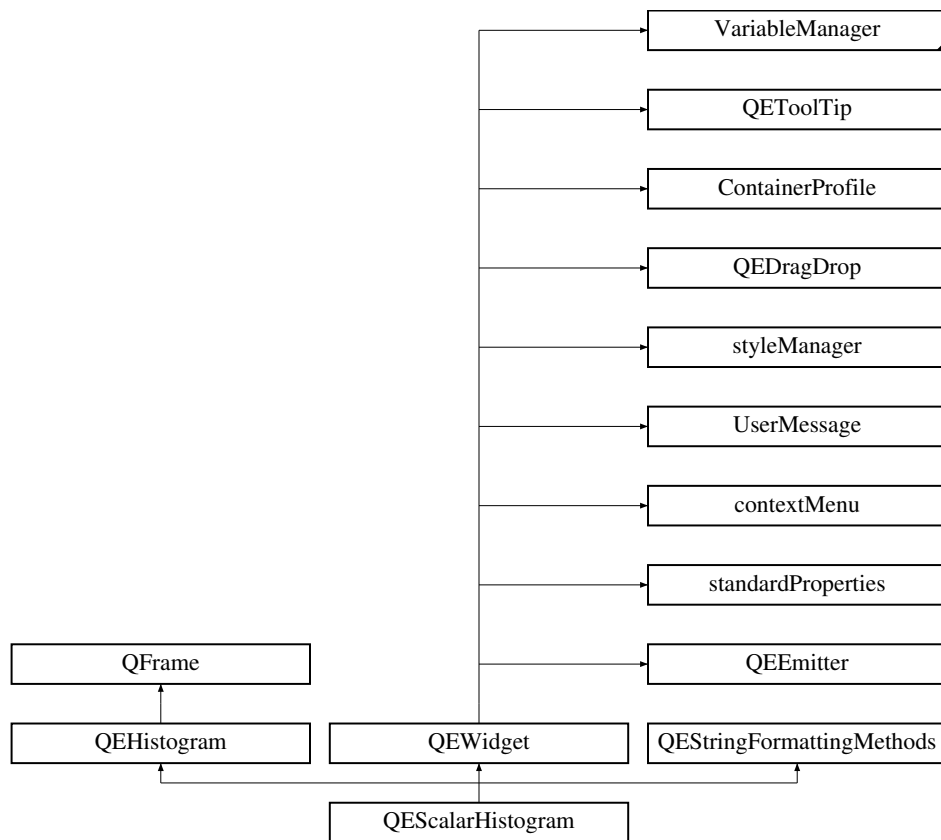
Nominated edge for the grabbing location. Defaults to `BottomEdge`.

The documentation for this class was generated from the following files:

- `qeframework/widgets/QEResizeableFrame/QEResizeableFrame.h`
- `qeframework/widgets/QEResizeableFrame/QEResizeableFrame.cpp`

9.258 QEScalarHistogram Class Reference

Inheritance diagram for QEScalarHistogram:



Public Types

- enum [ScaleModes](#) { [Manual](#) , [Auto](#) , [OperationalRange](#) }

Public Slots

- void [setManagedVisible](#) (bool v)

Public Member Functions

- **QEScalarHistogram** (QWidget *parent=0)
- void **setScaleMode** (const [ScaleModes](#) scaleMode)
- [ScaleModes](#) **getScaleMode** () const
- void **setReadoutPrecision** (const int readoutPrecision)
- int **getReadoutPrecision** () const
- void **setReadoutFormat** (const [QE::Formats](#) format)
- [QE::Formats](#) **getReadoutFormat** () const
- void **setReadoutNotation** (const [QE::Notations](#) notation)
- [QE::Notations](#) **getReadoutNotation** () const

Protected Member Functions

- [QEChannel](#) * [createQcaltem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- void **stringFormattingChange** ()
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant v)
- void **onMouseIndexChanged** (const int index)
- void **onMouseIndexPressed** (const int index, const Qt::MouseButton button)
- void **addPvName** (const QString &pvName)

Properties

- QString [variableSubstitutions](#)
- [ScaleModes](#) **scaleMode**
- int **readoutPrecision**
- [QE::Formats](#) **readoutFormat**
- [QE::Notations](#) **readoutNotation**
- QString [variable1](#)
EPICS variable names (CA PV).
- QString **variable2**
- QString **variable3**
- QString **variable4**
- QString **variable5**
- QString **variable6**
- QString **variable7**
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- QString **variable111**
- QString **variable112**
- QString **variable113**
- QString **variable114**
- QString **variable115**
- QString **variable116**
- QString **variable117**
- QString **variable118**
- QString **variable119**
- QString **variable120**
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- QE::UserLevels [userLevelVisibility](#)
- QE::UserLevels [userLevelEnabled](#)
- QE::DisplayAlarmStateOptions [displayAlarmStateOption](#)
- bool [oosAware](#)

Additional Inherited Members

9.258.1 Member Enumeration Documentation

9.258.1.1 ScaleModes

```
enum QEScalarHistogram::ScaleModes
```

Enumerator

Manual	Use property minimum/maximum to scale histogram.
Auto	Dynamically scale based on minimum/maximum displayed value.
OperationalRange	Use process variable operational range (LOPR/HOPR).

9.258.2 Member Function Documentation

9.258.2.1 setManagedVisible

```
void QEScalarHistogram::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.258.3 Property Documentation

9.258.3.1 allowDrop

```
bool QEScalarHistogram::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.258.3.2 defaultStyle

```
QString QEScalarHistogram::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.258.3.3 displayAlarmStateOption

`QE::DisplayAlarmStateOptions QEScalarHistogram::displayAlarmStateOption [read], [write]`

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.258.3.4 messageSourceId

`unsigned int QEScalarHistogram::messageSourceId [read], [write]`

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.258.3.5 oosAware

`bool QEScalarHistogram::oosAware [read], [write]`

Indicates whether the widget should respond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.258.3.6 styleSheet

`QString QEScalarHistogram::styleSheet [read], [write]`

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.258.3.7 userLevelEnabled

`QE::UserLevels QEScalarHistogram::userLevelEnabled [read], [write]`

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programmatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.258.3.8 userLevelEngineerStyle

`QString QEScalarHistogram::userLevelEngineerStyle [read], [write]`

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.258.3.9 userLevelScientistStyle

`QString QEScalarHistogram::userLevelScientistStyle [read], [write]`

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.258.3.10 userLevelUserStyle

`QString QEScalarHistogram::userLevelUserStyle [read], [write]`

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.258.3.11 userLevelVisibility

`QE::UserLevels QEScalarHistogram::userLevelVisibility [read], [write]`

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programmatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.258.3.12 variableAsToolTip

`bool QEScalarHistogram::variableAsToolTip [read], [write]`

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.258.3.13 variableSubstitutions

`QString QEScalarHistogram::variableSubstitutions [read]`

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'SAMPLE=SAM1, NAME = "Ref foil"' These substitutions are applied to all the variable names. //

9.258.3.14 visible

`bool QEScalarHistogram::visible [read], [write]`

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

The documentation for this class was generated from the following files:

- `qeframework/widgets/QEHistogram/QEScalarHistogram.h`
- `qeframework/widgets/QEHistogram/QEScalarHistogram.cpp`

9.259 QEScaling Class Reference

```
#include <QEScaling.h>
```

Static Public Member Functions

- static void [setScaling](#) (const int m, const int d)
- static void [getScaling](#) (int &m, int &d)
- static void [setFontScaling](#) (const int fm, const int fd)
- static void [getFontScaling](#) (int &fm, int &fd)
- static void [applyToWidget](#) (QWidget *widget)
- static void [rescaleWidget](#) (QWidget *widget, const int rm, const int rd)
- static void [rescaleWidget](#) (QWidget *widget, const double newScale)
- static void [getWidgetScaling](#) (const QWidget *widget, int &m, int &d)
- static int [scale](#) (const int v)
- static int [scaleFont](#) (const int v)
- *Font specific scale function.*
- static void [applyToPoint](#) (QPoint &point)
- static QString [scaleStyleSheet](#) (const QString &input)

9.259.1 Detailed Description

This class provide the means to scale a Qt application and/or an individual widget. When scaled, the size related attributes including size, min size, max size and font size are scaled (note: different from the standard Qt layout stretching). While not perfect, this can be useful and acceptable.

The scaling is defined using a rational number specified by two integers (m, d). The first (m) integer is the multiplier and the second (d) integer is the divisor. For example, if m = 4 and d = 5, then this specifies an 80%; and if m = 5 and d = 4, this specifies that a 125% scaling is required.

Additional font Scaling, above and beyond the overall scaling, may also be specified.

Scaling is deemed to be application wide, hence all scaling data (and functions) are static.

9.259.2 Member Function Documentation

9.259.2.1 [applyToPoint\(\)](#)

```
void QEScaling::applyToPoint (
    QPoint & point ) [static]
```

Scales a point.

9.259.2.2 applyToWidget()

```
void QEScaling::applyToWidget (
    QWidget * widget ) [static]
```

Adjust the geometry and font scaling of the widget and all child widgets by the defined global application scaling parameters (m, d). Unless m and d different, no scaling occurs. The function tree walks the hierarchy of widgets paranted by the specified widget. This function is idempotent.

9.259.2.3 getFontScaling()

```
void QEScaling::getFontScaling (
    int & fm,
    int & fd ) [static]
```

Extract currently applied font scaling - allows widgets to perform widget class specific scaling functionality.

9.259.2.4 getScaling()

```
void QEScaling::getScaling (
    int & m,
    int & d ) [static]
```

Extract currently applied scaling - allows widgets to perform widget class specific scaling functionality.

9.259.2.5 getWidgetScaling()

```
void QEScaling::getWidgetScaling (
    const QWidget * widget,
    int & m,
    int & d ) [static]
```

Get the current widget scaling. This may not be the global application scaling if rescaleWidget has been called, but does include the global scaling.

9.259.2.6 rescaleWidget()

```
void QEScaling::rescaleWidget (
    QWidget * widget,
    const int rm,
    const int rd ) [static]
```

Applies the specified scaling to the nominated widget and all child widgets. This is in addition to the application wide scaling defined by setScaling. The scaling is limited to 0.1 to 400.0 (10% to 400%)

9.259.2.7 scale()

```
static int QEScaling::scale (
    const int v ) [inline], [static]
```

Convenience functions for widget specific 'scaleBy' functions.

Scales a single value. Note: all other scaling functions take a object by reference and modify that object. Only this function returns a scaled value.

General scale function.

9.259.2.8 scaleStyleSheet()

```
QString QEScaling::scaleStyleSheet (
    const QString & input ) [static]
```

Scales styleSheet - looks for "... <number>px ..." or "... <number>pt ..." and scales the number.

9.259.2.9 setFontScaling()

```
void QEScaling::setFontScaling (
    const int fm,
    const int fd ) [static]
```

Set application wide additional font scaling values. The final font scaling is thus $(m * fm) / (d * fd)$ Both values default to 1, which is a null scaling. Only valid scaling values ($fm > 0$, $fd > 0$) are accepted.

9.259.2.10 setScaling()

```
void QEScaling::setScaling (
    const int m,
    const int d ) [static]
```

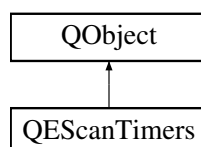
Set application wide general gui scaling values. Both values default to 1, which is a null scaling. Only valid scaling values ($m > 0$, $d > 0$) are accepted.

The documentation for this class was generated from the following files:

- qeframework/common/QEScaling.h
- qeframework/common/QEScaling.cpp

9.260 QEScanTimers Class Reference

Inheritance diagram for QEScanTimers:



Public Types

- enum **ScanRates** {
 VerySlow , **Slow** , **Medium** , **Fast** ,
 VeryFast }

Signals

- void **flipFlopVerySlow** (const bool)
- void **flipFlopSlow** (const bool)
- void **flipFlopMedium** (const bool)
- void **flipFlopFast** (const bool)
- void **flipFlopVeryFast** (const bool)

Static Public Member Functions

- static bool **attach** (QObject *target, const char *member, const ScanRates scanRate)
- static void **detach** (QObject *target, const char *member)

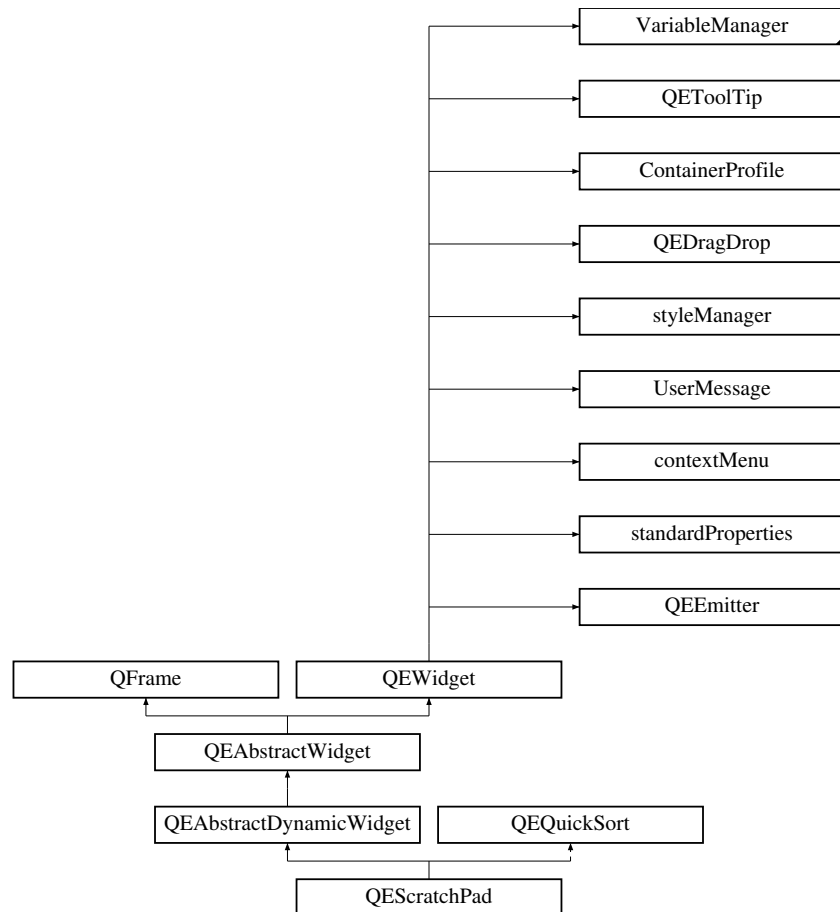
The documentation for this class was generated from the following files:

- qeframework/common/QEScanTimers.h
- qeframework/common/QEScanTimers.cpp

9.261 QEScratchPad Class Reference

```
#include <QEScratchPad.h>
```

Inheritance diagram for QEScratchPad:



Public Slots

- void **setSelection** (int value)
- void **setPvNameSet** (const QStringList &pvNameSet)

Signals

- void **selectionChanged** (int value)
- void **pvnNameSetChanged** (const QStringList &nameSet)

Public Member Functions

- **QEScratchPad** (QWidget *parent=0)
- QSize **sizeHint** () const
- void **setPvName** (const int slot, const QString &pvnName)
- QString **getPvName** (const int slot) const
- int **addPvName** (const QString &pvnName)
- void **clearAllPvNames** ()
- int **getSelection** () const
- QStringList **getPvNameSet** () const

Static Public Attributes

- static const int **NUMBER_OF_ITEMS** = 72

Protected Member Functions

- void **resizeEvent** (QResizeEvent *event)
- void **activated** ()
Do any post-all-widgets-constructed, i.e. activated stuff.
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)
- void **enableEditPvChanged** ()
- bool **itemLessThan** (const int a, const int b, QObject *context=NULL) const
- void **swapItems** (const int a, const int b, QObject *context=NULL)
- void **mousePressEvent** (QMouseEvent *event)
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dragMoveEvent** (QDragMoveEvent *event)
- void **dropEvent** (QDropEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)
- void **saveConfiguration** (PersistenceManager *pm)
- void **restoreConfiguration** (PersistenceManager *pm, restorePhases restorePhase)
- void **showEvent** (QShowEvent *event)
- void **keyPressEvent** (QKeyEvent *event)
- bool **eventFilter** (QObject *obj, QEvent *event)
- int **findSlot** (QObject *obj) const

Additional Inherited Members

9.261.1 Detailed Description

This class provides a flexible scratch pad form, to which any Process Variable may be added. It displays the PV Name, the Description, i.e. the content of the DESC field together with the value of PV itself.

This class is a direct re-implementation of TScratch_Pad_Form out of the Delphi OPI framework.

9.261.2 Member Function Documentation

9.261.2.1 restoreConfiguration()

```
void QEScratchPad::restoreConfiguration (
    PersistenceManager * ,
    restorePhases ) [protected], [virtual]
```

Service a request to restore the [QE](#) widget's configuration. A [QE](#) widget recover any configuration details from the [PersistenceManager](#). For example, a [QEStripChart](#) may restore the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method. This is called twice with an incrementing restorePhase. Most widgets will miss the first call as they don't exist yet (they are created as part of the first phase).

Reimplemented from [QEWidget](#).

9.261.2.2 saveConfiguration()

```
void QEScratchPad::saveConfiguration (
    PersistenceManager * ) [protected], [virtual]
```

Service a request to save the [QE](#) widget's current configuration. A widget may save any configuration details through the [PersistenceManager](#). For example, a [QEStriptChart](#) may save the variables being plotted. Many [QE](#) widgets do not have any persistent data requirements and do not implement this method.

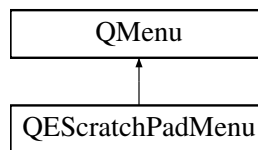
Reimplemented from [QEWidget](#).

The documentation for this class was generated from the following files:

- qeframework/widgets/QEScratchPad/QEScratchPad.h
- qeframework/widgets/QEScratchPad/QEScratchPad.cpp

9.262 QEScratchPadMenu Class Reference

Inheritance diagram for QEScratchPadMenu:



Public Types

- enum **ContextMenuOptions** {
SCRATCHPAD_NONE = QEAbstractDynamicWidget::ADWCM_SUB_CLASS_WIDGETS_START_HERE ,
SCRATCHPAD_SORT_PV_NAMES , **SCRATCHPAD_CLEAR_ALL** , **SCRATCHPAD_ADD_PV_NAME** ,
SCRATCHPAD_PASTE_PV_NAME , **SCRATCHPAD_EDIT_PV_NAME** , **SCRATCHPAD_DATA_CLEAR** }

Signals

- void **contextMenuSelected** (const int, const QEScratchPadMenu::ContextMenuOptions)

Public Member Functions

- **QEScratchPadMenu** (const int slot, QWidget *parent=0)
- void **setActionChecked** (const ContextMenuOptions option, const bool visible)
- void **setActionEnabled** (const ContextMenuOptions option, const bool visible)
- void **setActionVisible** (const ContextMenuOptions option, const bool visible)
- void **setIsInUse** (const bool isInUse)

Static Public Attributes

- static const ContextMenuOptions **ContextMenuFirst** = SCRATCHPAD_ADD_PV_NAME
- static const ContextMenuOptions **ContextMenuLast** = SCRATCHPAD_DATA_CLEAR
- static const int **NumberContextMenuItems** = ContextMenuLast - ContextMenuFirst + 1

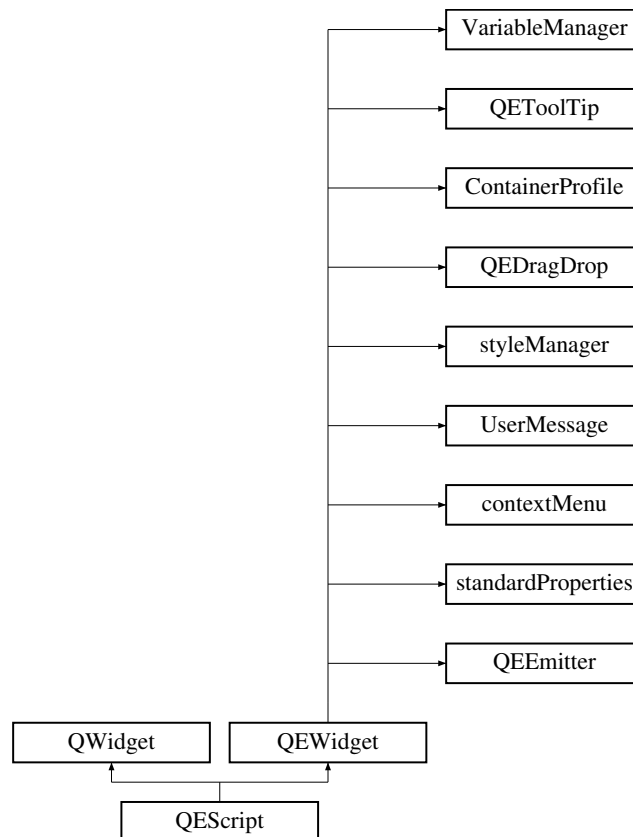
The documentation for this class was generated from the following files:

- qeframework/widgets/QEScratchPad/QEScratchPadMenu.h
- qeframework/widgets/QEScratchPad/QEScratchPadMenu.cpp

9.263 QEScript Class Reference

```
#include <QEScript.h>
```

Inheritance diagram for QEScript:



Public Slots

- void [setManagedVisible](#) (bool v)

Signals

- void **selected** (QString pFilename)

Public Member Functions

- **QEScript** (QWidget *pParent=0)
- void **setShowScriptList** (bool pValue)
- bool **getShowScriptList** ()
- void **setShowNew** (bool pValue)
- bool **getShowNew** ()
- void **setShowSave** (bool pValue)
- bool **getShowSave** ()
- void **setShowDelete** (bool pValue)
- bool **getShowDelete** ()
- void **setShowExecute** (bool pValue)
- bool **getShowExecute** ()
- void **setShowAbort** (bool pValue)
- bool **getShowAbort** ()
- void **setEditableTable** (bool pValue)
- bool **getEditableTable** ()
- void **setShowTable** (bool pValue)
- bool **getShowTable** ()
- void **setShowTableControl** (bool pValue)
- bool **getShowTableControl** ()
- void **setShowColumnNumber** (bool pValue)
- bool **getShowColumnNumber** ()
- void **setShowColumnEnable** (bool pValue)
- bool **getShowColumnEnable** ()
- void **setShowColumnProgram** (bool pValue)
- bool **getShowColumnProgram** ()
- void **setShowColumnParameters** (bool pValue)
- bool **getShowColumnParameters** ()
- void **setShowColumnWorkingDirectory** (bool pValue)
- bool **getShowColumnWorkingDirectory** ()
- void **setShowColumnTimeout** (bool pValue)
- bool **getShowColumnTimeout** ()
- void **setShowColumnStop** (bool pValue)
- bool **getShowColumnStop** ()
- void **setShowColumnLog** (bool pValue)
- bool **getShowColumnLog** ()
- void **setScriptType** (QE::SourceOptions pValue)
- QE::SourceOptions **getScriptType** ()
- void **setScriptFile** (QString pValue)
- QString **getScriptFile** ()
- void **setScriptText** (QString pValue)
- QString **getScriptText** ()
- void **setScriptDefault** (QString pValue)
- QString **getScriptDefault** ()
- void **setExecuteText** (QString pValue)
- QString **getExecuteText** ()
- void **setOptionsLayout** (QE::LayoutOptions pValue)
- QE::LayoutOptions **getOptionsLayout** ()
- void **insertRow** (bool pEnable, QString pProgram, QString pParameter, QString pWorkingDirectory, int pTimeout, bool pStop, bool pLog)
- bool **saveScriptList** ()
- void **refreshScriptList** ()
- void **refreshWidgets** ()

Protected Attributes

- QComboBox * **qComboBoxScriptList**
- QPushButton * **qPushButtonNew**
- QPushButton * **qPushButtonSave**
- QPushButton * **qPushButtonDelete**
- QPushButton * **qPushButtonExecute**
- QPushButton * **qPushButtonAbort**
- QPushButton * **qPushButtonAdd**
- QPushButton * **qPushButtonRemove**
- QPushButton * **qPushButtonUp**
- QPushButton * **qPushButtonDown**
- QPushButton * **qPushButtonCopy**
- QPushButton * **qPushButtonPaste**
- [_QTableWidgetScript](#) * **qTableWidgetScript**
- QString [scriptIdFile](#)
Define the file where to save the scripts (if not defined then the scripts will be saved in a file named "QEScript.xml")
- QString [scriptIdText](#)
Define the XML text that contains the scripts.
- QString [scriptIdDefault](#)
Define the script (previously saved by the user) that will be load as the default script when the widget starts.
- [QE::SourceOptions](#) [scriptIdType](#)
Select if the scripts are to be loaded/saved from an XML file or from an XML text.
- [QE::LayoutOptions](#) [optionsLayout](#)
Change the order of the widgets. Valid orders are: TOP, BOTTOM, LEFT and RIG.
- QDomDocument **document**
- QString **filename**
- QList< [_CopyPaste](#) * > **copyPasteList**
- bool [editableTable](#)
Enable/disable table edition.
- bool **isExecuting**

Properties

- bool [showScriptList](#)
Show/hide combobox that contains the list of existing scripts created by the user.
- bool [showNew](#)
Show/hide button to reset (initialize) the table that contains the sequence of programs to be executed.
- bool [showSave](#)
Show/hide button to save/overwrite a new/existing script.
- bool [showDelete](#)
Show/hide button to delete an existing script.
- bool [showExecute](#)
Show/hide button to execute a sequence of programs.
- bool [showAbort](#)
Show/hide button to abort the execution of a sequence of programs.
- bool [showTable](#)
Show/hide table that contains a sequence of programs to be executed.
- bool [showTableControl](#)
Show/hide the controls of the table that contains a sequence of programs to be executed.
- bool [showColumnNumber](#)

- *Show/hide the column '#' that displays the sequential number of programs.*
- bool [showColumnEnable](#)
Show/hide the column 'Enable' that enables the execution of programs.
- bool [showColumnProgram](#)
Show/hide the column 'Program' that contains the external programs to be executed.
- bool [showColumnParameters](#)
Show/hide the column 'Parameters' that contains the parameters that are passed to external programs to be executed.
- bool [showColumnWorkingDirectory](#)
Show/hide the column 'Directory' that defines the working directory to be used when external programs are executed.
- bool [showColumnTimeout](#)
Show/hide the column 'Timeout' that defines a time out period in seconds (if equal to 0 then the program runs until it finishes; otherwise if greater than 0 then the program will only run during this amount of seconds and will be aborted beyond this time)
- bool [showColumnStop](#)
Show/hide the column 'Stop' that enables stopping the execution of subsequent programs when the current one exited with an error code different from 0.
- bool [showColumnLog](#)
Show/hide the column 'Log' that enables the generation of log messages (these messages may be displayed using the [QELog](#) widget)
- QString [executeText](#)
Define the caption of the button responsible for starting the execution of external programs (if not defined then the caption will be "Execute")
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)

Additional Inherited Members

9.263.1 Detailed Description

This class is a EPICS aware widget. The [QEScript](#) widget allows the user to define a certain sequence of external programs to be executed. This sequence may be saved, modified or loaded for future usage.

9.263.2 Member Function Documentation

9.263.2.1 setManagedVisible

```
void QEScript::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.263.3 Property Documentation

9.263.3.1 defaultStyle

```
QString QEScript::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.263.3.2 messageSourceId

```
unsigned int QEScript::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.263.3.3 styleSheet

```
QString QEScript::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.263.3.4 userLevelEnabled

```
QE::UserLevels QEScript::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programmatically through `setUserLevel()`. Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.263.3.5 userLevelEngineerStyle

```
QString QEScript::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red'. This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.263.3.6 userLevelScientistStyle

```
QString QEScript::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.263.3.7 userLevelUserStyle

```
QString QEScript::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.263.3.8 userLevelVisibility

```
QE::UserLevels QEScript::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.263.3.9 visible

```
bool QEScript::visible [read], [write]
```

`variableAsToolTip` not applicale to V2 options. Class should ensure this option is initialised to false. `allowDrop` not applicale to V2 options. Class should ensure this option is initialised to false. Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

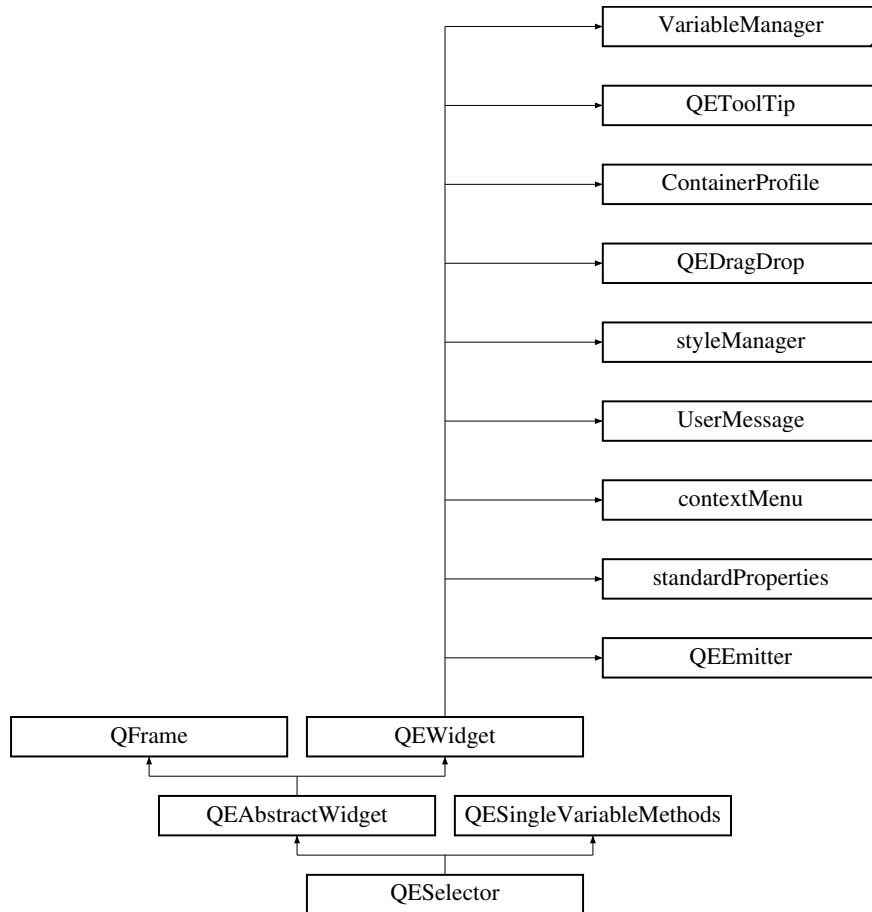
The documentation for this class was generated from the following files:

- qeframework/widgets/QEScript/QEScript.h
- qeframework/widgets/QEScript/QEScript-wip.cpp
- qeframework/widgets/QEScript/QEScript.cpp

9.264 QESelector Class Reference

```
#include <QESelector.h>
```

Inheritance diagram for QESelector:



Public Types

- enum **Delimiters** { **NoDelimiter** = 0 , **SpaceDelimiter** , **CommaDelimiter** }

Public Slots

- void **writeNow** ()

Signals

- void **dbValueChanged** ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void **dbConnectionChanged** (const bool &isConnected)

Public Member Functions

- **QESelector** (QWidget *parent=0)
- **QESelector** (const QString &variableName, QWidget *parent=0)
- void **setSource** (const [QE::SourceOptions](#) infoSource)
- [QE::SourceOptions](#) **getSource** () const
- void **setStringList** (const QStringList &userInfoList)
- QStringList **getStringList** () const
- void **setSourceFilename** (const QString &userInfoFile)
- QString **getSourceFilename** () const
- void **setDelimiter** (const Delimiters delimiter)
- Delimiters **getDelimiter** () const
- void **setSubscribe** (const bool [subscribe](#))
- bool **getSubscribe** () const
- void **setWriteOnChange** (const bool [writeOnChange](#))
- bool **getWriteOnChange** () const
- void **setAllowFocusUpdate** (const bool [allowFocusUpdate](#))
- bool **getAllowFocusUpdate** () const

Protected Member Functions

- QSize **sizeHint** () const
- void **focusInEvent** (QFocusEvent *event)
- bool **eventFilter** (QObject *watched, QEvent *event)
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- [QEChannel](#) * **createQcaltem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant v)

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- [QE::SourceOptions](#) **source**
- QStringList **stringList**
- QString **sourceFilename**
- int **maxVisibleItems**
- Delimiters **delimiter**
- bool [subscribe](#)
- bool [writeOnChange](#)
- bool [allowFocusUpdate](#)

Additional Inherited Members

9.264.1 Detailed Description

This widget allows one on a number of string values to be selected from a pre-determined list of strings and the selected value written to the associated PV. The underlying widget used for this is a QComboBox widget. A typical use case would be selecting a positioner PV name or detector PV name for use with the sscan record.

A delimiter may also be specified. Only text before the delimiter is written to the PV. In this way a user friendly comment can be added to the combo box drop down list. A typical delimiter would be a space or comma.

9.264.2 Member Function Documentation

9.264.2.1 dbConnectionChanged

```
void QESelector::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change.

9.264.2.2 dbValueChanged

```
void QESelector::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.264.3 Property Documentation

9.264.3.1 allowFocusUpdate

```
bool QESelector::allowFocusUpdate [read], [write]
```

Allow updated while widget has focus - defaults to false.

9.264.3.2 arrayIndex

```
int QESelector::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.264.3.3 elementsRequired

```
int QESelector::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.264.3.4 subscribe

```
bool QESelector::subscribe [read], [write]
```

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.264.3.5 variable

```
QString QESelector::variable [read], [write]
```

EPICS variable name (CA PV)

9.264.3.6 variableSubstitutions

```
QString QESelector::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.264.3.7 writeOnChange

```
bool QESelector::writeOnChange [read], [write]
```

Sets if this widget writes any changes as the user selects values (the QComboBox 'activated' signal is emitted). Default is 'true' (writes any changes when the QComboBox 'activated' signal is emitted).

The documentation for this class was generated from the following files:

- qeframework/widgets/QESelector/QESelector.h
- qeframework/widgets/QESelector/QESelector.cpp

9.265 QESettings Class Reference

```
#include <QESettings.h>
```

Public Member Functions

- **QESettings** (const QString &fileName)
- bool **isDefined** () const
- QString **getSettingFileName** () const
- QStringList **groupKeys** (const QString &group)
- QVariant **getValue** (const QString &key, const QVariant &defaultValue)
- bool **getBool** (const QString &key, const bool defaultValue)
- QString **getString** (const QString &key, const QString &defaultValue)
- int **getInt** (const QString &key, const int defaultValue)
- double **getFloat** (const QString &key, const double defaultValue)
- QString **getFilename** (const QString &key, const QString &defaultValue)
- [QESettings](#) * **getSettings** (const QString &key)

Static Public Member Functions

- static [QESettings](#) * **getConfigurationParameters** (const QString §ion)

9.265.1 Detailed Description

This class provides access to user defined settings. The setting file is expected to conform to the `QSettings::IniFormat`, i.e. like a windows style .ini file.

The types of data that may be extracted from an option are bool, QString, int and double; These are accessed by the associated `getXxx` () functions where Xxx is one of Bool, String, Int and Float.

Each of these function takes a default value of the appropriate type which is returned to the caller if the value is not defined or, as in the case of numerical values, is ill-defined.

Numeric validity is defined by `QString::toInt` () and `QString::toDouble` () Do note that `QString::toInt` () does not accept 0x... hexadecimal or 0... octal numbers.

The class also provides a number of additional convenience functions that are described below.

This class is one of several provided to access configuration data such as from environment variables, command line options and settings. They all provide a similar API.

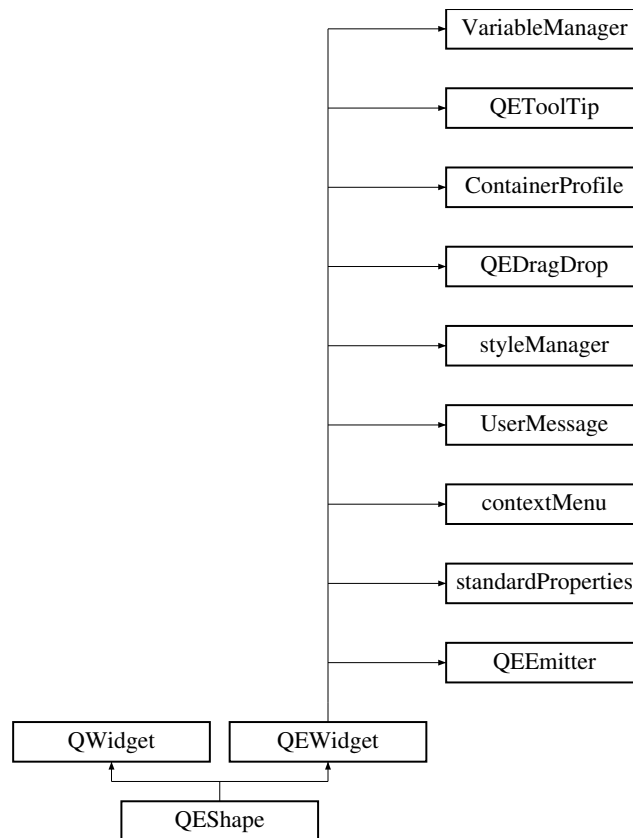
The documentation for this class was generated from the following files:

- qeframework/adaptation_parameters/QESettings.h
- qeframework/adaptation_parameters/QESettings.cpp

9.266 QEShape Class Reference

```
#include <QEShape.h>
```

Inheritance diagram for QEShape:



Public Types

- enum [shapeOptions](#) {
Line , **Points** , **Polyline** , **Polygon** ,
Rect , **RoundedRect** , **Ellipse** , **Arc** ,
Chord , **Pie** , **Path** }
- enum [animationOptions](#) {
Width , **Height** , **X** , **Y** ,
Transperency , **Rotation** , **ColourHue** , **ColourSaturation** ,
ColourValue , **ColourIndex** , **Penwidth** }

Public Slots

- void [setManagedVisible](#) (bool v)

Signals

- void [dbValueChanged1](#) (const qlonglong &out)
- void [dbValueChanged2](#) (const qlonglong &out)
- void [dbValueChanged3](#) (const qlonglong &out)
- void [dbValueChanged4](#) (const qlonglong &out)
- void [dbValueChanged5](#) (const qlonglong &out)
- void [dbValueChanged6](#) (const qlonglong &out)

Public Member Functions

- [QEShape](#) (QWidget *parent=0)
- [QEShape](#) (const QString &variableName, QWidget *parent=0)
- QSize [sizeHint](#) () const
- void [scaleBy](#) (const int m, const int d)
Scale the widgets my m/d.
- void [setAnimation](#) ([animationOptions](#) animation, const int index)
Access function for #animation' properties - refer to animation' properties for details.
- [animationOptions](#) [getAnimation](#) (const int index) const
Access function for #animation' properties - refer to animation' properties for details.
- void [setScale](#) (const double scale, const int index)
Access function for #scale' properties - refer to scale' properties for details.
- double [getScale](#) (const int index) const
Access function for #scale' properties - refer to scale' properties for details.
- void [setOffset](#) (const double offset, const int index)
Access function for #offset' properties - refer to offset' properties for details.
- double [getOffset](#) (const int index) const
Access function for #offset' properties - refer to offset' properties for details.
- void [setBorder](#) (const bool border)
Access function for #border' properties - refer to border' properties for details.
- bool [getBorder](#) () const
Access function for #border' properties - refer to border' properties for details.
- void [setFill](#) (const bool fill)
Access function for #fill' properties - refer to fill' properties for details.
- bool [getFill](#) () const
Access function for #fill' properties - refer to fill' properties for details.
- void [setShape](#) ([shapeOptions](#) shape)
Access function for #shape' properties - refer to shape' properties for details.
- [shapeOptions](#) [getShape](#) () const
Access function for #shape' properties - refer to shape' properties for details.
- void [setNumPoints](#) (const unsigned int numPoints)
Access function for #number of points' properties - refer to number of points' properties for details.
- unsigned int [getNumPoints](#) () const
Access function for #number of points' properties - refer to number of points' properties for details.
- void [setOriginTranslation](#) (const QPoint &originTranslation)
Access function for #origin translation' properties - refer to origin translation' properties for details.
- QPoint [getOriginTranslation](#) () const
Access function for #origin translation' properties - refer to origin translation' properties for details.
- void [setPoint](#) (const QPoint &point, const int index)
Access function for #point' properties - refer to point' properties for details.
- QPoint [getPoint](#) (const int index) const
Access function for #point' properties - refer to point' properties for details.
- void [setColor](#) (const QColor &color, const int index)
Access function for #colour' properties - refer to colour' properties for details.
- QColor [getColor](#) (const int index) const
Access function for #colour' properties - refer to colour' properties for details.
- void [setDrawBorder](#) (const bool drawBorder)
Access function for #draw border' properties - refer to draw border' properties for details.
- bool [getDrawBorder](#) () const
Access function for #draw border' properties - refer to draw border' properties for details.

- void [setLineWidth](#) (const unsigned int lineWidth)
Access function for #line width' properties - refer to line width' properties for details.
- unsigned int [getLineWidth](#) () const
Access function for #line width' properties - refer to line width' properties for details.
- void [setStartAngle](#) (const double startAngle)
Access function for #start angle' properties - refer to start angle' properties for details.
- double [getStartAngle](#) () const
Access function for #start angle' properties - refer to start angle' properties for details.
- void [setRotation](#) (const double rotation)
Access function for #rotation' properties - refer to rotation' properties for details.
- double [getRotation](#) () const
Access function for #rotation' properties - refer to rotation' properties for details.
- void [setArcLength](#) (const double arcLength)
Access function for #arc length' properties - refer to arc length' properties for details.
- double [getArcLength](#) () const
Access function for #arc length' properties - refer to arc length' properties for details.
- void [setVariableNameSubstitutionsProperty](#) (QString variableNameSubstitutions)
Property access function for [variableSubstitutions](#) property. This has special behaviour to work well within designer.
- QString [getVariableNameSubstitutionsProperty](#) ()
Property access function for [variableSubstitutions](#) property. This has special behaviour to work well within designer.

Properties

- QString [variable1](#)
- QString [variable2](#)
- QString [variable3](#)
- QString [variable4](#)
- QString [variable5](#)
- QString [variable6](#)
- QString [variableSubstitutions](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- bool [oosAware](#)
- [animationOptions](#) [animation1](#)
- [animationOptions](#) [animation2](#)
- [animationOptions](#) [animation3](#)
- [animationOptions](#) [animation4](#)
- [animationOptions](#) [animation5](#)
- [animationOptions](#) [animation6](#)
- double [scale1](#)
Scale factor applied to data from the 1st variable before it is used to animate the shape.
- double [scale2](#)

- double [scale3](#)
- double [scale4](#)
- double [scale5](#)
- double [scale6](#)
- double [offset1](#)
- double [offset2](#)
- double [offset3](#)
- double [offset4](#)
- double [offset5](#)
- double [offset6](#)
- QPoint [point1](#)
- QPoint [point2](#)
- QPoint [point3](#)
- QPoint [point4](#)
- QPoint [point5](#)
- QPoint [point6](#)
- QPoint [point7](#)
- QPoint [point8](#)
- QPoint [point9](#)
- QPoint [point10](#)
- QColor [color1](#)
- QColor [color2](#)
- QColor [color3](#)
- QColor [color4](#)
- QColor [color5](#)
- QColor [color6](#)
- QColor [color7](#)
- QColor [color8](#)
- QColor [color9](#)
- QColor [color10](#)

Additional Inherited Members

9.266.1 Detailed Description

This class is a EPICS aware shape widget based on the Qt widget. One of several shapes can be drawn within the widget, and up to 6 variables can be used to animate various attributes of the shape. For example to represent beam positino and size, an ellipse can be drawn with four variables animating its vertcal and horizontal size and position. It is tightly integrated with the base class [QEWidget](#) which provides generic support such as macro substitutions, drag/drop, and standard properties.

9.266.2 Member Enumeration Documentation

9.266.2.1 animationOptions

enum [QEShape::animationOptions](#)

Options for how a variable will animate the shape.

9.266.2.2 shapeOptions

enum [QEShape::shapeOptions](#)

Options for the type of shape.

9.266.3 Constructor & Destructor Documentation

9.266.3.1 QEShape() [1/2]

```
QEShape::QEShape (  
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use `setVariableNameProperty()` and `setSubstitutionsProperty()` to define a variable and, optionally, macro substitutions later.

9.266.3.2 QEShape() [2/2]

```
QEShape::QEShape (  
    const QString & variableName,  
    QWidget * parent = 0 ) [explicit]
```

Create with a single variable. (Note, the [QEShape](#) widget can use up to 6 variables) A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.266.4 Member Function Documentation

9.266.4.1 dbValueChanged1

```
void QEShape::dbValueChanged1 (  
    const qulonglong & out ) [signal]
```

Sent when the widget is updated following a data change for the first variable Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.266.4.2 dbValueChanged2

```
void QEShape::dbValueChanged2 (  
    const qulonglong & out ) [signal]
```

Sent when the widget is updated following a data change for the second variable Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.266.4.3 dbValueChanged3

```
void QEShape::dbValueChanged3 (
    const qlonglong & out ) [signal]
```

Sent when the widget is updated following a data change for the third variable Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.266.4.4 dbValueChanged4

```
void QEShape::dbValueChanged4 (
    const qlonglong & out ) [signal]
```

Sent when the widget is updated following a data change for the fourth variable Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.266.4.5 dbValueChanged5

```
void QEShape::dbValueChanged5 (
    const qlonglong & out ) [signal]
```

Sent when the widget is updated following a data change for the fifth variable Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.266.4.6 dbValueChanged6

```
void QEShape::dbValueChanged6 (
    const qlonglong & out ) [signal]
```

Sent when the widget is updated following a data change for the sixth variable Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.266.4.7 setManagedVisible

```
void QEShape::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.266.5 Property Documentation

9.266.5.1 allowDrop

```
bool QEShape::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.266.5.2 animation1

`animationOptions` QEShape::animation1 [read], [write]

Animation to be effected by the 1st variable. This is used to select what the effect changing data for the 1st variable will have on the shape.

9.266.5.3 animation2

`animationOptions` QEShape::animation2 [read], [write]

Animation to be effected by the 2nd variable. This is used to select what the effect changing data for the 2nd variable will have on the shape.

9.266.5.4 animation3

`animationOptions` QEShape::animation3 [read], [write]

Animation to be effected by the 3rd variable. This is used to select what the effect changing data for the 3rd variable will have on the shape.

9.266.5.5 animation4

`animationOptions` QEShape::animation4 [read], [write]

Animation to be effected by the 4th variable. This is used to select what the effect changing data for the 4th variable will have on the shape.

9.266.5.6 animation5

`animationOptions` QEShape::animation5 [read], [write]

Animation to be effected by the 5th variable. This is used to select what the effect changing data for the 5th variable will have on the shape.

9.266.5.7 animation6

`animationOptions` QEShape::animation6 [read], [write]

Animation to be effected by the 6th variable. This is used to select what the effect changing data for the 6th variable will have on the shape.

9.266.5.8 color1

`QColor` QEShape::color1 [read], [write]

Used by the color animation to determine the color based on a data value. The scaled and offset data is used as an index to select color properties 'color1' to 'color10'.

9.266.5.9 color10

```
QColor QEShape::color10 [read], [write]
```

Used by the color animation to determine the color based on a data value. The scaled and offset data is used as an index to select color properties 'color1' to 'color10'.

9.266.5.10 color2

```
QColor QEShape::color2 [read], [write]
```

Used by the color animation to determine the color based on a data value. The scaled and offset data is used as an index to select color properties 'color1' to 'color10'.

9.266.5.11 color3

```
QColor QEShape::color3 [read], [write]
```

Used by the color animation to determine the color based on a data value. The scaled and offset data is used as an index to select color properties 'color1' to 'color10'.

9.266.5.12 color4

```
QColor QEShape::color4 [read], [write]
```

Used by the color animation to determine the color based on a data value. The scaled and offset data is used as an index to select color properties 'color1' to 'color10'.

9.266.5.13 color5

```
QColor QEShape::color5 [read], [write]
```

Used by the color animation to determine the color based on a data value. The scaled and offset data is used as an index to select color properties 'color1' to 'color10'.

9.266.5.14 color6

```
QColor QEShape::color6 [read], [write]
```

Used by the color animation to determine the color based on a data value. The scaled and offset data is used as an index to select color properties 'color1' to 'color10'.

9.266.5.15 color7

```
QColor QEShape::color7 [read], [write]
```

Used by the color animation to determine the color based on a data value. The scaled and offset data is used as an index to select color properties 'color1' to 'color10'.

9.266.5.16 color8

```
QColor QEShape::color8 [read], [write]
```

Used by the color animation to determine the color based on a data value. The scaled and offset data is used as an index to select color properties 'color1' to 'color10'.

9.266.5.17 color9

```
QColor QEShape::color9 [read], [write]
```

Used by the color animation to determine the color based on a data value. The scaled and offset data is used as an index to select color properties 'color1' to 'color10'.

9.266.5.18 defaultStyle

```
QString QEShape::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.266.5.19 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEShape::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.266.5.20 messageSourceId

```
unsigned int QEShape::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.266.5.21 offset1

```
double QEShape::offset1 [read], [write]
```

Offset applied to data from the 1st variable before it is used to animate the shape

9.266.5.22 offset2

```
double QEShape::offset2 [read], [write]
```

Offset applied to data from the 2nd variable before it is used to animate the shape

9.266.5.23 offset3

```
double QEShape::offset3 [read], [write]
```

Offset applied to data from the 3rd variable before it is used to animate the shape

9.266.5.24 offset4

```
double QEShape::offset4 [read], [write]
```

Offset applied to data from the 4th variable before it is used to animate the shape

9.266.5.25 offset5

```
double QEShape::offset5 [read], [write]
```

Offset applied to data from the 5th variable before it is used to animate the shape

9.266.5.26 offset6

```
double QEShape::offset6 [read], [write]
```

Offset applied to data from the 6th variable before it is used to animate the shape

9.266.5.27 oosAware

```
bool QEShape::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.266.5.28 point1

```
QPoint QEShape::point1 [read], [write]
```

1st coordinate used when drawing the shape. Used for the following shapes: Line, Points, Polyline, Polygon, Rect, RoundedRect, Ellipse, Arc, Chord, Pie, Path, Text, Pixmap

9.266.5.29 point10

```
QPoint QEShape::point10 [read], [write]
```

10th coordinate used when drawing the shape. Used for the following shapes: Points, Polyline, Polygon, Path

9.266.5.30 point2

```
QPoint QEShape::point2 [read], [write]
```

2nd coordinate used when drawing the shape. Used for the following shapes: Line, Points, Polyline, Polygon, Rect, RoundedRect, Ellipse, Arc, Chord, Pie, Path, QPixmap

9.266.5.31 point3

```
QPoint QEShape::point3 [read], [write]
```

3rd coordinate used when drawing the shape. Used for the following shapes: Points, Polyline, Polygon, Path

9.266.5.32 point4

```
QPoint QEShape::point4 [read], [write]
```

4th coordinate used when drawing the shape. Used for the following shapes: Points, Polyline, Polygon, Path

9.266.5.33 point5

```
QPoint QEShape::point5 [read], [write]
```

5th coordinate used when drawing the shape. Used for the following shapes: Points, Polyline, Polygon, Path

9.266.5.34 point6

```
QPoint QEShape::point6 [read], [write]
```

6th coordinate used when drawing the shape. Used for the following shapes: Points, Polyline, Polygon, Path

9.266.5.35 point7

```
QPoint QEShape::point7 [read], [write]
```

7th coordinate used when drawing the shape. Used for the following shapes: Points, Polyline, Polygon, Path

9.266.5.36 point8

```
QPoint QEShape::point8 [read], [write]
```

8th coordinate used when drawing the shape. Used for the following shapes: Points, Polyline, Polygon, Path

9.266.5.37 point9

```
QPoint QEShape::point9 [read], [write]
```

9th coordinate used when drawing the shape. Used for the following shapes: Points, Polyline, Polygon, Path

9.266.5.38 scale2

```
double QEShape::scale2 [read], [write]
```

Scale factor applied to data from the 2nd variable before it is used to animate the shape

9.266.5.39 scale3

```
double QEShape::scale3 [read], [write]
```

Scale factor applied to data from the 3rd variable before it is used to animate the shape

9.266.5.40 scale4

```
double QEShape::scale4 [read], [write]
```

Scale factor applied to data from the 4th variable before it is used to animate the shape

9.266.5.41 scale5

```
double QEShape::scale5 [read], [write]
```

Scale factor applied to data from the 5th variable before it is used to animate the shape

9.266.5.42 scale6

```
double QEShape::scale6 [read], [write]
```

Scale factor applied to data from the 6th variable before it is used to animate the shape

9.266.5.43 styleSheet

```
QString QEShape::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.266.5.44 `userLevelEnabled`

```
QE::UserLevels QEShape::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.266.5.45 `userLevelEngineerStyle`

```
QString QEShape::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.266.5.46 `userLevelScientistStyle`

```
QString QEShape::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.266.5.47 `userLevelUserStyle`

```
QString QEShape::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.266.5.48 `userLevelVisibility`

```
QE::UserLevels QEShape::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.266.5.49 variable1

```
QString QEShape::variable1 [read], [write]
```

EPICS variable name (CA PV). This variable is read and used to animate an attribute of the shape. The value read is first scaled and offset by properties scale1 and offset1 then the attribute selected for animation is selected by the property animation1.

9.266.5.50 variable2

```
QString QEShape::variable2 [read], [write]
```

EPICS variable name (CA PV). This variable is read and used to animate an attribute of the shape. The value read is first scaled and offset by properties scale2 and offset2 then the attribute selected for animation is selected by the property animation2.

9.266.5.51 variable3

```
QString QEShape::variable3 [read], [write]
```

EPICS variable name (CA PV). This variable is read and used to animate an attribute of the shape. The value read is first scaled and offset by properties scale3 and offset3 then the attribute selected for animation is selected by the property animation3.

9.266.5.52 variable4

```
QString QEShape::variable4 [read], [write]
```

EPICS variable name (CA PV). This variable is read and used to animate an attribute of the shape. The value read is first scaled and offset by properties scale4 and offset4 then the attribute selected for animation is selected by the property animation4.

9.266.5.53 variable5

```
QString QEShape::variable5 [read], [write]
```

EPICS variable name (CA PV). This variable is read and used to animate an attribute of the shape. The value read is first scaled and offset by properties scale5 and offset5 then the attribute selected for animation is selected by the property animation5.

9.266.5.54 variable6

```
QString QEShape::variable6 [read], [write]
```

EPICS variable name (CA PV). This variable is read and used to animate an attribute of the shape. The value read is first scaled and offset by properties scale6 and offset6 then the attribute selected for animation is selected by the property animation6.

9.266.5.55 variableAsToolTip

```
bool QEShape::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.266.5.56 variableSubstitutions

```
QString QEShape::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'SAMPLE=SAM1, NAME = "Ref foil"' These substitutions are applied to all the variable names.

9.266.5.57 visible

```
bool QEShape::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

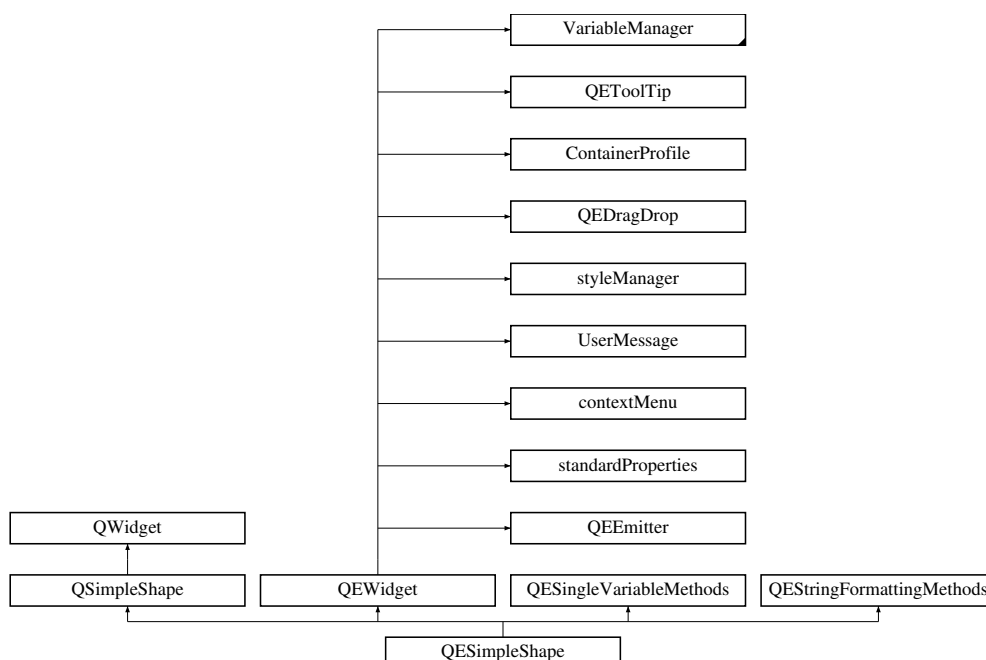
The documentation for this class was generated from the following files:

- qeframework/widgets/QEShape/QEShape.h
- qeframework/widgets/QEShape/QEShape.cpp

9.267 QESimpleShape Class Reference

```
#include <QESimpleShape.h>
```

Inheritance diagram for QESimpleShape:



Public Slots

- void [setManagedVisible](#) (bool v)

Signals

- void [dbValueChanged](#) ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void [dbConnectionChanged](#) (const bool &isConnected)

Public Member Functions

- **QESimpleShape** (QWidget *parent=0)
- **QESimpleShape** (const QString &variableName, QWidget *parent=0)
- void **setVariableNameSubstitutionsProperty** (const QString &substitutions)
- void **setEdgeVariableNameProperty** (const QString &variableName)
- QString **getEdgeVariableNameProperty** () const
- void **setEdgeElementsRequired** (const int [elementsRequired](#))
- int **getEdgeElementsRequired** () const
- void **setEdgeArrayIndex** (const int [arrayIndex](#))
- int **getEdgeArrayIndex** () const
- [QE::DisplayAlarmStateOptions](#) **getEdgeAlarmStateOptionProperty** () const
Access function for #edgeAlarmStateOption property - refer to #edgeAlarmStateOption property for details.
- void **setEdgeAlarmStateOptionProperty** ([QE::DisplayAlarmStateOptions](#) option)
Access function for #edgeAlarmStateOption property - refer to #edgeAlarmStateOption property for details.
- void **setEdgeAlarmStateOption** ([QE::DisplayAlarmStateOptions](#) option)
- [QE::DisplayAlarmStateOptions](#) **getEdgeAlarmStateOption** () const
- void **setUseStyleAlarmColours** (const bool [useStyleAlarmColours](#))
- bool **getUseStyleAlarmColours** () const
- void **setEdgeUsesStyleAlarmColours** (const bool [useStyleAlarmColours](#))
- bool **getEdgeUsesStyleAlarmColours** () const

Protected Member Functions

- void [activated](#) ()
Do any post-all-widgets-constructed, i.e. activated stuff.
- [QEChannel](#) * [createQcaltem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant v)
- QString **getItemText** ()
- QColor [getItemColour](#) ()
returns "" unless overridden.
- void **stringFormattingChange** ()

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- QString [edgeVariable](#)
- int [edgeElementsRequired](#)
- int [edgeArrayIndex](#)
- [QE::DisplayAlarmStateOptions](#) [edgeAlarmStateOption](#)
- bool [addUnits](#)
- QString [localEnumeration](#)
- int [value](#)
- bool [isActive](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- bool [oosAware](#)
- bool [useStyleAlarmColours](#)
- bool [edgeUseStyleAlarmColours](#)

Additional Inherited Members

9.267.1 Detailed Description

This class is an EPICS aware simple shape.

The shape is selected at design time using the shape property - refer to #Shapes for details, or may be set by the `setShape` function.

The colour is based on either the value of or the alarm state of a single PV.

When the (standard) `displayAlarmState` property is false, the PV value (modulo 16) is used to select one of the colours defined by the `colour0` to `colour15` properties.

When the `displayAlarmState` property is true, the colour is based on the alarm state, i.e. green when no alarm, yellow for minor alarm etc.

This class extends [QSimpleShape](#) in order to make it EPICS aware.

9.267.2 Member Function Documentation

9.267.2.1 dbConnectionChanged

```
void QESimpleShape::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.267.2.2 dbValueChanged

```
void QESimpleShape::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.267.2.3 setManagedVisible

```
void QESimpleShape::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.267.3 Property Documentation

9.267.3.1 addUnits

```
bool QESimpleShape::addUnits [read], [write]
```

If true (default), add engineering units supplied with the data.

9.267.3.2 allowDrop

```
bool QESimpleShape::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.267.3.3 arrayIndex

```
int QESimpleShape::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.267.3.4 defaultStyle

```
QString QESimpleShape::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.267.3.5 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QESimpleShape::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.267.3.6 edgeUseStyleAlarmColours

```
bool QESimpleShape::edgeUseStyleAlarmColours [read], [write]
```

Use "style"/label-like alarm colours. The default is false. Applies to the edge alarm colour only.

9.267.3.7 elementsRequired

```
int QESimpleShape::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.267.3.8 localEnumeration

```
QString QESimpleShape::localEnumeration [read], [write]
```

An enumeration list used to data values. Used only when the formatting option is 'local enumeration'. Value is converted to an integer and used to select a string from this list.

Format is:

```
[[<|<=|=|!=|>|=>]value1|*]: string1 , [[<|<=|=|!=|>|=>]value2|*]: string2 , [[<|<=|=|!=|>|=>]value3|*]: string3 , ...
```

Where: < Less than <= Less than or equal = Equal (default if no operator specified) >= Greather than or equal

Greater than

* Always match (used to specify default text)

Values may be numeric or textual Values do not have to be in any order, but first match wins Values may be quoted Strings may be quoted Consecutive values do not have to be present. Operator is assumed to be equality if not present. White space is ignored except within quoted strings. may be included in a string to indicate a line break

Examples are:

```
0:Off,1:On 0 : "Pump Running", 1 : "Pump not running" 0:" ", 1:"Warning!\nAlarm" <2:"Value is less than two", =2↵
:"Value is equal to two", >2:"Value is grater than 2" 3:"Beamline Available", *:"" "Pump Off": "OH NO!, the pump is
OFF!", "Pump On": "It's OK, the pump is on"
```

The data value is converted to a string if no enumeration for that value is available. For example, if the local enumeration is '0:off,1:on', and a value of 10 is processed, the text generated is '10'. If a blank string is required, this should be explicit. for example, '0:off,1:on,10: ""'

A range of numbers can be covered by a pair of values as in the following example: >=4:"Between 4 and 8", <=8↵
:"Between 4 and 8"

9.267.3.9 messageSourceId

```
unsigned int QESimpleShape::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.267.3.10 oosAware

```
bool QESimpleShape::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.267.3.11 styleSheet

```
QString QESimpleShape::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly setable per se. This also stops transient styles being saved to the ui file.

9.267.3.12 userLevelEnabled

```
QE::UserLevels QESimpleShape::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through setUserLevel() Widgets that are always accessable should be visible at 'User'. Widgets that are only accessable to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessable to engineers maintaining the facility should be visible at 'Engineer'.

9.267.3.13 userLevelEngineerStyle

```
QString QESimpleShape::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.267.3.14 userLevelScientistStyle

```
QString QESimpleShape::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.267.3.15 userLevelUserStyle

```
QString QESimpleShape::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.267.3.16 userLevelVisibility

```
QE::UserLevels QESimpleShape::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through setUserLevel() Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.267.3.17 useStyleAlarmColours

```
bool QESimpleShape::useStyleAlarmColours [read], [write]
```

Use "style"/label-like alarm colours. The default is false. Applies to the main alarm colour only.

9.267.3.18 variable

```
QString QESimpleShape::variable [read], [write]
```

EPICS variable name (CA PV)

9.267.3.19 variableAsToolTip

```
bool QESimpleShape::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.267.3.20 variableSubstitutions

```
QString QESimpleShape::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.267.3.21 visible

```
bool QESimpleShape::visible [read], [write]
```

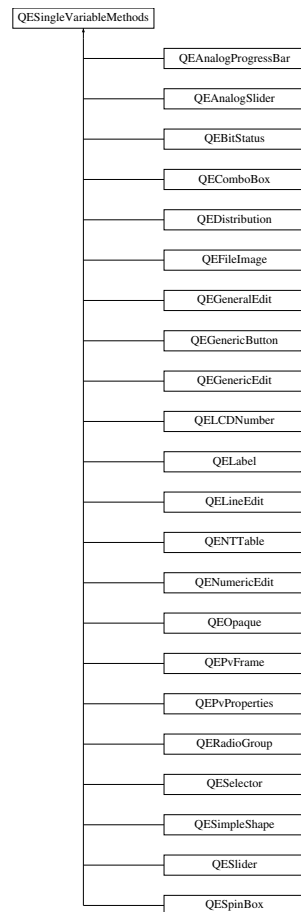
Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

The documentation for this class was generated from the following files:

- qeframework/widgets/QESimpleShape/QESimpleShape.h
- qeframework/widgets/QESimpleShape/QESimpleShape.cpp

9.268 QESingleVariableMethods Class Reference

Inheritance diagram for QESingleVariableMethods:



Public Member Functions

- **QESingleVariableMethods** ([QEWidget](#) *owner, unsigned int variableIndex=0)
- void [setVariableNameProperty](#) (const QString &variableName)
- QString [getVariableNameProperty](#) () const
- void [setVariableNameSubstitutionsProperty](#) (const QString &substitutions)
- QString [getVariableNameSubstitutionsProperty](#) () const
- void [setElementsRequired](#) (const int elementsRequired)
- int [getElementsRequired](#) () const
- void [setArrayIndex](#) (const int arrayIndex)
- int [getArrayIndex](#) () const
- void [connectNewVariableNameProperty](#) (const char *useNameSlot)
- void [connectPvNameProperties](#) (const char *useNameSlot)
- void [setSingleVariableQCaProperties](#) ([QEChannel](#) *qca)

9.268.1 Member Function Documentation

9.268.1.1 connectNewVariableNameProperty()

```
void QESingleVariableMethods::connectNewVariableNameProperty (
    const char * useNameSlot )
```

Connects internal variable name property manager's newVariableNameProperty signal to the specified slot.

Example: this->connectNewVariableNameProperty (SLOT (useNewVariableNameProperty (QString, QString, unsigned int)));

9.268.1.2 connectPvNameProperties()

```
void QESingleVariableMethods::connectPvNameProperties (
    const char * useNameSlot )
```

As above, save using new [QEPvNameProperties](#) structure.

Example: this->connectPvNameProperties (SLOT setPvNameProperties (const [QEPvNameProperties](#)&));

9.268.1.3 getArrayIndex()

```
int QESingleVariableMethods::getArrayIndex ( ) const
```

Property access function for #arrayIndex property. Array element to access if variable is an array variable. Defaults to 0, i.e. first element.

9.268.1.4 getElementsRequired()

```
int QESingleVariableMethods::getElementsRequired ( ) const
```

Property access function to get #elementsRequired property. Limits the the numer of elements requested to the given value.

9.268.1.5 getVariableNameProperty()

```
QString QESingleVariableMethods::getVariableNameProperty ( ) const
```

Property access function for #variable property. This has special behaviour to work well within designer.

9.268.1.6 getVariableNameSubstitutionsProperty()

```
QString QESingleVariableMethods::getVariableNameSubstitutionsProperty ( ) const
```

Property access function for #variableSubstitutions property. This has special behaviour to work well within designer.

9.268.1.7 setArrayIndex()

```
void QESingleVariableMethods::setArrayIndex (
    const int arrayIndex )
```

Property access function for #arrayIndex property. Array element to access if variable is an array variable. Defaults to 0, i.e. first element. arrayIndex value is restricted to be ≥ 0

If the associated QEChannel exists then calls its setArrayIndex function and then requests that the object resend last data. The function adjusts the #elementsRequired property value if necessary.

9.268.1.8 setElementsRequired()

```
void QESingleVariableMethods::setElementsRequired (
    const int elementsRequired )
```

Property access function to set #elementsRequired property. Limits the the numer of elements requested to the given value. Defaults to 0, which means no limit is applied to the subscription. The function adjusts the #arrayIndex property value if necessary.

9.268.1.9 setVariableNameProperty()

```
void QESingleVariableMethods::setVariableNameProperty (
    const QString & variableName )
```

Property access function for #variable property. This has special behaviour to work well within designer.

9.268.1.10 setVariableNameSubstitutionsProperty()

```
void QESingleVariableMethods::setVariableNameSubstitutionsProperty (
    const QString & substitutions )
```

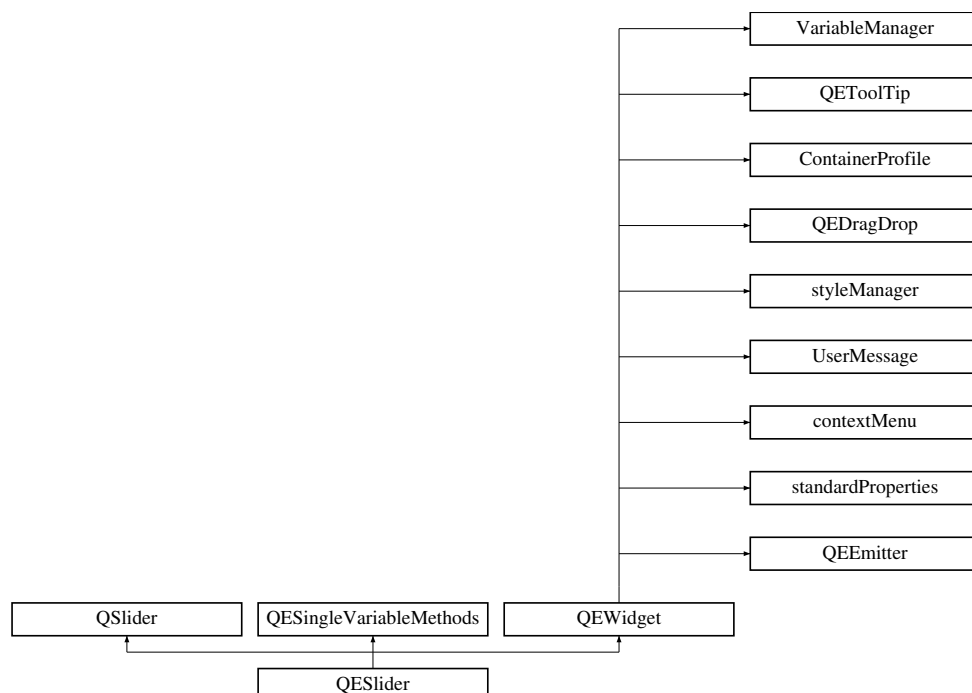
Property access function for #variableSubstitutions property. This has special behaviour to work well within designer.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/QESingleVariableMethods.h
- qeframework/widgets/QEWidget/QESingleVariableMethods.cpp

9.269 QESlider Class Reference

Inheritance diagram for QESlider:



Public Slots

- void **writeNow** ()
- void **setPvValue** (const QString &text)
- void **setPvValue** (const int value)
- void **setPvValue** (const double value)
- void **setPvValue** (const bool value)
- void **setDefaultStyle** (const QString &style)
Update the default style applied to this widget.
- void **setManagedVisible** (bool v)

Signals

- void **dbValueChanged** ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void **dbConnectionChanged** (const bool &isConnected)

Public Member Functions

- **QESlider** (QWidget *parent=0)
- **QESlider** (const QString &variableName, QWidget *parent=0)
- void **setWriteOnChange** (bool **writeOnChange**)
- bool **getWriteOnChange** () const
- void **setSubscribe** (bool **subscribe**)
- bool **getSubscribe** () const
- void **setScale** (double scaleIn)
- double **getScale** () const
- void **setOffset** (double offsetIn)
- double **getOffset** () const
- void **setAllowFocusUpdate** (bool **allowFocusUpdate**)
- bool **getAllowFocusUpdate** () const
- void **setAutoScale** (const bool autoScale)
- bool **getAutoScale** () const

Protected Member Functions

- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **setDrop** (QVariant drop)
- QVariant **getDrop** ()
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)
- **QEChannel** * **createQcaltem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- bool [subscribe](#)
- bool [writeOnChange](#)
- bool [allowFocusUpdate](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- bool [oosAware](#)
- double **scale**
- double **offset**
- bool **autoScale**
- double **value**
- int **sliderPosition**

Additional Inherited Members

9.269.1 Member Function Documentation

9.269.1.1 dbConnectionChanged

```
void QESlider::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to provary variable.

9.269.1.2 dbValueChanged

```
void QESlider::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.269.1.3 setManagedVisible

```
void QESlider::setManagedVisible (
    bool v ) [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.269.1.4 setPvValue

```
void QESlider::setPvValue (
    const QString & text ) [slot]
```

Update the widget and write to the main associated control process variable. Supplied values are converted to the appropriate widget value type. If conversion not possible or resultant value out of range, no widget/PV update occurs.

9.269.2 Property Documentation

9.269.2.1 allowDrop

```
bool QESlider::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.269.2.2 allowFocusUpdate

```
bool QESlider::allowFocusUpdate [read], [write]
```

Allow updated while widget has focus - defaults to false

9.269.2.3 arrayIndex

```
int QESlider::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.269.2.4 defaultStyle

```
QString QESlider::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.269.2.5 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QESlider::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.269.2.6 elementsRequired

```
int QESlider::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.269.2.7 messageSourceId

```
unsigned int QESlider::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.269.2.8 oosAware

```
bool QESlider::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.269.2.9 styleSheet

```
QString QESlider::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly setable per se. This also stops transient styles being saved to the ui file.

9.269.2.10 subscribe

```
bool QESlider::subscribe [read], [write]
```

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.269.2.11 `userLevelEnabled`

```
QE::UserLevels QESlider::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()`. Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.269.2.12 `userLevelEngineerStyle`

```
QString QESlider::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.269.2.13 `userLevelScientistStyle`

```
QString QESlider::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.269.2.14 `userLevelUserStyle`

```
QString QESlider::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.269.2.15 `userLevelVisibility`

```
QE::UserLevels QESlider::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()`. Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.269.2.16 variable

```
QString QESlider::variable [read], [write]
```

EPICS variable name (CA PV)

9.269.2.17 variableAsToolTip

```
bool QESlider::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.269.2.18 variableSubstitutions

```
QString QESlider::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.269.2.19 visible

```
bool QESlider::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

9.269.2.20 writeOnChange

```
bool QESlider::writeOnChange [read], [write]
```

Sets if this widget writes any changes as the user moves the slider (the QSlider 'valueChanged' signal is emitted). Default is 'true' (writes any changes when the QSlider 'valueChanged' signal is emitted).

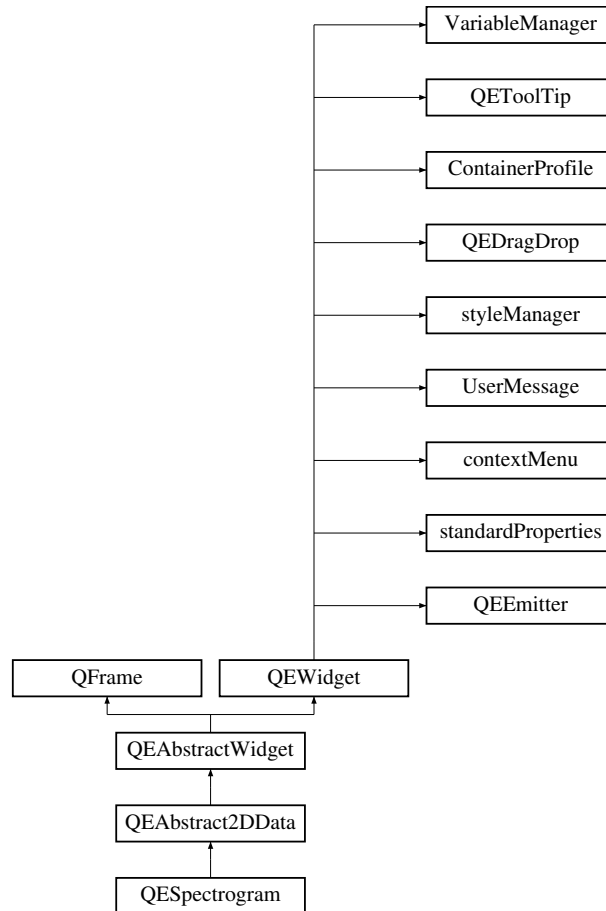
The documentation for this class was generated from the following files:

- qeframework/widgets/QESlider/QESlider.h
- qeframework/widgets/QESlider/QESlider.cpp

9.270 QESpectrogram Class Reference

```
#include <QESpectrogram.h>
```

Inheritance diagram for QESpectrogram:



Public Types

- enum **OwnContextMenuOptions** { **QESPEC_NONE** = A2DDCM_SUB_CLASS_WIDGETS_START_HERE , **QESPEC_USE_FALSE_COLOUR_FLIP** , **QESPEC_USE_LOG_SCALE_FLIP** , **QESPEC_SUB_CLASS_WIDGETS_START_HERE** }
- typedef void(* **CustomisePaintHandlers**) ([QESpectrogram](#) *spectrogram, QPainter &painter, const QRect rect, QObject *context)

Public Slots

- void **setUseFalseColour** (const bool [useFalseColour](#))
- void **setScaleWrap** (const int [scaleWrap](#))
- void **setMargin** (const int [margin](#))

Public Member Functions

- [QESpectrogram](#) (QWidget *parent=0)
duplicate
- [QESpectrogram](#) (const QString &dataVariableName, QWidget *parent=0)
- **QESpectrogram** (const QString &dataVariableName, const QString &widthVariableName, QWidget *parent=0)
- virtual [~QESpectrogram](#) ()
Destruction.
- void **setCustomisePaintHandler** (CustomisePaintHandlers paintExtraHandler, QObject *context=0)
- CustomisePaintHandlers **getCustomisePaintHandler** () const
- bool **getUseFalseColour** () const
- int **getScaleWrap** () const
- int **getMargin** () const

Protected Member Functions

- QSize **sizeHint** () const
- bool **eventFilter** (QObject *watched, QEvent *event)
- void **updateDataVisulation** ()
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)

Properties

- bool [useFalseColour](#)
- int [scaleWrap](#)
- int [margin](#)

Additional Inherited Members

9.270.1 Detailed Description

This class presents 2D data as a spectrogram. The value of each element is mapped to the range 0 to 255, and presented in gray scale (black to white) or with false colour (similar to [QEImage](#)).

9.270.2 Constructor & Destructor Documentation

9.270.2.1 QESpectrogram() [1/2]

```
QESpectrogram::QESpectrogram (
    QWidget * parent = 0 ) [explicit]
```

duplicate

Create without a variable. Use setDataPvName () and setVariableNameSubstitutions() to define a variable and, optionally, macro substitutions later.

9.270.2.2 QESpectrogram() [2/2]

```
QESpectrogram::QESpectrogram (
    const QString & dataVariableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable (s). A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.270.3 Property Documentation

9.270.3.1 margin

```
int QESpectrogram::margin [read], [write]
```

Margin 0 to 40 - default is 4 Margin area shows the alarm colour, provided not inhibited by the value of the display↔ AlarmStateOption property.

9.270.3.2 scaleWrap

```
int QESpectrogram::scaleWrap [read], [write]
```

Specifies the number of times around the gray/colour scale we map values in the range minimum to maximum. Allowed range: 1 .. 10 - note 10 is more than enough. Default: 1

9.270.3.3 useFalseColour

```
bool QESpectrogram::useFalseColour [read], [write]
```

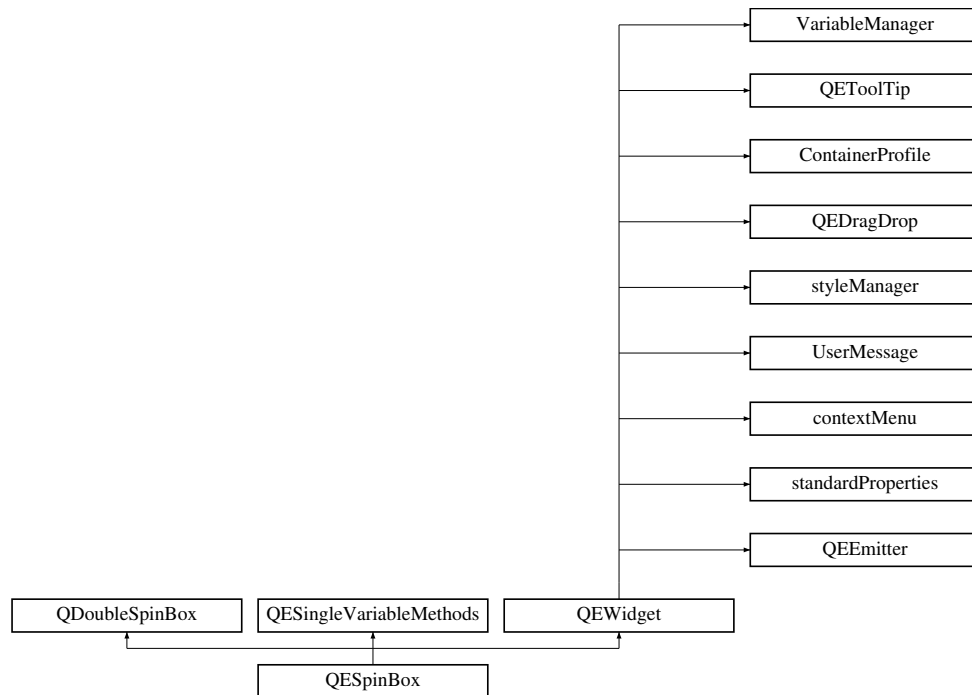
If true, the apply false colour to the data, othewrwise gray scale

The documentation for this class was generated from the following files:

- qeframework/widgets/QE2DDataVisualisation/QESpectrogram.h
- qeframework/widgets/QE2DDataVisualisation/QESpectrogram.cpp

9.271 QESpinBox Class Reference

Inheritance diagram for QESpinBox:



Public Slots

- void **writeNow** ()
- void **setPvValue** (const QString &text)
- void **setPvValue** (const int value)
- void **setPvValue** (const double value)
- void **setPvValue** (const bool value)
- void **setDefaultStyle** (const QString &style)
Update the default style applied to this widget.
- void **setManagedVisible** (bool v)

Signals

- void **dbValueChanged** ()
- void **dbValueChanged** (const QString &out)
- void **dbValueChanged** (const int &out)
- void **dbValueChanged** (const long &out)
- void **dbValueChanged** (const qlonglong &out)
- void **dbValueChanged** (const double &out)
- void **dbValueChanged** (const bool &out)
- void **dbConnectionChanged** (const bool &isConnected)
- void **userChange** (const QString &oldValue, const QString &newValue, const QString &lastValue)
Internal use only. Used by [QEConfiguredLayout](#) to be notified when one of its widgets has written something.

Public Member Functions

- **QESpinBox** (QWidget *parent=0)
- **QESpinBox** (const QString &variableName, QWidget *parent=0)
- void **setWriteOnChange** (bool writeOnChangeIn)
- bool **getWriteOnChange** () const
- void **setSubscribe** (bool [subscribe](#))
- bool **getSubscribe** () const
- void **setAddUnitsAsSuffix** (bool addUnitsAsSuffixIn)
- bool **getAddUnitsAsSuffix** () const
- void **setAutoScale** (const bool autoScaleIn)
- bool **getAutoScale** () const
- void **setUseDbPrecisionForDecimals** (bool useDbPrecisionForDecimalsIn)
- bool **getUseDbPrecisionForDecimals** () const
- void **setAllowFocusUpdate** (bool [allowFocusUpdate](#))
- bool **getAllowFocusUpdate** () const
- void **setAutoStepSize** (bool autoStepSize)
- bool **getAutoStepSize** () const

Protected Member Functions

- void **stepBy** (int steps) override
- bool **eventFilter** (QObject *watch, QEvent *event)
- [QEChannel](#) * **createQcItem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **setDrop** (QVariant drop)
- QVariant **getDrop** ()
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant s)
- QMenu * **getDefaultContextMenu** ()
Flag indicating this widget is running inside Qt's 'designer'.

Properties

- QString [variable](#)
- QString [variableSubstitutions](#)
- int [elementsRequired](#)
- int [arrayIndex](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)

- [QE::UserLevels userLevelVisibility](#)
- [QE::UserLevels userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions displayAlarmStateOption](#)
- bool [oosAware](#)
- bool [subscribe](#)
- bool [allowFocusUpdate](#)
- bool [autoScale](#)
- bool [useDbPrecision](#)
- bool [addUnits](#)
- bool [autoStepSize](#)
- double [value](#)

Additional Inherited Members

9.271.1 Member Function Documentation

9.271.1.1 dbConnectionChanged

```
void QESpinBox::dbConnectionChanged (
    const bool & isConnected ) [signal]
```

Sent when the widget state updated following a channel connection change Applied to primary variable.

9.271.1.2 dbValueChanged

```
void QESpinBox::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget.

9.271.1.3 setManagedVisible

```
void QESpinBox::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.271.1.4 setPvValue

```
void QESpinBox::setPvValue (
    const QString & text ) [slot]
```

Update the widget and write to the main associated control process variable. Supplied values are converted to the appropriate widget value type. If conversion not possible or resultant value out of range, no widget/PV update occurs.

9.271.2 Property Documentation

9.271.2.1 allowDrop

```
bool QESpinBox::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.271.2.2 allowFocusUpdate

```
bool QESpinBox::allowFocusUpdate [read], [write]
```

Allow updated while widget has focus - defaults to false

9.271.2.3 arrayIndex

```
int QESpinBox::arrayIndex [read], [write]
```

Index used to select a single item of data for processing. The default is 0.

9.271.2.4 defaultStyle

```
QString QESpinBox::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.271.2.5 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QESpinBox::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.271.2.6 elementsRequired

```
int QESpinBox::elementsRequired [read], [write]
```

The number of elements required to be subscribed for from the PV host (IOC). The default is 0 which means subscribed for all elements. Note: changing this value causes the unsubscribe/re-subscribe just as if the variable name changed.

9.271.2.7 messageSourceId

```
unsigned int QESpinBox::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.271.2.8 oosAware

```
bool QESpinBox::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.271.2.9 styleSheet

```
QString QESpinBox::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly setable per se. This also stops transient styles being saved to the ui file.

9.271.2.10 subscribe

```
bool QESpinBox::subscribe [read], [write]
```

Sets if this widget subscribes for data updates and displays current data. Default is 'true' (subscribes for and displays data updates)

9.271.2.11 userLevelEnabled

```
QE::UserLevels QESpinBox::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through setUserLevel() Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.271.2.12 userLevelEngineerStyle

```
QString QESpinBox::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.271.2.13 `userLevelScientistStyle`

```
QString QESpinBox::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.271.2.14 `userLevelUserStyle`

```
QString QESpinBox::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.271.2.15 `userLevelVisibility`

```
QE::UserLevels QESpinBox::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.271.2.16 `variable`

```
QString QESpinBox::variable [read], [write]
```

EPICS variable name (CA PV)

9.271.2.17 `variableAsToolTip`

```
bool QESpinBox::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.271.2.18 `variableSubstitutions`

```
QString QESpinBox::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is `NAME1=VALUE1[, NAME2=VALUE2...` Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all [QE](#) widgets. In some widgets are also used for other purposes.

9.271.2.19 visible

```
bool QESpinBox::visible [read], [write]
```

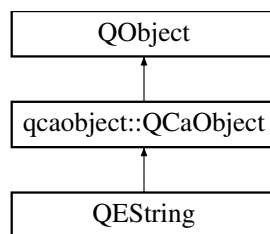
Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

The documentation for this class was generated from the following files:

- qeframework/widgets/QESpinBox/QESpinBox.h
- qeframework/widgets/QESpinBox/QESpinBox.cpp

9.272 QEStrng Class Reference

Inheritance diagram for QEStrng:



Public Slots

- void **writeString** (const QString &data)
- void **writeStringElement** (const QString &data)
- void **writeString** (const QVector< QString > &data)

Signals

- void **valueUpdated** (const [QEStrngValueUpdate](#) &value)
- void **arrayUpdated** (const [QEStrngArrayUpdate](#) &array)
- void **stringChanged** (const QString &value, [QCaAlarmInfo](#) &alarmInfo, [QCaDateTime](#) &timeStamp, const unsigned int &variableIndex)
- void **stringArrayChanged** (const QVector< QString > &values, [QCaAlarmInfo](#) &alarmInfo, [QCaDateTime](#) &timeStamp, const unsigned int &variableIndex)

Public Member Functions

- **QEStrng** (const QString &pvName, QObject *eventObject, [QEStrngFormatting](#) *stringFormattingIn, unsigned int variableIndexIn)
- **QEStrng** (const QString &pvName, QObject *eventObject, [QEStrngFormatting](#) *stringFormattingIn, unsigned int variableIndexIn, [UserMessage](#) *userMessageIn)
- bool **writeString** (const QString &data, QString &message)
- bool **writeStringElement** (const QString &data, QString &message)
- bool **writeString** (const QVector< QString > &data, QString &message)

Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/data/QCString.h
- qeframework/data/QCString.cpp

9.273 QCStringArrayUpdate Struct Reference

Public Attributes

- QVector< QCString > **values**
- [QCaAlarmInfo](#) **alarmInfo**
- [QCaDateTime](#) **timeStamp**
- unsigned int **variableIndex**
- bool **isMetaUpdate**

The documentation for this struct was generated from the following file:

- qeframework/data/QCString.h

9.274 QCStringFormatting Class Reference

Public Member Functions

- QCString **formatString** (const QVariant &value, int arrayIndex) const
- QVector< QCString > [formatStringArray](#) (const QVariant &value) const
- QVariant **formatValue** (const QCString &text, bool &ok) const
- QVariant **formatValue** (const QVector< QCString > &text, bool &ok) const
- void **setDbEgu** (const QCString egu)
- void **setDbEnumerations** (const QStringList enumerations)
- void **setDbPrecision** (const unsigned int dbPrecisionIn)
- void **setPrecision** (const int precision)
- void **setUseDbPrecision** (const bool useDbPrecision)
- void **setLeadingZero** (const bool leadingZero)
- void **setTrailingZeros** (const bool trailingZeros)
- void **setForceSign** (const bool forceSign)
- void **setFormat** (const [QE::Formats](#) format)
- void **setSeparator** (const [QE::Separators](#) separator)
- void **setRadix** (const int radix)
- void **setNotation** (const [QE::Notations](#) notation)
- void **setArrayAction** (const [QE::ArrayActions](#) arrayActionIn)
- void **setAddUnits** (const bool addUnits)
- void **setLocalEnumeration** (const QCString localEnumerationIn)
- void **setUseRadixPrefix** (const bool useRadixPrefix)
- void **setLeadingZeros** (const int leadingZeros)
- int **getPrecision** () const
- bool **getUseDbPrecision** () const

- bool **getLeadingZero** () const
- bool **getTrailingZeros** () const
- bool **getForceSign** () const
- [QE::Formats](#) **getFormat** () const
- [QE::Separators](#) **getSeparator** () const
- unsigned int **getRadix** () const
- [QE::Notations](#) **getNotation** () const
- [QE::ArrayActions](#) **getArrayAction** () const
- bool **getAddUnits** () const
- QString **getLocalEnumeration** () const
- [QELocalEnumeration](#) **getLocalEnumerationObject** () const
- bool **getUseRadixPrefix** () const
- int **getLeadingZeros** () const
- QString **toString** (const long value) const
- QString **toString** (const unsigned long value) const
- QString **toString** (const double value) const
- long **toInt** (const QString &image, bool &okay) const
- long **toLong** (const QString &image, bool &okay) const
- unsigned long **toULong** (const QString &image, bool &okay) const
- double **toDouble** (const QString &image, bool &okay) const

9.274.1 Member Function Documentation

9.274.1.1 formatStringArray()

```
QVector< QString > QEStringFormatting::formatStringArray (
    const QVariant & value ) const
```

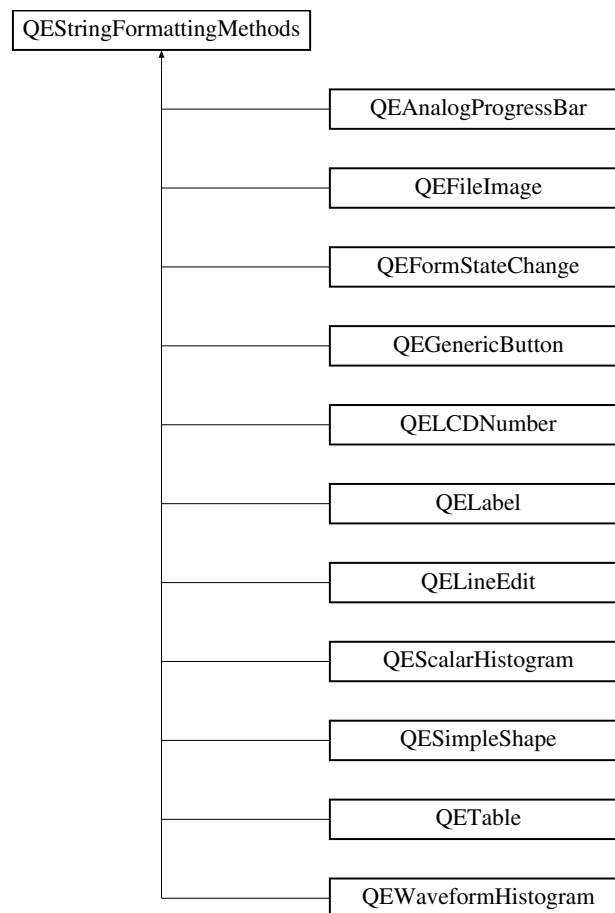
This is used to convert the QVariant value received from a QEChannel, which is still based on the data variable type, to a string array. Typically used where the input QVariant value is an array of data values, but will work for any QVariant type.

The documentation for this class was generated from the following files:

- qeframework/data/QCStringFormatting.h
- qeframework/data/QCStringFormatting.cpp

9.275 QStringFormattingMethods Class Reference

Inheritance diagram for QStringFormattingMethods:



Public Member Functions

- virtual void **stringFormattingChange** ()=0
- void **setPrecision** (int precision)
- int **getPrecision** () const
- void **setUseDbPrecision** (bool useDbPrecision)
- bool **getUseDbPrecision** () const
- void **setLeadingZeros** (const int leadingZeros)
- int **getLeadingZeros** () const
- Q_DECL_DEPRECATED void **setLeadingZero** (bool leadingZero)
- Q_DECL_DEPRECATED bool **getLeadingZero** () const
- Q_DECL_DEPRECATED void **setTrailingZeros** (bool trailingZeros)
- Q_DECL_DEPRECATED bool **getTrailingZeros** () const
- void **setForceSign** (bool forceSign)
- bool **getForceSign** () const
- void **setAddUnits** (bool addUnits)
- bool **getAddUnits** () const
- void **setLocalEnumeration** (QString localEnumeration)
- QString **getLocalEnumeration** () const
- void **setFormat** (QE::Formats format)

- [QE::Formats](#) **getFormat** () const
- void **setSeparator** (const [QE::Separators](#) separator)
- [QE::Separators](#) **getSeparator** () const
- void **setRadix** (const int radix)
- unsigned int **getRadix** () const
- void **setNotation** ([QE::Notations](#) notation)
- [QE::Notations](#) **getNotation** () const
- void **setArrayAction** ([QE::ArrayActions](#) arrayAction)
- [QE::ArrayActions](#) **getArrayAction** () const
- [QELocalEnumeration](#) **getLocalEnumerationObject** () const

Protected Attributes

- [QEStringFormatting](#) **stringFormatting**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/QEStringFormattingMethods.h
- qeframework/widgets/QEWidget/QEStringFormattingMethods.cpp

9.276 QEStringValueUpdate Struct Reference

Public Attributes

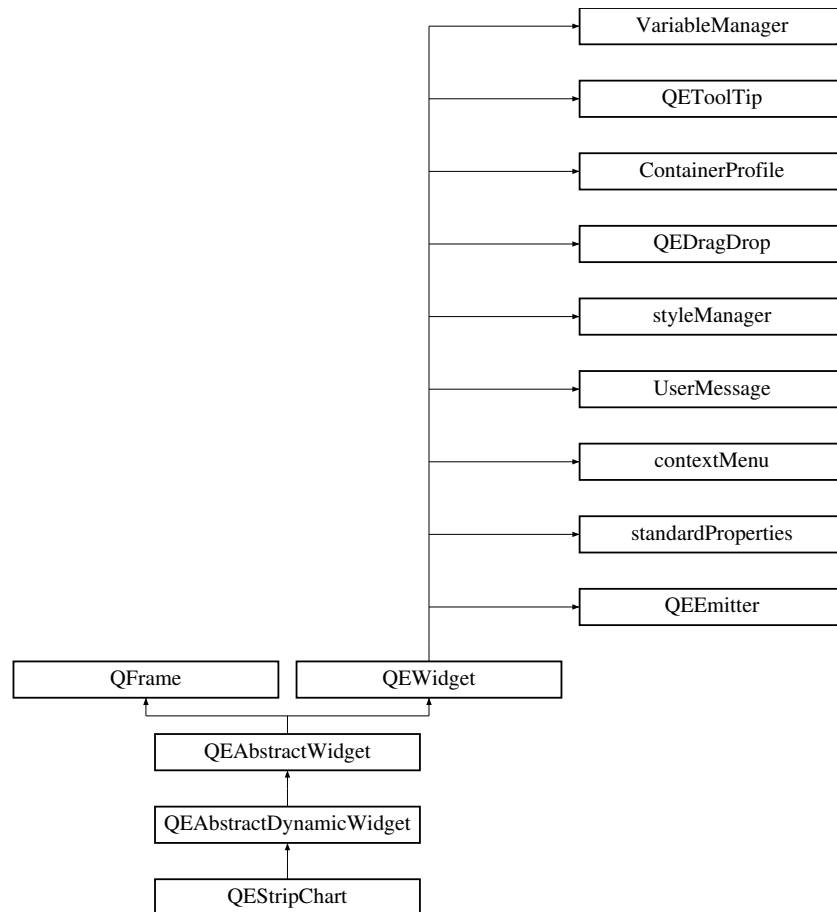
- QString **value**
- [QCaAlarmInfo](#) **alarmInfo**
- [QCaDateTime](#) **timeStamp**
- unsigned int **variableIndex**
- bool **isMetaUpdate**

The documentation for this struct was generated from the following file:

- qeframework/data/QEString.h

9.277 QEStripChart Class Reference

Inheritance diagram for QEStripChart:



Public Types

- enum **PropertyChartYRanges** { **manual** = QEStripChartNames::manual , **dynamic** = QEStripChartNames::dynamic }
- enum **Constants** { **NUMBER_OF_PVS** = 16 }

Public Slots

- void **videoModeSelected** (const [QE::VideoModes](#) mode)
- void **yRangeSelected** (const QEStripChartNames::ChartYRanges scale)
- void **yScaleModeSelected** (const QEStripChartNames::YScaleModes mode)
- void **writeTraceToFile** (const int slot)

Public Member Functions

- **QEStripChart** (QWidget *parent=0)
- QSize **sizeHint** () const
- QDateTime **getStartDateTime** () const
- QDateTime **getEndDateTime** () const
- void **setEndDateTime** (QDateTime endDateTimeIn)
- int **getDuration** () const
- void **setDuration** (int durationIn)
- double **getYMinimum** () const

- void **setYMinimum** (const double yMinimumIn)
- double **getYMaximum** () const
- void **setYMaximum** (const double yMaximumIn)
- void **setYRange** (const double yMinimumIn, const double yMaximumIn)
- void **setPvName** (const int slot, const QString &pvName)
- QString **getPvName** (const int slot) const
- int **addPvName** (const QString &pvName)
- void **clearAllPvNames** ()
- PropertyChartYRanges **getYRangeMode** () const
- void **setYRangeMode** (const PropertyChartYRanges scale)
- [QE::VideoModes](#) **getVideoMode** () const
- QEStripChartNames::YScaleModes **getYScaleMode** () const
- void **updateItemMenu** (const int slot, QAction *action, const bool inUseMenu)
- void **setEnabledContextMenu** (bool enable)
- bool **getEnabledContextMenu** () const
- void **setToolBarVisible** (bool [visible](#))
- bool **getToolBarVisible** () const
- void **setPvItemsVisible** (bool [visible](#))
- bool **getPvItemsVisible** () const
- void **setNumberPvsVisible** (int number)
- int **getNumberPvsVisible** () const
- void **setVariableNameProperty** (const int slot, const QString &pvName)
- QString **getVariableNameProperty** (const int slot) const
- void **setVariableNameSubstitutionsProperty** (const QString &variableNameSubstitutions)
- QString **getVariableNameSubstitutionsProperty** () const
- void **setAliasName** (const int slot, const QString &aliasName)
- QString **getAliasName** (const int slot) const
- void **setColourProperty** (const int slot, const QColor &color)
- QColor **getColourProperty** (const int slot) const
- void **menuSetYScale** (QEStripChartNames::ChartYRanges ys)

Public Attributes

- QString **variableNameSubstitutions**

Protected Member Functions

- bool **eventFilter** (QObject *watched, QEvent *event)
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)
- void **pvLabelModeChanged** ()
- void **mousePressEvent** (QMouseEvent *event)
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- QString **copyVariable** ()
- QVariant **copyData** ()
- [QEChannel](#) * **createQcItem** (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void **establishConnection** (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **saveConfiguration** ([PersistenceManager](#) *pm)
- void **restoreConfiguration** ([PersistenceManager](#) *pm, [restorePhases](#) restorePhase)
- void **addToPredefinedList** (const QString &pvName)
- QStringList **getPredefinedPVNameList** () const
- QString **getPredefinedItem** (int i) const
- void **setReplotsRequired** ()
- void **evaluateAllowDrop** ()

Properties

- int **duration**
- double **yMinimum**
- double **yMaximum**
- [QE::VideoModes](#) **videoMode**
- PropertyChartYRanges **chartRange**
- QEStripChartNames::YScaleModes **scaleMode**
- bool **enableContextMenu**
- bool **toolBarsVisible**
- bool **pvtItemsVisible**
- int **numberPvsVisible**
- QString **variable1**
- QString **variable2**
- QString **variable3**
- QString **variable4**
- QString **variable5**
- QString **variable6**
- QString **variable7**
- QString **variable8**
- QString **variable9**
- QString **variable10**
- QString **variable11**
- QString **variable12**
- QString **variable13**
- QString **variable14**
- QString **variable15**
- QString **variable16**
- QString **aliasName1**
- QString **aliasName2**
- QString **aliasName3**
- QString **aliasName4**
- QString **aliasName5**
- QString **aliasName6**
- QString **aliasName7**
- QString **aliasName8**
- QString **aliasName9**
- QString **aliasName10**
- QString **aliasName11**
- QString **aliasName12**
- QString **aliasName13**
- QString **aliasName14**
- QString **aliasName15**
- QString **aliasName16**
- QColor **colour1**
- QColor **colour2**
- QColor **colour3**
- QColor **colour4**
- QColor **colour5**
- QColor **colour6**
- QColor **colour7**
- QColor **colour8**
- QColor **colour9**
- QColor **colour10**
- QColor **colour11**

- QColor **colour12**
- QColor **colour13**
- QColor **colour14**
- QColor **colour15**
- QString [variableSubstitutions](#)

Friends

- class **QEStripChartItem**

Additional Inherited Members

9.277.1 Member Function Documentation

9.277.1.1 `restoreConfiguration()`

```
void QEStripChart::restoreConfiguration (
    PersistenceManager * ,
    restorePhases ) [protected], [virtual]
```

Service a request to restore the [QE](#) widget's configuration. A [QE](#) widget recover any configuration details from the [PersistenceManager](#). For example, a [QEStripChart](#) may restore the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method. This is called twice with an incrementing restorePhase. Most widgets will miss the first call as they don't exist yet (they are created as part of the first phase).

Reimplemented from [QEWidget](#).

9.277.1.2 `saveConfiguration()`

```
void QEStripChart::saveConfiguration (
    PersistenceManager * ) [protected], [virtual]
```

Service a request to save the [QE](#) widget's current configuration. A widget may save any configuration details through the [PersistenceManager](#). For example, a [QEStripChart](#) may save the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method.

Reimplemented from [QEWidget](#).

9.277.2 Property Documentation

9.277.2.1 variableSubstitutions

```
QString QEStripChart::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[,] NAME2=VALUE2... Values may be quoted strings. For example, 'SAMPLE=SAM1, NAME = "Ref foil"' These substitutions are applied to all the variable names.

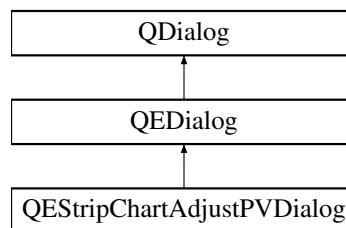
The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChart.h
- qeframework/widgets/QEStripChart/QEStripChart.cpp

9.278 QEStripChartAdjustPVDialo Class Reference

```
#include <QEStripChartAdjustPVDialo.h>
```

Inheritance diagram for QEStripChartAdjustPVDialo:



Public Member Functions

- **QEStripChartAdjustPVDialo** (QWidget *parent=0)
- void **setValueScaling** (const [QEValueScaling](#) &valueScale)
- [QEValueScaling](#) **getValueScaling** () const
- void **setSupport** (const double min, const double max, const [QEDisplayRanges](#) &loprHoppr, const [QEDisplayRanges](#) &plotted, const [QEDisplayRanges](#) &buffered)

Additional Inherited Members

9.278.1 Detailed Description

Manager class for the QEStripChartAdjustPVDialo.ui compiled form.

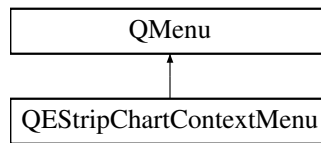
This dialog form allows the user to select a start datetime and end datetime pair for the strip chart.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartAdjustPVDialo.h
- qeframework/widgets/QEStripChart/QEStripChartAdjustPVDialo.cpp

9.279 QEStripChartContextMenu Class Reference

Inheritance diagram for QEStripChartContextMenu:



Signals

- void **contextMenuSelected** (const QEStripChartNames::ContextMenuOptions)

Public Member Functions

- [QEStripChartContextMenu](#) (bool inUse, QWidget *parent=0)
- void **setActionChecked** (const QEStripChartNames::ContextMenuOptions option, const bool checked)
- void **setActionEnabled** (const QEStripChartNames::ContextMenuOptions option, const bool enabled)
- void **setActionVisible** (const QEStripChartNames::ContextMenuOptions option, const bool visible)
- void **setActionText** (const QEStripChartNames::ContextMenuOptions option, const QString &caption)
- void **setIsCalculation** (const bool isCalculation)
- void **setPredefinedNames** (const QStringList &pvList)
- void **setUseReceiveTime** (const bool useReceiveTime)
- void **setArchiveReadHow** (const QEArchiveInterface::How how)
- void **setLineDrawMode** (const QEStripChartNames::LineDrawModes mode)
- void **setLinePlotMode** (const QEStripChartNames::LinePlotModes mode)

9.279.1 Constructor & Destructor Documentation

9.279.1.1 QEStripChartContextMenu()

```

QEStripChartContextMenu::QEStripChartContextMenu (
    bool inUse,
    QWidget * parent = 0 ) [explicit]
  
```

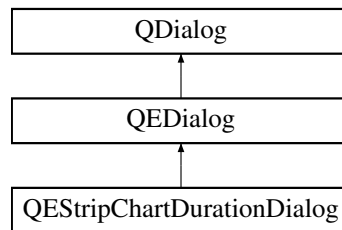
Construct strip chart item context menu. This menu item creates all required sub menu items. inUse set true for an inuse slot, i.e. already has a PV allocated. inUse set false for an empty slot.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartContextMenu.h
- qeframework/widgets/QEStripChart/QEStripChartContextMenu.cpp

9.280 QEStripChartDurationDialog Class Reference

Inheritance diagram for QEStripChartDurationDialog:



Public Member Functions

- **QEStripChartDurationDialog** (QWidget *parent=0)
- void **setDuration** (int secs)
- int **getDuration** () const

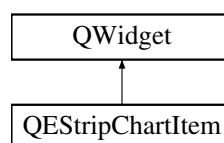
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartDurationDialog.h
- qeframework/widgets/QEStripChart/QEStripChartDurationDialog.cpp

9.281 QEStripChartItem Class Reference

Inheritance diagram for QEStripChartItem:



Public Types

- typedef double **CalcInputs**[QEStripChart::NUMBER_OF_PVS]

Public Slots

- void **setColour** (const QColor &colour)

Signals

- void **requestAction** (const [QEActionRequests](#) &)

Public Member Functions

- **QESTripChartItem** ([QESTripChart](#) *chart, const int slot, QWidget *parent)
- bool **isInUse** () const
- bool **isPvData** () const
- bool **isCalculation** () const
- void **setPvName** (const QString &pvName, const QString &substitutions)
- QString **getPvName** () const
- QString **getEgu** () const
- QString **getDescription** () const
- QString **getCaptionLabel** () const
- QPointF **dataPointToReal** (const [QCaDataPoint](#) &point) const
- bool **getUseReceiveTime** () const
- QArchiveInterface::How **getArchiveReadHow** () const
- QESTripChartNames::LineDrawModes **getLineDrawMode** () const
- QESTripChartNames::LinePlotModes **getLinePlotMode** () const
- void **setAliasName** (const QString &aliasName)
- QString **getAliasName** () const
- QColor **getColour** () const
- [QEDisplayRanges](#) **getLoprHopr** (bool doScale) const
- [QEDisplayRanges](#) **getDisplayedMinMax** (bool doScale) const
- [QEDisplayRanges](#) **getBufferedMinMax** (bool doScale) const
- [QCaDataPointList](#) **extractPlotPoints** (const bool doBuffered) const
- void **readArchive** ()
- void **recalcualteBufferedValues** ()
- void **normalise** ()
- void **plotData** ()
- void **setCaption** ()
- double **getCurrentValue** (bool &okay) const
- void **calculateAndUpdate** (const [QCaDateTime](#) &datetime, const CalcInputs values)
- const [QCaDataPoint](#) * **findNearestPoint** (const [QCaDateTime](#) &searchTime) const
- void **saveConfiguration** ([PMElement](#) &parentElement)
- void **restoreConfiguration** ([PMElement](#) &parentElement)
- void **updateMenu** (QAction *action, const bool inUseMenu)
- void **writeTraceToFile** ()

Static Public Member Functions

- static void **writeListToFile** ([QESTripChart](#) *chart, const QVector< [QESTripChartItem](#) * > &itemList, const QString &filename, const double interval, const bool isCSV)

Public Attributes

- [QCaVariableNamePropertyManager](#) **pvnNameProperyManager**

Protected Member Functions

- bool **eventFilter** (QObject *obj, QEvent *event)

Friends

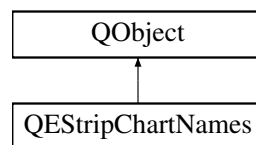
- class **QEStripChart**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartItem.h
- qeframework/widgets/QEStripChart/QEStripChartItem.cpp

9.282 QEStripChartNames Class Reference

Inheritance diagram for QEStripChartNames:



Public Types

- enum **ChartTimeModes** { **tmRealTime** , **tmPaused** , **tmHistorical** }
- enum **ChartYRanges** { **manual** , **operatingRange** , **plotted** , **buffered** , **dynamic** , **normalised** }
- enum **PlayModes** { **play** , **pause** , **forward** , **backward** , **selectTimes** }
- enum **StateModes** { **previous** , **next** }
- enum **YScaleModes** { **linear** , **log** }
- enum **LineDrawModes** { **ldmHide** , **ldmRegular** , **ldmBold** }
- enum **LinePlotModes** { **lpmRectangular** , **lpmSmooth** }
- enum **ContextMenuOptions** {
SCCM_NONE = QEAbstractDynamicWidget::ADWCM_SUB_CLASS_WIDGETS_START_HERE , **SCCM_↵**
_SHOW_HIDE_TOOLBAR , **SCCM_SHOW_HIDE_PV_ITEMS** , **SCCM_READ_ARCHIVE** ,
SCCM_RECALCULATE , **SCCM_SCALE_CHART_AUTO** , **SCCM_SCALE_CHART_PLOTTED** , **SCCM_↵**
_SCALE_CHART_BUFFERED ,
SCCM_SCALE_PV_RESET , **SCCM_SCALE_PV_GENERAL** , **SCCM_SCALE_PV_AUTO** , **SCCM_↵**
SCALE_PV_PLOTTED ,
SCCM_SCALE_PV_PLOTTED_UPPER , **SCCM_SCALE_PV_PLOTTED_CENTRE** , **SCCM_SCALE_↵**
PV_PLOTTED_LOWER , **SCCM_SCALE_PV_BUFFERED** ,
SCCM_SCALE_PV_FIRST_CENTRE , **SCCM_PLOT_RECTANGULAR** , **SCCM_PLOT_SMOOTH** ,
SCCM_PLOT_SERVER_TIME ,
SCCM_PLOT_CLIENT_TIME , **SCCM_ARCH_LINEAR** , **SCCM_ARCH_PLOTBIN** , **SCCM_ARCH_RAW** ,
SCCM_ARCH_SHEET , **SCCM_ARCH_AVERAGED** , **SCCM_LINE_HIDE** , **SCCM_LINE_REGULAR** ,
SCCM_LINE_BOLD , **SCCM_LINE_COLOUR** , **SCCM_PV_EDIT_NAME** , **SCCM_ADD_TO_PREDEFINED**
, **SCCM_PV_WRITE_TRACE** , **SCCM_PV_STATS** , **SCCM_PV_CLEAR** , **SCCM_PV_ADD_NAME** ,
SCCM_PV_PASTE_NAME , **SCCM_PREDEFINED_01** , **SCCM_PREDEFINED_02** , **SCCM_PREDEFINED_↵**
_03 ,
SCCM_PREDEFINED_04 , **SCCM_PREDEFINED_05** , **SCCM_PREDEFINED_06** , **SCCM_PREDEFINED_↵**
_07 ,
SCCM_PREDEFINED_08 , **SCCM_PREDEFINED_09** , **SCCM_PREDEFINED_10** }

Static Public Member Functions

- static QString **chartTimeModeStatus** (const ChartTimeModes mode)
- static QString **chartYRangeStatus** (const ChartYRanges yRange)

Static Public Attributes

- static const ContextMenuOptions **ContextMenuItemFirst** = SCCM_SHOW_HIDE_TOOLBAR
- static const ContextMenuOptions **ContextMenuItemLast** = SCCM_PREDEFINED_10
- static const int **NumberMenuItems** = (ContextMenuItemLast - ContextMenuItemFirst + 1)
- static const int **NumberPrefinedItems** = (SCCM_PREDEFINED_10 - SCCM_PREDEFINED_01 + 1)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartNames.h
- qeframework/widgets/QEStripChart/QEStripChartNames.cpp

9.283 QEStripChartPushButtonSpecifications Struct Reference

Public Attributes

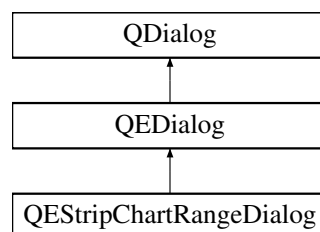
- int **gap**
- int **width**
- int **value**
- bool **isIcon**
- const QString **captionOrIcon**
- const QString **toolTip**
- const char * **member**

The documentation for this struct was generated from the following file:

- qeframework/widgets/QEStripChart/QEStripChartToolBar.cpp

9.284 QEStripChartRangeDialog Class Reference

Inheritance diagram for QEStripChartRangeDialog:



Public Member Functions

- **QEStripChartRangeDialog** (QWidget *parent=0)
- void **setRange** (const double min, const double max)
- double **getMinimum** ()
- double **getMaximum** ()

Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartRangeDialog.h
- qeframework/widgets/QEStripChart/QEStripChartRangeDialog.cpp

9.285 QEStripChartState Class Reference

Public Member Functions

- void **saveConfiguration** ([PMElement](#) &parentElement)
- void **restoreConfiguration** ([PMElement](#) &parentElement)

Public Attributes

- bool **isNormalVideo**
- QEStripChartNames::ChartTimeModes **chartTimeMode**
- QEStripChartNames::YScaleModes **yScaleMode**
- QEStripChartNames::ChartYRanges **chartYScale**
- double **yMinimum**
- double **yMaximum**
- int **duration**
- Qt::TimeSpec **timeZoneSpec**
- QDateTime **endDateTime**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartState.h
- qeframework/widgets/QEStripChart/QEStripChartState.cpp

9.286 QEStripChartStateList Class Reference

Public Member Functions

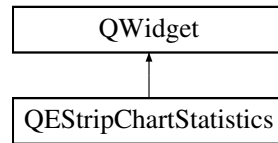
- void **clear** ()
- void **push** (const [QEStripChartState](#) &state)
- bool **prev** ([QEStripChartState](#) &state)
- bool **next** ([QEStripChartState](#) &state)
- bool **prevAvailable** ()
- bool **nextAvailable** ()

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartState.h
- qeframework/widgets/QEStripChart/QEStripChartState.cpp

9.287 QEStripChartStatistics Class Reference

Inheritance diagram for QEStripChartStatistics:



Public Member Functions

- **QEStripChartStatistics** (const QString &pvName, const QString &egu, const [QCaDataPointList](#) &dataList, [QEStripChartItem](#) *owner, QWidget *parent=0)

Protected Member Functions

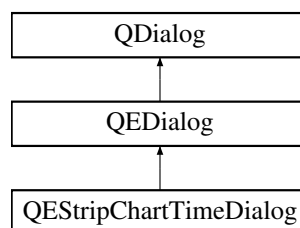
- bool **eventFilter** (QObject *watched, QEvent *event)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartStatistics.h
- qeframework/widgets/QEStripChart/QEStripChartStatistics.cpp

9.288 QEStripChartTimeDialog Class Reference

Inheritance diagram for QEStripChartTimeDialog:



Public Member Functions

- **QEStripChartTimeDialog** (QWidget *parent=0)
- void **setMaximumDateTime** (QDateTime datetime)
- void **setStartDateTime** (QDateTime datetime)
- QDateTime **getStartDateTime** ()
- void **setEndDateTime** (QDateTime datetime)
- QDateTime **getEndDateTime** ()

Additional Inherited Members

The documentation for this class was generated from the following files:

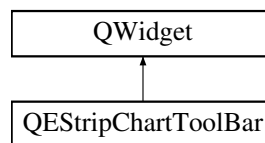
- qeframework/widgets/QEStripChart/QEStripChartTimeDialog.h
- qeframework/widgets/QEStripChart/QEStripChartTimeDialog.cpp

9.289 QEStripChartToolBar Class Reference

This class holds all the StripChart tool bar widgets.

```
#include <QEStripChartToolBar.h>
```

Inheritance diagram for QEStripChartToolBar:



Classes

- class [OwnTabWidget](#)

Signals

- void **stateSelected** (const QEStripChartNames::StateModes mode)
- void **videoModeSelected** (const [QE::VideoModes](#) mode)
- void **yScaleModeSelected** (const QEStripChartNames::YScaleModes mode)
- void **yRangeSelected** (const QEStripChartNames::ChartYRanges scale)
- void **durationSelected** (const int seconds)
- void **selectDuration** ()
- void **timeZoneSelected** (const Qt::TimeSpec timeSpec)
- void **playModeSelected** (const QEStripChartNames::PlayModes mode)
- void **writeAllSelected** ()
- void **readArchiveSelected** ()
- void **loadSelected** ()
- void **saveAsSelected** ()
- void **loadSelectedFile** (const QString &filename)

Public Member Functions

- **QEStripChartToolBar** (QWidget *parent=0)
- void **setYRangeStatus** (const QEStripChartNames::ChartYRanges yRange)
- void **setTimeStatus** (const QString &timeStatus)
- void **setDurationStatus** (const QString &durationStatus)
- void **setNOARStatus** (const int noar)
- void **setTimeModeStatus** (const QEStripChartNames::ChartTimeModes timeMode)
- void **setStateSelectionEnabled** (const QEStripChartNames::StateModes mode, const bool enabled)
- void **setTimeRefs** (const QDateTime &t1, const QDateTime &t2)
- void **setValue1Refs** (const double v1, const double v2)
- void **setValue2Refs** (const double v1, const double v2)

Static Public Member Functions

- static int **designHeight** ()

Friends

- class **OwnTabWidget**

9.289.1 Detailed Description

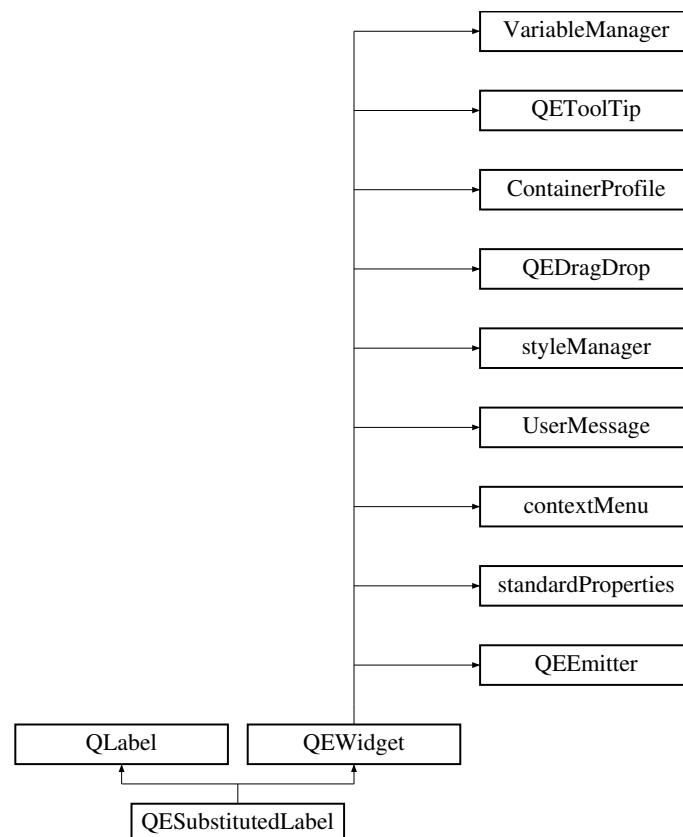
This class holds all the StripChart tool bar widgets.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartToolBar.h
- qeframework/widgets/QEStripChart/QEStripChartToolBar.cpp

9.290 QESubstitutedLabel Class Reference

Inheritance diagram for QESubstitutedLabel:



Public Member Functions

- **QESubstitutedLabel** (QWidget *parent=0)
- void **setLabelTextProperty** (QString labelTextIn)
- QString **getLabelTextProperty** ()
- void **setSubstitutionsProperty** (QString macroSubstitutionsIn)
- QString **getSubstitutionsProperty** ()
- QString **getLabelTextPropertyFormat** ()
- void **setLabelTextPropertyFormat** (QString labelTextIn)

Protected Attributes

- QString [labelText](#)

Properties

- QString **text**
- QString [textSubstitutions](#)

Additional Inherited Members

9.290.1 Member Data Documentation

9.290.1.1 labelText

```
QString QESubstitutedLabel::labelText [read], [write], [protected]
```

Label text to be substituted. This text will be copied to the label text after applying any macro substitutions from the textSubstitutions property

9.290.2 Property Documentation

9.290.2.1 textSubstitutions

```
QString QESubstitutedLabel::textSubstitutions [read], [write]
```

Text substitutions. These substitutions are applied to the 'labelText' property prior to copying it to the label text.

The documentation for this class was generated from the following files:

- qeframework/widgets/QESubstitutedLabel/QESubstitutedLabel.h
- qeframework/widgets/QESubstitutedLabel/QESubstitutedLabel.cpp

Public Member Functions

- [QESurface](#) (QWidget *parent=0)
- [QESurface](#) (const QString &dataVariableName, QWidget *parent=0)
- **QESurface** (const QString &dataVariableName, const QString &widthVariableName, QWidget *parent=0)
- virtual [~QESurface](#) ()
 - Destruction.*
- void **setGridStyle** (const Qt::PenStyle gridStyle)
- Qt::PenStyle **getGridStyle** () const
- void **setGridColour** (const QColor &gridColour)
- QColor **getGridColour** () const
- void **setAxisColour** (const QColor &axisColour)
- QColor **getAxisColour** () const
- void **setSurfaceStyle** (const Qt::BrushStyle surfaceStyle)
- Qt::BrushStyle **getSurfaceStyle** () const
- void **setClampData** (const bool clampData)
- bool **getClampData** () const
- void **setShowScaling** (const bool showScaling)
- bool **getShowScaling** () const
- bool **getShowGrid** () const
- bool **getShowSurface** () const
- double **getTheta** () const
- double **getPhi** () const
- double **getZoom** () const
- double **getXScale** () const
- double **getYScale** () const
- double **getZScale** () const

Protected Member Functions

- QSize **sizeHint** () const
- void **paintEvent** (QPaintEvent *event)
- void **wheelEvent** (QWheelEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- void **mouseMoveEvent** (QMouseEvent *event)
- void **mouseReleaseEvent** (QMouseEvent *event)
- void **updateDataVisulation** ()
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)

Properties

- bool **showGrid**
- Qt::PenStyle **gridStyle**
- QColor **gridColour**
- bool **showSurface**
- Qt::BrushStyle **surfaceStyle**
- QColor **axisColour**
- double **theta**
- double **phi**
- double **zoom**
- double **xScale**
- double **yScale**
- double **zScale**
- bool **clampData**
- bool **showScaling**

Additional Inherited Members

9.291.1 Detailed Description

The [QESurface](#) class is a non-EPICS aware surface plot widget.

9.291.2 Constructor & Destructor Documentation

9.291.2.1 QESurface() [1/2]

```
QESurface::QESurface (
    QWidget * parent = 0 ) [explicit]
```

Construction Create without a variable. Use setDataPvName () and setVariableNameSubstitutions() to define a variable and, optionally, macro substitutions later.

9.291.2.2 QESurface() [2/2]

```
QESurface::QESurface (
    const QString & dataVariableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable (s). A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

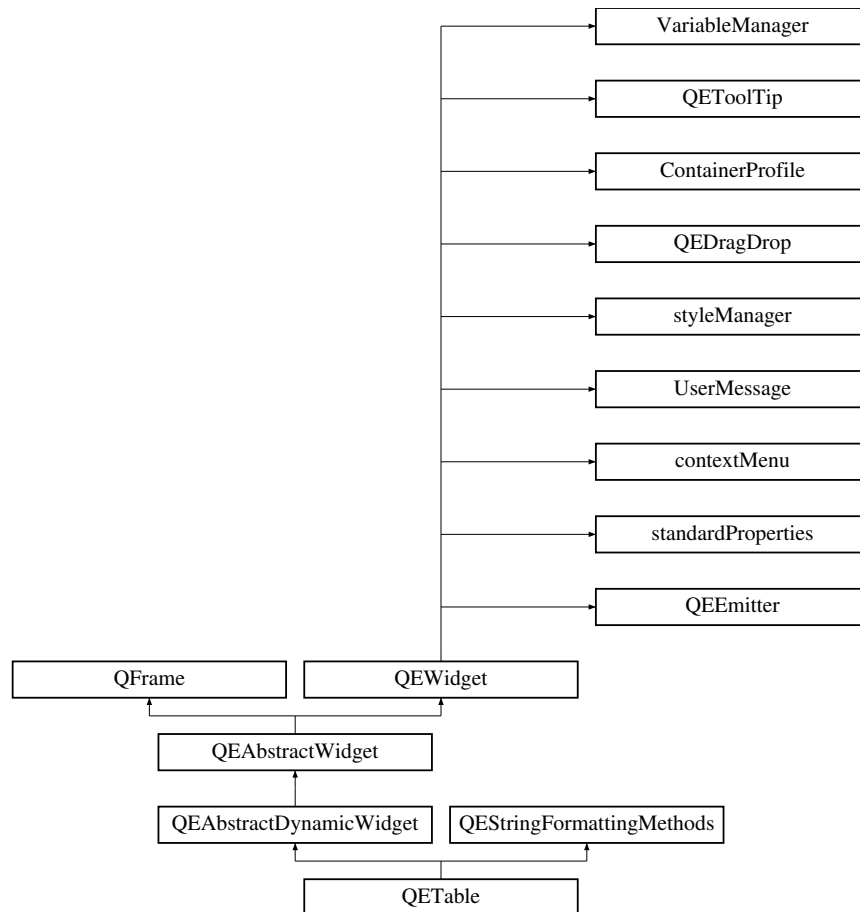
The documentation for this class was generated from the following files:

- qeframework/widgets/QE2DDataVisualisation/QESurface.h
- qeframework/widgets/QE2DDataVisualisation/QESurface.cpp

9.292 QETable Class Reference

```
#include <QETable.h>
```

Inheritance diagram for QETable:



Public Types

- enum **Constants** { **MAXIMUM_NUMBER_OF_VARIABLES** = 20 }

Public Slots

- void **setVariableName** (const int, const QString &)
- void **setTitles** (const QStringList &titles)
- void **setTitle** (const int slot, const QString &title)
- void **setSelection** (int value)
- void **setPvNameSet** (const QStringList &pvNameSet)
- void **setTableEntry** (const int slot, const QString &pvName, const QString &title)

Signals

- void **selectionChanged** (int value)
- void **pvNameSetChanged** (const QStringList &pvNameSet)
- void **titlesChanged** (const QStringList &pvNameSet)
- void **dbValueChanged** ()

Public Member Functions

- [QETable](#) (QWidget *parent=0)
- virtual [~QETable](#) ()
Destruction.
- QString [getVariableName](#) (const int) const
- void [setSubstitutions](#) (const QString &substitutions)
- QString [getSubstitutions](#) () const
- void [setColumnWidthMinimum](#) (const int columnWidthMinimum)
- int [getColumnWidthMinimum](#) () const
- void [setDisplayMaximum](#) (const int [displayMaximum](#))
- int [getDisplayMaximum](#) () const
- void [setOrientation](#) (const Qt::Orientation [orientation](#))
- Qt::Orientation [getOrientation](#) () const
- void [setFormatProperty](#) (const [QE::Formats](#) format)
- [QE::Formats](#) [getFormatProperty](#) () const
- void [setSeparatorProperty](#) (const [QE::Separators](#) separator)
- [QE::Separators](#) [getSeparatorProperty](#) () const
- void [setNotationProperty](#) (const [QE::Notations](#) notation)
- [QE::Notations](#) [getNotationProperty](#) () const
- int [addPvName](#) (const QString &pvName)
- void [clearAllPvNames](#) ()
- QStringList [getTitles](#) () const
- int [getSelection](#) () const
- QStringList [getPvNameSet](#) () const
- void [setSlotAddUnits](#) (const int slot, const bool [addUnits](#))
- void [setSlotForceSign](#) (const int slot, const bool [forceSign](#))
- void [setSlotFormat](#) (const int slot, const [QE::Formats](#) format)
- void [setSlotSeparator](#) (const int slot, const [QE::Separators](#) separator)
- void [setSlotNotation](#) (const int slot, const [QE::Notations](#) notation)

Protected Types

- enum [OwnContextMenuOptions](#) { [CM_HORIZONTAL_TABLE](#) = ADWCM_SUB_CLASS_WIDGETS_↵
START_HERE , [CM_VERTICAL_TABLE](#) }

Protected Member Functions

- QSize [sizeHint](#) () const
- bool [eventFilter](#) (QObject *watched, QEvent *event)
- void [resizeEvent](#) (QResizeEvent *event)
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- [QEChannel](#) * [createQcItem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void [activated](#) ()
Do any post-all-widgets-constructed, i.e. activated stuff.
- void [stringFormattingChange](#) ()
- QMenu * [buildContextMenu](#) ()
- void [contextMenuTriggered](#) (int selectedItemNum)
- void [mousePressEvent](#) (QMouseEvent *event)
- void [dragEnterEvent](#) (QDragEnterEvent *event)
- void [dropEvent](#) (QDropEvent *event)
- QString [copyVariable](#) ()
- QVariant [copyData](#) ()
- void [saveConfiguration](#) ([PersistenceManager](#) *pm)
- void [restoreConfiguration](#) ([PersistenceManager](#) *pm, [restorePhases](#) restorePhase)

Properties

- QString [variableName1](#)
EPICS variable names (CA PV)
- QString **variableName2**
- QString **variableName3**
- QString **variableName4**
- QString **variableName5**
- QString **variableName6**
- QString **variableName7**
- QString **variableName8**
- QString **variableName9**
- QString **variableName10**
- QString **variableName11**
- QString **variableName12**
- QString **variableName13**
- QString **variableName14**
- QString **variableName15**
- QString **variableName16**
- QString **variableName17**
- QString **variableName18**
- QString **variableName19**
- QString **variableName20**
- QString [variableSubstitutions](#)
- QStringList [titles](#)
- int [colWidthMinimum](#)
- int [displayMaximum](#)
- Qt::Orientation [orientation](#)
- bool [showGrid](#)
- Qt::PenStyle [gridStyle](#)
- bool [addUnits](#)
- bool [forceSign](#)
- QE::Formats [format](#)
- QE::Separators [separator](#)
- QE::Notations [notation](#)

Additional Inherited Members

9.292.1 Detailed Description

This class provides an EPICS aware table widget which is capable of displaying up to 20 array PVs in tabular form. When in the default vertical orientation each column displays a consecutive element from an array EPICS variable.

When in horizontal mode, the table and functionality is transposed.

It is tightly integrated, via the [QEAbstractWidget](#), with the base class [QEWidget](#) class which provides generic support such as macro substitutions, drag/drop, and standard properties. [QEAbstractWidget](#) provides all standard properties.

9.292.2 Constructor & Destructor Documentation

9.292.2.1 QETable()

```
QETable::QETable (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable(s). Use setVariableName functions.

9.292.3 Member Function Documentation

9.292.3.1 dbValueChanged

```
void QETable::dbValueChanged ( ) [signal]
```

Sent when the widget is updated following a data change Can be used to pass on EPICS data (as presented in this widget) to other widgets. For example a QList widget could log updates from this widget. Note: this widget emits the numeric enumeration value as opposed to the associated text.

9.292.3.2 restoreConfiguration()

```
void QETable::restoreConfiguration (
    PersistenceManager * ,
    restorePhases ) [protected], [virtual]
```

Service a request to restore the [QE](#) widget's configuration. A [QE](#) widget recover any configuration details from the [PersistenceManager](#). For example, a [QEStripChart](#) may restore the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method. This is called twice with an incrementing restorePhase. Most widgets will miss the first call as they don't exist yet (they are created as part of the first phase).

Reimplemented from [QEWidget](#).

9.292.3.3 saveConfiguration()

```
void QETable::saveConfiguration (
    PersistenceManager * ) [protected], [virtual]
```

Service a request to save the [QE](#) widget's current configuration. A widget may save any configuration details through the [PersistenceManager](#). For example, a [QEStripChart](#) may save the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method.

Reimplemented from [QEWidget](#).

9.292.4 Property Documentation

9.292.4.1 addUnits

```
bool QETable::addUnits [read], [write]
```

If true (default), add engineering units supplied with the data.

9.292.4.2 colWidthMinimum

```
int QETable::colWidthMinimum [read], [write]
```

Specified the minimum allow column width. The widget will shrink/expand the width of each column to as to exactly fit the with of the widget. However, columns will not shrink to less than the value provided by this property. Defaults to 80.

9.292.4.3 displayMaximum

```
int QETable::displayMaximum [read], [write]
```

The maximum number of array elements that will be displayed irrespective of the number of elements that the EPICS variable contains. Defaults to 4096.

9.292.4.4 forceSign

```
bool QETable::forceSign [read], [write]
```

If false (default), no "+" sign, when true always add a sign.

9.292.4.5 format

```
QE::Formats QETable::format [read], [write]
```

Format to apply to data. Default is 'Default' in which case the data type supplied with the data determines how the data is formatted. For all other options, an attempt is made to format the data as requested (whatever its native form).

9.292.4.6 gridStyle

```
Qt::PenStyle QETable::gridStyle [read], [write]
```

Sets table grid style. Defaults to SolidLine.

9.292.4.7 notation

```
QE::Notations QETable::notation [read], [write]
```

Notation used for numerical formatting. Default is fixed.

9.292.4.8 orientation

```
Qt::Orientation QETable::orientation [read], [write]
```

Determines if the variable values are displayed in rows (orientation is horizontal) or in columns (orientation is vertical). The default is vertical.

9.292.4.9 separator

```
QE::Separators QETable::separator [read], [write]
```

Separators used for interger and fixed point formatting. Default is None.

9.292.4.10 showGrid

```
bool QETable::showGrid [read], [write]
```

Controls if table grid is displayed. Default to true.

9.292.4.11 titles

```
QStringList QETable::titles [read], [write]
```

Allows specification of tables titles. If blank, the default, then out-of-the-box QTableWidgetItem heading are used, i.e. 1, 2, etc. If "<>" is specified, then this is replaced by the PV name. This is particularly useful when PV names are specified dynamically or by substitution.

9.292.4.12 variableSubstitutions

```
QString QETable::variableSubstitutions [read], [write]
```

Macro substitutions. The default is no substitutions. The format is NAME1=VALUE1[, NAME2=VALUE2... Values may be quoted strings. For example, 'PUMP=PMP3, NAME = "My Pump"' These substitutions are applied to variable names for all QE widgets. In some widgets are also used for other purposes.

The documentation for this class was generated from the following files:

- qeframework/widgets/QETable/QETable.h
- qeframework/widgets/QETable/QETable.cpp

9.293 QThreadSafeQueue< Type > Class Template Reference

```
#include <QThreadSafeQueue.h>
```

Public Member Functions

- void **clear** ()
- void **enqueue** (const Type &t)
- bool **dequeue** (Type &t)
- int **size** ()
- bool **isEmpty** ()

9.293.1 Detailed Description

```
template<typename Type>
class QThreadSafeQueue< Type >
```

[QThreadSafeQueue](#) is a wrapper around a QQueue with a mutex to make thread safe. If a queue of references, these may become un-referenced orphans when the queue cleared or deleted. Note: the dequeue API is different from the embedded queue's dequeue method.

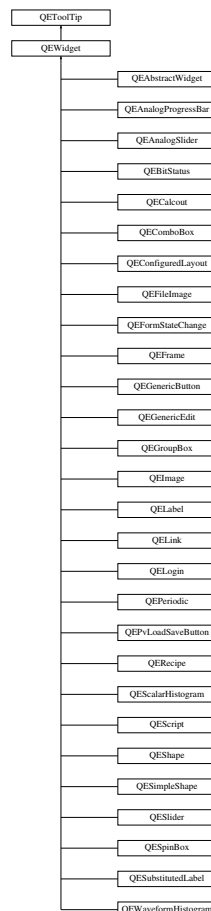
The documentation for this class was generated from the following file:

- qeframework/common/QThreadSafeQueue.h

9.294 QEToolTip Class Reference

```
#include <QEToolTip.h>
```

Inheritance diagram for QEToolTip:



Public Member Functions

- [QEToolTip](#) (QWidget *ownerIn)
- void **setNumberToolTipVariables** (const unsigned int number)
- void **updateToolTipVariable** (const QString &variable, const unsigned int variableIndex)
- void **updateToolTipAlarm** (const [QCaAlarmInfo](#) &alarmInfo, const unsigned int variableIndex)
- void **updateToolTipConnection** (bool connection, const unsigned int variableIndex=0)
- void **updateToolTipCustom** (const QString &custom)
- void **setVariableAsToolTip** (const bool variableAsToolTip)
- bool **getVariableAsToolTip** () const

Friends

- class **QEToolTipSingleton**

9.294.1 Detailed Description

[QE](#) widget tool tips may be set to display the names of the process variables supplying data to the widget and the alarm state and connectino status of those variables. The [QE](#) widget may also set some custom text to be displayed along with this information. The [QEToolTip](#) class manages building and setting the [QE](#) widget tool tips when this functino is required.

9.294.2 Constructor & Destructor Documentation

9.294.2.1 QEToolTip()

```
QEToolTip::QEToolTip (
    QWidget * ownerIn )
```

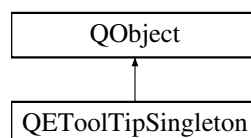
*** Fix this

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/QEToolTip.h
- qeframework/widgets/QEWidget/QEToolTip.cpp

9.295 QEToolTipSingleton Class Reference

Inheritance diagram for QEToolTipSingleton:



Friends

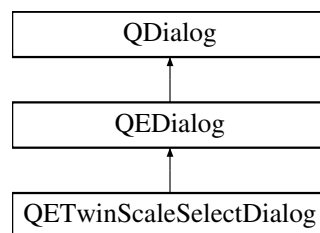
- class **QEToolTip**

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/QEToolTip.h
- qeframework/widgets/QEWidget/QEToolTip.cpp

9.296 QETwinScaleSelectDialog Class Reference

Inheritance diagram for QETwinScaleSelectDialog:



Classes

- struct [ScaleLimit](#)

Public Member Functions

- **QETwinScaleSelectDialog** (const QString &windowTitle, const QString &scaleOneName, const QString &scaleTwoName, QWidget *parent=0)
- void **setActiveMap** (const [ScaleLimit](#) &scaleOne, const [ScaleLimit](#) &scaleTwo)
- void **getActiveMap** ([ScaleLimit](#) &scaleOne, [ScaleLimit](#) &scaleTwo)

Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/QETwinScaleSelectDialog.h
- qeframework/common/QETwinScaleSelectDialog.cpp

9.297 QUtilities Class Reference

Static Public Member Functions

- static QColor [fontColour](#) (const QColor &backgroundColour)
- static QString [colourToStyle](#) (const QColor &backgroundColour, const QColor &foregroundColour)
- static QString [colourToStyle](#) (const QColor &backgroundColour)
- static QString [offBackgroundStyle](#) ()
- static QColor [darkColour](#) (const QColor &lightColour)
- static QColor [blandColour](#) (const QColor &vibrantColour)
- static int [roundToInt](#) (const double x, bool *ok=NULL)
- static QString [intervalToString](#) (const double time, const int precision, const bool showDays)
- static int [getTimeZoneOffset](#) (const QDateTime &atTime)
- static QString [getTimeZoneTLA](#) (const Qt::TimeSpec timeSpec, const QDateTime &atTime)
- static QString [getTimeZoneTLA](#) (const QDateTime &atTime)
- static QString [enumToString](#) (const QObject &object, const QString &enumTypeName, const int enumValue)
- static QString [enumToString](#) (const QMetaObject &mo, const QString &enumTypeName, const int enum↵
Value)
- static int [stringToEnum](#) (const QObject &object, const QString &enumTypeName, const QString &enum↵
Image, bool *ok=0)
- static int [stringToEnum](#) (const QMetaObject &mo, const QString &enumTypeName, const QString &enum↵
Image, bool *ok=0)
- static QWidget * [findWidget](#) (QWidget *parent, const QString &className)
- static QStringList [split](#) (const QString &s)
- static QStringList [variantToStringList](#) (const QVariant &v)
- static void [listPVNames](#) (QWidget *rootWidget, const QString &targetFile, const QString &comment)
- static void [activate](#) (QWidget *rootWidget)
- static void [deactivate](#) (QWidget *rootWidget)
- static QString [dirName](#) (const QString &pathName)
- static QMainWindow * [findMainWindow](#) (QWidget *widget)
- static void [setWindowTitle](#) (QWidget *widget, const QString &title)
- static QString [getWindowTitle](#) (QWidget *widget)
- static QRect [desktopGeometry](#) ()
- static QRect [screenGeometry](#) (QWidget *widget)
- static void [tagObject](#) (QObject *object, const int tag)
- static int [objectTag](#) (const QObject *object, const int defaultValue=-1)
- static void [setPaddedText](#) (QLabel *label, const QString &text, const QString &pad=" . ")
- static void [debugWidgetHierarchy](#) (const QWidget *root, const int instance=0, const int level=0)

9.297.1 Member Function Documentation

9.297.1.1 activate()

```
void QUtilities::activate (
    QWidget * rootWidget ) [static]
```

Performs deep search and initiates updates.

9.297.1.2 colourToStyle() [1/2]

```
QString QEUtilities::colourToStyle (
    const QColor & backgroundColour ) [static]
```

Converts a given colour to associated background-color style, and sets foreground (font) colour to black or white accordingly.

9.297.1.3 colourToStyle() [2/2]

```
QString QEUtilities::colourToStyle (
    const QColor & backgroundColour,
    const QColor & foregroundColour ) [static]
```

Converts a given colours to the associated style.

9.297.1.4 deactivate()

```
void QEUtilities::deactivate (
    QWidget * rootWidget ) [static]
```

Performs deep search and terminates updates.

9.297.1.5 debugWidgetHierarchy()

```
void QEUtilities::debugWidgetHierarchy (
    const QWidget * root,
    const int instance = 0,
    const int level = 0 ) [static]
```

Testing only - outputs the widget hierarchy to std out.

9.297.1.6 desktopGeometry()

```
QRect QEUtilities::desktopGeometry ( ) [static]
```

Returns the virtual desktop geometry.

9.297.1.7 dirName()

```
QString QEUtilities::dirName (
    const QString & pathName ) [static]
```

Returns the directory name as QString of the given pathname. This is essentially like the Linux `dirname` command. This convenience functions hides the faffing with `QFileInfo/QDir`

9.297.1.8 enumToString() [1/2]

```
QString QEUtilities::enumToString (
    const QMetaObject & mo,
    const QString & enumTypeName,
    const int enumValue ) [static]
```

This function returns the image, as a QString, of a enumeration value (cast as an integer). An invalid enumeration value returns a null string.

Note: This functions rely on the meta object compiler (moc) generated code. To use this functions, the enumeration type must be declared within a Q_OBJECT QObject and the enum type qualifier with the Q_ENUM, e.g. Q_ENUM (Shapes).

9.297.1.9 enumToString() [2/2]

```
QString QEUtilities::enumToString (
    const QObject & object,
    const QString & enumTypeName,
    const int enumValue ) [static]
```

This function returns the image, as a QString, of a enumeration value (cast as an integer). An invalid enumeration value returns a null string.

Note: This functions rely on the meta object compiler (moc) generated code. To use this functions, the enumeration type must be declared within a Q_OBJECT QObject and the enum type qualifier with the Q_ENUM, e.g. Q_ENUM (Shapes).

9.297.1.10 findMainWindow()

```
QMainWindow * QEUtilities::findMainWindow (
    QWidget * widget ) [static]
```

Find containing MainWindow object, or return NULL if it does not exist. The specified widget itself can be the MainWindow.

9.297.1.11 findWidget()

```
QWidget * QEUtilities::findWidget (
    QWidget * parent,
    const QString & className ) [static]
```

This function tree walks the given parent looking a widget of the specified class name. The function returns the first found or NULL.

9.297.1.12 fontColour()

```
QColor QEUtilities::fontColour (
    const QColor & backgroundColour ) [static]
```

For the specified background colour, returns font colour (black or white) with a suitable contrast. Currently based on rgb values, and ignores alpha.

9.297.1.13 getTimeZoneOffset()

```
int QEUtilities::getTimeZoneOffset (
    const QDateTime & atTime ) [static]
```

Get the local time zone offset (in seconds) for the nominated time. This is not fixed for the location and will depend on if the the time is a standard time or a daylight savings/summer time.

9.297.1.14 getTimeZoneTLA() [1/2]

```
QString QEUtilities::getTimeZoneTLA (
    const QDateTime & atTime ) [static]
```

Overloaded function that uses the timeSpec associated with atTime.

9.297.1.15 getTimeZoneTLA() [2/2]

```
QString QEUtilities::getTimeZoneTLA (
    const Qt::TimeSpec timeSpec,
    const QDateTime & atTime ) [static]
```

Extract the local time zone three letter acronym for the time. If the timeSpec parameter is Qt::UTC, then this simply returns "UTC". However, if the timeSpec parameter is Qt::LocalTime, then extract the TLA application of the time in question (and NOT for the current time). E.g. for the UK this would be GMT or BST, for Victoria Australia EST or EST.

9.297.1.16 getWindowTitle()

```
QString QEUtilities::getWindowTitle (
    QWidget * widget ) [static]
```

Gets the form title of the containing MainWindow if found, else empty string.

9.297.1.17 intervalToString()

```
QString QEUtilities::intervalToString (
    const double time,
    const int precision,
    const bool showDays ) [static]
```

Converts a time interval (in seconds) to format +d hh:mm:ss.fff

9.297.1.18 listPVNames()

```
void QEUtilities::listPVNames (
    QWidget * rootWidget,
    const QString & targetFile,
    const QString & comment ) [static]
```

Performs deep search and writes all PV names to nominated target file.

9.297.1.19 offBackgroundStyle()

```
QString QEUtilities::offBackgroundStyle ( ) [static]
```

Provides a slightly lighter colour than the standard form background colour. This is particularly useful for QLabel based widgets.

9.297.1.20 roundToInt()

```
int QEUtilities::roundToInt (
    const double x,
    bool * ok = NULL ) [static]
```

Rounds a float to the nearest integer. Not all compilers support lround. If x outside of integer range, the result is set to the min or max allowed integer value and the ok parameter (if defined) set false if x out of range.

9.297.1.21 screenGeometry()

```
QRect QEUtilities::screenGeometry (
    QWidget * widget ) [static]
```

Returns the screen geometry (of the specified widget) If specified widget is null, then returns desktopGeometry.

9.297.1.22 setPaddedText()

```
void QEUtilities::setPaddedText (
    QLabel * label,
    const QString & text,
    const QString & pad = " . " ) [static]
```

Repeatedly right pads text with pad, and then applies to the label. Does nothing if label is nullptr, no padding if pad is an empty string.

9.297.1.23 setWindowTitle()

```
void QEUtilities::setWindowTitle (
    QWidget * widget,
    const QString & title ) [static]
```

Sets the form title of the containing MainWindow if found.

9.297.1.24 split()

```
QStringList QEUtilities::split (
    const QString & s ) [static]
```

Split a string to a string list based on white space, skipping empty parts.

9.297.1.25 stringToEnum() [1/2]

```
int QEUtilities::stringToEnum (
    const QMetaObject & mo,
    const QString & enumTypeName,
    const QString & enumImage,
    bool * ok = 0 ) [static]
```

This function returns the enumeration value given an enumeration image. The caller must cast the result to the appropriate enumeration type. The image must be exact match including case. The only tolerance allowed for is that the image is trimmed. An invalid image cause this function to return -1. However, -1 is an uncommon but not invalid enumeration value. If -1 might be a valid value, then the caller should specify the and check the ok parameter.

Note: This functions rely on the meta object compiler (moc) generated code. To use this functions, the enumeration type must be declared within a Q_OBJECT QObject and the enum type qualifier with the Q_ENUM, e.g. Q_ENUM (Shapes).

9.297.1.26 stringToEnum() [2/2]

```
int QEUtilities::stringToEnum (
    const QObject & object,
    const QString & enumTypeName,
    const QString & enumImage,
    bool * ok = 0 ) [static]
```

This function returns the enumeration value given an enumeration image. The caller must cast the result to the appropriate enumeration type. The image must be exact match including case. The only tolerance allowed for is that the image is trimmed. An invalid image cause this function to return -1. However, -1 is an uncommon but not invalid enumeration value. If -1 might be a valid value, then the caller should specify the and check the ok parameter.

Note: This functions rely on the meta object compiler (moc) generated code. To use this functions, the enumeration type must be declared within a Q_OBJECT QObject and the enum type qualifier with the Q_ENUM, e.g. Q_ENUM (Shapes).

9.297.1.27 tagObject()

```
void QEUtilities::tagObject (
    QObject * object,
    const int tag ) [static]
```

Allows an object to be assigned an arbitartint tag value. Under the covers, an obkect property is created/updated.

9.297.1.28 variantToStringList()

```
QStringList QEUtilities::variantToStringList (
    const QVariant & v ) [static]
```

Converts QVariant to a QStringList, where each string item is further split using white space as delimiter. This function was principally designed for use within the paste/drop functions of widgets that support multiple PVs.

The documentation for this class was generated from the following files:

- qeframework/common/QECommon.h
- qeframework/common/QECommon.cpp

9.298 QEValueScaling Class Reference

Public Member Functions

- void **reset** ()
- void **assign** (const [QEValueScaling](#) &s)
- void **set** (const double d, const double m, const double c, const double t)
- void **get** (double &d, double &m, double &c, double &t) const
- double **getTimeOffset** () const
- void **map** (const double fromLower, const double fromUpper, const double toLower, const double toUpper)
- bool **isScaled** () const
- double **value** (const double x) const
- [QEDisplayRanges](#) **value** (const [QEDisplayRanges](#) &displayRange) const
- void **saveConfiguration** ([PMElement](#) &parentElement) const
- void **restoreConfiguration** ([PMElement](#) &parentElement)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEStripChart/QEStripChartUtilities.h
- qeframework/widgets/QEStripChart/QEStripChartUtilities.cpp

9.299 QEVariantUpdate Struct Reference

Public Attributes

- QVariant **value**
- [QCaAlarmInfo](#) **alarmInfo**
- [QCaDateTime](#) **timeStamp**
- unsigned int **variableIndex**
- bool **isMetaUpdate**

The documentation for this struct was generated from the following file:

- qeframework/data/QCaObject.h

9.300 QEVectorVariants Class Reference

Public Types

- enum **OwnTypes** {
Invalid , **DoubleVector** , **FloatVector** , **BoolVector** ,
Int8Vector , **Int16Vector** , **Int32Vector** , **Int64Vector** ,
UInt8Vector , **UInt16Vector** , **UInt32Vector** , **UInt64Vector** }

Static Public Member Functions

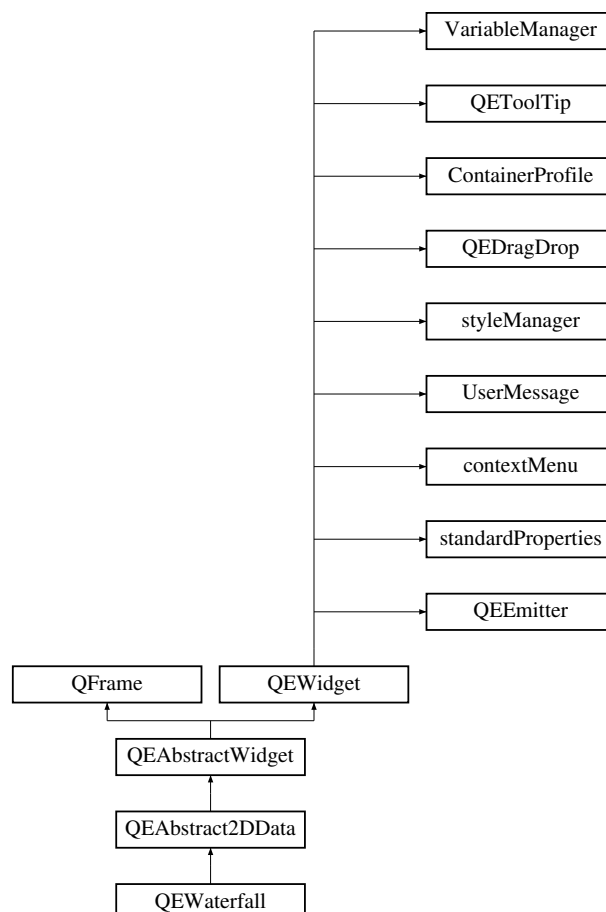
- static OwnTypes **getOwnType** (const QVariant &item)
- static bool **isVectorVariant** (const QVariant &item)
- static int **vectorCount** (const QVariant &vector)
- static QVector< double > **convertToFloatingVector** (const QVariant &vector, bool &okay)
- static QVector< long > **convertToIntegerVector** (const QVariant &vector, bool &okay)
- static QVariantList **convertToVariantList** (const QVariant &vector, bool &okay)
- static QByteArray **getAsByteArray** (const QVariant &vector, bool &okay)
- static unsigned **getElementSize** (const QVariant &vector)
- static double **getDoubleValue** (const QVariant &vector, const int index, const double &defaultValue)
- static long **getIntegerValue** (const QVariant &vector, const int index, const long &defaultValue)
- static QVariant **getVariantValue** (const QVariant &vector, const int index, const QVariant &defaultValue)
- static bool **replaceValue** (QVariant &vector, const int index, const QVariant &value)
- static bool **registerMetaTypes** ()

The documentation for this class was generated from the following files:

- qeframework/protocol/QEVectorVariants.h
- qeframework/protocol/QEVectorVariants.cpp

9.301 QEWaterfall Class Reference

Inheritance diagram for QEWaterfall:



Public Slots

- void **setAngle** (const int [angle](#))
- void **setTraceGap** (const int [traceGap](#))
- void **setTraceWidth** (const int [traceWidth](#))
- void **setTraceColour** (const QColor &[traceColour](#))
- void **setMutableHue** (const bool [mutableHue](#))
- void **setBackgroundColour** (const QColor &[traceColour](#))
- void **setMargin** (const int [margin](#))

Public Member Functions

- [QWaterfall](#) (QWidget *parent=0)
- [QWaterfall](#) (const QString &dataVariableName, QWidget *parent=0)
- **QWaterfall** (const QString &dataVariableName, const QString &widthVariableName, QWidget *parent=0)
- virtual [~QWaterfall](#) ()
Destruction.
- int **getAngle** () const
- int **getTraceGap** () const
- int **getTraceWidth** () const
- QColor **getTraceColour** () const
- bool **getMutableHue** () const
- QColor **getBackgroundColour** () const
- int **getMargin** () const

Protected Member Functions

- QSize **sizeHint** () const
- bool **eventFilter** (QObject *watched, QEvent *event)
- void **updateDataVisulation** ()

Properties

- int [angle](#)
- int [traceGap](#)
- int [traceWidth](#)
- QColor [backgroundColour](#)
- QColor [traceColour](#)
- bool [mutableHue](#)
- int [margin](#)

Additional Inherited Members

9.301.1 Constructor & Destructor Documentation

9.301.1.1 QWaterfall() [1/2]

```
QWaterfall::QWaterfall (
    QWidget * parent = 0 ) [explicit]
```

Create without a variable. Use setDataPvName () and setVariableNameSubstitutions() to define a variable and, optionally, macro substitutions later.

9.301.1.2 QWaterfall() [2/2]

```
QWaterfall::QWaterfall (
    const QString & dataVariableName,
    QWidget * parent = 0 ) [explicit]
```

Create with a variable (s). A connection is automatically established. If macro substitutions are required, create without a variable and set the variable and macro substitutions after creation.

9.301.2 Property Documentation**9.301.2.1 angle**

```
int QWaterfall::angle [read], [write]
```

Offset display angle from vertical in degrees. Allowed: 0 .. 90 Default: 30

9.301.2.2 backgroundColour

```
QColor QWaterfall::backgroundColour [read], [write]
```

Default: white

9.301.2.3 margin

```
int QWaterfall::margin [read], [write]
```

Margin 0 to 40 - default is 4 Margin area shows the alarm colour, provided not inhibited by the value of the display↔ AlarmStateOption property.

9.301.2.4 mutableHue

```
bool QWaterfall::mutableHue [read], [write]
```

When true, the hue of each trace is increased by 12 modulo 360. Default: false

9.301.2.5 traceColour

QColor QEWaterfall::traceColour [read], [write]

Default: dark blue

9.301.2.6 traceGap

int QEWaterfall::traceGap [read], [write]

Trace gap/separation, expressed in pixels. Allowed: 1 .. 40 – max is arbitrary but sufficient Default: 5

9.301.2.7 traceWidth

int QEWaterfall::traceWidth [read], [write]

Pen width - 0 best auto guess. Allowed: 0 .. 10 – max is arbitrary but sufficient Default: 1

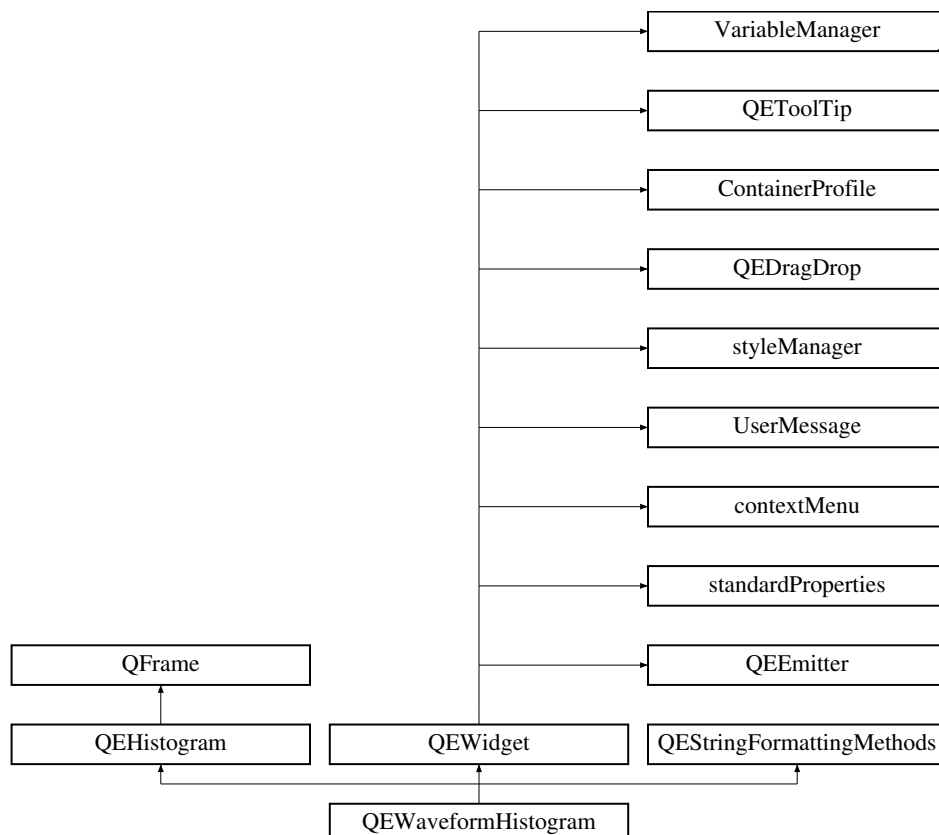
The documentation for this class was generated from the following files:

- qeframework/widgets/QE2DDataVisualisation/QEWaterfall.h
- qeframework/widgets/QE2DDataVisualisation/QEWaterfall.cpp

9.302 QEWaveformHistogram Class Reference

```
#include <QEWaveformHistogram.h>
```

Inheritance diagram for QEWaveformHistogram:



Public Types

- enum [ScaleModes](#) { [Manual](#) , [Auto](#) , [OperationalRange](#) }
- enum **OwnContextMenuOptions** { **QEWH_NONE** = CM_SPECIFIC_WIDGETS_START_HERE , **QEWH_PV_NAME_SELECT_DIALOG** , **QEWH_SUB_CLASS_WIDGETS_START_HERE** }

Public Slots

- void [setManagedVisible](#) (bool v)

Signals

- void [dbConnectionChanged](#) (const bool &isConnected)
Sent when the widget state updated following a channel connection change.
- void [dbValueChanged](#) ()
Sent when the widget is updated following a data change.

Public Member Functions

- **QEWaveformHistogram** (QWidget *parent=0)
- void **setScaleMode** (const [ScaleModes](#) scaleMode)
- [ScaleModes](#) **getScaleMode** () const
- void **setReadoutPrecision** (const int readoutPrecision)
- int **getReadoutPrecision** () const
- void **setReadoutFormat** (const [QE::Formats](#) format)
- [QE::Formats](#) **getReadoutFormat** () const
- void **setReadoutNotation** (const [QE::Notations](#) notation)
- [QE::Notations](#) **getReadoutNotation** () const
- void **setFullLengthArraySubscriptions** (const bool useFullLengthArraySubscriptions)
- bool **getFullLengthArraySubscriptions** () const

Protected Member Functions

- [QEChannel](#) * [createQcalItem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- void **dragEnterEvent** (QDragEnterEvent *event)
- void **dropEvent** (QDropEvent *event)
- void **mousePressEvent** (QMouseEvent *event)
- void **stringFormattingChange** ()
- QString **copyVariable** ()
- QVariant **copyData** ()
- void **paste** (QVariant v)
- QMenu * **buildContextMenu** ()
- void **contextMenuTriggered** (int selectedItemNum)
- void **onMouseIndexChanged** (const int index)
- void **onMouseIndexPressed** (const int index, const Qt::MouseButton button)
- void **setPvName** (const QString &pvName)

Properties

- QString **variable**
- QString [variableSubstitutions](#)
- [ScaleModes](#) **scaleMode**
- int **readoutPrecision**
- [QE::Formats](#) **readoutFormat**
- [QE::Notations](#) **readoutNotation**
- bool [fullLengthArraySubscriptions](#)
- bool [variableAsToolTip](#)
- bool [allowDrop](#)
- bool [visible](#)
- unsigned int [messageSourceId](#)
- QString [styleSheet](#)
- QString [defaultStyle](#)
- QString [userLevelUserStyle](#)
- QString [userLevelScientistStyle](#)
- QString [userLevelEngineerStyle](#)
- [QE::UserLevels](#) [userLevelVisibility](#)
- [QE::UserLevels](#) [userLevelEnabled](#)
- [QE::DisplayAlarmStateOptions](#) [displayAlarmStateOption](#)
- bool [oosAware](#)

Additional Inherited Members

9.302.1 Detailed Description

The [QEWaveformHistogram](#) class is a EPICS aware histogram widget. The value of, i.e. the length of each bar of the histogram is controlled by the corresponding element of an array, e.g. waveform, process variable. When the variable is defined (connected), the bar lengths are updated, and optionally the bar colours set to reflect the variable's severity status. The bar is 'grayed-out' when the variable is disconnected (although the bars retains their last known values/lengths). The histogram nature of the this widget is provided by a [QEHistogram](#) widget. The [QEWaveformHistogram](#) widget is tightly integrated with the base class [QEWidget](#), via [QEFrame](#), which provides generic support such as macro substitutions, drag/drop, and standard properties.

9.302.2 Member Enumeration Documentation

9.302.2.1 ScaleModes

```
enum QEWaveformHistogram::ScaleModes
```

Enumerator

Manual	Use property minimum/maximum to scale histogram.
Auto	Dynamically scale based on minimum/maximum displayed value.
OperationalRange	Use process variable operational range (LOPR/HOPR).

9.302.3 Member Function Documentation

9.302.3.1 setManagedVisible

```
void QEWaveformHistogram::setManagedVisible (
    bool v ) [inline], [slot]
```

Slot to set the visibility of a [QE](#) widget, taking into account the user level. Widget will be hidden if hidden by a call this slot, by will only be made visible by a call to this slot if the user level allows.

9.302.4 Property Documentation

9.302.4.1 allowDrop

```
bool QEWaveformHistogram::allowDrop [read], [write]
```

Allow drag/drops operations to this widget. Default is false. Any dropped text will be used as a new variable name.

9.302.4.2 defaultStyle

```
QString QEWaveformHistogram::defaultStyle [read], [write]
```

Style Sheet string to be applied before, i.e. lower priority than, any other style, e.g. alarm style and/or user level style. Default is an empty string.

9.302.4.3 displayAlarmStateOption

```
QE::DisplayAlarmStateOptions QEWaveformHistogram::displayAlarmStateOption [read], [write]
```

If 'Always' (default) widget will indicate the alarm state of any variable data it is displaying, including 'No Alarm' If 'WhenInAlarm' widget only indicate the alarm state of any variable data it is displaying if it is 'in alarm' or 'Out of Service'. If 'WhenInvalid' widget only indicate the alarm state of any variable data it is in the 'Invalid' alarm state or 'Out of Service'. If 'Never' widget will never indicate the alarm state of any variable data it is displaying. Typically the background colour is set to indicate the alarm state. Note, this property is included in the set of standard properties as it applies to most widgets. It will do nothing for widgets that don't display data.

9.302.4.4 fullLengthArraySubscriptions

```
bool QEWaveformHistogram::fullLengthArraySubscriptions [read], [write]
```

Older servers (3.14 era as I recall) do not support zero length subscriptions. Enable this property to work with ancient IOCs. Note: This applies to all arrays, no per plot control is provided.

9.302.4.5 messageSourceId

```
unsigned int QEWaveformHistogram::messageSourceId [read], [write]
```

Set the ID used by the message filtering system. Default is zero. Widgets or applications that use messages from the framework have the option of filtering on this ID. For example, by using a unique message source ID a [QELog](#) widget may be set up to only log messages from a select set of widgets.

9.302.4.6 oosAware

```
bool QEWaveformHistogram::oosAware [read], [write]
```

Indicates whether the widget should repond to the underlying PV(s) being declared Out of Service. The default oosAware value is most often true, however is false for some widgets.

9.302.4.7 styleSheet

```
QString QEWaveformHistogram::styleSheet [read], [write]
```

Hide style sheet from designer as style calculation by the [styleManager](#) and not directly settable per se. This also stops transient styles being saved to the ui file.

9.302.4.8 userLevelEnabled

```
QE::UserLevels QEWaveformHistogram::userLevelEnabled [read], [write]
```

Lowest user level at which the widget is enabled. Default is 'User'. Used when designing GUIs that allow access to more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programatically through setUserLevel() Widgets that are always accessible should be visible at 'User'. Widgets that are only accessible to scientists managing the facility should be visible at 'Scientist'. Widgets that are only accessible to engineers maintaining the facility should be visible at 'Engineer'.

9.302.4.9 userLevelEngineerStyle

```
QString QEWaveformHistogram::userLevelEngineerStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.302.4.10 userLevelScientistStyle

```
QString QEWaveformHistogram::userLevelScientistStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.302.4.11 `userLevelUserStyle`

```
QString QEWaveformHistogram::userLevelUserStyle [read], [write]
```

Style Sheet string to be applied when the widget is displayed in 'User' mode. Default is an empty string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red' This Style Sheet string will be applied by the [styleManager](#) class. Refer to the [styleManager](#) class for details about how this Style Sheet string will be merged with any pre-existing Style Sheet string and any Style Sheet strings generated during the display of data.

9.302.4.12 `userLevelVisibility`

```
QE::UserLevels QEWaveformHistogram::userLevelVisibility [read], [write]
```

Lowest user level at which the widget is visible. Default is 'User'. Used when designing GUIs that display more and more detail according to the user mode. The user mode is set application wide through the [QELogin](#) widget, or programmatically through `setUserLevel()` Widgets that are always visible should be visible at 'User'. Widgets that are only used by scientists managing the facility should be visible at 'Scientist'. Widgets that are only used by engineers maintaining the facility should be visible at 'Engineer'.

9.302.4.13 `variableAsToolTip`

```
bool QEWaveformHistogram::variableAsToolTip [read], [write]
```

Use the variable as the tool tip. Default is true. Tool tip property will be overwritten by the variable name.

9.302.4.14 `variableSubstitutions`

```
QString QEWaveformHistogram::variableSubstitutions [read]
```

Default macro substitutions. The default is no substitutions.

9.302.4.15 `visible`

```
bool QEWaveformHistogram::visible [read], [write]
```

Display the widget. Default is true. Setting this property false is usefull if widget is only used to provide a signal - for example, when supplying data to a [QELink](#) widget. Note, when false the widget will still be visible in Qt Designer.

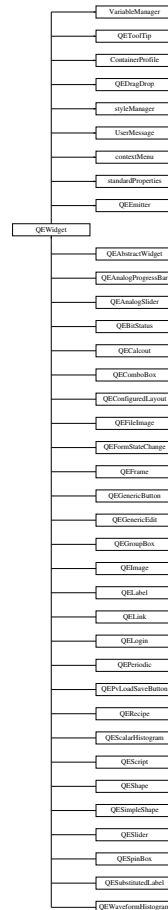
The documentation for this class was generated from the following files:

- `qeframework/widgets/QEHistogram/QEWaveformHistogram.h`
- `qeframework/widgets/QEHistogram/QEWaveformHistogram.cpp`

9.303 QEWidget Class Reference

```
#include <QEWidget.h>
```

Inheritance diagram for QEWidget:



Public Types

- enum [restorePhases](#) { **APPLICATION** = SaveRestoreSignal::RESTORE_APPLICATION , **FRAMEWORK** = SaveRestoreSignal::RESTORE_QEFRAMEWORK }
- Restore phases. When a widget's persistent data is restored, the restore occurs in two phases.*
- typedef QList< unsigned int > [ControlVariableIndicesSet](#)

Public Member Functions

- [QEWidget](#) (QWidget *ownerIn)
Constructor.
- virtual [~QEWidget](#) ()
Destructor.
- unsigned int [getMessageSourceId](#) () const
- void [setMessageSourceId](#) (unsigned int messageSourceId)
- void [processConnectionInfo](#) (bool isConnected, const unsigned int variableIndex=0)
- void [processAlarmInfo](#) (const [QCaAlarmInfo](#) &alarmInfo, const unsigned int variableIndex=0)

- virtual void [setVariableNameAndSubstitutions](#) (QString variableNameIn, QString variableName↔ SubstitutionsIn, unsigned int variableIndex)
- void [reestablishConnection](#) (unsigned int variableIndex)
- QFile * [openQEFile](#) (QString name, QIODevice::OpenModeFlag mode)
- QString [defaultFileLocation](#) () const
- QString [getFrameworkVersion](#) () const
- virtual void [saveConfiguration](#) (PersistenceManager *)
- virtual void [restoreConfiguration](#) (PersistenceManager *, restorePhases)
- QDomElement [getDetailedConfiguration](#) (const QString &tagName)
- void [setDetailedConfiguration](#) (const QDomElement &config)
- virtual void [scaleBy](#) (const int, const int)
- QWidget * [getQWidget](#) () const
- const QList< QCalInfo > [getQCalInfo](#) ()
- virtual QMenu * [getDefaultContextMenu](#) ()
Flag indicating this widget is running inside Qt's 'designer'.
- virtual int [getUserPrecision](#) ()
Return the Qt default context menu to add to the [QE](#) context menu.
- virtual int [getUserAlarmMin](#) ()
Return the widgets precision property if any (otherwise zero)
- virtual int [getUserAlarmMax](#) ()
Return the widget alarm minimum if any (otherwise zero)
- virtual bool [getAlarmSensitive](#) ()
Return the widget alarm maximum if any (otherwise zero)

Static Public Member Functions

- static QColor [getColor](#) (const QCaAlarmInfo &alarmInfo, const int saturation)
- static QFile * [findQEFile](#) (QString name, ContainerProfile *profile)
- static QFile * [findQEFile](#) (QString name)
- static void [doAction](#) (QWidget *searchPoint, QString widgetName, QString action, QStringList arguments, bool initialise, QAction *originator)
- static bool [inDesigner](#) ()

Protected Member Functions

- QEChannel * [createConnection](#) (unsigned int variableIndex)
Create a CA connection. Use default subscribe option. Return a QEChannel if successfull.
- QEChannel * [createConnection](#) (unsigned int variableIndex, const bool do_subscribe)
Create a CA connection. Use specified subscribe option. Return a QEChannel if successfull.
- QString [persistantName](#) (const QString &prefix) const
Returns a string that will not change between runs of the application (given the same configuration)
- virtual void [actionRequest](#) (QString, QStringList, bool, QAction *)
Perform a named action.
- void [setDrop](#) (QVariant drop)
Default get drop action.
- QVariant [getDrop](#) ()
Default set drop action.
- void [setControlIPV](#) (const unsigned int variableIndex)
- void [setControlPVs](#) (const ControlVariableIndicesSet &variableIndexList)
Nominate a single variable index as the control variable index.
- ControlVariableIndicesSet [getControlPVs](#) () const
Nominate a variable number of variable indicis as the control variable indies list.
- void [setAccessCursorStyle](#) ()
- void [startGui](#) (const QEActionRequests &request)
Update cursor style if all control variable indices are write inhibited.

Protected Attributes

- bool [useOwnPersistantName](#)

Support function for slot for launching a new gui. Used by [QE](#) buttons and [QEForm](#) as the default action for launching a gui.

9.303.1 Detailed Description

This class is used as a base for all CA aware wigets, such as [QELabel](#), [QESpinBox](#), etc. It manages common issues including creating a source of CA data updates, handling error, warning and status messages, and setting tool tips based on variable names.

Note, there is tight integration between the CA aware widget classes, this class, and its base classes, especially [VariableManager](#), [VariableNameManager](#) and [QEToolTip](#).

In particular, the [VariableManager](#) class manages QEChannel classes that stream updates to the CA aware widgets. But the [VariableManager](#) class, however, doesn't know how to format the data, or how the updates will be used. To resolve this, the [VariableManager](#) class asks a parent class (such as [QELabel](#)) to create the QEChannel class in what ever flavour it wants, by calling the virtual function `createQcalttem`. A [QELabel](#), for example, wants string updates so it creates a [QEString](#) which is based on a QEChannel class and formats all updates as strings.

The CA aware parent class (such as [QELabel](#)) defines a variable by calling `VariableNameManager::setVariableName()`. The `VariableNamePropertyManager` class calls the `establishConnection` function of the CA aware parent class, such as [QELabel](#) when it has a new variable name.

This class uses its base [QEToolTip](#) class to format tool tips. that class in turn calls the CA aware parent class (such as [QELabel](#)) directly to make use of a new tool tip.

After construction, a CA aware widget is activated (starts updating) by [VariableManager](#) calling its `establishConnection()` function in one of two ways:

1) The variable name or variable name substitutions is changed by calling `setVariableName` or `setVariableNameSubstitutions` respectively. These functions are in the [VariableNameManager](#) class. The `VariableNamePropertyManager` calls a virtual function `establishConnection()` which is implemented by the CA aware widget. This is how a CA aware widget is activated in 'designer'. It occurs when 'designer' updates the variable name property or variable name substitution property.

2) When an [QEForm](#) widget is created, resulting in a set of CA aware widgets being created by loading a UI file containing plugin definitions. After loading the plugin widgets, code in the [QEForm](#) class calls the `activate()` function in the [VariableManager](#) base of this class ([QEWidget](#)). The `activate()` function calls `establishConnection()` in the CA aware widget for each variable. This simulates what the `VariableNamePropertyManager` does as each variable name is entered (see 1, above, for details)

No matter which way a CA aware widget is activated, the `establishConnection()` function in the CA aware widget is called for each variable. The `establishConnection()` function asks this [QEWidget](#) base class, by calling the `createConnection()` function, to perform the tasks common to all CA aware widgets for establishing a stream of CA data.

The `createConnection()` function sets up the widget 'tool tip', then immedietly calls (through [VariableManager::createVariable](#)) the CA aware widget back asking it to create an object based on QEChannel. This object will supply a stream of CA update signals to the CA aware object in a form that it needs. For example a [QELabel](#) creates a [QEString](#) object. The [QEString](#) class is based on the QEChannel class and converts all update data to a strings which is required for updating a Qt label widget. This class stores the QEChannel based class.

After the `establishConnection()` function in the CA aware widget has called `createConnection()`, the remaining task of the `establishConnection()` function is to connect the signals of the newly created QEChannel based classes to its own slots so that data updates can be used. For example, a [QELabel](#) connects the 'stringChanged' signal from the [QEString](#) object to its `setLabelText` slot.

9.303.2 Member Typedef Documentation

9.303.2.1 ControlVariableIndicesSet

```
typedef QList<unsigned int> QEWidget::ControlVariableIndicesSet
```

Defines a list or set of variable indices. Used to indicate those variables that are considered control variables, and thus have their cursor updated to the forbidden cursor if/when write access is denied.

9.303.3 Member Function Documentation

9.303.3.1 defaultFileLocation()

```
QString QEWidget::defaultFileLocation ( ) const
```

Returns the default location to create files. Use this to create files in a consistent location

9.303.3.2 doAction()

```
void QEWidget::doAction (
    QWidget * searchPoint,
    QString widgetName,
    QString action,
    QStringList arguments,
    bool initialise,
    QAction * originator ) [static]
```

Find a [QE](#) widget and request an action. The widget hierarchy under a supplied widget is searched for a [QE](#) widget with a given name. If found the [QE](#) widget will attempt to carry out the requested action which consists of an action string and an argument list. This method allows an application to initiate [QE](#) widget activity. The QEGui application uses this mechanism when providing custom menus defined in XML files. The method returns true if the named widget was found. (The action was not necessarily performed, or even recognised by the widget)

9.303.3.3 findQEFile() [1/2]

```
QFile * QEWidget::findQEFile (
    QString name ) [static]
```

Static method that looks for a file in a standard set of locations and assumes a current published profile. Returns a pointer to a QFile which is the caller's responsibility to delete, or NULL if the file was not found.

9.303.3.4 findQEFile() [2/2]

```
QFile * QEWidget::findQEFile (
    QString name,
    ContainerProfile * profile ) [static]
```

Static method that looks for a file in a standard set of locations Returns a pointer to a QFile which is the caller's responsibility to delete, or NULL if the file was not found.

9.303.3.5 getColor()

```
QColor QEWidget::getColor (
    const QCaAlarmInfo & alarmInfo,
    const int saturation ) [static]
```

Static method to return a colour to update the widget's look to reflect the current alarm state Note, the color is determined by the alarmInfo class, but since that class is used in non gui applications, it can't return a QColor

9.303.3.6 getDetailedConfiguration()

```
QDomElement QEWidget::getDetailedConfiguration (
    const QString & tagName )
```

This method returns an xml dom element that reflects the detailed configuration of the widget. It leverages off saveConfiguration, however it provides a "raw" QDomElement elements; i.e. no file reading or writing. Note: this is back to front, i.e. saveConfiguration should leverage off this method.

9.303.3.7 getFrameworkVersion()

```
QString QEWidget::getFrameworkVersion ( ) const
```

Returns the [QE](#) framework that built this instance of the widget. On windows, the QEFramework DLL may be loaded twice with potentially different versions of it.

9.303.3.8 getMessageSourceId()

```
unsigned int QEWidget::getMessageSourceId ( ) const [inline]
```

Get the message source ID. The message source ID is used as part of the system where [QE](#) widgets can emit a message and have the right [QE](#) widget in the right form catch the message. Refer to the [UserMessage](#) class for further details.

9.303.3.9 getQCaInfo()

```
const QList< QCaInfo > QEWidget::getQCaInfo ( )
```

Return information about the data sources for this widget

9.303.3.10 getQWidget()

```
QWidget * QEWidget::getQWidget ( ) const
```

Get the QWidget that the parent of this [QEWidget](#) instance is based on. For example, the parent of a [QEWidget](#) might be a [QELabel](#), which is based on QLabel which is based on QWidget.

9.303.3.11 openQEFile()

```
QFile * QEWidget::openQEFile (
    QString name,
    QIODevice::OpenModeFlag mode )
```

Looks for a file in a standard set of locations (and opens the file)

9.303.3.12 processAlarmInfo()

```
void QEWidget::processAlarmInfo (
    const QCaAlarmInfo & alarmInfo,
    const unsigned int variableIndex = 0 )
```

This convenience function updates the alarm tool tip, and alarm status style if the displayAlarmState property is set to true - assumes the widget uses standard properties. This function is perhaps most usefull for single-variable widgets.

9.303.3.13 processConnectionInfo()

```
void QEWidget::processConnectionInfo (
    bool isConnected,
    const unsigned int variableIndex = 0 )
```

This convenience function updates the connection style and also clears/re-initialises the last saved severity and display alarm info state, as used by processAlarmInfo.

9.303.3.14 reestablishConnection()

```
void QEWidget::reestablishConnection (
    unsigned int variableIndex )
```

Essentially like setVariableNameAndSubstitutions, but without resetting the variable name or substitutions. This function calls establishConnection.

9.303.3.15 restoreConfiguration()

```
virtual void QEWidget::restoreConfiguration (
    PersistenceManager * ,
    restorePhases ) [inline], [virtual]
```

Service a request to restore the [QE](#) widget's configuration. A [QE](#) widget recover any configuration details from the [PersistenceManager](#). For example, a [QEStriptChart](#) may restore the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method. This is called twice with an incrementing restorePhase. Most widgets will miss the first call as they don't exist yet (they are created as part of the first phase).

Reimplemented in [QETable](#), [QEStriptChart](#), [QEScratchPad](#), [QEPvProperties](#), [QEPlotter](#), [QEFormGroupBox](#), [QEDynamicFormGrid](#), [QEDistribution](#), and [QECorrelation](#).

9.303.3.16 saveConfiguration()

```
virtual void QEWidget::saveConfiguration (
    PersistenceManager * ) [inline], [virtual]
```

Service a request to save the [QE](#) widget's current configuration. A widget may save any configuration details through the [PersistenceManager](#). For example, a [QEStriptChart](#) may save the variables being plotted. Many [QE](#) widgets do not have any persistant data requirements and do not implement this method.

Reimplemented in [QETable](#), [QEStriptChart](#), [QEScratchPad](#), [QEPvProperties](#), [QEPlotter](#), [QEFormGroupBox](#), [QEDynamicFormGrid](#), [QEDistribution](#), and [QECorrelation](#).

9.303.3.17 scaleBy()

```
virtual void QEWidget::scaleBy (
    const int ,
    const int ) [inline], [virtual]
```

Any [QEWidget](#) that requires additional scaling, i.e. above and beyond the standard scaling applied to size, minimum size, maximum size and font size, may override this function in order to perform any bespoke scaling need by the widget (for example see [QEShape](#)). The scaling is defined using a rational number specified by two integers (m, d). The first (m) parameter is the multiplier and the second (d) parameter is the divisor. For example, if m = 4 and d = 5, then an 80% scaling should be applied. And if m = 5 and d = 4, and a 125% scaling is required.

Reimplemented in [QEShape](#), and [QEPvProperties](#).

9.303.3.18 setDetailedConfiguration()

```
void QEWidget::setDetailedConfiguration (
    const QDomElement & config )
```

This method uses the given an xml dom element to configure the widget. Note: it leverages off restoreConfiguration to acheive this.

9.303.3.19 setMessageSourceId()

```
void QEWidget::setMessageSourceId (
    unsigned int messageSourceId ) [inline]
```

Set the message source ID. The message source ID is used as part of the system where [QE](#) widgets can emit a message and have the right [QE](#) widget in the right form catch the message. Refer to the [UserMessage](#) class for further details.

9.303.3.20 setVariableNameAndSubstitutions()

```
void QEWidget::setVariableNameAndSubstitutions (
    QString variableNameIn,
    QString variableNameSubstitutionsIn,
    unsigned int variableIndex ) [virtual]
```

Virtual function that may be implimented by users of [QEWidget](#) to update variable names and macro substitutions. A default is provided that is suitable in most cases.

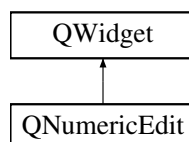
The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/QEWidget.h
- qeframework/widgets/QEWidget/QEWidget.cpp

9.304 QNumericEdit Class Reference

```
#include <QNumericEdit.h>
```

Inheritance diagram for QNumericEdit:



Public Slots

- void **setValue** (const double [value](#))
- void **setValue** (const int [value](#))

Signals

- void **valueChanged** (const double [value](#))
- void **valueChanged** (const int [value](#))
- void **valueEdited** (const double [value](#))
- void **valueEdited** (const int [value](#))
- void **returnPressed** ()
- void **editingFinished** ()

Public Member Functions

- [QNumericEdit](#) (QWidget *parent=0)
Construction.
- virtual [~QNumericEdit](#) ()
Destruction.
- double **getValue** () const
- void **setPrefix** (const QString &prefix)
- QString **getPrefix** () const
- void **setSuffix** (const QString &suffix)
- QString **getSuffix** () const
- QString **getCleanText** () const
- void **setLeadingZeros** (const int value)
- int **getLeadingZeros** () const
- void **setPrecision** (const int value)
- int **getPrecision** () const
- void **setForceSign** (const bool value)
- bool **getForceSign** () const
- void **setWrapValue** (const bool value)
- bool **getWrapValue** () const
- void **setNotation** (const [QE::Notations](#) notation)
- [QE::Notations](#) **getNotation** () const
- void **setMinimum** (const double value)
- double **getMinimum** () const
- void **setMaximum** (const double value)
- double **getMaximum** () const
- void **setRadix** (const [QE::Radices](#) value)
- [QE::Radices](#) **getRadix** () const
- void **setSeparator** (const [QE::Separators](#) value)
- [QE::Separators](#) **getSeparator** () const

Protected Member Functions

- void **focusInEvent** (QFocusEvent *event)
- bool **eventFilter** (QObject *obj, QEvent *event)

Protected Attributes

- [QFixedPointRadix](#) fpr

Properties

- bool [frame](#)
- QString [suffix](#)
- QString [prefix](#)
- Qt::Alignment [alignment](#)
- QString [cleanText](#)
- [QE::Notations](#) [notation](#)
- [QE::Radices](#) [radix](#)
- [QE::Separators](#) [separator](#)
- int [leadingZeros](#)
- int [precision](#)
- bool [forceSign](#)
- bool [wrapValue](#)
- double [minimum](#)
- double [maximum](#)
- double [value](#)

Friends

- class **QNumericEdit**

9.304.1 Detailed Description

This class provides a non EPICS aware numeric edit widget, somewhat akin to QDoubleSpin. However [QNumericEdit](#) provides a number of advantages over QDoubleSpin: a/ the former is restricted to a single spin value, where as [QNumericEdit](#) allows any digit to be selected and used as the spin value; b/ the former is restricted to decimal representation where as [QNumericEdit](#) allows for hexadecimal, octal and binary representations; c/ this [QNumericEdit](#) allow allows a 'thousands' separator character to be specified; and d/ by setting the precision to 0, it effectively becems akin to a QSpinBox.

9.304.2 Property Documentation

9.304.2.1 alignment

```
Qt::Alignment QNumericEdit::alignment [read], [write]
```

This property holds the alignment of the numeric edit. Both horizontal and vertical alignment is allowed here, Qt::AlignJustify will map to Qt::AlignLeft. By default, this property contains a combination of Qt::AlignRight and Qt::AlignVCenter.

9.304.2.2 cleanText

```
QString QNumericEdit::cleanText [read]
```

This property holds the displayed text. Not a property available to designer. It exludes any prefix/suffix.

9.304.2.3 forceSign

```
bool QNumericEdit::forceSign [read], [write]
```

When true, there is always a sign caractere "+" or "-". When false (default), no "+" sign when range of valued non-negative.

9.304.2.4 frame

```
bool QNumericEdit::frame [read], [write]
```

This property holds whether the numeric edit draws itself with a frame. If enabled (the default) the numeric edit draws itself inside a frame, otherwise the line edit draws itself without any frame.

9.304.2.5 leadingZeros

```
int QNumericEdit::leadingZeros [read], [write]
```

Specifies the number of leading zeros. Strictly speaking, this should be an unsigned int, but designer int properties editor much 'nicer'.

9.304.2.6 maximum

```
double QNumericEdit::maximum [read], [write]
```

Specify the maximum allowed value.

9.304.2.7 minimum

```
double QNumericEdit::minimum [read], [write]
```

Specify the minimum allowed value.

9.304.2.8 notation

```
QE::Notations QNumericEdit::notation [read], [write]
```

Notation used for formatting/editing. Default is fixed. The widget does not allow Automatic

9.304.2.9 precision

```
int QNumericEdit::precision [read], [write]
```

Precision used for the display and editing of numbers. The default is 4. Strictly speaking, this should be an unsigned int, but designer int properties editor much 'nicer'.

9.304.2.10 prefix

```
QString QNumericEdit::prefix [read], [write]
```

This property holds any fixed text (default is "") displayed before the numeric value.

9.304.2.11 radix

```
QE::Radices QNumericEdit::radix [read], [write]
```

Specify radix, default is Decimal.

9.304.2.12 separator

`QE::Separators QNumericEdit::separator [read], [write]`

Specify digit 'thousands' separator character, default is NoSeparator.

9.304.2.13 suffix

`QString QNumericEdit::suffix [read], [write]`

This property holds any fixed text (default is "") displayed after the numeric value.

9.304.2.14 value

`double QNumericEdit::value [read], [write]`

Specify the value after min/max

9.304.2.15 wrapValue

`bool QNumericEdit::wrapValue [read], [write]`

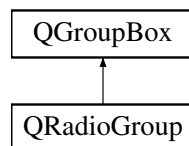
When set true (default is false) the numeric edit widget will wrap around from the max/min to min/max value.

The documentation for this class was generated from the following files:

- qeframework/widgets/QNumericEdit/QNumericEdit.h
- qeframework/widgets/QNumericEdit/QNumericEdit.cpp

9.305 QRadioGroup Class Reference

Inheritance diagram for QRadioGroup:



Public Types

- enum `ButtonStyles` { `Radio` , `Push` }

Public Slots

- void **setValue** (const int value)

Signals

- void **valueChanged** (const int value)

Public Member Functions

- [QRadioGroup](#) (QWidget *parent=0)
- [QRadioGroup](#) (const QString &title, QWidget *parent=0)
- virtual [~QRadioGroup](#) ()
Destruction.
- int **getMaximumButtons** () const
- int **getValue** () const
- void **setOwnTitle** (const QString &title)
- QString **getOwnTitle** () const
- void **setStrings** (const QStringList &strings)
- QStringList **getStrings** () const
- void **setColumns** (int columns)
- int **getColumns** () const
- void **setSpacing** (int spacing)
- int **getSpacing** () const
- void **setButtonStyle** (const [ButtonStyles](#) buttonStyle)
- [ButtonStyles](#) **getButtonStyle** () const
- void **setButtonOrder** (const [QE::GridOrders](#) buttonOrder)
- [QE::GridOrders](#) **getButtonOrder** () const
- int **findText** (const QString &text) const

Protected Member Functions

- QSize **sizeHint** () const
- bool **eventFilter** (QObject *watched, QEvent *event)

Properties

- QString **title**
- [QRadioGroup::ButtonStyles](#) **buttonStyle**
- [QE::GridOrders](#) **buttonOrder**
- int **columns**
- int **spacing**
- QStringList **strings**
- int **value**

9.305.1 Member Enumeration Documentation

9.305.1.1 ButtonStyles

enum [QRadioGroup::ButtonStyles](#)

Enumerations values used to select the button style. Whereas check box buttons can/do work, this option not provided as check boxes are not associated with the radio button, i.e. one and only one selected, paradigm.

Enumerator

Radio	Use radio buttons - the default.
Push	Use push buttons.

9.305.2 Constructor & Destructor Documentation

9.305.2.1 QRadioGroup() [1/2]

```
QRadioGroup::QRadioGroup (  
    QWidget * parent = 0 ) [explicit]
```

Create with default title.

9.305.2.2 QRadioGroup() [2/2]

```
QRadioGroup::QRadioGroup (  
    const QString & title,  
    QWidget * parent = 0 ) [explicit]
```

Create with a group title. Note "-" is interpreted as an empty title.

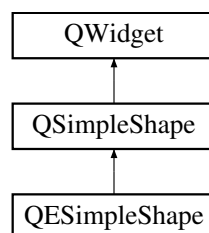
The documentation for this class was generated from the following files:

- qeframework/widgets/QERadioGroup/QRadioGroup.h
- qeframework/widgets/QERadioGroup/QRadioGroup.cpp

9.306 QSimpleShape Class Reference

```
#include <QSimpleShape.h>
```

Inheritance diagram for QSimpleShape:



Public Types

- enum **Shapes** {
noShape , **circle** , **ellipse** , **rectangle** ,
roundRectangle , **roundSquare** , **square** , **triangleUp** ,
triangleDown , **triangleLeft** , **triangleRight** , **triangleTopRight** ,
triangleBottomRight , **triangleBottomLeft** , **triangleTopLeft** , **diamond** ,
equalDiamond , **arrowUp** , **arrowDown** , **arrowLeft** ,
arrowRight , **crossHorizontal** , **crossVertical** , **hexagon** ,
octagon , **snakeHorizontal** , **snakeVertical** , **pentagon** ,
star , **plus** , **tick** , **cross** ,
roundpie , **pie** , **heart** }
- enum **TextFormats** { **FixedText** , **StateSet** , **PvText** , **LocalEnumeration** }

Public Slots

- void **setValue** (const bool [value](#))
- void **setValue** (const int [value](#))
- void **setShape** (const int [value](#))

Public Member Functions

- [QSimpleShape](#) (QWidget *parent=0)
- virtual [~QSimpleShape](#) ()
Destruction.
- int **getValue** () const
- void **setModulus** (const int [value](#))
- int **getModulus** () const
- void **setStateSet** (const QStringList &[stateSet](#))
- QStringList **getStateSet** () const
- void **setShape** (const Shapes [value](#))
- Shapes **getShape** () const
- void **setEdgeWidth** (const int [value](#))
- int **getEdgeWidth** () const
- void **setSemiCycles** (const int [value](#))
- int **getSemiCycles** () const
- void **setPercentSize** (const int [value](#))
- int **getPercentSize** () const
- void **setCentreAngle** (const int angle)
- int **getCentreAngle** () const
- void **setSpanAngle** (const int span)
- int **getSpanAngle** () const
- void **setEdgeStyle** (const Qt::PenStyle penStyle)
- Qt::PenStyle **getEdgeStyle** () const
- void **setEdgeColour** (const QColor [value](#))
- QColor **getEdgeColour** () const
- void **setTextFormat** (const [TextFormats](#) [value](#))
- [TextFormats](#) **getTextFormat** () const
- void **setFixedText** (const QString &[value](#))
- QString **getFixedText** () const
- void **setAlignment** (const Qt::Alignment [alignment](#))
- Qt::Alignment **getAlignment** () const
- void **setIndent** (const int [indent](#))

- int **getIndent** () const
- void **setFlashRate** (const QEScanTimers::ScanRates)
- QEScanTimers::ScanRates **getFlashRate** () const
- void **setFlashOffColour** (const QColor colour)
- QColor **getFlashOffColour** () const
- void **setIsActive** (const bool value)
- bool **getIsActive** () const
- void **setColourProperty** (const int slot, const QColor color)
- QColor **getColourProperty** (const int slot) const
- void **setFlashProperty** (int slot, const bool isFlashing)
- bool **getFlashProperty** (int slot) const

Protected Member Functions

- void **paintEvent** (QPaintEvent *event)
- virtual QString **getItemText** ()
- virtual QColor **getItemColour** ()

returns "" unless overridden.

Properties

- Shapes [shape](#)
- int [edgeWidth](#)
- Qt::PenStyle [edgeStyle](#)
Edge style - default is solid line.
- int [semiCycles](#)
- int [percentSize](#)
- int [centreAngle](#)
- int [spanAngle](#)
The span of the pie in degrees.
- int [value](#)
Shape value - range 0 to 15 - default is zero.
- int [modulus](#)
Shape value modulus - range 2 to 16 - default is 16.
- [TextFormats](#) [format](#)
- QString [fixedText](#)
User specified text - defaults to null string.
- QStringList [stateSet](#)
Text values used when format is LocalEnumeration.
- Qt::Alignment [alignment](#)
Text alignment.
- int [indent](#)
Text indent - the default is 6. Constrained to be ≥ -1 .
- QEScanTimers::ScanRates [flashRate](#)
- bool [isActive](#)
When isActive, the widgers is grayed-out, as if disabled. The default is true.
- QColor [edgeColour](#)
Edge colour - default is black.
- QColor [flashOffColour](#)
Flash offColour - default is clear.
- QColor [colour0](#)

Value (modulo modulus) used to select colour.

- QColor **colour1**
- QColor **colour2**
- QColor **colour3**
- QColor **colour4**
- QColor **colour5**
- QColor **colour6**
- QColor **colour7**
- QColor **colour8**
- QColor **colour9**
- QColor **colour10**
- QColor **colour11**
- QColor **colour12**
- QColor **colour13**
- QColor **colour14**
- QColor **colour15**
- bool **flash0**
- bool **flash1**
- bool **flash2**
- bool **flash3**
- bool **flash4**
- bool **flash5**
- bool **flash6**
- bool **flash7**
- bool **flash8**
- bool **flash9**
- bool **flash10**
- bool **flash11**
- bool **flash12**
- bool **flash13**
- bool **flash14**
- bool **flash15**

9.306.1 Detailed Description

This class is an basic simple shape widget. It has been refactored from [QESimpleShape](#) and will become the bases for [QESimpleShape](#).

9.306.2 Member Enumeration Documentation

9.306.2.1 TextFormats

```
enum QSimpleShape::TextFormats
```

Enumerator

FixedText	Use user specified fixed text (default)
StateSet	Use one of the stae Set values.
PvText	Use EPICS value agumented with units if selected.
GlobalEnumeration	Use specied enumeration values, PV value used to select item.

9.306.3 Constructor & Destructor Documentation

9.306.3.1 QSimpleShape()

```
QSimpleShape::QSimpleShape (
    QWidget * parent = 0 )
```

Construction

9.306.4 Property Documentation

9.306.4.1 centreAngle

```
int QSimpleShape::centreAngle [read], [write]
```

The pie centre angle in degrees. Zero represents a vertical centerline and angles increment clockwise.

9.306.4.2 edgeWidth

```
int QSimpleShape::edgeWidth [read], [write]
```

Edge width - range 0 to 20 - default is 1. If edge width set to 0, then shape colour used for edge/boarder colour.

9.306.4.3 flash0

```
bool QSimpleShape::flash0 [read], [write]
```

When the widget's state corresponds to N and flashN is set true, the widget will alternate its normal colour with the specified flashOffColour.

9.306.4.4 flashRate

```
QEScanTimers::ScanRates QSimpleShape::flashRate [read], [write]
```

Flash rate. The default value is Medium, i.e. 1Hz flash rate.

9.306.4.5 format

```
TextFormats QSimpleShape::format [read], [write]
```

Nominated text format

9.306.4.6 percentSize

```
int QSimpleShape::percentSize [read], [write]
```

Breadth of serpentine line as a percentage of widget height or width depending on wheather horizontal or vertical orientatiion. Range 1 to 50, default 10% Only applies to serpentine items.

9.306.4.7 semiCycles

```
int QSimpleShape::semiCycles [read], [write]
```

Number of semi-cycles - range 1 to 30 - default is 8. Only applies to serpentine items.

9.306.4.8 shape

```
Shapes QSimpleShape::shape [read], [write]
```

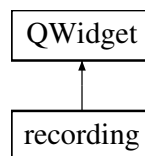
Nominated shape

The documentation for this class was generated from the following files:

- qeframework/widgets/QSimpleShape/QSimpleShape.h
- qeframework/widgets/QSimpleShape/QSimpleShape.cpp

9.307 recording Class Reference

Inheritance diagram for recording:



Signals

- void **byteArrayChanged** (const QByteArray &value, unsigned long dataSize, const [QCaAlarmInfo](#) &alarmInfo, const [QCaDateTime](#) &timeStamp, const unsigned int &variableIndex)
- void **playingBack** (bool playing)

Public Member Functions

- **recording** (QWidget *parent=0)
- bool **isRecording** ()
- void **recordImage** (const QByteArray &image, unsigned long dataSize, const [QCaAlarmInfo](#) &alarmInfo, const [QCaDateTime](#) &time)
- void **nextFrameDue** ()

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/recording.h
- qeframework/widgets/QEImage/recording.cpp

9.308 PersistenceManager::ResourceLocker Class Reference

Public Member Functions

- **ResourceLocker** (const QString &)
- **tryLock** (int=0)

The documentation for this class was generated from the following file:

- qeframework/widgets/QEWidget/persistenceManager.cpp

9.309 QEArchiveInterface::ResponseValues Struct Reference

Public Attributes

- QString **pvName**
- double **displayLow**
- double **displayHigh**
- int **precision**
- QString **units**
- unsigned int **elementCount**
- [QCaDataPointList](#) **dataPoints**

The documentation for this struct was generated from the following file:

- qeframework/archive/QEArchiveInterface.h

9.310 imageDisplayProperties::rgbPixel Struct Reference

Public Attributes

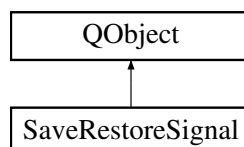
- unsigned char **p** [4]

The documentation for this struct was generated from the following file:

- qeframework/widgets/QEImage/brightnessContrast.h

9.311 SaveRestoreSignal Class Reference

Inheritance diagram for SaveRestoreSignal:



Public Types

- enum **saveRestoreOptions** { **SAVE** , **RESTORE_APPLICATION** , **RESTORE_QEFRAMEWORK** }

Signals

- void **saveRestore** (SaveRestoreSignal::saveRestoreOptions option)

Public Member Functions

- void **setOwner** ([PersistenceManager](#) *ownerIn)
- void [save](#) ()
- void [restore](#) ()

9.311.1 Member Function Documentation

9.311.1.1 [restore\(\)](#)

```
void SaveRestoreSignal::restore ( )
```

!! signal must be blocking

9.311.1.2 [save\(\)](#)

```
void SaveRestoreSignal::save ( )
```

!! signal must be blocking

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/persistenceManager.h
- qeframework/widgets/QEWidget/persistenceManager.cpp

9.312 QETwinScaleSelectDialog::ScaleLimit Struct Reference

Public Attributes

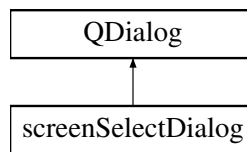
- double **min**
- double **max**

The documentation for this struct was generated from the following file:

- qeframework/common/QETwinScaleSelectDialog.h

9.313 screenSelectDialog Class Reference

Inheritance diagram for screenSelectDialog:



Public Types

- enum **screens** { **PRIMARY_SCREEN** = -3 , **THIS_SCREEN** = -2 , **ALL_SCREEN** = -1 }

Public Member Functions

- screenSelectDialog** (int numScreens, QWidget *parent=0)
- int **getScreenNum** ()

Static Public Member Functions

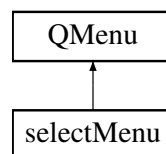
- static bool **getFullscreenGeometry** (QWidget *target, QRect &geom)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/screenSelectDialog.h
- qeframework/widgets/QEImage/screenSelectDialog.cpp

9.314 selectMenu Class Reference

Inheritance diagram for selectMenu:



Public Member Functions

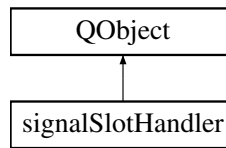
- selectMenu** (QWidget *parent=0)
- imageContextMenu::imageContextMenuOptions **getSelectOption** (const QPoint &pos)
- void **enable** (imageContextMenu::imageContextMenuOptions option, bool state)
- bool **isEnabled** (imageContextMenu::imageContextMenuOptions option)
- void **setChecked** (const int mode)
- void **setItemText** (imageContextMenu::imageContextMenuOptions option, QString title)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/selectMenu.h
- qeframework/widgets/QEImage/selectMenu.cpp

9.315 signalSlotHandler Class Reference

Inheritance diagram for signalSlotHandler:



Public Slots

- void **saveRestore** (SaveRestoreSignal::saveRestoreOptions option)

Public Member Functions

- void **setOwner** (QEWidget *ownerIn)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/QEWidget.h
- qeframework/widgets/QEWidget/QEWidget.cpp

9.316 SourceSpec Class Reference

Public Types

- enum **Constants** { **noKey** = -1 , **numberOfKeys** = 4 }

Public Member Functions

- QList< int > **keys** () const
- bool **contains** (const int key) const
- KeyTimeSpec **value** (const int key) const
- bool **insert** (const KeyTimeSpec spec)

Public Attributes

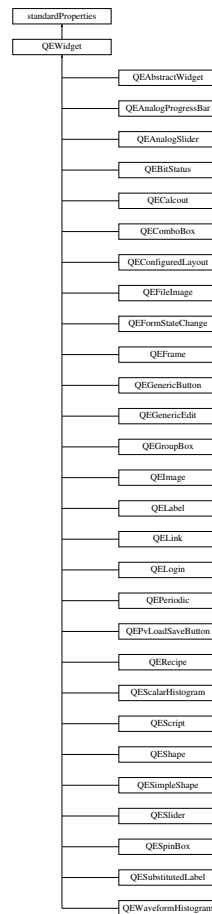
- QEArchiveInterfaceManager * **interfaceManager**
- KeyTimeSpec **keyToTimeSpecLookUp** [numberOfKeys]

The documentation for this class was generated from the following file:

- qeframework/archive/QEArchiveManager.cpp

9.317 standardProperties Class Reference

Inheritance diagram for standardProperties:



Public Member Functions

- **standardProperties** (QWidget *ownerIn)
- [QE::UserLevels](#) **getUserLevelVisibility** () const
- void **setUserLevelVisibility** ([QE::UserLevels](#) level)
- [QE::UserLevels](#) **getUserLevelEnabled** () const
- void **setUserLevelEnabled** ([QE::UserLevels](#) level)
- bool **getApplicationEnabled** () const
- void **setApplicationEnabled** (bool state)
- void **setRunVisible** (bool visibleIn)
- bool **getRunVisible** () const
- void **setDisplayAlarmStateOption** ([QE::DisplayAlarmStateOptions](#) displayAlarmStateIn)
- [QE::DisplayAlarmStateOptions](#) **getDisplayAlarmStateOption** () const
- void **setOosAware** (const bool oosAware)
- bool **getOosAware** () const
- bool **getUseAlarmState** (const [QCaAlarmInfo](#) &alarmInfo) const

Protected Member Functions

- void **checkVisibilityEnabledLevel** ([QE::UserLevels](#) level)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/standardProperties.h
- qeframework/widgets/QEWidget/standardProperties.cpp

9.318 QCaDataPointList::Statistics Struct Reference

Public Attributes

- bool **isDefined**
- double **mean**
- double **stdDeviation**
- double **slope**
- double **integral**
- double **minimum**
- double **maximum**
- double **initialValue**
- double **finalValue**

The documentation for this struct was generated from the following file:

- qeframework/data/QCaDataPoint.h

9.319 QEArchiveAccess::Status Struct Reference

Public Attributes

- QString **hostName**
- int **portNumber**
- QString **endPoint**
- QEArchiveInterface::States **state**
- int **available**
- int **read**
- int **numberPVs**
- int **pending**

The documentation for this struct was generated from the following file:

- qeframework/archive/QEArchiveAccess.h

9.320 QEPvLoadSaveCommon::StatusSummary Struct Reference

Public Attributes

- int **isEqualCount**
- int **isNotEqualCount**
- int **isNotApplicableCount**

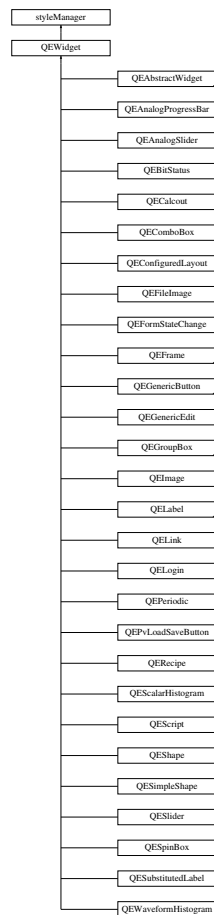
The documentation for this struct was generated from the following file:

- qeframework/widgets/QEPvLoadSave/QEPvLoadSaveCommon.h

9.321 styleManager Class Reference

```
#include <styleManager.h>
```

Inheritance diagram for styleManager:



Public Member Functions

- **styleManager** (QWidget *ownerIn)
 - void [setStyleDefault](#) (QString style)
 - QString [getStyleDefault](#) () const
 - void [setStyleUser](#) (QString style)
 - QString [getStyleUser](#) () const
 - void [setStyleScientist](#) (QString style)
 - QString [getStyleScientist](#) () const
 - void [setStyleEngineer](#) (QString style)
 - QString [getStyleEngineer](#) () const
 - void [updateDataStyle](#) (QString style)
 - void [updateStatusStyle](#) (QString style)
 - void [updatePropertyStyle](#) (QString style)
- For example, a style string is used to set [QE](#) button text alignment.*
- void [updateConnectionStyle](#) (bool connected)
 - void [styleUserLevelChanged](#) ([QE::UserLevels](#) levelIn)

Friends

- class **changeEventFilter**

9.321.1 Detailed Description

This class adds common style support to all [QE](#) widgets if required.

Standard properties for all [QE](#) widgets specify a style to be applied for user, scientist, and engineer mode. Also [QE](#) widgets can specify data or status related style changes.

The syntax for all Style Sheet strings used by this class is the standard Qt Style Sheet syntax. For example, 'background-color: red'. Refer to Qt Style Sheets Reference for full details.

Note, as well as the large number of defined Style Sheet properties, the Style Sheet syntax allows setting any widget property using the 'qproperty' property. For example 'qproperty-geometry:rect(10 10 100 100);' Caution, any amount of weird behaviour can be effected using Style Sheet strings. Without carefull consideration they should only be used for simple visualisation effects such as altering the background color.

All [QE](#) widgets (eg, [QELabel](#), [QELineEdit](#)) have an instance of this class as they based on [QEWidget](#) which itself uses this class as a base class.

To use the functionality provided by this class, [QE](#) widgets must include the following:

- Q_PROPERTY(QString styleSheet READ styleSheet WRITE setStyleSheet DESIGNABLE false)
- Q_PROPERTY(QString defaultStyle READ getStyleDefault WRITE setStyleDefault)
- Q_PROPERTY(QString userLevelUserStyle READ getStyleUser WRITE setStyleUser)
- Q_PROPERTY(QString userLevelScientistStyle READ getStyleScientist WRITE setStyleScientist)
- Q_PROPERTY(QString userLevelEngineerStyle READ getStyleEngineer WRITE setStyleEngineer)

To use this class to manage style changes related to data and status the [QE](#) widget must include the following:

- On presentation of new data call [updateDataStyle\(\)](#) with the Style Sheet string related to the new data (if any).
- On change of data status, call [updateStatusStyle\(\)](#) with the Style Sheet string related to the new status (if any).

Note, this class notes the initial style when instantiated and uses that style as the base style for all style changes. This means any style changes not performed through this class will be lost the next time this class changes the style.

Note, the stylesheet built by this class is not actually applied if the widget being managed is disabled. Instead it is noted and applied if and when the widget becomes enabled. Changes that affect the style will still cause a regeneration of the style while the widget is disabled, but the updated style will not be applied until the widget is enabled.

9.321.2 Member Function Documentation

9.321.2.1 `getStyleDefault()`

```
QString styleManager::getStyleDefault ( ) const
```

Get the default Style Sheet string.

9.321.2.2 `getStyleEngineer()`

```
QString styleManager::getStyleEngineer ( ) const
```

Get the Style Sheet string to be applied when the widget is displayed in 'Engineer' mode.

9.321.2.3 `getStyleScientist()`

```
QString styleManager::getStyleScientist ( ) const
```

Get the Style Sheet string to be applied when the widget is displayed in 'Scientist' mode.

9.321.2.4 `getStyleUser()`

```
QString styleManager::getStyleUser ( ) const
```

Get the Style Sheet string to be applied when the widget is displayed in 'User' mode.

9.321.2.5 `setStyleDefault()`

```
void styleManager::setStyleDefault (
    QString style )
```

Set the default Style Sheet string. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red'.

9.321.2.6 setStyleEngineer()

```
void styleManager::setStyleEngineer (
    QString style )
```

Set the Style Sheet string to be applied when the widget is displayed in 'Engineer' mode. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red'

9.321.2.7 setStyleScientist()

```
void styleManager::setStyleScientist (
    QString style )
```

Set the Style Sheet string to be applied when the widget is displayed in 'Scientist' mode. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red'

9.321.2.8 setStyleUser()

```
void styleManager::setStyleUser (
    QString style )
```

Set the Style Sheet string to be applied when the widget is displayed in 'User' mode. The syntax is the standard Qt Style Sheet syntax. For example, 'background-color: red'.

9.321.2.9 styleUserLevelChanged()

```
void styleManager::styleUserLevelChanged (
    QE::UserLevels levelIn )
```

Set the current user level.

9.321.2.10 updateConnectionStyle()

```
void styleManager::updateConnectionStyle (
    bool connected )
```

Set the Style Sheet string to be applied to reflect the current connection state (connected or disconnected) of the current data. For example, a disconnected value may be greyed out.

9.321.2.11 updateDataStyle()

```
void styleManager::updateDataStyle (
    QString style )
```

Set the Style Sheet string to be applied to reflect an aspect of the current data. For example, a value over a high limit may be displayed in red.

9.321.2.12 updatePropertyStyle()

```
void styleManager::updatePropertyStyle (
    QString style )
```

For example, a style string is used to set [QE](#) button text alignment.

Set the Style Sheet string to be applied to implement a widget property.

9.321.2.13 updateStatusStyle()

```
void styleManager::updateStatusStyle (
    QString style )
```

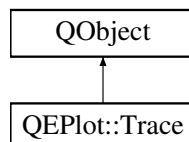
Set the Style Sheet string to be applied to reflect an aspect of the current status. For example, invalid data may be displayed with a white background.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/styleManager.h
- qeframework/widgets/QEWidget/styleManager.cpp

9.322 QEPlot::Trace Class Reference

Inheritance diagram for QEPlot::Trace:



Public Member Functions

- **Trace** (const int instanceIn, [QEPlot](#) *parent)
- void **reset** ()
- void **requestArchiveData** ()
- void **setArchiveData** (const [QCaDataPointList](#) &archiveData)

Public Attributes

- const [QEPlot](#) * **owner**
- const int **instance**
- int **width**
- bool **isVisible**
- [QEArchiveAccess](#) **archiveAccess**
- [QCaDataPointList](#) **scalarData**
- `QVector< double >` **ydata**
- `QColor` **color**
- `QwtPlotCurve::CurveStyle` **style**
- `QString` **legend**
- bool **isInUse**
- bool **isConnected**
- bool **isFirstUpdate**
- bool **isWaveform**
- bool **dataSizeDefined**
- int **dataSize**
- [QCaVariableNamePropertyManager](#) **dnpm**
- [QCaVariableNamePropertyManager](#) **snpm**

The documentation for this class was generated from the following file:

- `qeframework/widgets/QEPlot/QEPlot.cpp`

9.323 `userInfoStruct` Class Reference

Public Attributes

- bool **enable**
- double **value1**
- double **value2**
- `QString` **elementText**

The documentation for this class was generated from the following file:

- `qeframework/widgets/QEPeriodic/QEPeriodic.h`

9.324 `QEPeriodic::userInfoStructArray` Struct Reference

Public Attributes

- [userInfoStruct](#) **array** [NUM_ELEMENTS]

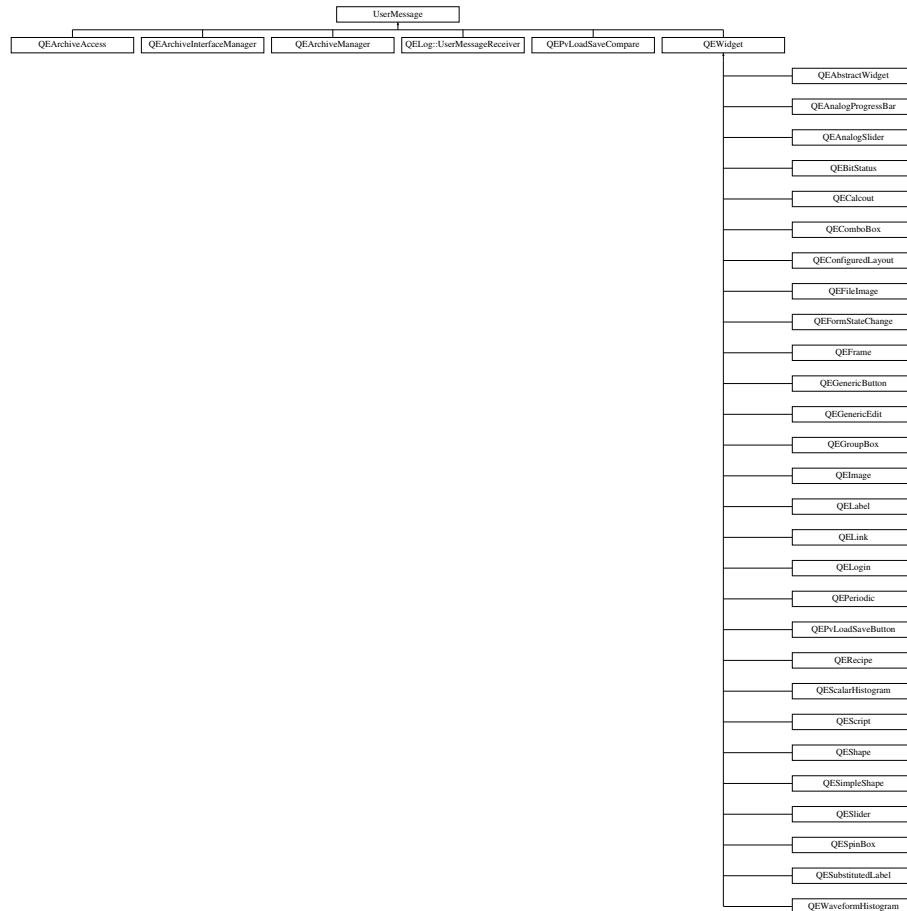
The documentation for this struct was generated from the following file:

- `qeframework/widgets/QEPeriodic/QEPeriodic.h`

9.325 UserMessage Class Reference

```
#include <UserMessage.h>
```

Inheritance diagram for UserMessage:



Public Member Functions

- void [setSourceId](#) (unsigned int sourceId)
Set the source ID (the ID set up by the GUI designer, usually matched to the source ID of logging widgets)
- void [setFormId](#) (unsigned int formId)
Set the form ID (the the same ID for all sibling widgets within an [QEForm](#) widget)
- void [setFormFilter](#) ([QE::MessageFilterOptions](#) formFilterIn)
Set the message filtering applied to the form ID.
- void [setSourceFilter](#) ([QE::MessageFilterOptions](#) sourceFilterIn)
Set the message filtering applied to the source ID.
- unsigned int [getSourceId](#) () const
Get the source ID (the ID set up by the GUI designer, usually matched to the source ID of logging widgets).
- unsigned int [getFormId](#) () const
Get the form ID (the the same ID for all sibling widgets within an [QEForm](#) widget)
- [QE::MessageFilterOptions](#) [getFormFilter](#) () const
Get the message filtering applied to the form ID.
- [QE::MessageFilterOptions](#) [getSourceFilter](#) () const

- *Get the message filtering applied to the source ID.*
- void [setChildFormId](#) (unsigned int)
 - *Set the for ID of all widgets that are children of this widget.*
- unsigned int [getChildFormId](#) () const
 - *Get the for ID of all widgets that are children of this widget.*
- unsigned int [getNextMessageFormId](#) () const
 - *Generate a new form ID for all widgets in a new form.*
- void [sendMessage](#) (QString message, [message_types](#) type=[message_types](#)(MESSAGE_TYPE_INFO))
 - *Send a message to the user.*
- void [sendMessage](#) (QString message, QString source, [message_types](#) type=[message_types](#)(MESSAGE_TYPE_INFO))
 - *Send a message to the user with a source reference.*
- QString [getMessageTypeName](#) ([message_types](#) type)
 - *Convenience function to provide string names for each message type.*
- virtual void [newMessage](#) (QString, [message_types](#))
 - *Virtual function to pass messages to derived classes (typically logging widgets or application windows)*

Friends

- class **UserMessageSlot**
- class **UserMessageSignal**

9.325.1 Detailed Description

A class to manage user messages.

This class passes messages between widgets and application code

This class is used as a base class.

Messages are sent by calling [sendMessage\(\)](#) Messages are received by implementing [newMessage\(\)](#) in the derived class.

Messages can be filtered based on a source ID or a form ID

The derived widget is free to set the source ID to any value

Derived form widgets ([QEForm](#)) get a unique form ID using [getNextMessageFormId\(\)](#) (as well as being able to set a source ID like any other [QE](#) widget) and pass this unique form ID to all widgets within the form using the [ContainerProfile](#) class.

Messages sent by a [QE](#) widget are received by all [QE](#) widgets and can filter the messages required by form ID and source ID. The form ID is under the management of the [QEForm](#) widget, the source ID is under the control of the GUI designer.

The [QEForm](#) widget does not display messages, but re-send them using its own form ID. Read on to see how this can be used.

Widgets that generate messages, and widgets (or application code) that use messages can be set up as follows:

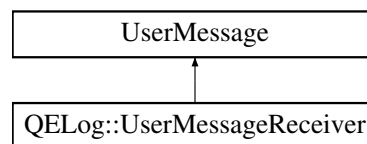
- Application wide logging: An application with a single log window can base a class on the [UserMessage](#) class and set up filtering to receive all messages. An application with log messages for separate windows containing [QForm](#) widgets (such as [QEGui](#)) can base each window class on the [UserMessage](#) class, then set up filtering for the appropriate form ID.
- Logging within a [QForm](#). A logging widget can be set to filter matching on the current form and so will pick up messages from any sibling widget. This includes messages from a sibling widget which is a nested [QForm](#). Whatever messages that nested form is set to receive, it will resend to its siblings. For example, if it is set to receive messages from the widgets it contains, these are resent up one level to the main form. If messages are dealt with within the nested [QForm](#) (for example, it may have its own logging [QE](#) widget) then the nested [QForm](#) could be set up not to filter and resend any messages.

The documentation for this class was generated from the following files:

- `qeframework/widgets/QEWidget/UserMessage.h`
- `qeframework/widgets/QEWidget/UserMessage.cpp`

9.326 QLog::UserMessageReceiver Class Reference

Inheritance diagram for `QLog::UserMessageReceiver`:



Public Member Functions

- void **registerLogWidget** ([QLog](#) *logWidget)
- void **deregisterLogWidget** ([QLog](#) *logWidget)

Protected Member Functions

- void **newMessage** (QString message, [message_types](#) mt)
Virtual function to pass messages to derived classes (typically logging widgets or application windows)

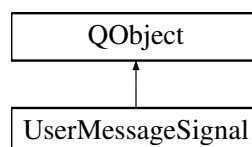
The documentation for this class was generated from the following file:

- `qeframework/widgets/QLog/QLog.cpp`

9.327 UserMessageSignal Class Reference

```
#include <UserMessage.h>
```

Inheritance diagram for `UserMessageSignal`:



Signals

- void [message](#) (QString msg, [message_types](#) type, unsigned int formId, unsigned int sourceId, [UserMessage](#) *originator)

Emit a message signal. Any widget based on the [UserMessage](#) class can receive these messages, filtered on formId and sourceId.

Public Member Functions

- void [sendMessage](#) (QString msg, [message_types](#) type, unsigned int formId, unsigned int sourceId, [UserMessage](#) *originator)

Send a message to all widgets based on the [UserMessage](#) class.

9.327.1 Detailed Description

Class used to send message signals. Used only within UserMessage.cpp A single instance of this class is shared by all instances of the [UserMessage](#) class. This allows every [UserMessage](#) class instance to connect to a single source of messages

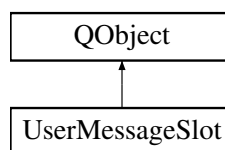
The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/UserMessage.h
- qeframework/widgets/QEWidget/UserMessage.cpp

9.328 UserMessageSlot Class Reference

```
#include <UserMessage.h>
```

Inheritance diagram for UserMessageSlot:



Public Slots

- void [message](#) (QString msg, [message_types](#) type, unsigned int formId, unsigned int sourceId, [UserMessage](#) *originator)

A message has been received.

Public Member Functions

- void [setOwner](#) ([UserMessage](#) *ownerIn)

Set the [UserMessage](#) class this is a part of.

9.328.1 Detailed Description

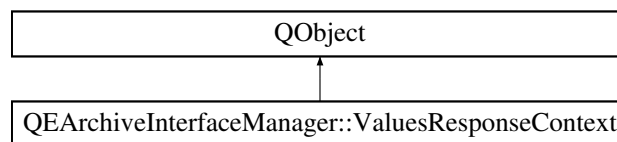
Class used to receive message signals. Used only within UserMessage.cpp An instance of this class is created by all instances of the [UserMessage](#) class. The [UserMessage](#) class uses an instance of this class to receive messages so it does not have to be based on QObject itself. This is required as derived classes generally need to be also based on another object derived from QObject (and QObject can only be the base of a single base class)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/UserMessage.h
- qeframework/widgets/QEWidget/UserMessage.cpp

9.329 QEArchiveInterfaceManager::ValuesResponseContext Class Reference

Inheritance diagram for QEArchiveInterfaceManager::ValuesResponseContext:



Public Member Functions

- **ValuesResponseContext** (const [QEArchiveInterfaceManager](#) *aim, const qint64 uniqueIn)

Public Attributes

- const [QEArchiveInterfaceManager](#) * **archiveInterfaceManager**
- const qint64 **unique**

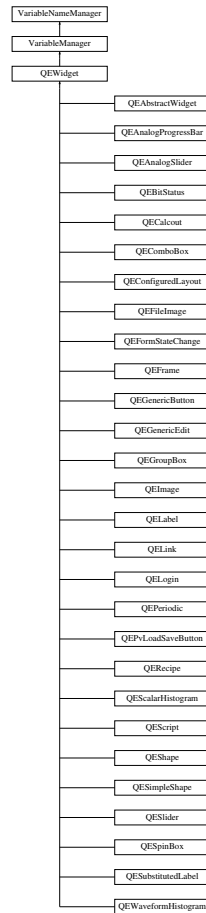
The documentation for this class was generated from the following file:

- qeframework/archive/QEArchiveInterfaceManager.cpp

9.330 VariableManager Class Reference

```
#include <VariableManager.h>
```

Inheritance diagram for VariableManager:



Public Member Functions

- [VariableManager](#) ()
Constructor.
- virtual [~VariableManager](#) ()
Destructor.
- void [activate](#) ()
- void [deactivate](#) ()
- [QEChannel](#) * [getQcaltem](#) (unsigned int variableIndex) const
- void [readNow](#) ()
- virtual void [writeNow](#) ()
- int * [getDisconnectedCountRef](#) () const
- int * [getConnectedCountRef](#) () const

Protected Member Functions

- void [setNumVariables](#) (const unsigned int numVariablesIn)
Set the number of variables that will stream data updates to the widget. Default of 1 if not called.
- unsigned int [getNumVariables](#) () const
Get the number of variables streaming data updates to the widget.
- [QEChannel](#) * [createVariable](#) (const unsigned int variableIndex, const bool do_subscribe)
Return a QEChannel if successfull.
- virtual [QEChannel](#) * [createQcaltem](#) (unsigned int variableIndex)
Function to create a appropriate superclass of QEChannel to stream data updates.
- virtual void [establishConnection](#) (unsigned int variableIndex)
Create a CA connection and initiates updates if required.
- virtual void [activated](#) ()
Do any post-all-widgets-constructed, i.e. activated stuff.
- virtual void [deactivated](#) ()
Do any post deactivated stuff.
- void [deleteQcaltem](#) (const unsigned int variableIndex, const bool disconnect)

Protected Attributes

- bool [subscribe](#)
Flag if data updates should be requested (default value used by [QEWidget](#))

9.330.1 Detailed Description

This class manages CA process variables for the [QEWidget](#) class. The [QEWidget](#) class provides a lot of generic support for all [QE](#) widgets (such as [QELabel](#), [QESpinBox](#), etc), including managing CA data sources. It manages CA and PVA data sources through this class.

Refer to the [QEwidget](#) class description for an overview of how [QE](#) widgets use [QEWidget](#).

The [VariableManager](#) class manages [QEChannel](#) classes that stream updates to the CA aware widgets. But the [VariableManager](#) class, however, doesn't know how to format the data, or how the updates will be used.

To resolve this, the [VariableManager](#) class asks a parent class (such as [QELabel](#)) to create the [QEChannel](#) class in what ever flavour it wants, by calling the virtual function [createQcaltem](#). A [QELabel](#), for example, wants string updates so it creates a [QEString](#) which is based on a [QEChannel](#) class and formats all updates as strings.

The CA aware parent class (such as [QELabel](#)) defines a variable by calling [VariableNameManager::setVariableName\(\)](#). The [VariableNamePropertyManager](#) class calls the [establishConnection](#) function of the CA aware parent class, such as [QELabel](#) when it has a new variable name.

After construction, a CA/PVA aware widget is activated (starts updating) by [VariableManager](#) calling its [establishConnection\(\)](#) function in one of two ways:

1) The variable name or variable name substitutions is changed by calling [setVariableName](#) or [setVariableNameSubstitutions](#) respectively. These functions are in the [VariableNameManager](#) class. The [VariableNamePropertyManager](#) calls a virtual function [establishConnection\(\)](#) which is implemented by the CA aware widget. This is how a CA aware widget is activated in 'designer'. It occurs when 'designer' updates the variable name property or variable name substitution property.

2) When an [QEForm](#) widget is created, resulting in a set of CA/PVA aware widgets being created by loading a UI file containing plugin definitions. After loading the plugin widgets, code in the [QEForm](#) class calls the [activate\(\)](#)

function in the [VariableManager](#) base of this class (QEWiget). The [activate\(\)](#) function calls [establishConnection\(\)](#) in the CA aware widget for each variable. This simulates what the VariableNamePropertyManager does as each variable name is entered (see 1, above, for details)

No matter which way a CA/PVA aware widget is activated, the [establishConnection\(\)](#) function in the CA aware widget is called for each variable. The [establishConnection\(\)](#) function asks this [QEWWidget](#) base class, by calling the [createConnection\(\)](#) function, to perform the tasks common to all CA/PVA aware widgets for establishing a stream of CA data.

The [createVariable\(\)](#) function calls the CA/PVA aware widget back asking it to create an object based on [QEChannel](#). This object will supply a stream of CA/PVA update signals to the CA/PVA aware object in a form that it needs. For example a [QELabel](#) creates a [QEString](#) object. The [QEString](#) class is based on the [QEChannel](#) class and converts all update data to a strings which is required for updating a Qt label widget. This class stores the [QEChannel](#) based class.

After the [establishConnection\(\)](#) function in the CA/PVA aware widget has called [createConnection\(\)](#), the remaining task of the [establishConnection\(\)](#) function is to connect the signals of the newly created [QEChannel](#) based classes to its own slots so that data updates can be used. For example, a [QELabel](#) connects the 'stringChanged' signal from the [QEString](#) object to its [setLabelText](#) slot.

9.330.2 Member Function Documentation

9.330.2.1 activate()

```
void VariableManager::activate ( )
```

Initiate updates. Called after all configuration is complete. Note: This function invokes the virtual [activated\(\)](#) function.

9.330.2.2 createVariable()

```
QEChannel * VariableManager::createVariable (
    const unsigned int variableIndex,
    const bool do_subscribe )    [protected]
```

Return a [QEChannel](#) if successfull.

Parameters

<i>variableIndex</i>	Create a CA/PVA connection. <i>do_subscribe</i> indicated if updates should be requested.
----------------------	---

9.330.2.3 deactivate()

```
void VariableManager::deactivate ( )
```

Terminates updates. This has been provided for third party (non QEGui) applications using the framework. Note: This function invokes the virtual [deactivated\(\)](#) function.

9.330.2.4 deleteQcaItem()

```
void VariableManager::deleteQcaItem (
    const unsigned int variableIndex,
    const bool disconnect ) [protected]
```

Note: we use deleteLater() here (and above). This is because if a widget calls (re-)establishConnection from within a slot function responding to a signal from this QEChannel object (as does [QEImage](#)), this can often cause a segmentation fault.

Parameters

<i>variableIndex</i>	Delete a stream of CA/PVA updates
----------------------	-----------------------------------

9.330.2.5 getConnectedCountRef()

```
int * VariableManager::getConnectedCountRef ( ) const
```

Return references to the current count of connections. The plugin library (and therefore the static connection and disconnection counts) can be mapped twice (on Windows at least). So it is no use just referencing these static variables from an application if the widgets of interest have been created by the UI Loader. This function can be called on any widget loaded by the UI loader and the reference returned can be used to get counts for all widgets loaded by the UI loader.

9.330.2.6 getDisconnectedCountRef()

```
int * VariableManager::getDisconnectedCountRef ( ) const
```

Return references to the current count of disconnections. The plugin library (and therefore the static connection and disconnection counts) can be mapped twice (on Windows at least). So it is no use just referencing these static variables from an application if the widgets of interest have been created by the UI Loader. This function can be called on any widget loaded by the UI loader and the reference returned can be used to get counts for all widgets loaded by the UI loader.

9.330.2.7 getQcaItem()

```
QEChannel * VariableManager::getQcaItem (
    unsigned int variableIndex ) const
```

Return a reference to one of the QEChannels used to stream CA updates

9.330.2.8 readNow()

```
void VariableManager::readNow ( )
```

Perform a single shot read on all variables (Usefull when not subscribing by default)

9.330.2.9 writeNow()

```
void VariableManager::writeNow ( ) [virtual]
```

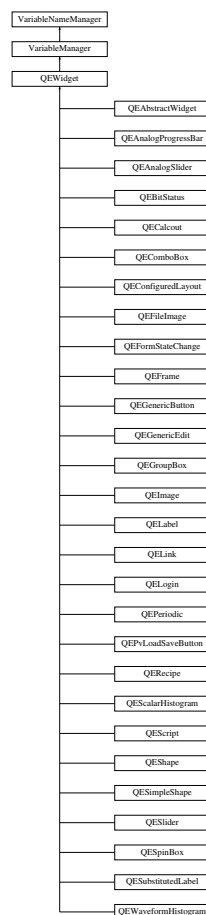
(Control widgets only - such as [QLineEdit](#)) Write the value now. Used when writeOnChange, writeOnEnter, etc. are all false.

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/VariableManager.h
- qeframework/widgets/QEWidget/VariableManager.cpp

9.331 VariableNameManager Class Reference

Inheritance diagram for VariableNameManager:



Public Member Functions

- void [variableNameManagerInitialise](#) (unsigned int numVariables)
- int **getNumberVariables** () const
- QString **getOriginalVariableName** (unsigned int variableIndex) const
- QString **getVariableNameSubstitutions** () const
- QString **getSubstitutedVariableName** (unsigned int variableIndex) const
- void **setVariableNameSubstitutionsOverride** (const QString &substitutions)
- void **setVariableName** (const QString &variableName, unsigned int variableIndex)
- void **setVariableNameSubstitutions** (const QString &substitutions)
- QString [substituteThis](#) (const QString &string) const

9.331.1 Member Function Documentation

9.331.1.1 substituteThis()

```
QString VariableNameManager::substituteThis (
    const QString & string ) const
```

!! for efficiency, should this be done when substitutions are added or removed?? !! Build a list of keys and values...

9.331.1.2 variableNameManagerInitialise()

```
void VariableNameManager::variableNameManagerInitialise (
    unsigned int numVariables )
```

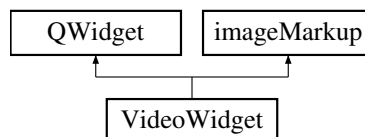
WHY?

The documentation for this class was generated from the following files:

- qeframework/widgets/QEWidget/VariableNameManager.h
- qeframework/widgets/QEWidget/VariableNameManager.cpp

9.332 VideoWidget Class Reference

Inheritance diagram for VideoWidget:



Public Types

- typedef void(* **CustomisePaintHandlers**) ([QEImage](#) *image, QPainter &painter, const QRect rect, QObject *context)

Signals

- void **userSelection** (imageMarkup::markupIds mode, bool complete, bool clearing, QPoint point1, QPoint point2, unsigned int thickness)
- void **zoomInOut** (int zoomAmount)
- void **currentPixelInfo** (QPoint pos)
- void **pan** (QPoint pos)
- void **redraw** ()

Public Member Functions

- **VideoWidget** ([QEImage](#) *parent)
- void **setNewImage** (QImage image, [QCaDateTime](#) &time)
- void **setPanning** (bool panningIn)
- bool **getPanning** ()
- QPoint **scalePoint** (QPoint pnt)
- int **scaleOrdinate** (int ord)
- QPoint **scaleImagePoint** (QPoint pnt)
- QRect **scaleImageRectangle** (QRect r)
- int **scaleImageOrdinate** (int ord)
- QImage **getImage** ()
- QSize **getImageSize** ()
- bool **hasCurrentImage** ()
- void **markupChange** ()
- void **setCustomisePaintHandler** (CustomisePaintHandlers paintExtraHandler, QObject *context=0)
- CustomisePaintHandlers **getCustomisePaintHandler** () const

Protected Member Functions

- void **paintEvent** (QPaintEvent *)
- void **mousePressEvent** (QMouseEvent *event)
- void **mouseReleaseEvent** (QMouseEvent *event)
- void **mouseMoveEvent** (QMouseEvent *event)
- void **wheelEvent** (QWheelEvent *event)
- void **keyPressEvent** (QKeyEvent *event)
- void **markupChange** (QVector< QRect > &changedAreas)
- void **resizeEvent** (QResizeEvent *event)
- void **markupSetCursor** (QCursor cursor)
- void **markupAction** (markupIds mode, bool complete, bool clearing, QPoint point1, QPoint point2, unsigned int thickness)

Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/videowidget.h
- qeframework/widgets/QEImage/videowidget.cpp

9.333 windowCreationListItem Class Reference

Public Member Functions

- **windowCreationListItem** ([windowCreationListItem](#) *item)

Public Attributes

- QString **uiFile**
- QString **macroSubstitutions**
- QString **customisationName**
- QEFormMapper::FormHandles **formHandle**
- [QE::CreationOptions](#) **creationOption**
- bool **hidden**
- QString **title**

The documentation for this class was generated from the following files:

- qeframework/common/QEActionRequests.h
- qeframework/common/QEActionRequests.cpp

9.334 windowCustomisation Class Reference

Public Member Functions

- **windowCustomisation** (const QString nameIn)
- void **addItem** ([windowCustomisationMenuItem](#) *menuItem, QStringList preMenuHierarchy=QStringList())
- void **addItem** ([windowCustomisationButtonItem](#) *button)
- QList< [windowCustomisationMenuItem](#) * > **getMenuItems** () const
- QList< [windowCustomisationButtonItem](#) * > **getButtons** () const
- QString **getName** ()

Static Public Member Functions

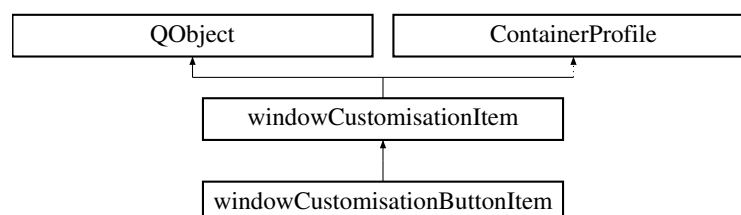
- static [QE::CreationOptions](#) **translateCreationOption** (QString creationOption)

The documentation for this class was generated from the following files:

- qeframework/common/windowCustomisation.h
- qeframework/common/windowCustomisation.cpp

9.335 windowCustomisationButtonItem Class Reference

Inheritance diagram for windowCustomisationButtonItem:



Public Member Functions

- **windowCustomisationButtonItem** (const QString buttonGroupIn, const QString buttonToolBarIn, const Qt::ToolBarArea buttonLocationIn, const QString buttonTextIn, const QString buttonIconIn, const QObject *launchRequestReceiver, const QList< [windowCreationListItem](#) > &windowsIn, const QString programIn, const QStringList argumentsIn, const [QEPvInfo](#) &pvInfo)
- **windowCustomisationButtonItem** (const QString buttonGroupIn, const QString buttonToolBarIn, const Qt::ToolBarArea buttonLocationIn, const QString buttonTextIn, const QString buttonIconIn, const QString builtIn, const QString widgetNameIn)
- **windowCustomisationButtonItem** (const QString buttonGroupIn, const QString buttonToolBarIn, const Qt::ToolBarArea buttonLocationIn, const QString buttonTextIn, const QString buttonIconIn)
- **windowCustomisationButtonItem** ([windowCustomisationButtonItem](#) *buttonItem)
- QString **getButtonGroup** () const
- QString **getButtonToolBar** () const
- Qt::ToolBarArea **getButtonLocation** () const
- QString **getButtonText** () const
- QString **getButtonIcon** () const

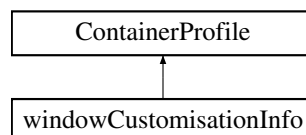
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/windowCustomisation.h
- qeframework/common/windowCustomisation.cpp

9.336 windowCustomisationInfo Class Reference

Inheritance diagram for windowCustomisationInfo:



Public Member Functions

- void **userLevelChangedGeneral** ([QE::UserLevels](#))

Public Attributes

- QMap< QString, QMenu * > **placeholderMenus**
- QMap< QString, QMenu * > **menus**
- QMap< QString, QToolBar * > **toolbars**
- QList< [windowCustomisationItem](#) * > **items**

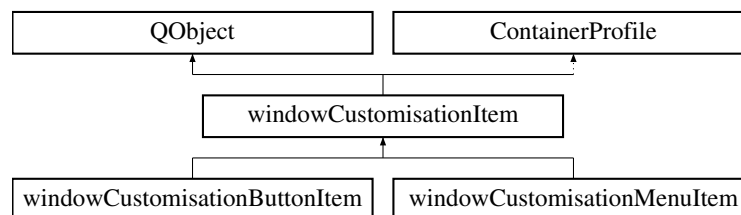
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/windowCustomisation.h
- qeframework/common/windowCustomisation.cpp

9.337 windowCustomisationItem Class Reference

Inheritance diagram for windowCustomisationItem:



Public Slots

- void **itemAction** ()

Signals

- void **newGui** (const [QEActionRequests](#) &request)

Public Member Functions

- **windowCustomisationItem** (const QObject *launchRequestReceiver, const QList< [windowCreationListItem](#) > &windows, const QString program, const QStringList arguments, const [QEPvInfo](#) &pvinfos)
- **windowCustomisationItem** ([windowCustomisationItem](#) *item)
- **windowCustomisationItem** (const QString builtinActionIn)
- **windowCustomisationItem** (const QString builtinActionIn, const QString widgetNameIn)
- **windowCustomisationItem** (const QString dockTitleIn, bool unused)
- void **commonInit** ()
- QString **getProgram** () const
- QStringList **getArguments** () const
- [QEPvInfo](#) **getPvinfos** () const
- QString **getBuiltinAction** () const
- QString **getDockTitle** () const
- QString **getGUITitle** () const
- bool **createsDocks** ()
- void **initialise** ()
- void **logItem** ([customisationLog](#) &log)
- void **addUserLevelAccess** (QDomElement element, [customisationLog](#) &log)
- void **setUserLevelState** ([QE::UserLevels](#) currentUserLevel)
- QAction * **getAction** () const

Protected Attributes

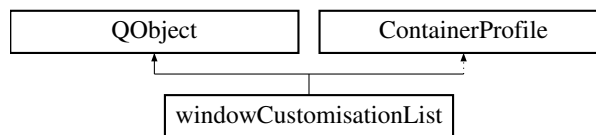
- [ContainerProfile](#) **profile**
- QAction * **iAction**

The documentation for this class was generated from the following files:

- qeframework/common/windowCustomisation.h
- qeframework/common/windowCustomisation.cpp

9.338 windowCustomisationList Class Reference

Inheritance diagram for windowCustomisationList:



Public Types

- typedef QMap< QString, QDockWidget * > **dockMap**

Public Member Functions

- bool **loadCustomisation** (QString xmlFile)
- void **applyCustomisation** (QMainWindow *mw, QString customisationName, [windowCustomisationInfo](#) *customisationInfo, dockMap dockedComponents=dockMap())
- [windowCustomisation](#) * **getCustomisation** (QString name)
- void **initialise** ([windowCustomisationInfo](#) *customisationInfo)

Public Attributes

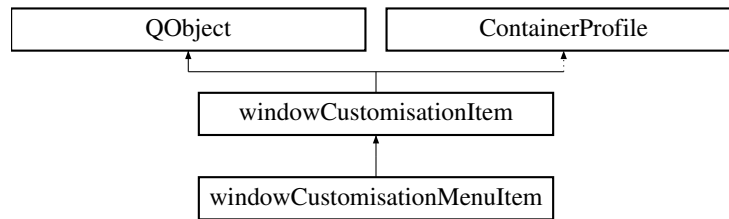
- [customisationLog](#) **log**

The documentation for this class was generated from the following files:

- qeframework/common/windowCustomisation.h
- qeframework/common/windowCustomisation.cpp

9.339 windowCustomisationMenuItem Class Reference

Inheritance diagram for windowCustomisationMenuItem:



Public Types

- enum **menuObjectTypes** { MENU_UNKNOWN , MENU_ITEM , MENU_PLACEHOLDER , MENU_BUILT_IN }

Public Member Functions

- **windowCustomisationMenuItem** ([customisationLog](#) &log, const QStringList menuHierarchyIn, const QString titleIn, const menuObjectTypes type, const bool separatorIn, const [itemCheckInfo](#) &checkInfoIn, const QObject *launchRequestReceiver, const QList< [windowCreationListItem](#) > &windowsIn, const QString programIn, const QStringList argumentsIn, const [QEPvInfo](#) &pvinfos)
- **windowCustomisationMenuItem** ([customisationLog](#) &log, const QStringList menuHierarchyIn, const QString titleIn, const menuObjectTypes typeIn, const bool separatorIn, const [itemCheckInfo](#) &checkInfoIn)
- **windowCustomisationMenuItem** ([customisationLog](#) &log, const QStringList menuHierarchyIn, const QString titleIn, const menuObjectTypes typeIn, const bool separatorIn, const [itemCheckInfo](#) &checkInfoIn, const QString builtIn, const QString widgetNameIn)
- **windowCustomisationMenuItem** ([customisationLog](#) &log, const QStringList menuHierarchyIn, const QString titleIn, const menuObjectTypes typeIn, const bool separatorIn, const [itemCheckInfo](#) &checkInfoIn, const QString dockTitleIn)
- **windowCustomisationMenuItem** ([windowCustomisationMenuItem](#) *menuitem)
- QStringList **getMenuHierarchy** ()
- void **prependMenuHierarchy** (QStringList preMenuHierarchy)
- QString **getTitle** () const
- menuObjectTypes **getType** () const
- bool **hasSeparator** () const
- const [itemCheckInfo](#) & **getCheckInfo** () const

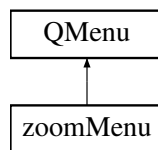
Additional Inherited Members

The documentation for this class was generated from the following files:

- qeframework/common/windowCustomisation.h
- qeframework/common/windowCustomisation.cpp

9.340 zoomMenu Class Reference

Inheritance diagram for zoomMenu:



Public Member Functions

- **zoomMenu** (QWidget *parent=0)
- void **enableAreaSelected** (bool enable)
- imageContextMenu::imageContextMenuOptions **getZoom** (const QPoint &pos)

The documentation for this class was generated from the following files:

- qeframework/widgets/QEImage/zoomMenu.h
- qeframework/widgets/QEImage/zoomMenu.cpp

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